A catalogue of 166 candidate new impact sites on Mars from repeat Context Camera imaging

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Abstract

Small meteoroids still strike Mars, at a rate of a few hundred craters larger than about 4 m each year. Each leaves a crater only metres across but strips or disturbs the bright dust around it over tens of metres, so the resulting dark "blast zone" is visible in orbital images whose pixels are several metres wide even when the crater itself is not resolved. Comparing images of the same ground taken at different times therefore reveals new impacts and brackets their formation dates. We present 166 candidate new impact sites found by comparing public images from the Context Camera (CTX) on Mars Reconnaissance Orbiter, about 5.5 m per pixel, taken between 2006 and 2026. Each candidate is a dark spot, typically 35–70 m across (median 50 m), that is absent from at least one earlier, co-registered image and present in at least one later image, and that we could not find in the published catalogues of dated new impacts or among the impact sites targeted by the camera teams. We rate 80 candidates as strong (seen on at least two orbits or by a second camera, with a well-matched earlier image showing nothing there), 81 as promising and 5 as weak. The estimated probabilities that the spots are new impacts range from 20% to 90% and sum to about 116. The formation windows range from 51 days to 17 years (median 6 years). Six sites were imaged by chance at high resolution by HiRISE after their spots formed; at two of them small pits 2–3 m across lie at the centre of the dark zone, and at a third HiRISE images taken before and after show the darkening appear. As a test, the same method found the dark blast zone left by NASA's InSight lander in 2018 without being told where to look. All 166 sites remain unconfirmed candidates; targeted HiRISE images would test each of them. For every candidate we give its position, formation window, image identifiers, before-and-after images and an account of the evidence.

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1 Introduction

Mars is still being cratered. Repeated imaging from orbit has shown that new craters form every year, from less than a metre to a few tens of metres across. The first of them were found by Malin et al. (2006), who compared images of the same areas taken years apart by the Mars Orbiter Camera (MOC) on Mars Global Surveyor and found 20 new impact sites. The search continued with the Context Camera (CTX) on Mars Reconnaissance Orbiter (Malin et al., 2007; Zurek and Smrekar, 2007), and candidate sites found in CTX images have routinely been imaged by the High Resolution Imaging Science Experiment (HiRISE) on the same spacecraft (McEwen et al., 2007). Daubar et al. (2013) used 248 dated new impact sites to measure the present-day cratering rate, 1.65×10^{-6} craters with effective diameter of at least 3.9 m per square kilometre per year, or roughly 240 such events per year over the whole planet. The current catalogue of Daubar et al. (2022) lists 1,203 new impacts dated by repeat imaging, with crater diameters from less than 1 m to 58 m; more than half of them are clusters of craters made by meteoroids that broke up in the atmosphere (Daubar et al., 2019, 2022). A systematic search around NASA's InSight lander found 123 impacts that formed between December 2018 and December 2022 and suggested a rate 1.6 to 2.5 times higher than earlier estimates (Bickel et al., 2025).

The craters themselves are mostly too small to be seen in CTX images, whose pixels are about 5–6 m across. What makes new impacts detectable is the disturbance around them. On dusty ground the blast of the impact removes or redistributes the bright surface dust over an area many times wider than the crater, producing a dark halo, often with rays or arcs (Daubar et al., 2022, 2016; Malin et al., 2006). A crater 2–3 m across can thus leave a dark spot 40–50 m across, several CTX pixels wide (candidates 21 and 57 below are examples). Such spots fade over years to decades as dust settles back (Daubar et al., 2016). A dark spot on its own proves little, since dark spots of many origins are common on Mars. What identifies a new impact is a change: an earlier image of the same ground under comparable conditions shows nothing, and a later one shows the spot. The pair of images also dates the event, from the last image without the spot to the first image with it.

New impacts matter for several reasons. They measure the present impact flux and its size distribution, which underpin crater-based ages of Martian surfaces (Daubar et al., 2013; Hartmann, 2005). Their clusters record how meteoroids break up in the thin atmosphere (Daubar et al., 2019; Hartmann et al., 2018). Impacts at middle latitudes have exposed shallow ground ice (Byrne et al., 2009; Dundas et al., 2014), their airblasts can trigger dust avalanches on nearby slopes (Burleigh et al., 2012), and impacts that occurred while InSight was operating have been matched with seismic signals, constraining the structure of the Martian crust (Bickel et al., 2025; Garcia et al., 2022; Posiolova et al., 2022). Each new site also adds a fresh crater whose morphology can be studied (Daubar et al., 2014).

Finding new impacts is laborious. The CTX team examines new images for fresh dark spots and the HiRISE team images many of the resulting candidates (Daubar et al., 2022, 2013); machine-learning methods have been used to reduce the biases of manual searches (Bickel et al., 2025; Wagstaff et al., 2022); and compilations bring the results together (Podobnaya et al., 2026). Because CTX has now imaged much of Mars more than once, and many places many times, the archive certainly contains new impact sites that have not yet been recognised.

This paper presents 166 such sites, found by a systematic comparison of public CTX images of the same ground taken in different years. None of them appears in the catalogues we could search or among the impact sites targeted by the camera teams. All are *candidates*: only six have been imaged by HiRISE, all by chance, and only two of those show small pits at the centre of the dark zone. We describe the data (Section 2) and the method, including the ways in which look-alike changes were rejected and the independent re-examination of 38 sites (Section 3). Section 4 summarises the candidates in tables and figures, and Section 5 gives a catalogue entry for each, with its before-and-after images, a plain account of what changed and why it may be an impact, and its weaknesses. Section 6 discusses the limitations and how the candidates could be confirmed.

2 Data

2.1 Context Camera images

CTX is a push-broom camera with a single broad visible band centred near 0.65 µm, on Mars Reconnaissance Orbiter, which has been in a near-polar, near-circular orbit at about 250–320 km altitude since 2006 (Zurek and Smrekar, 2007). Its 5,000-pixel line array images a swath about 30 km wide at about 6 m per pixel (Malin et al., 2007). Images are identified by product names such as `W09_092578_1739_XN_06S129W`, in which the six-digit field is the orbit number and the last field gives the approximate latitude and west longitude of the image centre; in tables we shorten them to their first fifteen characters. The images used for the candidates were taken between 3 October 2006 and 26 May 2026 at pixel scales of 5.1 to 12.2 m (median 5.5 m), with the Sun 30° to 78° from the zenith and emission angles up to 31°. The figures and image tables of the 166 candidates alone draw on 791 distinct CTX images.

The search worked with the raw Experiment Data Records (EDRs; Malin, 2007) distributed by the Imaging Node of NASA's Planetary Data System (PDS). CTX data are companded to 8 bits with a square-root lookup table, so a contrast of 1% in raw data numbers corresponds to roughly 2% in radiance. Rather than downloading whole images, the lines around each site were read from the archive with HTTP range requests. Image dates, illumination and viewing angles, pixel scales, footprints and the stated purpose of each image (its targeting rationale, for example "Candidate impact site formed between . . . ") were taken from the cumulative CTX index, through images released by 31 May 2026.

For the independent re-examination (Section 3.8) and for the four before-and-after figures that we made for this paper (Section 3.9), map-projected versions of CTX images distributed by Arizona State University were used. These are produced with the USGS Integrated Software for Imagers and Spectrometers (ISIS) from the same EDRs, calibrated and projected onto a planetocentric simple-cylindrical grid at the native pixel scale.

2.2 Other images and data

HiRISE. HiRISE images a swath about 6 km wide at 0.25–0.3 m per pixel, or 0.5 m per pixel when binned (McEwen et al., 2007). The footprints and targeting rationales of all 102,819 HiRISE red-filter products in the PDS cumulative index of reduced data records through 31 August 2026 (McEwen, 2007) were used to test whether any HiRISE image covers each candidate or was aimed at it. Six candidates (4, 21, 57, 69, 85 and 94) were imaged by HiRISE after their spots formed, all on images targeted at something else; two more (100 and 106) were imaged only before. The HiRISE titles listed on the HiRISE team's website were also searched (Section 3.8).

MOC. Narrow-angle images from the Mars Orbiter Camera, at 1.5–12 m per pixel from 1997 to 2006 (Malin and Edgett, 2001), were sought for each site through the PDS Orbital Data Explorer. Only four candidates, 42, 55, 97 and 112, are known to have such images; for all but candidate 97 they predate CTX images that already show the ground without the spot, so they add nothing.

Catalogues of new impacts. The candidates were compared with the catalogue of 1,203 dated new impacts of Daubar et al. (2022) (data in Daubar, 2022), which includes the 248 sites of Daubar et al. (2013); for the 38 re-examined sites also with the compilation of Podobnaya et al. (2026), version 1.5 with 1,335 entries, and with the 69 confirmed sites of the machine-learning search of Wagstaff et al. (2022). The catalogue of Bickel et al. (2025), 123 impacts within 50° of InSight that formed between December 2018 and December 2022, could not be retrieved from its data portal; Section 3.5 explains what can be inferred for the candidates in its area. Impacts known from InSight's seismic and orbital co-investigation, in particular the two largest of 2021 (Posiolova et al., 2022), were also considered.

Topography. The map in Figure 4 uses topography from the Mars Orbiter Laser Altimeter (MOLA; Smith et al., 2001), gridded at 16 pixels per degree.

Table 1 summarises the data.

Table 1: Data used in this study.

Data	Pixel scale	Use
CTX EDRs, PDS Imaging Node	5.1–12.2 m	Change detection, measurements, published figures; 791 images of the candidate sites, October 2006 to May 2026
CTX cumulative index (to 2026-05-31)	—	Dates, illumination and viewing angles, footprints, targeting rationales
CTX map-projected images (Arizona State University)	native	Independent re-examination; four figures made for this paper
HiRISE RDR index and images (to 2026-08-31)	0.25–0.5 m	Coverage tests; views of six candidates
MOC narrow-angle images (1997–2006)	1.5–12 m	Earlier coverage (four candidates)
Catalogues of new impacts	—	Daubar et al. (2022); Podobnaya et al. (2026); Wagstaff et al. (2022); Posiolova et al. (2022); Bickel et al. (2025) (not retrievable)
MOLA gridded topography	16 pixels per degree	Base map

3 Methods

3.1 Overview

The method has five stages (Figure 1): compare pairs of CTX images of the same ground taken at different times and rank the places that became darker; re-measure the most promising places in every CTX image that covers them; inspect the best of them by eye and reject the many changes that are not new impacts; check the survivors against the archives and catalogues; and rate each remaining candidate and estimate the probability that it is a new impact. The method was validated on known events, and 38 of the candidates were re-examined independently from the original archive data. We describe each stage below as far as the records of the search allow; where those records give only a summary, we give only a summary.

3.2 Image pairs and change detection

The search compared 8,587 pairs, each consisting of an earlier and a later CTX image of the same ground. In each pair the two images were co-registered and compared pixel by pixel, and the places where the later image had become darker than the earlier one were ranked by the strength of the darkening; the strongest such changes in each pair were passed on, so that every detection is identified by its later image and its rank within the pair. The highest-ranked detections, 9,221 in all, were then re-measured in every CTX image that covers them (Section 3.3). This re-measurement is what turns a single image pair into a history of the site: it shows whether a spot was already present in older images, when it first appears, and whether it persists.

3.3 Co-registration and measurements

Common grid. For each site, a window from every covering CTX image was cut straight from the raw EDR and resampled onto the same north-up grid of 4 m per pixel centred on the site. Because no map projection is involved, no artefact of projection can be introduced.

Registration. Each image was registered to a reference image, normally the one in which the spot was found, in two steps: phase correlation over a window 6 km on a side, then a local refinement that uses only the surroundings of the site. The site itself is masked during the refinement, so that a real change cannot bias the alignment. The quality of the result is the correlation coefficient r between the surroundings in each image and in the reference, which we call the alignment score; values of about 0.3 or more mean that the same landforms line up.

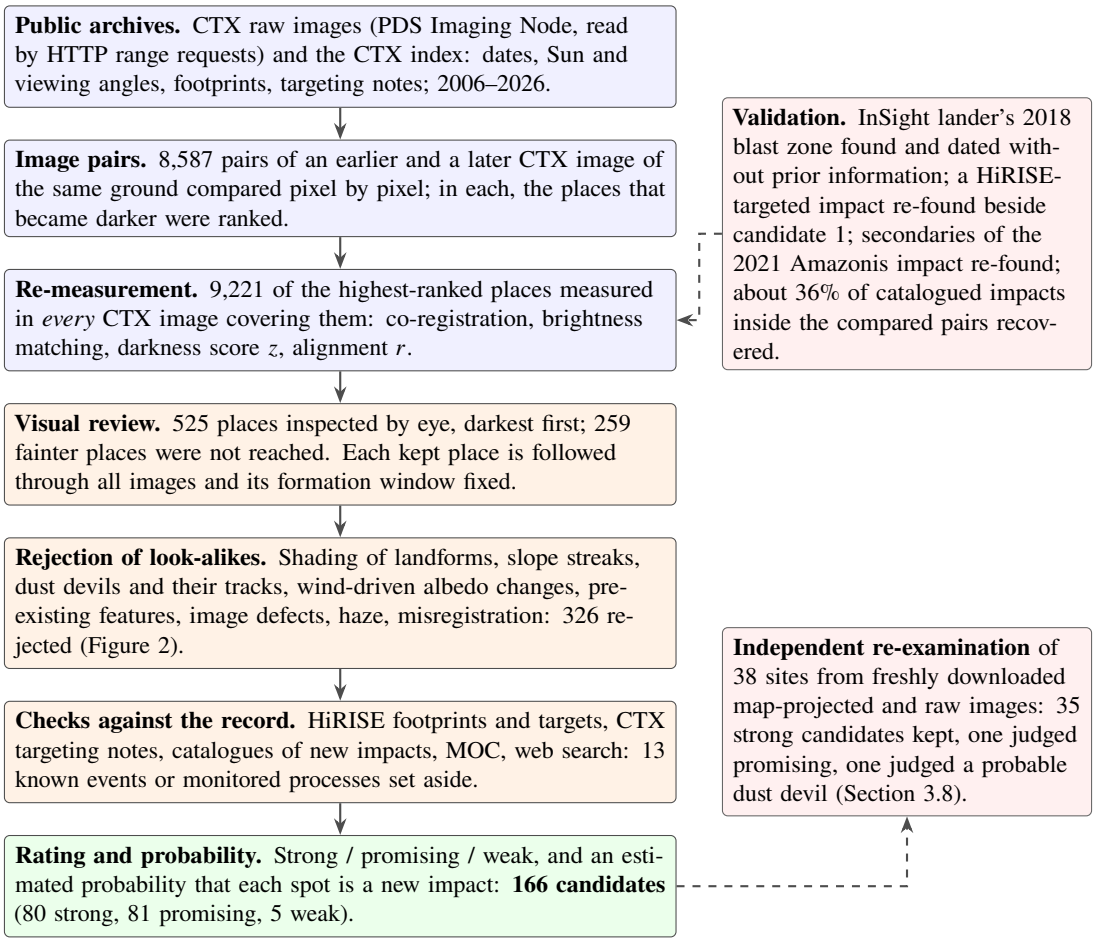

Figure 1: The method, from the public archives to the catalogue. Numbers are those of the search reported here. The validation tests (Section 3.7) and the independent re-examination of 38 sites (Section 3.8) are shown at the side.

Brightness matching. The brightness and contrast of each image were matched linearly to the reference using only pixels more than three spot radii from the centre, and all images of a site are displayed with one common grey stretch, so that a display choice cannot create or remove a spot.

Darkness score. The darkness of a spot in an image is measured by the score

$$z = \frac{\bar{I}_{\mathrm{spot}} - \mathrm{median}_k\, \bar{I}_k}{\sigma_{\mathrm{ring}}}, \tag{1}$$

where $\bar{I}_{\mathrm{spot}}$ is the mean brightness (in raw data numbers) of a disc at the spot, $\bar{I}_k$ are the means of equal discs placed in a ring 2 to 5 radii from the centre, and σ_{ring} is the robust spread of those disc means. Values of $|z|$ below about 1 are ordinary texture; a spot counts as present in an image when $z \leq -2$. Darkness scores are quoted from the image-by-image measurements; the scores printed on the published time-series figures come from those figures' own alignment and can differ slightly (by at most 0.2 for 90% of the images without the spot), but no image changes between showing and not showing the spot.

Size, contrast and texture. Sizes are measured at half the depth of the darkening, on a 2 m grid, and quoted as the extent of the dark core and, where present, of a fainter halo. As a test for a pre-existing landform at the site, the local high-pass energy at the spot was compared with its surroundings in the earlier images: for genuine absences this ratio is close to 1 (for example 0.2–2.1 for candidate 1), while the pre-existing small craters rejected on this ground scored 8 to 22.

Formation window. A spot is dated from the last informative image without it to the first image with it. An earlier image is informative only if it is well aligned and could have shown the spot. Where possible this was tested with a control: a pre-existing dark feature of comparable size nearby, typically 100–300 m away. If the

control is clearly dark in an earlier image while the candidate's site is not, that image would have shown the spot. Such a control was available for 100 of the 166 candidates.

Registered difference. For each site, the brightness-matched earlier image was also subtracted from the later one, at two smoothing scales, to see whether the spot is the strongest change in the surrounding 2×2 km or whether comparable changes occur elsewhere, as happens when the illumination differs.

Illumination and viewing geometry. Shading changes with the height and direction of the Sun, so an absence counts most when an earlier image was taken under a similar Sun; a spot absent under both a higher and a lower Sun than the image showing it cannot be the shading of a landform. In the captions of the before-and-after figures, two images are said to have a similar Sun when their solar incidence angles differ by at most 5° and their solar azimuths by at most 20°. Solar azimuths are quoted as the CTX labels give them (`SUB_SOLAR_AZIMUTH`): measured clockwise from the line joining the image centre to the middle of its right-hand edge, not from north. North lies at about 277° in that frame, so the compass bearing of the Sun is about 83° more than the quoted value; comparisons between images are not affected.

Dust storms and season. Global dust storms in 2007 and 2018 veiled the surface for months (Guzewich et al., 2019; Smith, 2009); a spot hidden by haze in such an image could appear "new" afterwards. The search encountered exactly this false positive: a "new" spot in Elysium was an old dark patch that the 2007 storm had hidden, and a MOC image of 2003 shows it. No absence of any candidate rests only on an image taken during the global dust storms of June to October 2007 or June to September 2018. Seasonal frost brightens rather than darkens the ground and is not a plausible source of new dark spots at the latitudes of the candidates.

3.4 Rejecting look-alikes

Almost every change found by comparing two images is something other than a new impact. During the visual review each detection was followed through all its images and was rejected if a mundane explanation fits. The recorded reasons for the 326 rejected changes fall into the broad groups of Figure 2:

- *Shading of landforms.* A small crater, pit, knob, ridge or scarp casts a different shadow when the Sun is lower or comes from another direction. Such changes follow the Sun: they are seen in all images with a similar Sun, before and after, and the landform is visible in every image.
- *Slope streaks.* Dark streaks formed by dust avalanches on steep slopes are common in dusty regions such as Arabia Terra and Tharsis (Schorghofer et al., 2007; Sullivan et al., 2001). They are elongated downslope, often fan- or finger-shaped and hundreds of metres long, and appear beside older faded streaks; they were rejected unless the spot was compact and on level ground. An impact airblast can itself trigger such streaks (Burleigh et al., 2012).
- *Dust devils.* Dust devils leave long, narrow, often curving tracks (Balme and Greeley, 2006), and an active dust devil caught in an image casts a dark shadow on the side away from the Sun beside a bright dust column. Changes with tracks, or with a bright patch on the sunward side, were rejected.
- *Wind-driven albedo changes.* Wind can remove or deposit dust over irregular, diffuse patches, change wind streaks behind craters, or gather dark sand between ripples. Such changes are usually diffuse, irregular or aligned with ripples and lack a compact core.
- *Pre-existing features.* A dark feature that is visible in an older image, perhaps only faintly or under a different Sun, is not new.
- *Image defects, noise and haze.* CTX images can have line dropouts, column striping, bit errors and compression blocks, and low-Sun or hazy images can be noisy or washed out. Defects are straight lines, single pixels or square blocks rather than smooth spots with graded edges spanning many lines and samples; a spot seen in images from two orbits cannot be a defect of one image.
- *Misregistration.* Poorly aligned images can make a dark feature appear to move. The alignment score, the masking of the site during registration and the comparison of control features guard against this.

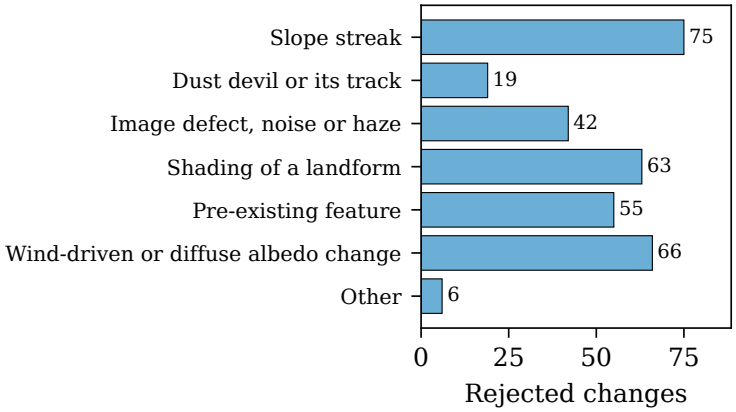

Figure 2: Reasons for rejecting 326 of the changes inspected by eye, grouped by keyword from the recorded reasons (approximate). A further 13 changes were set aside as known events or monitored processes (Section 3.7).

In addition, 13 changes were set aside because they belong to known events or monitored processes (Section 3.7). The groups in Figure 2 were assigned by keyword from the recorded reasons and are approximate.

3.5 Checks against the record

A candidate was kept only if it is absent from every public record that could be searched:

- the footprints of all HiRISE red-filter products in the PDS index through August 2026 were tested for containing the spot, and the targeting rationales of nearby HiRISE images were read;
- the targeting rationales of all CTX images covering the spot were read, to exclude sites that the camera teams had already targeted as candidate impacts;
- the distance to the nearest new impact in the catalogue of Daubar et al. (2022) was computed (for the candidates the median is 85 km and the smallest 9.5 km, for candidate 36);
- MOC coverage was sought, and read-only web searches were made for each site.

The catalogue of Bickel et al. (2025) could not be retrieved. Its area, within 50° of InSight, contains 31 of the candidates (2, 5, 9, 11, 16, 30, 34, 40, 48, 52, 59, 74, 83, 86, 89, 94, 99, 102, 106, 108, 115, 117, 120, 123, 128, 134, 139, 142, 157, 159 and 164). Seven of these (30, 40, 59, 102, 106, 134 and 142) formed after its period. Most impacts in that catalogue have been imaged by HiRISE, and none of the others except candidate 94 has a HiRISE image, so the others are probably not in it; candidate 94, which HiRISE imaged twice in 2021, may well be.

3.6 Rating and probability

Each candidate was rated by the strength of its evidence:

- **Strong** requires all of the following: the spot is in at least two images from different orbits, or in a second instrument; at least one well-aligned earlier image under comparable lighting shows nothing there; the spot is compact, or a halo with a distinct dark core; and no pre-existing landform could explain it.
- **Promising**: a real change is likely, but one link is weaker, most often that only one image shows the spot, or that the earlier images are weak or poorly aligned, the contrast is low, or the shape is one that the wind could equally make.
- **Weak**: an artefact or a pre-existing feature explains the data about as well as an impact.

Each candidate was also given a probability that the spot is the blast zone of a new impact (not that it is unknown to the camera teams). These probabilities are judgements that weigh the evidence of each case: the

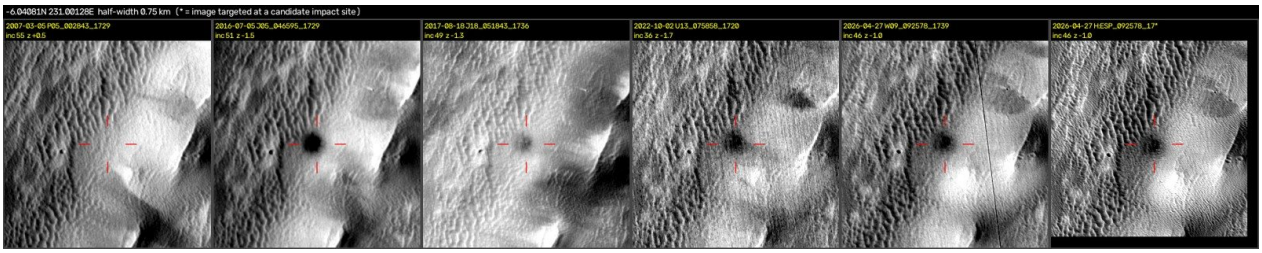

Figure 3: Validation beside candidate 1: the same comparison applied to a candidate recent impact site 16 km to the south, targeted by HiRISE, at 6.0408° S, 231.0013° E. Each panel is 1.5 km across, north up, with the image date, product and darkness score z above it; red ticks mark the site. Its dark spot is absent in 2007 and present from 2016 on. The last panel is a reduced-resolution view of HiRISE image `ESP_092578_1740` (27 April 2026), which was targeted at this site as a candidate recent impact. CTX: NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

number and quality of the images, the strength of the earlier absences, the shape and setting of the spot, and the plausibility of the alternatives listed in its entry. They are not calibrated statistics. They lie between about 70% and 90% for strong candidates, 40% and 75% for promising ones and 20% and 40% for weak ones. The probabilities used throughout this paper are those published with the candidates (Section 7).

3.7 Validation

The InSight lander. NASA's InSight lander touched down in Elysium Planitia on 26 November 2018, and its retrorockets left a dark blast zone about 20 m in radius around it (Golombek et al., 2020). Without being told where to look, the search found a new dark spot at 4.5007° N, 135.6195° E, 254 m from the lander's position measured in HiRISE images (4.5024° N, 135.6234° E), an offset within the errors of CTX geolocation. The spot is absent in five images from July 2014 to December 2016, faint on 16 December 2018 ($z = -1.7$), clear on 4 May 2019 ($z = -3.9$) and strong from September 2019 ($z = -9.7$), fading afterwards. The method thus detected and correctly dated a known disturbance of the right size.

Known impacts. Several dark marks found by the search lie in the field of secondary craters of the 150 m impact crater of 24 December 2021 in Amazonis Planitia, detected by InSight as event S1094b (Posiolova et al., 2022), and one lies 23 km from the crater cluster of 18 September 2021 in Tempe Terra (event S1000a); these were set aside as parts of known events. In the image pair in which candidate 1 was found, the second-ranked detection is a site 16 km to the south that HiRISE imaged in April 2026 as a candidate recent impact (Figure 3). Of the catalogued new impacts that lie inside the compared image pairs, the search recovered about 36%. Missed impacts are expected: many are faint, small or in low-contrast images, and the visual review did not reach the faintest detections.

Monitored processes. The search also found a new slope streak at a HiRISE slope-streak monitoring site and changing wind streaks at a CTX wind-streak monitoring site, both set aside as known processes.

3.8 Independent re-examination

To test the strongest candidates, 38 sites were re-examined independently: the 36 rated strong among candidates 19 to 85 at the time of the check, together with candidate 43 (because of its proximity to candidate 42) and candidate 92 (to test its 51-day window). The re-examination was deliberately sceptical and did not rely on the original figures, which were used only to learn which places and images to check. It proceeded as follows.

1. *Inventory.* Every CTX and HiRISE image containing each spot was found from the footprints of the PDS Orbital Data Explorer and compared with the original list; a few informative earlier images that had not been cited were added.
2. *Fresh data and co-registration.* The map-projected CTX images were downloaded again, resampled onto a common north-up grid of 2.5 m per pixel centred on the spot and registered to one reference image by matching ground texture, normalised for local contrast, over about 3 km.

3. *Detectability test.* For each earlier image B and each later image A showing the spot, both were blurred to the coarser of their two resolutions and B was fitted to A as $B = aA + b$ over stable ground within 600 m, excluding the spot. The fit predicts how dark the spot would look in B had it already been there; haze, a lower Sun or blur reduce a and increase the scatter, so the prediction is weakened accordingly. An earlier image counts as evidence of absence only if the predicted darkness is at least 4 noise units and B shows nothing.
4. *Raw data.* For the key images the raw EDR lines around the spot were downloaded and the contrast of the spot measured in raw data numbers, so that no artefact of map projection or processing could be involved; the windows were also checked for zero or saturated pixels and for compression blocks. For the first 24 sites examined, every earlier frame is within ±0.6% of its surroundings at the spot and every later frame 1.0–6.3% darker.
5. *HiRISE.* Where HiRISE images exist they were registered to the CTX image of the same orbit, and the darkest compact feature at the spot was tested for the shading of a pit: along the line towards the Sun, a bowl is dark on its sunward wall and bright on its far wall, while a boulder or knob shows the reverse.
6. *Alternatives and records.* Shadows, frost, haze, dust devils, slope streaks, wind-made dark patches, compression artefacts, noise, cosmic rays and misregistration were weighed for each site. The catalogues of Daubar et al. (2022), Podobnaya et al. (2026) and Wagstaff et al. (2022), the 4,504 observations returned by the HiRISE website for searches on new, recent, fresh or candidate impact sites through October 2026, the CTX targeting rationales and the captioned image releases of the CTX team were all searched.
7. *Positions.* After co-registration the spot sits at the same ground position in every image, and the positions given by the images' own map projections agree to within 55–350 m, the normal geolocation error of CTX products. This made it possible to separate errors in the reported coordinates from errors of the images.

Each site received one of four verdicts: kept (it passes all tests), downgraded (the change is real but one link is weaker than a strong rating requires), rejected (the change is not real or another process explains it) or already known. The outcome is summarised in Section 4.5 and Table 3, and in the entries of the 38 sites in Section 5.

3.9 Figures made for this paper

Every candidate except 1, 2, 3 and 62 had a published close-up of one earlier and one later image. For these four we made new figures from the map-projected images of Arizona State University, using the image pairs or triples named in their captions. Each image was resampled onto a north-up grid of 4 m per pixel centred on the catalogued position, registered to the last image by phase correlation of band-pass-filtered ground texture over 4 km and then 2 km with the site masked, and matched in brightness and contrast to it outside three spot radii. To make sure that each figure shows exactly the ground of the published time series, the reference panel of that series was located in our grid by normalised cross-correlation (correlation 0.42–0.57); for candidates 1 and 3 the position agrees to within 25 m with that of the strongest new darkening in the registered difference. In the map-projected frame the spots lie 134 m, 551 m, 298 m and 1.47 km from the catalogued positions of candidates 1, 2, 3 and 62. For candidate 62 the three images agree with one another to within about 100 m, so its catalogued coordinate is probably in error (Section 5).

4 Results

4.1 Overview

The search yielded 166 candidate new impact sites: 80 rated strong, 81 promising and 5 weak (Table 2). Of the 505 changes recorded after visual review, the other 339 were rejected as look-alikes (326) or set aside as known events and processes (13). The mean estimated probability of a new impact is 83% for the strong candidates, 59% for the promising ones and 31% for the weak ones; 25 candidates are rated at 90% (37, 40, 51, 53, 56, 57, 58, 59, 63, 68, 69, 71, 72, 85, 124, 125, 133, 136, 137, 140, 141, 147, 155, 158 and 162). The probabilities sum to 116 (Figure 5): if the judgements are sound and independent, about 116 of the 166 sites

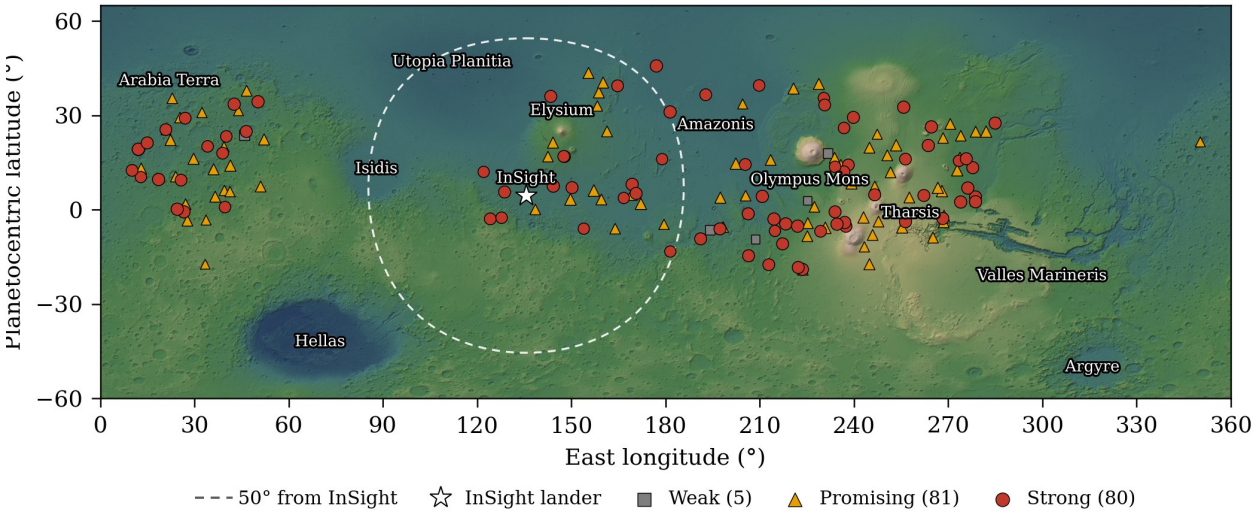

Figure 4: Locations of the 166 candidates on shaded MOLA topography (Smith et al., 2001), by rating; symbol size increases with the estimated probability. The dashed curve encloses the area within 50° of the InSight lander (star) searched by Bickel et al. (2025).

are new impacts, give or take about 6. Ninety-seven of the spots are seen in at least two images (from different orbits, or by HiRISE as well as CTX); 69 rest on a single CTX image.

4.2 Where the candidates are

The candidates lie between 19.1° S and 45.7° N, 121 of them north of the equator (Figure 4). They cluster on the dust-mantled volcanic provinces and plains: 70 in and around Tharsis (220–290° E), 42 in Arabia Terra and western Terra Sabaea, 28 in Elysium Planitia and the Cerberus and Aeolis regions (120–175° E) and 26 in Amazonis Planitia, the Medusae Fossae Formation and Memnonia (175–220° E). None lies in the southern highlands south of 19° S or at high northern latitudes. This distribution resembles that of the catalogued new impacts, which are found mostly on dusty, high-albedo ground where blast zones show well (Daubar et al., 2022, 2013); in regions of low dust cover (Ruff and Christensen, 2002) a new impact may leave no detectable dark spot. It also reflects where CTX has imaged the same ground repeatedly. The median MOLA elevation of the candidates is 0.4 km, with a range of −4.1 to +7.4 km.

4.3 Probabilities, formation windows and sizes

The formation windows, from the last image without the spot to the first image with it, have a median of 6.1 years (interquartile range 2.0–11.1 years; Figure 7). Eighteen are shorter than one year and 42 shorter than two years; the shortest are 51 days (candidate 92), 56 days (18), 84 days (158), 88 days (33) and 110 days (40). At the other extreme, 52 windows exceed ten years, mostly because only one old image precedes the first image with the spot; the longest is 17.1 years (candidate 46). The first images with the spots were taken between January 2012 and May 2026, 129 of them since the start of 2023; the last images without them date from February 2007 to June 2025. Figure 8 shows every window.

The dark spots have a median size of 50 m (interquartile range 35–70 m; Figure 6); 96 are 50 m or smaller and 23 larger than 100 m. The largest, candidate 123 on the yardangs of Zephyria Planum, is a rayed zone about 1.3 km across with a dark core about 500 m wide. The craters inside such blast zones are usually only a few metres across, as at candidates 21 and 57, below the resolution of CTX. Several candidates are clusters of spots, as expected for meteoroids that broke up in the atmosphere (Daubar et al., 2019): for example candidates 23, 32, 59, 118, 127, 133, 139 and 145.

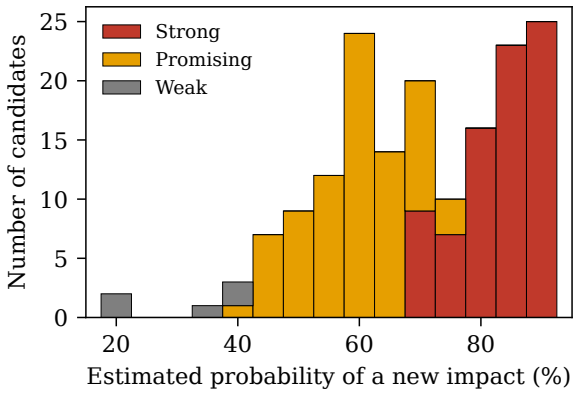

Figure 5: Estimated probabilities that the spots are new impacts, by rating. They sum to 116.

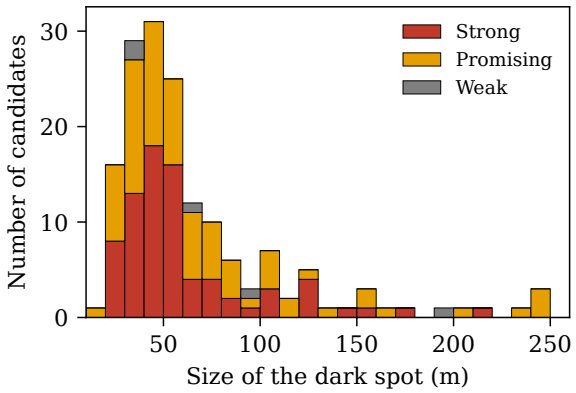

Figure 6: Sizes of the dark spots, by rating. Candidate 123, a rayed zone about 1.3 km across, lies off the scale.

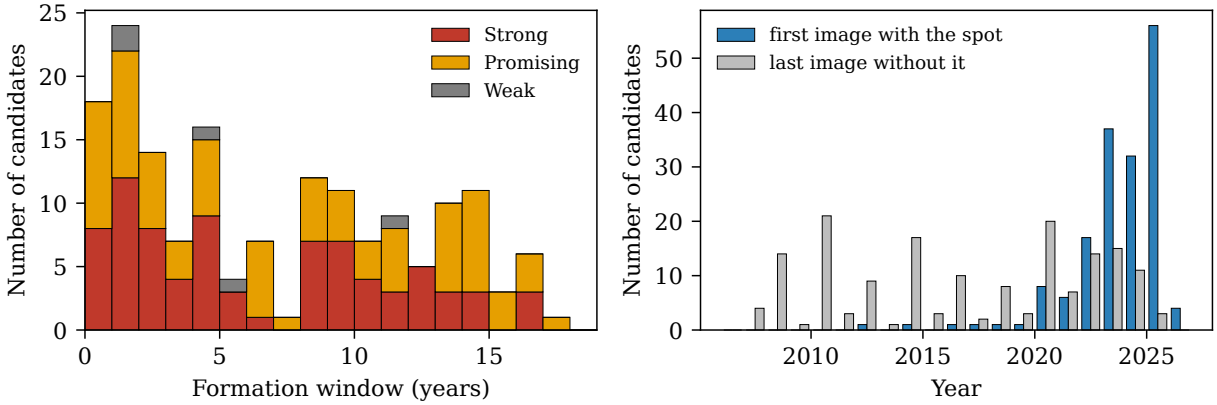

Figure 7: Left: lengths of the formation windows, by rating. Right: years of the last CTX image without the spot (grey) and of the first image with it (blue).

4.4 Candidates seen by HiRISE

Six candidates happen to lie inside HiRISE images taken after their spots formed, in each case on an image targeted at something else and taken together with one of the candidate's CTX images.

- Candidate 21, west of Cyane Fossae: HiRISE shows dark rays spreading from at least three small pits 2–3 m across, shaded like bowls (Figures 54 and 56).
- Candidate 57, western Arabia Terra: a pit about 2.5 m wide lies at the centre of a 40×46 m dark patch, dark on its sunward wall and bright on its far wall (Figures 144 and 146). Candidate 72, 5.2 km away, formed between the same two CTX images.
- Candidate 69, among the Erebus Montes: HiRISE images taken before (April 2023) and after (March 2025) the spot formed show a darkening about 80 m across appear, with a new compact dark feature about 2.5 m across inside it, although that feature shows no bowl-like shading (Figures 177 and 179).
- Candidate 4, north-west of Pasteur crater: a sharp-edged dark patch with finger-like margins and a dark point 1–2 m across at its edge (Figure 17).
- Candidate 85, on the Medusae Fossae Formation: a dark zone about 80 m across; the 2 m dark dot inside it is not distinguishable from similar dots nearby (Figure 221).
- Candidate 94, between Aeolis Dorsa and Zephyria Planum: a dark zone about 90×107 m but no definite crater, in two images of 2021 (Figure 242).

Candidates 21 and 57 are therefore the closest to confirmation. Because HiRISE imaged these six sites, its

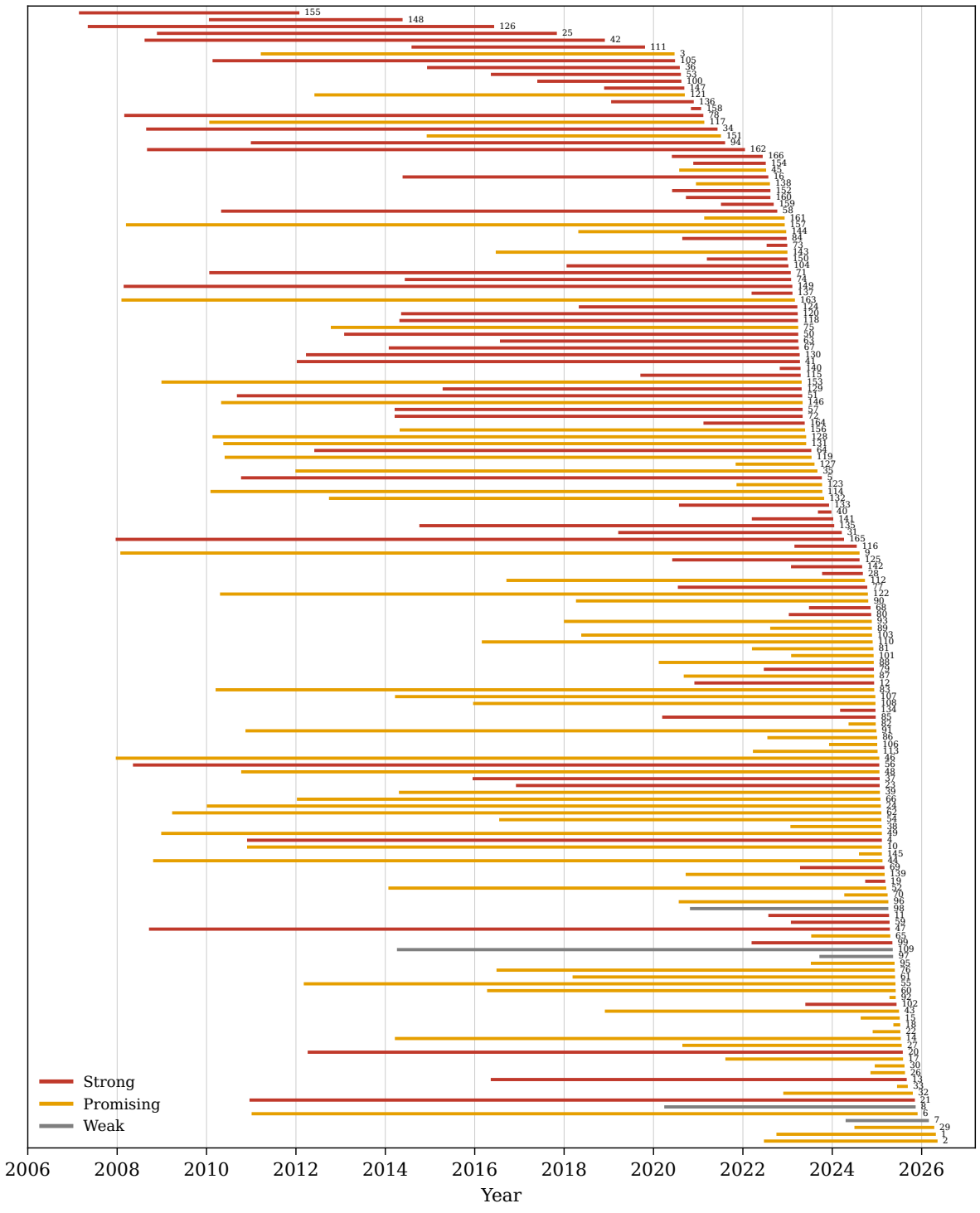

Figure 8: The formation window of every candidate, from the last CTX image without the spot to the first image with it, sorted by the date of the first image with the spot; numbers are those of the candidates.

team may already have noticed them, although none is listed in the catalogues we could search.

4.5 Outcome of the independent re-examination

Table 3 summarises the re-examination of 38 sites. No site was found to be already documented, and at every one of them the ground really changed between the dates given. Of the 36 sites then rated strong, 35 were kept as strong candidates, with the re-examination's own probabilities between 65% and 95%. One, candidate 73, was judged to be promising rather than strong (about 60%): its change is real and well dated but diffuse, without a compact core, and it formed in the dusty season, so dust stripped by wind explains it nearly as well. The re-examination agreed with the promising rating of candidate 43 (about 60%). It judged candidate 92 most probably a dust devil and its track rather than an impact (about 5%), because the same image shows a new bright patch on the sunward side of the spot and a new dark band extending from it. In this paper we use the probabilities published with the candidates, 75% for candidate 73 and 65% for candidate 92, but we regard candidate 92 as doubtful. Excluding candidate 92, the re-examination's probabilities differ from ours by −0.9 percentage points on average (mean absolute difference 4.7 points).

The re-examination also found errors in the original descriptions, none of which removes a site. It narrowed or revised the formation windows of candidates 20, 36, 43, 50 and 67, as described in their entries. Its measured positions of the spots differ from the catalogued positions by 45 m to 1.5 km (median 229 m); nine differ by more than 400 m, and for candidate 23 the catalogued position is 1.5 km west of the cluster in all eleven images, each in its own projection, so the catalogued coordinate is in error. Similar errors may affect candidates that were not re-examined; for candidate 62 our own figure suggests one of about 1.5 km (Section 3.9). These offsets are small compared with the 6 km width of a HiRISE image, but they matter when a follow-up image is planned, and the re-measured positions in Table 3 should be preferred where they exist.

Table 2: All 166 candidates. Positions are planetocentric latitude and east longitude. P is the estimated probability that the spot is a new impact. "Last without" and "first with" are the dates of the last CTX image that shows the ground without the spot and of the first image that shows it, which bracket its formation. N_-/N_+ counts the images without and with the spot (the latter from different orbits, including HiRISE where it exists). The two image columns give the before-and-after pair shown in the catalogue (shortened product identifiers). Sizes are those of the dark spot, in metres; the catalogue gives fuller descriptions. Candidate numbers link to the catalogue entries.

No.	Lat.	Lon.	Rating	P (%)	Last without	First with	N_-/N_+	Before image	Date	After image	Date	Size (m)
1	5.7715° S	230.9347° E	Promising	75	2022-10-02	2026-04-27	4/1	U13_075858_1720	2022-10-02	W09_092578_1739	2026-04-27	70 core
2	24.8860° N	161.4195° E	Promising	60	2022-06-22	2026-05-11	5/1	J09_048417_2047	2016-11-24	W10_092765_2031	2026-05-11	70
3	0.9058° N	227.4609° E	Promising	55	2011-03-20	2020-06-21	1/4	G09_021778_1830	2011-03-20	W08_092209_1803	2026-03-29	40
4	25.4522° N	20.9077° E	Strong	80	2010-11-28	2025-02-10	1/4	G05_020335_2066	2010-11-28	W05_090832_2057	2025-12-12	21 core
5	3.7234° N	166.7553° E	Strong	70	2010-10-10	2023-10-08	1/4	G04_019710_1836	2010-10-10	W04_090484_1843	2025-11-14	100-150
6	14.2867° N	235.8894° E	Promising	55	2011-01-04	2025-11-29	2/1	G05_020314_1935	2010-11-26	W04_090666_1922	2025-11-29	100-150
7	2.8777° N	225.1876° E	Weak	20	2024-04-19	2026-03-01	4/1	V08_083111_1832	2024-04-19	W08_091853_1832	2026-03-01	60-80
8	23.7579° N	45.8964° E	Weak	40	2020-03-29	2025-11-13	2/1	N04_064087_2042	2020-03-29	W04_090462_2049	2025-11-13	30-40
9	6.0891° N	157.2199° E	Promising	75	2008-01-27	2024-08-11	3/3	P15_007052_1871	2008-01-27	V16_086278_1886	2024-12-22	two spots ~20-25 m
10	25.3879° N	20.9581° E	Promising	45	2010-11-28	2025-02-10	1/3	G05_020335_2066	2010-11-28	W05_090832_2057	2025-12-12	20-25
11	16.8842° N	147.8693° E	Strong	80	2022-07-28	2025-04-10	2/4	F02_036655_1950	2014-05-22	W01_089311_1976	2025-08-15	25-30
12	6.8683° S	229.4246° E	Strong	80	2020-11-30	2024-12-08	2/2	N12_067246_1729	2020-11-30	V16_086104_1729	2024-12-08	50-60
13	35.5125° N	230.4463° E	Strong	70	2016-05-11	2025-09-01	3/2	J03_045896_2183	2016-05-11	W02_089532_2161	2025-09-01	100-120 diffuse
14	7.3638° N	51.0985° E	Promising	60	2014-03-20	2025-07-15	2/1	D22_035841_1874	2014-03-20	V23_088919_1873	2025-07-15	30-35
15	3.3061° S	33.7150° E	Promising	45	2024-08-20	2025-07-06	10/1	V02_080916_1769	2023-10-31	V23_088801_1769	2025-07-06	250 × 150 oval
16	16.9493° N	147.6420° E	Strong	80	2014-05-22	2022-07-28	1/7	F02_036655_1950	2014-05-22	W01_089311_1976	2025-08-15	35-40
17	35.4151° N	22.9629° E	Promising	55	2021-08-11	2025-08-02	3/1	N21_070512_2146	2021-08-11	W01_089144_2145	2025-08-02	60-70
18	8.0460° S	245.9977° E	Promising	60	2025-05-15	2025-07-10	5/1	V21_088134_1722	2025-05-15	V23_088846_1721	2025-07-10	20-25
19	4.7343° N	246.6535° E	Strong	85	2024-09-26	2025-03-10	6/4	U23_079840_1846	2023-08-08	W02_089703_1846	2025-09-15	25-30
20	6.7466° S	214.8862° E	Strong	70	2012-04-06	2025-07-31	2/2	G22_026697_1737	2012-04-06	V23_089124_1730	2025-07-31	40-50 × 30
21	25.9771° N	236.8899° E	Strong	75	2010-12-18	2025-11-07	1/2	G06_020604_2065	2010-12-18	W04_090389_2061	2025-11-07	40-50 rayed dark zone
22	24.0029° N	247.4640° E	Promising	45	2024-11-26	2025-07-11	6/1	P20_008802_2030	2008-06-12	V23_088859_2045	2025-07-11	250 × 200 irregular
23	5.2784° S	237.4930° E	Strong	85	2016-12-03	2025-01-23	7/3	B01_010187_1759	2008-09-28	V23_088899_1764	2025-07-14	3-4 dots of 15-30 m
24	30.9059° N	32.4196° E	Promising	50	2010-01-03	2025-02-02	1/2	B17_016115_2137	2010-01-03	V18_086823_2129	2025-02-02	30-40
25	20.1303° N	34.2063° E	Strong	75	2008-11-21	2017-11-03	2/3	B03_010880_2023	2008-11-21	V02_080784_2019	2023-10-21	80-100 core
26	27.2350° N	270.7245° E	Promising	60	2024-11-08	2025-08-18	4/1	F02_036690_2079	2014-05-25	W01_089346_2075	2025-08-18	35-40
27	1.7153° N	27.1971° E	Promising	60	2020-08-24	2025-07-22	8/1	F21_044097_1814	2015-12-23	V23_088999_1804	2025-07-22	100 × 150 core
28	10.3730° N	12.9418° E	Strong	70	2023-10-10	2024-09-07	2/2	V02_080653_1896	2023-10-10	V23_088894_1895	2025-07-13	40 × 60
29	17.2763° N	250.6787° E	Promising	60	2024-06-30	2026-04-14	3/1	P07_003607_1995	2007-05-04	W09_092419_1990	2026-04-14	25-30
30	16.8749° N	142.6064° E	Promising	55	2024-12-13	2025-08-14	6/1	F02_036642_1970	2014-05-21	W01_089298_1973	2025-08-14	20 × 35
31	14.2270° N	238.2743° E	Strong	80	2019-03-19	2024-03-19	2/5	B17_016424_1929	2010-01-27	V20_087765_1941	2025-04-17	70 × 55
32	14.0079° N	41.4657° E	Promising	70	2022-11-27	2025-10-21	3/1	J06_047103_1946	2016-08-13	W03_090172_1959	2025-10-21	60 main spot + 4-5 dots
33	17.3398° S	33.4530° E	Promising	40	2025-06-14	2025-09-10	3/1	V22_088511_1628	2025-06-14	W02_089645_1628	2025-09-10	80-100
34	7.0015° N	150.2711° E	Strong	85	2008-08-25	2021-06-08	1/5	P22_009755_1878	2008-08-25	N19_069690_1882	2021-06-08	25-30 m core
35	4.4544° N	205.5598° E	Promising	55	2011-12-28	2023-09-02	2/4	P20_008830_1838	2008-06-14	W03_090021_1853	2025-10-09	150-200 diffuse
36	29.0978° N	27.0612° E	Strong	80	2014-12-08	2020-08-03	3/4	B01_010155_2094	2008-09-25	V21_088221_2087	2025-05-22	~90 m core
37	4.0687° S	237.1175° E	Strong	90	2015-12-15	2025-01-23	3/2	F21_043997_1753	2015-12-15	V21_088253_1764	2025-05-25	60 × 45 core
38	14.6517° N	202.4361° E	Promising	70	2023-01-24	2025-02-08	4/1	U16_077323_1964	2023-01-24	V18_086896_1965	2025-02-08	100-130, rayed
39	7.3902° N	246.6840° E	Promising	55	2014-04-22	2025-01-25	2/1	B19_016938_1888	2010-03-08	V17_086723_1880	2025-01-25	85 × 130 core
40	7.4966° N	144.3039° E	Strong	90	2023-09-06	2023-12-25	4/2	V01_080213_1870	2023-09-06	V20_087874_1870	2025-04-25	55 × 75 core
41	33.2901° N	230.7321° E	Strong	80	2012-01-08	2023-04-12	1/3	G19_025549_2126	2012-01-08	U19_078324_2118	2023-04-12	~50 core

Table 2 (continued)

No.	Lat.	Lon.	Rating	P (%)	Last without	First with	N_-/N_+	Before image	Date	After image	Date	Size (m)
42	18.9845° S	223.6547° E	Strong	85	2008-08-12	2018-11-30	3/2	P22_009581_1610	2008-08-12	V22_088715_1611	2025-06-30	~40 core
43	19.0840° S	223.7612° E	Promising	60	2018-11-30	2025-06-30	3/1	P22_009581_1610	2008-08-12	V22_088715_1611	2025-06-30	60 × 70 irregular
44	31.4688° N	44.0356° E	Promising	45	2008-10-21	2025-02-15	1/2	B02_010484_2122	2008-10-21	V18_086994_2136	2025-02-15	~70 irregular
45	24.7886° N	278.8201° E	Promising	60	2020-07-29	2022-07-10	2/2	P19_008379_2060	2008-05-10	V20_087658_2042	2025-04-08	45 × 35 core
46	22.1339° N	22.3971° E	Promising	65	2007-12-21	2025-01-20	1/1	P14_006569_2000	2007-12-21	V17_086652_2026	2025-01-20	50 × 60 core
47	2.8114° S	268.3592° E	Strong	85	2008-09-17	2025-04-16	1/2	B01_010054_1768	2008-09-17	V20_087764_1754	2025-04-16	30 × 35 core
48	1.8749° N	172.1951° E	Promising	70	2010-10-11	2025-01-21	1/1	G04_019723_1821	2010-10-11	V17_086673_1822	2025-01-21	50 × 40 core
49	5.8922° N	268.0812° E	Promising	70	2008-12-26	2025-02-10	2/1	B02_010410_1861	2008-10-15	V18_086920_1842	2025-02-10	45 × 35 core
50	4.0621° N	278.7650° E	Strong	85	2013-01-30	2023-03-29	3/4	G21_026510_1854	2012-03-23	V20_087724_1848	2025-04-13	~50 core
51	0.6905° S	26.6850° E	Strong	90	2010-09-06	2023-05-01	1/2	G03_019280_1784	2010-09-06	U20_078569_1782	2023-05-01	55 × 65 core
52	21.3396° N	144.1996° E	Promising	65	2014-01-26	2025-03-18	1/1	D20_035165_1993	2014-01-26	V19_087386_1985	2025-03-18	65 × 45 core
53	26.3306° N	264.7174° E	Strong	90	2016-05-12	2020-08-12	2/3	J03_045908_2089	2016-05-12	V16_086063_2065	2024-12-05	35 × 30 core
54	16.6374° N	273.8008° E	Promising	50	2016-07-19	2025-02-06	2/1	J05_046778_1953	2016-07-19	V18_086867_1960	2025-02-06	200-300 diffuse
55	10.5054° N	24.0527° E	Promising	55	2012-03-05	2025-06-01	3/1	G21_026282_1910	2012-03-05	V22_088353_1903	2025-06-01	~80 cluster
56	6.8861° N	276.3125° E	Strong	90	2008-05-09	2025-01-20	1/2	P19_008366_1853	2008-05-09	V21_088001_1867	2025-05-05	55 × 45 core
57	19.1431° N	12.1955° E	Strong	90	2014-03-18	2023-05-04	2/4	D22_035816_1981	2014-03-18	U20_078609_1978	2023-05-04	40 × 46 dark patch
58	11.9906° N	236.7925° E	Strong	90	2010-04-30	2022-10-09	1/4	B20_017624_1906	2010-04-30	V21_088187_1917	2025-05-19	~45 core
59	8.0391° N	169.4077° E	Strong	90	2023-01-28	2025-04-14	4/2	N17_068977_1869	2021-04-14	V20_087728_1884	2025-04-14	cluster ~120 × 100 of 4-5 blobs
60	5.6411° S	255.4212° E	Promising	70	2016-04-11	2025-06-03	2/1	P21_009171_1743	2008-07-11	V22_088371_1745	2025-06-03	~45 core
61	4.1341° N	36.6353° E	Promising	60	2018-03-11	2025-05-29	2/1	K03_054474_1830	2018-03-11	V21_088313_1833	2025-05-29	~60
62	3.7684° S	247.8767° E	Promising	60	2009-03-26	2025-02-06	1/2	B07_012481_1756	2009-03-26	V18_086934_1771	2025-02-11	~40-50 core
63	2.5252° N	278.7758° E	Strong	90	2016-07-25	2023-03-29	3/2	J05_046857_1823	2016-07-25	V18_086946_1848	2025-02-12	~47
64	0.6852° S	234.0286° E	Strong	85	2012-05-30	2023-07-14	1/2	G23_027382_1803	2012-05-30	V20_087818_1810	2025-04-21	~26
65	20.4810° N	253.4915° E	Promising	70	2023-07-13	2025-04-20	3/1	U22_079510_2005	2023-07-13	V20_087804_2005	2025-04-20	~34
66	6.9479° N	266.7737° E	Promising	70	2012-01-09	2025-01-29	2/1	G19_025561_1887	2012-01-09	V17_086775_1869	2025-01-29	~42
67	9.2688° S	191.2269° E	Strong	85	2014-01-29	2023-04-03	3/2	B19_016861_1687	2010-03-02	V18_087002_1682	2025-02-16	~34
68	34.3303° N	50.2662° E	Strong	90	2023-06-25	2024-11-09	4/2	U21_079280_2140	2023-06-25	V17_086796_2139	2025-01-31	~42
69	31.0754° N	181.4958° E	Strong	90	2023-04-13	2025-03-03	2/2	U19_078339_2114	2023-04-13	V20_087767_2106	2025-04-17	diffuse ~70
70	3.8713° N	257.6935° E	Promising	60	2024-04-08	2025-03-29	4/1	G20_026102_1823	2012-02-20	V19_087527_1819	2025-03-29	~19
71	2.4389° N	274.0536° E	Strong	90	2010-01-23	2023-01-27	1/3	B17_016370_1834	2010-01-23	U16_077360_1843	2023-01-27	~33
72	19.2191° N	12.1485° E	Strong	90	2014-03-18	2023-05-04	2/4	D22_035816_1981	2014-03-18	U20_078609_1978	2023-05-04	~55
73	39.4963° N	209.8523° E	Strong	75	2022-07-14	2022-12-30	4/9	U01_071494_2210	2021-10-27	U19_078272_2209	2023-04-08	diffuse ~210
74	16.0736° N	178.9461° E	Strong	70	2014-06-09	2023-01-30	2/3	F03_036878_1964	2014-06-09	W03_089956_1957	2025-10-04	~55
75	2.8034° N	278.7676° E	Promising	50	2012-10-12	2023-03-29	2/2	P02_001905_1812	2006-12-22	V18_086946_1848	2025-02-12	~45
76	3.4926° S	27.7793° E	Promising	60	2016-06-27	2025-05-27	2/1	G01_018502_1769	2010-07-08	V21_088287_1770	2025-05-27	~55
77	13.3505° N	277.9001° E	Strong	85	2020-07-18	2024-10-13	3/2	P15_006797_1921	2008-01-08	V14_085390_1952	2024-10-13	~120-140
78	16.0710° N	256.4611° E	Strong	80	2008-02-28	2021-02-12	1/2	P16_007457_1962	2008-02-28	V17_086538_1946	2025-01-11	~70
79	24.8734° N	46.5175° E	Strong	85	2022-06-21	2024-12-06	2/2	G04_019622_2034	2010-10-03	W03_090119_2034	2025-10-17	~55-60
80	9.3943° N	25.7511° E	Strong	80	2023-01-11	2024-11-14	5/2	B17_016445_1890	2010-01-28	V17_086441_1884	2025-01-03	~25
81	11.8779° N	251.7127° E	Promising	60	2022-03-16	2024-12-02	7/1	D17_033803_1920	2013-10-12	V16_086024_1907	2024-12-02	~60
82	8.3239° N	239.2944° E	Promising	65	2024-05-13	2024-12-21	3/1	B19_016991_1885	2010-03-12	V16_086275_1884	2024-12-21	~40
83	4.6753° S	179.5549° E	Promising	50	2010-03-16	2024-12-09	1/3	B19_017046_1762	2010-03-16	V18_087042_1759	2025-02-19	~80-100 diffuse
84	29.3380° N	239.8719° E	Strong	85	2020-08-23	2022-12-25	3/4	P14_006561_2112	2007-12-20	U15_076939_2109	2022-12-25	~80
85	4.2241° N	210.8642° E	Strong	90	2020-03-12	2024-12-21	2/3	B09_013168_1827	2009-05-18	V23_089045_1816	2025-07-25	~60
86	6.0430° S	164.0042° E	Promising	55	2022-07-20	2025-01-03	4/2	N06_064518_1759	2020-05-01	V21_087992_1762	2025-05-04	~110 diffuse
87	22.3387° N	52.2009° E	Promising	65	2020-09-04	2024-12-07	5/1	G14_023815_2011	2011-08-26	V16_086084_2037	2024-12-07	~150-200 rayed blast zone

Table 2 (continued)

No.	Lat.	Lon.	Rating	P (%)	Last without	First with	N_-/N_+	Before image	Date	After image	Date	Size (m)
88	3.9974° S	268.6457° E	Promising	60	2020-02-13	2024-12-05	3/1	P12_005914_1745	2007-10-31	V16_086063_1743	2024-12-05	~35
89	0.0674° N	138.5886° E	Promising	55	2022-08-12	2024-11-20	6/1	B11_013975_1785	2009-07-20	V15_085870_1787	2024-11-20	~30
90	39.9957° N	228.8713° E	Promising	65	2018-04-08	2024-10-21	4/1	K04_054836_2198	2018-04-08	V14_085484_2205	2024-10-21	~50
91	33.6203° N	204.6124° E	Promising	45	2010-11-15	2024-12-27	2/1	P15_006971_2158	2008-01-21	V16_086342_2159	2024-12-27	~50
92	38.5305° N	220.8085° E	Promising	65	2025-04-13	2025-06-03	7/1	V20_087726_2183	2025-04-13	V22_088372_2191	2025-06-03	~65
93	17.3508° S	244.9885° E	Promising	65	2017-12-31	2024-11-19	3/1	P10_004926_1620	2007-08-15	V15_085853_1611	2024-11-19	~75
94	6.0510° S	154.0089° E	Strong	70	2010-12-29	2021-08-09	6/7	G02_019104_1740	2010-08-24	U21_079250_1736	2023-06-23	~100
95	1.9174° S	268.7421° E	Promising	65	2023-07-10	2025-05-25	4/1	U22_079470_1793	2023-07-10	V21_088265_1793	2025-05-25	~25
96	12.8483° N	36.1185° E	Promising	65	2020-07-25	2025-04-04	3/1	B17_016260_1928	2010-01-14	V20_087601_1927	2025-04-04	~40
97	9.3950° S	208.6036° E	Weak	20	2023-09-18	2025-05-13	6/1	V01_080369_1704	2023-09-18	V21_088109_1710	2025-05-13	~90 × 30 teardrop
98	17.9059° N	231.7729° E	Weak	35	2020-10-25	2025-04-04	2/1	N11_066784_1955	2020-10-25	V20_087607_1967	2025-04-04	~35
99	12.0131° N	122.1235° E	Strong	70	2022-03-13	2025-05-07	4/2	F20_043711_1927	2015-11-23	V21_088033_1912	2025-05-07	~25
100	27.5278° N	285.0236° E	Strong	85	2017-05-27	2020-08-16	4/3	J15_050786_2076	2017-05-27	V16_086260_2078	2024-12-20	~55
101	16.4977° N	233.9210° E	Promising	60	2023-01-30	2024-12-04	5/1	U16_077401_1951	2023-01-30	V16_086051_1951	2024-12-04	~25
102	13.3031° S	181.5627° E	Strong	70	2023-05-26	2025-06-10	2/2	U20_078893_1651	2023-05-26	V22_088466_1665	2025-06-10	~40 × 25 dark head
103	24.8247° N	282.1434° E	Promising	70	2018-05-21	2024-11-21	3/1	B19_016963_2053	2010-03-10	V15_085891_2040	2024-11-21	~55
104	14.3619° N	205.4454° E	Strong	80	2018-01-21	2023-01-08	3/2	K01_053848_1957	2018-01-21	U16_077125_1947	2023-01-08	~36
105	15.5441° N	273.4954° E	Strong	70	2010-02-19	2020-06-25	1/2	B18_016726_1975	2010-02-19	V18_087078_1971	2025-02-22	~69
106	37.2860° N	158.8058° E	Promising	65	2023-12-06	2025-01-03	4/1	B17_016229_2167	2010-01-12	V17_086436_2162	2025-01-03	~130
107	29.2900° N	25.4220° E	Promising	55	2014-03-21	2024-12-18	1/1	D22_035855_2081	2014-03-21	V16_086230_2092	2024-12-18	~35
108	4.7761° N	169.2420° E	Promising	70	2015-12-19	2024-12-19	1/1	F21_044039_1855	2015-12-19	V16_086238_1843	2024-12-19	~110
109	6.4592° S	194.1982° E	Weak	40	2014-04-06	2025-05-10	1/1	F01_036060_1732	2014-04-06	V21_088070_1743	2025-05-10	~190
110	16.1878° N	29.7458° E	Promising	65	2016-02-28	2024-11-26	3/1	P16_007373_1974	2008-02-21	V15_085953_1951	2024-11-26	~27
111	17.4903° S	212.9255° E	Strong	80	2014-08-03	2019-10-24	2/2	B20_017625_1641	2010-04-30	K22_062077_1630	2019-10-24	~40
112	8.8793° S	265.2273° E	Promising	60	2016-09-16	2024-09-24	5/1	P10_005123_1714	2007-08-30	V13_085140_1711	2024-09-24	~160
113	19.9814° N	39.4433° E	Promising	50	2022-03-24	2025-01-06	4/1	P16_007399_1996	2008-02-24	V17_086480_1980	2025-01-06	~230
114	37.8280° N	46.6507° E	Promising	60	2010-02-02	2023-10-12	1/3	B18_016510_2166	2010-02-02	V18_086928_2181	2025-02-10	~35
115	5.6256° N	128.7919° E	Strong	85	2019-09-16	2023-04-18	2/2	K21_061592_1855	2019-09-16	U19_078407_1853	2023-04-18	~120 halo
116	17.9591° N	39.0062° E	Strong	80	2023-02-26	2024-07-19	5/2	U17_077751_1999	2023-02-26	V16_086335_1999	2024-12-26	~75
117	40.4237° N	160.1682° E	Promising	65	2010-01-23	2021-02-20	1/3	B17_016374_2218	2010-01-23	U20_078643_2223	2023-05-07	~50
118	21.2241° N	14.9784° E	Strong	85	2014-04-26	2023-03-27	1/2	F01_036317_1997	2014-04-26	U22_079466_2023	2023-07-10	cluster of 6+ spots, largest ~40
119	22.8028° N	268.4438° E	Promising	55	2010-05-29	2023-07-16	1/1	B21_017992_2025	2010-05-29	U22_079549_2034	2023-07-16	~35
120	5.1641° N	170.6673° E	Strong	85	2014-05-10	2023-03-25	1/2	F02_036496_1861	2014-05-10	U21_079223_1848	2023-06-21	~50
121	21.5593° N	350.3671° E	Promising	45	2012-05-31	2020-09-13	2/2	G23_027404_2027	2012-05-31	V16_086139_2037	2024-12-11	~100-150 irregular patch
122	12.5218° N	272.9512° E	Promising	70	2010-04-21	2024-10-18	1/2	B20_017504_1905	2010-04-21	V16_086366_1917	2024-12-28	~50
123	3.2605° N	149.7742° E	Promising	75	2021-11-10	2023-10-10	4/1	U02_071681_1818	2021-11-10	V02_080648_1844	2023-10-10	~1300 rayed blast zone
124	33.5460° N	42.6671° E	Strong	90	2018-05-01	2023-03-22	2/3	B16_015930_2137	2009-12-19	V17_086493_2131	2025-01-07	~40
125	20.3980° N	263.7310° E	Strong	90	2020-06-02	2024-08-11	2/3	G20_026194_2003	2012-02-27	V14_085562_2004	2024-10-27	~45
126	13.6357° N	234.0298° E	Strong	75	2007-05-06	2016-06-08	1/3	P07_003634_1955	2007-05-06	V07_082847_1933	2024-03-29	~60 head
127	3.8206° N	197.5165° E	Promising	65	2021-11-03	2023-08-09	2/1	U02_071587_1860	2021-11-03	U23_079855_1859	2023-08-09	~40
128	43.4682° N	155.5156° E	Promising	65	2010-02-19	2023-06-02	1/1	B18_016717_2221	2010-02-19	U21_078986_2219	2023-06-02	~40
129	6.1020° S	197.3791° E	Strong	85	2015-04-14	2023-04-26	2/2	G23_027146_1723	2012-05-11	U19_078510_1720	2023-04-26	~45
130	36.5465° N	192.8396° E	Strong	80	2012-03-23	2023-04-10	2/3	G21_026513_2182	2012-03-23	U19_078299_2181	2023-04-10	~30
131	15.8187° N	213.5247° E	Promising	60	2010-05-18	2023-06-02	1/2	B21_017849_1970	2010-05-18	U21_078984_1972	2023-06-02	~90 × 25 wedge
132	6.1215° N	39.5546° E	Promising	50	2012-09-27	2023-10-27	2/1	D04_028932_1862	2012-09-27	V02_080863_1862	2023-10-27	two spots ~30 and ~20
133	9.6103° N	18.5988° E	Strong	90	2020-07-27	2023-12-06	2/2	J06_046972_1899	2016-08-03	V04_081378_1898	2023-12-06	cluster of 8+ spots

Table 2 (continued)

No.	Lat.	Lon.	Rating	P (%)	Last without	First with	N_-/N_+	Before image	Date	After image	Date	Size (m)
134	39.3839° N	164.7615° E	Strong	80	2024-03-05	2024-12-19	4/5	U18_078155_2214	2023-03-30	V21_088084_2213	2025-05-11	~40
135	12.4959° N	10.1536° E	Strong	80	2014-10-06	2024-01-17	2/3	F07_038414_1937	2014-10-06	V07_082842_1917	2024-03-29	~50 with two dark cores
136	3.7982° S	256.4596° E	Strong	90	2019-01-20	2020-11-25	3/2	F03_036928_1765	2014-06-12	U03_072310_1761	2021-12-29	~170 dark blast zone
137	45.7065° N	177.1405° E	Strong	90	2022-03-13	2023-02-10	4/8	U02_071693_2245	2021-11-11	V01_080449_2245	2023-09-24	~30
138	2.4732° S	243.1012° E	Promising	70	2020-12-13	2022-08-10	5/1	N11_066599_1778	2020-10-10	U11_075185_1777	2022-08-10	~50
139	32.9770° N	158.3317° E	Promising	50	2020-09-20	2025-03-05	3/1	B16_016084_2131	2009-12-31	V19_087214_2112	2025-03-05	chain of 5 patches
140	32.5821° N	255.8855° E	Strong	90	2022-10-29	2023-04-17	4/2	P16_007246_2145	2008-02-12	U19_078389_2141	2023-04-17	~45
141	0.1494° N	24.5323° E	Strong	90	2022-03-15	2024-01-09	3/2	U06_073281_1793	2022-03-15	V05_081813_1793	2024-01-09	~30
142	36.0228° N	143.4838° E	Strong	85	2023-01-29	2024-09-01	7/2	P03_002279_2161	2007-01-21	V13_084841_2152	2024-09-01	~50
143	19.7189° N	245.0063° E	Promising	60	2016-06-21	2022-12-31	2/1	J04_046423_2014	2016-06-21	U15_077018_1986	2022-12-31	~35
144	23.5276° N	274.0757° E	Promising	55	2018-04-27	2022-12-21	2/1	P13_006006_2027	2007-11-07	U15_076885_2049	2022-12-21	~35
145	9.1389° N	240.2842° E	Promising	45	2024-08-06	2025-02-10	2/1	V12_084508_1880	2024-08-06	V18_086921_1889	2025-02-10	three compact spots
146	5.7363° N	41.3369° E	Promising	50	2010-04-30	2023-05-04	2/1	B20_017618_1859	2010-04-30	U20_078608_1851	2023-05-04	~40
147	4.5183° S	218.3412° E	Strong	90	2018-11-24	2020-09-09	4/3	K11_057791_1757	2018-11-24	U12_075542_1757	2022-09-07	~50
148	14.6501° S	206.4798° E	Strong	85	2010-01-20	2014-05-22	1/3	B17_016333_1643	2010-01-20	F02_036653_1649	2014-05-22	~45
149	1.3076° S	206.3880° E	Strong	85	2008-02-23	2023-02-09	1/3	P16_007393_1788	2008-02-23	U17_077534_1788	2023-02-09	~30
150	18.3646° S	222.3114° E	Strong	85	2021-03-13	2022-12-31	3/2	N16_068566_1602	2021-03-13	U16_077296_1602	2023-01-22	~40
151	13.1083° N	13.0583° E	Promising	60	2014-12-05	2021-07-06	3/1	F05_037583_1912	2014-08-03	N20_070051_1937	2021-07-06	~40
152	4.5084° N	262.3156° E	Strong	85	2020-06-01	2022-08-14	2/2	N07_064910_1848	2020-06-01	U11_075237_1858	2022-08-14	~30
153	4.2734° S	225.3506° E	Promising	60	2008-12-28	2023-04-26	1/1	B04_011361_1737	2008-12-28	U19_078509_1752	2023-04-26	~35
154	23.2932° N	40.1699° E	Strong	75	2020-11-21	2022-07-07	5/3	N11_066514_2050	2020-10-04	U10_074744_2050	2022-07-07	~140
155	4.5992° S	234.6512° E	Strong	90	2007-02-23	2012-01-29	1/3	P04_002711_1747	2007-02-23	G19_025826_1744	2012-01-29	~120
156	8.4308° S	225.2100° E	Promising	60	2014-04-28	2023-05-24	4/1	G22_026934_1731	2012-04-25	U20_078865_1728	2023-05-24	~30
157	3.2439° N	159.6141° E	Promising	60	2008-03-13	2022-12-09	1/1	P17_007632_1835	2008-03-13	U15_076731_1823	2022-12-09	~70
158	5.2335° S	222.1199° E	Strong	90	2020-11-02	2021-01-25	2/3	N12_066890_1745	2020-11-02	U12_075621_1738	2022-09-13	~35
159	2.9275° S	124.2268° E	Strong	80	2021-07-06	2022-09-10	3/3	N13_067329_1766	2020-12-06	U12_075585_1758	2022-09-10	~50
160	0.8251° N	39.7276° E	Strong	75	2020-09-22	2022-08-14	2/2	N10_066369_1825	2020-09-22	U11_075232_1825	2022-08-14	~50
161	11.8158° S	243.3738° E	Promising	50	2021-02-19	2022-12-08	3/1	P12_005915_1706	2007-10-31	U15_076728_1687	2022-12-08	~70
162	2.9806° S	214.5769° E	Strong	90	2008-09-01	2022-01-18	1/2	B01_009845_1770	2008-09-01	U04_072562_1769	2022-01-18	~150 core
163	5.5014° S	198.3926° E	Promising	60	2008-02-05	2023-03-02	1/4	P16_007156_1758	2008-02-05	V16_086382_1757	2024-12-30	~250, faint core ~30
164	2.5584° S	127.8512° E	Strong	75	2021-02-13	2023-05-21	5/2	P18_008121_1776	2008-04-20	U20_078829_1784	2023-05-21	~25
165	10.8609° S	217.3246° E	Strong	85	2007-12-20	2024-04-06	2/3	P06_003371_1699	2007-04-16	V08_082940_1705	2024-04-06	~35
166	16.1984° N	275.8236° E	Strong	85	2020-05-30	2022-06-12	2/2	N06_064883_1971	2020-05-30	U09_074419_1967	2022-06-12	~40

Table 3: Outcome of the independent re-examination of 38 sites (Section 3.8). The first two data columns repeat the rating and probability used in this paper; "outcome" and "P_{chk}" are the re-examination's own verdict and probability. Its formation windows are marked with an asterisk where they differ from ours. The last two columns give its measured position of the spot, in the map projection of one of the images, and the distance of that position from ours.

No.	Rating	P (%)	Outcome	P_{chk} (%)	Window (check)	Re-measured position	Δ (m)
19	Strong	85	kept as strong	90	2024-09-26 – 2025-03-10	4.7346° N, 246.6520° E	92
20	Strong	70	kept as strong	80	2018-04-20 – 2022-11-12*	6.7444° S, 214.8835° E	203
21	Strong	75	kept as strong	95	2010-12-18 – 2025-11-07	25.9794° N, 236.8864° E	232
23	Strong	85	kept as strong	85	2016-12-03 – 2025-01-23	5.2771° S, 237.5184° E	1502
25	Strong	75	kept as strong	65	2008-11-21 – 2017-11-03	20.1310° N, 34.2074° E	72
28	Strong	70	kept as strong	70	2023-10-10 – 2024-09-07	10.3751° N, 12.9457° E	259
31	Strong	80	kept as strong	90	2019-03-19 – 2024-03-19	14.2288° N, 238.2695° E	295
34	Strong	85	kept as strong	85	2008-08-25 – 2021-06-08	7.0033° N, 150.2789° E	474
36	Strong	80	kept as strong	85	2018-09-11 – 2020-08-03*	29.0991° N, 27.0655° E	235
37	Strong	90	kept as strong	80	2015-12-15 – 2025-01-23	4.0683° S, 237.1256° E	480
40	Strong	90	kept as strong	90	2023-09-06 – 2023-12-25	7.4968° N, 144.3132° E	544
41	Strong	80	kept as strong	85	2012-01-08 – 2023-04-12	33.2894° N, 230.7350° E	151
42	Strong	85	kept as strong	90	2008-08-12 – 2018-11-30	18.9845° S, 223.6560° E	75
43	Promising	60	promising (agreed)	60	2008-08-12 – 2025-06-30*	19.0836° S, 223.7619° E	45
47	Strong	85	kept as strong	85	2008-09-17 – 2025-04-16	2.8147° S, 268.3595° E	198
50	Strong	85	kept as strong	85	2012-03-23 – 2023-03-29*	4.0624° N, 278.7623° E	161
51	Strong	90	kept as strong	90	2010-09-06 – 2023-05-01	0.6922° S, 26.6886° E	235
53	Strong	90	kept as strong	90	2016-05-12 – 2020-08-12	26.3330° N, 264.7201° E	200
56	Strong	90	kept as strong	80	2008-05-09 – 2025-01-20	6.8882° N, 276.3156° E	220
57	Strong	90	kept as strong	95	2014-03-18 – 2023-05-04	19.1428° N, 12.1880° E	421
58	Strong	90	kept as strong	90	2010-04-30 – 2022-10-09	11.9922° N, 236.7905° E	148
59	Strong	90	kept as strong	90	2023-01-28 – 2025-04-14	8.0426° N, 169.4178° E	628
63	Strong	90	kept as strong	90	2016-07-25 – 2023-03-29	2.5240° N, 278.7785° E	174
64	Strong	85	kept as strong	80	2012-05-30 – 2023-07-14	0.6833° S, 234.0278° E	121
67	Strong	85	kept as strong	85	2014-01-29 – 2018-12-16*	9.2691° S, 191.2234° E	203
68	Strong	90	kept as strong	80	2023-06-25 – 2024-11-09	34.3352° N, 50.2677° E	302
69	Strong	90	kept as strong	85	2023-04-13 – 2025-03-03	31.0735° N, 181.4839° E	614
71	Strong	90	kept as strong	85	2010-01-23 – 2023-01-27	2.4370° N, 274.0570° E	229
72	Strong	90	kept as strong	85	2014-03-18 – 2023-05-04	19.2186° N, 12.1405° E	449
73	Strong	75	promising, not strong	60	2022-07-14 – 2022-12-30	39.4960° N, 209.8607° E	383
74	Strong	70	kept as strong	65	2014-06-09 – 2023-01-30	16.0743° N, 178.9482° E	124
77	Strong	85	kept as strong	85	2020-07-18 – 2024-10-13	13.3537° N, 277.8947° E	367
78	Strong	80	kept as strong	85	2008-02-28 – 2021-02-12	16.0735° N, 256.4609° E	149
79	Strong	85	kept as strong	85	2022-06-21 – 2024-12-06	24.8751° N, 46.5177° E	103
80	Strong	80	kept as strong	80	2023-01-11 – 2024-11-14	9.3959° N, 25.7512° E	93
84	Strong	85	kept as strong	75	2020-08-23 – 2022-12-25	29.3396° N, 239.8731° E	113
85	Strong	90	kept as strong	75	2020-03-12 – 2024-12-21	4.2268° N, 210.8519° E	745
92	Promising	65	probably a dust devil	5	2025-04-13 – 2025-06-03	38.5289° N, 220.8006° E	377

5 Catalogue of the candidates

This section has one entry for each candidate, numbered as on the public pages (Section 7). The numbers follow the order in which the candidates were found, so neighbouring numbers are often nearby sites. Each entry gives the facts of the case (position, rating and probability, formation window, the CTX images without and with the spot, size, HiRISE coverage and the distance to the nearest catalogued new impact), a plain account of what changed and why it may be a new impact, its caveats, and, for the 38 re-examined sites, the outcome of the independent check.

Each entry has two or more figures. The first shows the same patch of ground, usually 500 to 800 m across, in an earlier image without the spot and a later image with it, usually under a similar Sun; the second shows every CTX image of the surrounding 2×2 km in time order. Some entries add a HiRISE view or the figures of the independent check. All figures are north up. Positions are planetocentric latitude and east longitude; dates are those of the images; the darkness score z and alignment score r printed on the figures are defined in Section 3.3,

as is the convention for the solar azimuths. “Formed between” gives the dates of the last image without the spot and the first image with it. The “absent” and “present” lists in the facts name the images described in the text, which may be fewer than all the images counted. Image credits: CTX, NASA/JPL-Caltech/MSSS; HiRISE, NASA/JPL-Caltech/UArizona; MOC, NASA/JPL/MSSS.

Candidate 1: Arsia Sulci

Location	5.7715° S, 230.9347° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 75%
Formed between	2 October 2022 and 27 April 2026
Images	4 without the spot (absent: four earlier images, the last on 2 October 2022); 1 with it (present: 27 April 2026)
Size	Round dark spot about 70 m across, inside a fainter halo about 150 m across
HiRISE	none covers the spot (nearest footprint edge 12 km)
Nearest catalogued new impact	47 km (Daubar et al., 2022)

What changed. A round dark spot about 70 m across, inside a fainter halo about 150 m across, appeared in Arsia Sulci between October 2022 and April 2026. So far it is seen in only one CTX image.

Why it may be a new impact. Four earlier images, aligned with the April 2026 one, show nothing at the spot, not even a landform. They were taken with the sun both higher and lower in the sky than in April 2026, so the spot is not a shadow. A round dark spot with a soft halo is how the blast zone of a small new crater looks.

Caveats. Only one image shows it so far, and a flaw in that single image cannot be fully ruled out. A small slope streak, a dark streak left by dust sliding down a slope, is unlikely, as the spot is round. No image from the high-resolution HiRISE camera covers it; the nearest ends 12 km away.

Figure note. This before-and-after figure was made for this paper (Section 3.9) from the map-projected versions of the two images; no close-up had been published for this candidate. In the map-projected frame the spot lies 134 m from the catalogued position, at 5.7712° S, 230.9369° E.

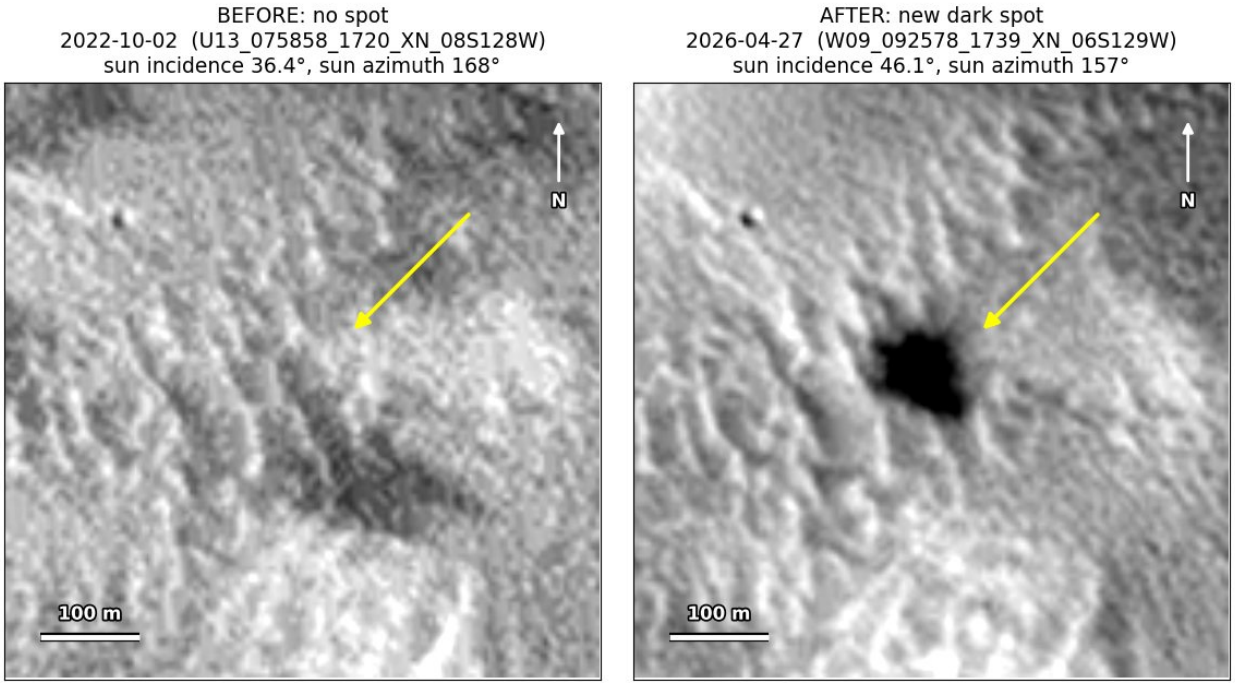

Figure 9: Candidate 1, before and after: the same 600 × 600 m of ground in 2 October 2022 (`U13_075858_1720_XN_08S128W`) and 27 April 2026 (`W09_092578_1739_XN_06S129W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS. Made for this paper from map-projected images distributed by Arizona State University.

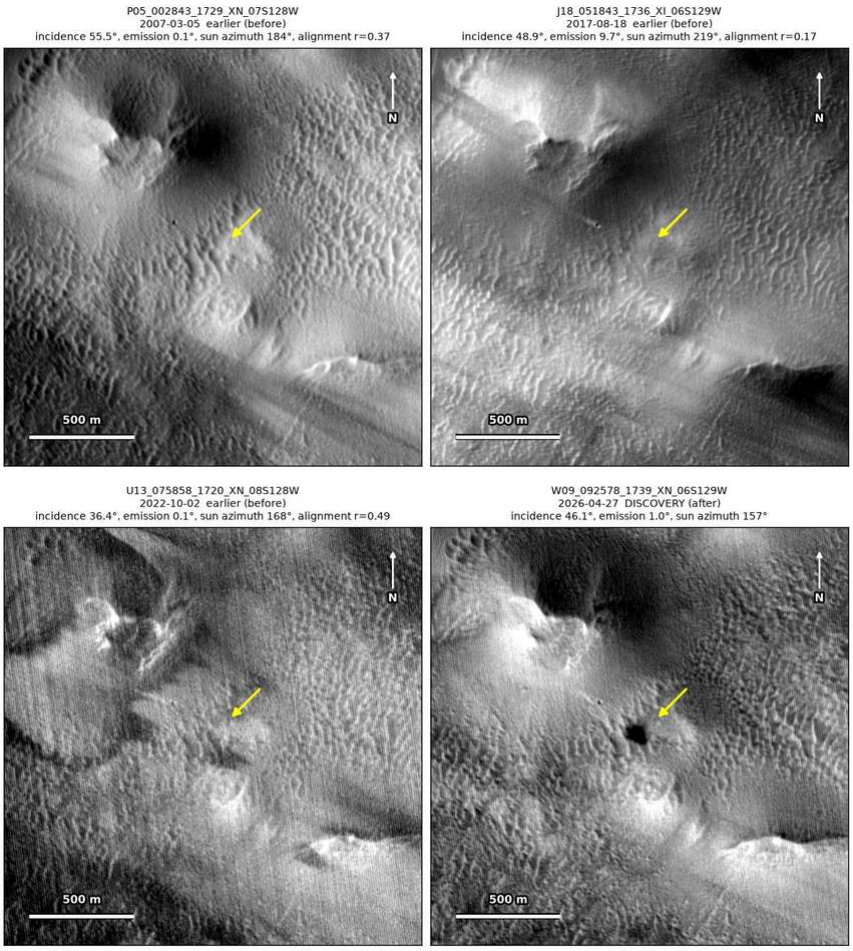

Figure 10: Candidate 1: four of the five CTX images of the same 2×2 km patch, 2007 to 2026. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 2: Plains at Phlegra Dorsa, eastern Elysium

Location	24.8860° N, 161.4195° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	22 June 2022 and 11 May 2026
Images	5 without the spot (absent: five earlier images, from 2016 to 22 June 2022); 1 with it (present: 11 May 2026)
Size	Compact dark spot about 70 m across
HiRISE	none covers the spot (nearest footprint edge 6 km)
Nearest catalogued new impact	10 km (Daubar et al., 2022)

What changed. A compact dark spot about 70 m across appeared on plains at Phlegra Dorsa, in eastern Elysium, between June 2022 and May 2026. So far it is seen in only one CTX image.

Why it may be a new impact. Five earlier images, from 2016 to 2022, show nothing there, three of them taken under a sun matching that of the May 2026 image. A known cluster of new impacts about 10 km away formed between 2016 and 2018, but this spot is still absent in the images from 2018 to 2022, so it is a separate, later event. A compact new dark spot fits the blast zone of a small new crater.

Caveats. Only one image shows it, and that image is noisy and was taken with the sun low in the sky, so a flaw in it cannot be fully ruled out. No image from the high-resolution HiRISE camera covers it; the nearest ends 6 km away.

Figure note. This figure was made for this paper (Section 3.9) from the map-projected versions of the two images. Its earlier image, of November 2016, is the one taken under nearly the same sun as the May 2026 image; the later images of 2018 to June 2022, shown in the time series, also show nothing there. In the map-projected frame the spot lies 551 m from the catalogued position, at 24.8882° N, 161.4295° E.

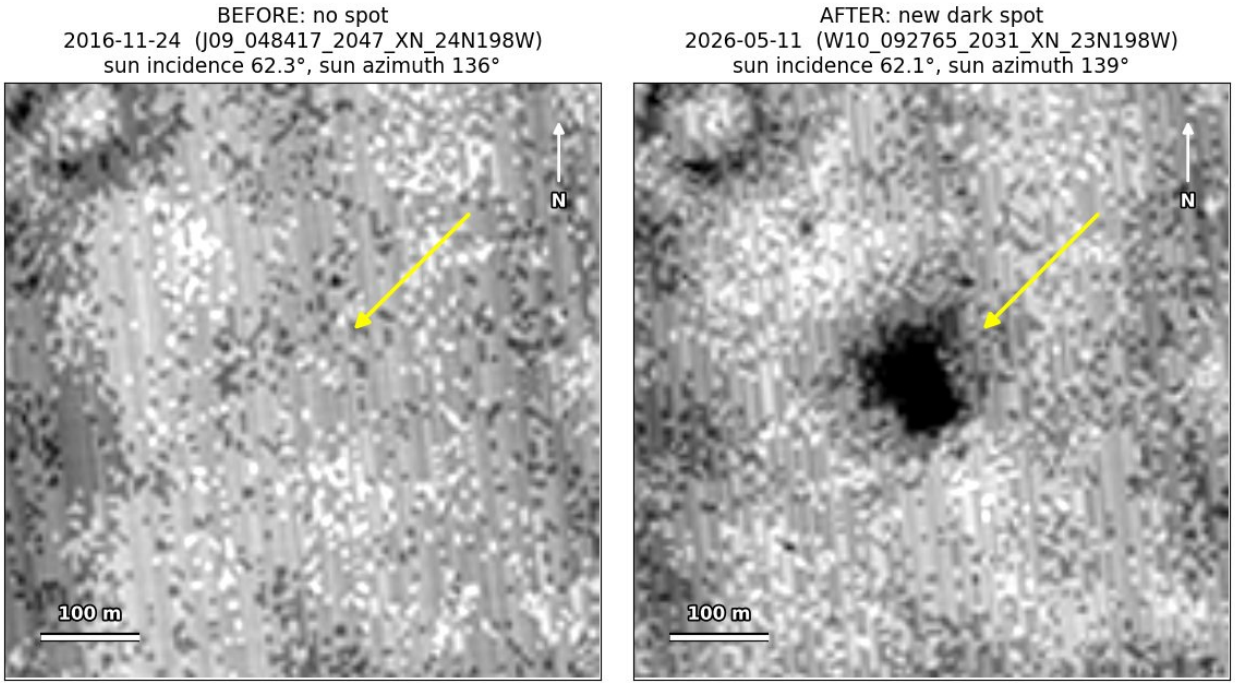

Figure 11: Candidate 2, before and after: the same 600 × 600 m of ground in 24 November 2016 (`J09_048417_2047_XN_24N198W`) and 11 May 2026 (`W10_092765_2031_XN_23N198W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS. Made for this paper from map-projected images distributed by Arizona State University.

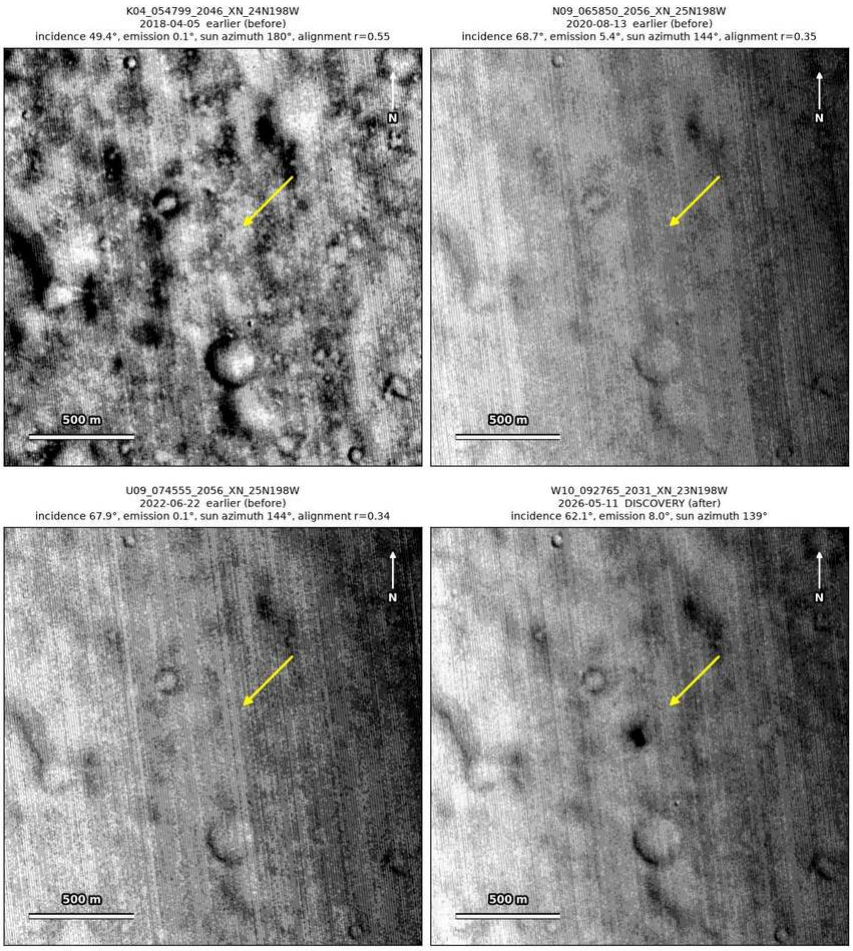

Figure 12: Candidate 2: four of the six CTX images of the same 2×2 km patch: April 2018, August 2020, June 2022 and May 2026. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 3: Wind-carved terrain south-east of Gordii Dorsum, south of Olympus Mons

Location	0.9058° N, 227.4609° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 55%
Formed between	20 March 2011 and 21 June 2020
Images	1 without the spot (absent: 20 March 2011); 4 with it (present: 21 June 2020 and three later images up to 2026)
Size	Compact dark spot about 40 m across
HiRISE	none covers the spot (nearest footprint edge 38 km)
Nearest catalogued new impact	54 km (Daubar et al., 2022)

What changed. A compact dark spot about 40 m across appeared on wind-carved terrain south-east of Gordii Dorsum between March 2011 and June 2020. Four CTX images from 2020 to 2026 show it fading steadily.

Why it may be a new impact. It shows in the same place in all four later images, so it is not a flaw in one image. It also shows under a higher sun than in the March 2011 image, in which it is absent, so it is not a shadow. A compact dark spot that appears and then fades steadily fits the blast zone of a small new crater.

Caveats. The sole earlier image lines up only modestly with the later ones, and the nearest dark feature shows only weakly in it, so it is less certain that this image would have shown the spot. A small change in the brightness of the ground caused by wind is also possible. No image from the high-resolution HiRISE camera covers it; the nearest ends 38 km away.

Figure note. This figure was made for this paper (Section 3.9) from the map-projected versions of the three images. It shows the spot absent in 2011, darkest in 2020 and fading by 2026. In the map-projected frame the spot lies 298 m from the catalogued position, at 0.9105° N, 227.4590° E.

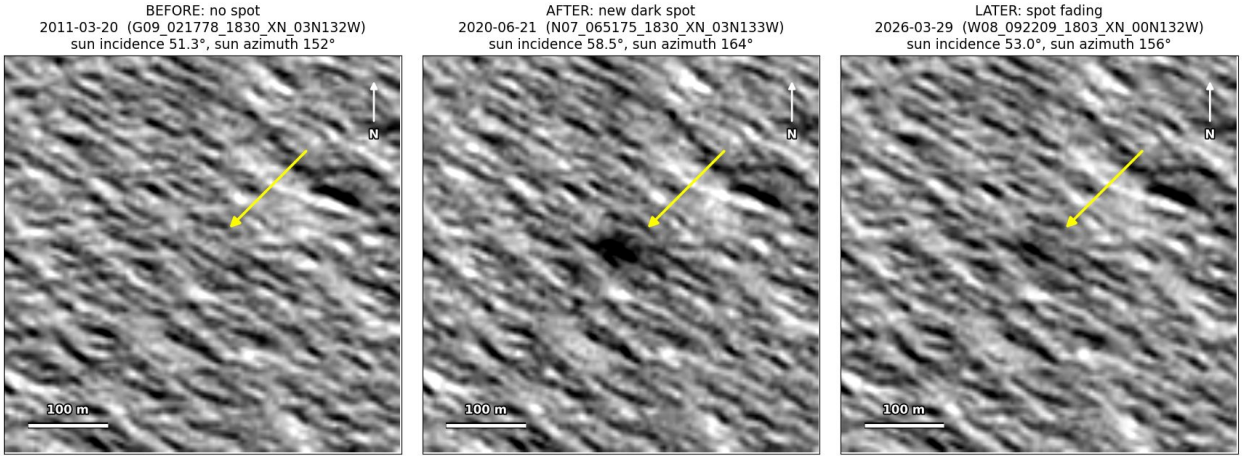

Figure 13: Candidate 3, before and after: the same 500 × 500 m of ground in 20 March 2011 (`G09_021778_1830_XN_03N132W`), 21 June 2020 (`N07_065175_1830_XN_03N133W`) and 29 March 2026 (`W08_092209_1803_XN_00N132W`), under the suns given above the panels; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS. Made for this paper from map-projected images distributed by Arizona State University.

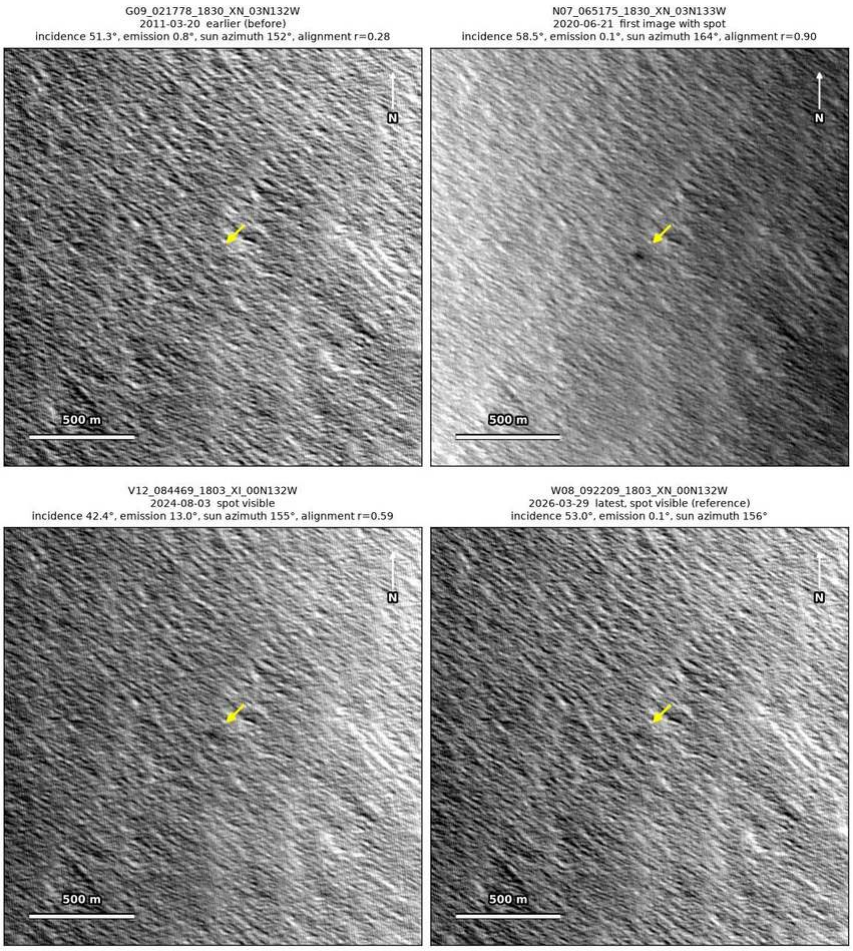

Figure 14: Candidate 3: four of the five CTX images of the same 2×2 km patch: 2011, 2020, 2024 and 2026. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 4: Plains in Arabia Terra, north-west of Pasteur crater

Location	25.4522° N, 20.9077° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	28 November 2010 and 10 February 2025
Images	1 without the spot (absent: 28 November 2010); 4 with it (present: 10 February 2025, 12 December 2025 (two images, CTX and HiRISE) and 24 April 2026)
Size	Dark core about 21 m across, 47 × 29 m including its tail
HiRISE	ESP_090832_2060 (2025-12-12, not impact-targeted)
Nearest catalogued new impact	131 km (Daubar et al., 2022)
Nearby	Candidate 10, 4.6 km away, appeared in the same time window.

What changed. A small dark spot with a tail, its core about 21 m across, appeared on plains in Arabia Terra, north-west of Pasteur crater, between November 2010 and February 2025. It shows in three CTX images up to April 2026 and in one HiRISE image.

Why it may be a new impact. The only earlier image, from November 2010, was taken under almost the same sun as the December 2025 image and shows nothing there. In the HiRISE image the spot is a sharp dark patch with finger-like edges on flat, dusty ground, with a dark point 1 to 2 m across at its edge, which is how the blast zone of a very small new crater looks.

Caveats. By chance, NASA's HiRISE camera photographed the spot on 12 December 2025, after it formed. Only one earlier image, from November 2010, shows the ground as it was before. The main alternative is a patch where wind has stripped away the dust, but that is unlikely, as the spot is compact and has a dark point at its edge.

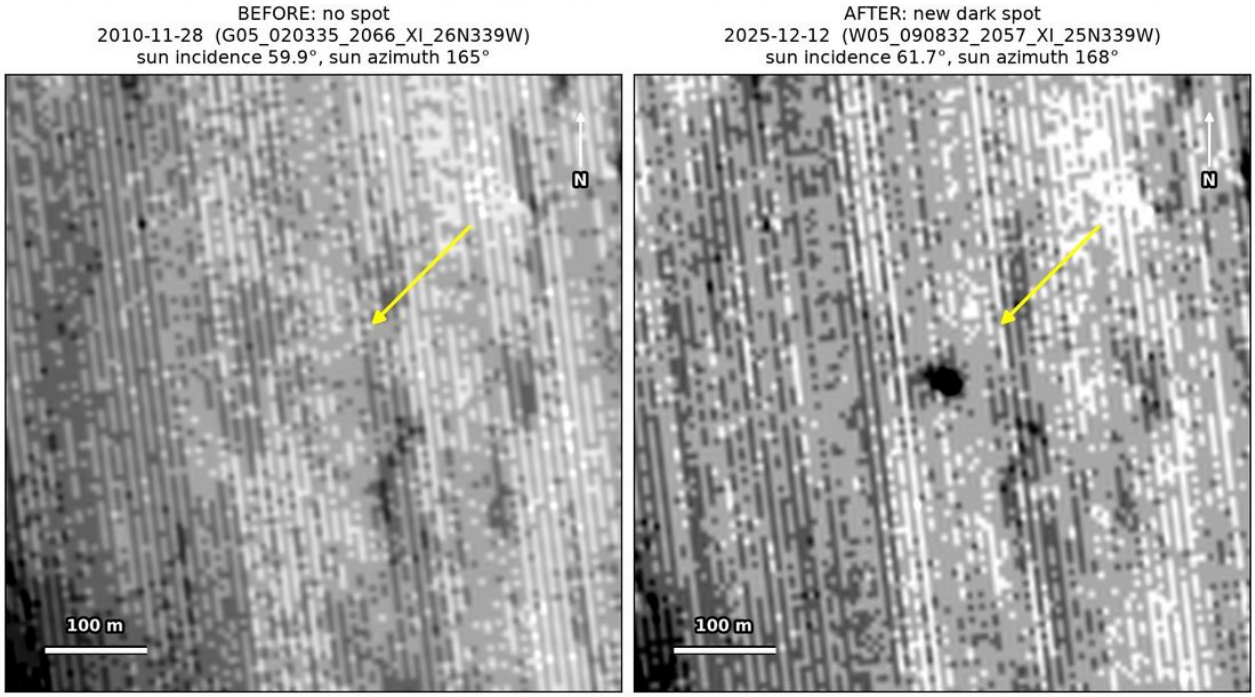

Figure 15: Candidate 4, before and after: the same 600 × 600 m of ground in 28 November 2010 (`G05_020335_2066_XI_26N339W`) and 12 December 2025 (`W05_090832_2057_XI_25N339W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

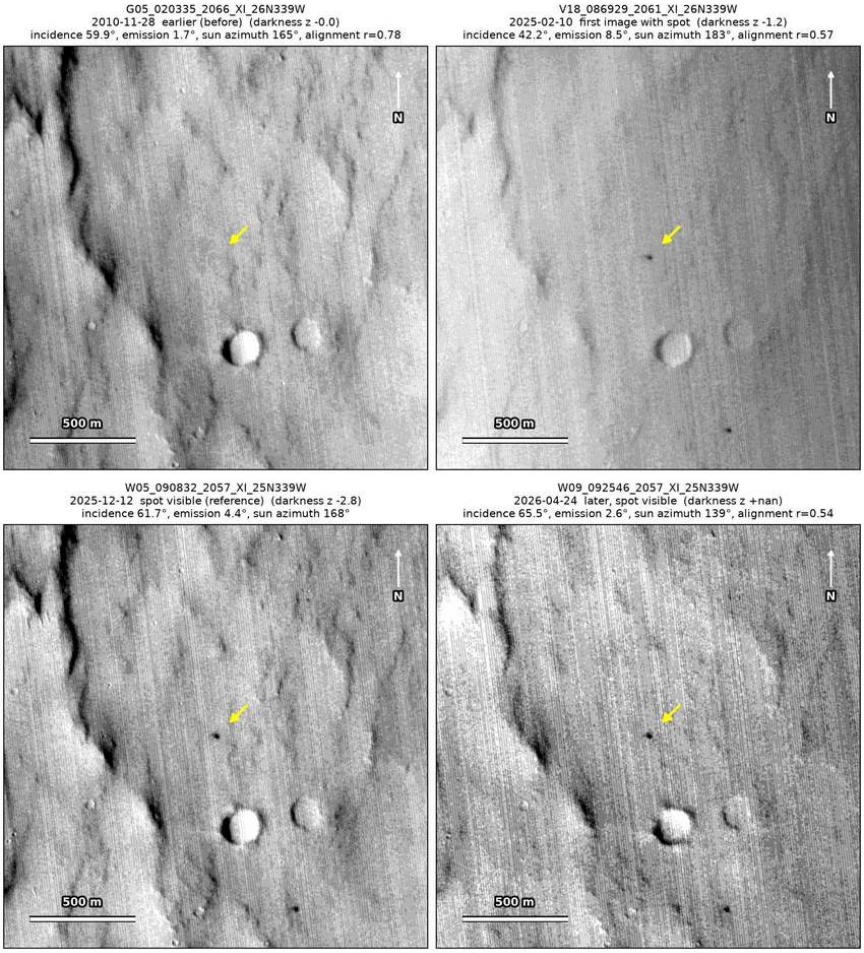

Figure 16: Candidate 4: four CTX images of the same 2×2 km patch, 2010 to 2026. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

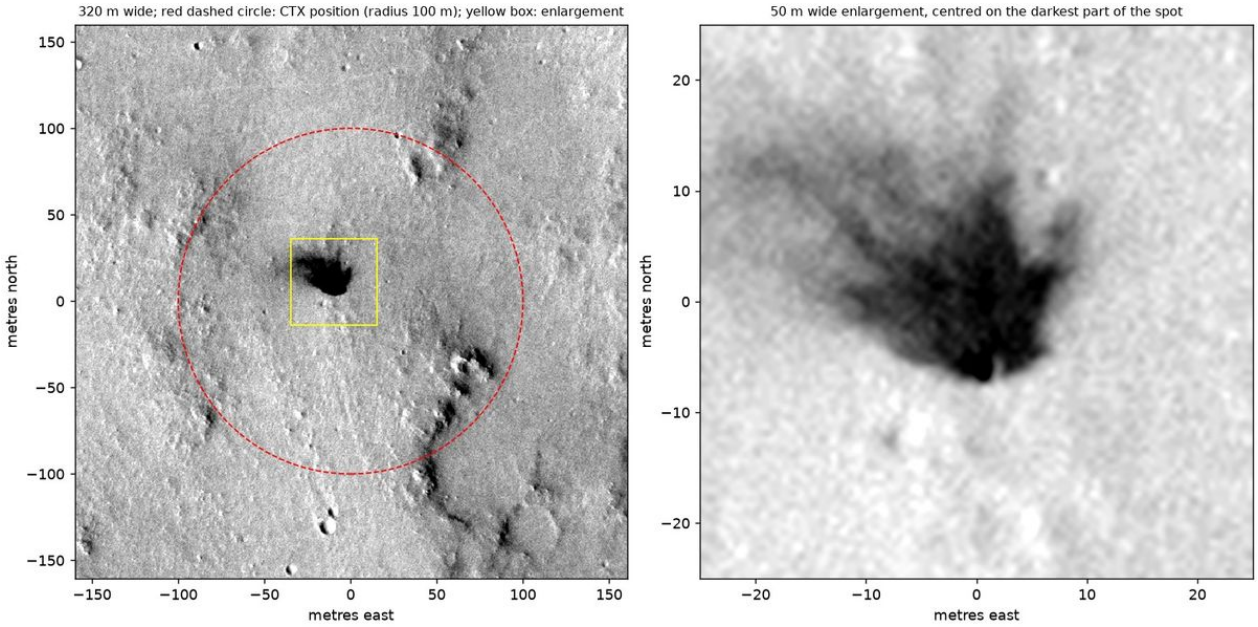

Figure 17: Candidate 4, HiRISE image `ESP_090832_2060`: HiRISE view of the spot (December 2025, 0.5 m per pixel), with a 50 m enlargement. NASA/JPL-Caltech/UArizona.

Candidate 5: Cerberus plains

Location	3.7234° N, 166.7553° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 70%
Formed between	10 October 2010 and 8 October 2023
Images	1 without the spot (absent: 10 October 2010); 4 with it (present: 8 October 2023, 7 December 2023, 4 January 2024 and 14 November 2025)
Size	Rayed dark spot 100 to 150 m across
HiRISE	none covers the spot (nearest footprint edge 39 km)
Nearest catalogued new impact	157 km (Daubar et al., 2022)

What changed. A rayed dark spot 100 to 150 m across, with dark fingers reaching out from a core of several dark blobs, appeared on the flat lava plains of Cerberus between October 2010 and October 2023. It looks identical in four CTX images taken over two years.

Why it may be a new impact. The only earlier image, from October 2010, was taken with the sun at almost the same height as in the four later images, from October 2023 to November 2025, and shows nothing there. Dark fingers spreading out from a dark core are how the blast zone of a small new crater looks.

Caveats. Wind stripping dust from a patch of ground could also make such a mark, while a slope streak (a dark streak left by dust sliding down a slope) on the narrow fracture beside it is unlikely. The spot might already be in a catalogue of new impacts from the era of NASA's InSight lander (Bickel et al. 2025) that could not be downloaded, though that is unlikely: every entry there has a HiRISE image, and none covers this spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 39 km away.

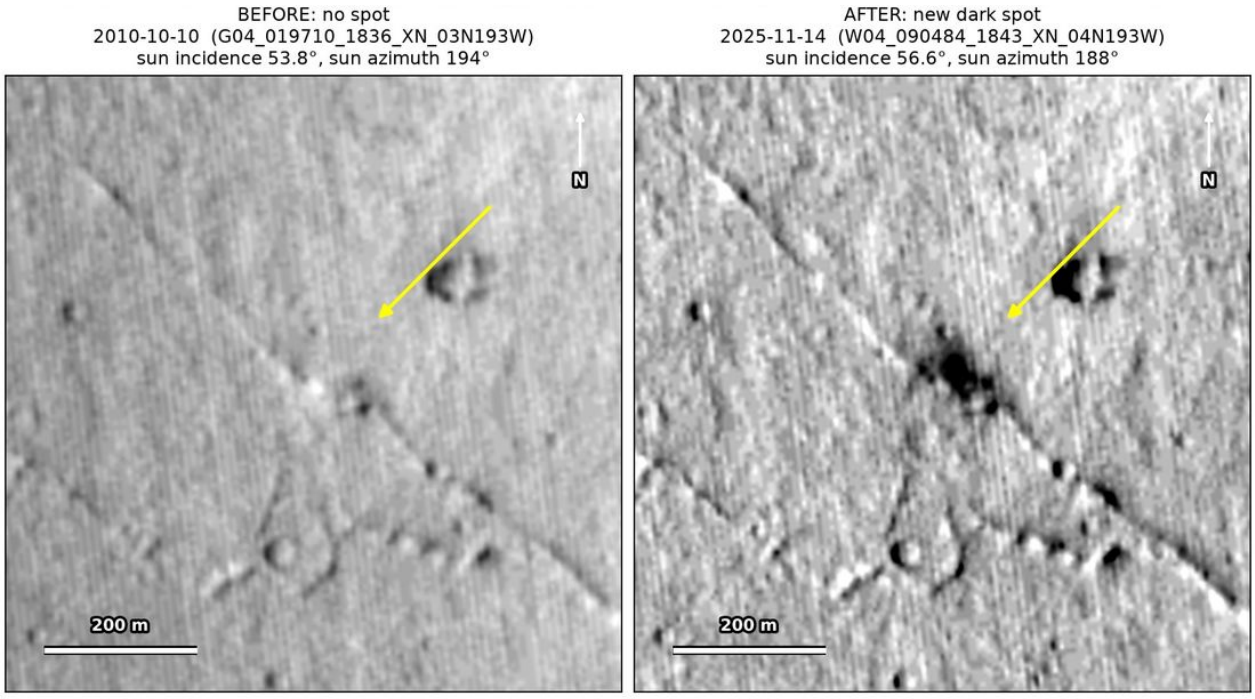

Figure 18: Candidate 5, before and after: the same 800 × 800 m of ground in 10 October 2010 (`G04_019710_1836_XN_03N193W`) and 14 November 2025 (`W04_090484_1843_XN_04N193W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

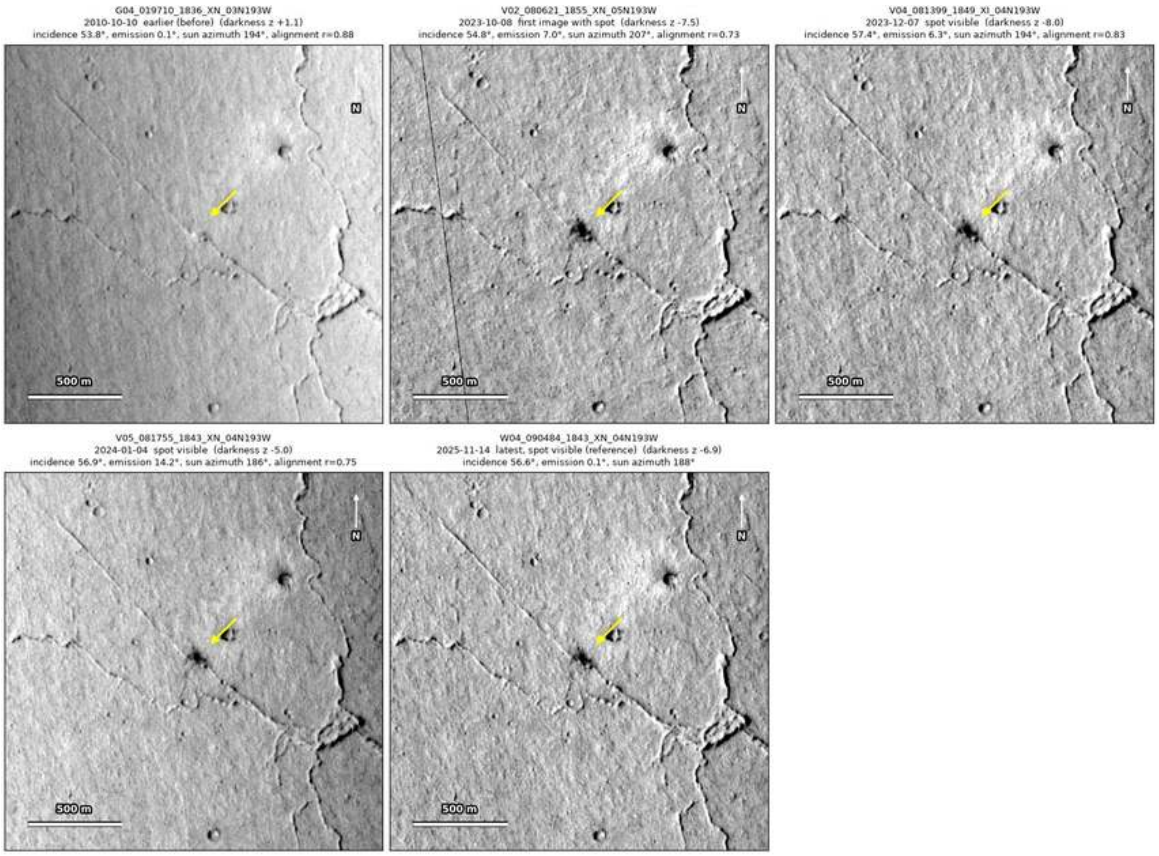

Figure 19: Candidate 5: five CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 6: Lava plains at Ulysses Fossae, between Olympus Mons and Ascraeus Mons

Location	14.2867° N, 235.8894° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 55%
Formed between	4 January 2011 and 29 November 2025
Images	2 without the spot (absent: 26 November 2010 and 4 January 2011); 1 with it (present: 29 November 2025)
Size	Diffuse dark patch 100 to 150 m across, with two or three darker blobs inside
HiRISE	none covers the spot (nearest footprint edge 24 km)
Nearest catalogued new impact	101 km (Daubar et al., 2022)

What changed. A diffuse dark patch 100 to 150 m across, with two or three darker blobs inside, appeared on the very dusty lava plains of Tharsis, at Ulysses Fossae, between January 2011 and November 2025. So far it is seen in only one CTX image.

Why it may be a new impact. Two earlier images, from November 2010 and January 2011, both taken under almost the same sun as the November 2025 image, show nothing at the spot, not even a landform. A dark patch appearing on very dusty plains can be the blast zone of a small new crater.

Caveats. Only one image shows it so far, so a flaw in that image cannot be fully ruled out. Because the patch is diffuse, darkening by wind or a dust devil is also possible. No image from the high-resolution HiRISE camera covers it; the nearest ends 24 km away.

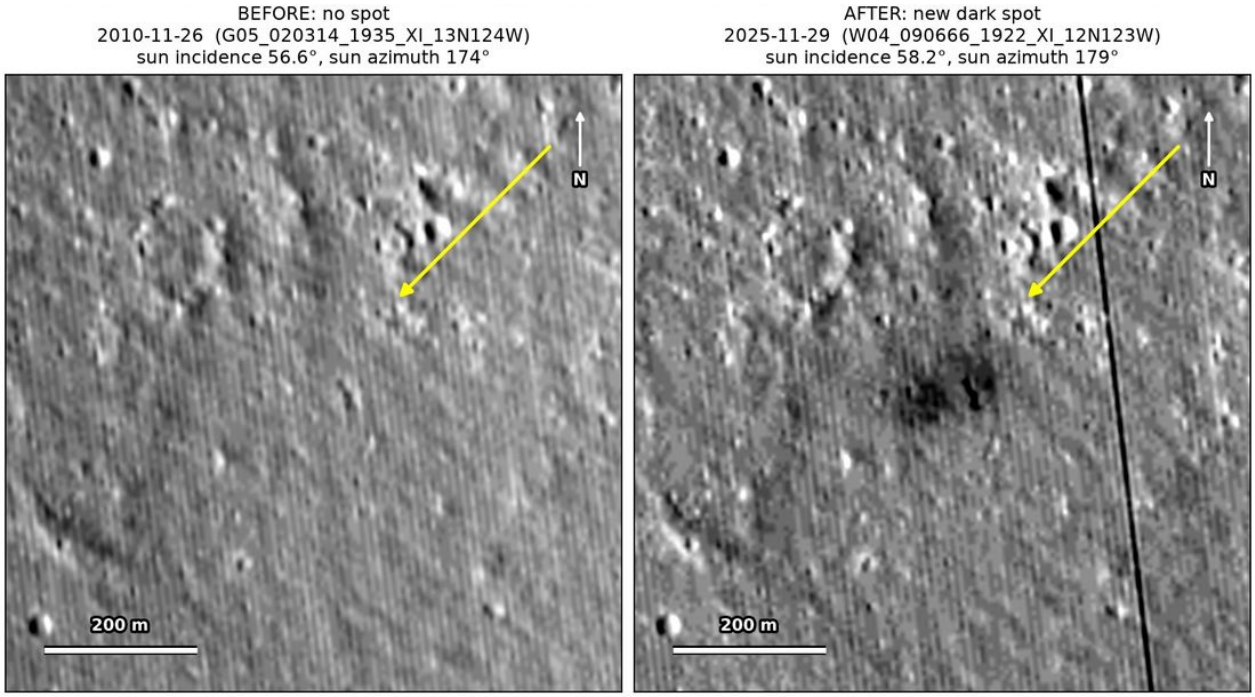

Figure 20: Candidate 6, before and after: the same 800 × 800 m of ground in 26 November 2010 (`G05_020314_1935_XI_13N124W`) and 29 November 2025 (`W04_090666_1922_XI_12N123W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

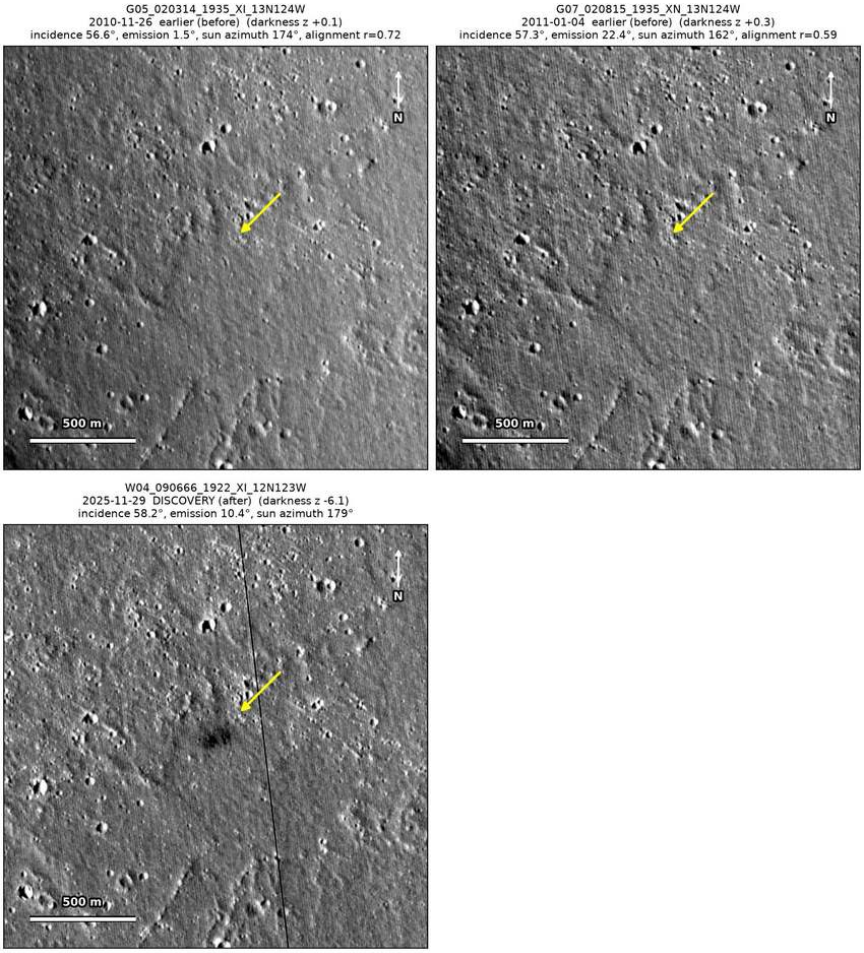

Figure 21: Candidate 6: three CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 7: Ridged terrain east of Gordii Dorsum, south of Olympus Mons

Location	2.8777° N, 225.1876° E (planetocentric)
Assessment	weak candidate; probability of a new impact about 20%
Formed between	19 April 2024 and 1 March 2026
Images	4 without the spot (absent: 11 April 2011, 21 September 2020, 12 August 2022 and 19 April 2024); 1 with it (present: 1 March 2026)
Size	Elongated dark spot 60 to 80 m in size, stretched along the ridges
HiRISE	none covers the spot (nearest footprint edge 15 km)
Nearest catalogued new impact	68 km (Daubar et al., 2022)

What changed. An elongated dark spot 60 to 80 m in size appeared on ridged terrain east of Gordii Dorsum between April 2024 and March 2026, on top of a faint dark mark that was already there. So far it is seen in only one CTX image.

Why it may be a new impact. The darkening is real: the April 2024 image, taken under nearly the same sun as the March 2026 image, does not show it. A new dark spot of this size could be the blast zone of a small new crater.

Caveats. It is a weak candidate: the faint mark beneath it was already visible in 2011, 2020 and 2022, in ridged terrain full of ridge shadows; the spot is stretched along the ridges; and only one image shows it. It could instead be wind darkening a small hollow, or shading made stronger by clearer air, as the April 2024 image was hazy and taken at a slant of 17°. No image from the high-resolution HiRISE camera covers it; the nearest ends 15 km away.

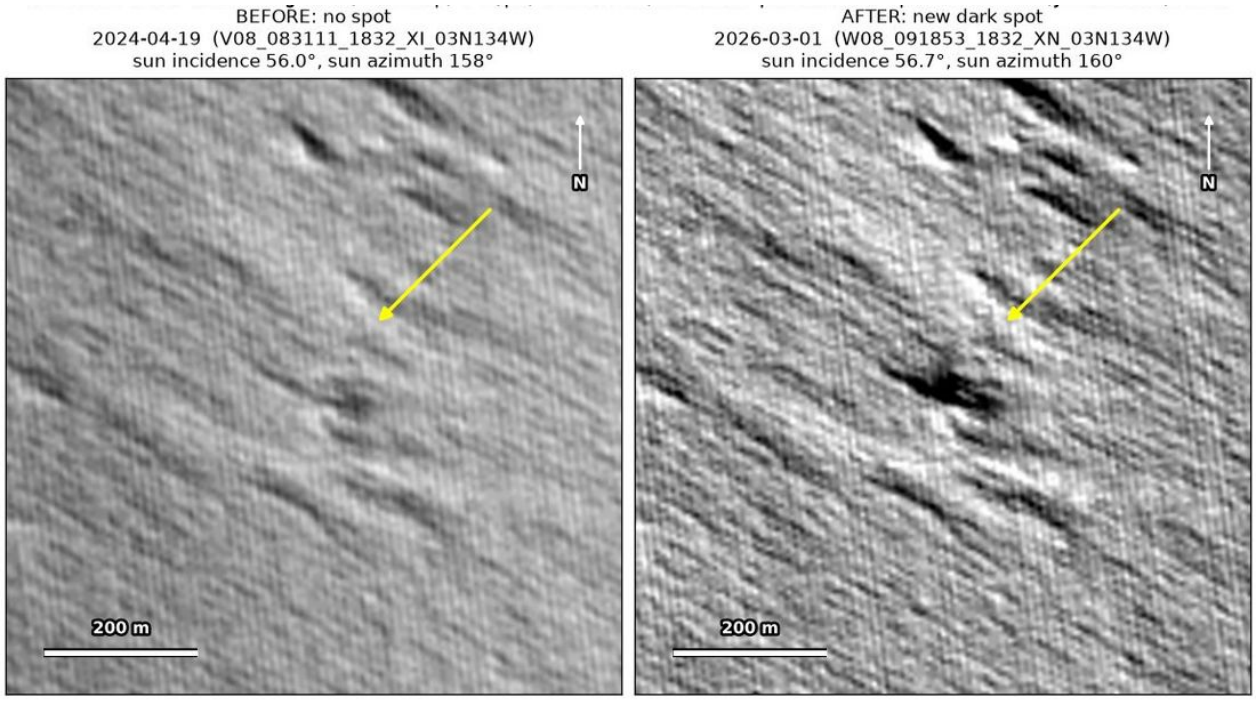

Figure 22: Candidate 7, before and after: the same 800 × 800 m of ground in 19 April 2024 (`V08_083111_1832_XI_03N134W`) and 1 March 2026 (`W08_091853_1832_XN_03N134W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

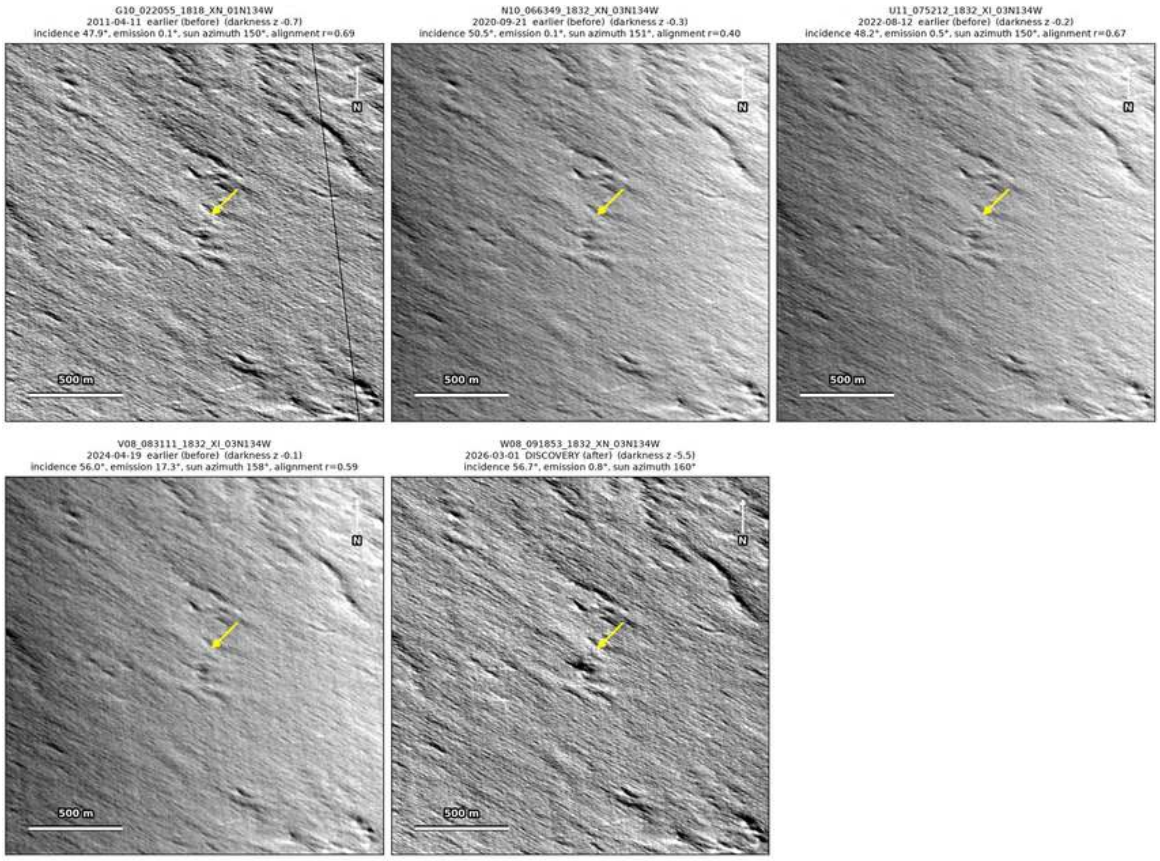

Figure 23: Candidate 7: five CTX images of the same 2×2 km patch, 2011 to 2026. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 8: Plains in Arabia Terra, west of Flammarion crater

Location	23.7579° N, 45.8964° E (planetocentric)
Assessment	weak candidate; probability of a new impact about 40%
Formed between	29 March 2020 and 13 November 2025
Images	2 without the spot (absent: 21 November 2010 and 29 March 2020); 1 with it (present: 13 November 2025)
Size	Compact dark dot 30 to 40 m across
HiRISE	none covers the spot (nearest footprint edge 38 km)
Nearest catalogued new impact	240 km (Daubar et al., 2022)

What changed. A compact dark dot 30 to 40 m across appeared on dusty plains of Arabia Terra, west of Flammarion crater, between March 2020 and November 2025. So far it is seen in only one CTX image.

Why it may be a new impact. Two earlier images, from November 2010 and March 2020, both taken under almost the same sun as the November 2025 image, show nothing at the spot, not even a landform. A compact dark dot appearing on dusty plains is how the blast zone of a small new crater can look.

Caveats. Only one image shows it, and that image is heavily compressed, so the dot may be a cluster of noise or a compression flaw rather than a real mark on the ground. No image from the high-resolution HiRISE camera covers it; the nearest ends 38 km away.

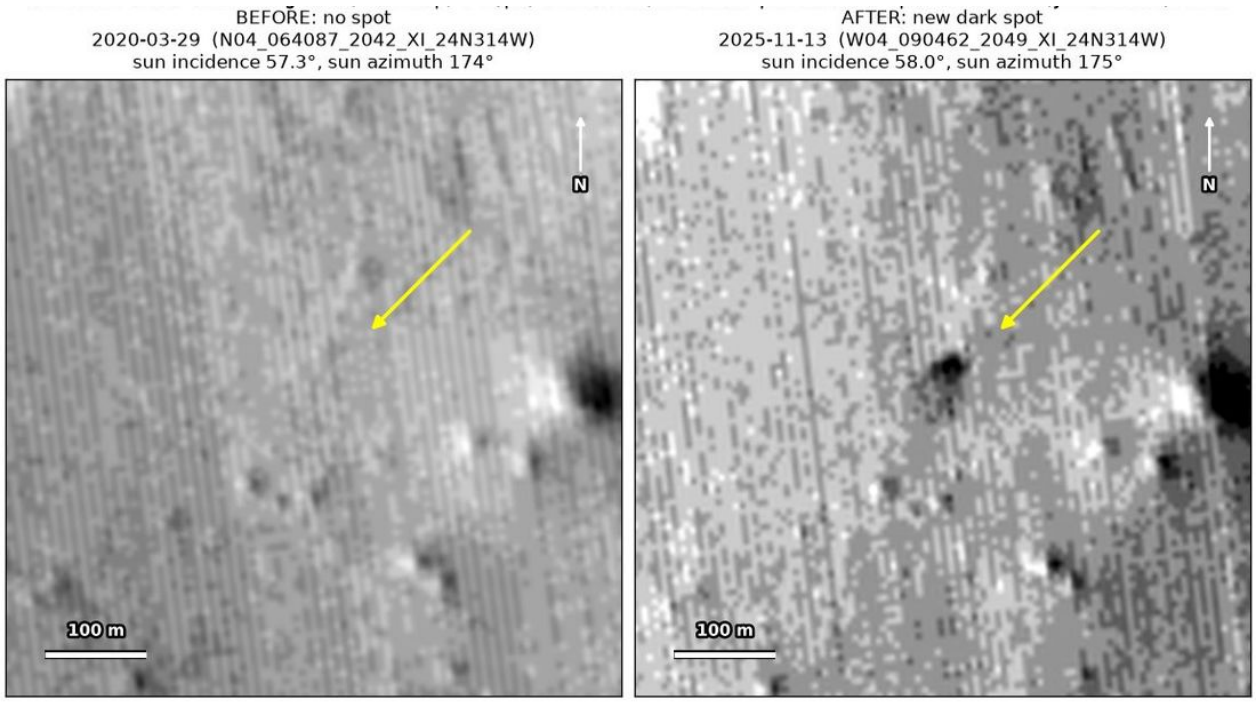

Figure 24: Candidate 8, before and after: the same 600 × 600 m of ground in 29 March 2020 (`N04_064087_2042_XI_24N314W`) and 13 November 2025 (`W04_090462_2049_XI_24N314W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

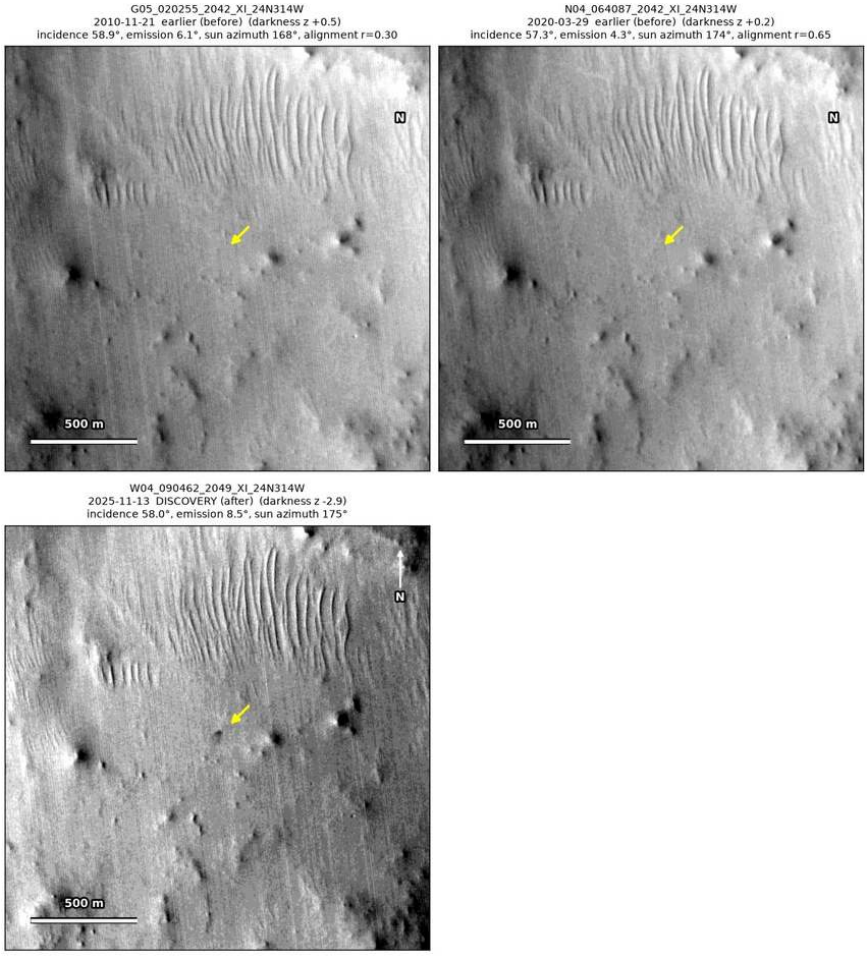

Figure 25: Candidate 8: three CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 9: Plains of Elysium Planitia

Location	6.0891° N, 157.2199° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 75%
Formed between	27 January 2008 and 11 August 2024
Images	3 without the spot (absent: 3 October 2006, 4 December 2006 and 27 January 2008); 3 with it (present: 11 August 2024, 22 December 2024 and 16 December 2025)
Size	Two dark spots, each about 20 to 25 m across, 35 m apart, one north of the other
HiRISE	none covers the spot (nearest footprint edge 37 km)
Nearest catalogued new impact	50 km (Daubar et al., 2022)

What changed. Two compact dark spots, each about 20 to 25 m across and 35 m apart, appeared on the smooth plains of Elysium Planitia between January 2008 and August 2024. They look identical in three CTX images from August 2024 to December 2025.

Why it may be a new impact. All three earlier images, from 2006 to 2008, show nothing there, while an older dark spot 190 m away shows in each of them and lines up in every image, so they would have shown the pair. Two compact spots that stay unchanged for 16 months fit the blast zones of small new impacts better than dark ground uncovered by wind.

Caveats. Only one thing keeps it from being a strong candidate: on these almost featureless plains, none of the earlier images lines up with the later ones as closely as a strong rating requires. No CTX image of the site was taken between January 2008 and August 2024, so it could have formed at any time in that gap of more than 16 years. No image from the high-resolution HiRISE camera covers it; the nearest ends 37 km away.

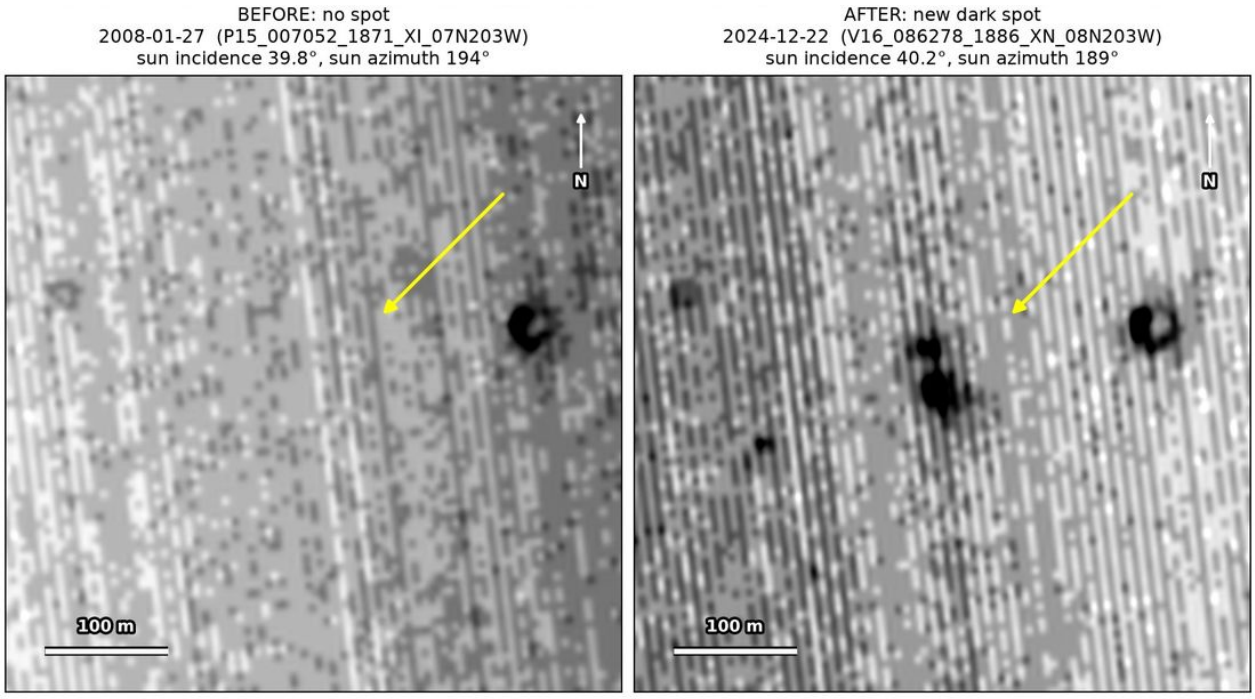

Figure 26: Candidate 9, before and after: the same 500 × 500 m of ground in 27 January 2008 (`P15_007052_1871_XI_07N203W`) and 22 December 2024 (`V16_086278_1886_XN_08N203W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

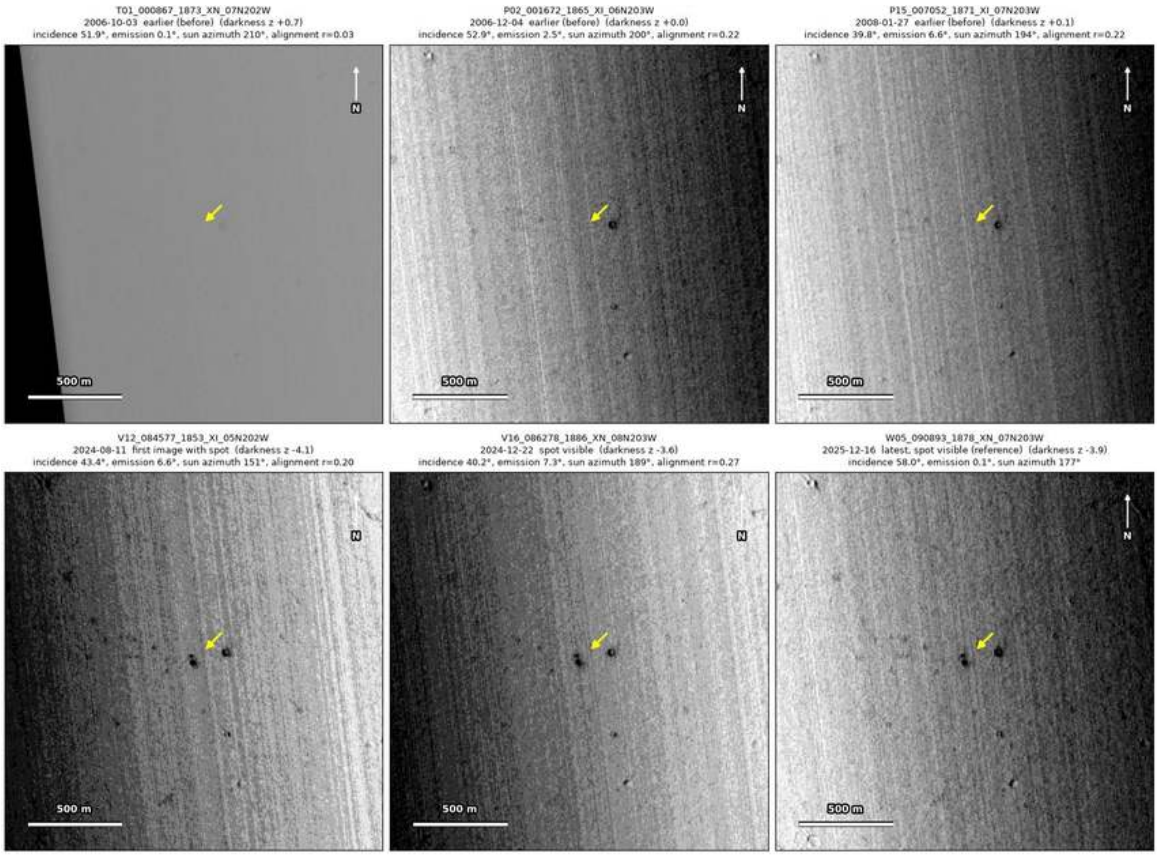

Figure 27: Candidate 9: six CTX images of the same 2×2 km patch, 2006 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 10: Plains in Arabia Terra, 4.6 km south-south-east of candidate 4

Location	25.3879° N, 20.9581° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 45%
Formed between	28 November 2010 and 10 February 2025
Images	1 without the spot (absent: 28 November 2010); 3 with it (present: 10 February 2025, 12 December 2025 and 24 April 2026 (faint))
Size	Small dark spot about 20 to 25 m across
HiRISE	none covers the spot
Nearest catalogued new impact	132 km (Daubar et al., 2022)
Nearby	Candidate 4, 4.6 km away, appeared in the same time window.

What changed. A small, compact dark spot about 20 to 25 m across appeared on the plains of Arabia Terra, just south of an older dark streak, between November 2010 and February 2025. It shows in three CTX images, though only faintly in the last, from April 2026.

Why it may be a new impact. The November 2010 image shows nothing at the spot, yet the older streak beside it shows clearly, so that image would have shown the spot too. A compact new dark spot fits the blast zone of a small new crater, and since it appeared in the same time window as candidate 4, 4.6 km away, it may mark a second fragment of the same impactor.

Caveats. It is faint, with weak contrast in noisy, compressed images, and it lies just south of an older dark streak, so it may simply be that streak growing longer. No image from the high-resolution HiRISE camera covers it; the nearest ends 1.8 km away.

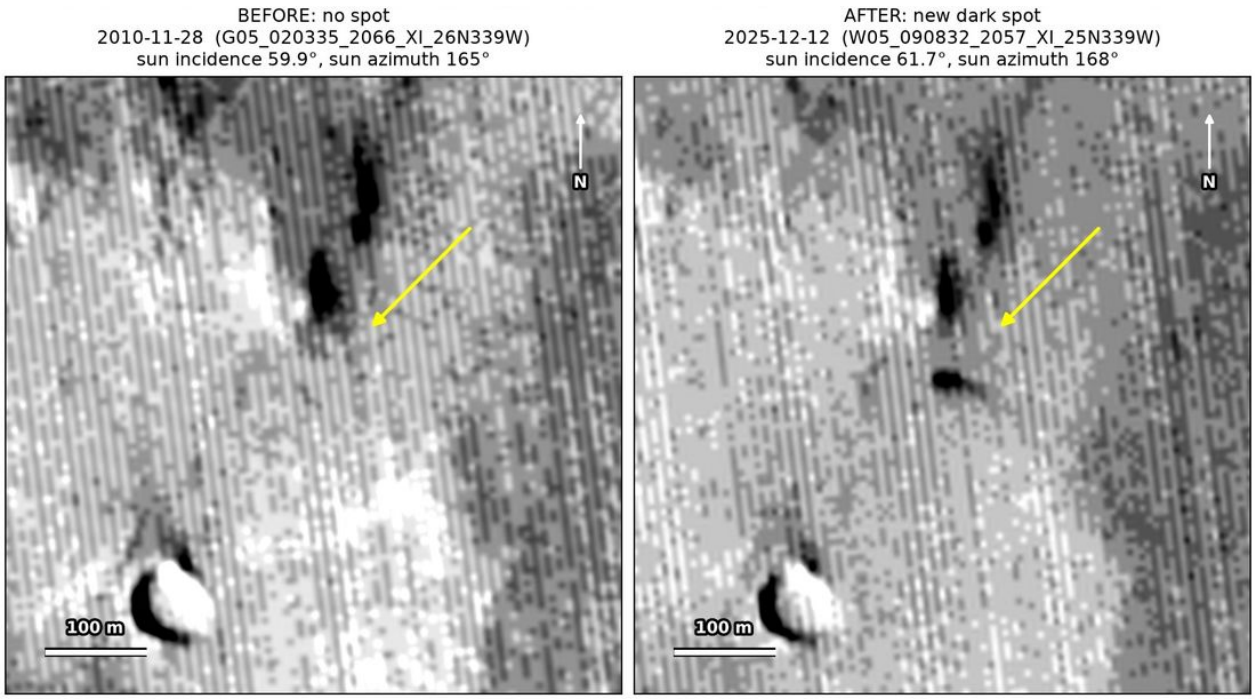

Figure 28: Candidate 10, before and after: the same 600 × 600 m of ground in 28 November 2010 (`G05_020335_2066_XI_26N339W`) and 12 December 2025 (`W05_090832_2057_XI_25N339W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

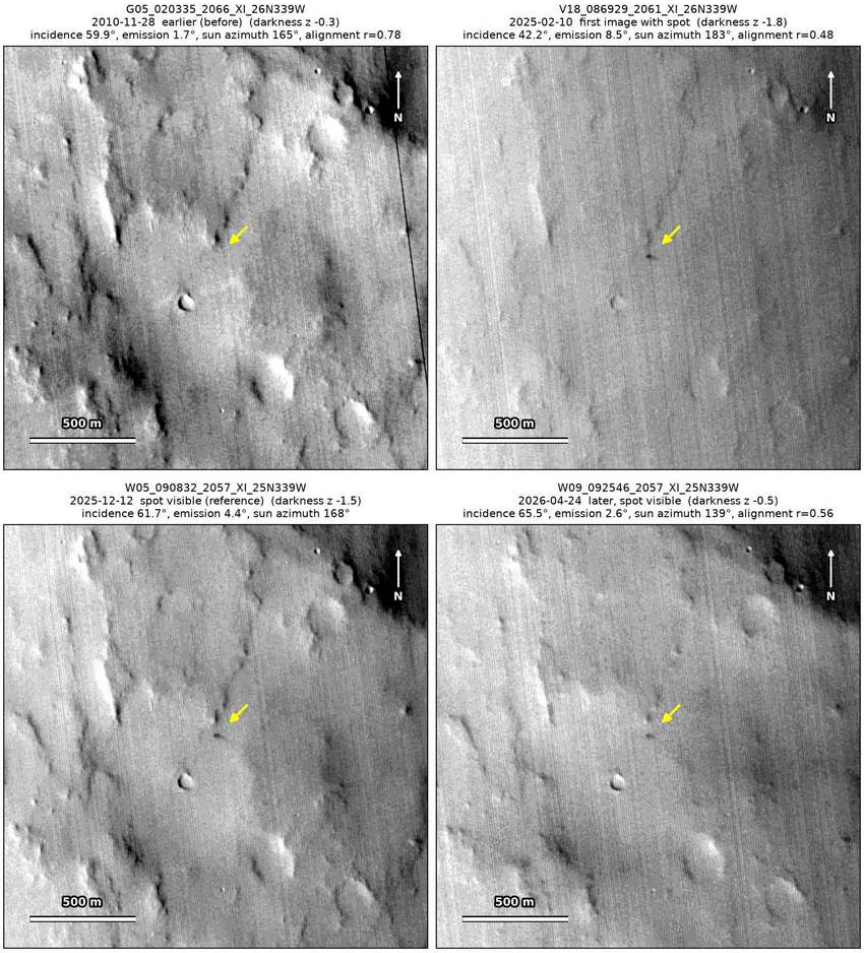

Figure 29: Candidate 10: four CTX images of the same 2×2 km patch, 2010 to 2026. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 11: Lava plains in Elysium Planitia, west of Albor Tholus

Location	16.8842° N, 147.8693° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	28 July 2022 and 10 April 2025
Images	2 without the spot (absent: 22 May 2014 and 28 July 2022); 4 with it (present: 10 April 2025, 18 July 2025, 15 August 2025 and 28 November 2025)
Size	Dark spot 25 to 30 m across, plus a fainter new companion 195 m to the north-north-west
HiRISE	none covers the spot (nearest footprint edge 6.2 km)
Nearest catalogued new impact	117 km (Daubar et al., 2022)
Nearby	Candidate 16, 13 km to the west-north-west, appeared earlier, between May 2014 and July 2022.

What changed. A compact dark spot 25 to 30 m across appeared on lava plains west of Albor Tholus between July 2022 and April 2025, and is unchanged in four CTX images over seven months. A fainter companion 195 m to the north-north-west is new too.

Why it may be a new impact. Neither the May 2014 image, taken under the same sun as the August 2025 image, nor the July 2022 image shows anything there, and the craters around the spot line up between the images. A compact spot that stays unchanged for seven months is not a dust-devil track or a wind streak; it looks like the blast zone of a small new crater, and with its new companion it forms a small cluster.

Caveats. The July 2022 image, the last one without the spot, is less well aligned than the May 2014 one, and an image from May 2024 does not reveal whether the spot was there yet. No image from the high-resolution HiRISE camera covers it; the nearest ends 6.2 km away.

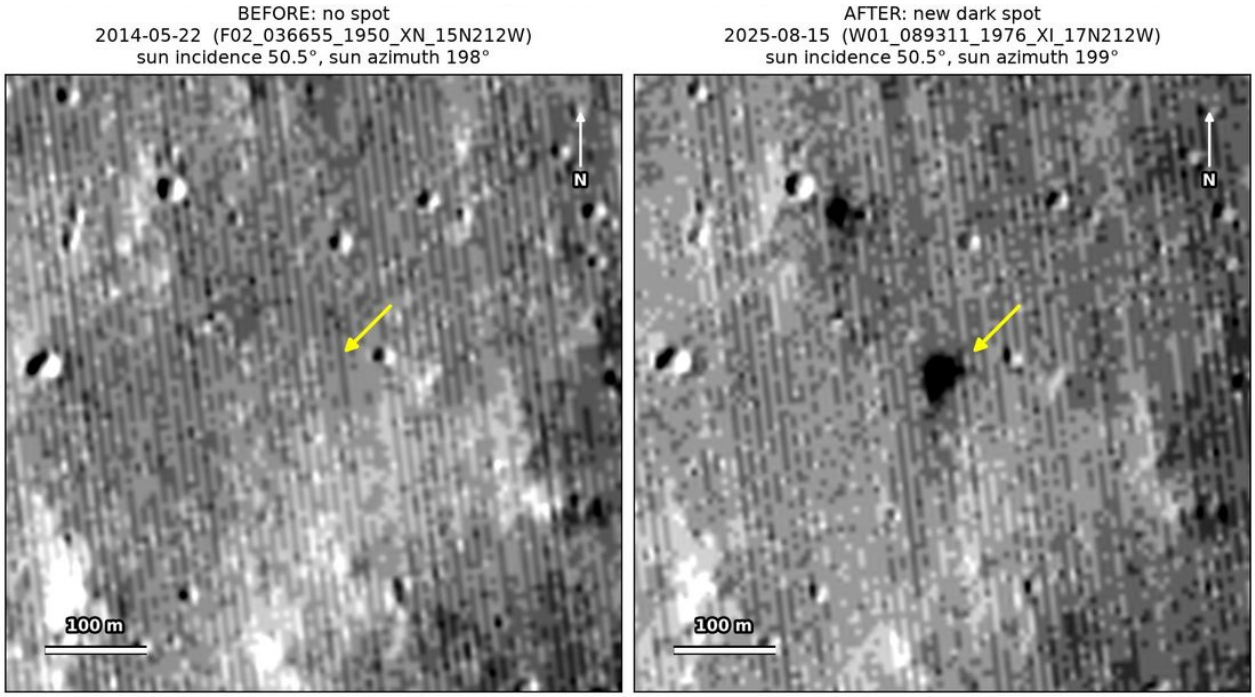

Figure 30: Candidate 11, before and after: the same 600 × 600 m of ground in 22 May 2014 (`F02_036655_1950_XN_15N212W`) and 15 August 2025 (`W01_089311_1976_XI_17N212W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

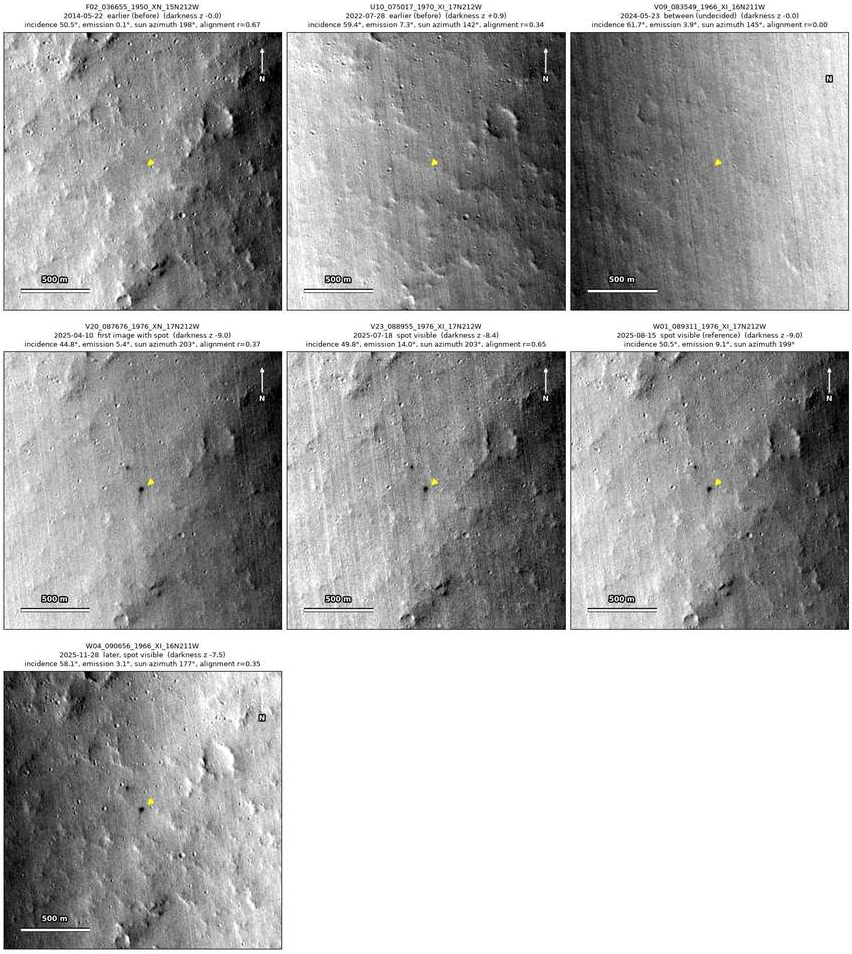

Figure 31: Candidate 11: seven CTX images of the same 2×2 km patch, 2014 to 2025. The May 2024 image is shown but not used to date the spot, as it does not reveal whether the spot was there yet. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 12: Lava plains of Daedalia Planum, west of Aganippe Fossa

Location	6.8683° S, 229.4246° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	30 November 2020 and 8 December 2024
Images	2 without the spot (absent: 22 February 2008 and 30 November 2020); 2 with it (present: 8 December 2024 and 10 November 2025)
Size	Very dark, lobed spot 50 to 60 m across
HiRISE	none covers the spot (nearest footprint edge 21.7 km)
Nearest catalogued new impact	51 km (Daubar et al., 2022)

What changed. A very dark, lobed spot 50 to 60 m across appeared on the lava plains of Daedalia Planum, west of Aganippe Fossa, between November 2020 and December 2024. It keeps the same shape in two CTX images 11 months apart.

Why it may be a new impact. Two earlier images, from February 2008 and November 2020, show nothing there. The spot keeps its shape in December 2024 and November 2025 with the sun at quite different heights, so it is not a shadow, and a very dark spot appearing on plains fits the blast zone of a small new crater.

Caveats. Only two images show it clearly; a hazy September 2025 image shows only a faint smudge and is not used. A slope streak, a dark streak left by dust sliding down a slope, is unlikely, as the spot lies on plains. No image from the high-resolution HiRISE camera covers it; the nearest ends 21.7 km away.

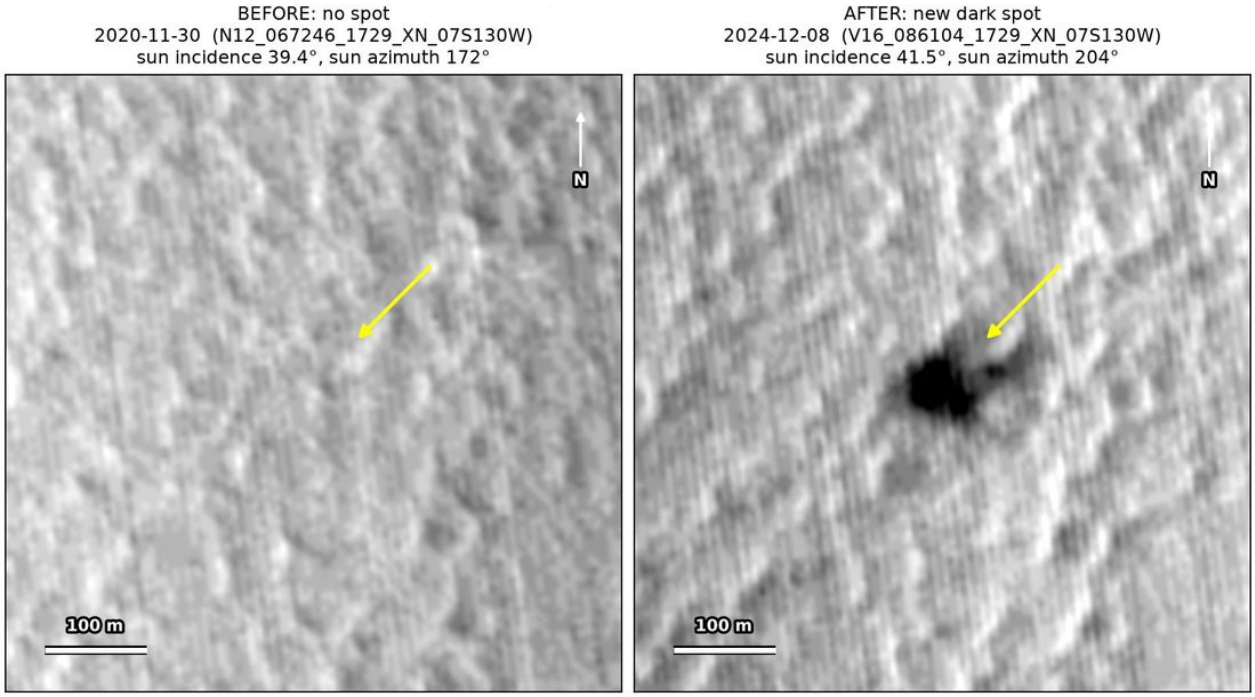

Figure 32: Candidate 12, before and after: the same 600 × 600 m of ground in 30 November 2020 (`N12_067246_1729_XN_07S130W`) and 8 December 2024 (`V16_086104_1729_XN_07S130W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

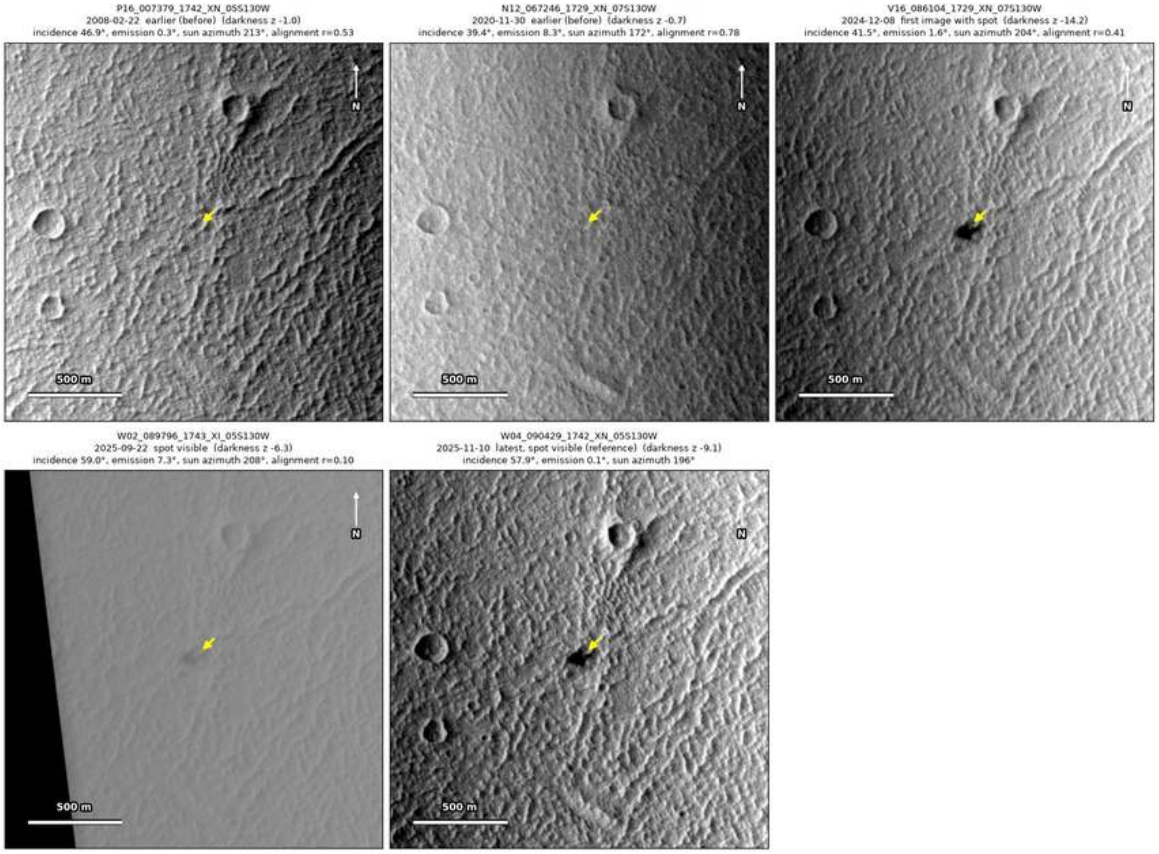

Figure 33: Candidate 12: five CTX images of the same 2×2 km patch, 2008 to 2025. The hazy September 2025 image, which shows only a faint smudge, is shown but not used. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 13: Plains south of Acheron Fossae

Location	35.5125° N, 230.4463° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 70%
Formed between	11 May 2016 and 1 September 2025
Images	3 without the spot (absent: 23 March 2007, 25 August 2009 and 11 May 2016); 2 with it (present: 1 September 2025 and 26 October 2025)
Size	Diffuse dark patch 100 to 120 m across, with a darker core
HiRISE	none covers the spot (nearest footprint edge 15.2 km)
Nearest catalogued new impact	30 km (Daubar et al., 2022)

What changed. A diffuse dark patch 100 to 120 m across, with a darker core, appeared on textured plains south of Acheron Fossae between May 2016 and September 2025. It looks the same in two CTX images from September and October 2025.

Why it may be a new impact. Three earlier images, from 2007, 2009 and 2016, show nothing there, although an older feature 300 m away shows in each; the May 2016 one was taken under the same sun as the September 2025 image. In October 2025 the patch looks the same under a different sun and viewing angle. Dark rather than bright, and seen in northern summer, it is not frost, and a round dark patch with a darker core fits the blast zone of a small new crater.

Caveats. It could have formed at any time in the nine years between May 2016 and September 2025. A slope streak, a dark streak left by dust sliding down a slope, is unlikely: the patch is round and there is no suitable slope. No image from the high-resolution HiRISE camera covers it; the nearest ends 15.2 km away.

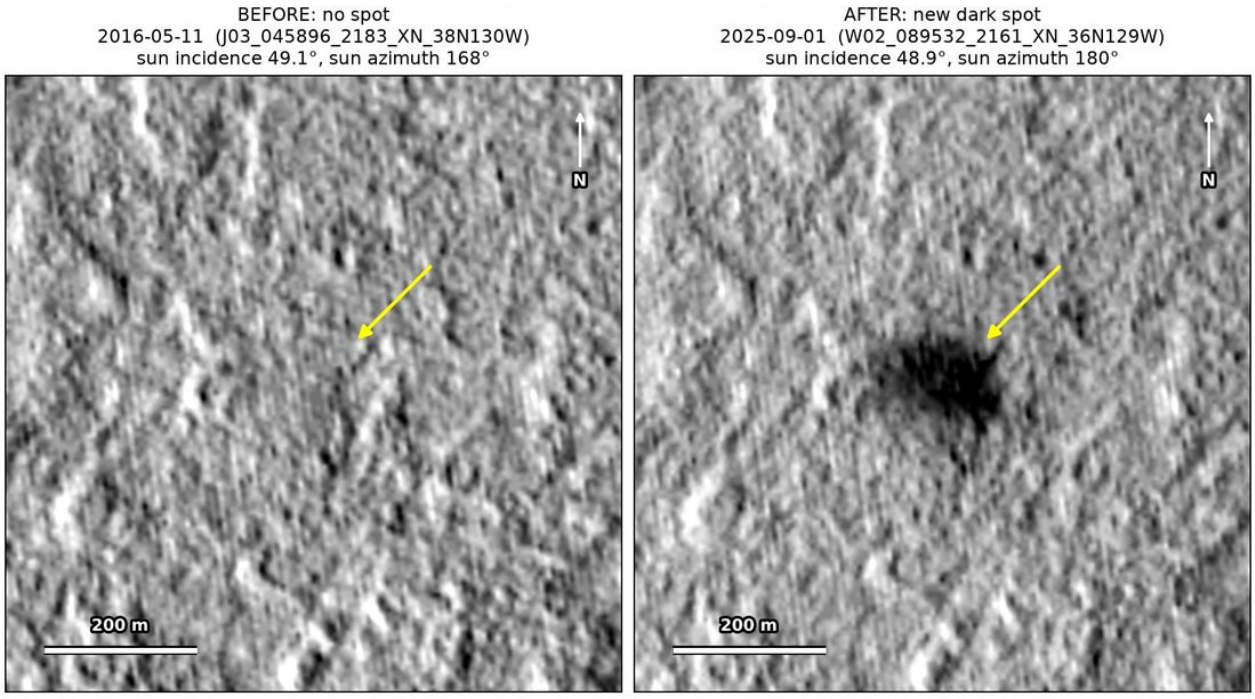

Figure 34: Candidate 13, before and after: the same 800 × 800 m of ground in 11 May 2016 (`J03_045896_2183_XN_38N130W`) and 1 September 2025 (`W02_089532_2161_XN_36N129W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

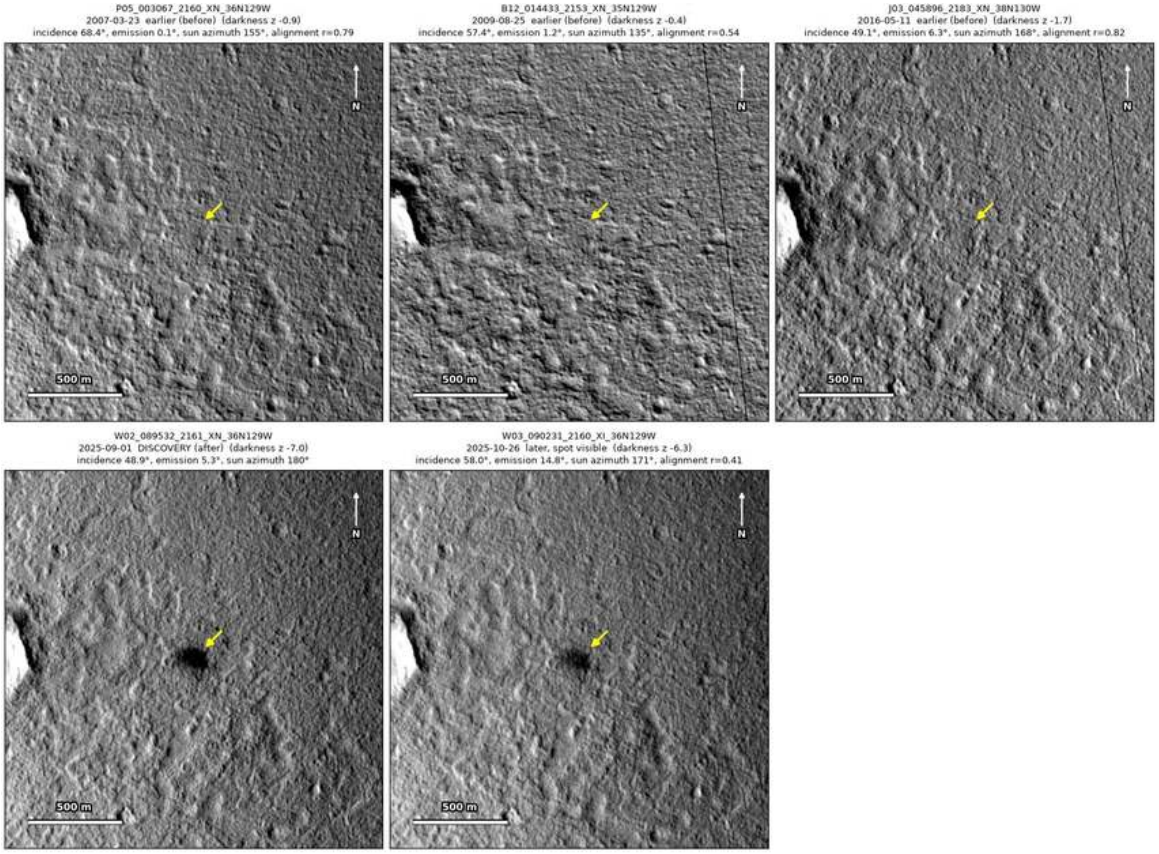

Figure 35: Candidate 13: five CTX images of the same 2×2 km patch, 2007 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 14: Plains in Arabia Terra, north-west of Schroeter crater

Location	7.3638° N, 51.0985° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	20 March 2014 and 15 July 2025
Images	2 without the spot (absent: 2 April 2012 and 20 March 2014); 1 with it (present: 15 July 2025)
Size	Sharp, very dark dot 30 to 35 m across
HiRISE	none covers the spot (nearest footprint edge 58.9 km)
Nearest catalogued new impact	72 km (Daubar et al., 2022)

What changed. A sharp, very dark dot 30 to 35 m across appeared on the plains of Arabia Terra, north-west of Schroeter crater, between March 2014 and July 2025. So far it is seen in only one CTX image.

Why it may be a new impact. The two earlier images, from April 2012 and March 2014, were taken under much the same sun as the July 2025 image and show nothing at the spot, although a dark feature 204 m away shows in both, so they would have shown the dot. A sharp, compact dark dot appearing on plains fits the blast zone of a small new crater.

Caveats. Only one image shows it, and its data are heavily compressed. A flaw from compression or transmission is unlikely, as such flaws look like lines or blocks rather than a round dot six pixels across, but it cannot be ruled out without a second image. No image from the high-resolution HiRISE camera covers it; the nearest ends 58.9 km away.

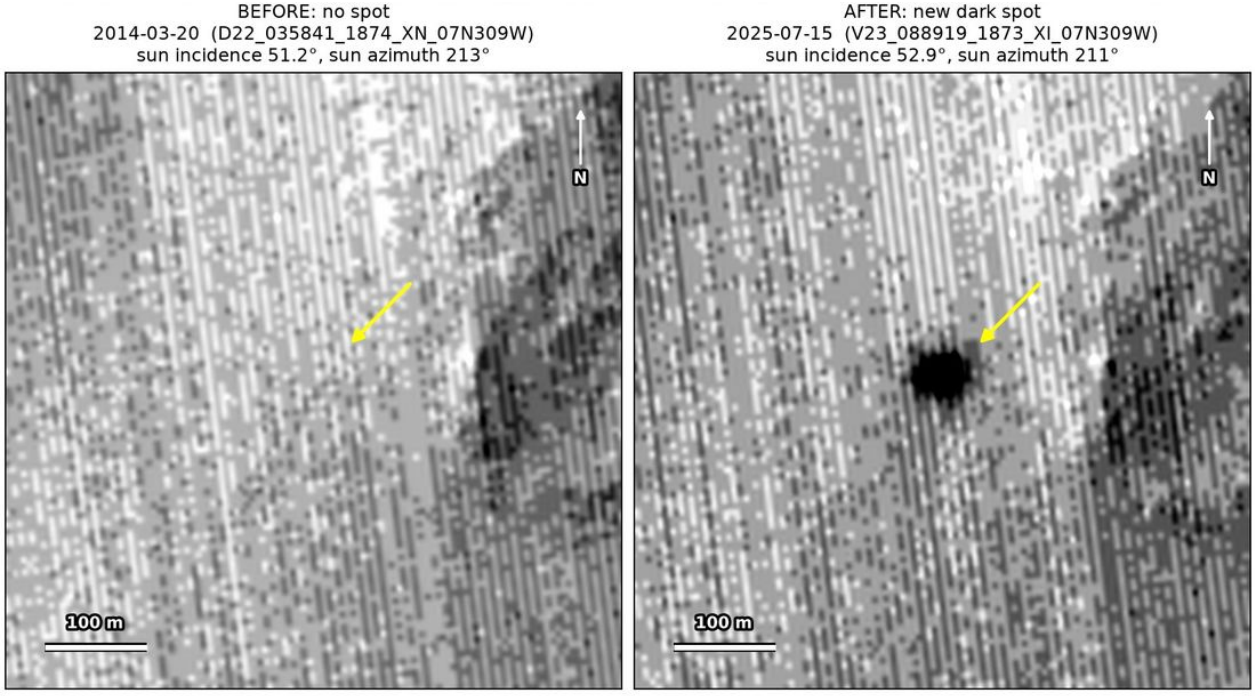

Figure 36: Candidate 14, before and after: the same 600 × 600 m of ground in 20 March 2014 (`D22_035841_1874_XN_07N309W`) and 15 July 2025 (`V23_088919_1873_XI_07N309W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

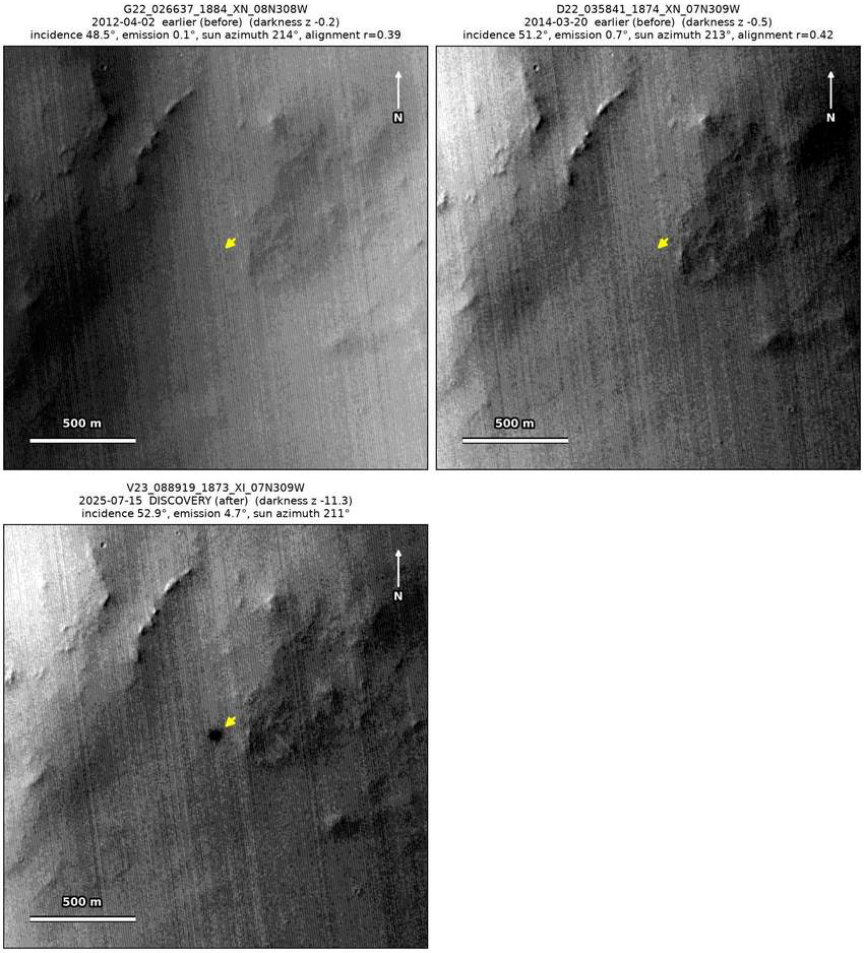

Figure 37: Candidate 14: three CTX images of the same 2×2 km patch, 2012 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 15: Plains in Arabia Terra, north-west of Dawes crater

Location	3.3061° S, 33.7150° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 45%
Formed between	20 August 2024 and 6 July 2025
Images	10 without the spot (absent: 26 February 2010, 9 April 2012, 2 July 2020, 17 February 2022, 10 May 2022, 7 January 2023, 3 October 2023, 31 October 2023, 19 June 2024 and 20 August 2024); 1 with it (present: 6 July 2025)
Size	Large, diffuse dark oval about 250 × 150 m, with a darker core
HiRISE	none covers the spot (nearest footprint edge 15.1 km)
Nearest catalogued new impact	31 km (Daubar et al., 2022)

What changed. A large, diffuse dark oval about 250 × 150 m, with a darker core, appeared on the dusty plains of Arabia Terra, north-west of Dawes crater, between August 2024 and July 2025, a window of 10.5 months. So far it is seen in only one CTX image.

Why it may be a new impact. All 10 earlier images, from February 2010 to August 2024, show nothing there. Four of them line up well with the July 2025 image, and an oval this large would be obvious anywhere in the 2 km patch around the spot in each image. Of the possible explanations, the blast zone of a new impact is slightly favoured.

Caveats. Only one image shows it, and dust devils are active on these plains, so it could also be a dust-devil scar, ground scoured by wind, or a change in brightness after a dust event. No image from the high-resolution HiRISE camera covers it; the nearest ends 15.1 km away.

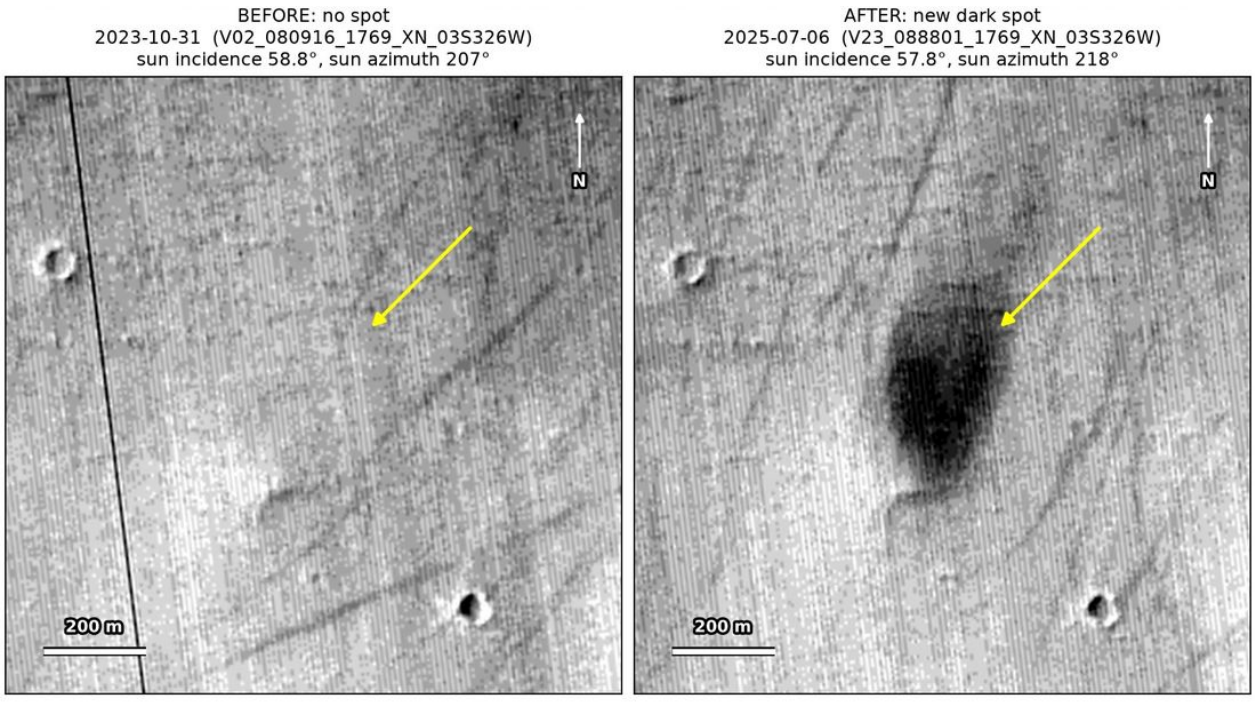

Figure 38: Candidate 15, before and after: the same 1200 × 1200 m of ground in 31 October 2023 (`V02_080916_1769_XN_03S326W`) and 6 July 2025 (`V23_088801_1769_XN_03S326W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

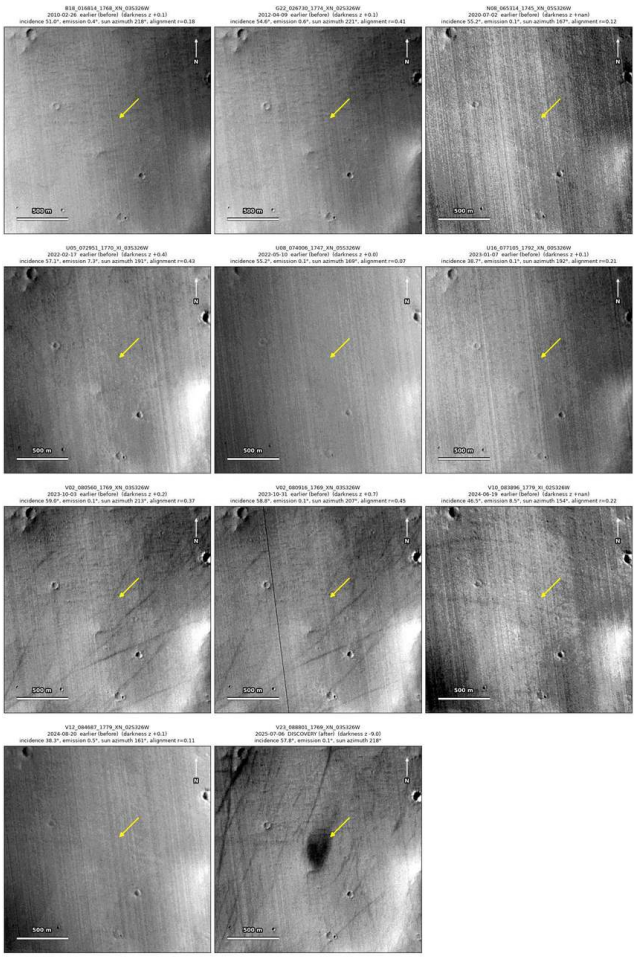

Figure 39: Candidate 15: eleven CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 16: Lava plains in Elysium Planitia, 13 km west-north-west of candidate 11

Location	16.9493° N, 147.6420° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	22 May 2014 and 28 July 2022
Images	1 without the spot (absent: 22 May 2014); 7 with it (present: 28 July 2022, 15 September 2022, 10 December 2023, 10 April 2025, 18 July 2025, 15 August 2025 and 28 September 2025)
Size	Compact, round dark spot 35 to 40 m across
HiRISE	none covers the spot (nearest footprint edge 1.0 km)
Nearest catalogued new impact	106 km (Daubar et al., 2022)
Nearby	Candidate 11, 13 km to the east-south-east, appeared later, between July 2022 and April 2025.

What changed. A compact, round dark spot 35 to 40 m across appeared on lava plains in Elysium Planitia between May 2014 and July 2022. It is unchanged in seven CTX images up to September 2025.

Why it may be a new impact. The May 2014 image, taken under the same sun as the August 2025 image, shows nothing there, although a shadow 92 m away shows in it. Since July 2022 the spot has been present under a range of sun heights and directions, so it is not a shadow. A compact, round dark spot that lasts for years is how the blast zone of a small new crater looks.

Caveats. Only one earlier image, from May 2014, shows the ground before it formed. The spot may already be in the catalogue of new impacts by Bickel et al. (2025), as its place and time window qualify, though no HiRISE image of it exists. No image from the high-resolution HiRISE camera covers it; the nearest ends 1.0 km away.

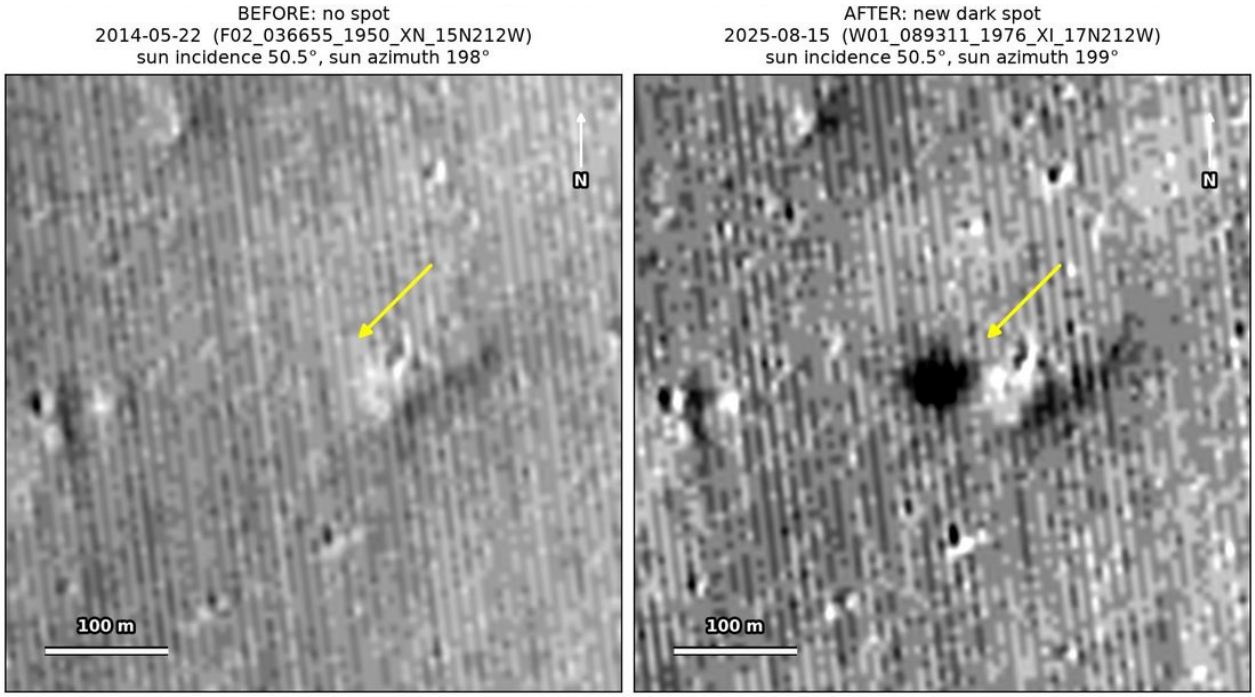

Figure 40: Candidate 16, before and after: the same 500 × 500 m of ground in 22 May 2014 (`F02_036655_1950_XN_15N212W`) and 15 August 2025 (`W01_089311_1976_XI_17N212W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

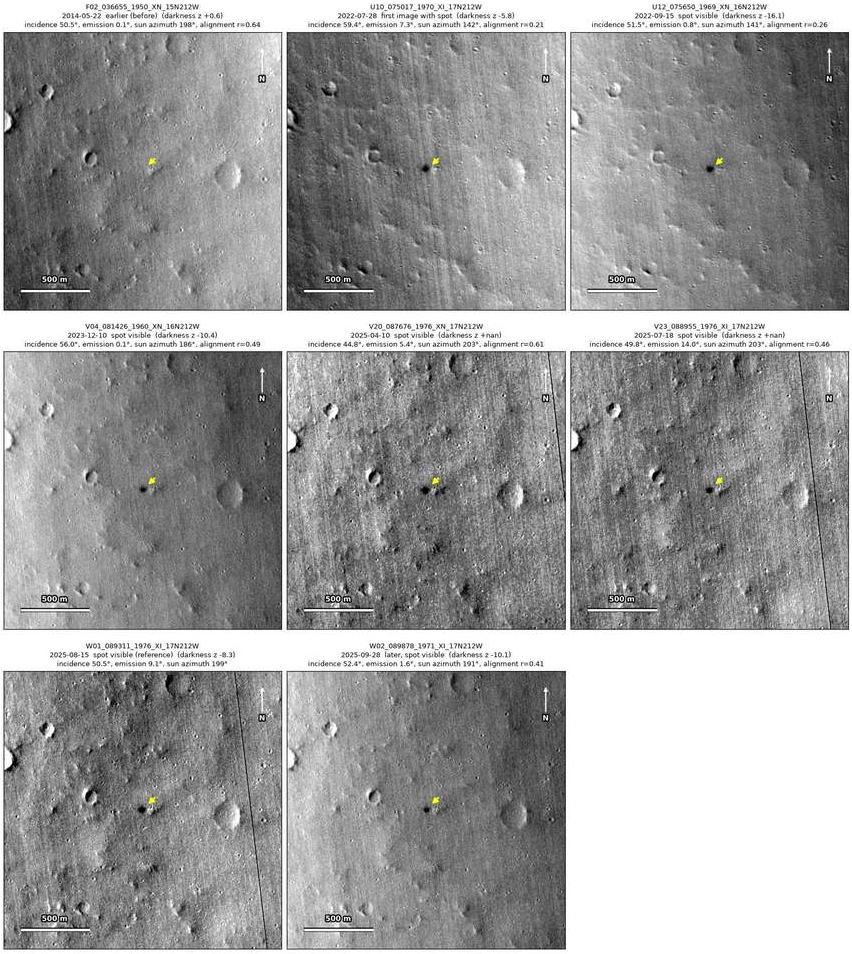

Figure 41: Candidate 16: eight CTX images of the same 2×2 km patch, 2014 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 17: Plains in northern Arabia Terra, south of Deuteronilus Mensae

Location	35.4151° N, 22.9629° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 55%
Formed between	11 August 2021 and 2 August 2025
Images	3 without the spot (absent: 28 January 2010, 31 May 2021 and 11 August 2021); 1 with it (present: 2 August 2025)
Size	Irregular dark spot 60 to 70 m across, with a darker core
HiRISE	none covers the spot (nearest footprint edge 26.4 km)
Nearest catalogued new impact	159 km (Daubar et al., 2022)

What changed. An irregular dark spot 60 to 70 m across, with a darker core, appeared on cratered plains of northern Arabia Terra between August 2021 and August 2025. So far it is seen in only one CTX image.

Why it may be a new impact. Three earlier images, from January 2010, May 2021 and August 2021, were taken with the sun at nearly the same height as in the August 2025 image and show nothing there, while a dark feature 207 m away shows in all three, so they would have shown the spot. A dark spot with a darker core appearing on plains fits the blast zone of a small new crater.

Caveats. Only one image shows it so far, though a flaw in that image is unlikely, as the spot is a smooth, irregular patch rather than a line or block. A change in the brightness of the ground caused by wind is possible. No image from the high-resolution HiRISE camera covers it; the nearest ends 26.4 km away.

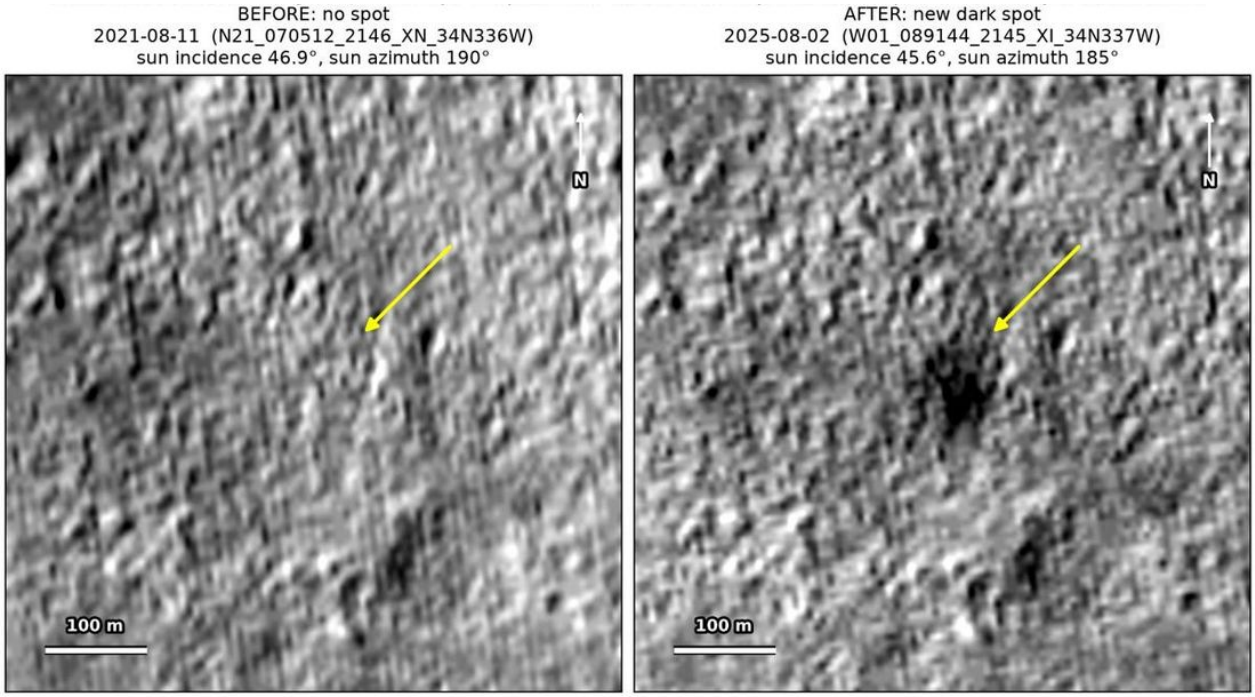

Figure 42: Candidate 17, before and after: the same 600 × 600 m of ground in 11 August 2021 (`N21_070512_2146_XN_34N336W`) and 2 August 2025 (`W01_089144_2145_XI_34N337W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

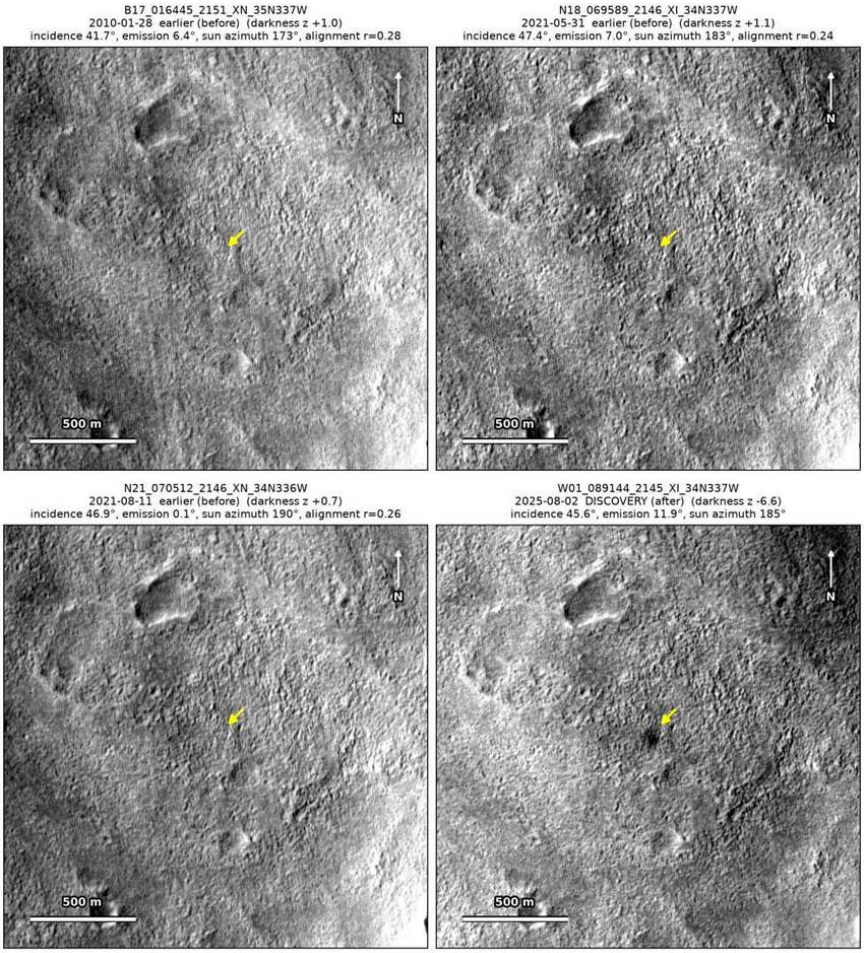

Figure 43: Candidate 17: four CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 18: Lava plains east of Arsia Mons

Location	8.0460° S, 245.9977° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	15 May 2025 and 10 July 2025
Images	5 without the spot (absent: 29 March 2010, 29 July 2016, 24 March 2018, 15 June 2018 and 15 May 2025); 1 with it (present: 10 July 2025)
Size	Compact dark dot 20 to 25 m across
HiRISE	none covers the spot (nearest footprint edge 16.5 km)
Nearest catalogued new impact	188 km (Daubar et al., 2022)

What changed. A compact dark dot 20 to 25 m across appeared on lava plains east of Arsia Mons between May 2025 and July 2025, a window of only 56 days. So far it is seen in one CTX image.

Why it may be a new impact. The May 2025 image, taken only 56 days earlier under almost the same sun and closely aligned with the July 2025 image, shows nothing there, although a dark feature 225 m away shows clearly in it; images from 2010 to 2018 show nothing either. A compact dark dot appearing on lava plains within two months fits the blast zone of a small new crater.

Caveats. Only one image shows it so far, so a flaw in that image cannot be fully ruled out until a second image is taken. No image from the high-resolution HiRISE camera covers it; the nearest ends 16.5 km away.

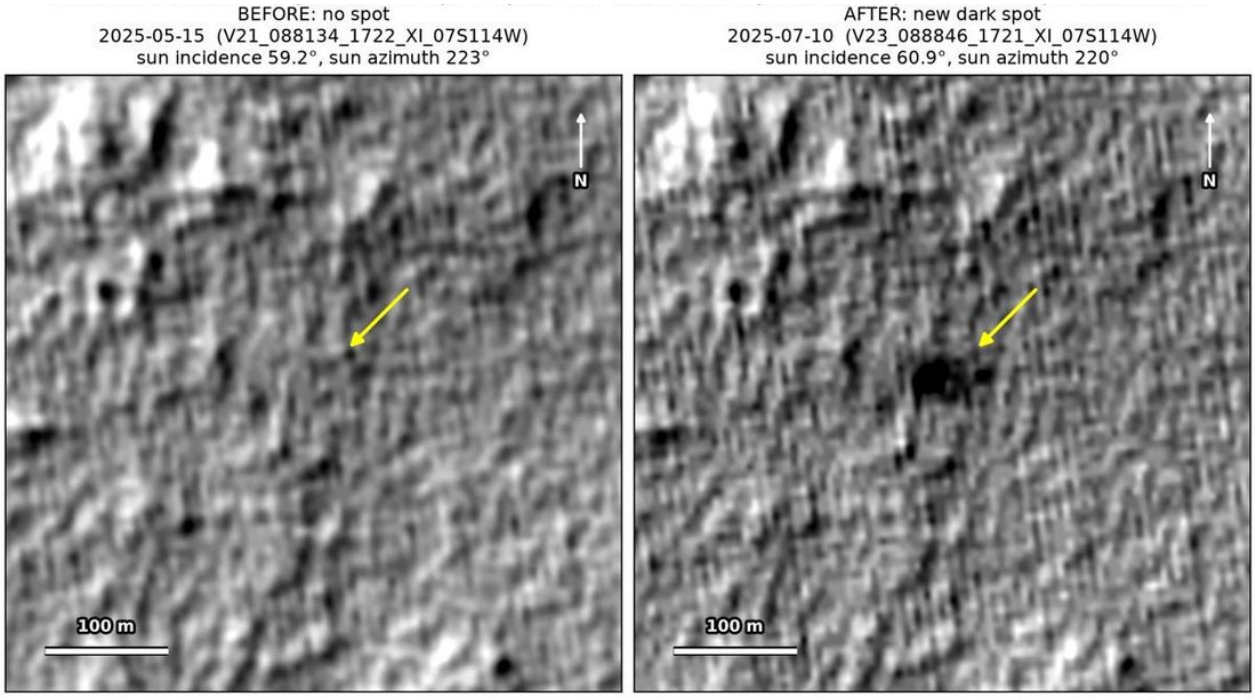

Figure 44: Candidate 18, before and after: the same 500 × 500 m of ground in 15 May 2025 (`V21_088134_1722_XI_07S114W`) and 10 July 2025 (`V23_088846_1721_XI_07S114W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

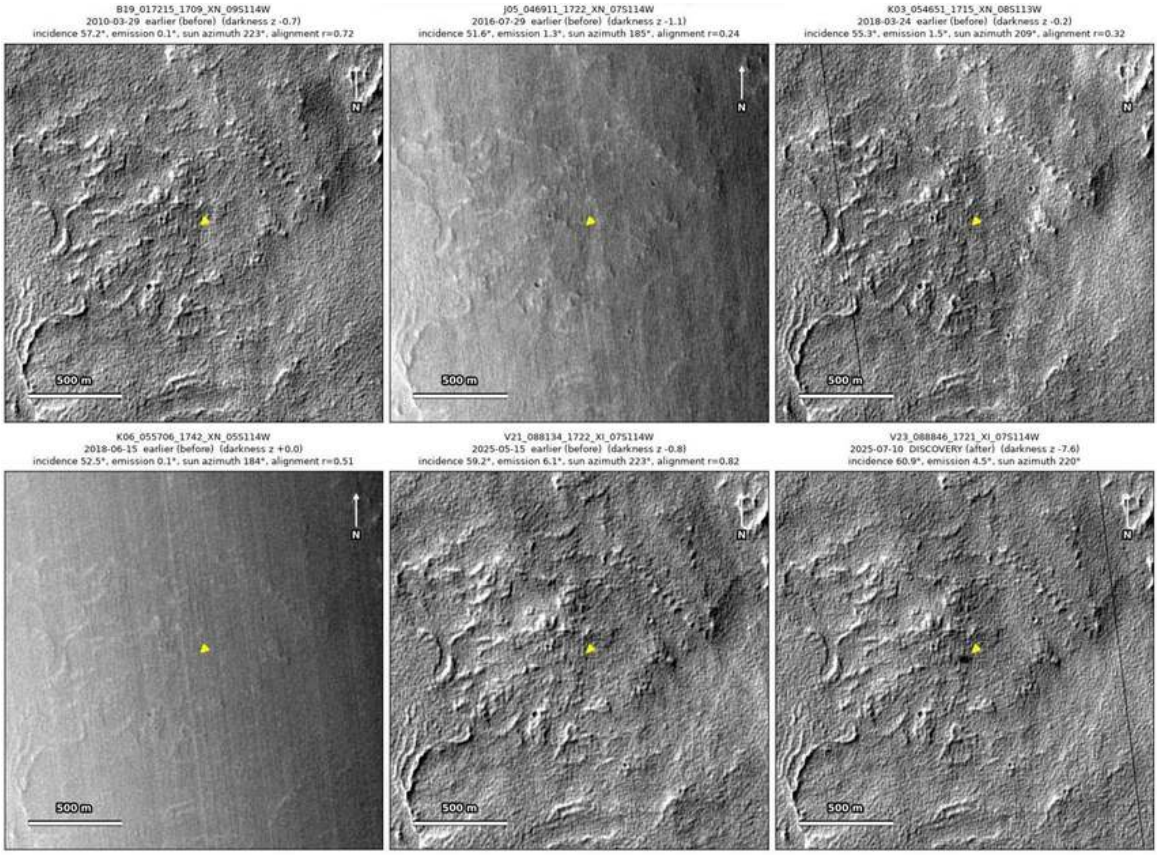

Figure 45: Candidate 18: six CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 19: Giant lobe north of Pavonis Mons

Location	4.7343° N, 246.6535° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	26 September 2024 and 10 March 2025
Images	6 without the spot (absent: 23 September 2019, 25 July 2021, 8 August 2023, 12 December 2023, 23 April 2024 and 26 September 2024); 4 with it (present: 10 March 2025, 10 May 2025, 15 September 2025 and 31 January 2026)
Size	Compact dark spot 25 to 30 m across
HiRISE	none covers the spot (nearest footprint edge 0.6 km)
Nearest catalogued new impact	21 km (Daubar et al., 2022)

What changed. A compact dark spot 25 to 30 m across appeared on the giant lobe north of Pavonis Mons between September 2024 and March 2025, a window of 165 days. Four CTX images from four orbits, up to January 2026, show it slowly fading.

Why it may be a new impact. Six earlier images, from September 2019 to September 2024, show nothing there, while a dark feature 250 m away shows in each, so they would have shown the spot. It is missing at every sun angle in the images up to 2024 and present at every sun angle since, so it is not a shadow, and a compact spot that slowly fades is how the blast zone of a small new crater looks.

Caveats. Shadows and camera flaws are both ruled out; what it still lacks is a view at higher resolution. No image from the high-resolution HiRISE camera covers it; the nearest ends 0.6 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 90% that the spot is a new impact. It found nine earlier images, from August 2014 to 26 September 2024, each of which would have shown the spot and none of which does, so the spot formed within 165 days. It placed the spot at 4.7346° N, 246.6520° E, 92 m from the position given above.

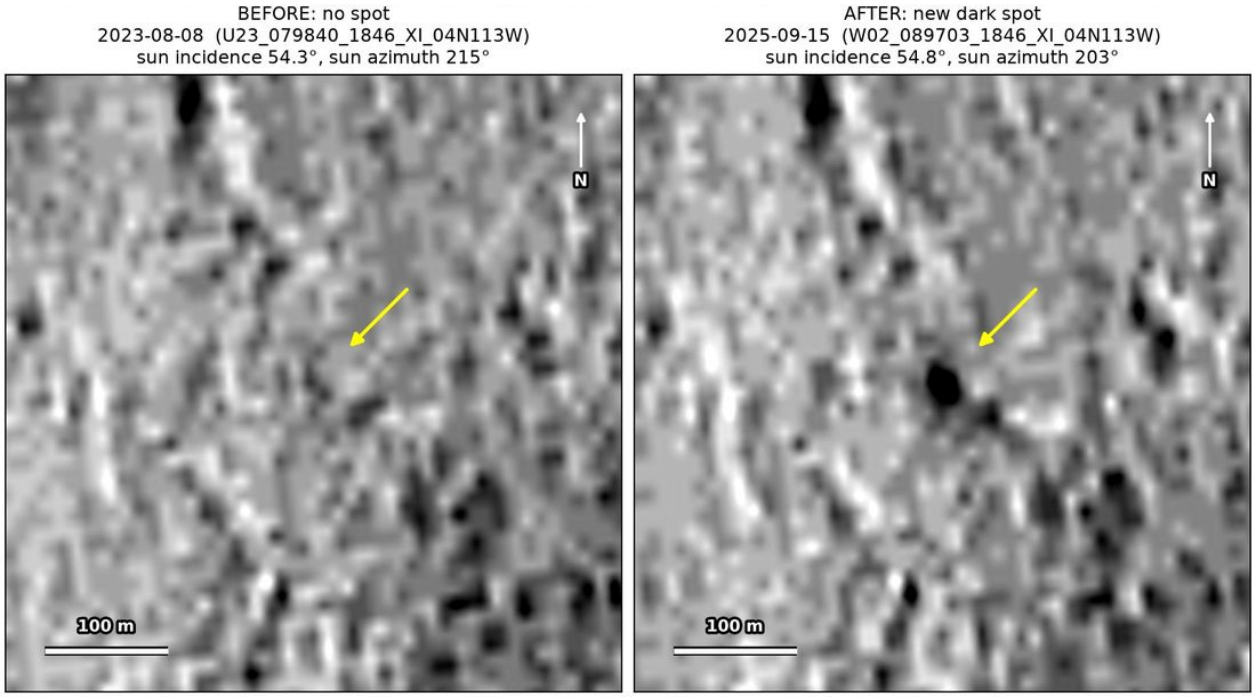

Figure 46: Candidate 19, before and after: the same 500 × 500 m of ground in 8 August 2023 (`U23_079840_1846_XI_04N113W`) and 15 September 2025 (`W02_089703_1846_XI_04N113W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

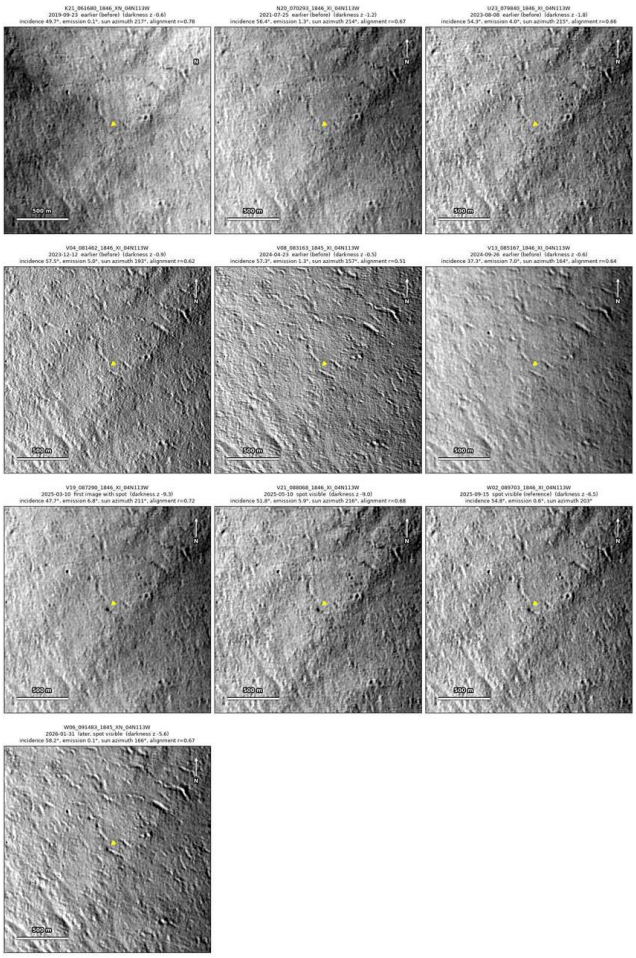

Figure 47: Candidate 19: ten of the 34 CTX images that cover the site, 2019 to 2026: the same 2×2 km patch at the same scale and contrast. The spot first appears in March 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

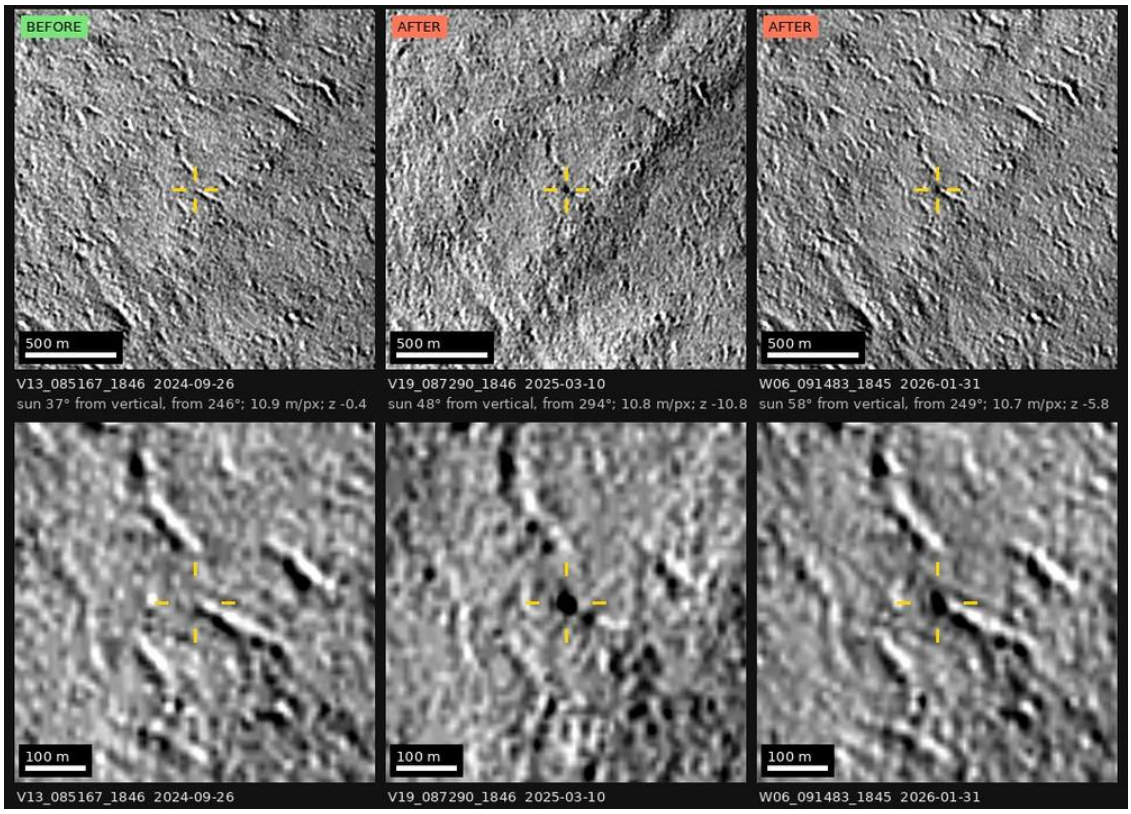

Figure 48: Candidate 19, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 20: Medusae Fossae Formation, south of Amazonis Mensa

Location	6.7466° S, 214.8862° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 70%
Formed between	6 April 2012 and 31 July 2025
Images	2 without the spot (absent: 16 August 2008 and 6 April 2012); 2 with it (present: 31 July 2025 and 7 November 2025)
Size	Compact, slightly elongated dark spot 40 to 50 × 30 m
HiRISE	none covers the spot (nearest footprint edge 4.7 km)
Nearest catalogued new impact	46 km (Daubar et al., 2022)

What changed. A compact, slightly elongated dark spot 40 to 50 × 30 m appeared on wind-streaked ground of the Medusae Fossae Formation, south of Amazonis Mensa, between April 2012 and July 2025. It shows in two CTX images and was still there in November 2025.

Why it may be a new impact. Two earlier images, from August 2008 and April 2012, were taken with the sun at nearly the same height as the July and November 2025 images and show nothing there, although a dark feature 300 m away shows in the April 2012 image. A compact new dark spot that persists is how the blast zone of a small new crater looks.

Caveats. The ground is crumbly, wind-carved terrain, so wind uncovering darker material beneath is possible. The April 2018 and November 2022 images show no detail at the spot, so it could have formed at any time in the 13 years between April 2012 and July 2025. No image from the high-resolution HiRISE camera covers it; the nearest ends 4.7 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 80% that the spot is a new impact. It found the spot already present on 12 November 2022 (`U14_076386_1754`) and absent on 20 April 2018 (`K04_054995_1748`), which would narrow the formation window to April 2018 to November 2022. Wind exposing darker material on the soft Medusae Fossae Formation remains its main alternative. It placed the spot at 6.7444° S, 214.8835° E, 203 m from the position given above.

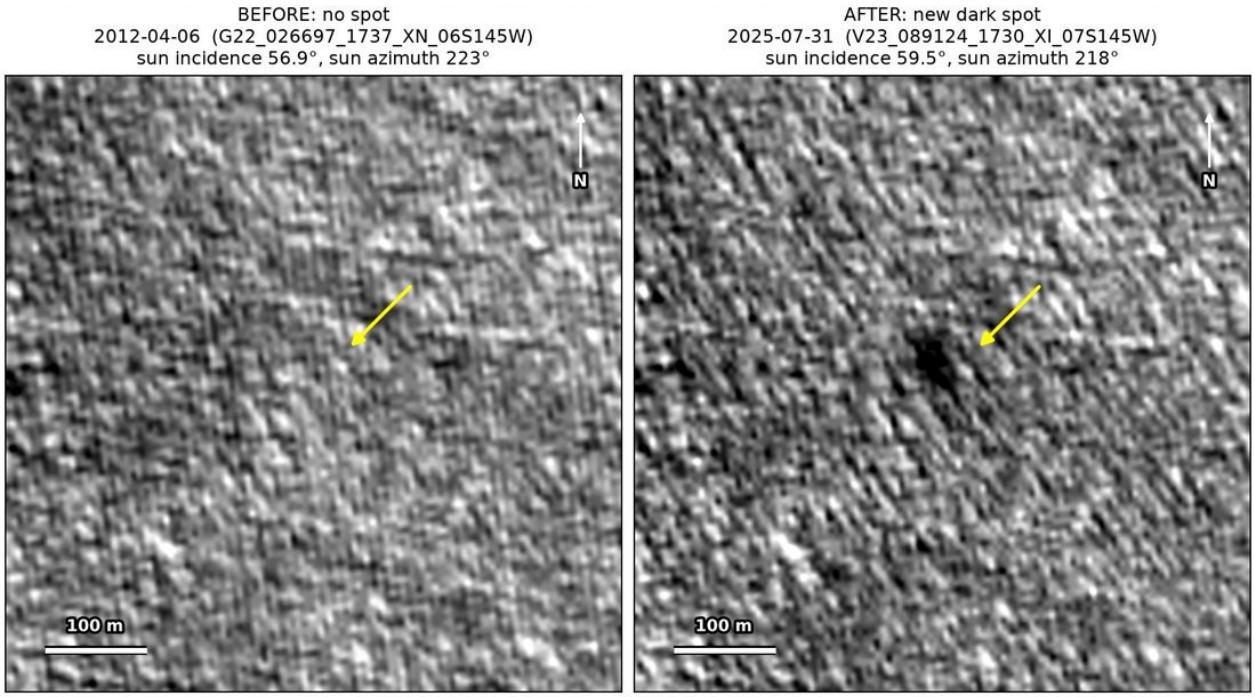

Figure 49: Candidate 20, before and after: the same 600 × 600 m of ground in 6 April 2012 (`G22_026697_1737_XN_06S145W`) and 31 July 2025 (`V23_089124_1730_XI_07S145W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

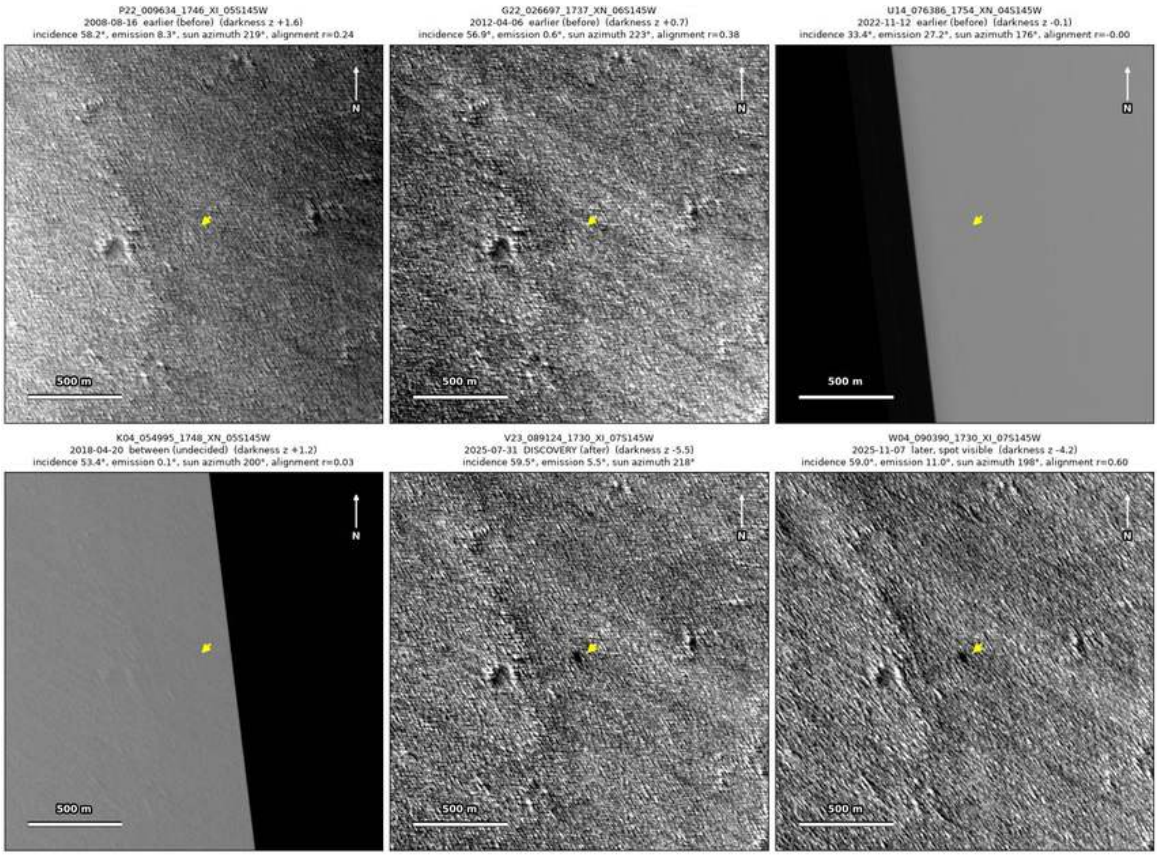

Figure 50: Candidate 20: six CTX images of the same 2×2 km patch, 2008 to 2025. The April 2018 and November 2022 images, which show no detail at the spot, are shown but not used to date it. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

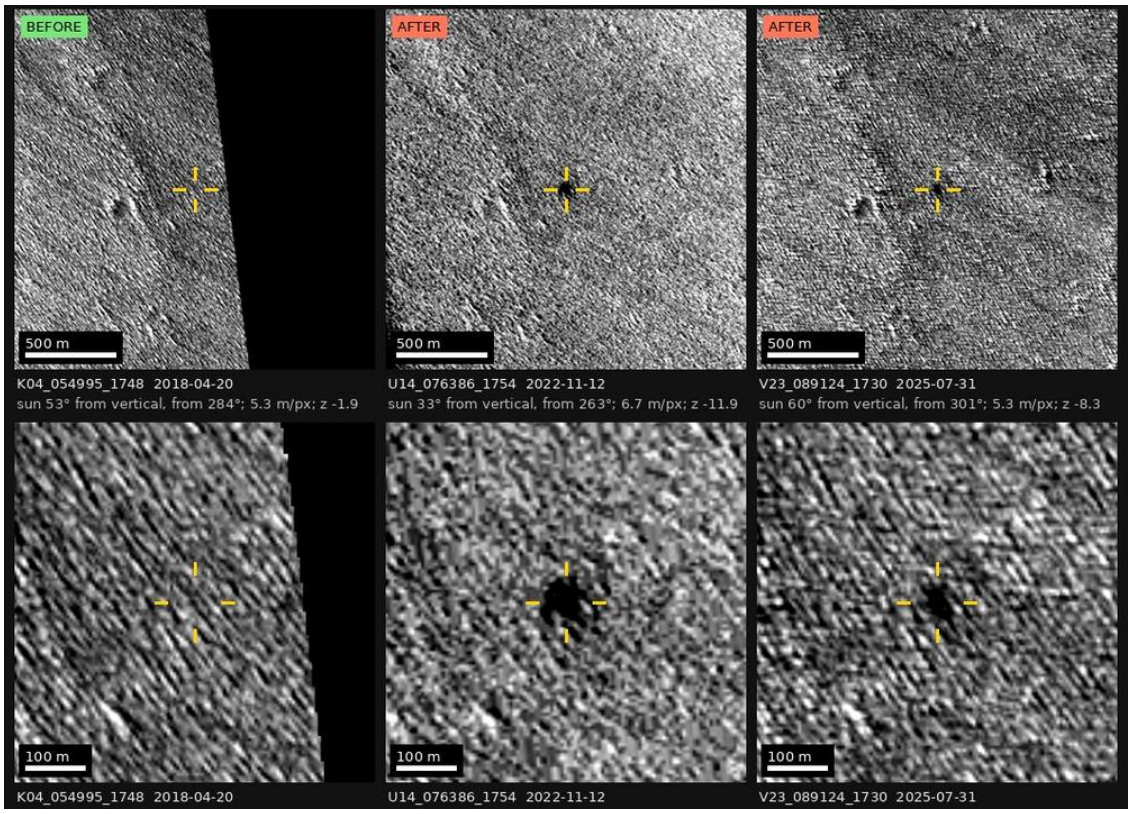

Figure 51: Candidate 20, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 21: Terrain west of Cyane Fossae

Location	25.9771° N, 236.8899° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 75%
Formed between	18 December 2010 and 7 November 2025
Images	1 without the spot (absent: 18 December 2010); 2 with it (present: 7 November 2025 (two images, CTX and HiRISE))
Size	Rayed dark zone 40 to 50 m across around at least three small craters, each 2 to 3 m across, seen by HiRISE
HiRISE	ESP_090389_2060 (2025-11-07, same orbit, not impact-targeted)
Nearest catalogued new impact	82 km (Daubar et al., 2022)

What changed. A black blob 40 to 50 m across appeared inside a hollow about 100 m wide west of Cyane Fossae between December 2010 and November 2025. It is seen in one CTX image and in a HiRISE image taken on the same orbit.

Why it may be a new impact. The only earlier image, from December 2010, shows a bright, sunlit patch where the blob now lies, while a dark feature 280 m away shows clearly in it. In the HiRISE image the blob is a pattern of dark rays spreading in several directions around at least three dark ovals 2 to 3 m across, whose bright eastern rims show that they are small craters lit from the west. That is how a small new impact looks.

Caveats. By chance, NASA's HiRISE camera photographed the spot on 7 November 2025, after it formed, on the same orbit as the CTX image that shows it. A dark streak left by a dust avalanche on the hollow's wall is possible, as such streaks occur in pits of Cyane Fossae, but the craters at its centre and the rays pointing upslope argue against it, and an impact can also set off such a streak. Only one earlier image, from December 2010, shows the ground before, so when it formed is known only to within about 15 years.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 95% that the spot is a new impact. Registering the HiRISE image to the CTX image of the same orbit, it found at least three fresh pits 2–3 m across with dark rays at the spot, shaded like small bowls rather than boulders (Figure 56). It placed the spot at 25.9794° N, 236.8864° E, 232 m from the position given above.

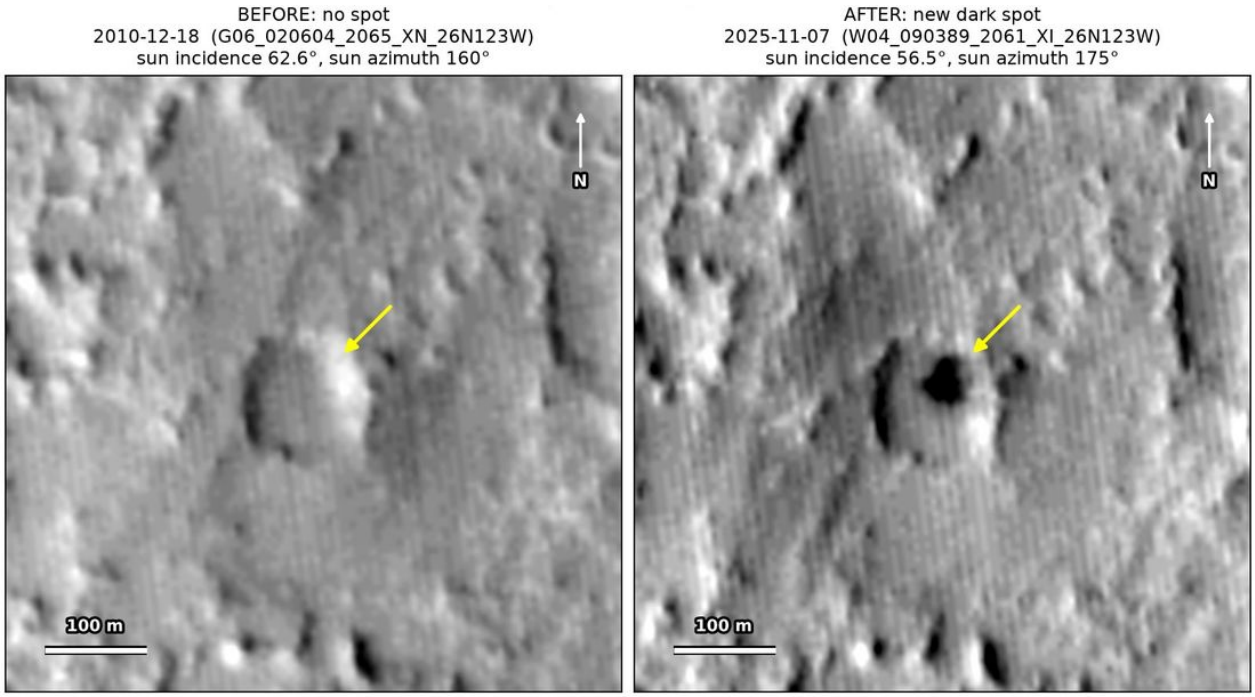

Figure 52: Candidate 21, before and after: the same 600 × 600 m of ground in 18 December 2010 (`G06_020604_2065_XN_26N123W`) and 7 November 2025 (`W04_090389_2061_XI_26N123W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

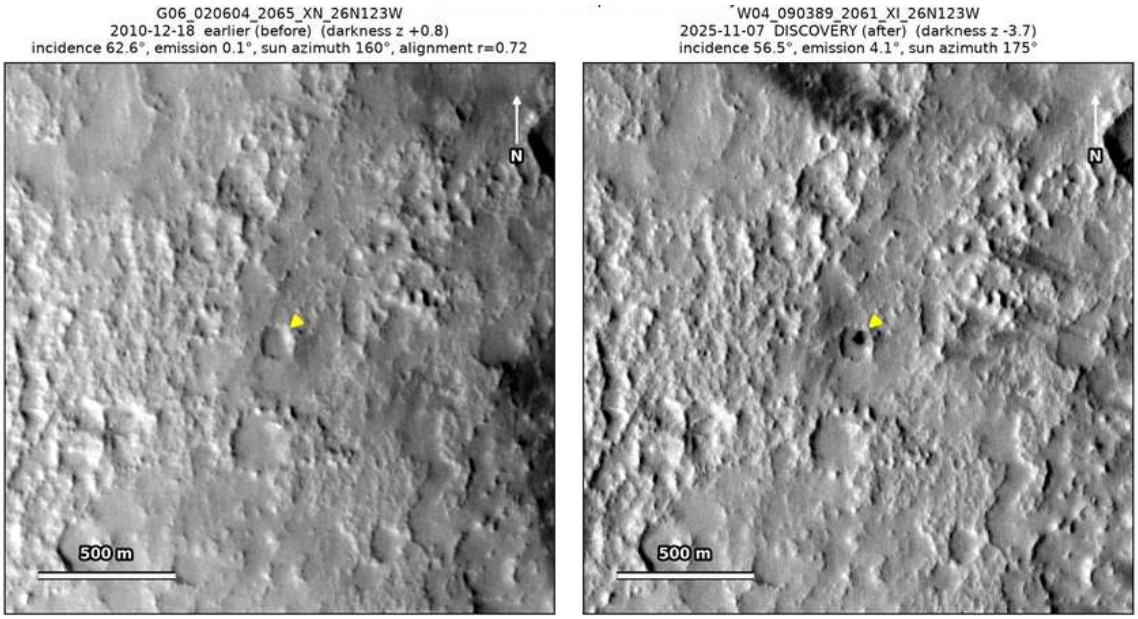

Figure 53: Candidate 21: both CTX images of the same 2 × 2 km patch, December 2010 and November 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

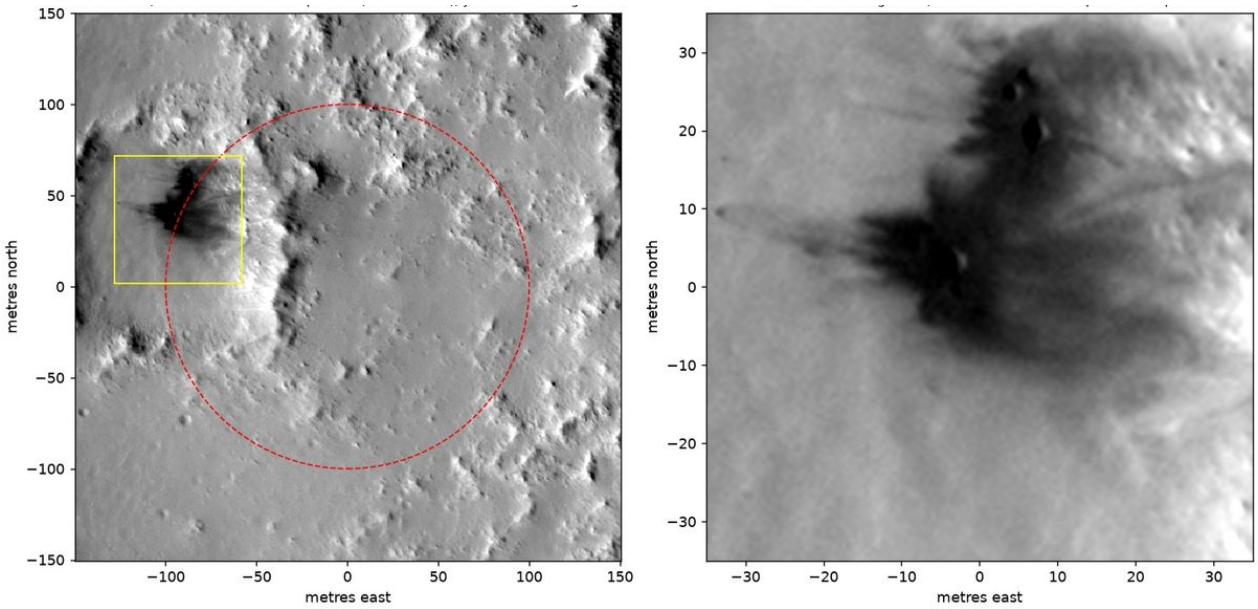

Figure 54: Candidate 21, HiRISE image `ESP_090389_2060`: HiRISE view of the spot (7 November 2025, 0.5 m per pixel), taken by chance on the same orbit as the CTX image, with a 70 m enlargement of its darkest part: dark rays spread from small pits. The red dashed circle marks the CTX position; the spot is at its north-west edge. NASA/JPL-Caltech/UArizona.

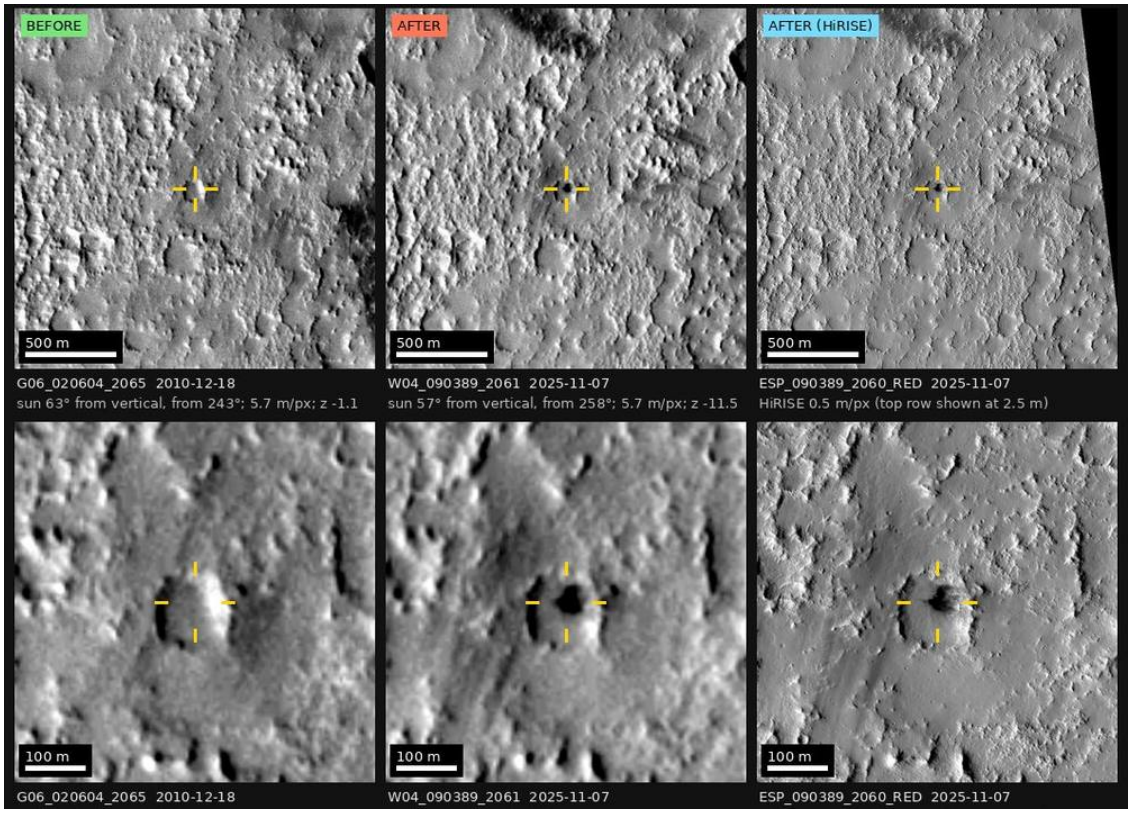

Figure 55: Candidate 21, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

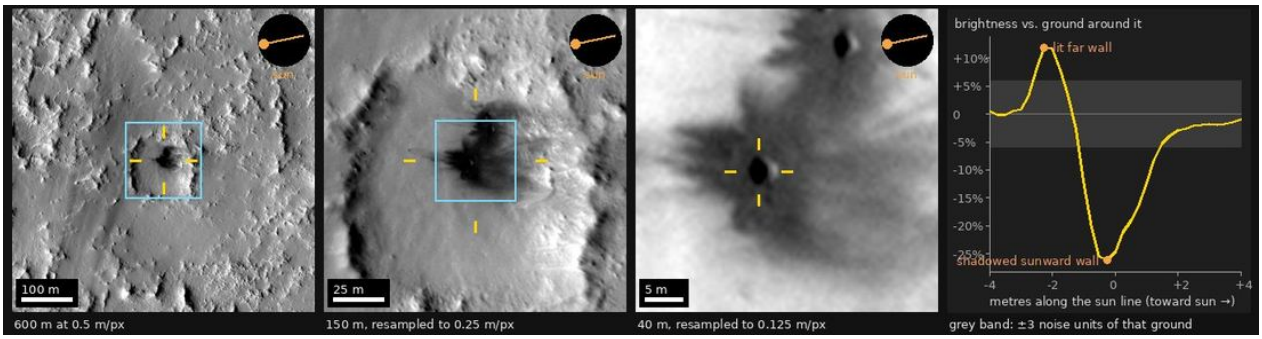

Figure 56: Candidate 21, HiRISE image `ESP_090389_2060`. The independent check: HiRISE close-ups, 600 m, 150 m and 40 m wide, and the brightness along the sun line through the darkest point: dark on the side towards the sun and bright on the far side, the shading of a small pit rather than a boulder. NASA/JPL-Caltech/UArizona.

Candidate 22: Olympica Fossae

Location	24.0029° N, 247.4640° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 45%
Formed between	26 November 2024 and 11 July 2025
Images	6 without the spot (absent: 9 July 2007, 12 June 2008, 8 November 2022, 11 January 2023, 24 October 2024 and 26 November 2024); 1 with it (present: 11 July 2025)
Size	Irregular dark patch 250 × 200 m, with several darker blobs
HiRISE	none covers the spot (nearest footprint edge 11.9 km)
Nearest catalogued new impact	65 km (Daubar et al., 2022)

What changed. A large, irregular dark patch 250 × 200 m, with several darker blobs, appeared on smooth Tharsis plains at Olympica Fossae in the seven and a half months between November 2024 and July 2025. So far, one CTX image shows it.

Why it may be a new impact. Six earlier images, from 2007 to November 2024, show plain ground there, among them two from October and November 2024 that line up well with the July 2025 image. The darker blobs are sharp-edged and there are no tracks, which slightly favours the blast zone of a cluster of small impacts over a scar left by a dust devil or ground scoured by wind.

Caveats. Only one image shows it, and it could still be a scar left by a dust devil or ground scoured by wind. No image from the high-resolution HiRISE camera covers it; the nearest ends 11.9 km away.

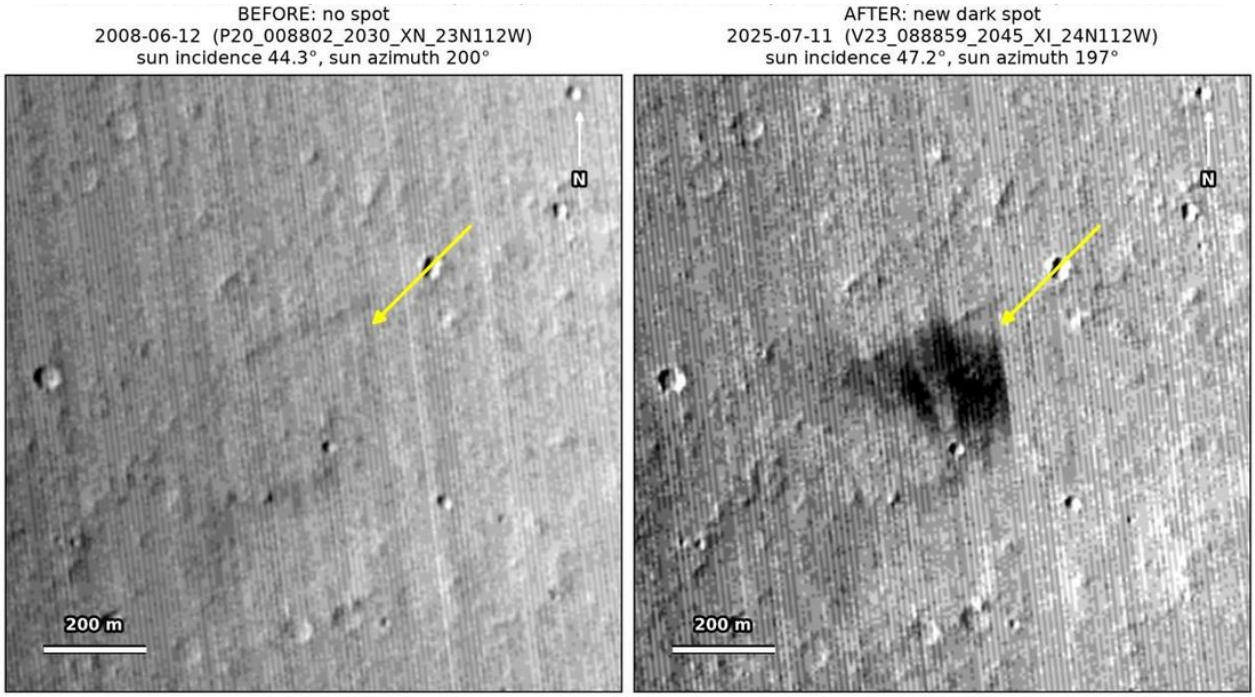

Figure 57: Candidate 22, before and after: the same 1200 × 1200 m of ground in 12 June 2008 (`P20_008802_2030_XN_23N112W`) and 11 July 2025 (`V23_088859_2045_XI_24N112W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

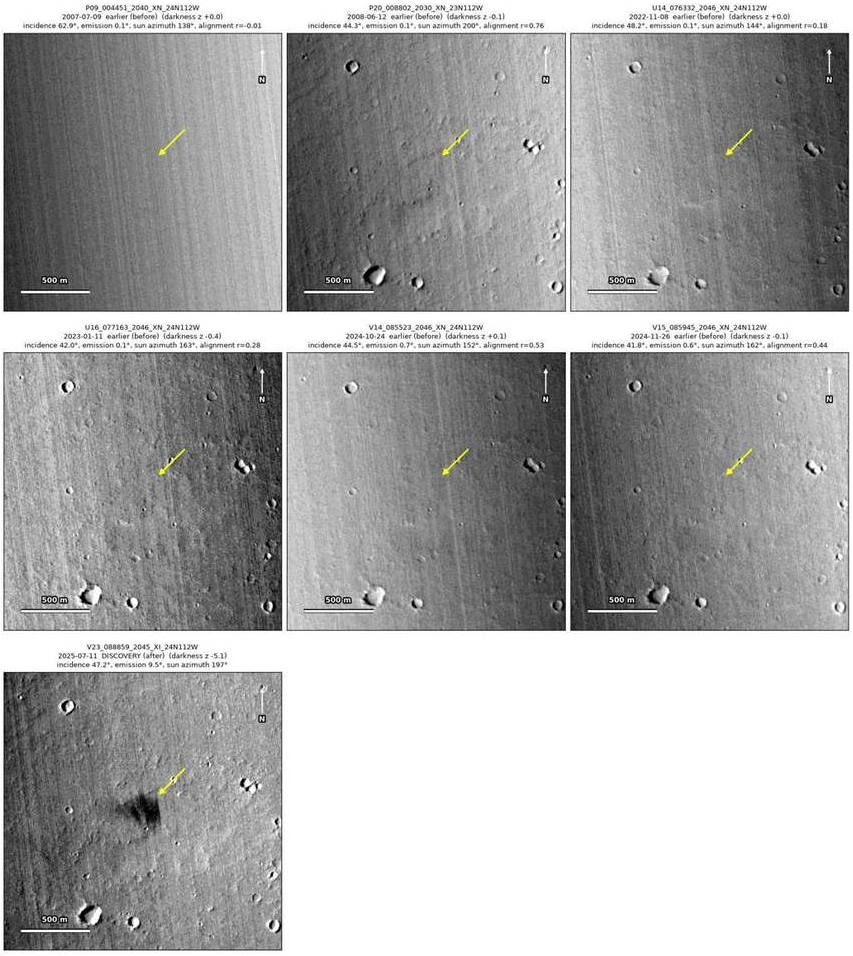

Figure 58: Candidate 22: seven CTX images of the same 2×2 km patch, 2007 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 23: North-west flank of Arsia Mons

Location	5.2784° S, 237.4930° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	3 December 2016 and 23 January 2025
Images	7 without the spot (absent: 17 July 2007, 15 August 2008, 28 September 2008, 15 December 2015, 27 December 2015, 13 October 2016 and 3 December 2016); 3 with it (present: 23 January 2025, 25 May 2025 and 14 July 2025)
Size	Three or four dark dots, each 15 to 30 m across, spread over 150 m
HiRISE	none covers the spot (nearest footprint edge 31.9 km)
Nearest catalogued new impact	36 km (Daubar et al., 2022)

What changed. Three or four compact dark dots, each 15 to 30 m across and spread over 150 m, appeared on bright plains on the north-west flank of Arsia Mons between December 2016 and January 2025. Three CTX images from three orbits, up to July 2025, show them in the same pattern.

Why it may be a new impact. Seven earlier images, from 2007 to 2016, show no dots there; the three that line up best, from September 2008, December 2015 and December 2016, were taken with the sun at heights that bracket those of the later images. A tight cluster of separate, compact dark dots that keeps its pattern is how a cluster of small new impact craters looks, and no other credible explanation is apparent.

Caveats. With no comparable dark mark nearby, the evidence that the earlier images would have shown the dots is weak, and they are seen only in images from January to July 2025. No image from the high-resolution HiRISE camera covers it; the nearest ends 31.9 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 85% that the spot is a new impact. It found the cluster 1.5 km east of the position given above in all 11 CTX images, each in its own map projection, so the catalogued position is in error; any follow-up should use the re-measured position. It placed the spot at 5.2771° S, 237.5184° E, 1502 m from the position given above.

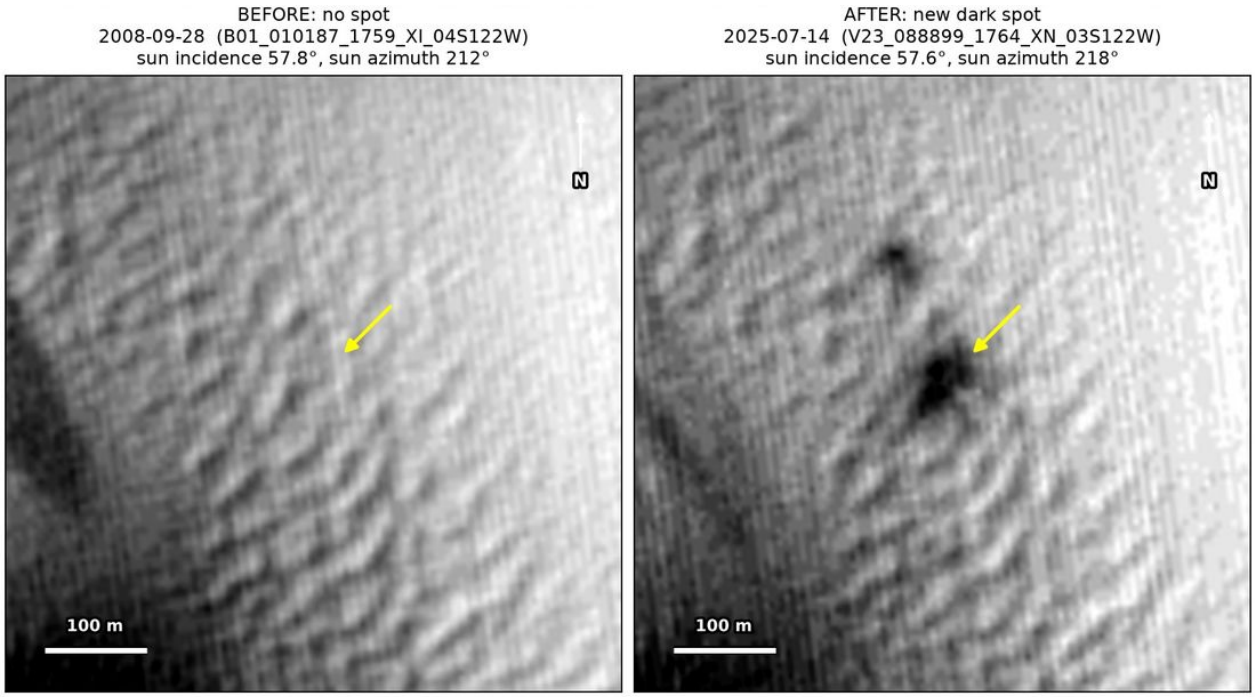

Figure 59: Candidate 23, before and after: the same 600 × 600 m of ground in 28 September 2008 (`B01_010187_1759_XI_04S122W`) and 14 July 2025 (`V23_088899_1764_XN_03S122W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

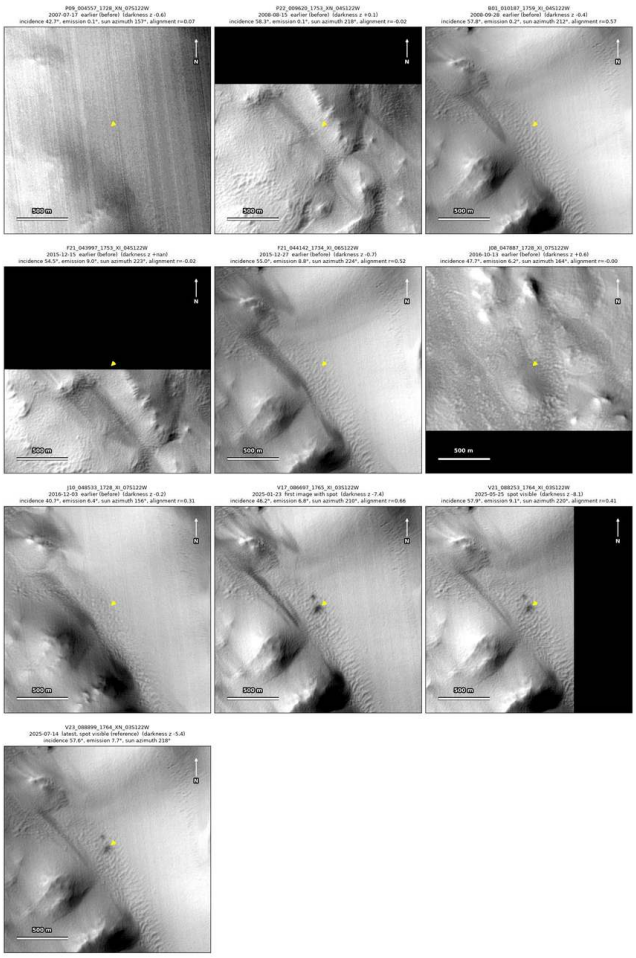

Figure 60: Candidate 23: ten CTX images of the same 2×2 km patch, 2007 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

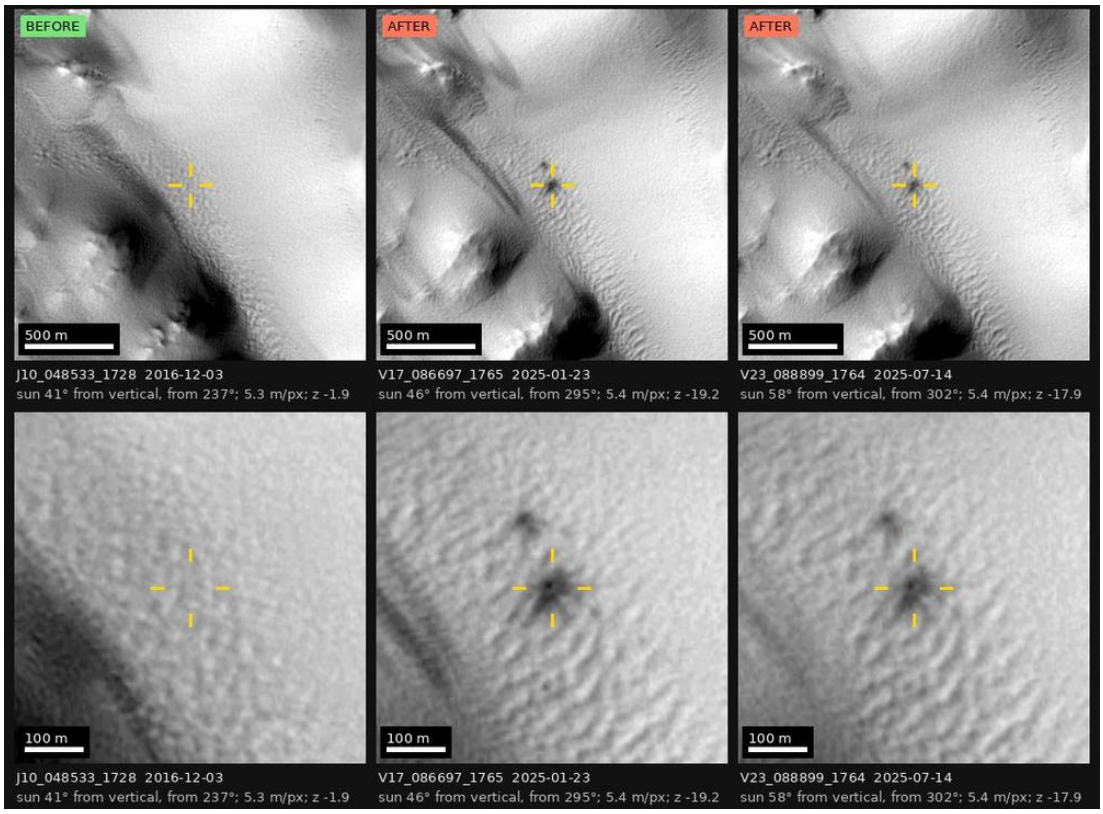

Figure 61: Candidate 23, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 24: Northern Arabia Terra, south-west of Ismeniae Fossae

Location	30.9059° N, 32.4196° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 50%
Formed between	3 January 2010 and 2 February 2025
Images	1 without the spot (absent: 3 January 2010); 2 with it (present: 2 February 2025 and 11 September 2025)
Size	Black, compact blob 30 to 40 m across
HiRISE	none covers the spot (nearest footprint edge 72.8 km)
Nearest catalogued new impact	123 km (Daubar et al., 2022)

What changed. A black, compact blob 30 to 40 m across appeared on knobby plains of northern Arabia Terra, south-west of Ismeniae Fossae, between January 2010 and February 2025. Two CTX images from two orbits, seven months apart, show it; in the second, of September 2025, it is smaller and fainter.

Why it may be a new impact. The only earlier image, of 3 January 2010, was taken under the same sun as the February 2025 image and shows no spot there, although a dark mark 150 m away does show in it. A compact black spot appearing on these plains could be the blast zone of a small new crater.

Caveats. With only one earlier image, the spot could have formed at any time in a span of 15 years, and by September 2025 it is already smaller and fainter: unusually fast fading for a blast zone. The ground is knobby, and in some images the shadows of knobs lie everywhere, making a small dark spot harder to single out. No image from the high-resolution HiRISE camera covers it; the nearest ends 72.8 km away.

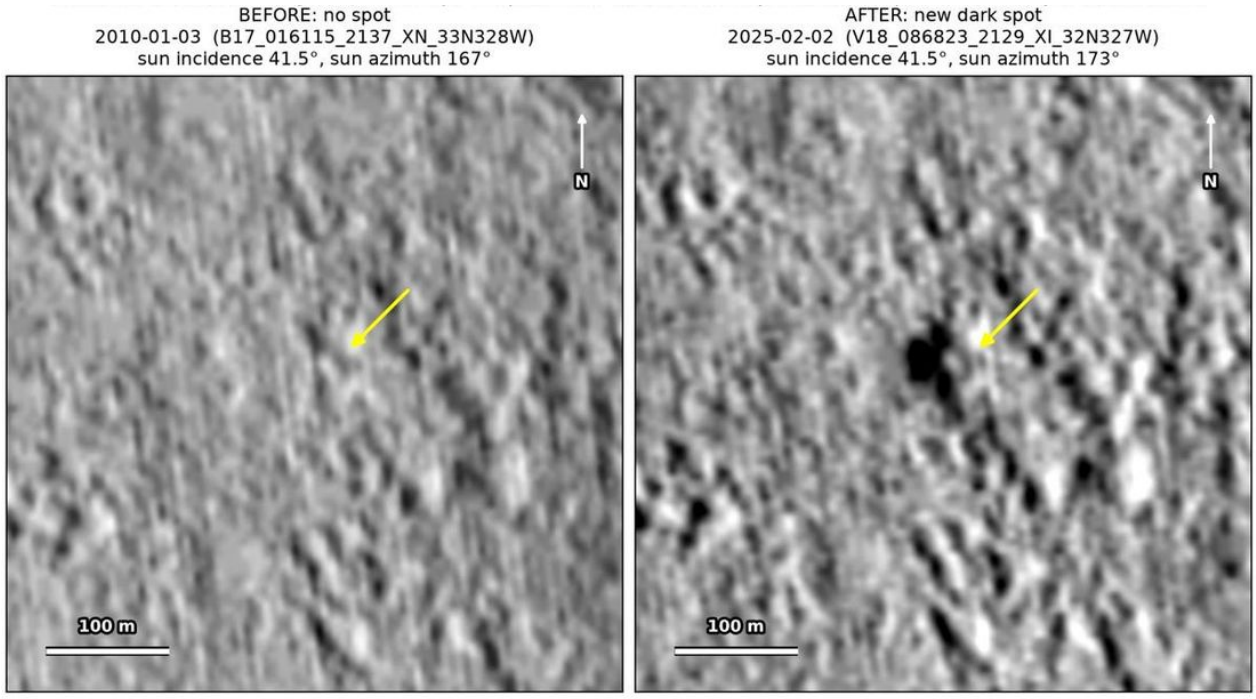

Figure 62: Candidate 24, before and after: the same 500 × 500 m of ground in 3 January 2010 (`B17_016115_2137_XN_33N328W`) and 2 February 2025 (`V18_086823_2129_XI_32N327W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

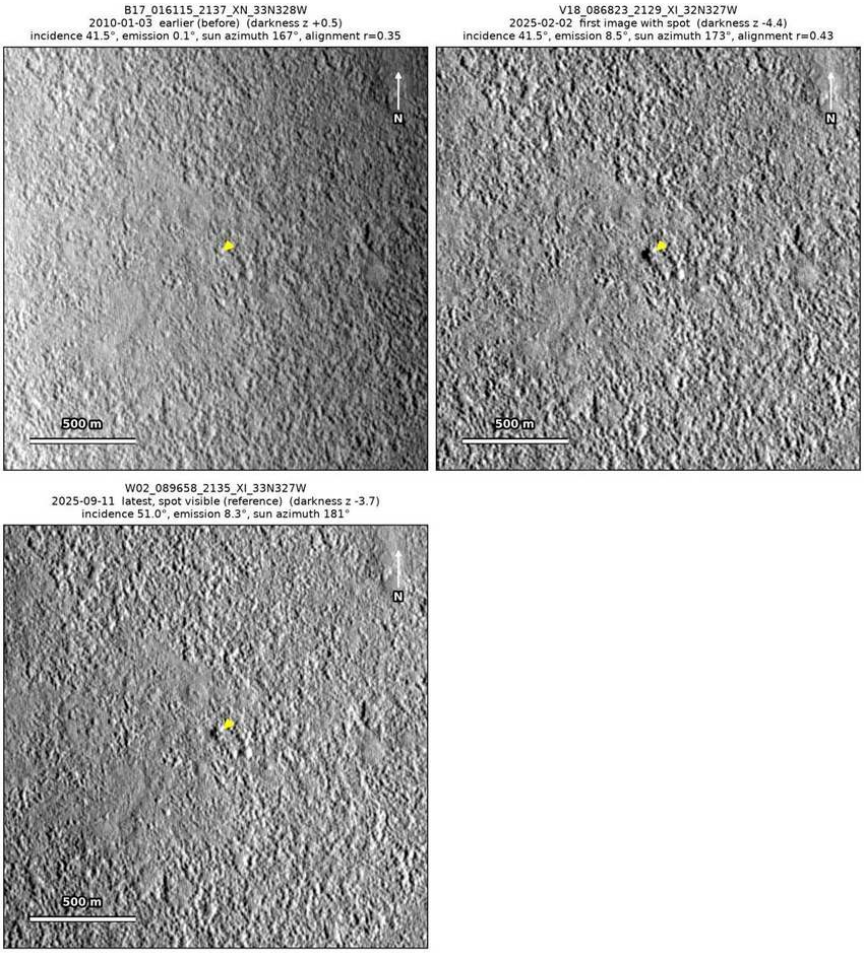

Figure 63: Candidate 24: three CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 25: Floor of Cassini crater, Arabia Terra

Location	20.1303° N, 34.2063° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 75%
Formed between	21 November 2008 and 3 November 2017
Images	2 without the spot (absent: 2 April 2008 and 21 November 2008); 3 with it (present: 3 November 2017, 21 October 2023 and 8 February 2026)
Size	Very dark round core 80 to 100 m across, with a tail 200 to 300 m long to the south
HiRISE	none covers the spot (nearest footprint edge 24.5 km)
Nearest catalogued new impact	51 km (Daubar et al., 2022)

What changed. A very dark, round spot 80 to 100 m across, with a dark tail 200 to 300 m long pointing south, appeared on the flat floor of Cassini crater between November 2008 and November 2017. Three CTX images from three orbits show it, darkest in October 2023 and fading by February 2026.

Why it may be a new impact. Two earlier images, from April and November 2008, show plain ground there, while a dark mark 300 m away shows in both, so they would have shown a spot this dark. A very dark, round spot whose core is itself new, on a crater floor too flat for a slope streak (a dark streak left by dust sliding down a slope), fits the blast zone of a small new crater.

Caveats. A dark wind streak trailing behind a small obstacle could look like this, although that is unlikely because the round core is itself new. The spot could also have formed at any time in the nine years between November 2008 and November 2017. No image from the high-resolution HiRISE camera covers it; the nearest ends 24.5 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 65% that the spot is a new impact. It considered a slope streak the main alternative on this crater floor and, without HiRISE or topography, estimated the probability at about 65%. It placed the spot at 20.1310° N, 34.2074° E, 72 m from the position given above.

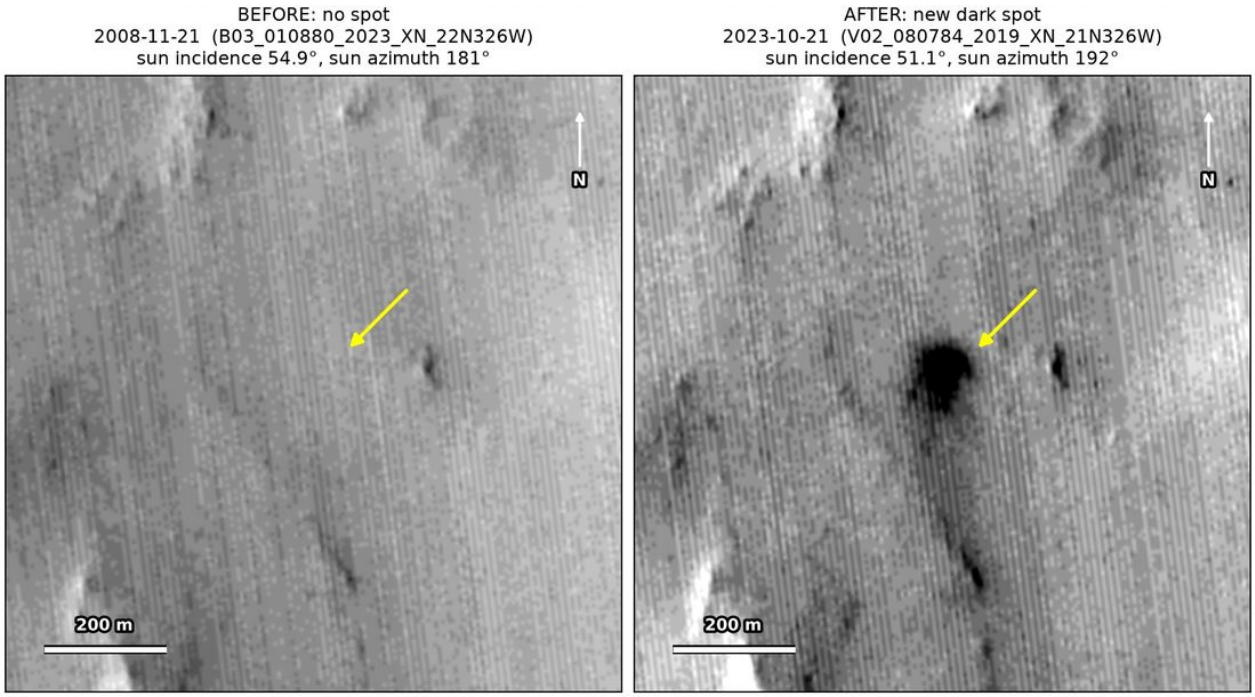

Figure 64: Candidate 25, before and after: the same 1000×1000 m of ground in 21 November 2008 (`B03_010880_2023_XN_22N326W`) and 21 October 2023 (`V02_080784_2019_XN_21N326W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

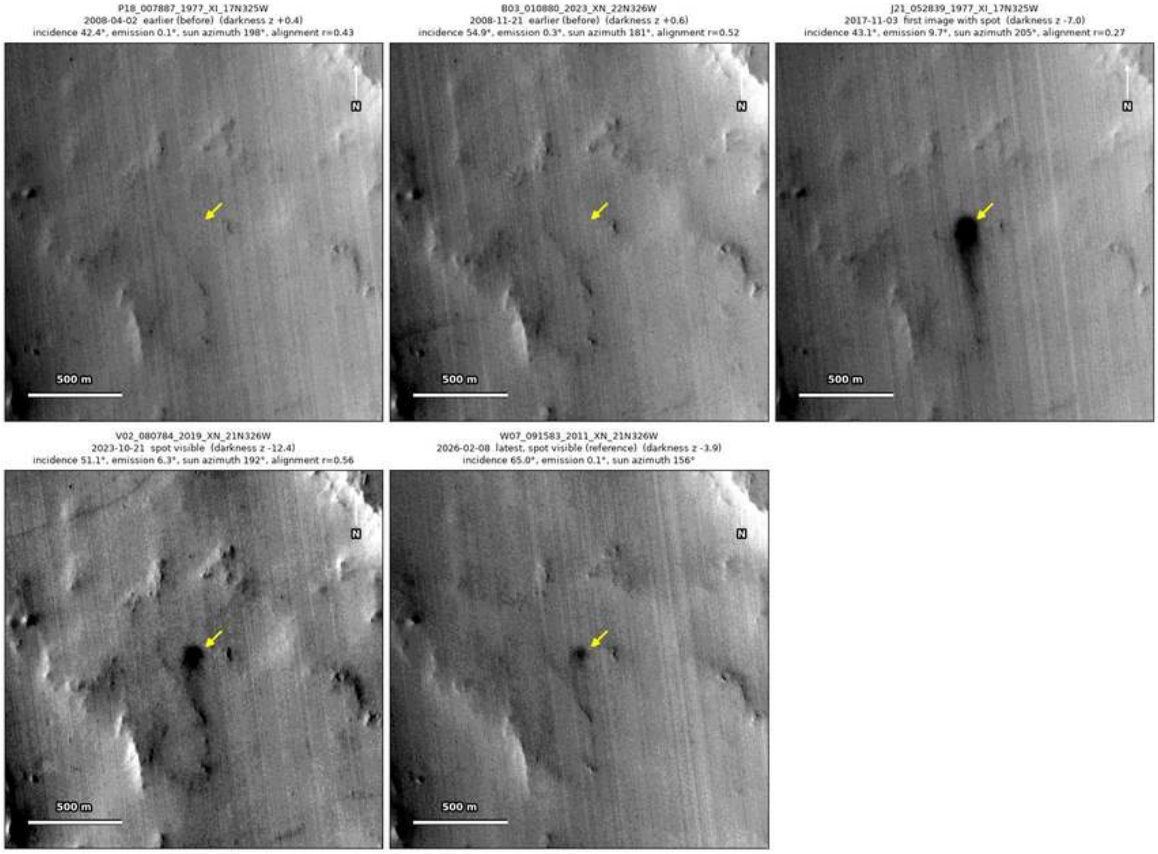

Figure 65: Candidate 25: five CTX images of the same 2×2 km patch, 2008 to 2026. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

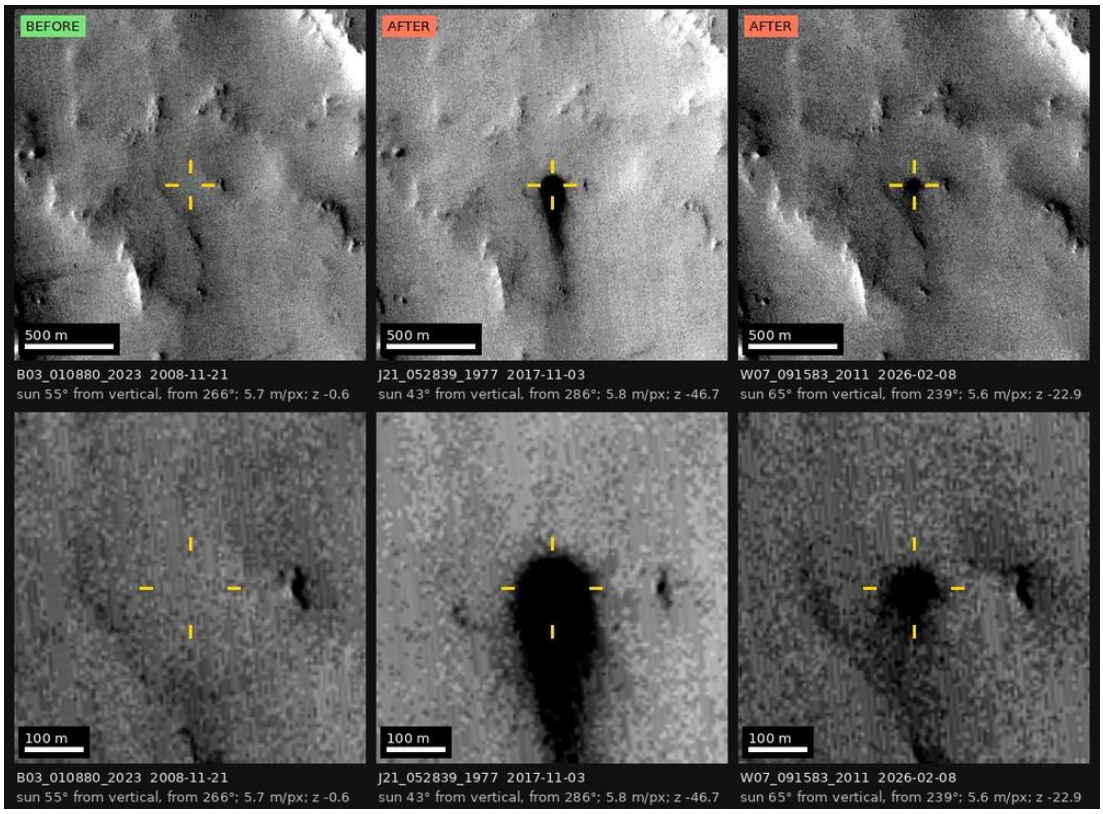

Figure 66: Candidate 25, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 26: Terrain north of Uranius Fossae

Location	27.2350° N, 270.7245° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	8 November 2024 and 18 August 2025
Images	4 without the spot (absent: 6 February 2012, 25 May 2014, 29 August 2024 and 8 November 2024); 1 with it (present: 18 August 2025)
Size	Round black spot 35 to 40 m across
HiRISE	none covers the spot (nearest footprint edge 4.9 km)
Nearest catalogued new impact	68 km (Daubar et al., 2022)

What changed. A round black spot 35 to 40 m across appeared on the plains north of Uranius Fossae in the nine months between November 2024 and August 2025. So far, one CTX image shows it.

Why it may be a new impact. Four earlier images, from 2012 to November 2024, show plain ground there, among them one of 25 May 2014 taken under the same sun as the August 2025 image. A round black spot that appeared on these plains within nine months could be the blast zone of a small new crater.

Caveats. Only one image shows it, so a flaw in that image cannot be fully ruled out, and the evidence that the earlier images would have shown a spot is weak, because the dark mark nearby is only faint. No image from the high-resolution HiRISE camera covers it; the nearest ends 4.9 km away.

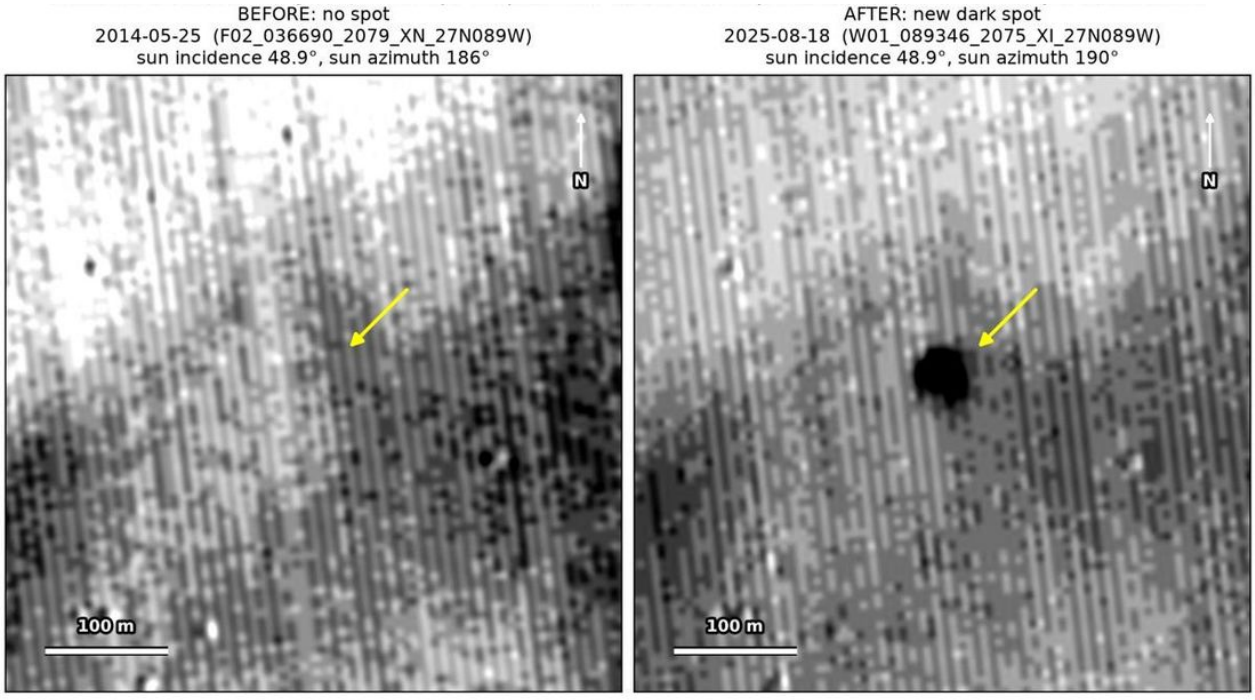

Figure 67: Candidate 26, before and after: the same 500 × 500 m of ground in 25 May 2014 (`F02_036690_2079_XN_27N089W`) and 18 August 2025 (`W01_089346_2075_XI_27N089W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

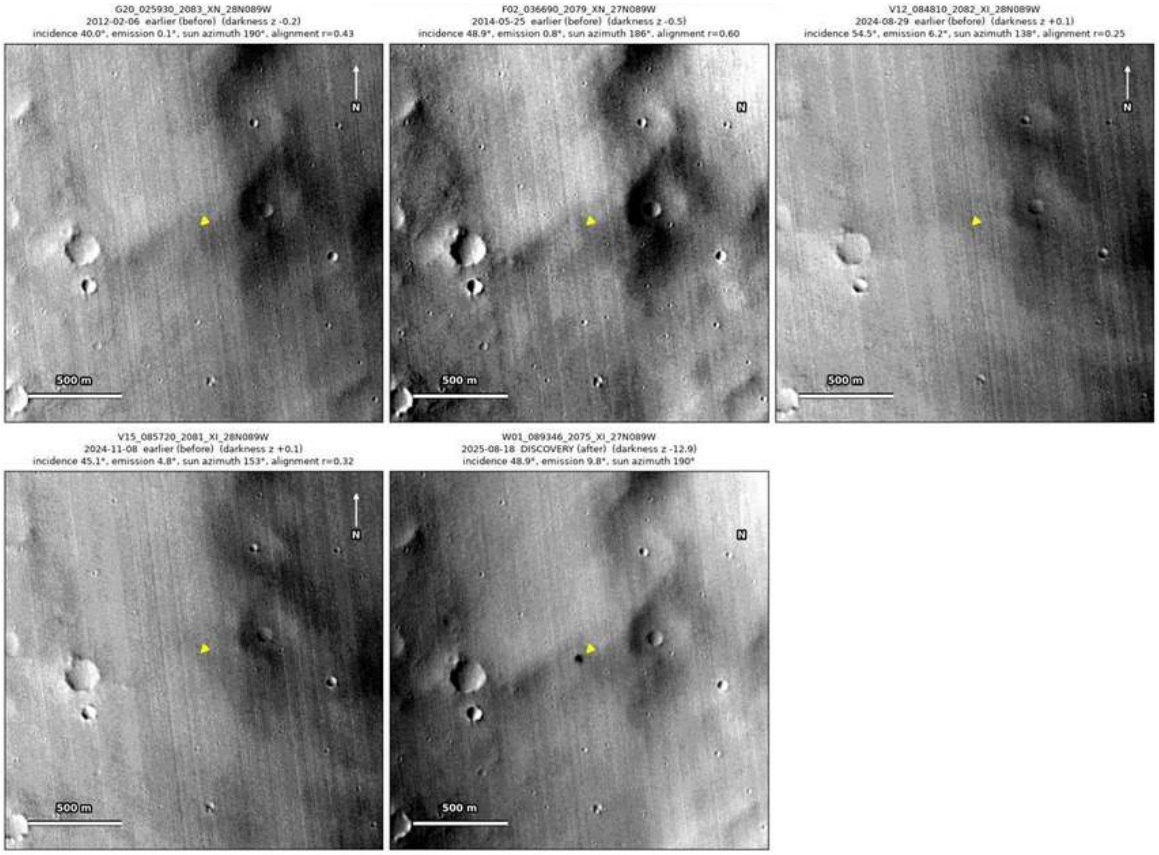

Figure 68: Candidate 26: five CTX images of the same 2×2 km patch, 2012 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 27: Southern Arabia Terra, north-east of Schiaparelli crater

Location	1.7153° N, 27.1971° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	24 August 2020 and 22 July 2025
Images	8 without the spot (absent: 9 May 2007, 2 October 2015, 18 October 2015, 13 December 2015, 23 December 2015, 23 July 2020, 13 August 2020 and 24 August 2020); 1 with it (present: 22 July 2025)
Size	Dark core 100 × 150 m with radiating lobes, inside a halo 250 m across
HiRISE	none covers the spot (nearest footprint edge 11.7 km)
Nearest catalogued new impact	113 km (Daubar et al., 2022)

What changed. A large, irregular dark spot with radiating lobes, its core 100 × 150 m inside a halo 250 m across, appeared on bright plains of southern Arabia Terra, north-east of Schiaparelli crater, between August 2020 and July 2025. So far, one CTX image shows it.

Why it may be a new impact. Eight earlier images, from 2007 to 2020, show plain ground there. In four of them, from October and December 2015, the sun shines from the same direction as in the July 2025 image, and a dark speck 290 m away shows in all four. A compact core with a halo and radiating lobes is how the blast zone of a small new crater can look, and is unlike the tracks of a dust devil.

Caveats. Only one image shows it, and that image has strong striping, so a flaw in it cannot be fully ruled out. No image from the high-resolution HiRISE camera covers it; the nearest ends 11.7 km away.

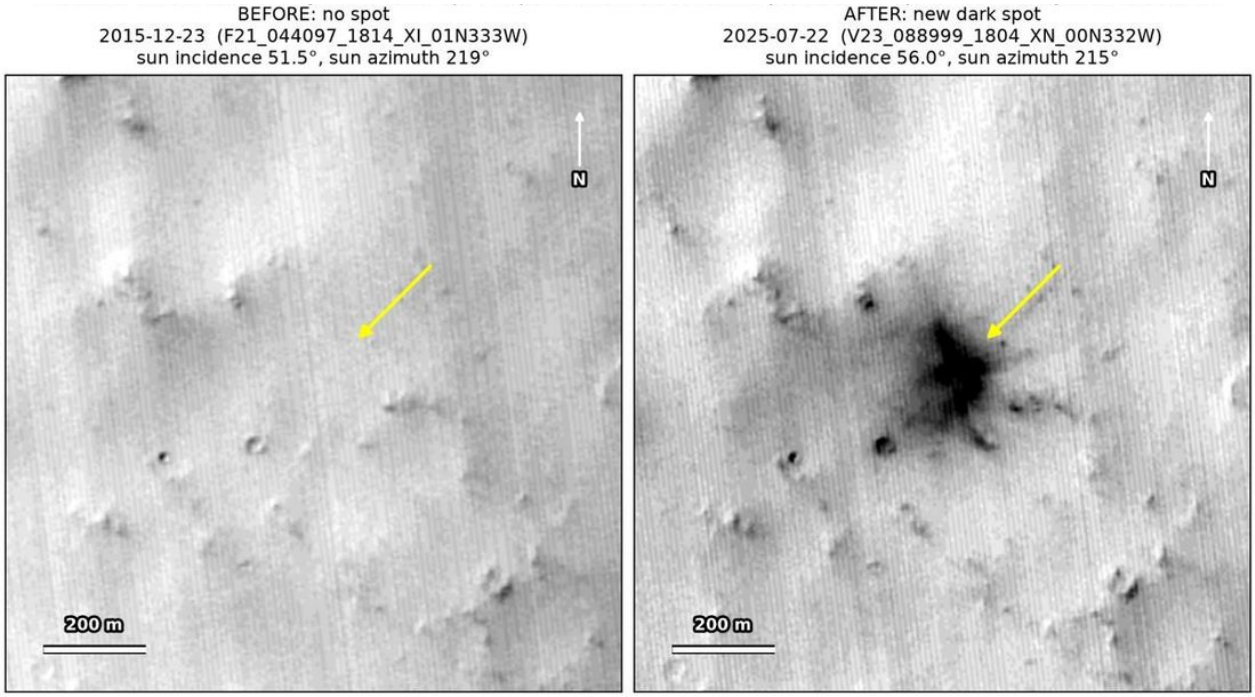

Figure 69: Candidate 27, before and after: the same 1200 × 1200 m of ground in 23 December 2015 (`F21_044097_1814_XI_01N333W`) and 22 July 2025 (`V23_088999_1804_XN_00N332W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

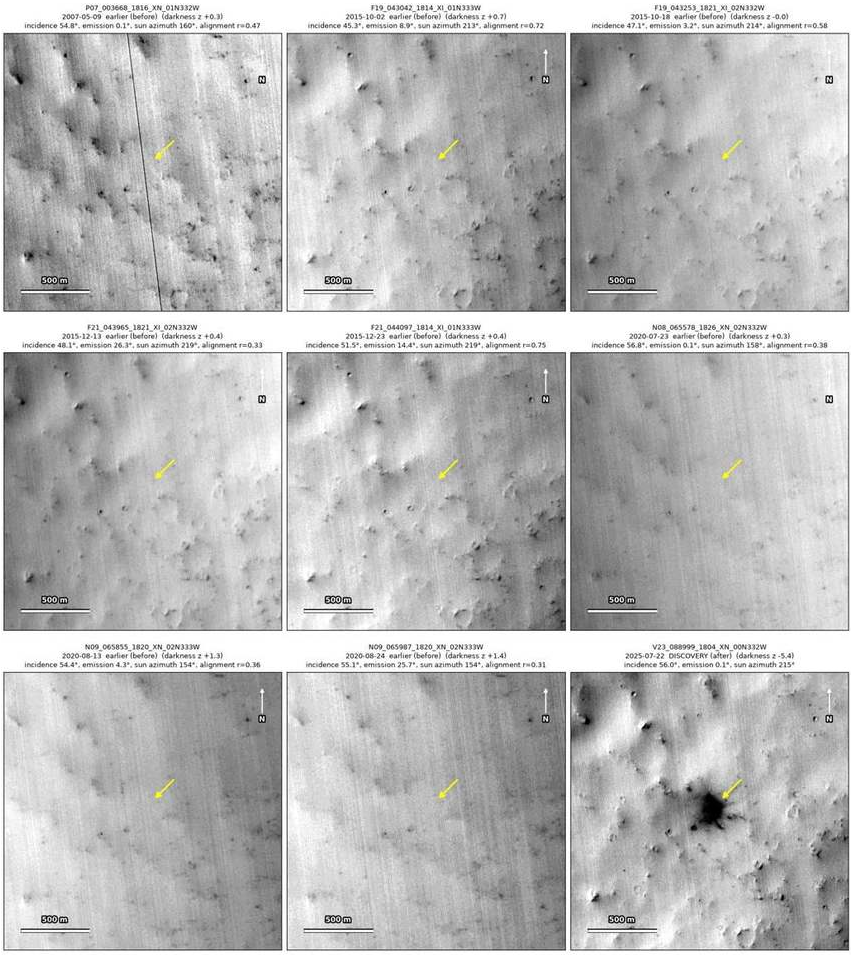

Figure 70: Candidate 27: nine CTX images of the same 2×2 km patch, 2007 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 28: Arabia Terra, north of Schiaparelli crater

Location	10.3730° N, 12.9418° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 70%
Formed between	10 October 2023 and 7 September 2024
Images	2 without the spot (absent: 19 December 2013 and 10 October 2023); 2 with it (present: 7 September 2024 and 13 July 2025)
Size	Compact dark spot 40×60 m
HiRISE	none covers the spot (nearest footprint edge 19.7 km)
Nearest catalogued new impact	128 km (Daubar et al., 2022)

What changed. A compact dark spot 40×60 m appeared on flat, bright plains of Arabia Terra, north of Schiaparelli crater, in the 11 months between October 2023 and September 2024. Two CTX images from two orbits, of September 2024 and July 2025, show it.

Why it may be a new impact. Two earlier images, of December 2013 and October 2023, line up well and show plain ground there, and the second was taken under the same sun as the July 2025 image. In the two images that show the spot, the sun shines from directions 57° apart, so it is not a shadow, and a compact new dark spot on flat plains is how the blast zone of a small new crater looks.

Caveats. No comparable dark mark lies within 300 m to show that the earlier images would have caught a spot like this, and the September 2024 image is low in contrast. No image from the high-resolution HiRISE camera covers it; the nearest ends 19.7 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 70% that the spot is a new impact. It noted that the head of a short new slope streak cannot be excluded at the resolution of CTX and estimated the probability at about 70%. It placed the spot at 10.3751° N, 12.9457° E, 259 m from the position given above.

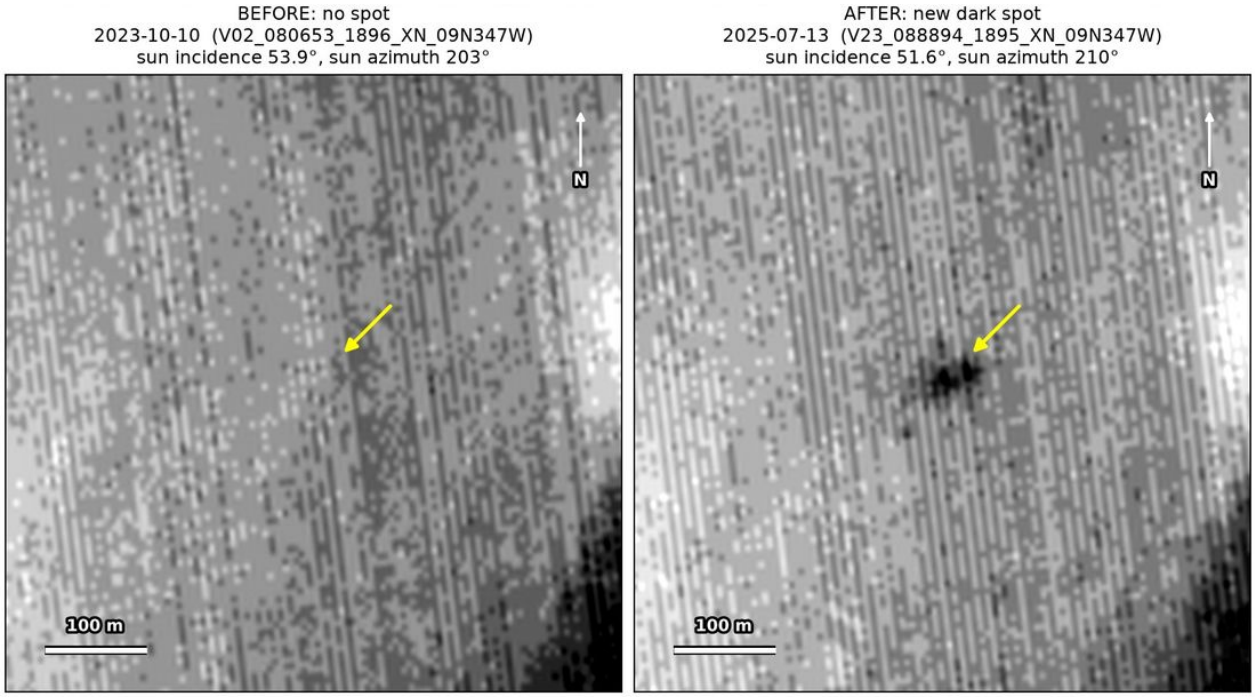

Figure 71: Candidate 28, before and after: the same 600×600 m of ground in 10 October 2023 (`V02_080653_1896_XN_09N347W`) and 13 July 2025 (`V23_088894_1895_XN_09N347W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

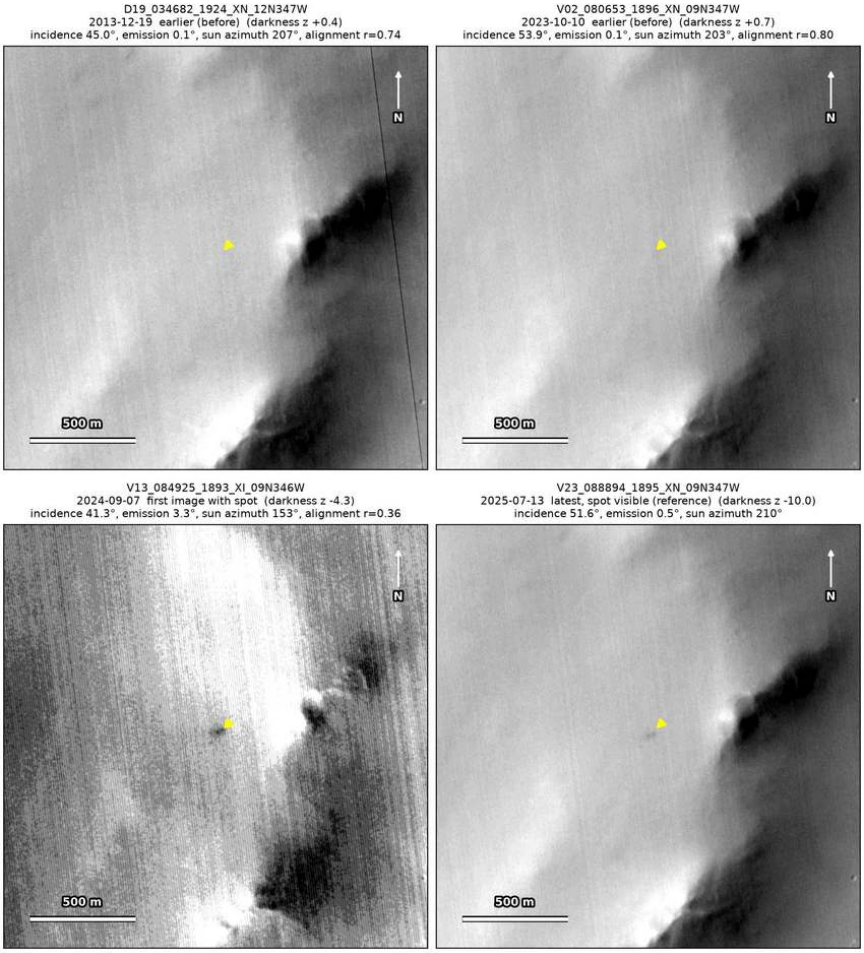

Figure 72: Candidate 28: four CTX images of the same 2×2 km patch, 2013 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

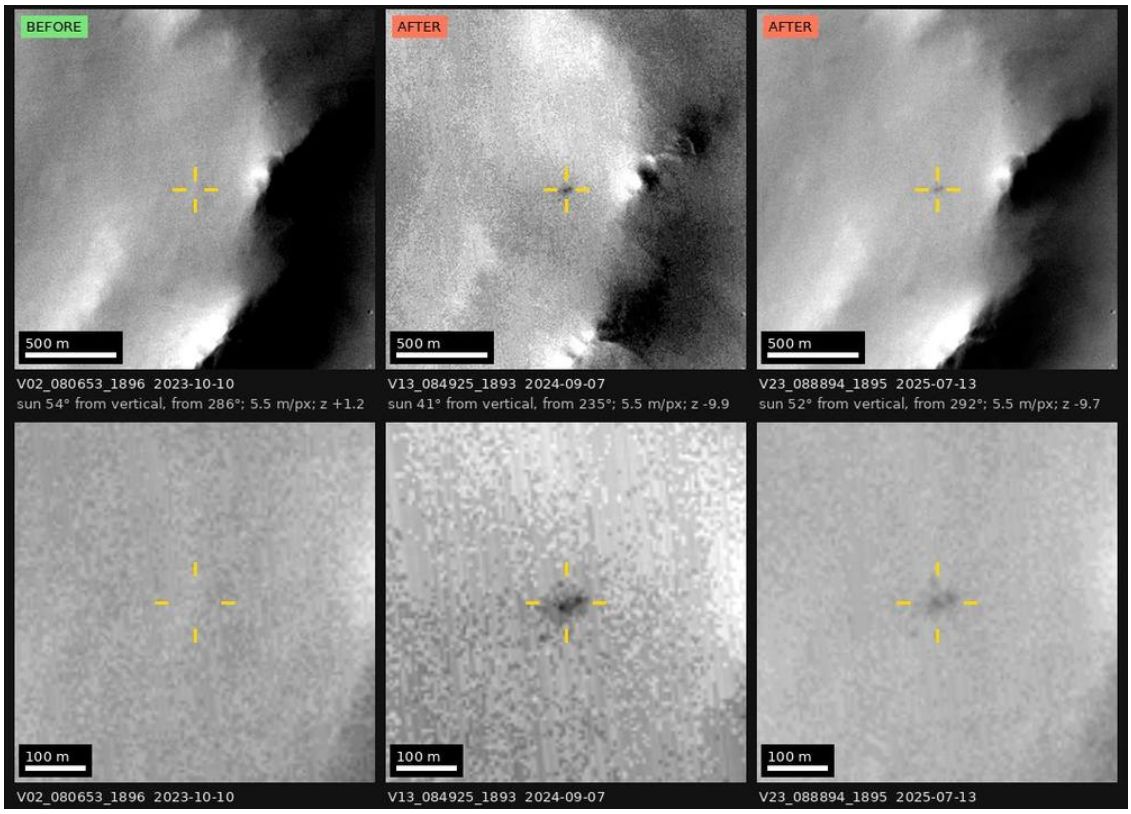

Figure 73: Candidate 28, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 29: Lava plains between Ceraunius Fossae and Ascraeus Mons

Location	17.2763° N, 250.6787° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	30 June 2024 and 14 April 2026
Images	3 without the spot (absent: 4 May 2007, 26 September 2013 and 30 June 2024); 1 with it (present: 14 April 2026)
Size	Black dot 25 to 30 m across
HiRISE	none covers the spot (nearest footprint edge 18.7 km)
Nearest catalogued new impact	109 km (Daubar et al., 2022)

What changed. A black dot 25 to 30 m across appeared on the lava plains between Ceraunius Fossae and Ascraeus Mons some time between June 2024 and April 2026. So far, one CTX image shows it.

Why it may be a new impact. Three earlier images, from 2007, 2013 and 2024, show plain ground there. Two of them, of 4 May 2007 and 30 June 2024, were taken under the same sun as the April 2026 image, and nearby pits cast the same shadows in them, so the dot is not a pit's shadow; a dark mark nearby also shows in the 2007 and 2013 images. A black dot appearing on lava plains could be the blast zone of a small new crater.

Caveats. Only one image shows it, so a flaw in that image cannot be fully ruled out, and in the last image without it, of June 2024, its absence can be judged only by eye. No image from the high-resolution HiRISE camera covers it; the nearest ends 18.7 km away.

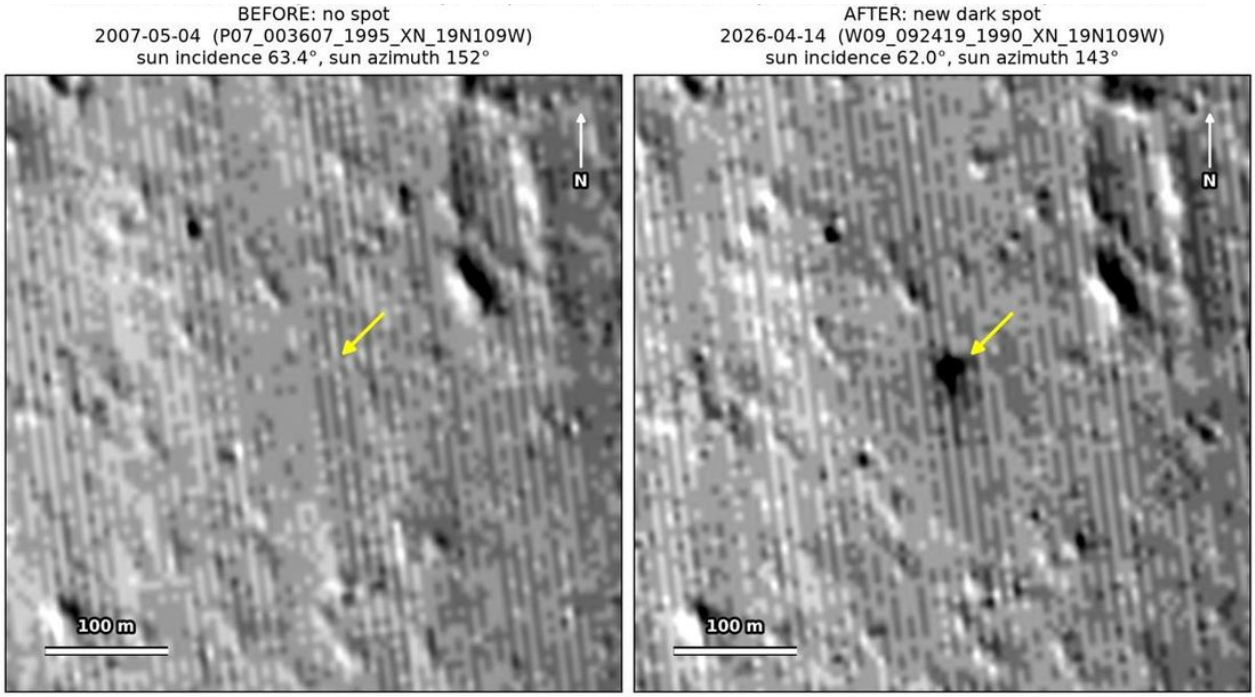

Figure 74: Candidate 29, before and after: the same 500 × 500 m of ground in 4 May 2007 (`P07_003607_1995_XN_19N109W`) and 14 April 2026 (`W09_092419_1990_XN_19N109W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

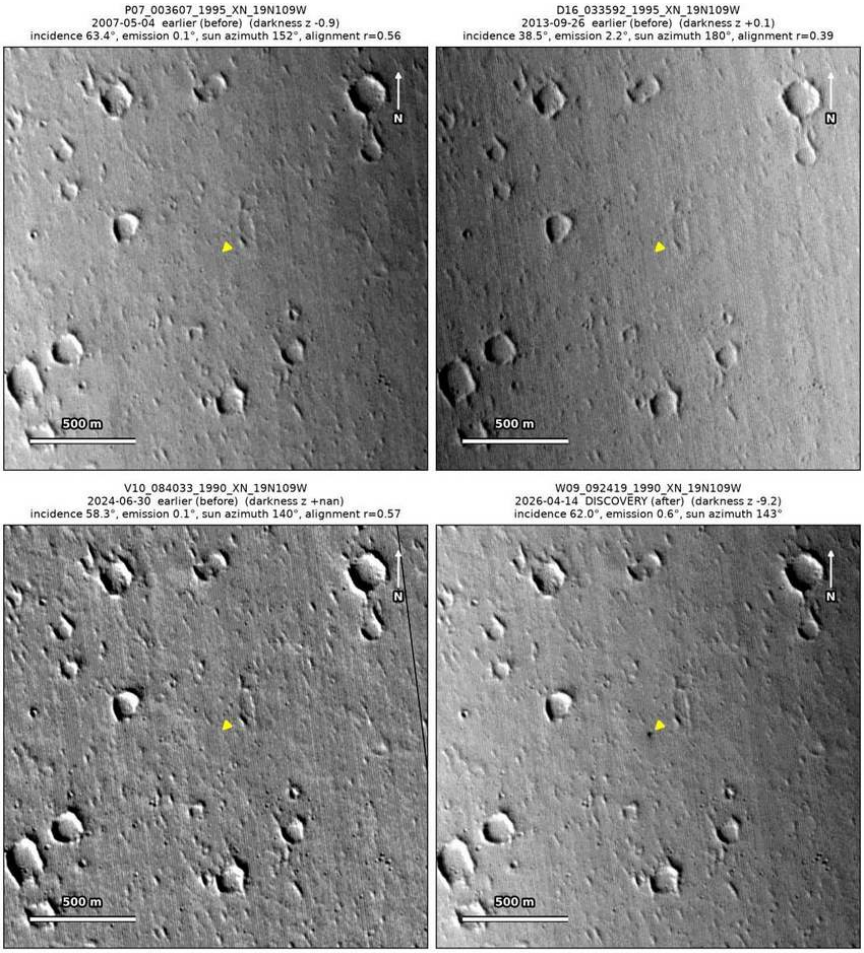

Figure 75: Candidate 29: four CTX images of the same 2×2 km patch, 2007 to 2026. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 30: Elysium Planitia, south-west of Elysium Mons

Location	16.8749° N, 142.6064° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 55%
Formed between	13 December 2024 and 14 August 2025
Images	6 without the spot (absent: 4 January 2014, 21 May 2014, 15 May 2018, 6 June 2018, 5 May 2022 and 13 December 2024); 1 with it (present: 14 August 2025)
Size	Compact dark dot 20×35 m
HiRISE	none covers the spot (nearest footprint edge 3.7 km)
Nearest catalogued new impact	54 km (Daubar et al., 2022)

What changed. A compact dark dot 20×35 m appeared on smooth plains of Elysium Planitia, south-west of Elysium Mons, in the eight months between December 2024 and August 2025. So far, one CTX image shows it.

Why it may be a new impact. Six earlier images, from 2014 to December 2024, show plain ground there, among them one of 21 May 2014 taken under the same sun as the August 2025 image, while a dark mark 240 m away shows clearly in each of the five that could be measured. A compact dark dot appearing on smooth plains could be the blast zone of a small new crater.

Caveats. Only one image shows it, and the dot is drawn out along the lines of that image, so a flaw in the image cannot be fully ruled out. No image from the high-resolution HiRISE camera covers it; the nearest ends 3.7 km away.

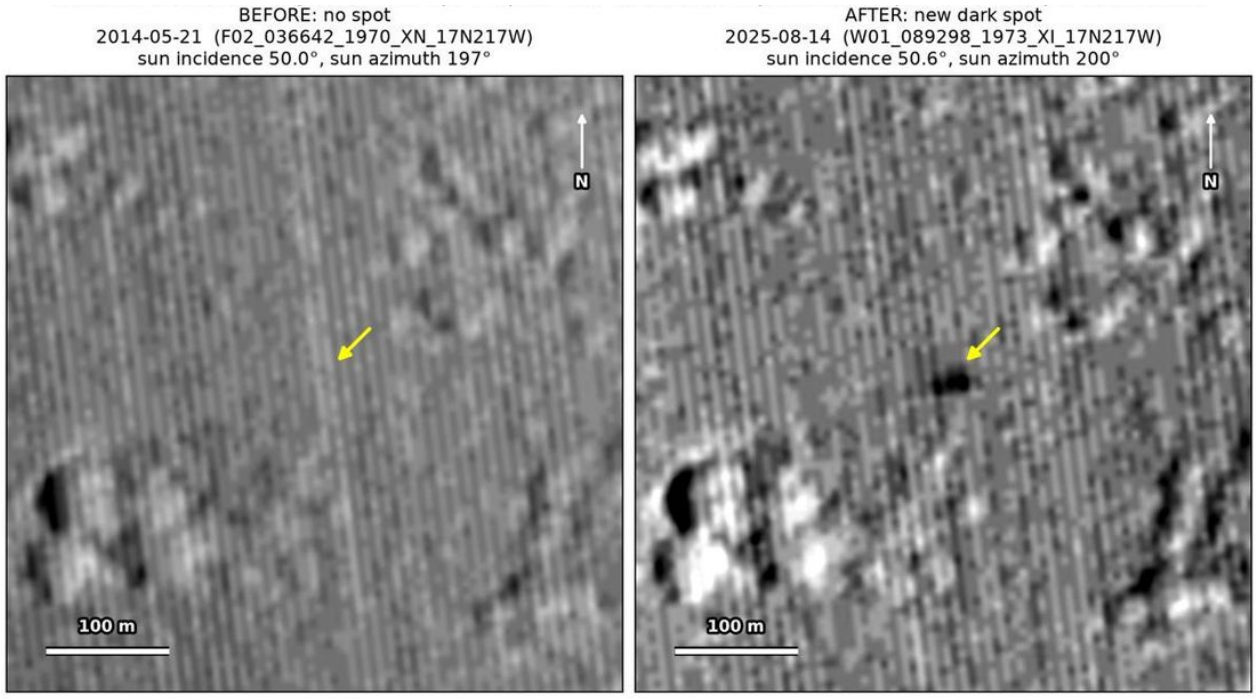

Figure 76: Candidate 30, before and after: the same 500×500 m of ground in 21 May 2014 (`F02_036642_1970_XN_17N217W`) and 14 August 2025 (`W01_089298_1973_XI_17N217W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

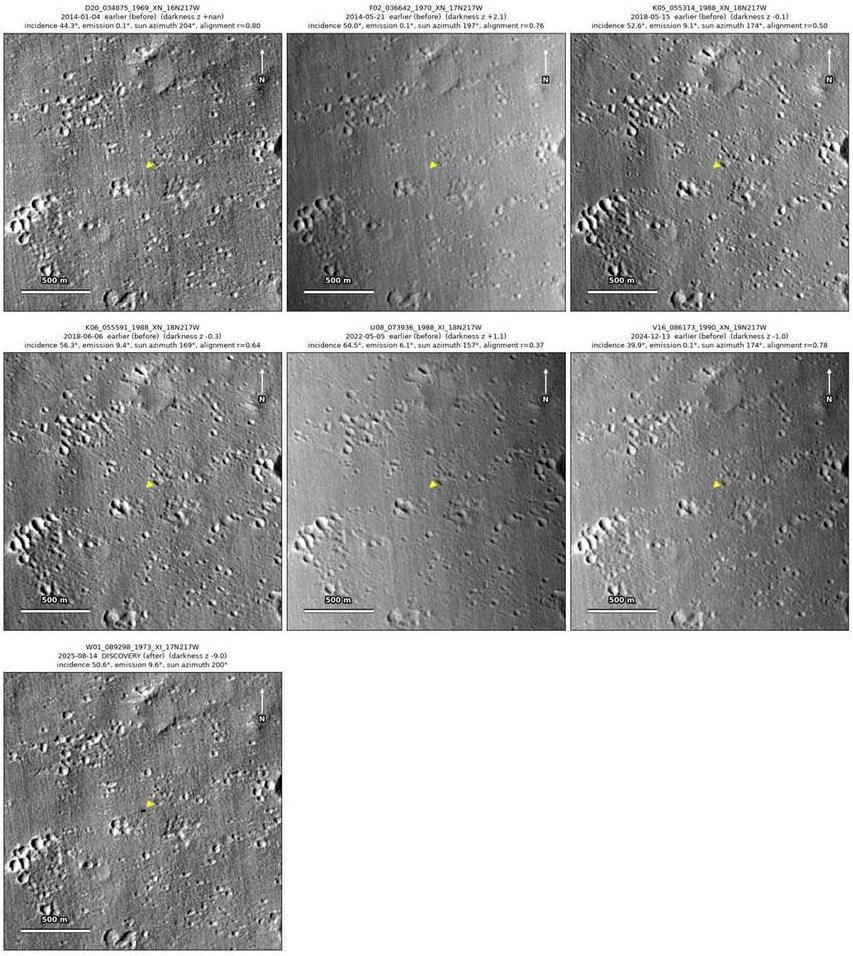

Figure 77: Candidate 30: seven CTX images of the same 2×2 km patch, 2014 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 31: Tharsis lava plains in eastern Ulysses Fossae

Location	14.2270° N, 238.2743° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	19 March 2019 and 19 March 2024
Images	2 without the spot (absent: 27 January 2010 and 19 March 2019); 5 with it (present: 19 March 2024, 2 January 2025, 17 April 2025, 11 June 2025 and 9 July 2025)
Size	Compact dark spot 70×55 m
HiRISE	none covers the spot (nearest footprint edge 10.2 km)
Nearest catalogued new impact	18 km (Daubar et al., 2022)

What changed. A compact dark spot 70×55 m appeared between March 2019 and March 2024 on smooth Tharsis lava plains lying between north-south fault troughs in eastern Ulysses Fossae. Five CTX images from five orbits show it unchanged up to July 2025.

Why it may be a new impact. An image of 27 January 2010, which lines up well and was taken with the sun at almost the same height as in the 2 January 2025 image, shows neither a spot nor any landform there, although a dark mark 192 m to the west does show. In the five images with the spot the sun shines from directions spanning 47°, so it is not a shadow, and a compact spot that stays unchanged fits a small new impact.

Caveats. Because the March 2019 image does not show the dark mark to the west, the spot can only be dated firmly to between January 2010 and March 2024. Wind could have exposed darker ground around a tiny pit, but that is unlikely: the well-aligned 2010 image shows no landform there, and the spot has stayed compact and unchanged for 16 months. No image from the high-resolution HiRISE camera covers it; the nearest ends 10.2 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 90% that the spot is a new impact. It placed the spot at 14.2288° N, 238.2695° E, 295 m from the position given above.

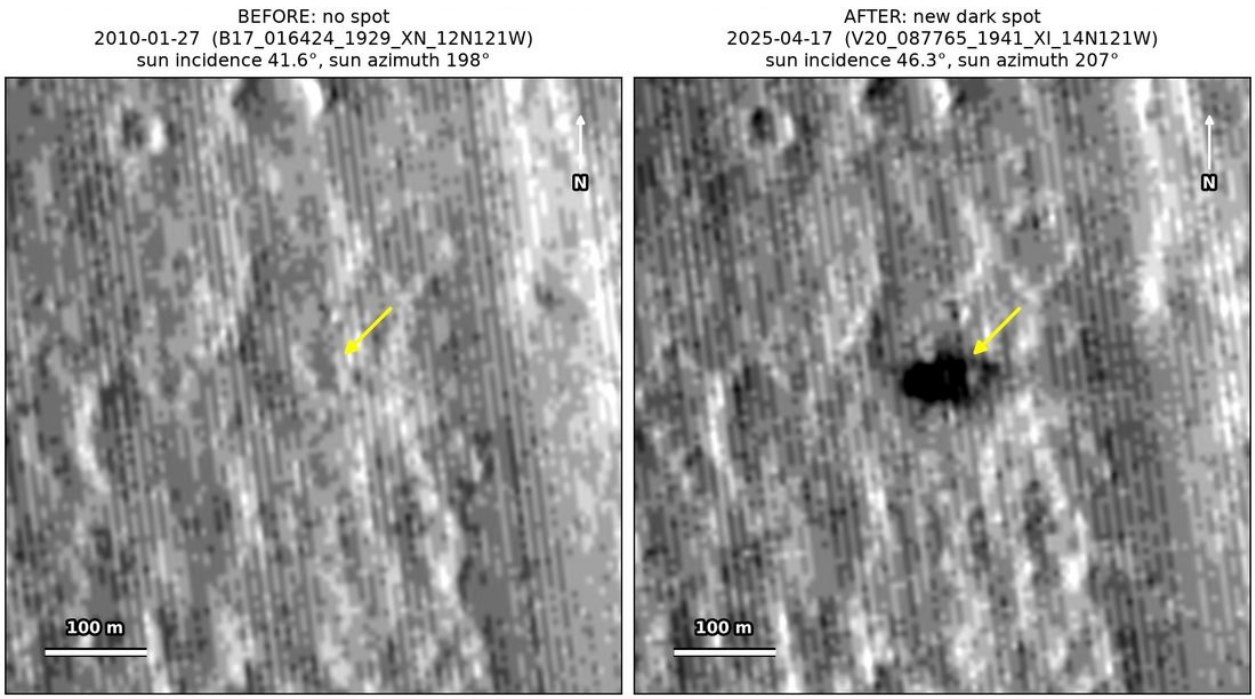

Figure 78: Candidate 31, before and after: the same 600 × 600 m of ground in 27 January 2010 (`B17_016424_1929_XN_12N121W`) and 17 April 2025 (`V20_087765_1941_XI_14N121W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

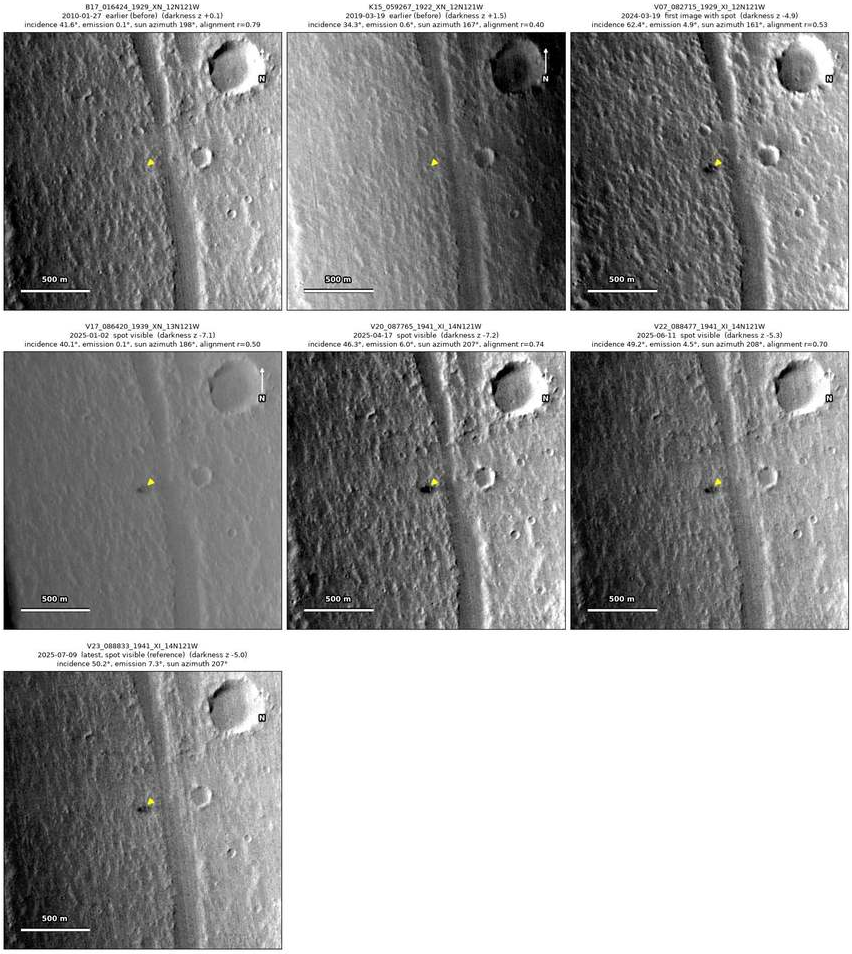

Figure 79: Candidate 31: seven CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

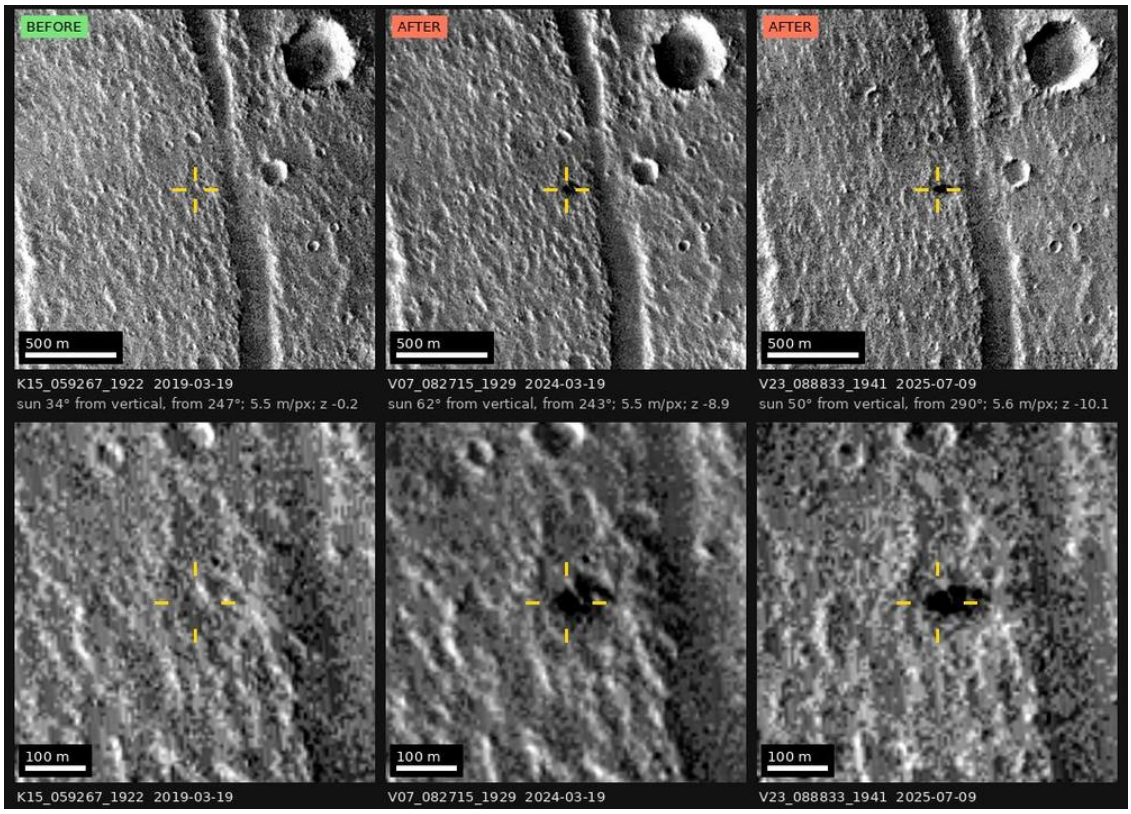

Figure 80: Candidate 31, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 32: Central Arabia Terra, east of Tikhonravov crater

Location	14.0079° N, 41.4657° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 70%
Formed between	27 November 2022 and 21 October 2025
Images	3 without the spot (absent: 30 December 2007, 13 August 2016 and 27 November 2022); 1 with it (present: 21 October 2025)
Size	A main spot about 60 m across and four or five dots 15 to 30 m across, spread over 300 m
HiRISE	none covers the spot (nearest footprint edge 22.3 km)
Nearest catalogued new impact	54 km (Daubar et al., 2022)

What changed. A cluster of dark spots, a main spot about 60 m across with four or five dots of 15 to 30 m within 300 m, appeared on the floor of a hollow in central Arabia Terra, east of Tikhonravov crater, between November 2022 and October 2025. So far, one CTX image shows it.

Why it may be a new impact. Three earlier images, from 2007, 2016 and 2022, show plain ground there. Dots of different sizes in a natural cluster pattern argue against a flaw in the image, separate round dots with no streak through them argue against dark ground exposed by wind, and the pattern fits a cluster of small new impacts.

Caveats. Only one image shows it, and there is no clearly dark mark nearby to show that the earlier images would have caught the dots: the nearest, a scarp (a steep slope) 255 m to the east, shows only faintly in them. No image from the high-resolution HiRISE camera covers it; the nearest ends 22.3 km away.

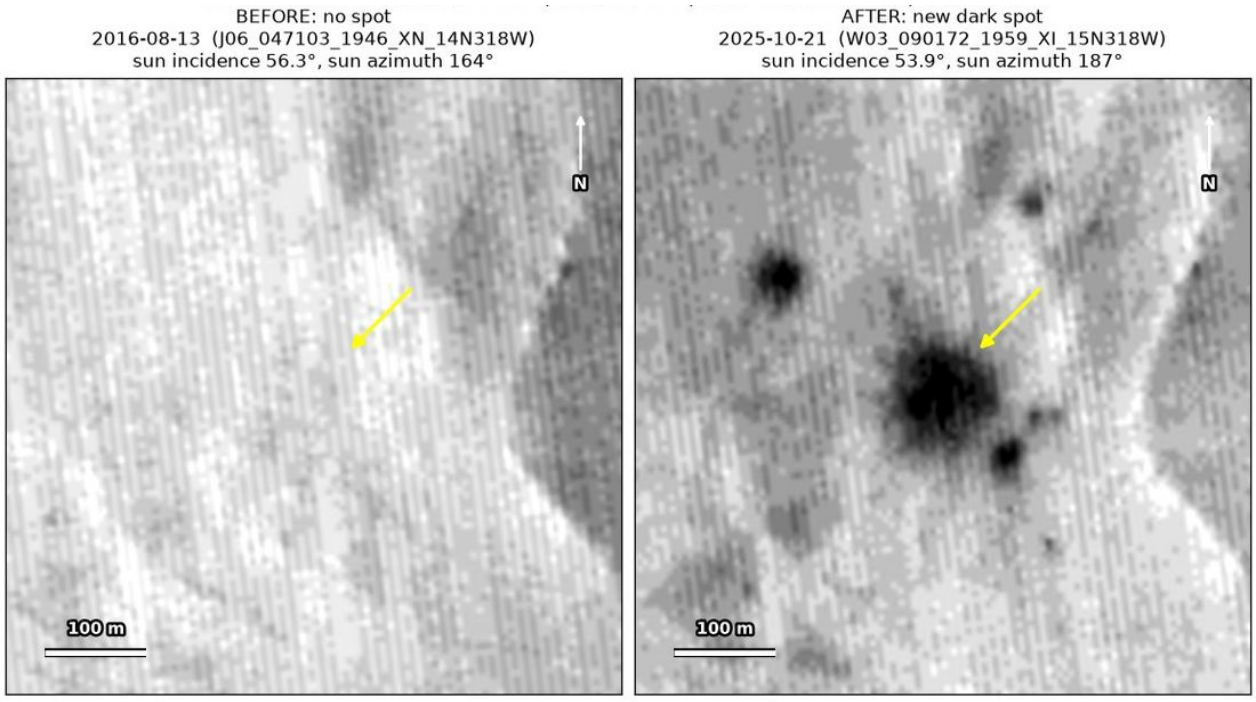

Figure 81: Candidate 32, before and after: the same 600 × 600 m of ground in 13 August 2016 (`J06_047103_1946_XN_14N318W`) and 21 October 2025 (`W03_090172_1959_XI_15N318W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

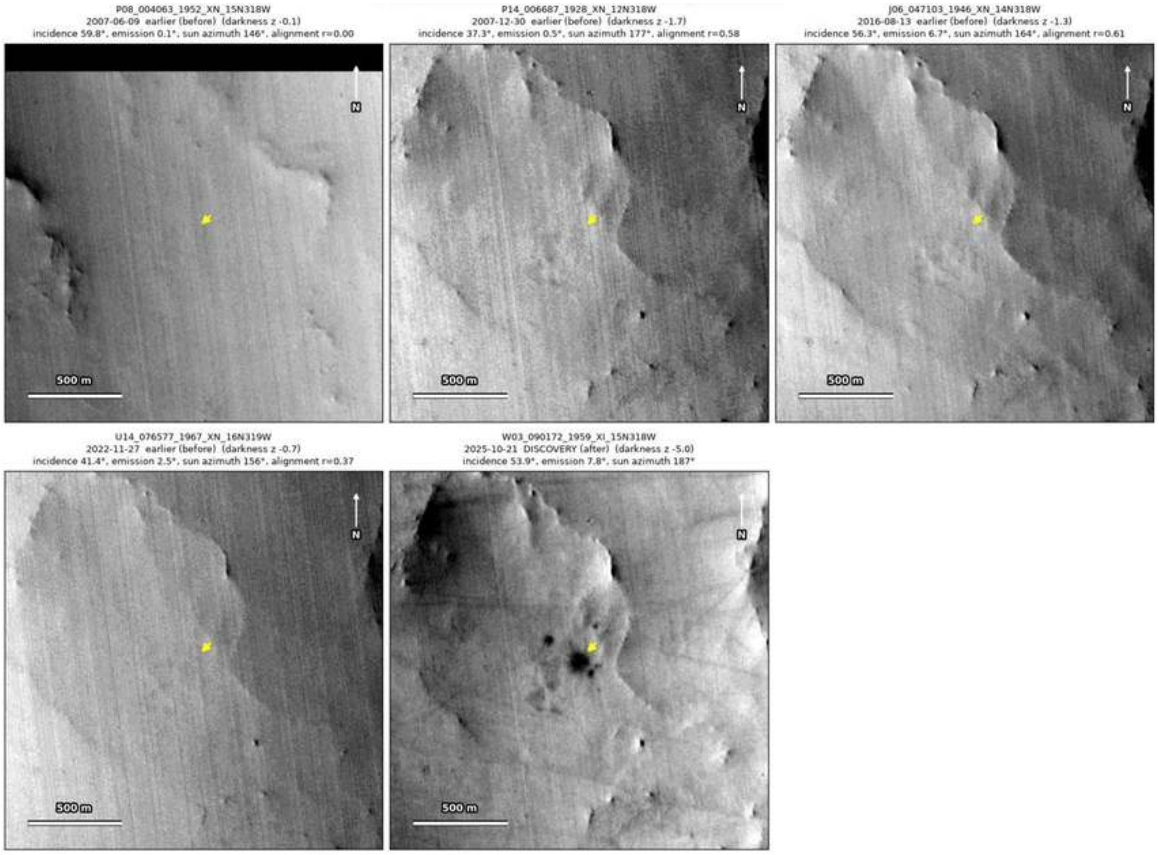

Figure 82: Candidate 32: five CTX images of the same 2×2 km patch, 2007 to 2025. The June 2007 image, which does not line up with the others, is shown but not used to date the spot. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 33: Floor of Denning crater

Location	17.3398° S, 33.4530° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 40%
Formed between	14 June 2025 and 10 September 2025
Images	3 without the spot (absent: 19 June 2012, 29 July 2020 and 14 June 2025); 1 with it (present: 10 September 2025)
Size	Very dark compact blob 80 to 100 m across
HiRISE	none covers the spot (nearest footprint edge 13.9 km)
Nearest catalogued new impact	68 km (Daubar et al., 2022)

What changed. A very dark, compact blob 80 to 100 m across appeared on the floor of Denning crater, beside a small older crater, in the space of only 88 days between June and September 2025. So far, one CTX image shows it.

Why it may be a new impact. Three earlier images, from 2012, 2020 and June 2025, show plain ground there, while a feature 295 m away shows in them; the last, of 14 June 2025, was taken under nearly the same sun as the September 2025 image. A very dark, compact blob that appeared within 88 days could be the blast zone of a small new crater.

Caveats. A lingering dust devil explains it nearly as well: dozens of new dust-devil tracks cross the scene in the same 88 days, and one extends from the blob to the north-east. Wind scouring beside the small crater is another possibility, and only one image shows the blob. No image from the high-resolution HiRISE camera covers it; the nearest ends 13.9 km away.

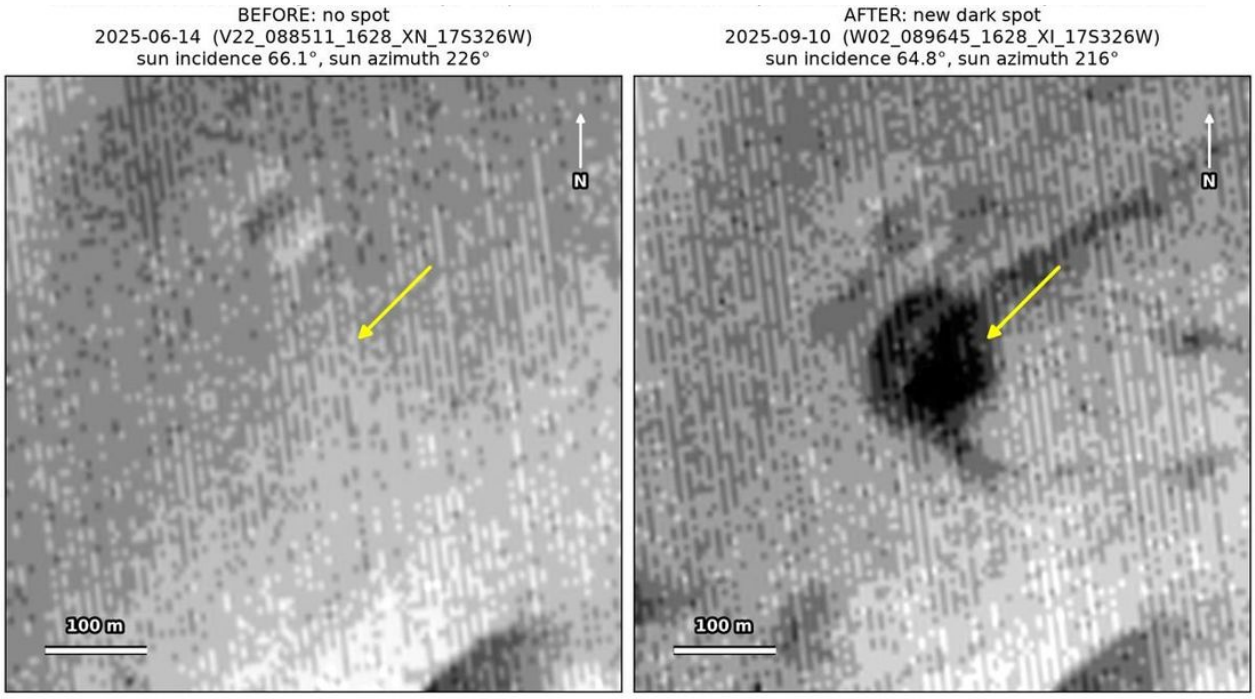

Figure 83: Candidate 33, before and after: the same 600 × 600 m of ground in 14 June 2025 (`V22_088511_1628_XN_17S326W`) and 10 September 2025 (`W02_089645_1628_XI_17S326W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

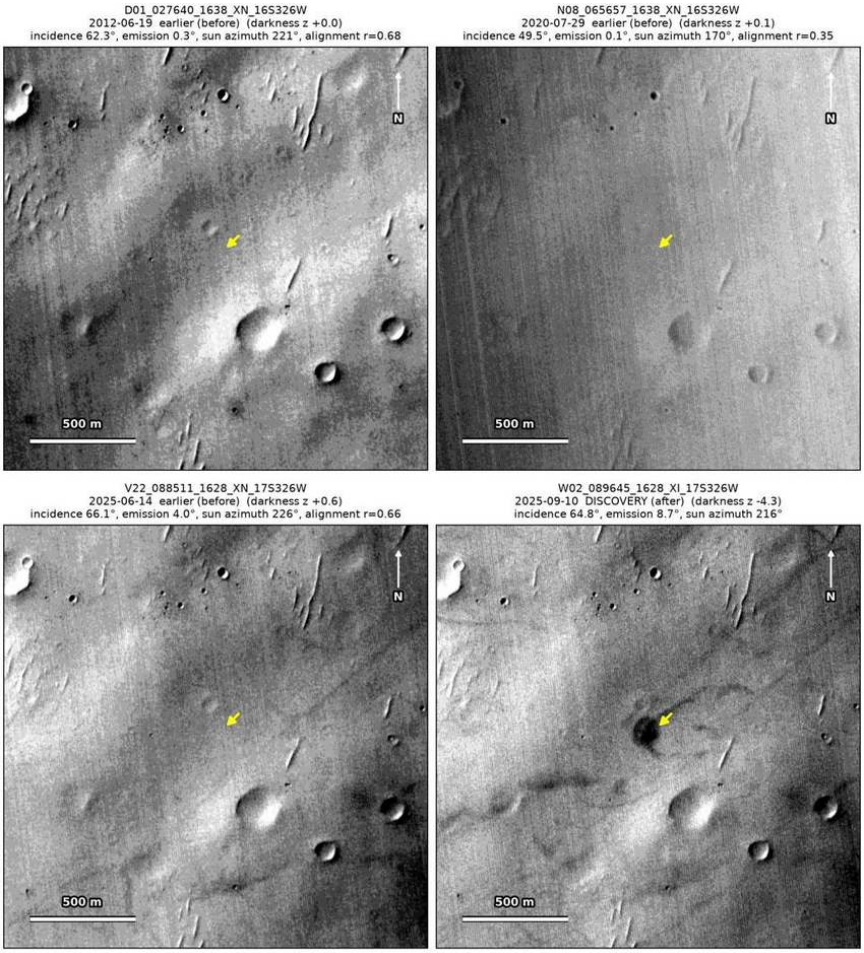

Figure 84: Candidate 33: four CTX images of the same 2×2 km patch, 2012 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 34: Platy lava plains of Elysium Planitia, west of Athabasca Valles

Location	7.0015° N, 150.2711° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	25 August 2008 and 8 June 2021
Images	1 without the spot (absent: 25 August 2008); 5 with it (present: 8 June 2021, 7 July 2024, 6 January 2025, 13 July 2025 and 3 May 2026)
Size	Dark core 25 to 30 m across, inside a halo about 100 m across
HiRISE	none covers the spot (nearest footprint edge 14.3 km)
Nearest catalogued new impact	113 km (Daubar et al., 2022)

What changed. A dark core 25 to 30 m across, inside a diffuse halo about 100 m across, appeared on platy lava plains west of Athabasca Valles between August 2008 and June 2021. Five CTX images from five orbits show it up to May 2026, very dark until 2025 and fainter in 2026.

Why it may be a new impact. The August 2008 image, taken under the same sun as the June 2021 one, lines up well and shows nothing at the spot, while a dark mark 226 m to the south-east shows in it, so it would have shown the spot. In the five later images the sun shines from directions up to 65° apart, so it is not a shadow, and a round core-and-halo spot that keeps its shape for five years is how the blast zone of a small new crater looks.

Caveats. Only one earlier image, from August 2008, shows the ground without it, so the spot could have formed at any time in the nearly 13 years before June 2021. Darker ground exposed by wind is the remaining alternative, but it is unlikely: the spot is round, has no streak and has kept the same shape for five years. No image from the high-resolution HiRISE camera covers it; the nearest ends 14.3 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 85% that the spot is a new impact. It noted that the site lies inside the area and period of the survey of Bickel et al. (2025), whose catalogue could not be retrieved, so a prior listing there cannot be excluded. It placed the spot at 7.0033° N, 150.2789° E, 474 m from the position given above.

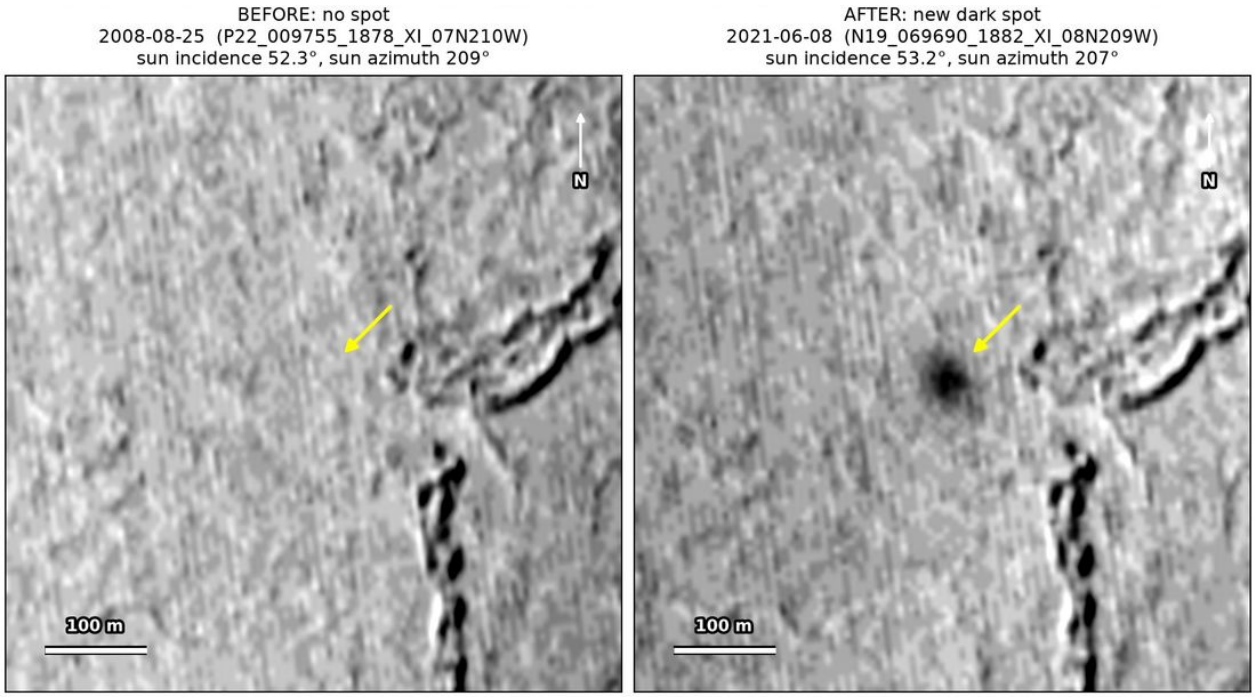

Figure 85: Candidate 34, before and after: the same 600 × 600 m of ground in 25 August 2008 (`P22_009755_1878_XI_07N210W`) and 8 June 2021 (`N19_069690_1882_XI_08N209W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

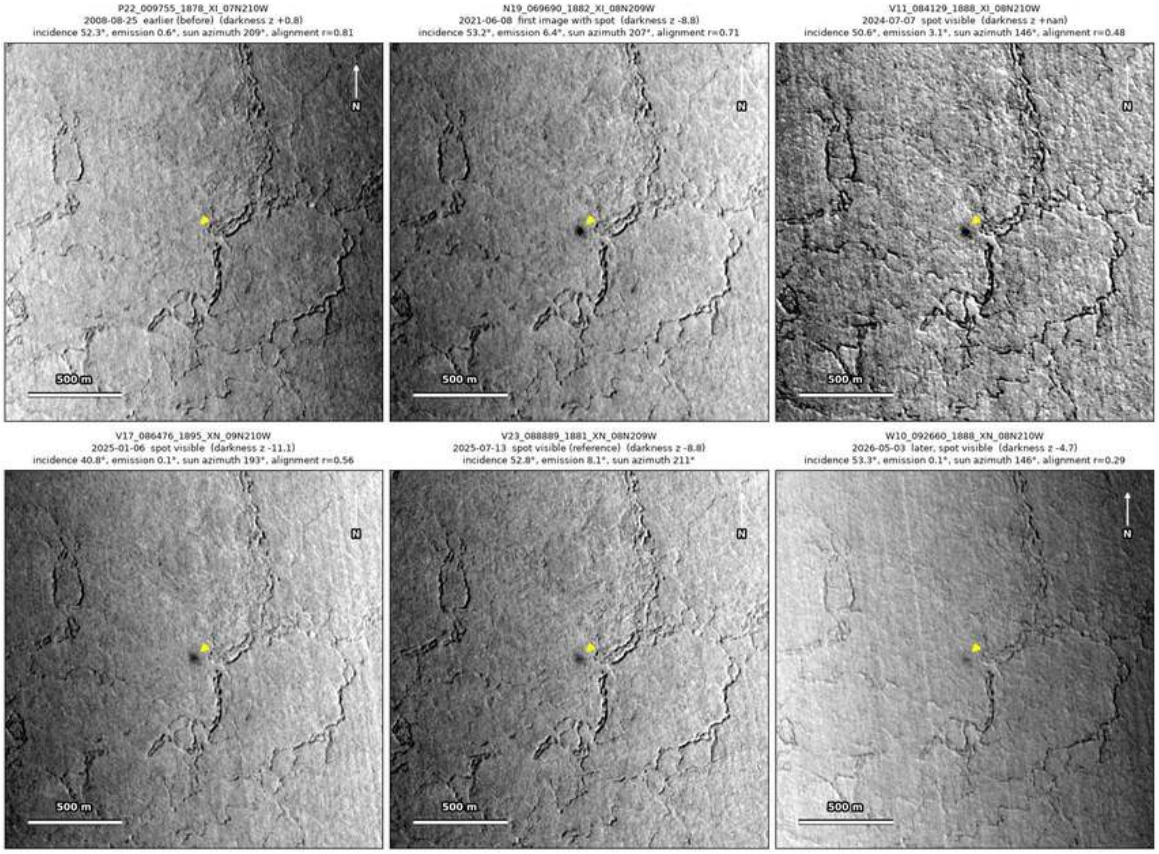

Figure 86: Candidate 34: six CTX images of the same 2×2 km patch, 2008 to 2026. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

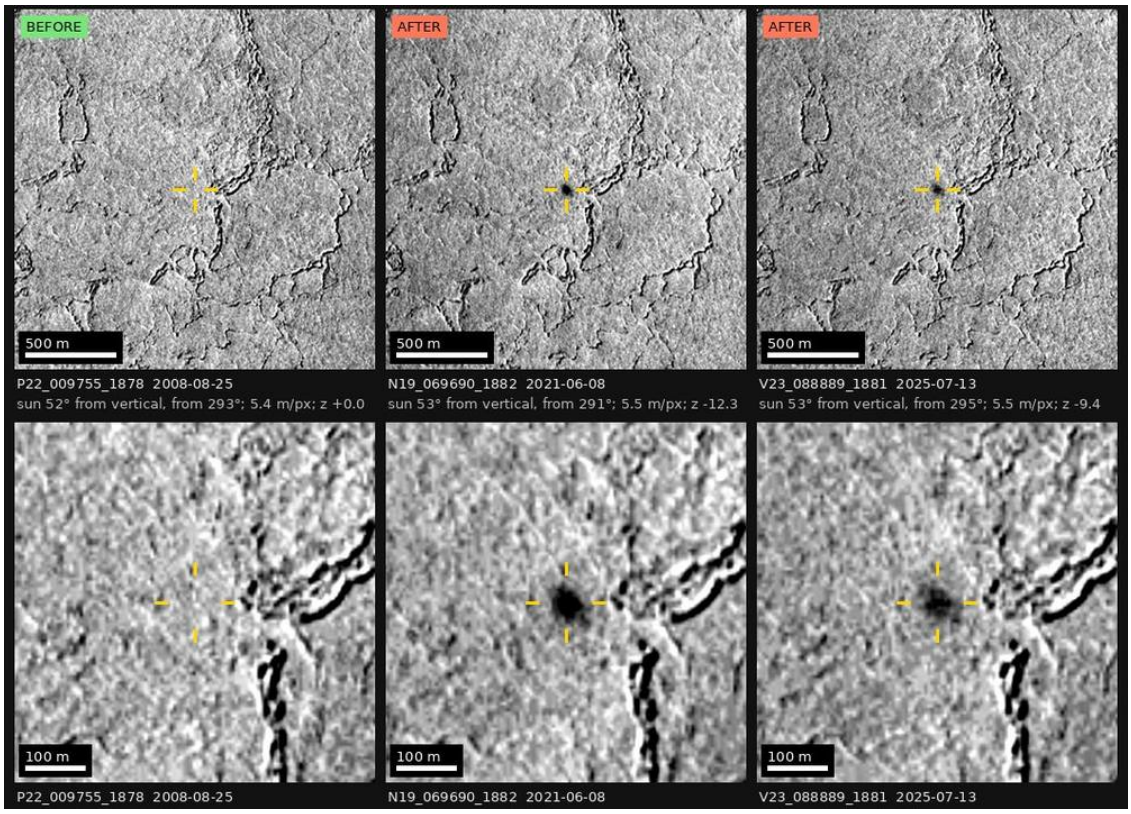

Figure 87: Candidate 34, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 35: Medusae Fossae Formation near Eumenides Dorsum

Location	4.4544° N, 205.5598° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 55%
Formed between	28 December 2011 and 2 September 2023
Images	2 without the spot (absent: 14 June 2008 and 28 December 2011); 4 with it (present: 2 September 2023, 19 March 2025, 18 May 2025 and 9 October 2025)
Size	Diffuse dark patch 150 to 200 m across, with a faint darker centre
HiRISE	none covers the spot (nearest footprint edge 28.1 km)
Nearest catalogued new impact	52 km (Daubar et al., 2022)

What changed. A diffuse dark patch 150 to 200 m across, with a faint darker centre, appeared on the wind-carved ground of the Medusae Fossae Formation near Eumenides Dorsum between December 2011 and September 2023. Four CTX images from four orbits show it, up to October 2025.

Why it may be a new impact. The June 2008 image, which lines up well and was taken with the sun at almost the same height as in the October 2025 image, shows no patch there, and neither does an image of December 2011. A dark patch that appeared and has lasted for two years, with no dust-devil track in any of the four images that show it, could be the blast zone of a small new crater.

Caveats. A change in surface brightness driven by wind on this dusty ground explains it nearly as well, since the patch is diffuse and low in contrast, with no distinct core. No clearly dark mark lies near the patch to prove that the earlier images would have shown it: the darkest, 253 m to the south-east, is only faint in 2008. No image from the high-resolution HiRISE camera covers it; the nearest ends 28.1 km away.

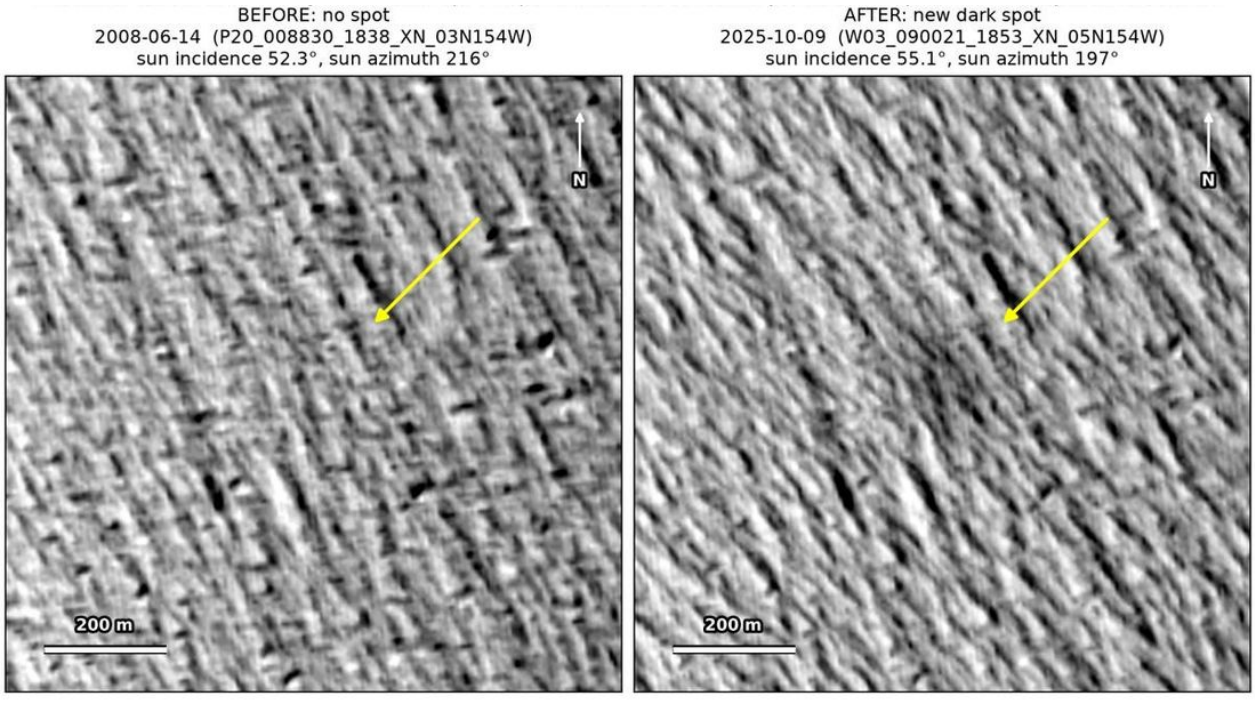

Figure 88: Candidate 35, before and after: the same 1000 × 1000 m of ground in 14 June 2008 (`P20_008830_1838_XN_03N154W`) and 9 October 2025 (`W03_090021_1853_XN_05N154W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

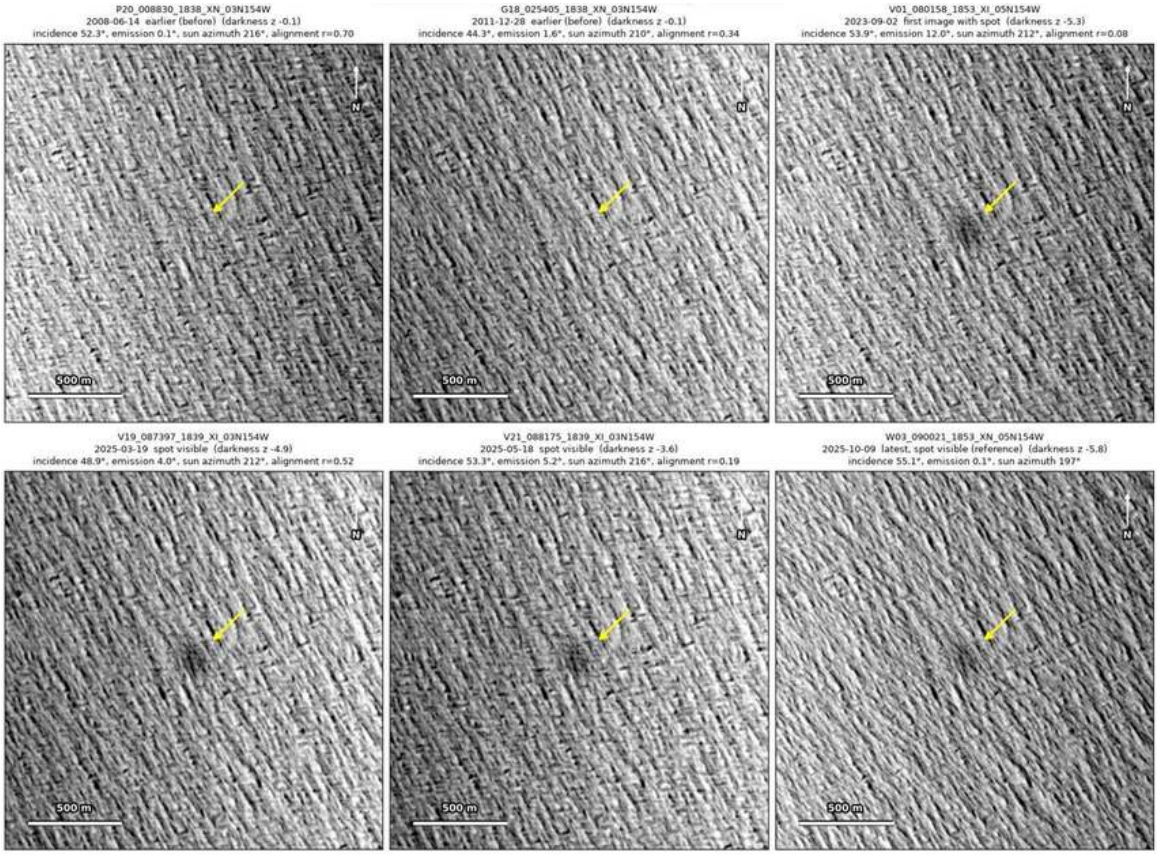

Figure 89: Candidate 35: six CTX images of the same 2×2 km patch, 2008 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 36: Northern Arabia Terra, north-west of Cassini crater

Location	29.0978° N, 27.0612° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	8 December 2014 and 3 August 2020
Images	3 without the spot (absent: 25 September 2008, 8 December 2014 and September 2018 (the last of these three images was taken during the global dust storm)); 4 with it (present: 3 August 2020, 17 June 2022, 22 March 2025 and 22 May 2025)
Size	Diffuse dark spot about 150 m across, with a darker core about 90 m across
HiRISE	none covers the spot (nearest footprint edge 4.7 km)
Nearest catalogued new impact	10 km (Daubar et al., 2022)

What changed. A diffuse dark spot about 150 m across, with a darker core about 90 m across, appeared in northern Arabia Terra, north-west of Cassini crater, between December 2014 and August 2020. Four CTX images from four orbits show it, unchanged up to May 2025.

Why it may be a new impact. The September 2008 image, under the same sun as the May 2025 image, shows nothing at the spot, though a dark mark 116 m to the north shows. Neither the December 2014 image, under the same sun as the August 2020 one, nor an image of September 2018, taken during the global dust storm, shows the spot. A round spot with a darker core that stays the same from 2020 to 2025 fits the blast zone of a small new crater.

Caveats. The December 2014 image lines up poorly, and without it the spot could have formed at any time after September 2008, since the September 2018 image, taken during the global dust storm, is not used to date the spot. A wind-driven change in surface brightness is less likely, as the spot is round, has a darker core and has been stable from 2020 to 2025, and no dust-devil track shows in the four images. No image from the high-resolution HiRISE camera covers it; the nearest ends 4.7 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 85% that the spot is a new impact. It found the September 2018 image, taken at the end of the global dust storm, informative once haze is allowed for, which narrows the window to September 2018 to August 2020. It also noted that the CTX images of this ground were taken alongside HiRISE monitoring of an older impact 9.7 km away, so the camera teams may have seen the spot. It placed the spot at 29.0991° N, 27.0655° E, 235 m from the position given above.

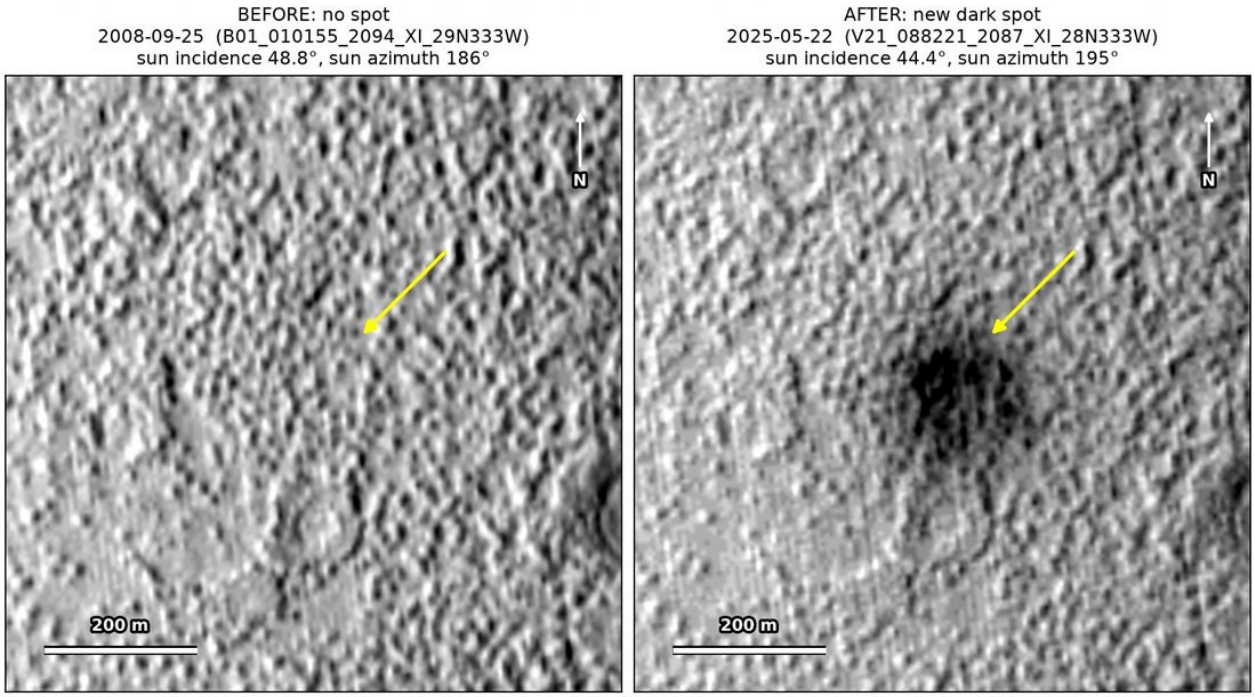

Figure 90: Candidate 36, before and after: the same 800 × 800 m of ground in 25 September 2008 (`B01_010155_2094_XI_29N333W`) and 22 May 2025 (`V21_088221_2087_XI_28N333W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

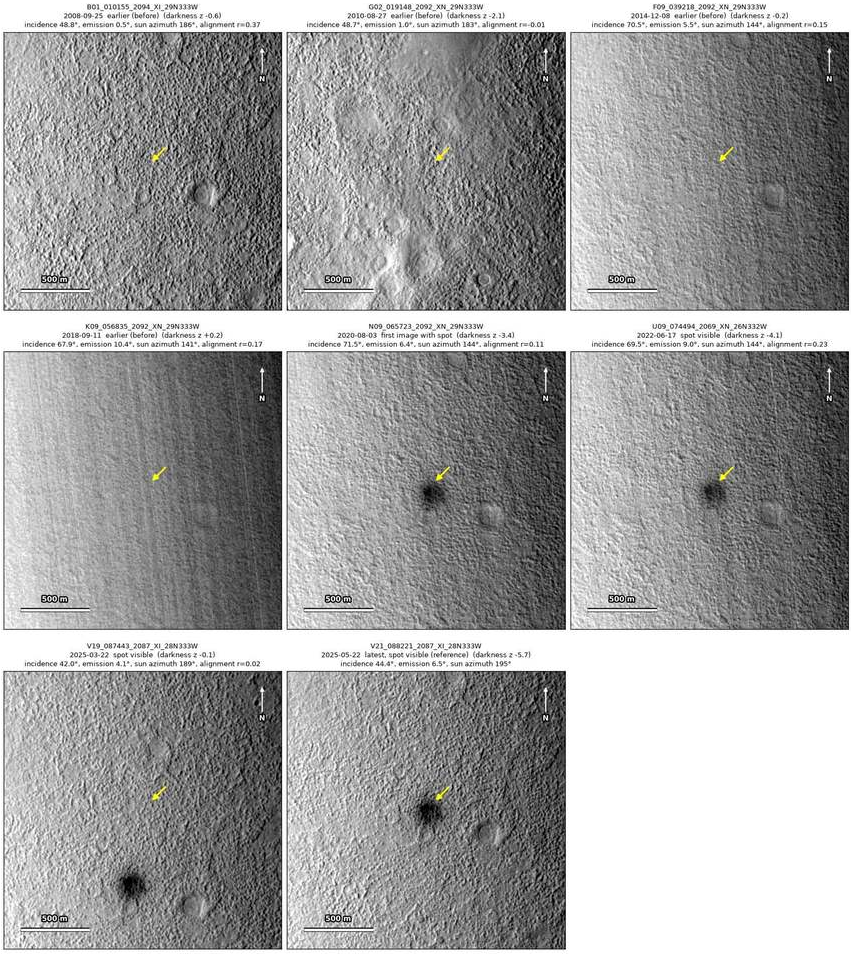

Figure 91: Candidate 36: eight CTX images of the same 2×2 km patch, 2008 to 2025. The August 2010 image is shown but not used to date the spot, and the March 2025 image is about 600 m out of alignment, so in that panel the spot lies below and to the left of the arrow. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

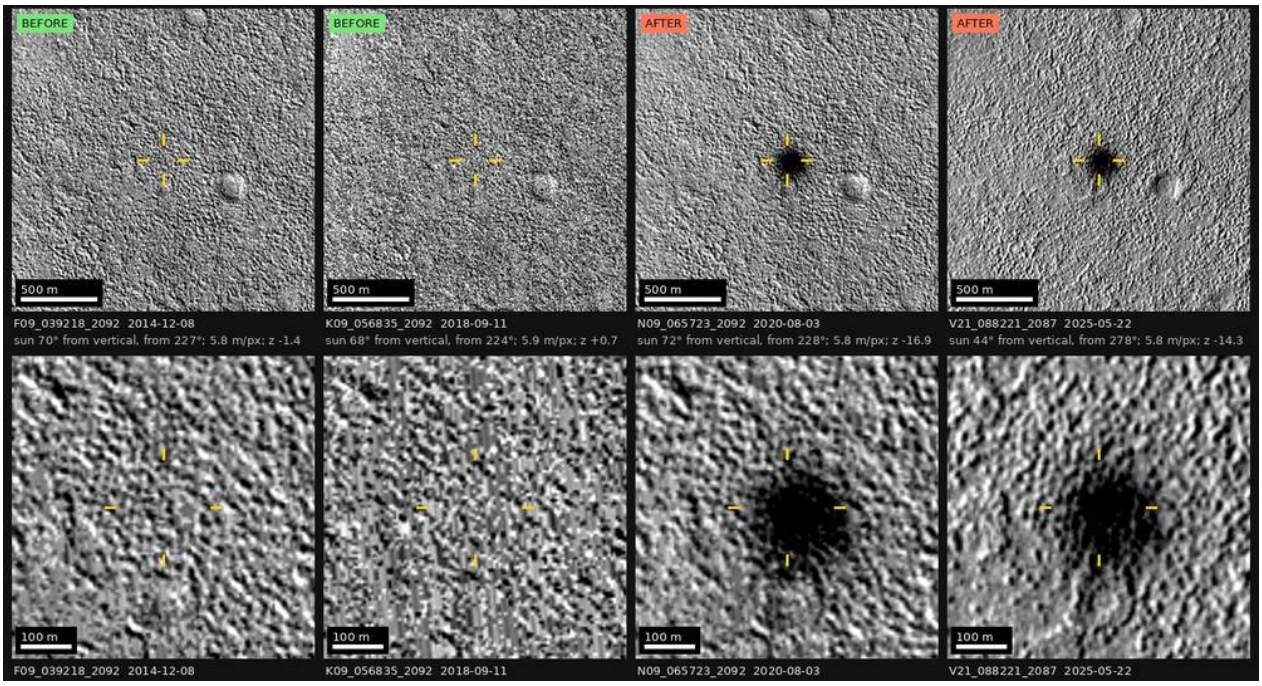

Figure 92: Candidate 36, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 37: Plains north of Arsia Mons, south of Biblis Patera

Location	4.0687° S, 237.1175° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	15 December 2015 and 23 January 2025
Images	3 without the spot (absent: 15 August 2008, 14 October 2008 and 15 December 2015); 2 with it (present: 23 January 2025 and 25 May 2025)
Size	Compact dark spot with a core about 60×45 m
HiRISE	none covers the spot (nearest footprint edge 15.5 km)
Nearest catalogued new impact	28 km (Daubar et al., 2022)

What changed. A compact dark spot with a core about 60×45 m appeared on the plains north of Arsia Mons, south of Biblis Patera, between December 2015 and January 2025. Two CTX images from two orbits, of January and May 2025, show it.

Why it may be a new impact. Three earlier images, from 2008 and 2015, show plain ground there, and each would have shown the spot, since a dark mark 300 m to the north-west shows in every image, earlier and later. One of them, of 15 August 2008, lines up well and was taken under the same sun as the May 2025 image, so the spot is not a shadow, and a compact new dark spot seen on two orbits is how the blast zone of a small new crater looks.

Caveats. So far only two images, four months apart, show it. Darker ground exposed by wind is the remaining alternative, though an unlikely one, since the spot is compact and the 2008 image under the same sun shows nothing there. No image from the high-resolution HiRISE camera covers it; the nearest ends 15.5 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 80% that the spot is a new impact. It noted that the spot had been followed for only four months, so a short-lived wind-made dark patch is not excluded, and estimated the probability at about 80%. It placed the spot at 4.0683° S, 237.1256° E, 480 m from the position given above.

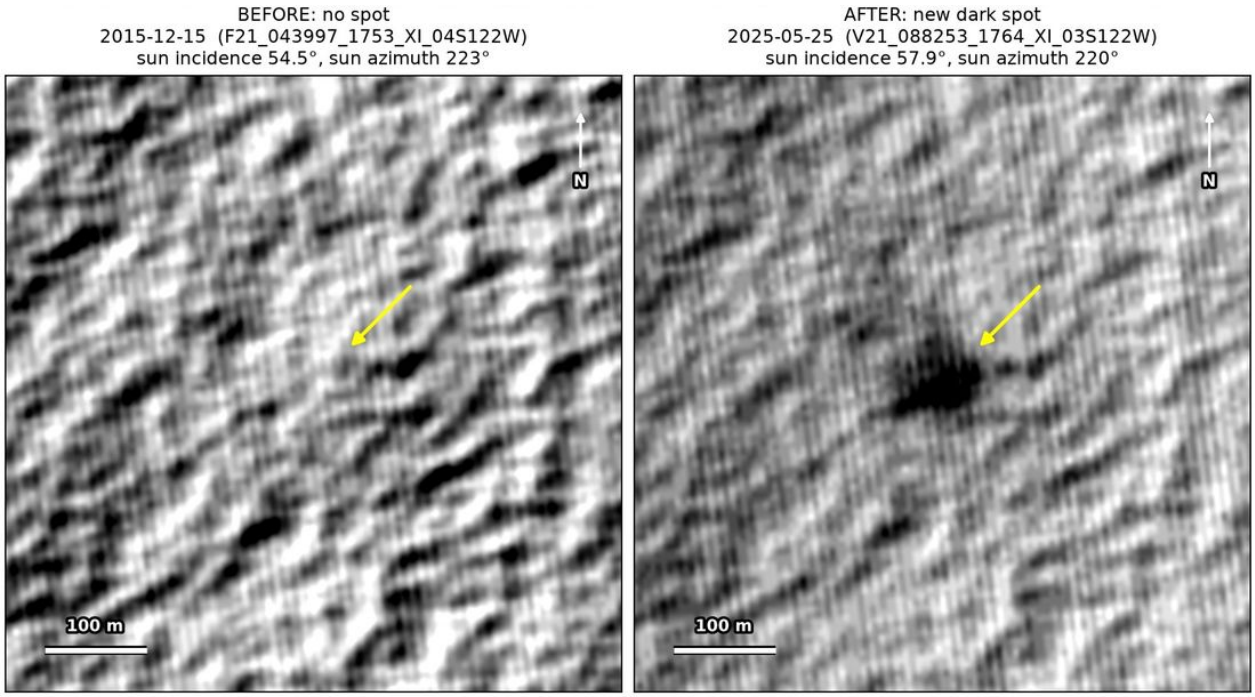

Figure 93: Candidate 37, before and after: the same 600 × 600 m of ground in 15 December 2015 (`F21_043997_1753_XI_04S122W`) and 25 May 2025 (`V21_088253_1764_XI_03S122W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

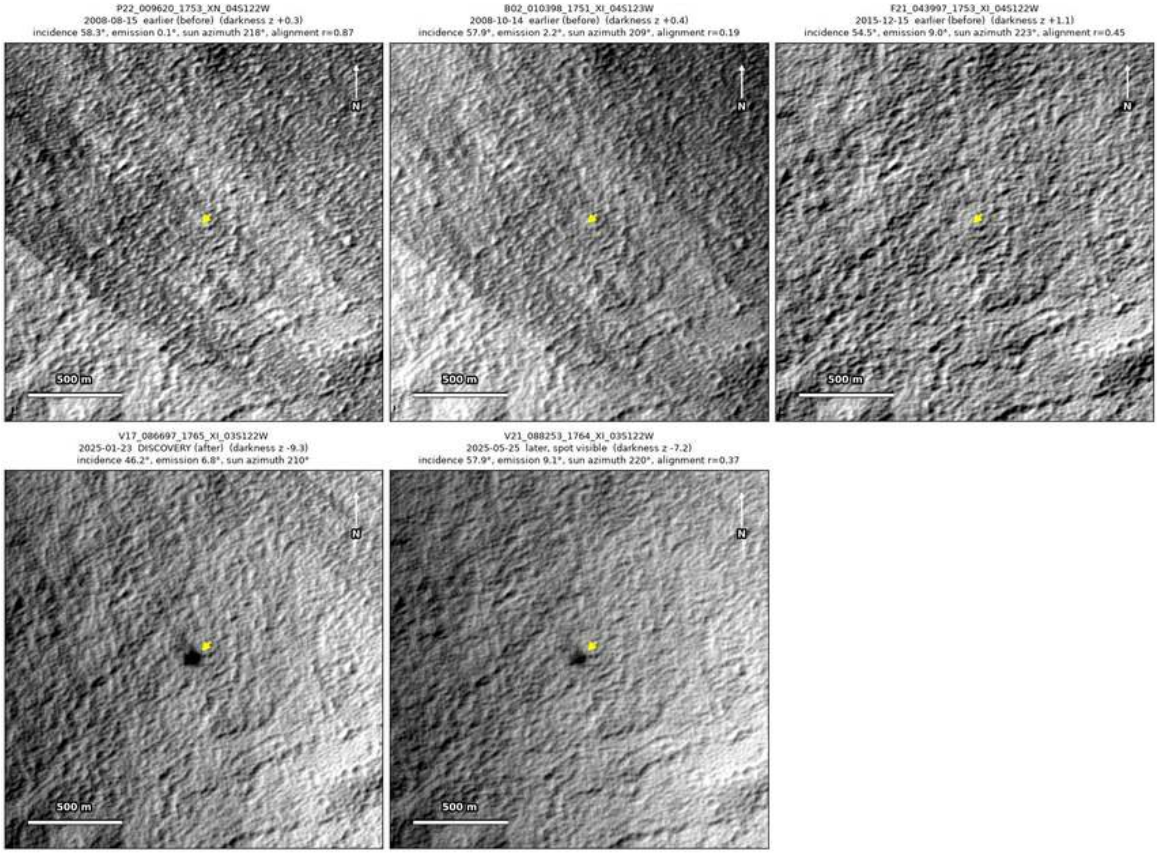

Figure 94: Candidate 37: five CTX images of the same 2×2 km patch, 2008 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

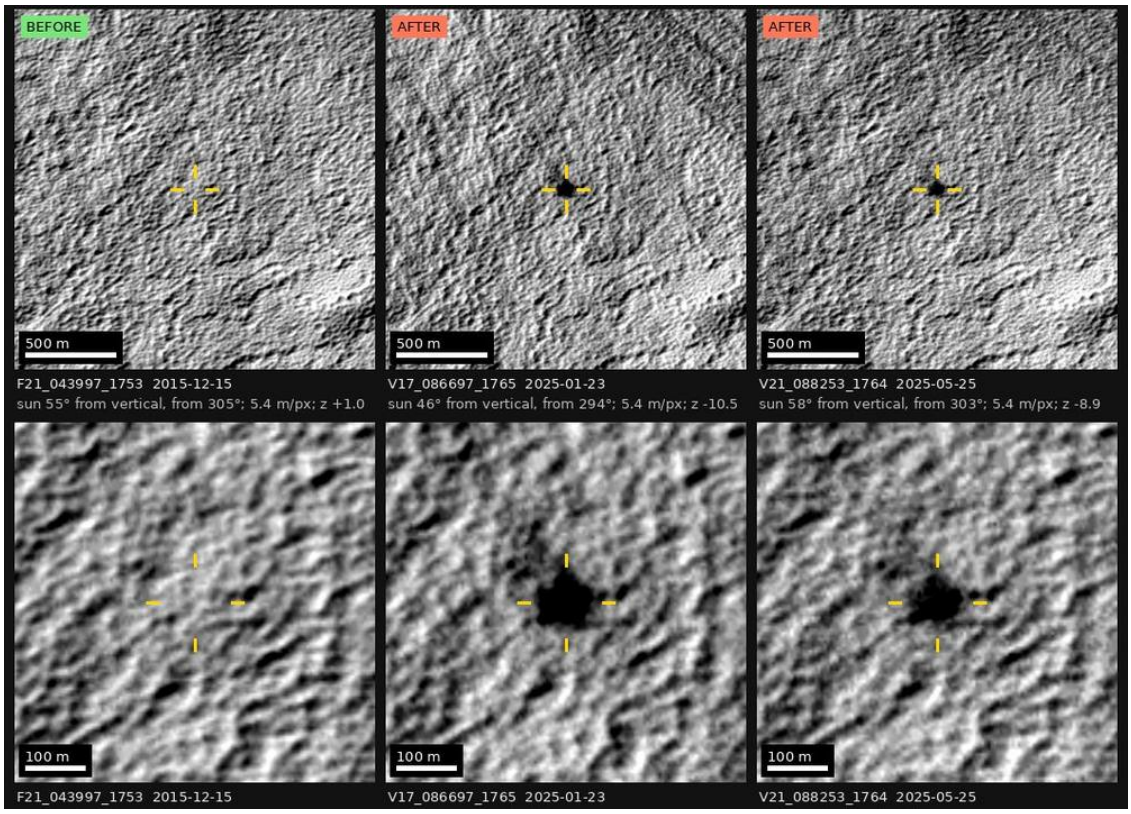

Figure 95: Candidate 37, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 38: Amazonis Planitia

Location	14.6517° N, 202.4361° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 70%
Formed between	24 January 2023 and 8 February 2025
Images	4 without the spot (absent: 30 June 2008, 11 May 2014, 30 June 2022 and 24 January 2023); 1 with it (present: 8 February 2025)
Size	Dark spot 100 to 130 m across, with short rays and a scatter of small satellite dots
HiRISE	none covers the spot (nearest footprint edge 6.7 km)
Nearest catalogued new impact	169 km (Daubar et al., 2022)

What changed. A dark spot 100 to 130 m across, with short rays and a scatter of small dark satellite dots, appeared on Amazonis Planitia between January 2023 and February 2025. So far, one CTX image shows it.

Why it may be a new impact. Four earlier images, from 2008 to January 2023, show plain ground there; two of them, from June 2008 and May 2014, were taken with the sun only a little lower than in the February 2025 image. Short rays and small dots arranged around a central spot, with no track, are how the blast zone of a small new crater can look, and they argue against an image flaw or a dust devil.

Caveats. Only one image shows it, and there is no clearly dark mark nearby to prove that the earlier images would have shown it: the darkest is barely darker than its surroundings in them. No image from the high-resolution HiRISE camera covers it; the nearest ends 6.7 km away.

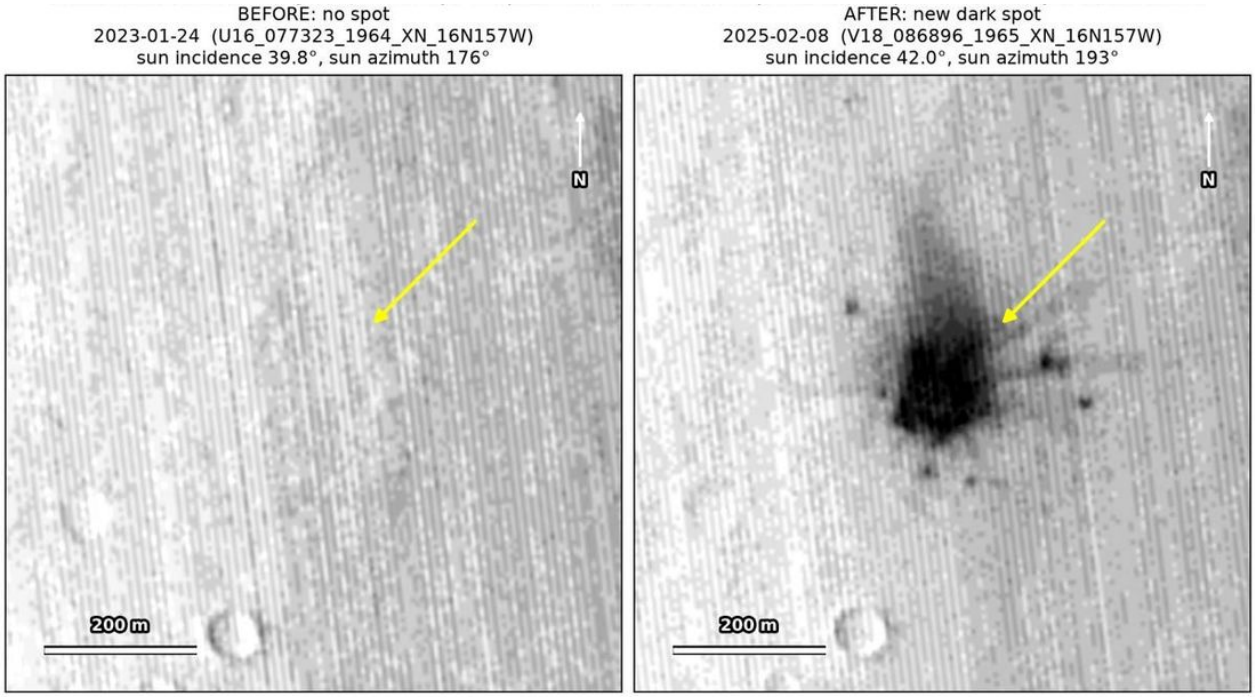

Figure 96: Candidate 38, before and after: the same 800 × 800 m of ground in 24 January 2023 (`U16_077323_1964_XN_16N157W`) and 8 February 2025 (`V18_086896_1965_XN_16N157W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

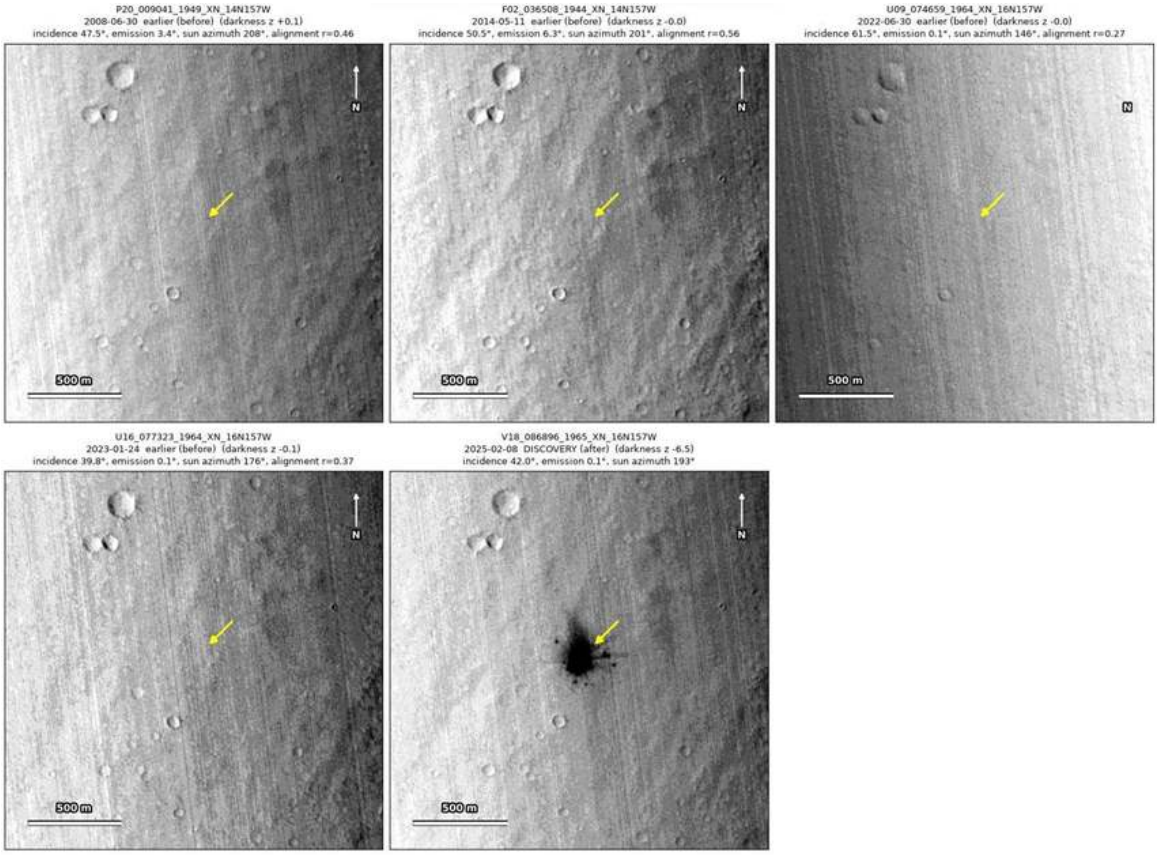

Figure 97: Candidate 38: five CTX images of the same 2×2 km patch, 2008 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 39: Plains north of Pavonis Mons

Location	7.3902° N, 246.6840° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 55%
Formed between	22 April 2014 and 25 January 2025
Images	2 without the spot (absent: 8 March 2010 and 22 April 2014); 1 with it (present: 25 January 2025)
Size	Dark core about 85 × 130 m, with a broad dark tail about 600 m long to the east-south-east
HiRISE	none covers the spot (nearest footprint edge 6.3 km)
Nearest catalogued new impact	76 km (Daubar et al., 2022)

What changed. A dark spot with a core about 85 × 130 m and a broad dark tail about 600 m long, pointing east-south-east, appeared on the plains north of Pavonis Mons between April 2014 and January 2025. So far, one CTX image shows it.

Why it may be a new impact. Two earlier images, of March 2010 and April 2014, show plain ground there, while a dark mark 300 m to the east-south-east shows in both. The compact head of the spot is better explained by a small new impact, and the tail could be its dark blast deposit, blown downwind.

Caveats. Only one image shows it, and a dust devil or a wind streak explains the tail nearly as well, though it explains the compact head less well. No image from the high-resolution HiRISE camera covers it; the nearest ends 6.3 km away.

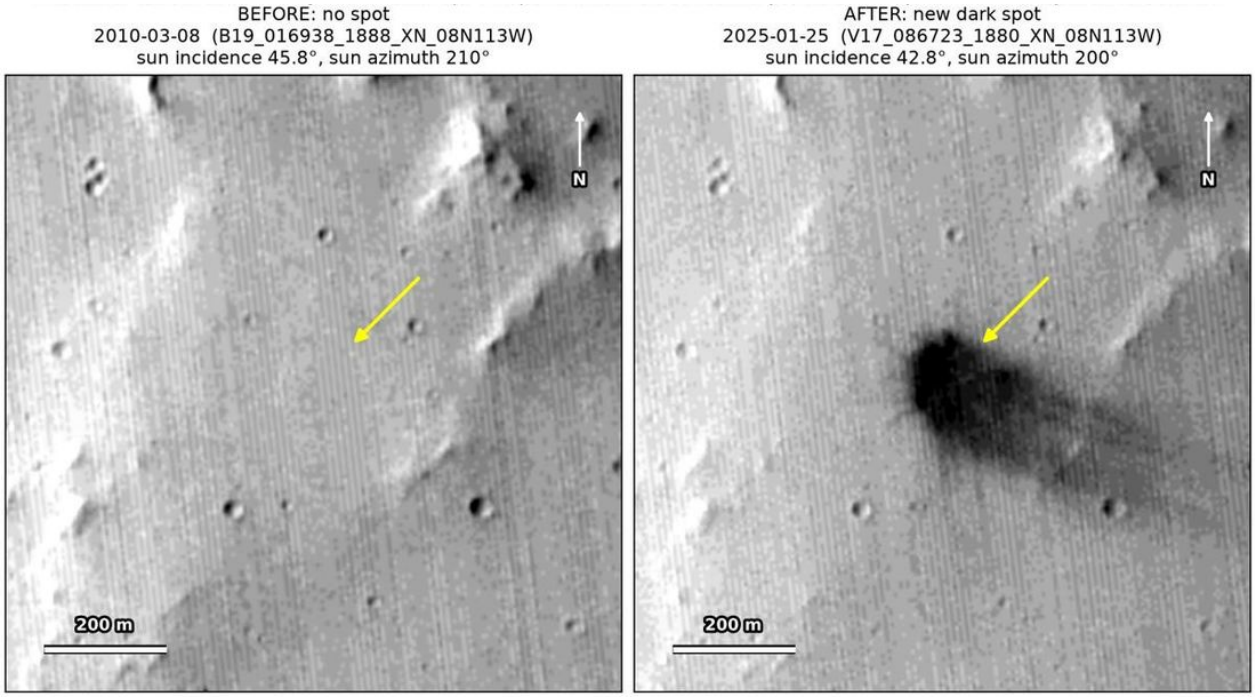

Figure 98: Candidate 39, before and after: the same 1000 × 1000 m of ground in 8 March 2010 (`B19_016938_1888_XN_08N113W`) and 25 January 2025 (`V17_086723_1880_XN_08N113W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

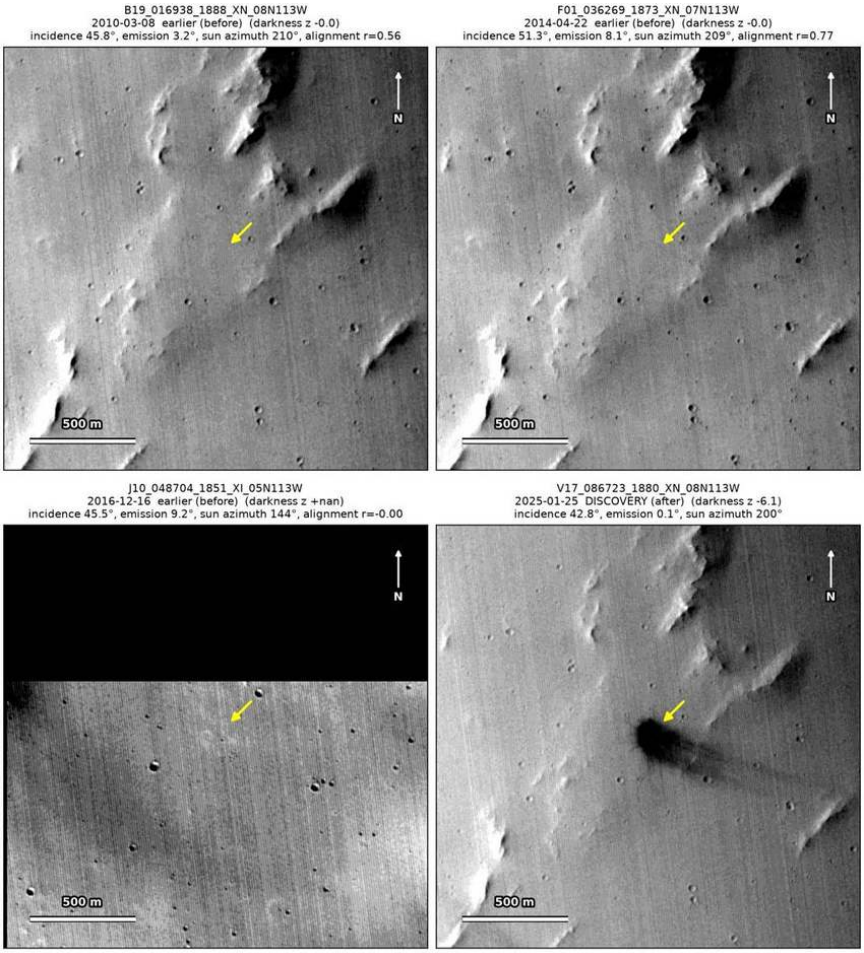

Figure 99: Candidate 39: four CTX images of the same 2×2 km patch, 2010 to 2025. The December 2016 image is shown but not used to date the spot. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 40: Elysium Planitia

Location	7.4966° N, 144.3039° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	6 September 2023 and 25 December 2023
Images	4 without the spot (absent: 3 June 2012, 3 May 2020, 7 December 2021 and 6 September 2023); 2 with it (present: 25 December 2023 and 25 April 2025)
Size	Compact dark spot with a core about 55×75 m
HiRISE	none covers the spot (nearest footprint edge 21.8 km)
Nearest catalogued new impact	198 km (Daubar et al., 2022)

What changed. A compact dark spot with a core about 55×75 m appeared on Elysium Planitia in the 110 days between September and December 2023. Two CTX images from two orbits, of December 2023 and April 2025, show it.

Why it may be a new impact. Four earlier images, from June 2012 to September 2023, show plain ground at the spot, while a dark mark 220 m to the west shows in them, so they would have shown it. A compact dark spot that appeared within 110 days and has stayed unchanged for 16 months fits a small new impact rather than darker ground exposed by wind.

Caveats. So far only two images, 16 months apart, show it, and darker ground exposed by wind, though unlikely for a compact spot, is not completely ruled out. It lies inside the area around the InSight lander studied by Bickel et al. (2025), but formed after that study. No image from the high-resolution HiRISE camera covers it; the nearest ends 21.8 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 90% that the spot is a new impact. It confirmed a formation window of 110 days, between 6 September and 25 December 2023. It placed the spot at 7.4968° N, 144.3132° E, 544 m from the position given above.

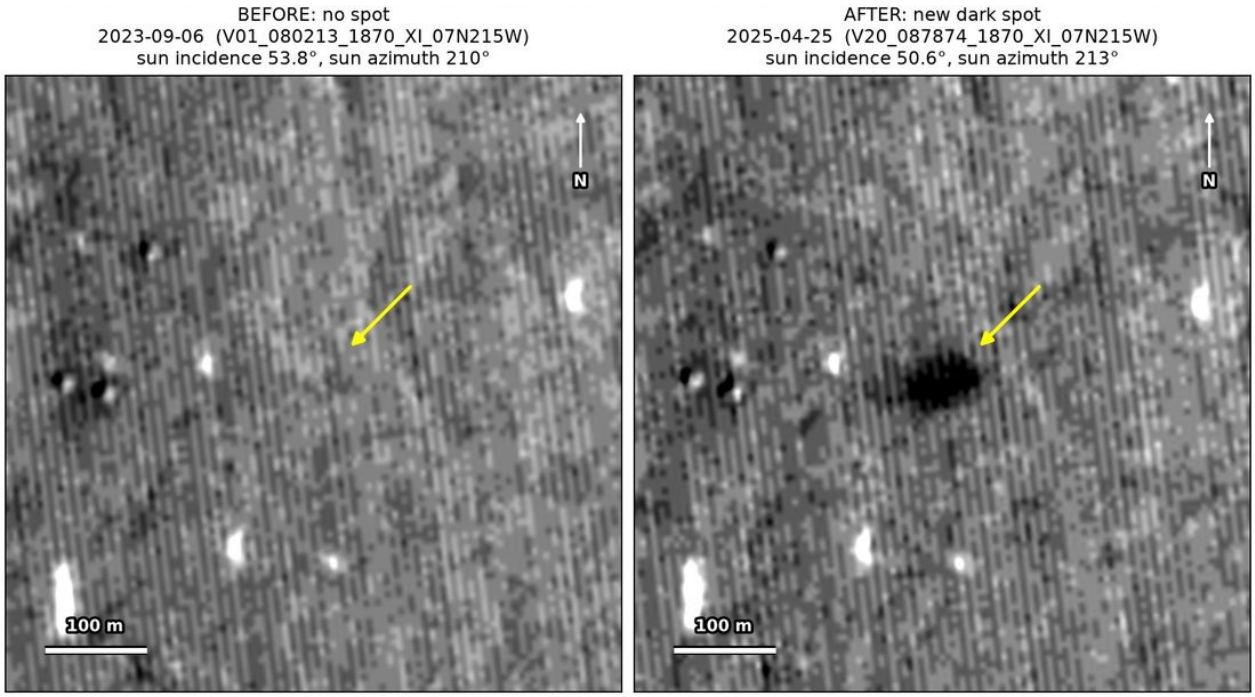

Figure 100: Candidate 40, before and after: the same 600 × 600 m of ground in 6 September 2023 (`V01_080213_1870_XI_07N215W`) and 25 April 2025 (`V20_087874_1870_XI_07N215W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

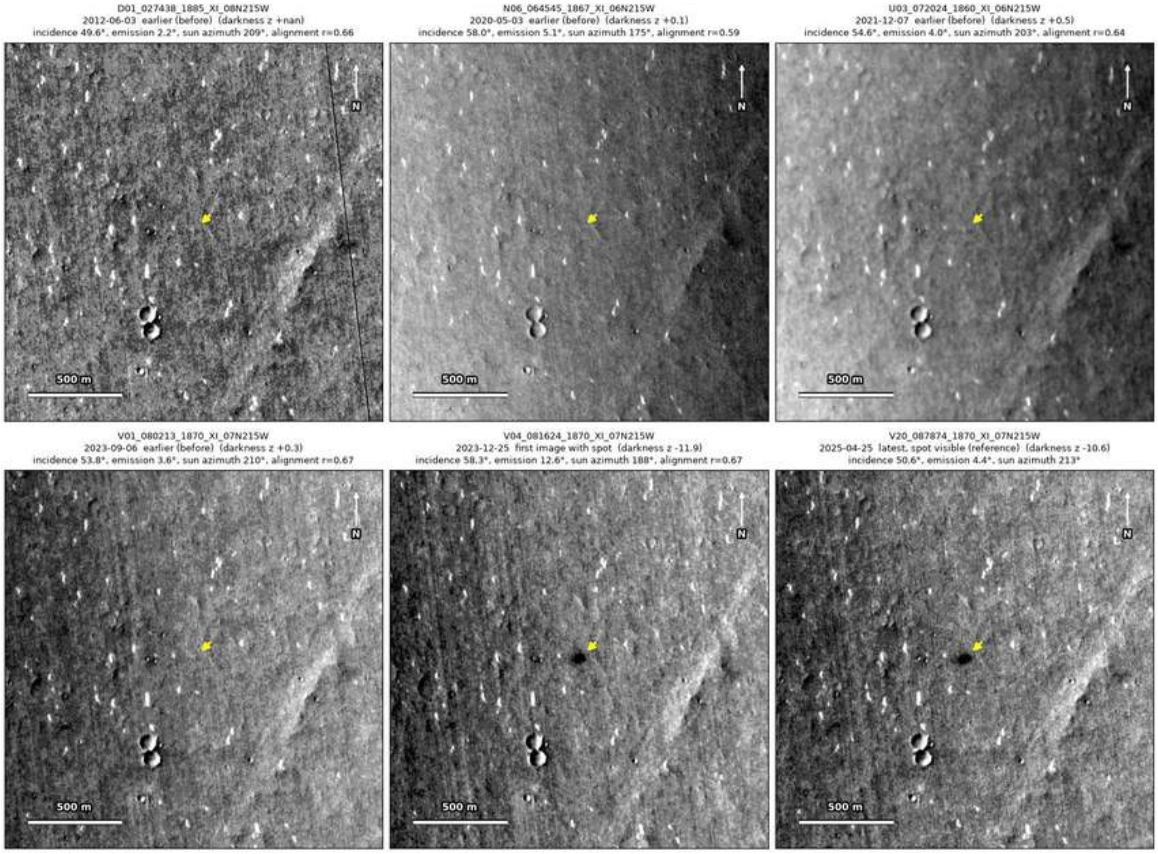

Figure 101: Candidate 40: six CTX images of the same 2×2 km patch, 2012 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

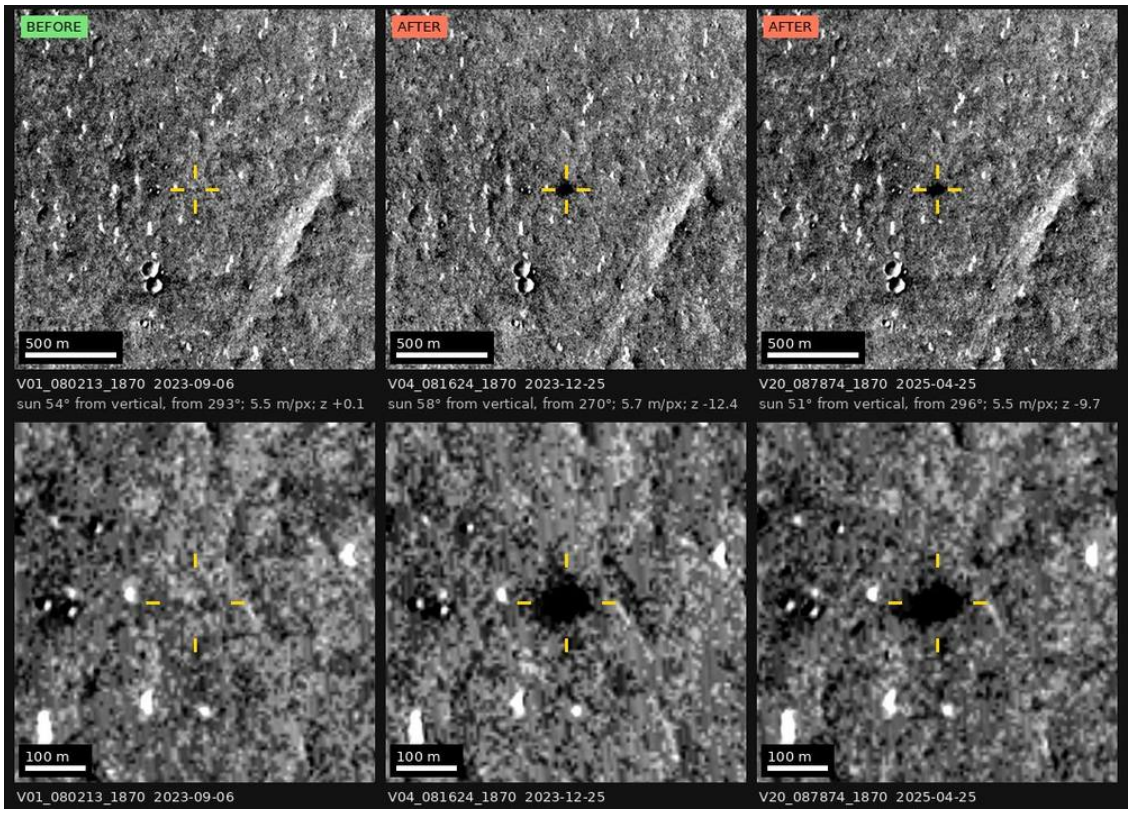

Figure 102: Candidate 40, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 41: Acheron Fossae region

Location	33.2901° N, 230.7321° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	8 January 2012 and 12 April 2023
Images	1 without the spot (absent: 8 January 2012); 3 with it (present: 12 April 2023, 11 June 2023 and 28 January 2025)
Size	Compact dark dot with a core about 50 m across
HiRISE	none covers the spot (nearest footprint edge 20.9 km)
Nearest catalogued new impact	58 km (Daubar et al., 2022)

What changed. A compact dark dot with a core about 50 m across appeared 70 m west of an old small crater in the Acheron Fossae region between January 2012 and April 2023. Three CTX images from three orbits, from April 2023 to January 2025, show it.

Why it may be a new impact. An image of 8 January 2012, taken under the same sun as the April 2023 image, shows the old crater but no dot, so the dot is not the shadow of a crater that was already there. A dark mark 170 m to the south-west shows clearly in that image, so it would have shown the dot. A compact dark dot that appears and lasts is how the blast zone of a small new crater looks.

Caveats. Only one earlier image shows the ground without it, so the dot could have formed at any time in an 11-year window. Darker ground exposed by wind is unlikely for so compact a dot, but it is not ruled out. No image from the high-resolution HiRISE camera covers it; the nearest ends 20.9 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 85% that the spot is a new impact. It placed the spot at 33.2894° N, 230.7350° E, 151 m from the position given above.

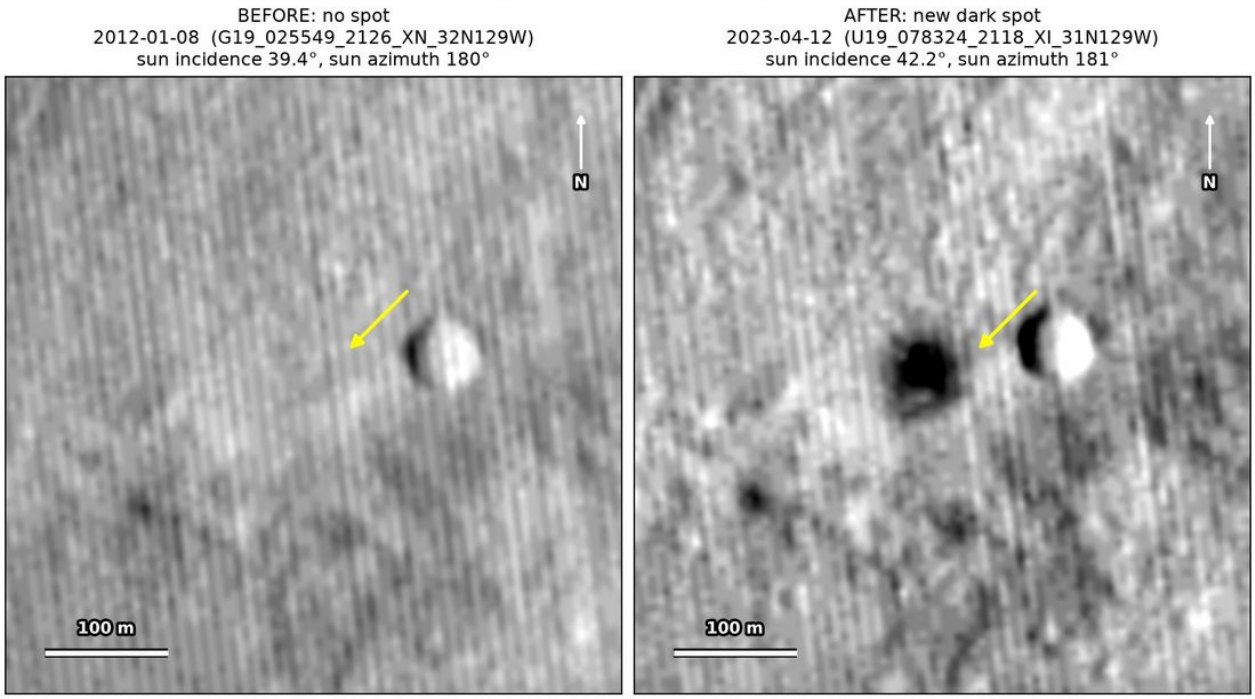

Figure 103: Candidate 41, before and after: the same 500 × 500 m of ground in 8 January 2012 (`G19_025549_2126_XN_32N129W`) and 12 April 2023 (`U19_078324_2118_XI_31N129W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

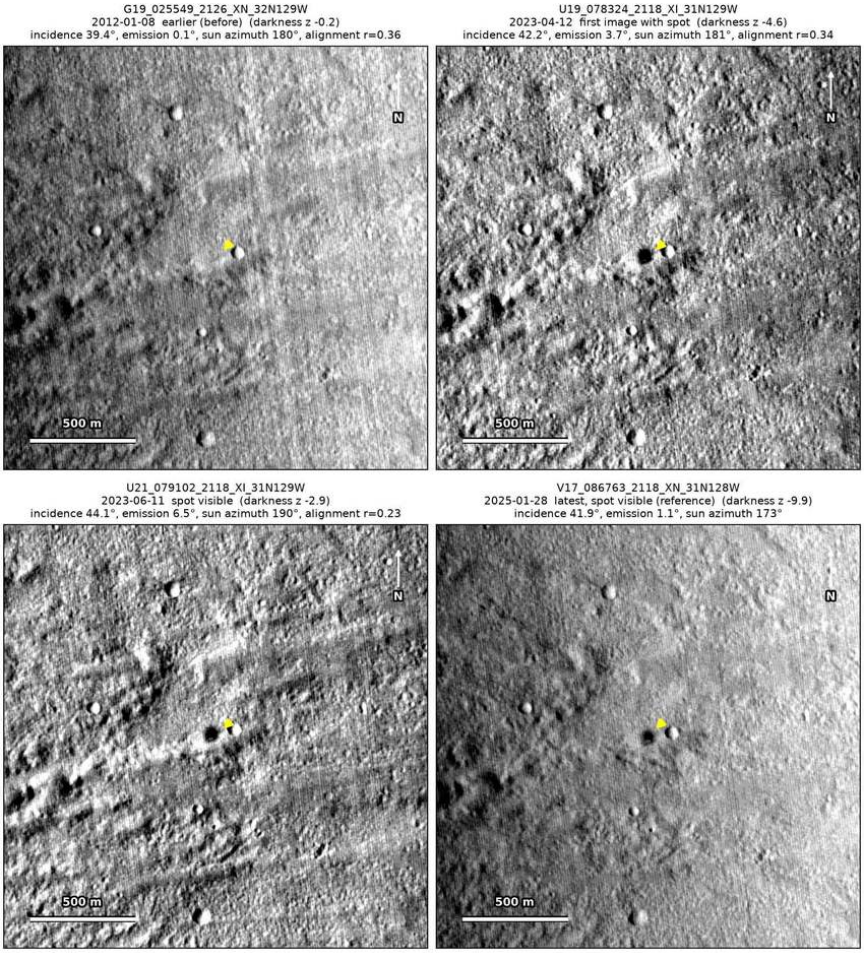

Figure 104: Candidate 41: four CTX images of the same 2×2 km patch, 2012 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

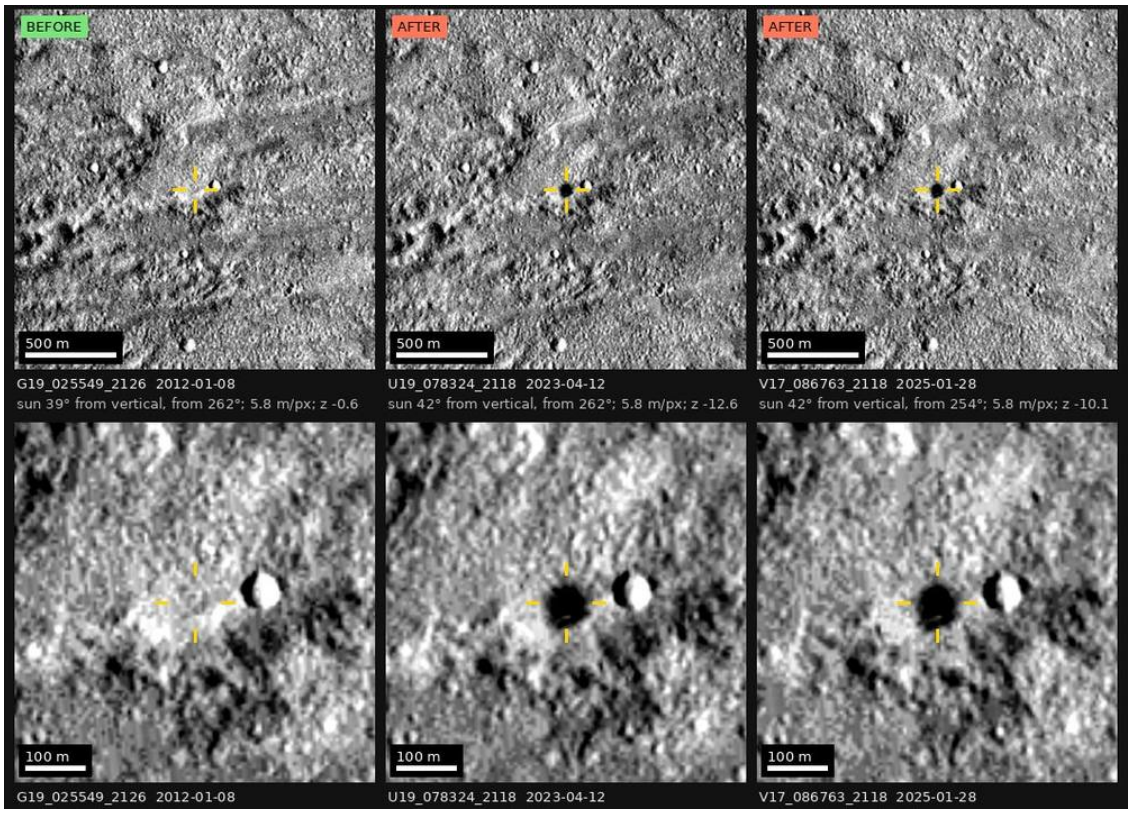

Figure 105: Candidate 41, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 42: Lava plains of Daedalia Planum

Location	18.9845° S, 223.6547° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	12 August 2008 and 30 November 2018
Images	3 without the spot (absent: 24 November 2007, 21 December 2007 and 12 August 2008); 2 with it (present: 30 November 2018 and 30 June 2025)
Size	Compact dark spot with a core about 40 m across and a faint halo
HiRISE	none covers the spot (nearest footprint edge 6.7 km)
Nearest catalogued new impact	91 km (Daubar et al., 2022)
Nearby	Candidate 43, 8 km to the south-east, appeared later, between November 2018 and June 2025.

What changed. A compact dark spot with a core about 40 m across and a faint halo appeared on the lava plains of Daedalia Planum between August 2008 and November 2018. Two CTX images from two orbits, of November 2018 and June 2025, show it.

Why it may be a new impact. Three earlier images, from 2007 and 2008, show plain ground there, while a dark mark 170 m to the west shows clearly in all three. The last of them, of 12 August 2008, was taken under the same sun as the June 2025 image and lines up almost perfectly, so the spot is not a shadow. A compact spot that stays unchanged for six and a half years is how the blast zone of a small new crater looks.

Caveats. Only two images show it, and it could have formed at any time in the 10 years between August 2008 and November 2018. Darker ground exposed by wind is the remaining alternative, though it is unlikely for a compact spot that has stayed unchanged from November 2018 to June 2025. No image from the high-resolution HiRISE camera covers it; the nearest ends 6.7 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 90% that the spot is a new impact. It placed the spot at 18.9845° S, 223.6560° E, 75 m from the position given above.

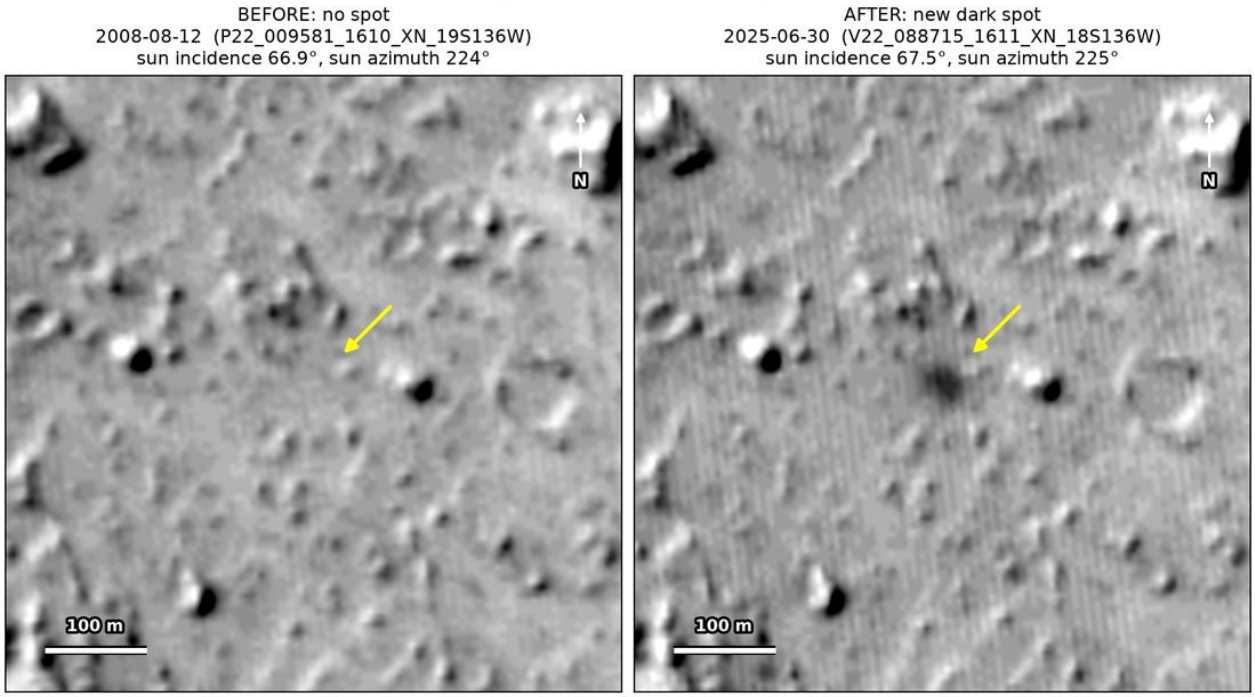

Figure 106: Candidate 42, before and after: the same 600 × 600 m of ground in 12 August 2008 (`P22_009581_1610_XN_19S136W`) and 30 June 2025 (`V22_088715_1611_XN_18S136W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

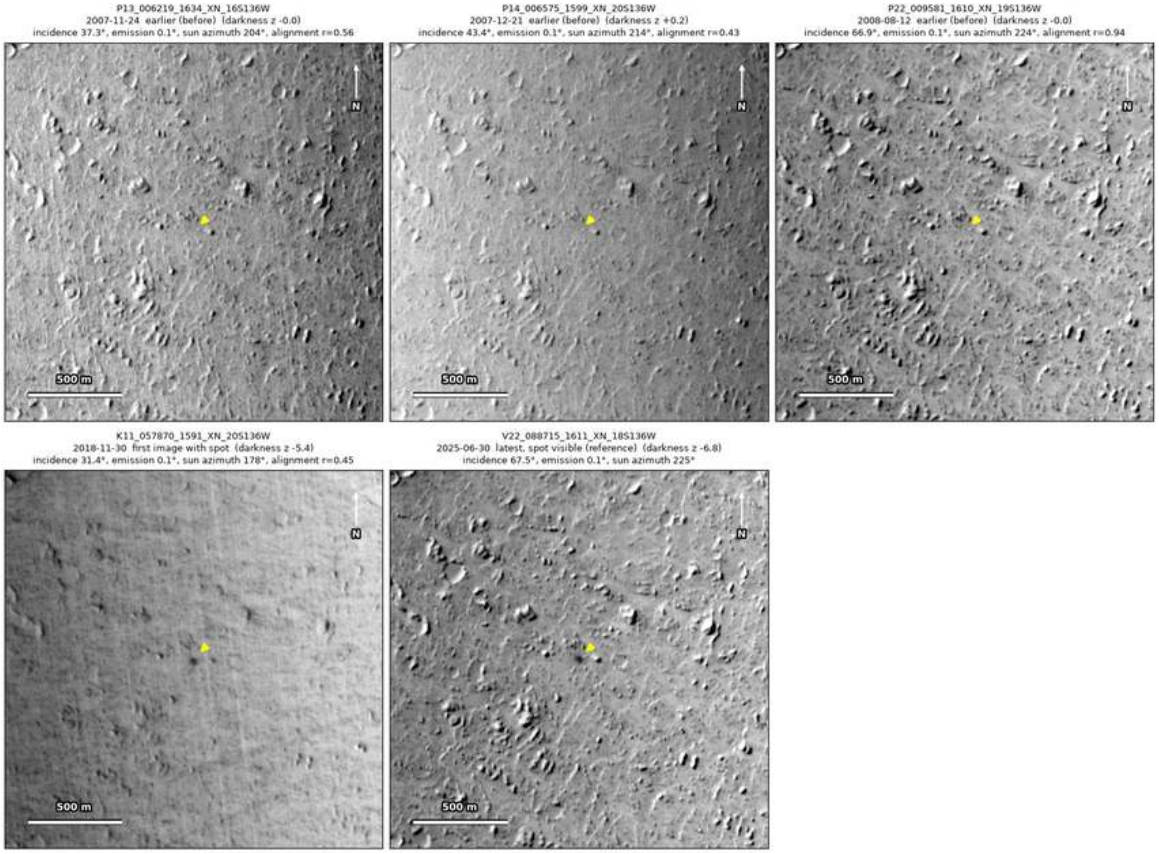

Figure 107: Candidate 42: five CTX images of the same 2×2 km patch, 2007 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

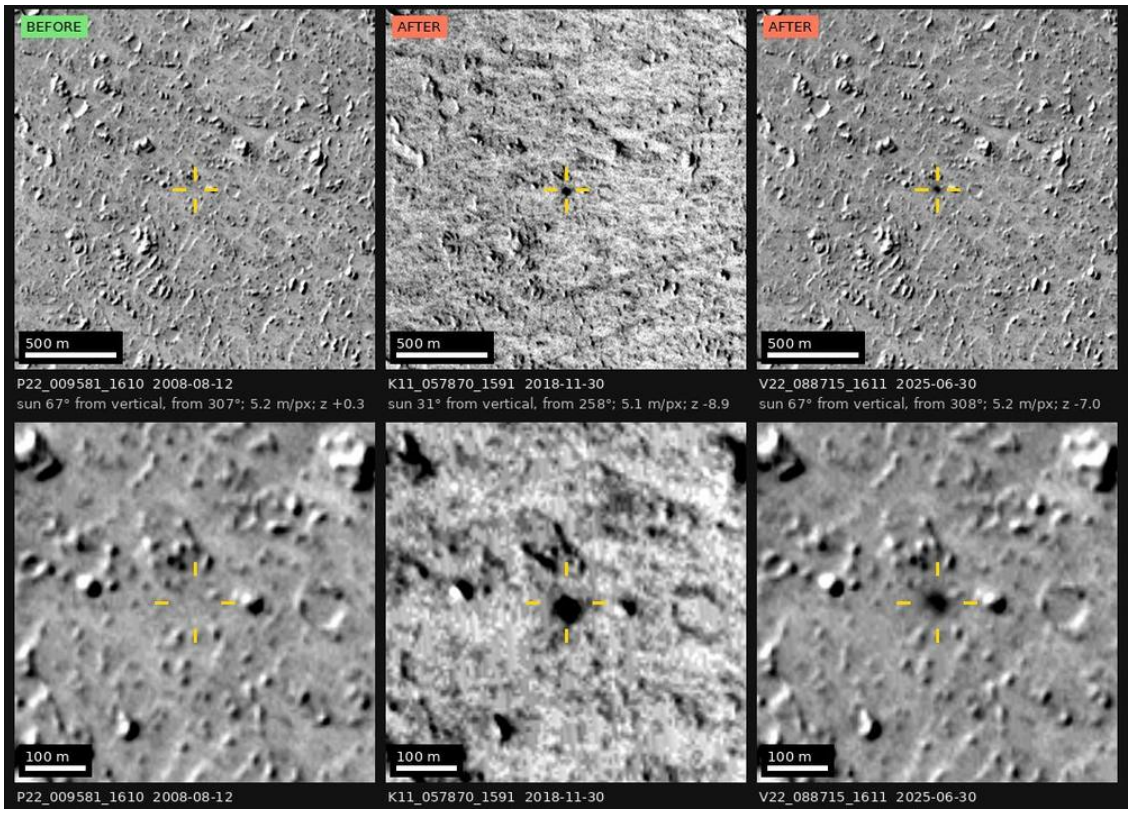

Figure 108: Candidate 42, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 43: Daedalia Planum, 8 km south-east of candidate 42

Location	19.0840° S, 223.7612° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	30 November 2018 and 30 June 2025
Images	3 without the spot (absent: 21 December 2007, 12 August 2008 and 30 November 2018); 1 with it (present: 30 June 2025)
Size	Irregular dark spot about 60×70 m
HiRISE	none covers the spot (nearest footprint edge 5.2 km)
Nearest catalogued new impact	85 km (Daubar et al., 2022)
Nearby	Candidate 42, 8 km to the north-west, already shows in the November 2018 image, which does not show this spot.

What changed. An irregular dark spot about 60×70 m appeared on Daedalia Planum, 8 km south-east of candidate 42, between November 2018 and June 2025; so far, one CTX image shows it.

Why it may be a new impact. Two earlier images, from December 2007 and August 2008, line up well and show no spot, while a dark feature 145 m to the south-east shows in both; the August 2008 one was taken under the same sun as the June 2025 image. A noisier November 2018 image, which does show candidate 42, does not show the spot either. A smooth blob without the line pattern of an image defect, the spot may be the blast zone of a small new impact.

Caveats. Only one image shows it, and with its irregular outline the spot could also be darker ground exposed by the wind. The November 2018 image is noisy and was taken under a high sun; if it is set aside, the spot could have formed at any time between August 2008 and June 2025. No image from the high-resolution HiRISE camera covers it; the nearest ends 5.2 km away.

Independent check. The independent re-examination (Section 3.8) agreed with the promising rating, with its own estimate of about 60% that the spot is a new impact. It agreed with the promising rating (about 60%) but found the November 2018 image uninformative at this site, so the firm window extends from August 2008 to June 2025. It placed the spot at 19.0836° S, 223.7619° E, 45 m from the position given above.

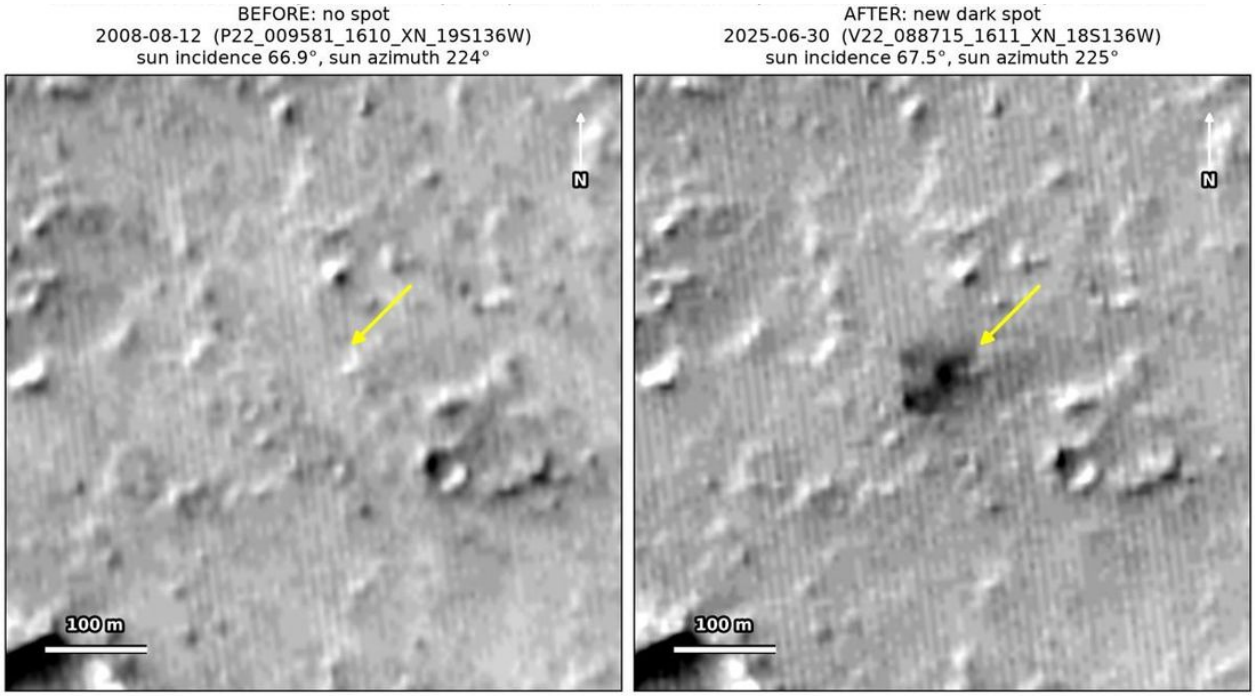

Figure 109: Candidate 43, before and after: the same 600 × 600 m of ground in 12 August 2008 (`P22_009581_1610_XN_19S136W`) and 30 June 2025 (`V22_088715_1611_XN_18S136W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

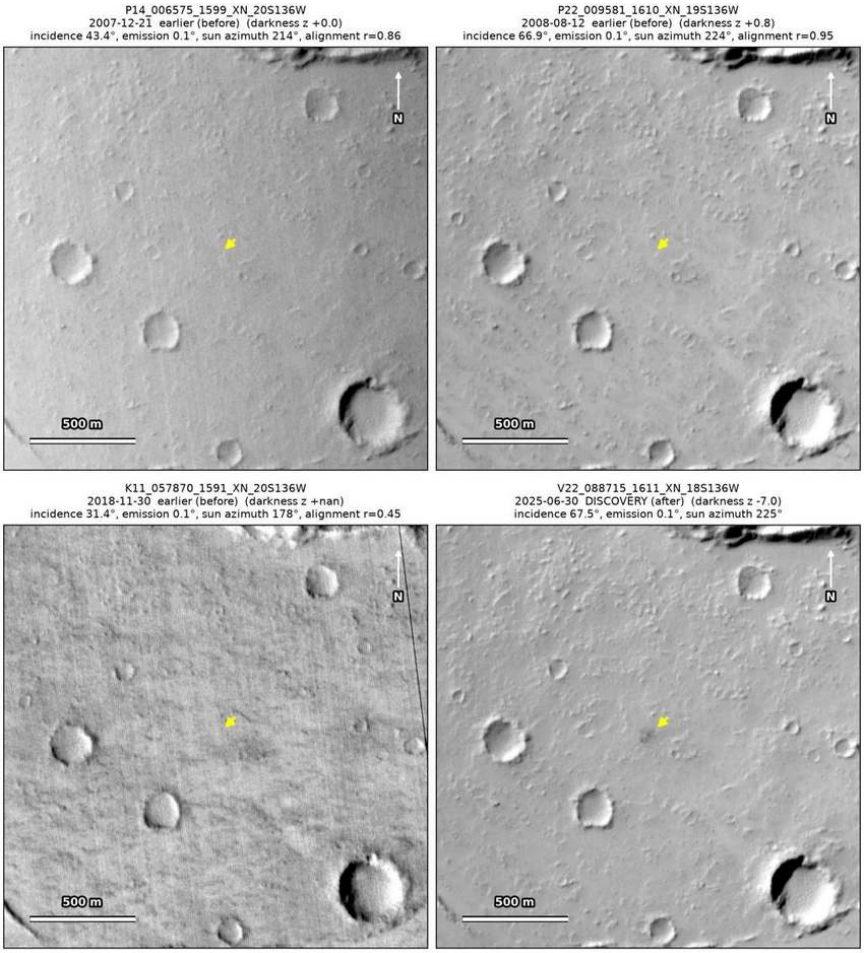

Figure 110: Candidate 43: four CTX images of the same 2×2 km patch, 2007 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 44: Northern Arabia Terra

Location	31.4688° N, 44.0356° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 45%
Formed between	21 October 2008 and 15 February 2025
Images	1 without the spot (absent: 21 October 2008); 2 with it (present: 15 February 2025 and 17 April 2025)
Size	Irregular dark patch about 70 m across
HiRISE	none covers the spot (nearest footprint edge 15.7 km)
Nearest catalogued new impact	111 km (Daubar et al., 2022)

What changed. An irregular dark patch about 70 m across appeared at the edge of a brighter ridge in northern Arabia Terra between October 2008 and February 2025. Two CTX images from two orbits, two months apart, show it.

Why it may be a new impact. The only earlier image, from October 2008, lines up well and shows no patch there, while a dark feature 300 m to the north shows in it. The sun is higher in the February and April 2025 images than in the October 2008 one, and since a higher sun weakens shadows, the patch is unlikely to be shading; it may be the blast zone of a small new impact.

Caveats. A new slope streak, a dark streak left by dust sliding down a slope, fits nearly as well as an impact: the ground is rugged, and the irregular patch lies beside a slope. With only one earlier image, the patch could have formed at any time in a span of more than 16 years. No image from the high-resolution HiRISE camera covers it; the nearest ends 15.7 km away.

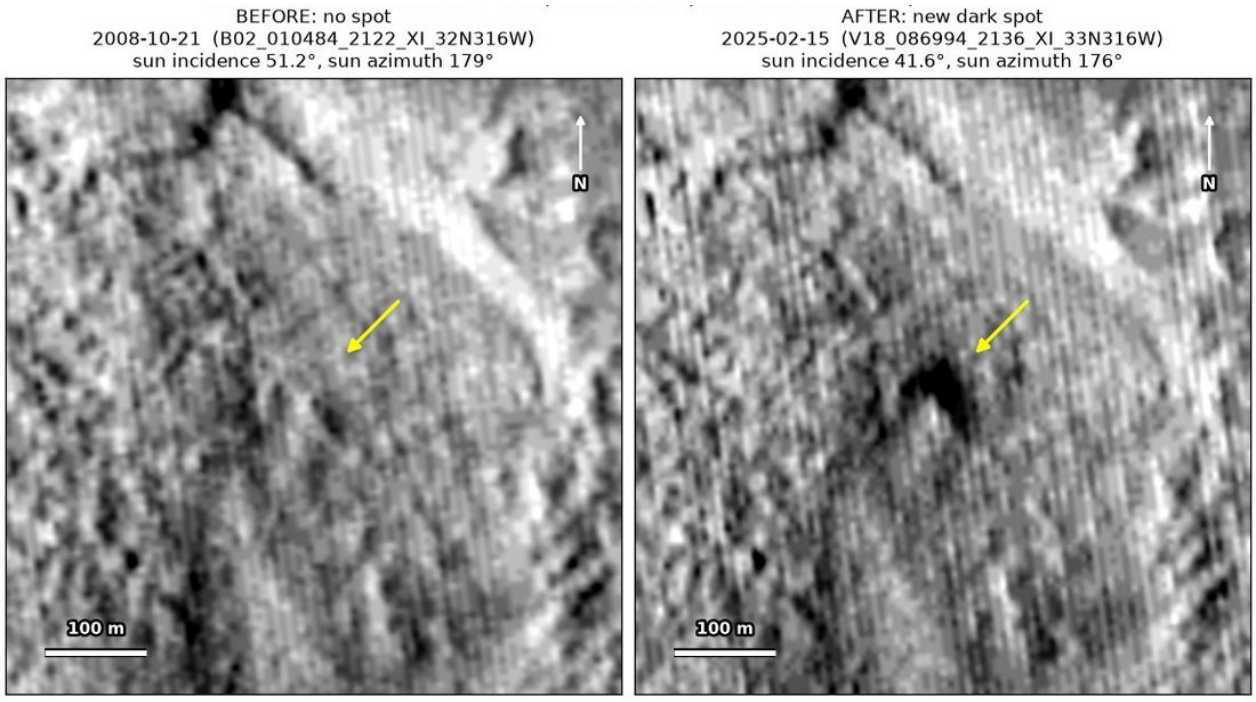

Figure 111: Candidate 44, before and after: the same 600 × 600 m of ground in 21 October 2008 (`B02_010484_2122_XI_32N316W`) and 15 February 2025 (`V18_086994_2136_XI_33N316W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

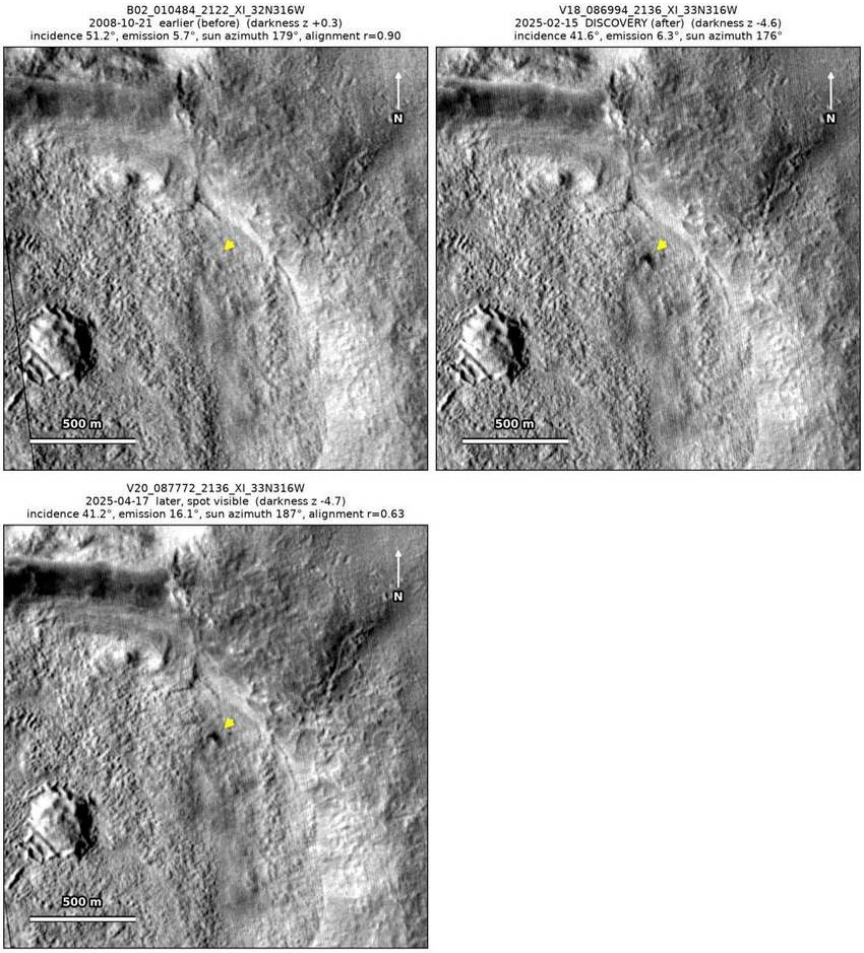

Figure 112: Candidate 44: three CTX images of the same 2×2 km patch, 2008 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 45: Terrain north-east of Fesenkov crater

Location	24.7886° N, 278.8201° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	29 July 2020 and 10 July 2022
Images	2 without the spot (absent: 10 May 2008 and 29 July 2020); 2 with it (present: 10 July 2022 and 8 April 2025)
Size	Main dark core about 45 × 35 m, with smaller satellite dots to the west and north-east
HiRISE	none covers the spot (nearest footprint edge 36.6 km)
Nearest catalogued new impact	66 km (Daubar et al., 2022)

What changed. A cluster of dark spots, a main core about 45 × 35 m with smaller dots to the west and north-east, appeared on terrain north-east of Fesenkov crater between July 2020 and July 2022. Two CTX images from two orbits show it: faintly in July 2022 and more clearly in April 2025.

Why it may be a new impact. A May 2008 image, taken under the same sun as the April 2025 image, shows nothing there, while a dark feature 275 m to the north-north-east shows clearly in it; a July 2020 image shows nothing either. A main spot with smaller dots to the west and north-east is most naturally read as a cluster impact: a group of small impacts made at the same time.

Caveats. In the July 2022 image the spots are faint and low in contrast, so darker ground exposed by the wind is possible. The July 2020 image lines up poorly with the others; if it is set aside, the spots could have formed at any time between May 2008 and July 2022. No image from the high-resolution HiRISE camera covers it; the nearest ends 36.6 km away.

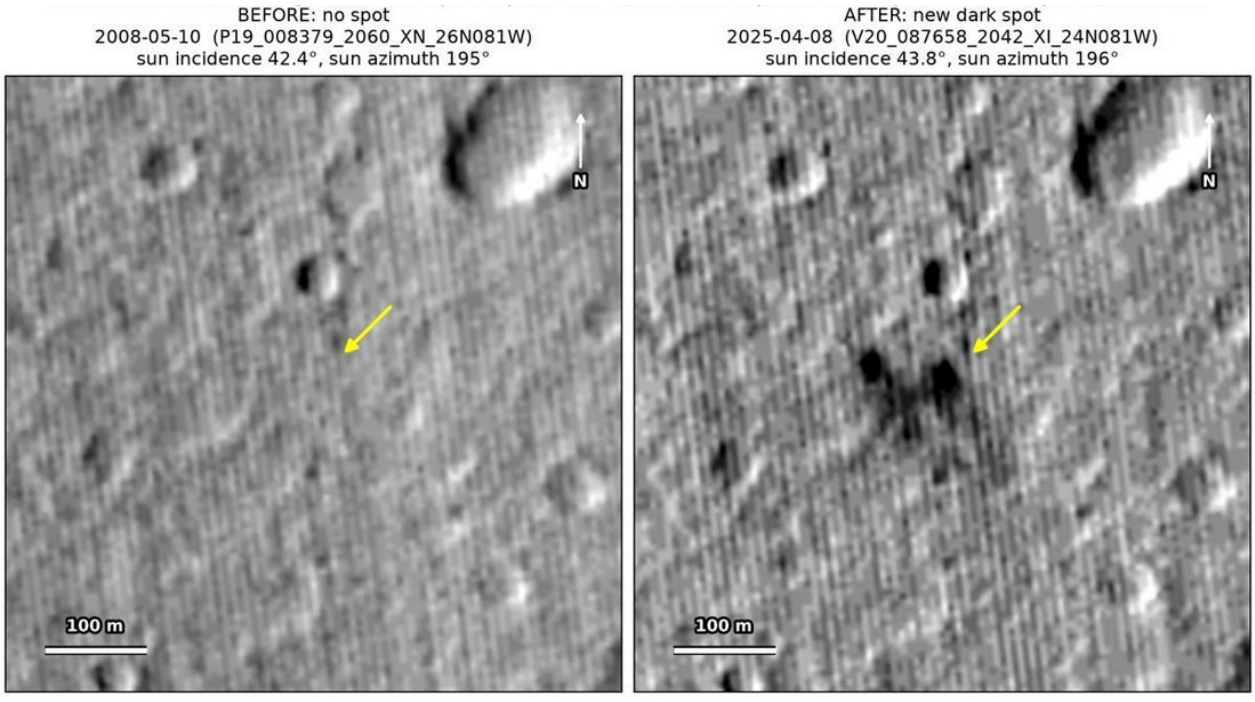

Figure 113: Candidate 45, before and after: the same 600 × 600 m of ground in 10 May 2008 (`P19_008379_2060_XN_26N081W`) and 8 April 2025 (`V20_087658_2042_XI_24N081W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

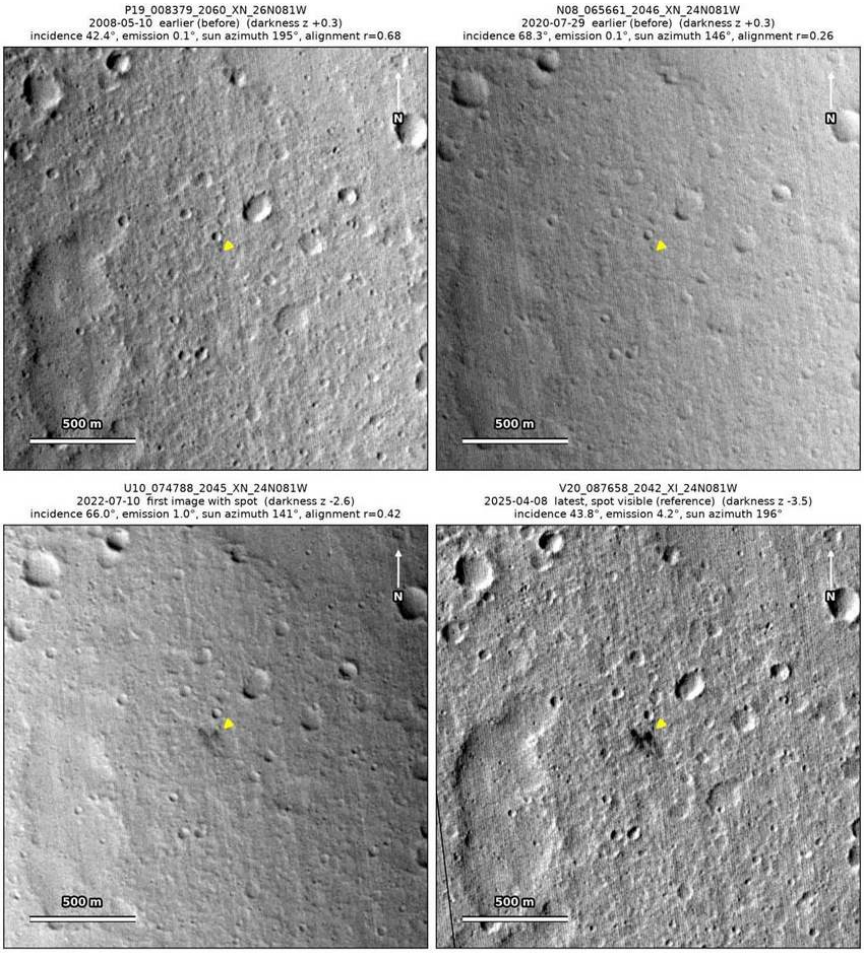

Figure 114: Candidate 45: four CTX images of the same 2×2 km patch, 2008 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 46: Central Arabia Terra

Location	22.1339° N, 22.3971° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	21 December 2007 and 20 January 2025
Images	1 without the spot (absent: 21 December 2007); 1 with it (present: 20 January 2025)
Size	Compact dark spot with a core about 50×60 m
HiRISE	none covers the spot (nearest footprint edge 11.8 km)
Nearest catalogued new impact	43 km (Daubar et al., 2022)

What changed. A compact, round dark spot with a core about 50×60 m appeared in central Arabia Terra between December 2007 and January 2025; so far, one CTX image shows it.

Why it may be a new impact. The only earlier image, from December 2007, was taken under the same sun as the January 2025 image and shows no spot there, while a dark feature 275 m to the north-north-east shows clearly in it. The spot is a smooth, round blob, unlike an image defect, and a compact dark spot like this may be the blast zone of a small new impact.

Caveats. Only one image shows it, and the earlier image was taken 17 years before, so darker ground exposed by the wind in that time is possible, though less likely for a round spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 11.8 km away.

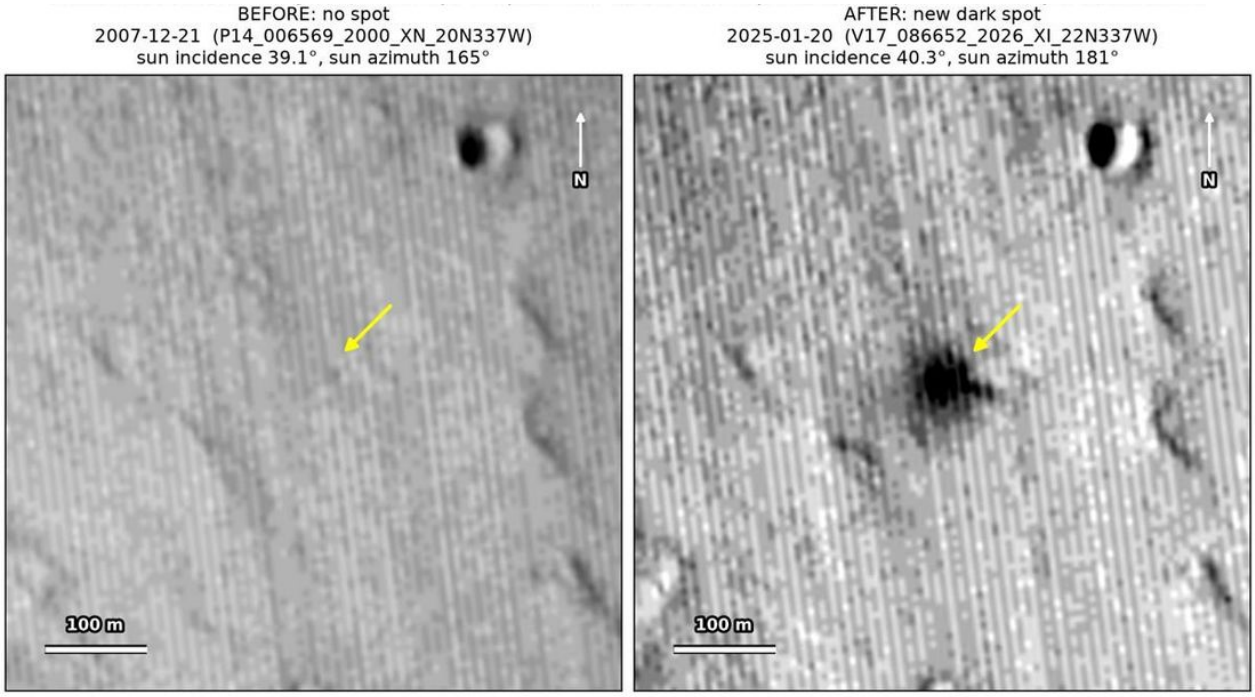

Figure 115: Candidate 46, before and after: the same 600×600 m of ground in 21 December 2007 (`P14_006569_2000_XN_20N337W`) and 20 January 2025 (`V17_086652_2026_XI_22N337W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

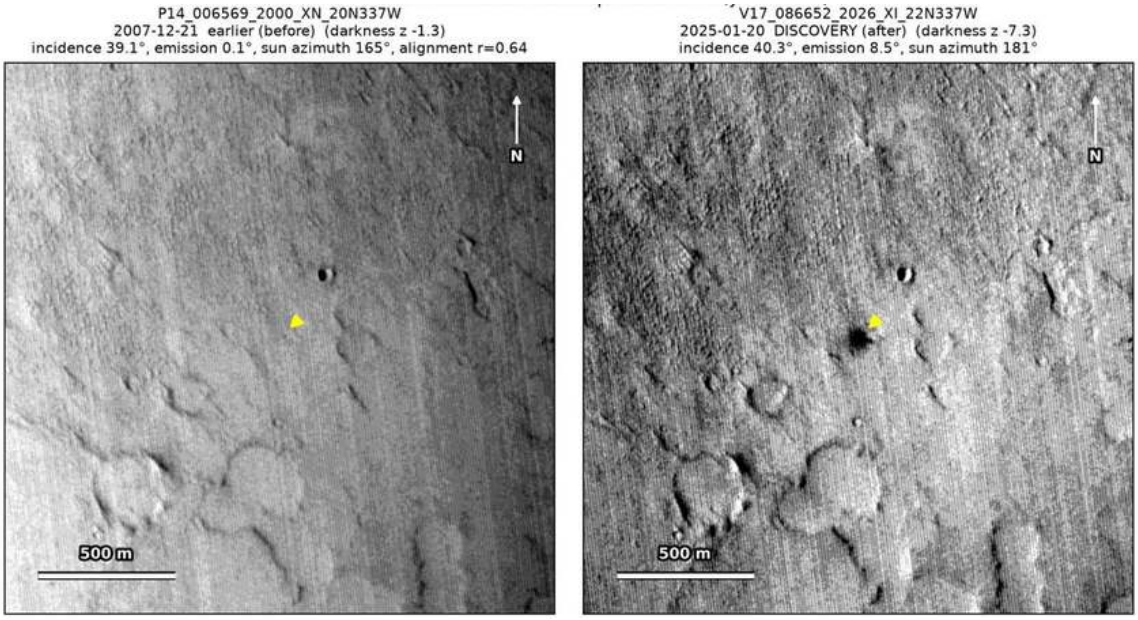

Figure 116: Candidate 46: both CTX images of the same 2×2 km patch, December 2007 and January 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 47: Plains north of Tithonium Chasma

Location	2.8114° S, 268.3592° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	17 September 2008 and 16 April 2025
Images	1 without the spot (absent: 17 September 2008); 2 with it (present: 16 April 2025 and 4 March 2026)
Size	Compact dark spot with a core about 30×35 m
HiRISE	none covers the spot (nearest footprint edge 16.9 km)
Nearest catalogued new impact	64 km (Daubar et al., 2022)

What changed. A compact dark spot with a core about 30×35 m appeared just west of a small knob on the plains north of Tithonium Chasma between September 2008 and April 2025, and is still there in March 2026; two CTX images from two orbits show it.

Why it may be a new impact. A September 2008 image, taken under the same sun as the April 2025 image, shows nothing there, while a dark feature 225 m to the east-north-east shows clearly in it. The spot lies on the side of the knob facing the sun and shows with the sun shining from two directions 58° apart, so it is not the knob's shadow; its compact, dark look is typical of a small new impact.

Caveats. With only one earlier image, the spot could have formed at any time in a span of nearly 17 years. The March 2026 image is a poor overall match to the others, although the knob and a crater 1 km to the south-south-west line up in it. No image from the high-resolution HiRISE camera covers it; the nearest ends 16.9 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 85% that the spot is a new impact. It noted that the small relief feature beside the spot is shaded like a small crater rather than a knob; this does not affect the assessment. It placed the spot at 2.8147° S, 268.3595° E, 198 m from the position given above.

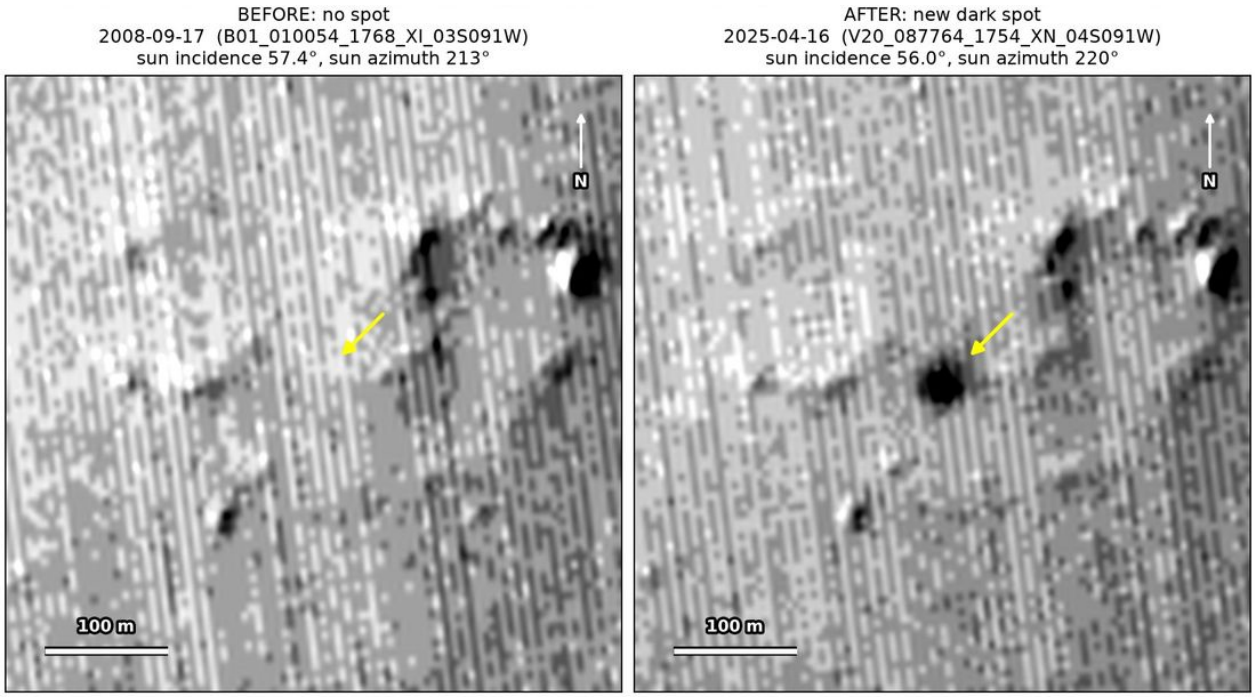

Figure 117: Candidate 47, before and after: the same 500 × 500 m of ground in 17 September 2008 (`B01_010054_1768_XI_03S091W`) and 16 April 2025 (`V20_087764_1754_XN_04S091W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

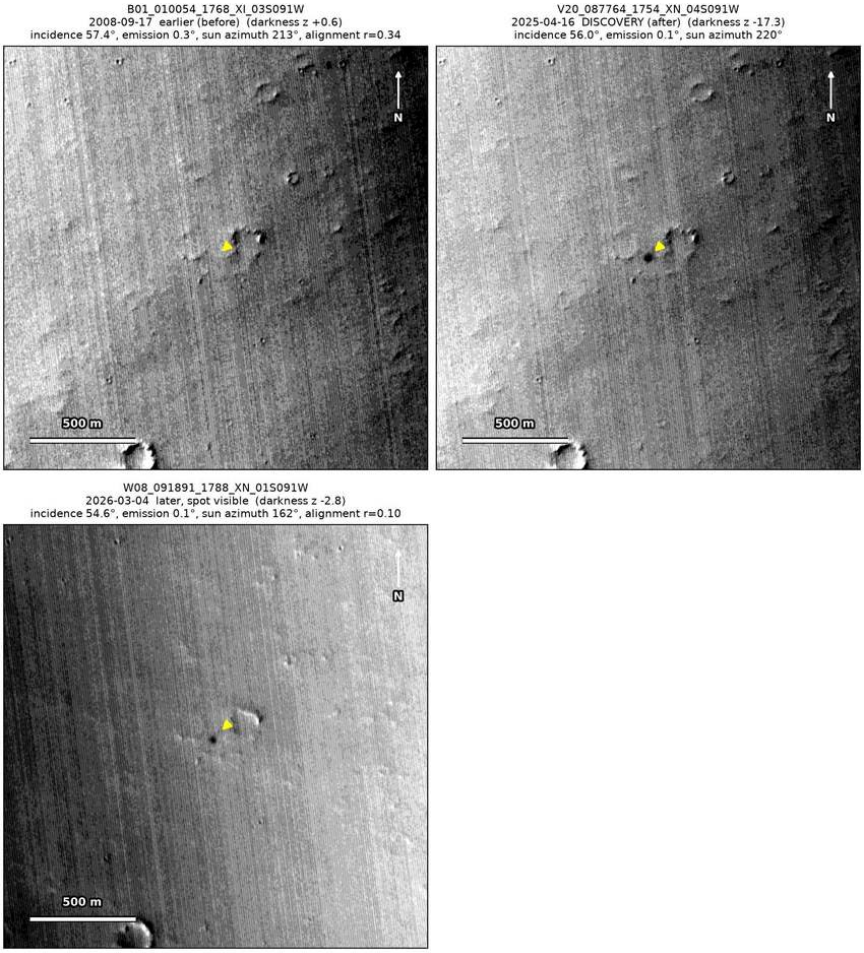

Figure 118: Candidate 47: three CTX images of the same 2×2 km patch, 2008 to 2026. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

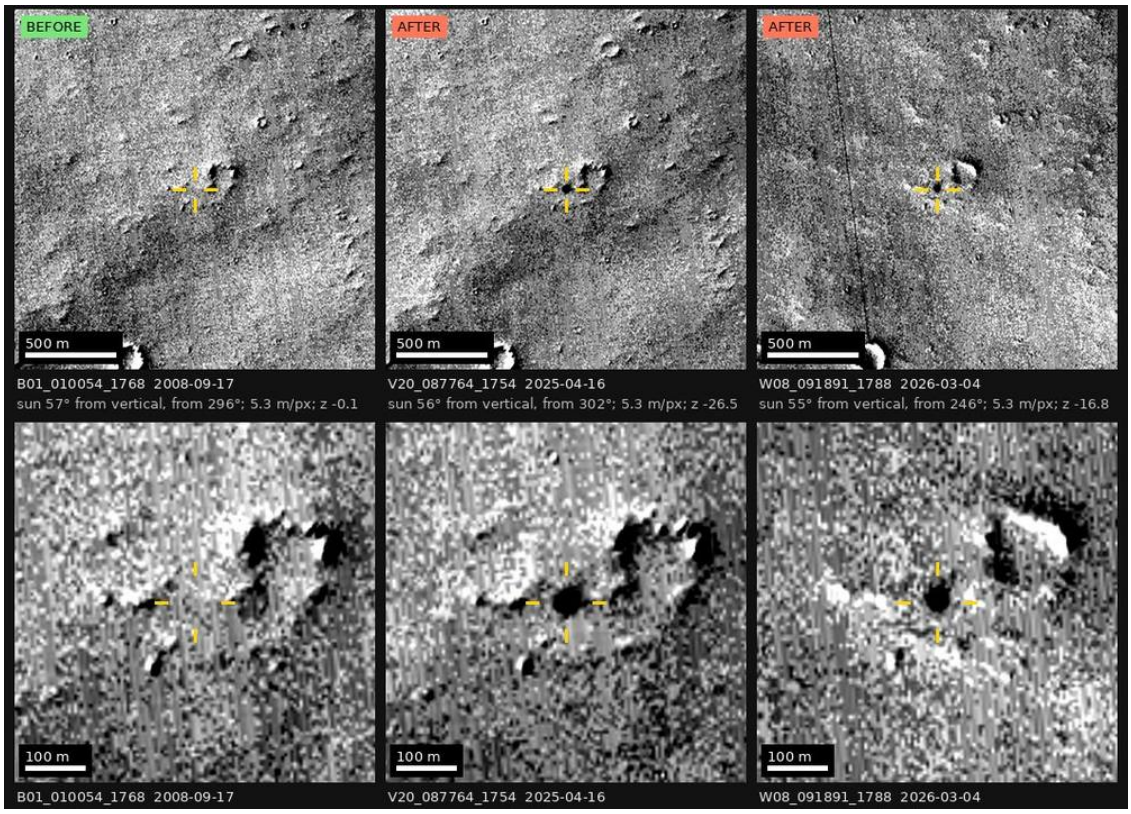

Figure 119: Candidate 47, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 48: Plains north of Avernus Colles

Location	1.8749° N, 172.1951° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 70%
Formed between	11 October 2010 and 21 January 2025
Images	1 without the spot (absent: 11 October 2010); 1 with it (present: 21 January 2025)
Size	Compact, very dark spot with a core about 50 × 40 m
HiRISE	none covers the spot (nearest footprint edge 41.1 km)
Nearest catalogued new impact	140 km (Daubar et al., 2022)

What changed. A compact, very dark spot with a core about 50 × 40 m appeared on the plains north of Avernus Colles between October 2010 and January 2025; so far, one CTX image shows it.

Why it may be a new impact. The only earlier image, from October 2010, lines up well and shows no spot there, while a dark feature 170 m to the west shows in it. Compact, very dark, smooth and round, the spot fits a small new impact better than darker ground exposed by the wind or an image defect. It lies inside the InSight study area of Bickel et al. (2025) but cannot be one of its 123 impacts, which all have HiRISE images; this spot has none.

Caveats. Only one image shows it, and the one earlier image is from more than 14 years before, so it could have formed at any time in that span. No image from the high-resolution HiRISE camera covers it; the nearest ends 41.1 km away.

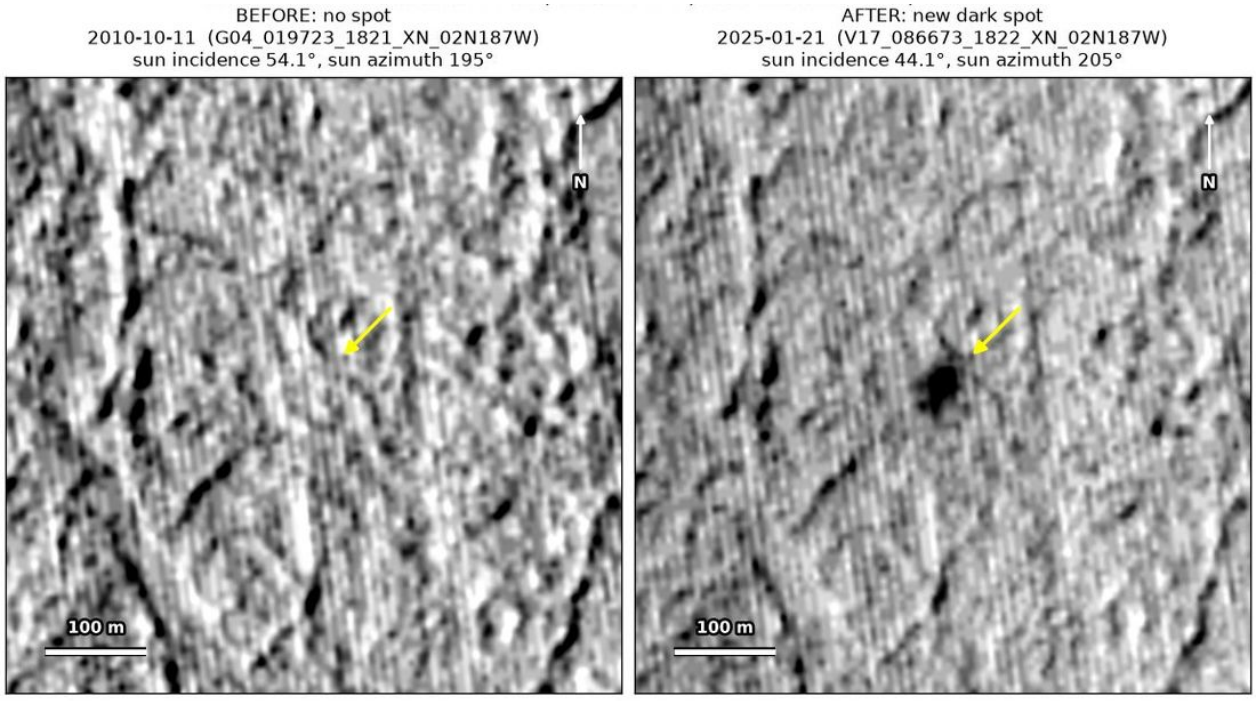

Figure 120: Candidate 48, before and after: the same 600 × 600 m of ground in 11 October 2010 (`G04_019723_1821_XN_02N187W`) and 21 January 2025 (`V17_086673_1822_XN_02N187W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

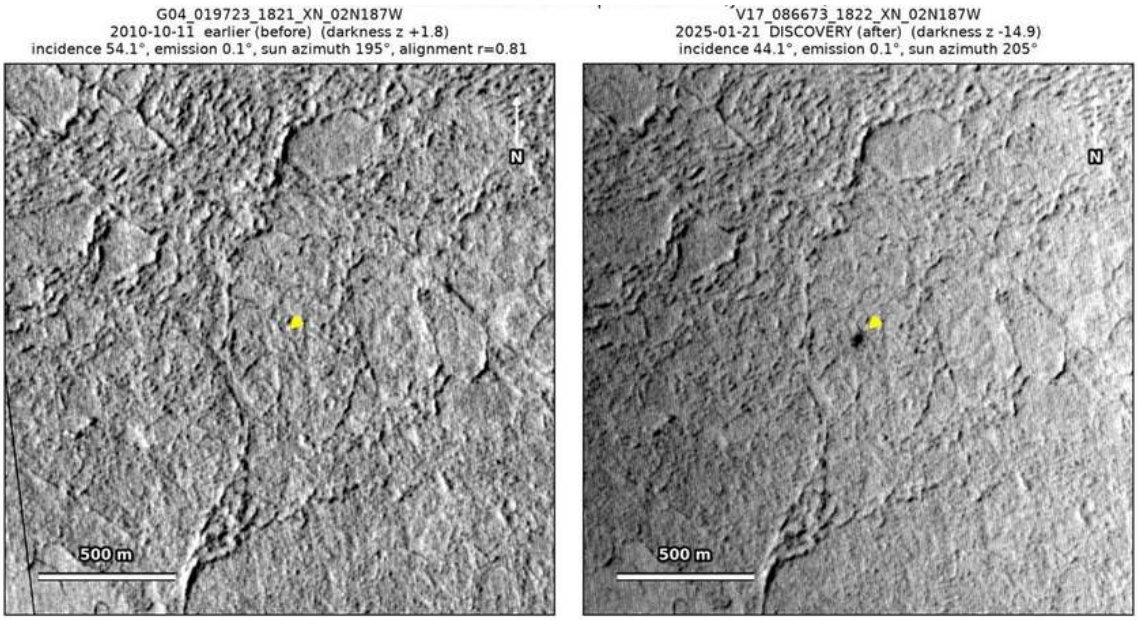

Figure 121: Candidate 48: both CTX images of the same 2×2 km patch, October 2010 and January 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 49: Fortuna Fossae region, north of Noctis Labyrinthus

Location	5.8922° N, 268.0812° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 70%
Formed between	26 December 2008 and 10 February 2025
Images	2 without the spot (absent: 15 October 2008 and 26 December 2008); 1 with it (present: 10 February 2025)
Size	Compact, very dark spot with a core about 45×35 m
HiRISE	none covers the spot (nearest footprint edge 38.0 km)
Nearest catalogued new impact	215 km (Daubar et al., 2022)

What changed. A compact, very dark spot with a core about 45×35 m appeared in the Fortuna Fossae region north of Noctis Labyrinthus between December 2008 and February 2025; so far, one CTX image shows it.

Why it may be a new impact. Two earlier images, from October and December 2008, show no spot there, while a dark feature 245 m to the north shows in both. Compact, very dark, smooth and round, the spot is unlikely to be darker ground exposed by the wind or an image defect, and may be the blast zone of a small new impact.

Caveats. Only one image shows it, and both earlier images are from 2008, so it could have formed at any time in a span of more than 16 years. No image from the high-resolution HiRISE camera covers it; the nearest ends 38.0 km away.

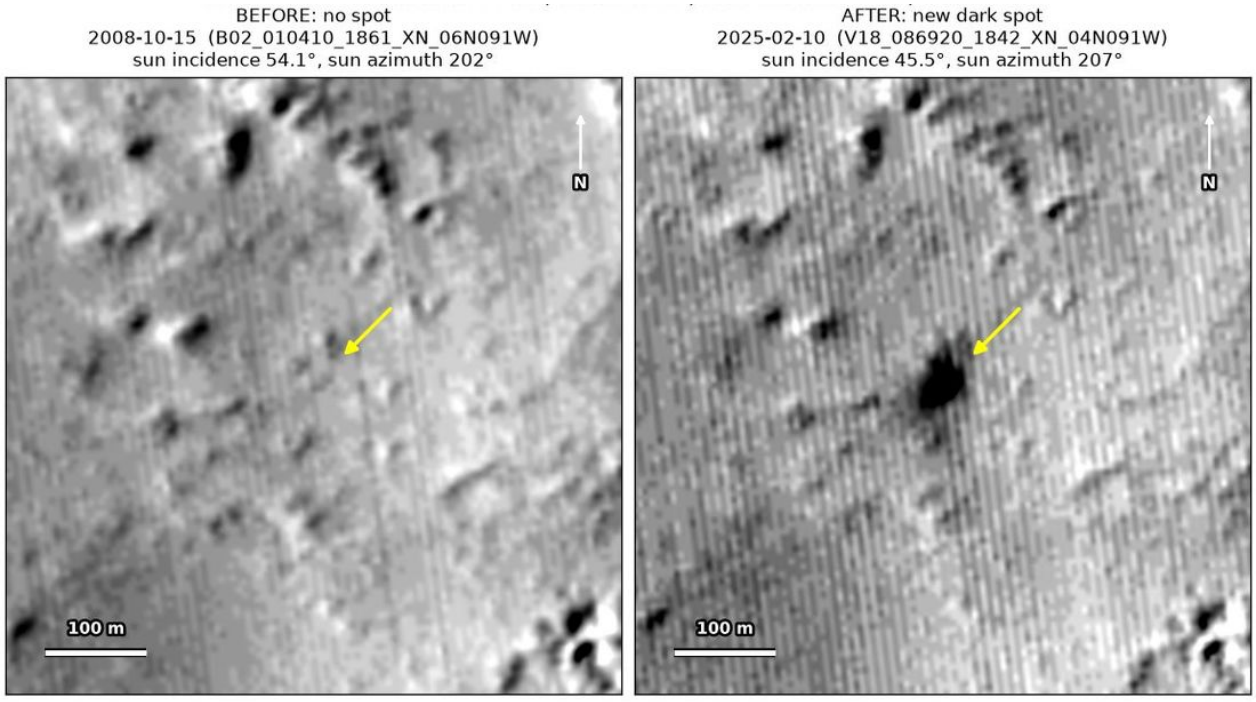

Figure 122: Candidate 49, before and after: the same 600×600 m of ground in 15 October 2008 (`B02_010410_1861_XN_06N091W`) and 10 February 2025 (`V18_086920_1842_XN_04N091W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

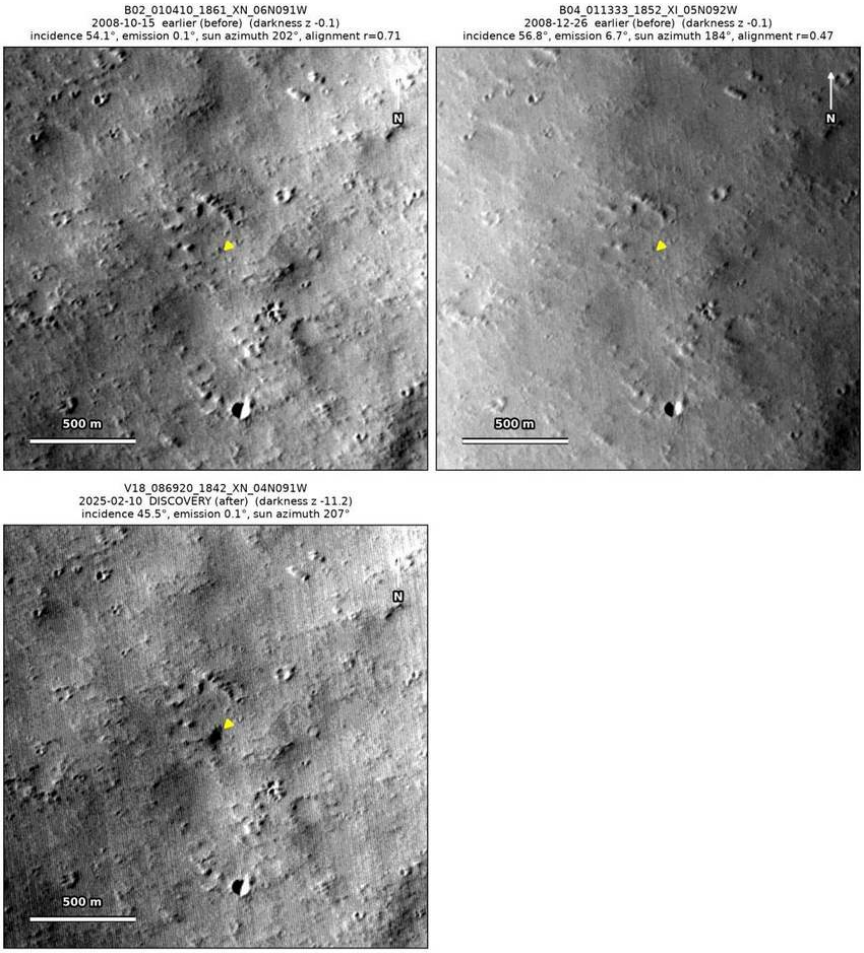

Figure 123: Candidate 49: three CTX images of the same 2×2 km patch, 2008 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 50: Rippled ground west of Echus Chasma

Location	4.0621° N, 278.7650° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	30 January 2013 and 29 March 2023
Images	3 without the spot (absent: 6 April 2007, 23 March 2012 and 30 January 2013); 4 with it (present: 29 March 2023, 12 February 2025, 13 April 2025 and 6 July 2025)
Size	Compact dark spot with a core about 50 m across
HiRISE	none covers the spot (nearest footprint edge 42.7 km)
Nearest catalogued new impact	59 km (Daubar et al., 2022)

What changed. A compact dark spot about 50 m across appeared on rippled ground west of Echus Chasma between January 2013 and March 2023, and is still there in July 2025; four CTX images from four orbits show it.

Why it may be a new impact. Three earlier images, from 2007, 2012 and 2013, show nothing there, while a dark feature 75 m to the south-west shows in them; the March 2012 image was taken under the same sun as the April 2025 image, so the spot is not a shadow. A compact, round dark spot that has not moved in more than two years is how the blast zone of a small new crater looks.

Caveats. The earlier images are a poor overall match to the later ones, the January 2013 one most of all, although the ripple crests line up in every image. Dark sand gathered between the ripples is the main alternative, though it is unlikely for a round spot that has stayed in place. No image from the high-resolution HiRISE camera covers it; the nearest ends 42.7 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 85% that the spot is a new impact. It found the January 2013 image too oblique to be informative and placed the start of the window at 23 March 2012. It placed the spot at 4.0624° N, 278.7623° E, 161 m from the position given above.

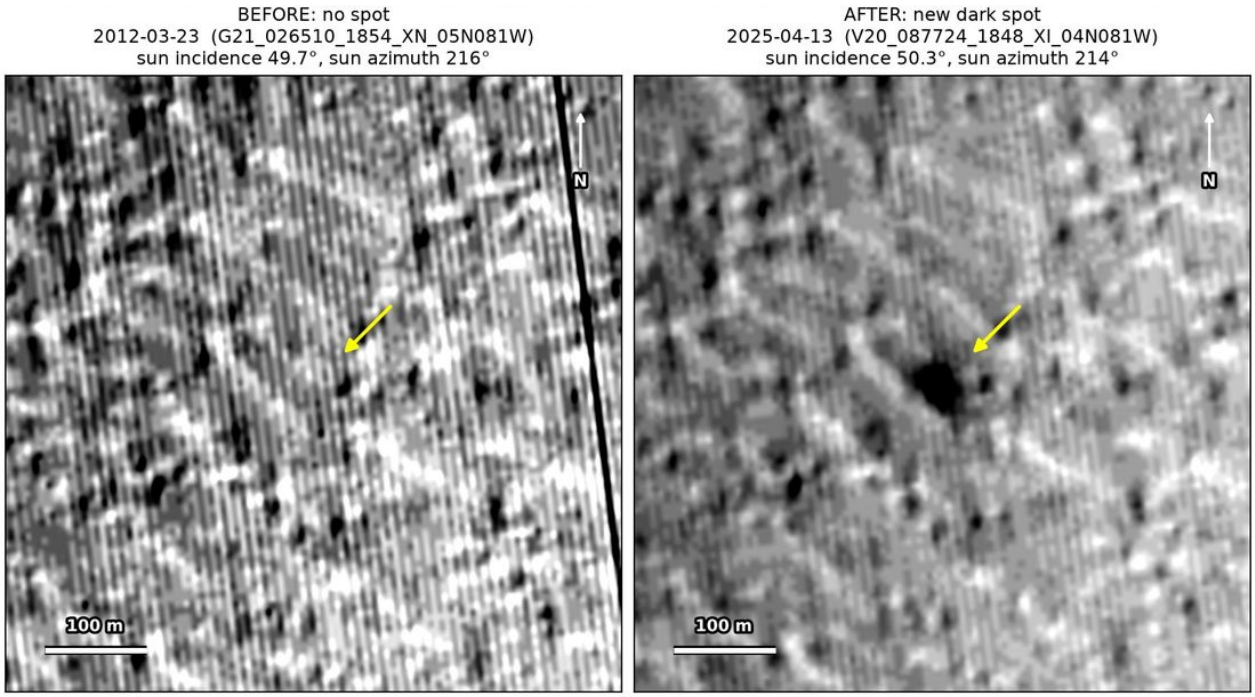

Figure 124: Candidate 50, before and after: the same 600 × 600 m of ground in 23 March 2012 (`G21_026510_1854_XN_05N081W`) and 13 April 2025 (`V20_087724_1848_XI_04N081W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

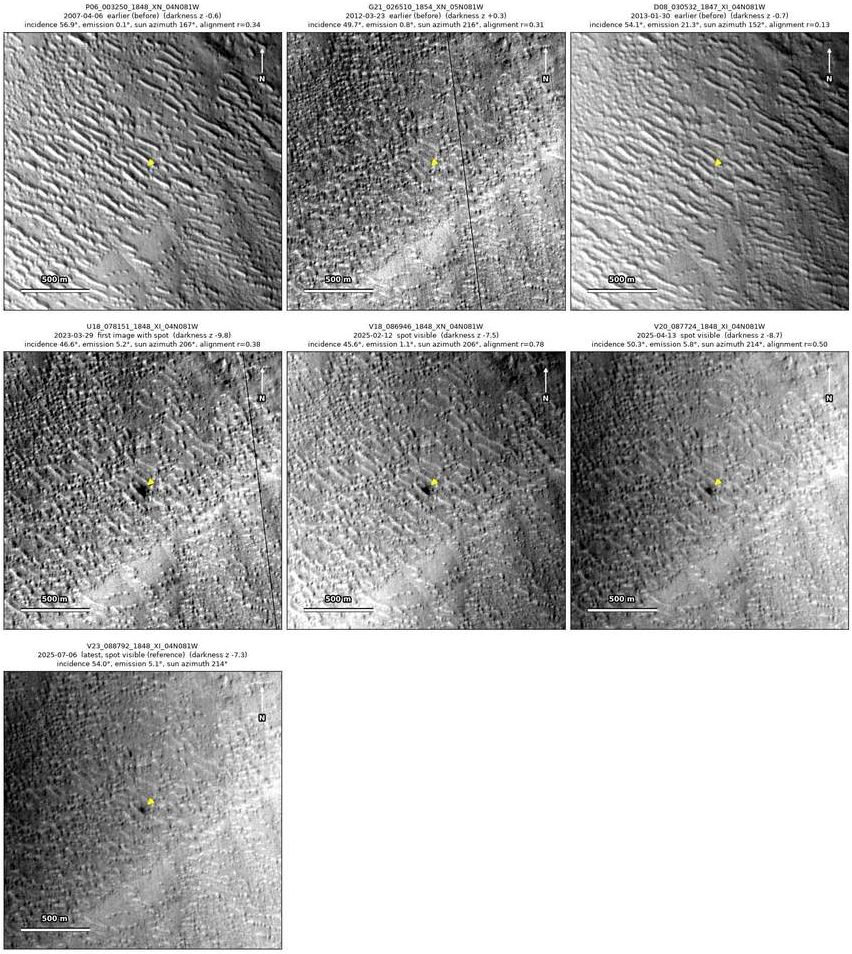

Figure 125: Candidate 50: seven CTX images of the same 2×2 km patch, 2007 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

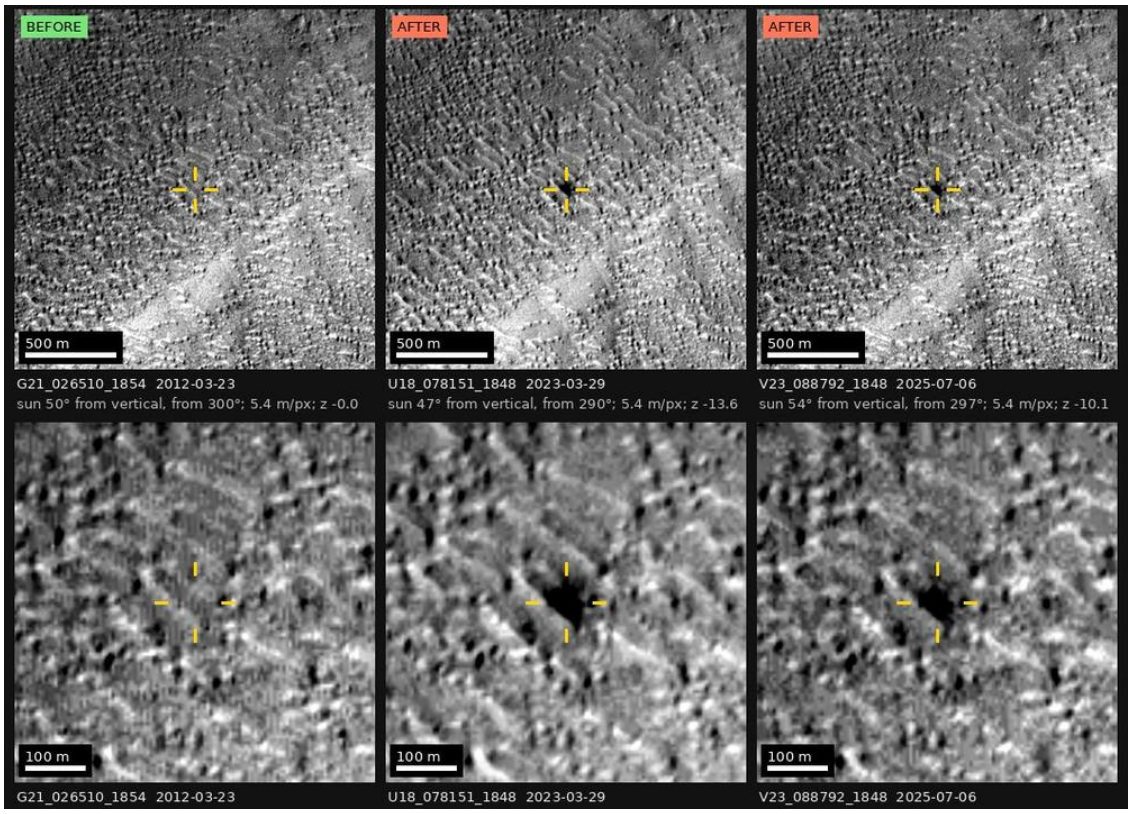

Figure 126: Candidate 50, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 51: Southern Arabia Terra, east of Schiaparelli crater

Location	0.6905° S, 26.6850° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	6 September 2010 and 1 May 2023
Images	1 without the spot (absent: 6 September 2010); 2 with it (present: 1 May 2023 and 17 February 2025)
Size	Compact, very dark spot with a core about 55 × 65 m
HiRISE	none covers the spot (nearest footprint edge 46.4 km)
Nearest catalogued new impact	122 km (Daubar et al., 2022)

What changed. A compact, very dark spot with a core about 55 × 65 m appeared in southern Arabia Terra, east of Schiaparelli crater, between September 2010 and May 2023, and is still there in February 2025; two CTX images from two orbits show it.

Why it may be a new impact. The only earlier image, from September 2010, lines up well and shows nothing there; it was taken under the same sun as the May 2023 image, so the spot is not a shadow. A compact, very dark spot unchanged for 21 months is unlikely to be darker ground exposed by the wind, and is the typical mark of a small new impact.

Caveats. With only one earlier image, the spot could have formed at any time in a span of almost 13 years. No image from the high-resolution HiRISE camera covers it; the nearest ends 46.4 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 90% that the spot is a new impact. It placed the spot at 0.6922° S, 26.6886° E, 235 m from the position given above.

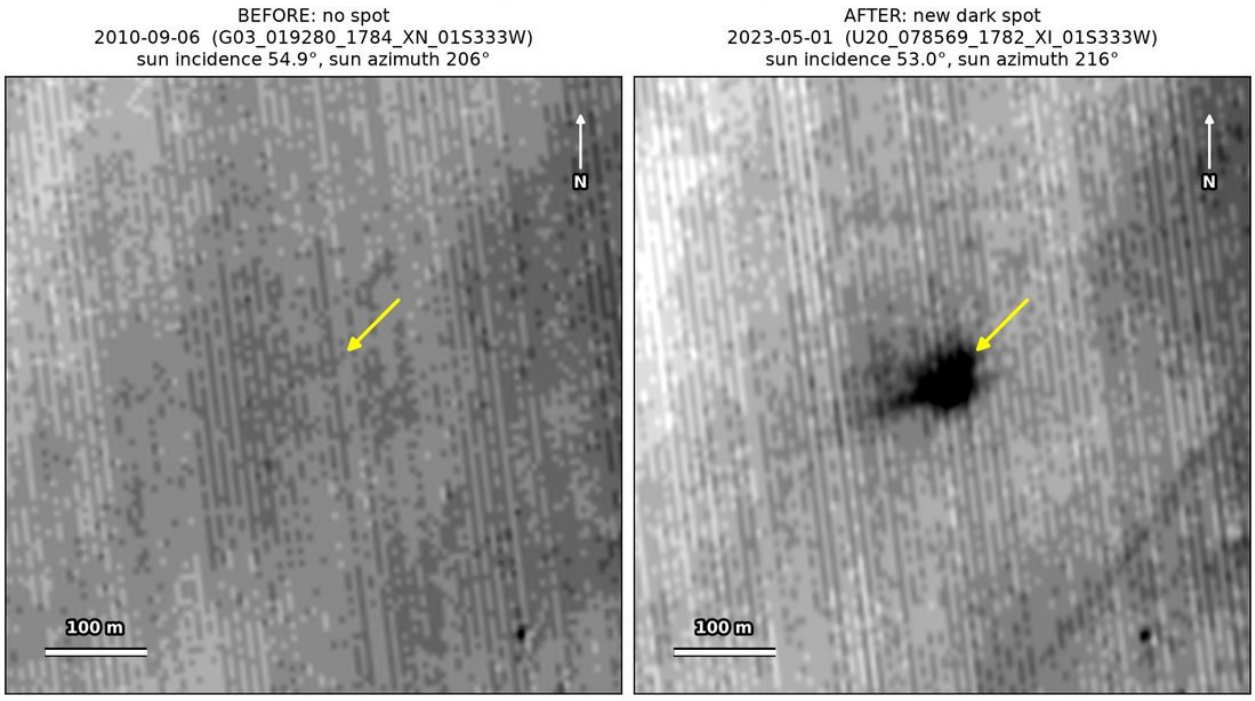

Figure 127: Candidate 51, before and after: the same 600 × 600 m of ground in 6 September 2010 (`G03_019280_1784_XN_01S333W`) and 1 May 2023 (`U20_078569_1782_XI_01S333W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

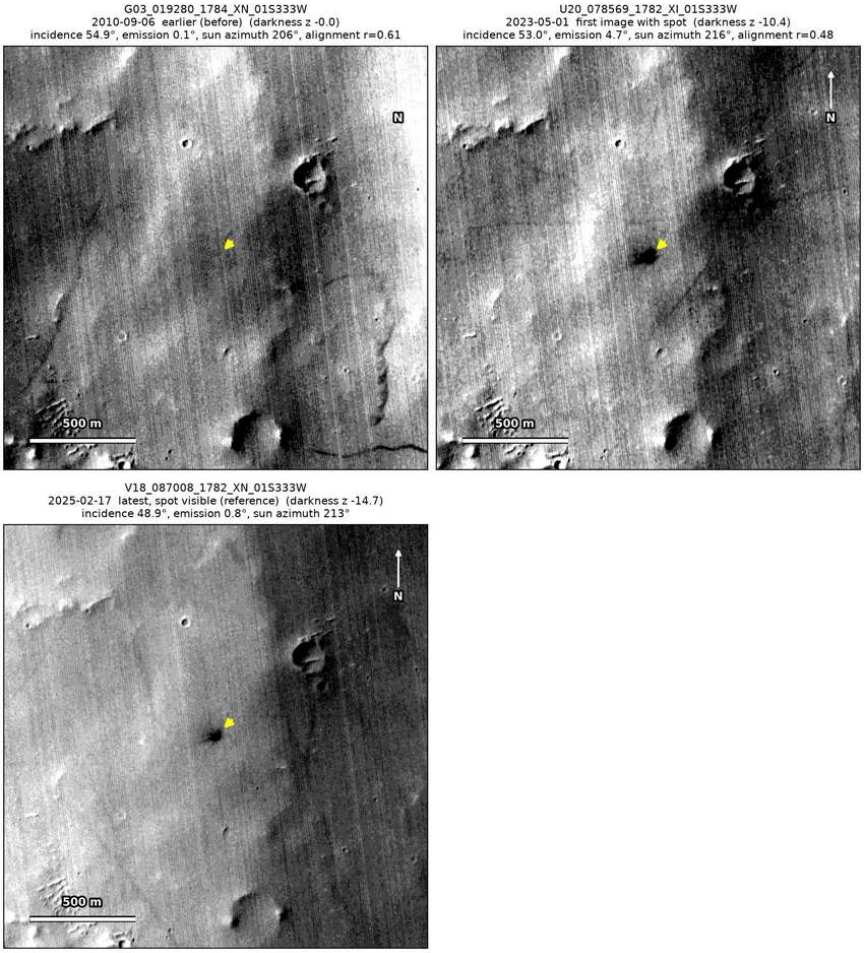

Figure 128: Candidate 51: three CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

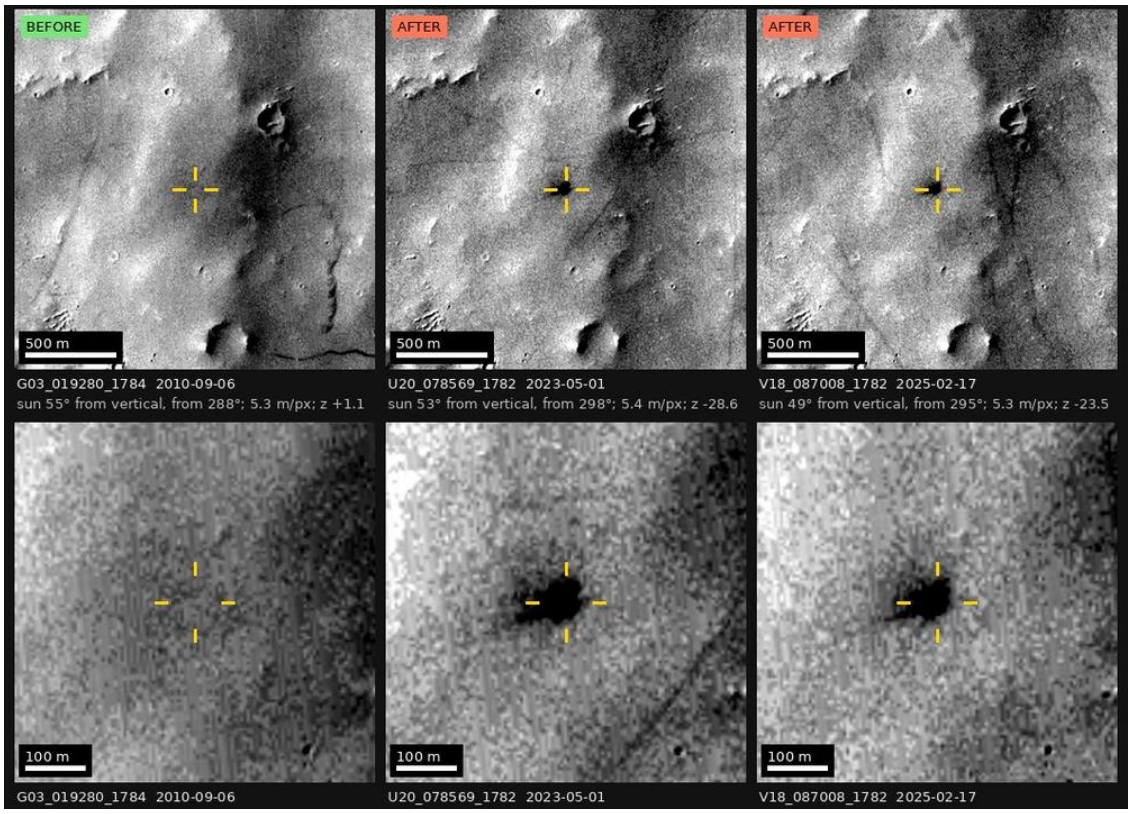

Figure 129: Candidate 51, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 52: Elysium rise west of Albor Tholus, south-west of Elysium Mons

Location	21.3396° N, 144.1996° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	26 January 2014 and 18 March 2025
Images	1 without the spot (absent: 26 January 2014); 1 with it (present: 18 March 2025)
Size	Compact dark spot with a core about 65 × 45 m
HiRISE	none covers the spot (nearest footprint edge 33.3 km)
Nearest catalogued new impact	46 km (Daubar et al., 2022)

What changed. A compact dark spot with a core about 65 × 45 m appeared on the Elysium rise west of Albor Tholus between January 2014 and March 2025; so far, one CTX image shows it.

Why it may be a new impact. The only earlier image, from January 2014, was taken under the same sun as the March 2025 image and shows no spot there, while a dark feature 230 m to the south-west shows in it. A compact, smooth blob, unlike an image defect, the spot may be the blast zone of a small new impact.

Caveats. Only one image shows it, and in the 11 years between the two images the wind could have exposed darker ground, though that is less likely for a compact spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 33.3 km away.

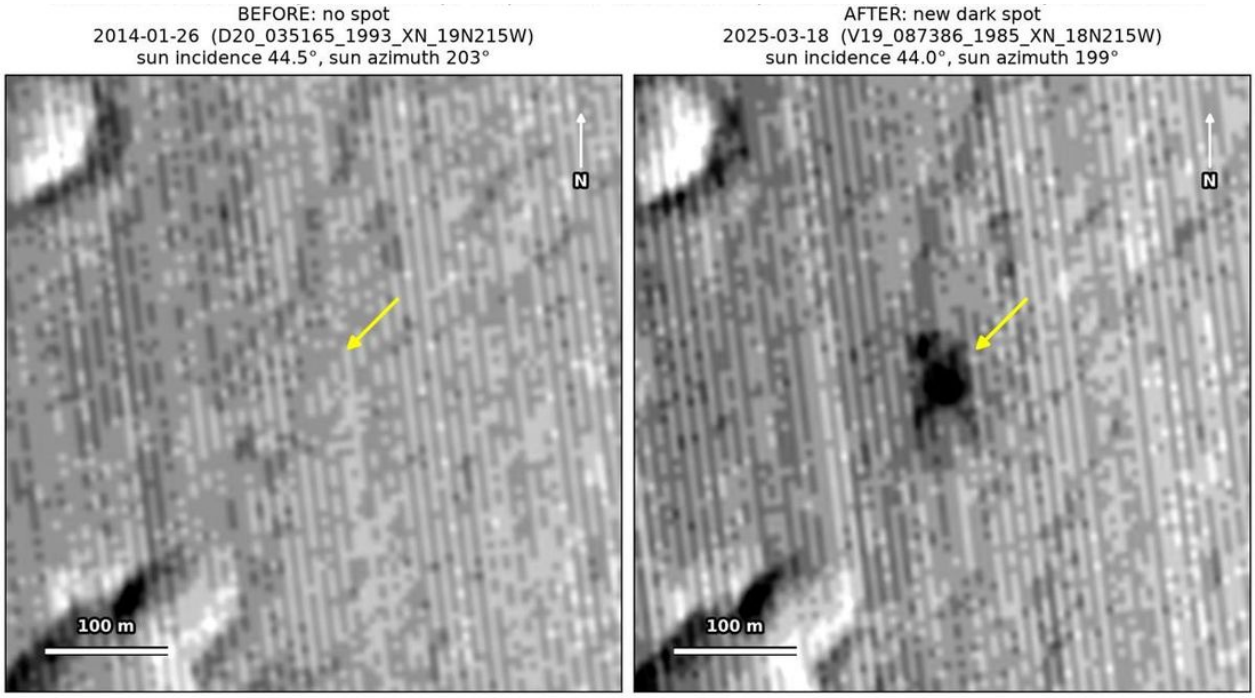

Figure 130: Candidate 52, before and after: the same 500 × 500 m of ground in 26 January 2014 (`D20_035165_1993_XN_19N215W`) and 18 March 2025 (`V19_087386_1985_XN_18N215W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

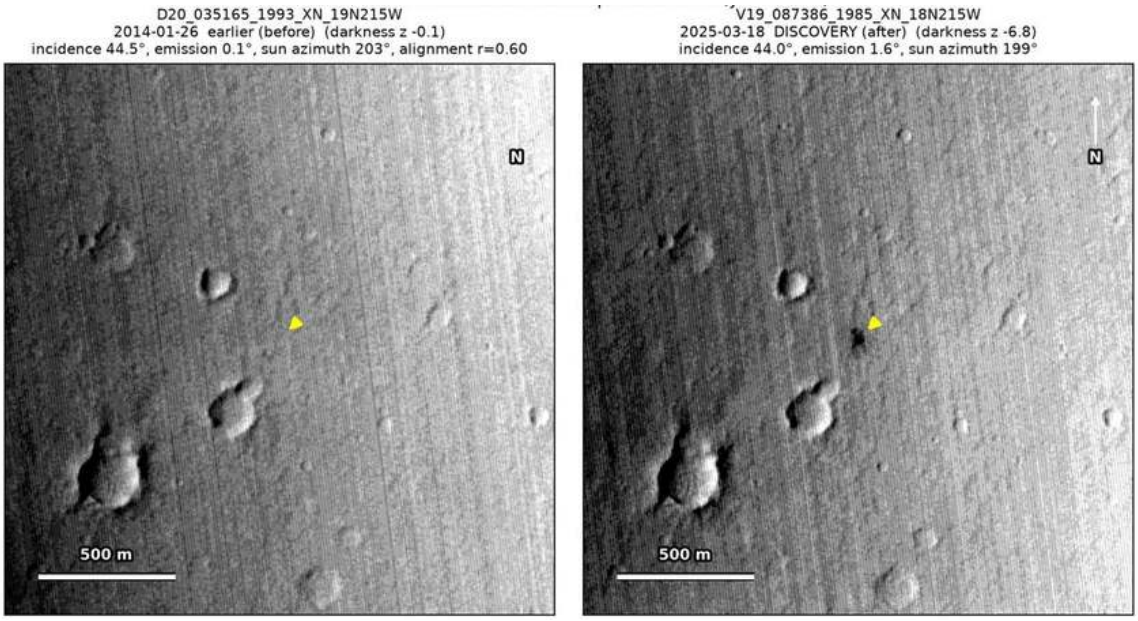

Figure 131: Candidate 52: both CTX images of the same 2×2 km patch, January 2014 and March 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 53: Plains west of Uranius Mons, between Uranius Tholus and Uranius Mons

Location	26.3306° N, 264.7174° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	12 May 2016 and 12 August 2020
Images	2 without the spot (absent: 9 August 2014 and 12 May 2016); 3 with it (present: 12 August 2020, 5 December 2024 and 10 February 2025)
Size	Compact, very dark spot with a core about 35 × 30 m
HiRISE	none covers the spot (nearest footprint edge 39.0 km)
Nearest catalogued new impact	42 km (Daubar et al., 2022)

What changed. A compact, very dark spot with a core about 35 × 30 m appeared on the plains west of Uranius Mons between May 2016 and August 2020, and is still there in February 2025; three CTX images from three orbits show it.

Why it may be a new impact. Images from 2014 and 2016 show nothing there, while two small dark dots nearby show, and line up, in every image. The spot is present under suns of very different heights, so it is not a shadow, and a compact spot unchanged for four and a half years is the typical mark of a small new impact.

Caveats. The two earlier images are a poor overall match to the later ones, although the two small dark dots nearby line up in every image. No image from the high-resolution HiRISE camera covers it; the nearest ends 39.0 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 90% that the spot is a new impact. It placed the spot at 26.3330° N, 264.7201° E, 200 m from the position given above.

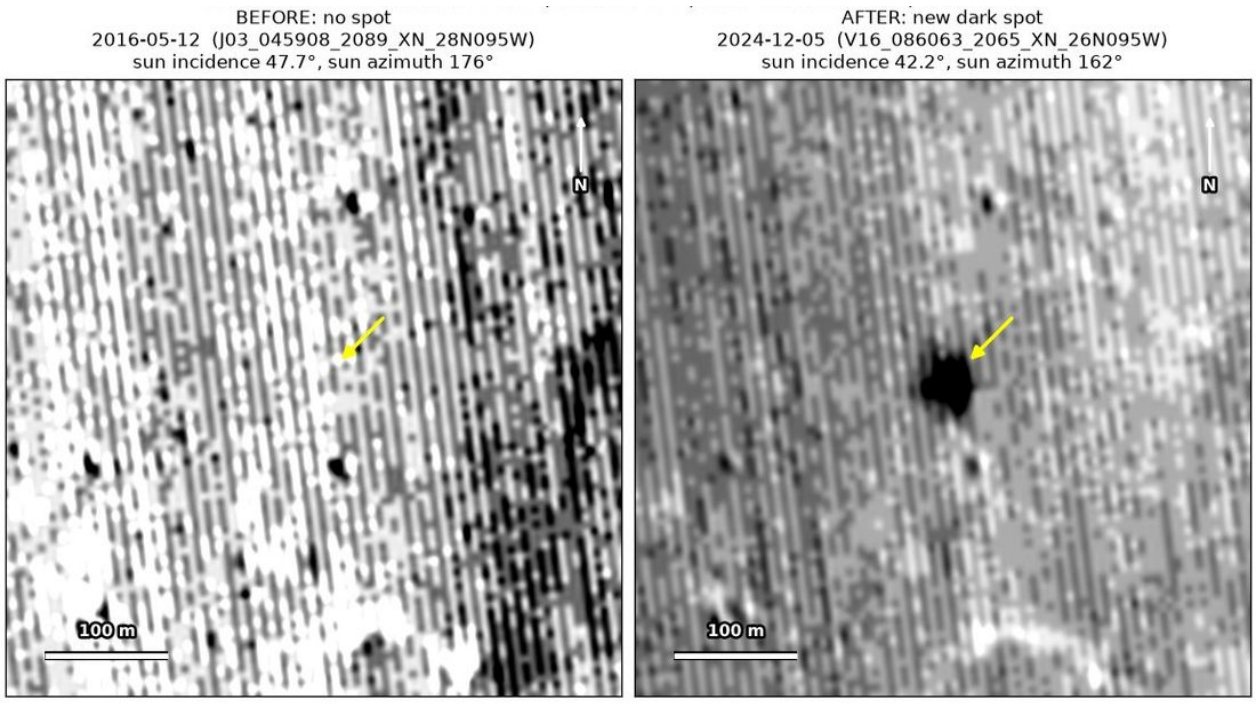

Figure 132: Candidate 53, before and after: the same 500 × 500 m of ground in 12 May 2016 (`J03_045908_2089_XN_28N095W`) and 5 December 2024 (`V16_086063_2065_XN_26N095W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

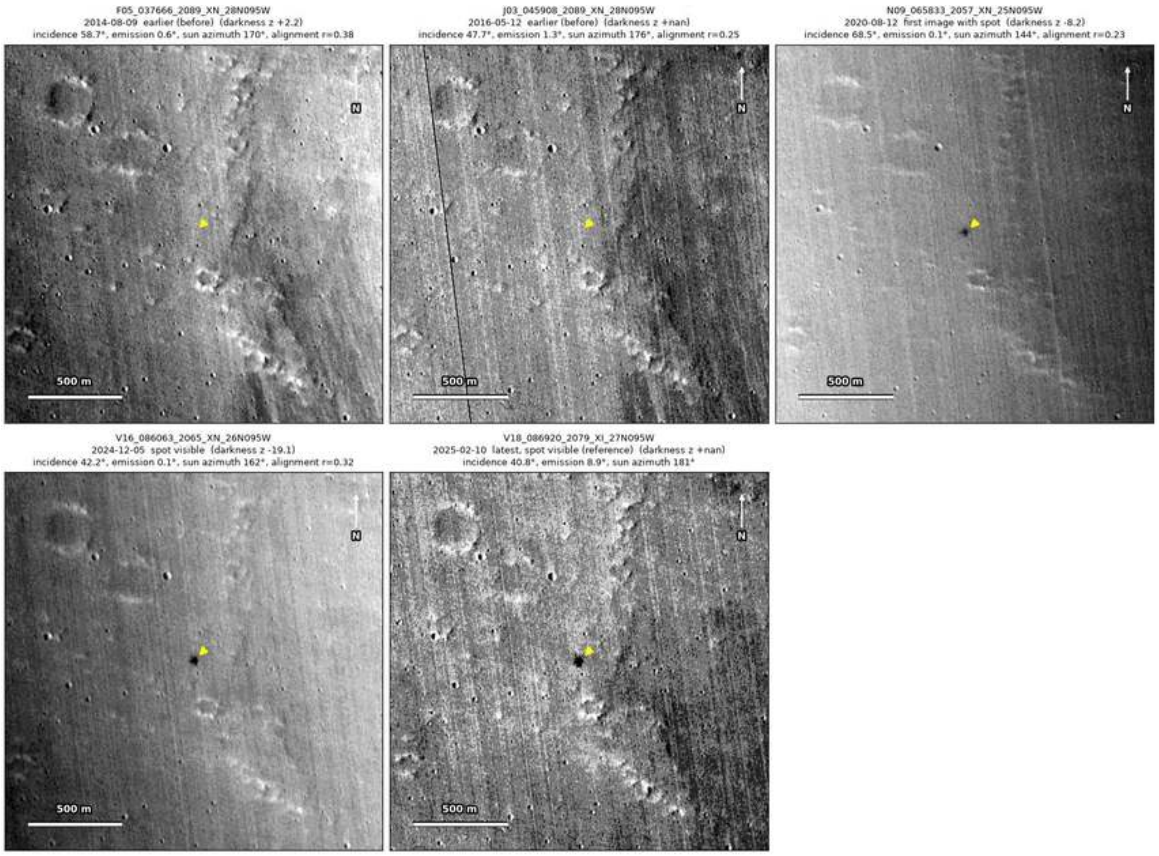

Figure 133: Candidate 53: five CTX images of the same 2×2 km patch, 2014 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

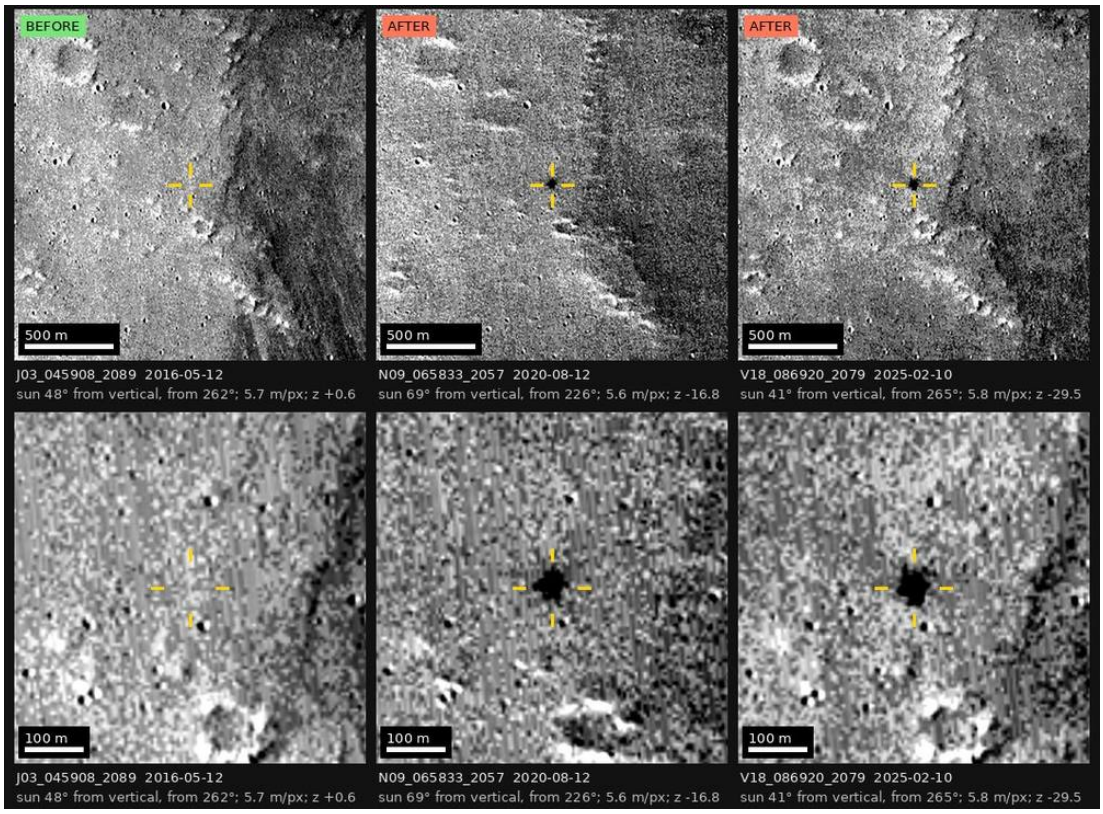

Figure 134: Candidate 53, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 54: Lava flows of eastern Tharsis, north-east of Tharsis Tholus

Location	16.6374° N, 273.8008° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 50%
Formed between	19 July 2016 and 6 February 2025
Images	2 without the spot (absent: 12 September 2007 and 19 July 2016); 1 with it (present: 6 February 2025)
Size	Diffuse dark patch about 200 to 300 m across, with a darker centre about 100 m across
HiRISE	none covers the spot (nearest footprint edge 54.0 km)
Nearest catalogued new impact	169 km (Daubar et al., 2022)

What changed. A diffuse dark patch about 200 to 300 m across, with darker blotches about 100 m across at its centre, appeared over the ridged surface of lava flows north-east of Tharsis Tholus between July 2016 and February 2025; so far, one CTX image shows it.

Why it may be a new impact. An image from July 2016 shows no patch there, while a dark feature 200 m to the north shows in it; an earlier image, from September 2007, is hazed by a dust storm. Laid over the lava ridges, the new patch with its darker centre may be the blast zone of a small new impact, and its size and sharpness make a cloud shadow unlikely.

Caveats. Only one image shows it, so a cloud shadow, though unlikely, cannot yet be ruled out, and darker ground exposed by the wind fits this diffuse patch nearly as well as an impact. No image from the high-resolution HiRISE camera covers it; the nearest ends 54.0 km away.

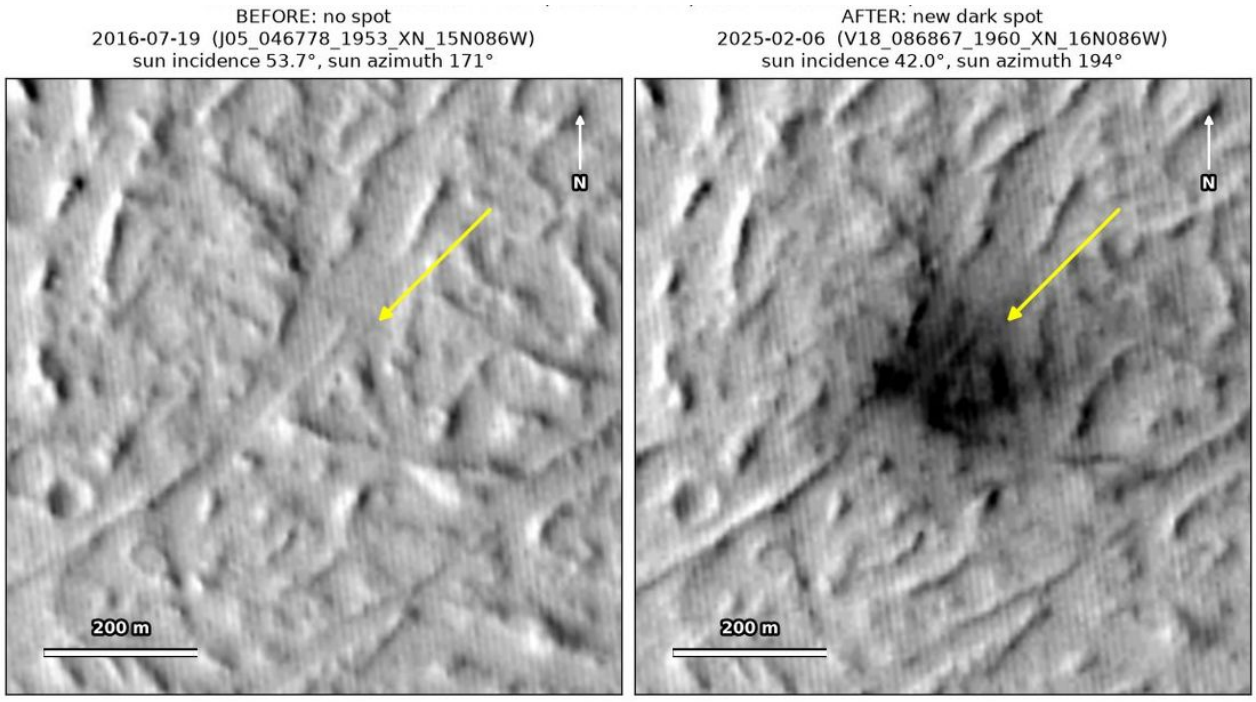

Figure 135: Candidate 54, before and after: the same 800 × 800 m of ground in 19 July 2016 (`J05_046778_1953_XN_15N086W`) and 6 February 2025 (`V18_086867_1960_XN_16N086W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

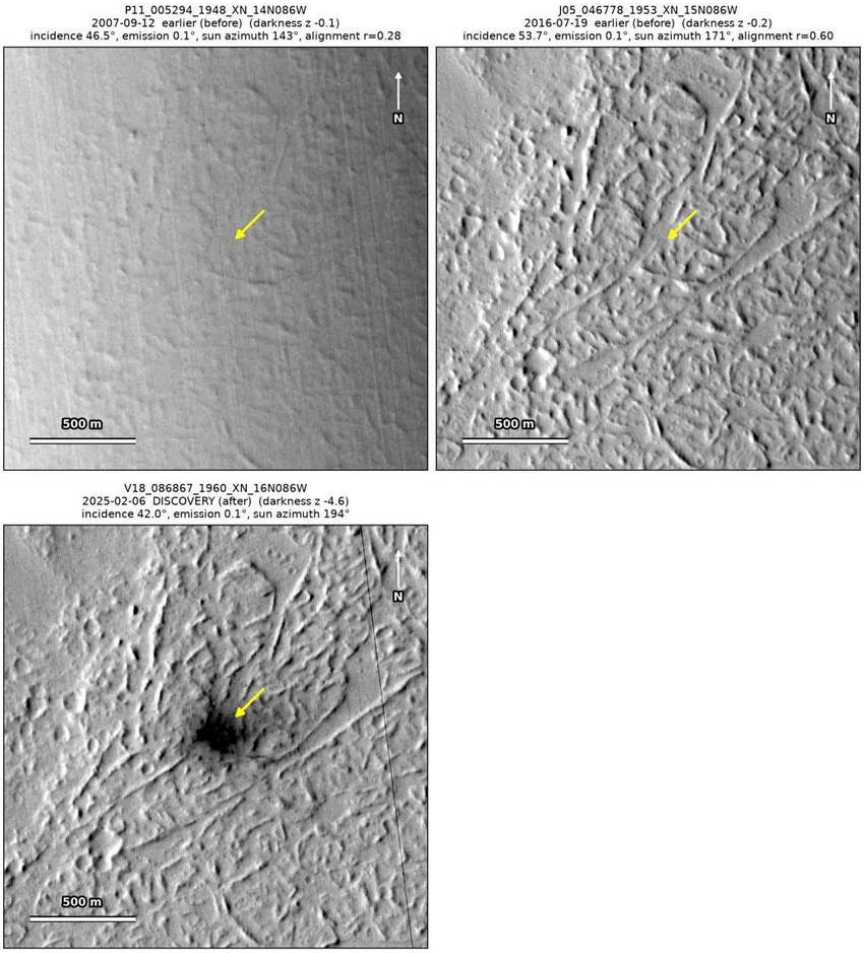

Figure 136: Candidate 54: three CTX images of the same 2×2 km patch, 2007 to 2025. The September 2007 image is hazed by a dust storm. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 55: Floor of Henry crater, Arabia Terra

Location	10.5054° N, 24.0527° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 55%
Formed between	5 March 2012 and 1 June 2025
Images	3 without the spot (absent: 16 December 2006, 16 November 2011 and 5 March 2012); 1 with it (present: 1 June 2025)
Size	Compact cluster of dark blobs about 80 m across, the largest about 30 m
HiRISE	none covers the spot (nearest footprint edge 6.6 km)
Nearest catalogued new impact	94 km (Daubar et al., 2022)

What changed. A compact cluster of dark blobs about 80 m across, the largest about 30 m, appeared on the floor of Henry crater in Arabia Terra between March 2012 and June 2025; so far, one CTX image shows it.

Why it may be a new impact. Three earlier images, from 2006, 2011 and 2012, show nothing there, while a dark feature 250 m to the east-south-east shows clearly in all three; the March 2012 one was taken under the same sun as the June 2025 image. Several dark blobs appearing together are most naturally read as a cluster impact: a group of small impacts made at the same time.

Caveats. Only one image shows it, and in the 13 years since the March 2012 image, dark sand or darker ground could have been exposed on the ridge where it lies. No image from the high-resolution HiRISE camera covers it; the nearest ends 6.6 km away.

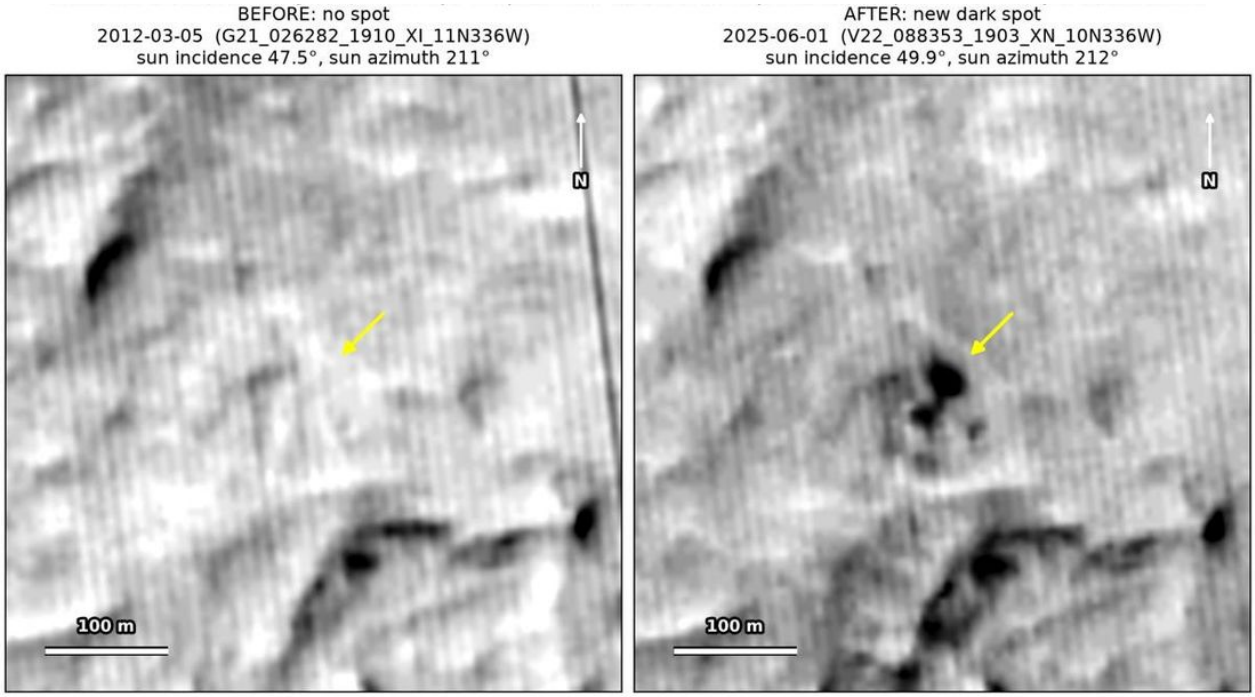

Figure 137: Candidate 55, before and after: the same 500 × 500 m of ground in 5 March 2012 (`G21_026282_1910_XI_11N336W`) and 1 June 2025 (`V22_088353_1903_XN_10N336W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

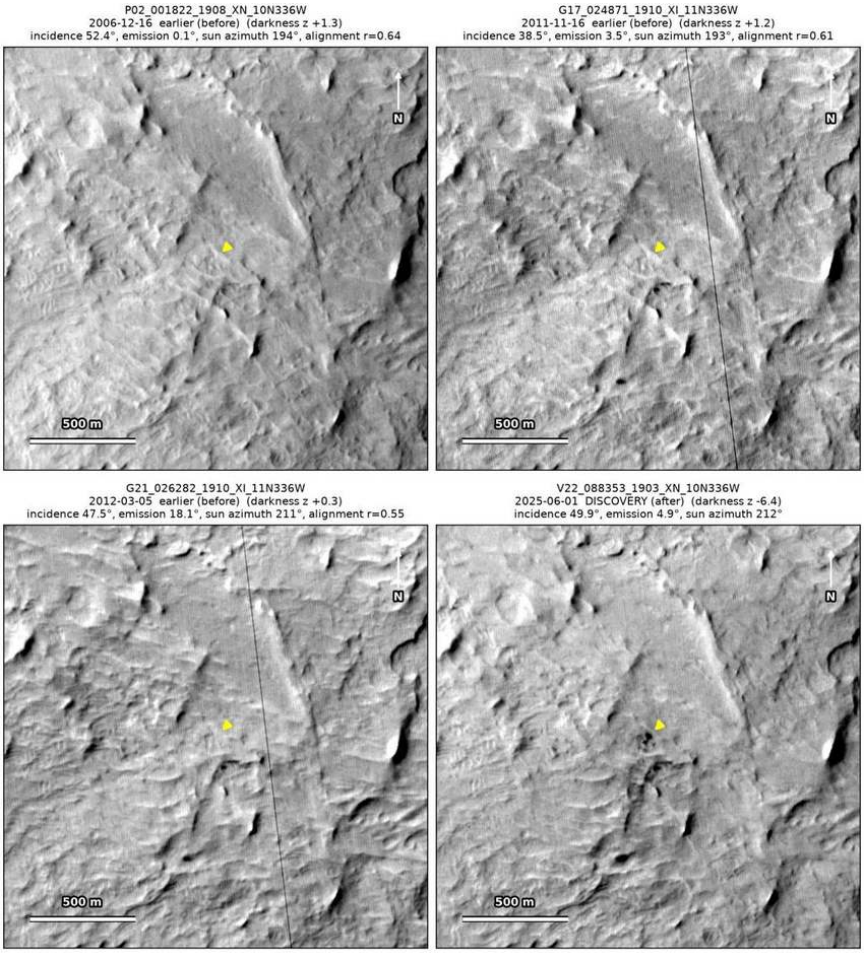

Figure 138: Candidate 55: four CTX images of the same 2×2 km patch, 2006 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 56: Plains north-west of Echus Chasma

Location	6.8861° N, 276.3125° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	9 May 2008 and 20 January 2025
Images	1 without the spot (absent: 9 May 2008); 2 with it (present: 20 January 2025 and 5 May 2025)
Size	Compact dark core about 55 × 45 m, inside a soft halo
HiRISE	none covers the spot (nearest footprint edge 5.5 km)
Nearest catalogued new impact	38 km (Daubar et al., 2022)

What changed. A compact dark core about 55 × 45 m, inside a soft halo, appeared among small knobs north-west of Echus Chasma between May 2008 and January 2025, and is still there in May 2025; two CTX images from two orbits show it.

Why it may be a new impact. A May 2008 image, taken under the same sun as the May 2025 image and with every knob lined up, shows nothing there, so the spot is not a shadow; a dark feature 300 m to the north-east shows clearly in it. A compact dark core with a halo is the classic look of a small fresh impact.

Caveats. It has been seen for only three and a half months, so darker ground exposed by the wind is not ruled out, although a compact core with a halo makes that unlikely. With only one earlier image, it could have formed at any time in a span of nearly 17 years. No image from the high-resolution HiRISE camera covers it; the nearest ends 5.5 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 80% that the spot is a new impact. It placed the spot at 6.8882° N, 276.3156° E, 220 m from the position given above.

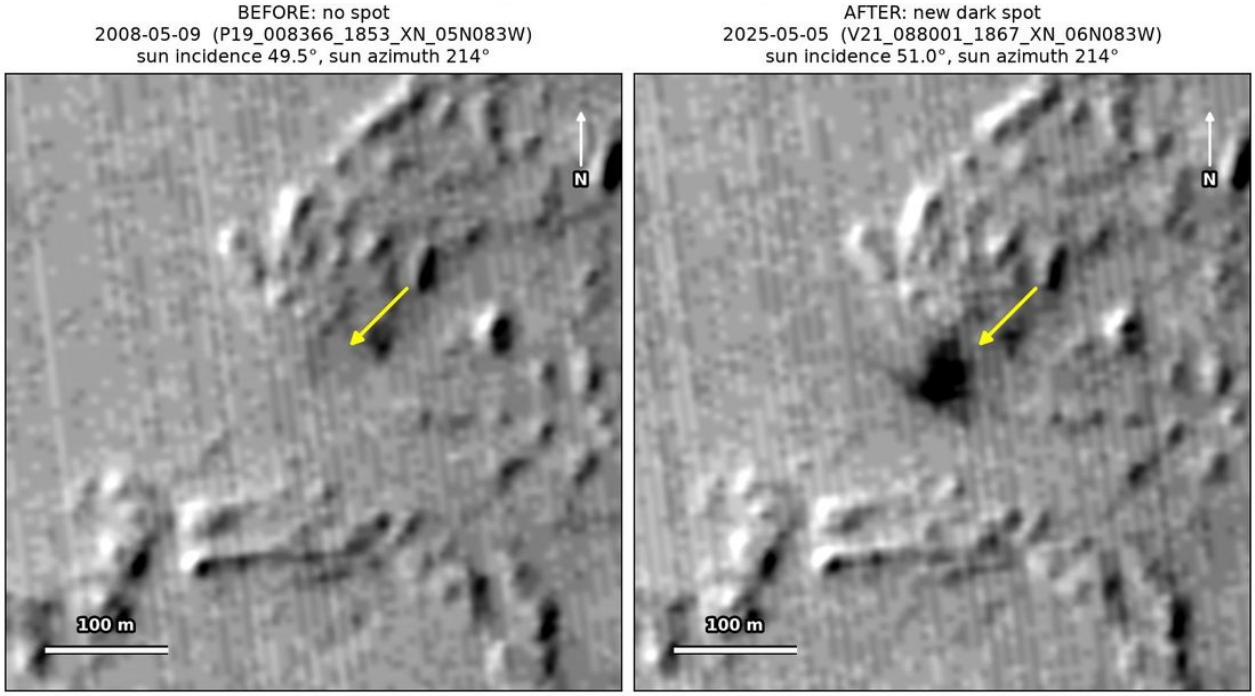

Figure 139: Candidate 56, before and after: the same 500 × 500 m of ground in 9 May 2008 (`P19_008366_1853_XN_05N083W`) and 5 May 2025 (`V21_088001_1867_XN_06N083W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

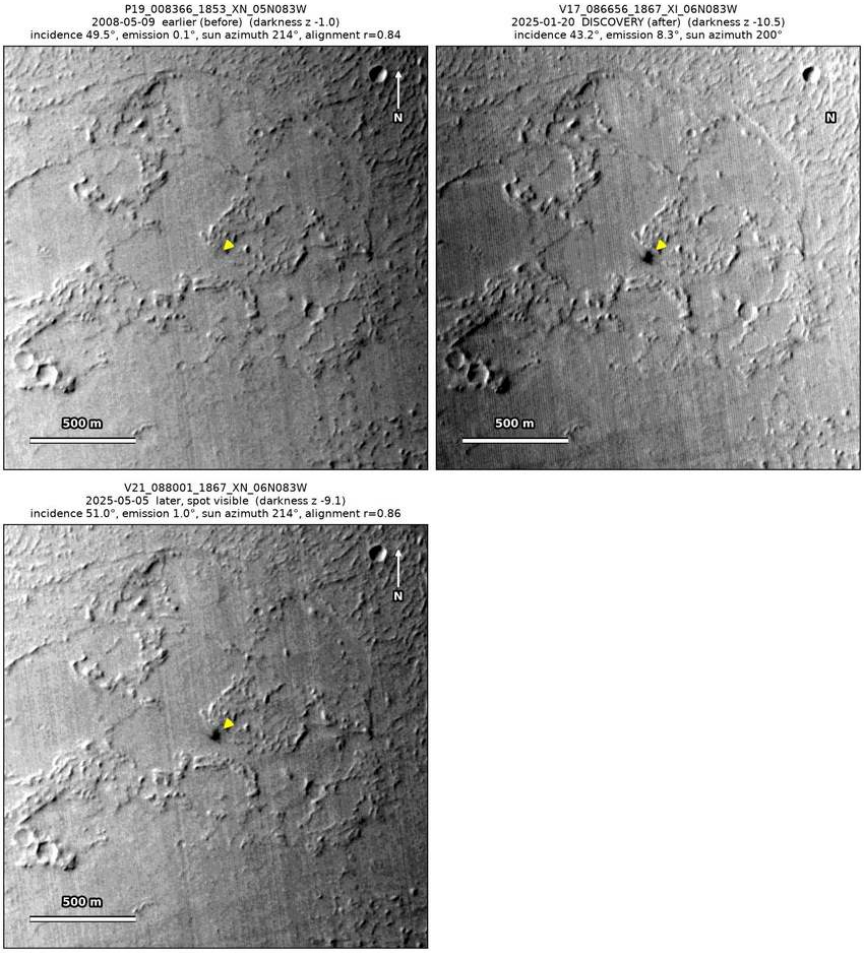

Figure 140: Candidate 56: three CTX images of the same 2×2 km patch, 2008 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

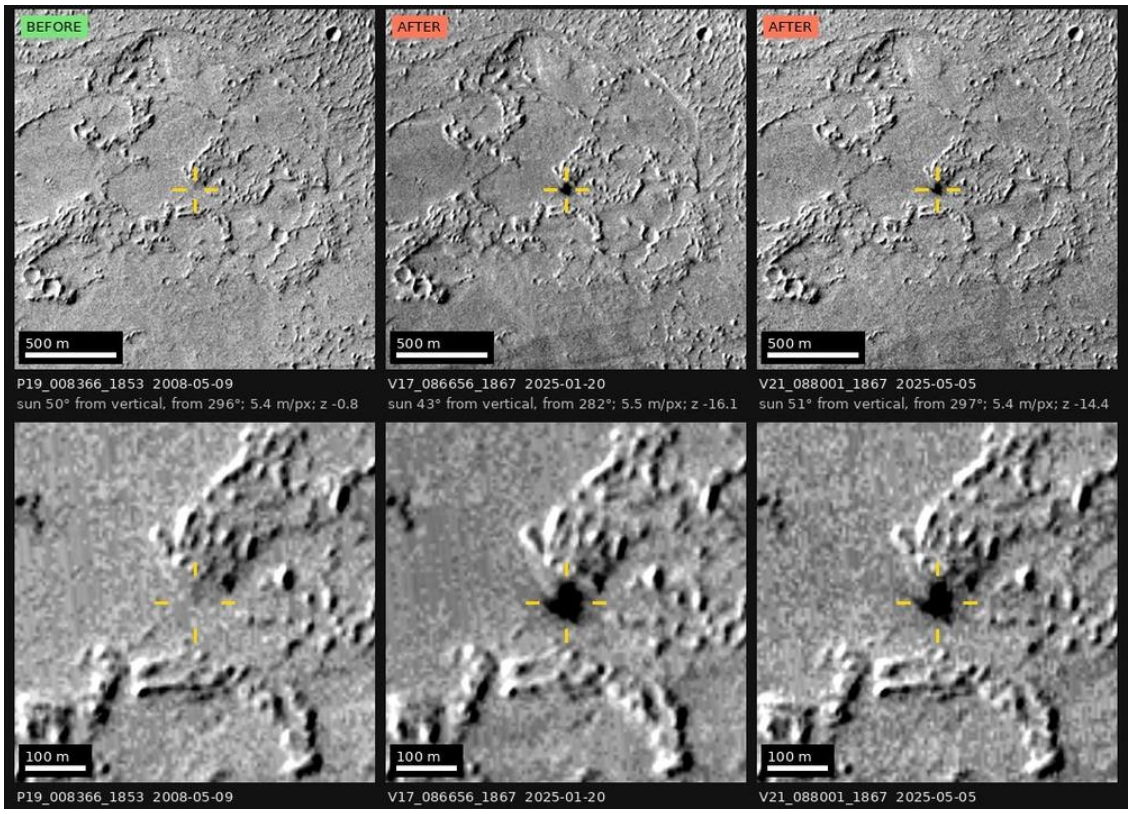

Figure 141: Candidate 56, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 57: Western Arabia Terra, about 700 km west of Pasteur crater

Location	19.1431° N, 12.1955° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	18 March 2014 and 4 May 2023
Images	2 without the spot (absent: 17 February 2010 and 18 March 2014); 4 with it (present: 4 May 2023, 6 January 2025, 3 February 2025 and 25 March 2025 (CTX and HiRISE))
Size	Dark patch 40 × 46 m, measured in the HiRISE image, around a crater about 2.5 m across
HiRISE	ESP_087470_1990 (2025-03-25, same orbit as V19, not impact-targeted)
Nearest catalogued new impact	187 km (Daubar et al., 2022)
Nearby	Candidate 72, 5.2 km to the north-north-west, formed between the same two images of March 2014 and May 2023.

What changed. A dark patch about 40 m across appeared beside an older crater 80 m wide in western Arabia Terra between March 2014 and May 2023, and is still there in March 2025; four CTX images from four orbits show it. A dark band stretches about 70 m south-south-east from it along the older crater's west rim.

Why it may be a new impact. Images from 2010 and 2014 line up well and show nothing there; the March 2014 one was taken under the same sun as the May 2023 image. A high-resolution HiRISE image shows a small crater at the centre of the patch, just what a small fresh impact leaves.

Caveats. By chance, NASA's HiRISE camera photographed the spot on 25 March 2025, after it formed, on the same orbit as the March 2025 CTX image. It shows a crater about 2.5 m across at the centre of a 40 × 46 m dark patch, dark on its west wall and bright on its east wall, the shading of a small pit lit from the west-north-west. The patch could be a slope streak left by dust sliding down from the older crater's rim, but the small crater at its centre argues against that, and the dark band may be a small avalanche set off by the impact. The small crater could be an old one stripped of dust by the wind or a dust devil, but that is unlikely for a round patch with no track.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 95% that the spot is a new impact. It examined the HiRISE image directly: a pit 2.5–3 m wide lies less than 3 m from the centre of the new dark patch, darker on its sunward wall and brighter on its far wall, the shading of a small bowl (Figure 146). It estimated the probability at about 95%. It placed the spot at 19.1428° N, 12.1880° E, 421 m from the position given above.

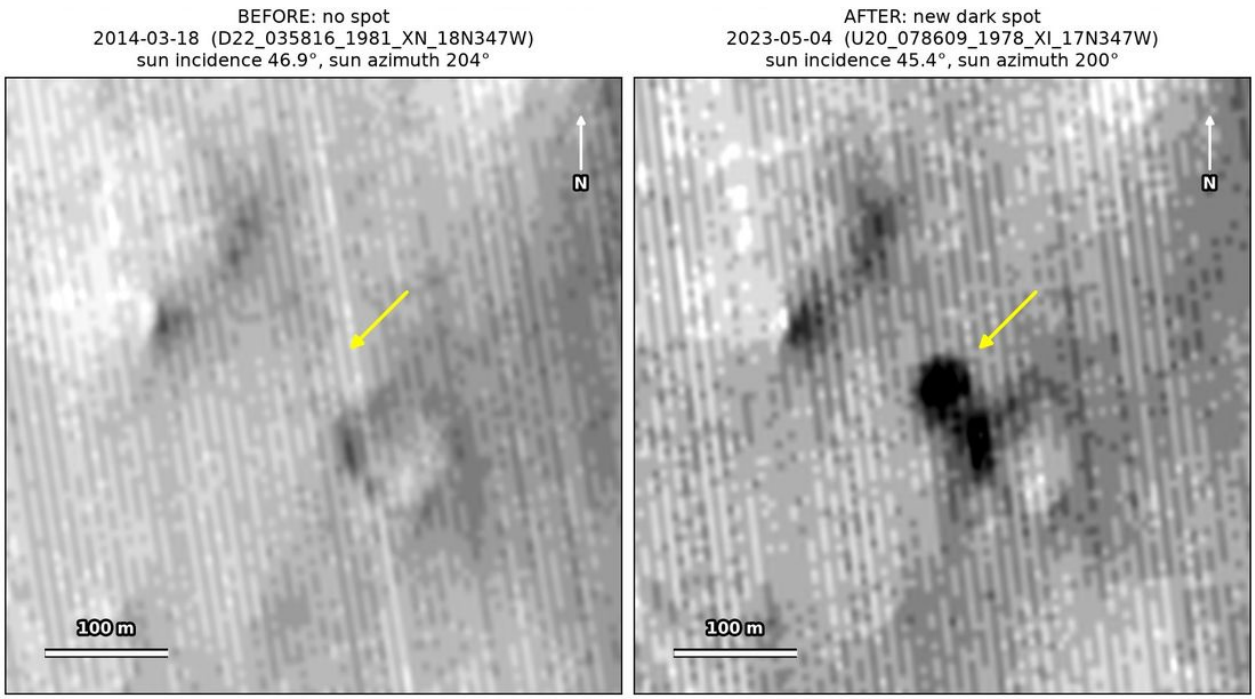

Figure 142: Candidate 57, before and after: the same 500 × 500 m of ground in 18 March 2014 (`D22_035816_1981_XN_18N347W`) and 4 May 2023 (`U20_078609_1978_XI_17N347W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

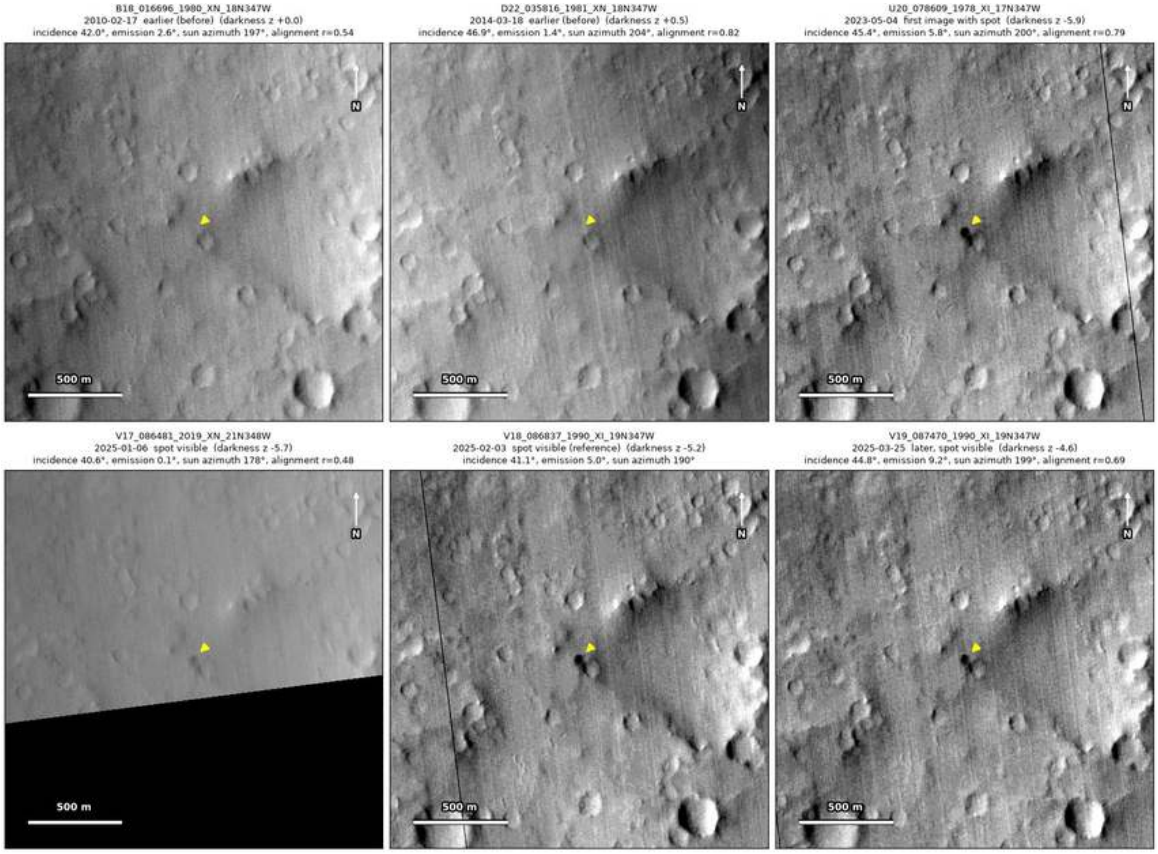

Figure 143: Candidate 57: six CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

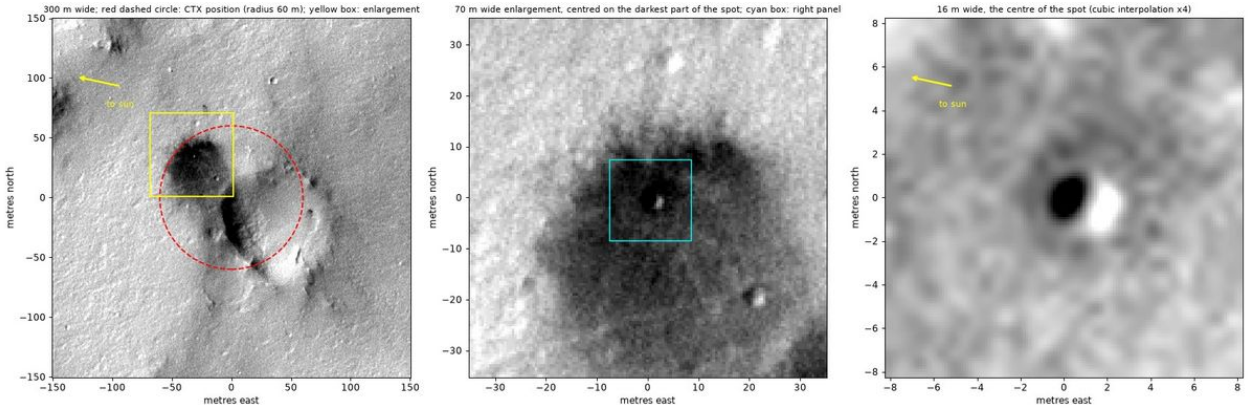

Figure 144: Candidate 57, HiRISE image `ESP_087470_1990`: HiRISE view of the spot (25 March 2025, 0.5 m per pixel), taken by chance on the same orbit as a CTX image: 300 m wide, then a 70 m enlargement, then the central 16 m. The dark patch lies beside the older crater, at its north-west edge, and a dark band extends about 70 m south-south-east from it along the crater's west rim; at the centre of the patch is a pit about 2.5 m wide, dark on the side towards the sun and bright on the far side. The red dashed circle marks the CTX position. NASA/JPL-Caltech/UArizona.

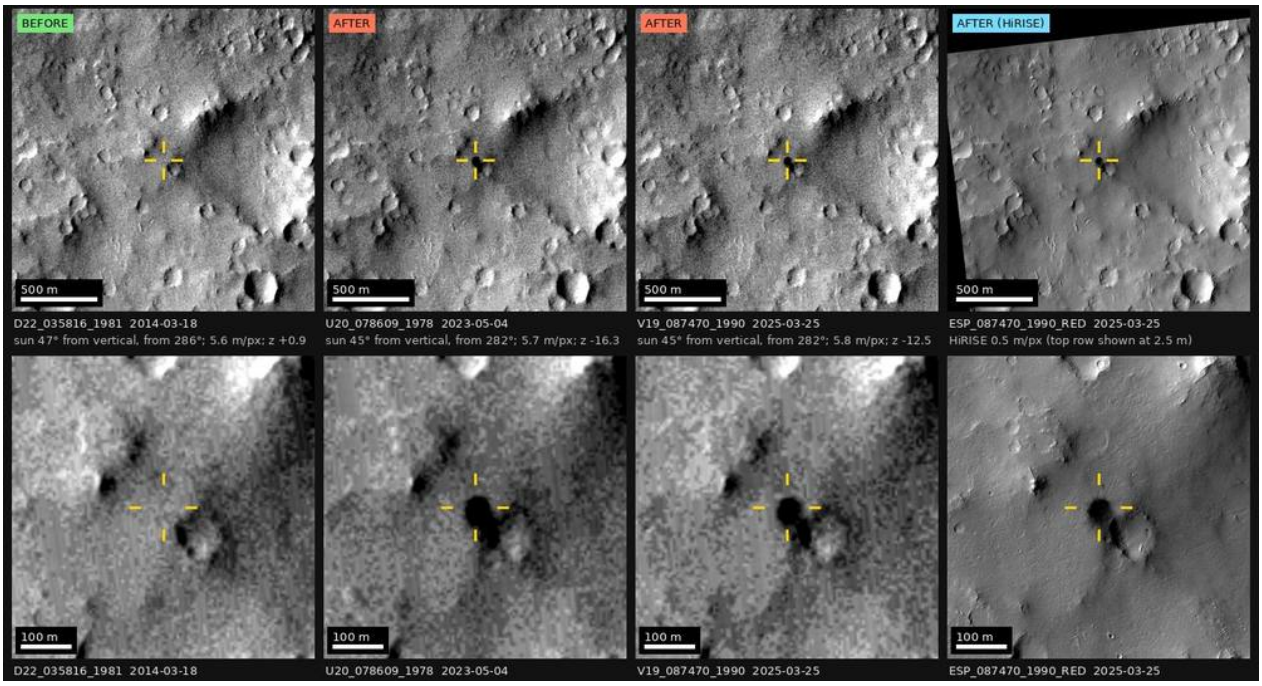

Figure 145: Candidate 57, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

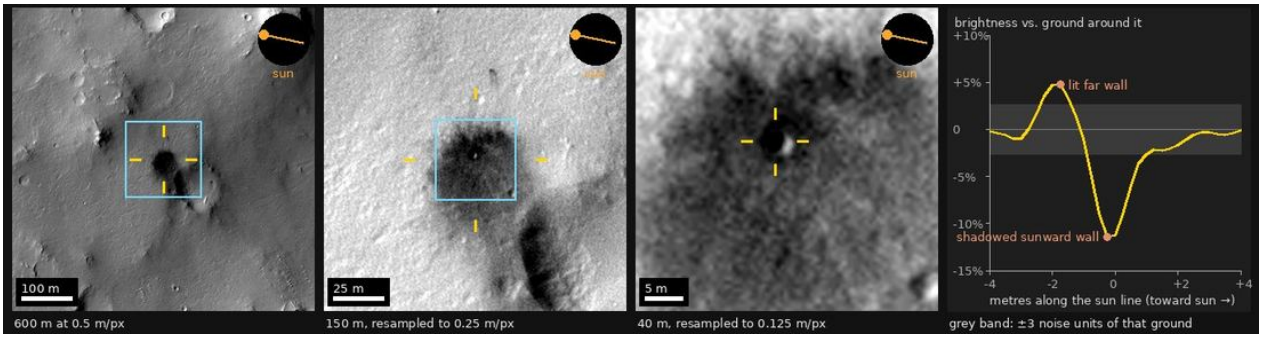

Figure 146: Candidate 57, HiRISE image `ESP_087470_1990`. The independent check: HiRISE close-ups, 600 m, 150 m and 40 m wide, and the brightness along the sun line through the darkest point: dark on the side towards the sun and bright on the far side, the shading of a small pit rather than a boulder. NASA/JPL-Caltech/UArizona.

Candidate 58: Lava plains beside a graben in central Ulysses Fossae, Tharsis

Location	11.9906° N, 236.7925° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	30 April 2010 and 9 October 2022
Images	1 without the spot (absent: 30 April 2010); 4 with it (present: 9 October 2022, 2 December 2022, 20 March 2025 and 19 May 2025)
Size	Compact, very dark spot with a core about 45 m across
HiRISE	none covers the spot (nearest footprint edge 18.9 km)
Nearest catalogued new impact	48 km (Daubar et al., 2022)

What changed. A compact, very dark spot about 45 m across appeared on smooth lava plains beside a graben (a long fault trough) in central Ulysses Fossae between April 2010 and October 2022, and is unchanged up to May 2025; four CTX images from four orbits show it.

Why it may be a new impact. An April 2010 image, taken under the same sun as the May 2025 image, shows nothing there, so the spot is not a shadow; a dark feature 230 m to the south-south-east shows clearly in it. A compact, very dark spot unchanged for more than two and a half years under four different suns is unlikely to be darker ground exposed by the wind, and is the typical mark of a small new impact.

Caveats. With only one earlier image, the spot could have formed at any time in a span of more than 12 years. No image from the high-resolution HiRISE camera covers it; the nearest ends 18.9 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 90% that the spot is a new impact. It placed the spot at 11.9922° N, 236.7905° E, 148 m from the position given above.

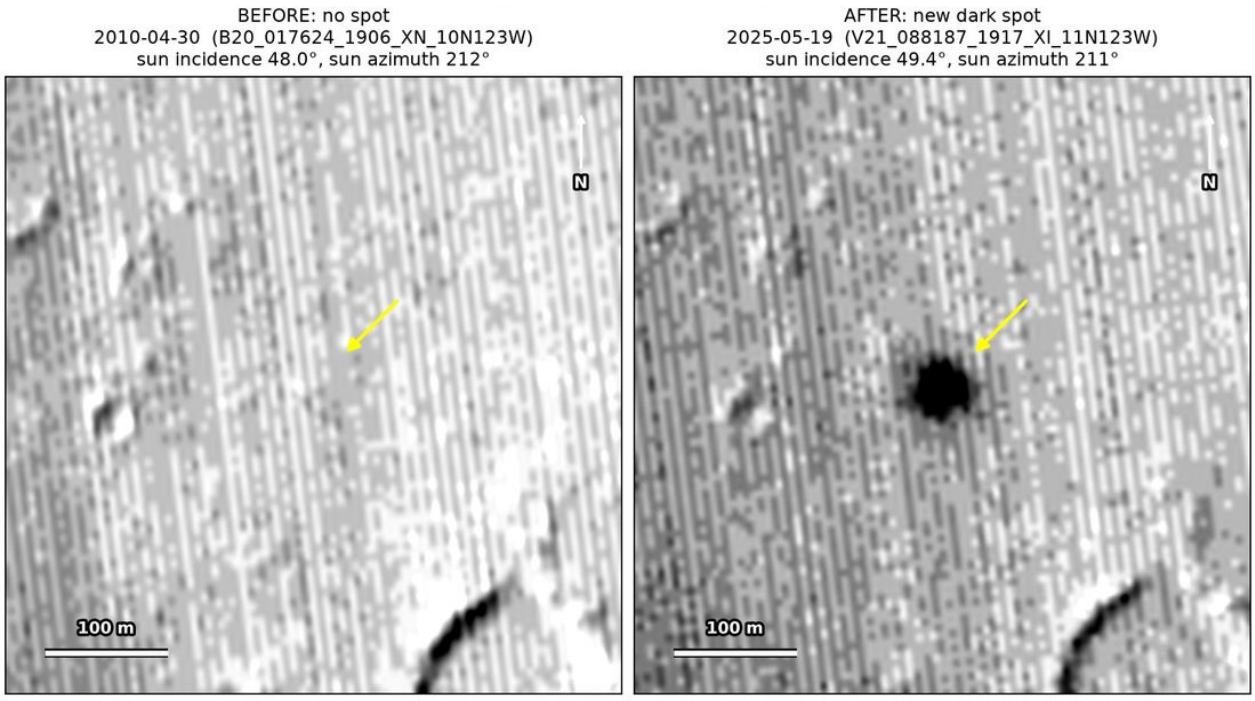

Figure 147: Candidate 58, before and after: the same 500 × 500 m of ground in 30 April 2010 (`B20_017624_1906_XN_10N123W`) and 19 May 2025 (`V21_088187_1917_XI_11N123W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

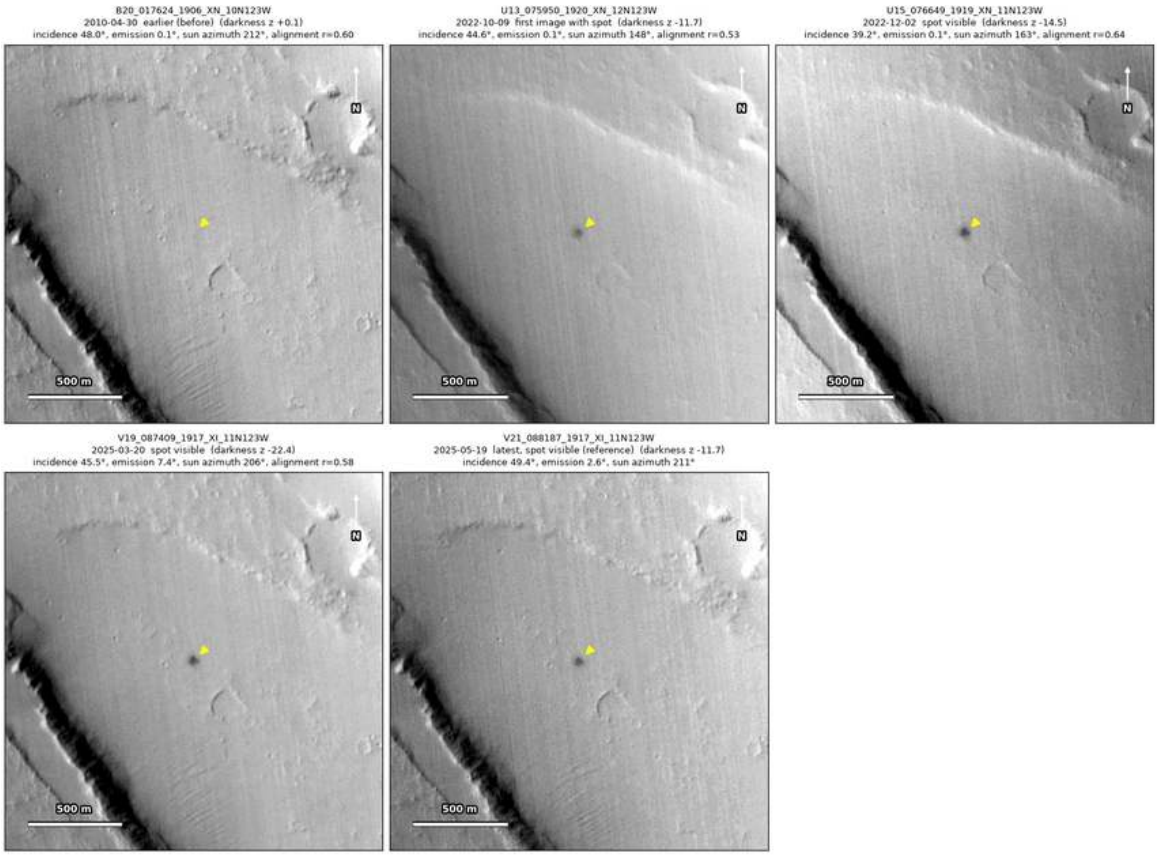

Figure 148: Candidate 58: five CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

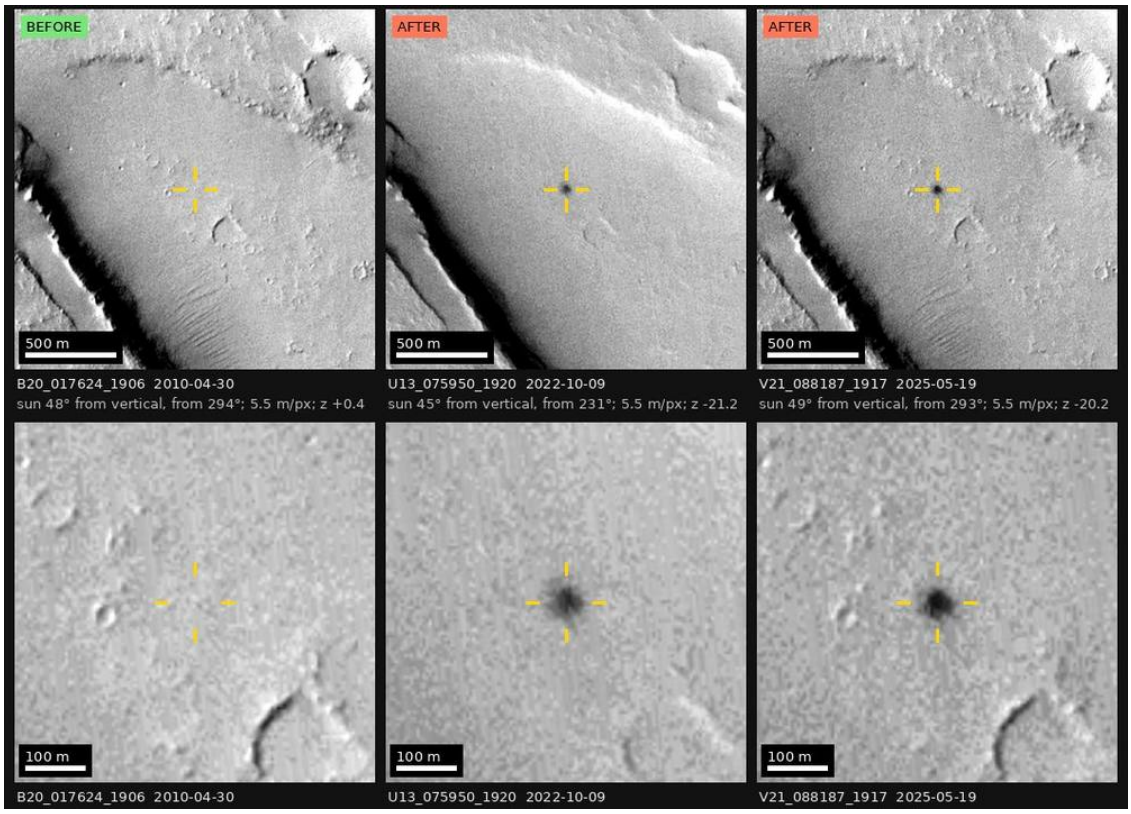

Figure 149: Candidate 58, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 59: Plains of Elysium Planitia near Cerberus Fossae, 190 km east of Zunil crater

Location	8.0391° N, 169.4077° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	28 January 2023 and 14 April 2025
Images	4 without the spot (absent: 23 April 2014, 6 February 2016, 14 April 2021 and 28 January 2023); 2 with it (present: 14 April 2025 and 6 July 2025)
Size	Cluster of four or five dark blobs within about 120 × 100 m, the largest about 50 m across
HiRISE	none covers the spot (nearest footprint edge 17.3 km)
Nearest catalogued new impact	141 km (Daubar et al., 2022)

What changed. A cluster of four or five dark blobs within about 120 × 100 m, the largest about 50 m across, appeared on the plains of Elysium Planitia, 190 km east of Zunil crater, between January 2023 and April 2025, and is still there in July 2025; two CTX images from two orbits show it.

Why it may be a new impact. Four earlier images, from 2014 to January 2023, show no cluster, while the small dark features around it show and line up in them; the April 2021 one was taken with the sun at the same height as in the April 2025 image. Several compact blobs appearing at once, with no dust devil track, are the usual signature of a meteoroid that broke up in the atmosphere.

Caveats. It has been seen for less than three months, so dark patches exposed by the wind are not ruled out, although several compact blobs appearing together make that unlikely. It formed after the end of InSight's mission, so it cannot be among the InSight-era impacts that Bickel et al. (2025) confirmed with HiRISE images. No image from the high-resolution HiRISE camera covers it; the nearest ends 17.3 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 90% that the spot is a new impact. It found six further earlier images of the site, all of which would have shown the cluster and none of which does. It placed the spot at 8.0426° N, 169.4178° E, 628 m from the position given above.

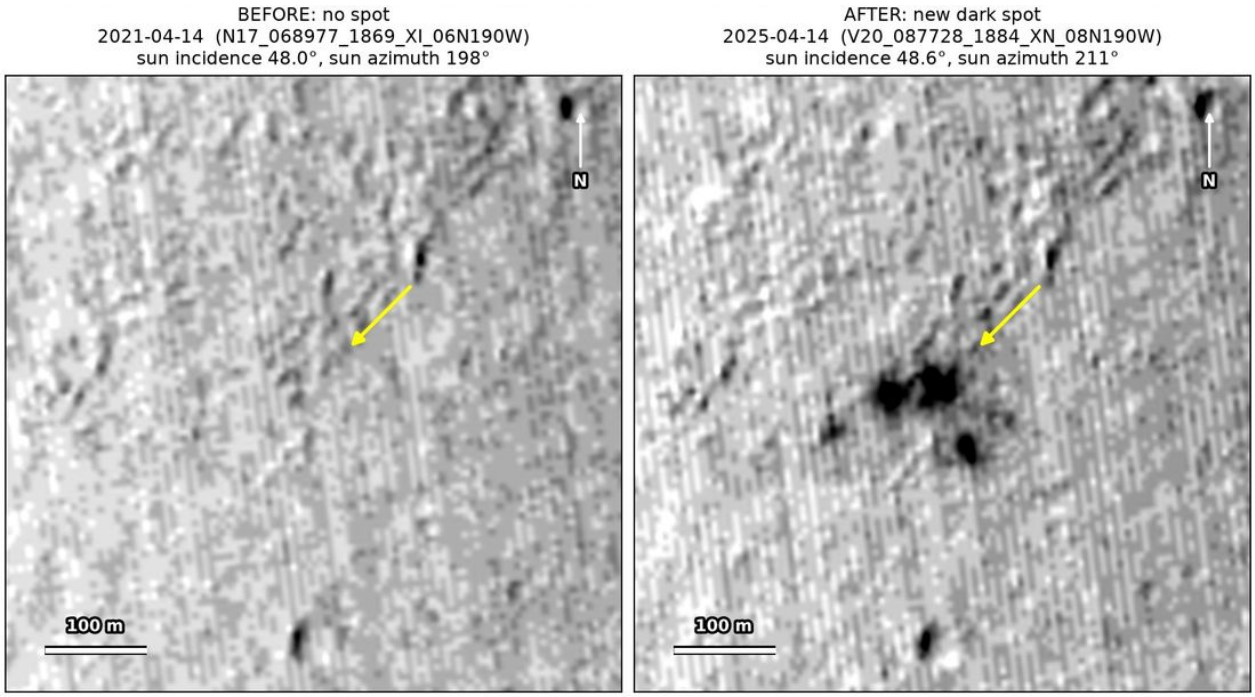

Figure 150: Candidate 59, before and after: the same 600 × 600 m of ground in 14 April 2021 (`N17_068977_1869_XI_06N190W`) and 14 April 2025 (`V20_087728_1884_XN_08N190W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

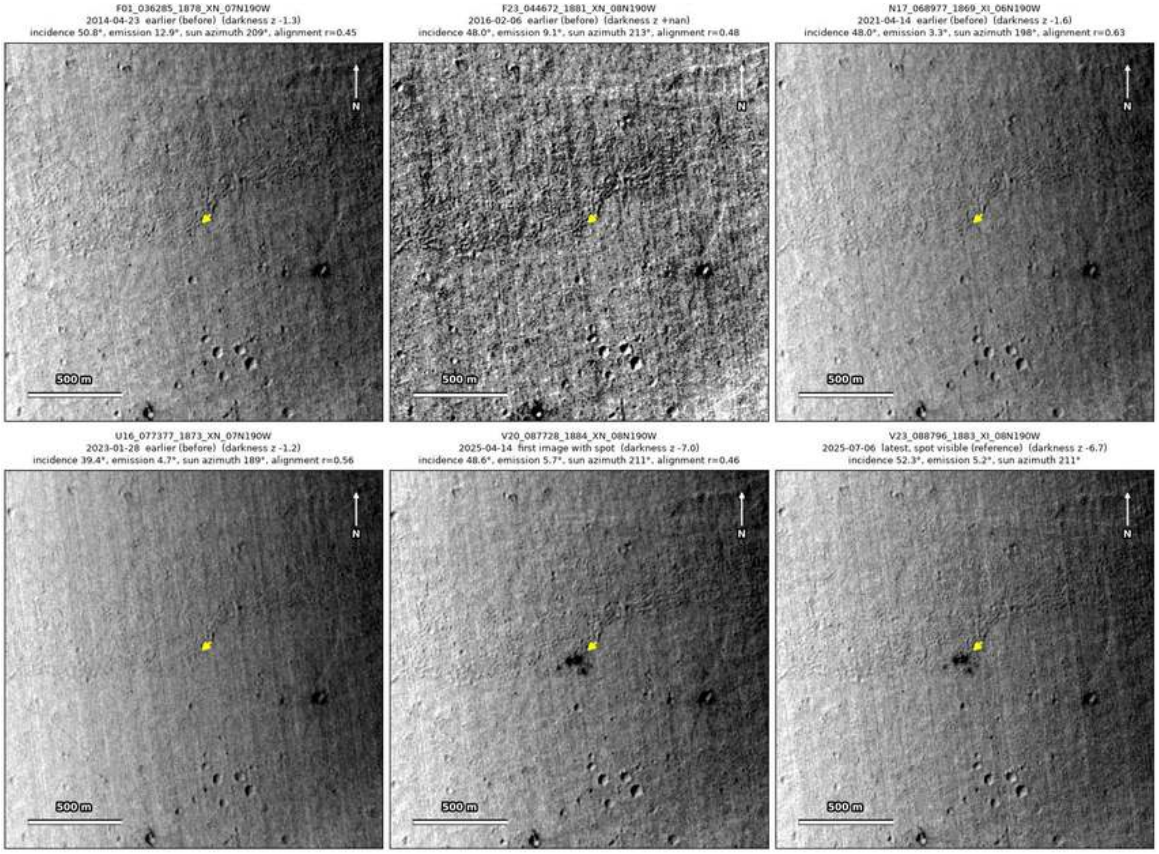

Figure 151: Candidate 59: six CTX images of the same 2×2 km patch, 2014 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

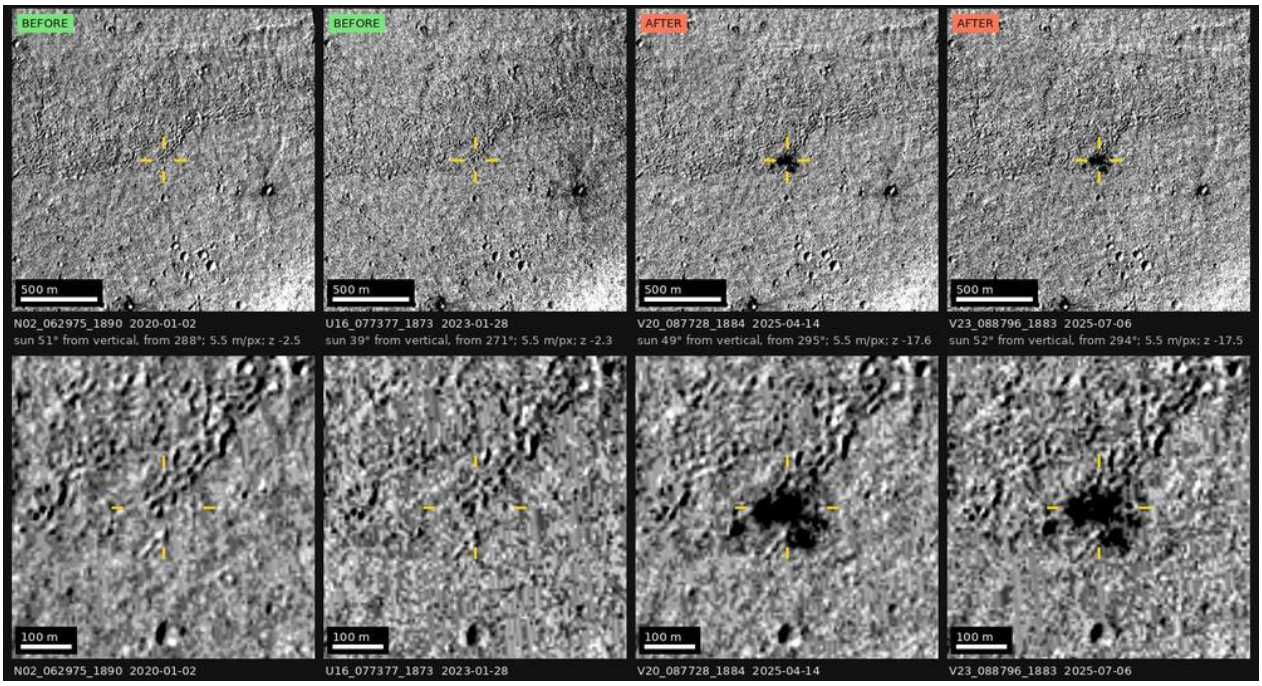

Figure 152: Candidate 59, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 60: Plains between the troughs of western Noctis Labyrinthus

Location	5.6411° S, 255.4212° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 70%
Formed between	11 April 2016 and 3 June 2025
Images	2 without the spot (absent: 11 July 2008 and 11 April 2016); 1 with it (present: 3 June 2025)
Size	Compact, very dark spot with a core about 45 m across, and a faint dark smudge about 60 m to its north-west
HiRISE	none covers the spot (nearest footprint edge 9.5 km)
Nearest catalogued new impact	37 km (Daubar et al., 2022)

What changed. Between April 2016 and June 2025, a compact, very dark spot with a core about 45 m across, and a faint dark smudge about 60 m to its north-west, appeared on the smooth plains between the troughs of western Noctis Labyrinthus; so far, one CTX image shows it.

Why it may be a new impact. Two earlier images, from July 2008 (under the same sun as the June 2025 image) and April 2016, show nothing there and line up well, with the knobs and a 500 m crater about 800 m to the south-west matching. Smooth, with a halo, and dark with no bright column or track, the spot is unlikely to be an image defect or a dust devil caught in the act, and may be the blast zone of a small new impact.

Caveats. Only one image shows it, and only a second image can rule out a short-lived feature. No image from the high-resolution HiRISE camera covers it; the nearest ends 9.5 km away.

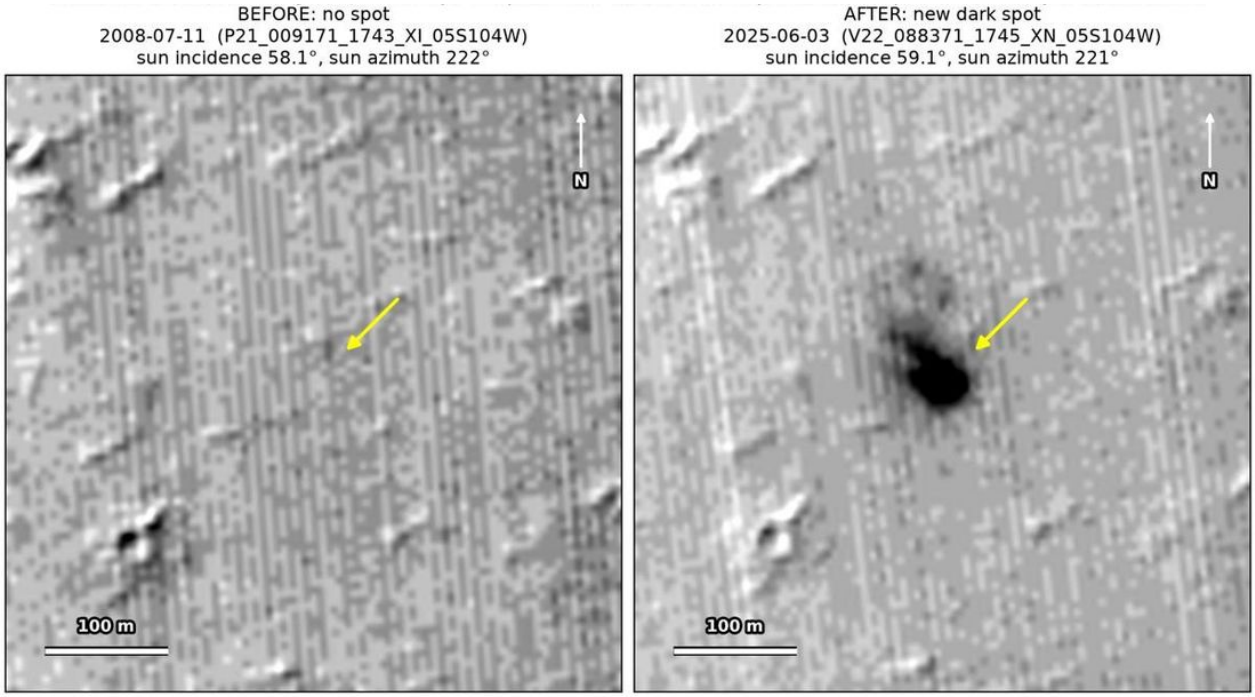

Figure 153: Candidate 60, before and after: the same 500 × 500 m of ground in 11 July 2008 (`P21_009171_1743_XI_05S104W`) and 3 June 2025 (`V22_088371_1745_XN_05S104W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

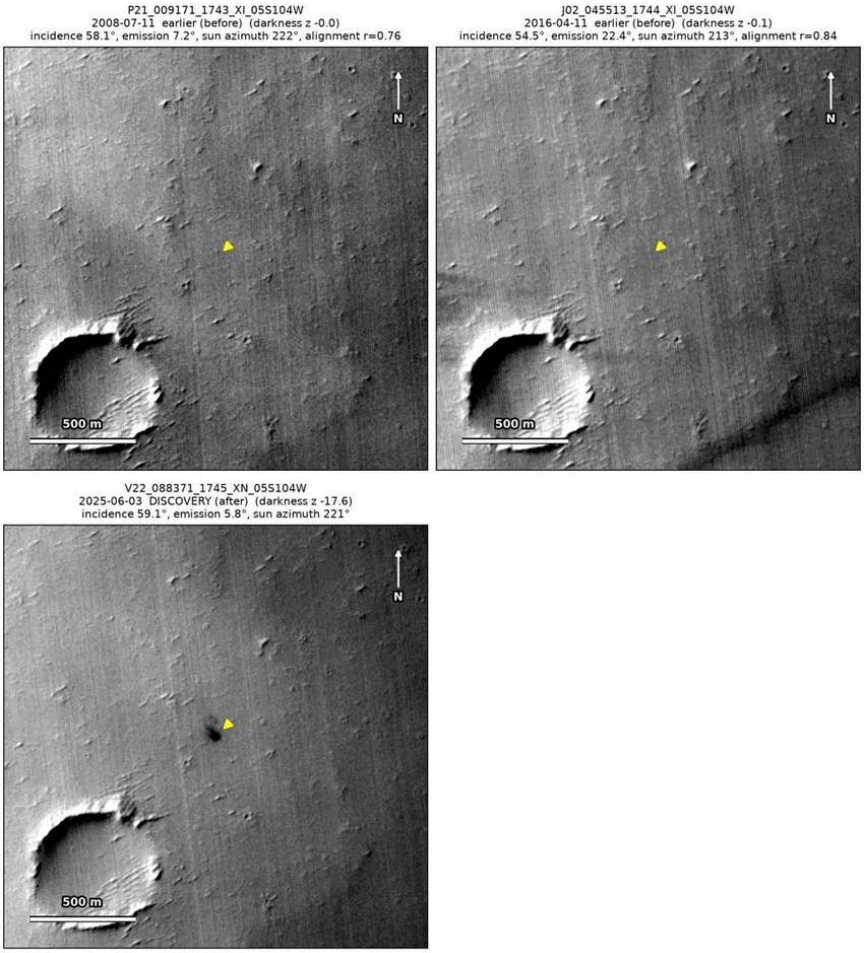

Figure 154: Candidate 60: three CTX images of the same 2×2 km patch, 2008 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 61: Arabia Terra, 26 km north-west of the rim of Janssen crater

Location	4.1341° N, 36.6353° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	11 March 2018 and 29 May 2025
Images	2 without the spot (absent: 5 March 2008 and 11 March 2018); 1 with it (present: 29 May 2025)
Size	Compact dark spot about 60 m across
HiRISE	none covers the spot (nearest footprint edge 22.6 km)
Nearest catalogued new impact	83 km (Daubar et al., 2022)

What changed. A compact dark spot about 60 m across appeared at the edge of a brighter, rougher area in Arabia Terra, 26 km north-west of the rim of Janssen crater, between March 2018 and May 2025; so far, one CTX image shows it.

Why it may be a new impact. Two earlier images, from March 2008 and March 2018, show nothing there, while a dark feature 225 m to the north shows in both; the March 2018 one was taken under a sun close to that of the May 2025 image. The spot is round rather than a tapering streak, and may be the blast zone of a small new impact.

Caveats. Only one image shows it, so an image defect is not ruled out. It sits at a low scarp at the edge of the brighter ground, so it could be a slope streak (a dark streak left by dust sliding down a slope), and in the seven years since the March 2018 image a dust devil or the wind could have uncovered darker ground. No image from the high-resolution HiRISE camera covers it; the nearest ends 22.6 km away.

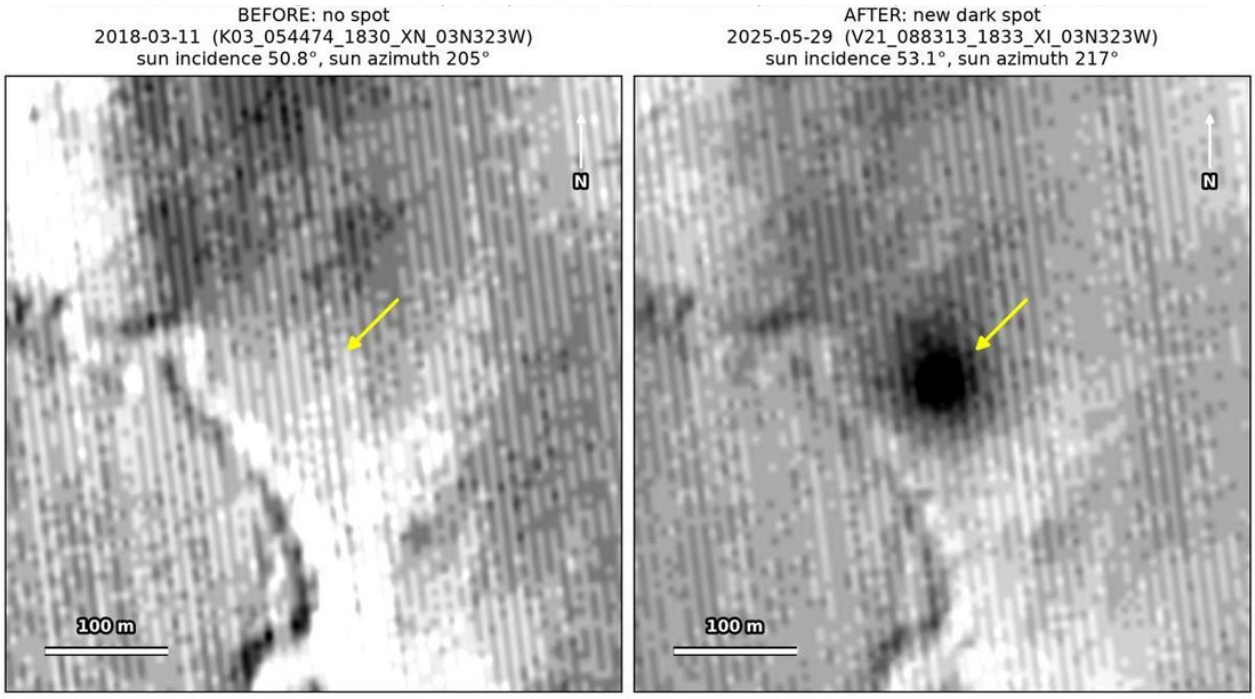

Figure 155: Candidate 61, before and after: the same 500 × 500 m of ground in 11 March 2018 (`K03_054474_1830_XN_03N323W`) and 29 May 2025 (`V21_088313_1833_XI_03N323W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

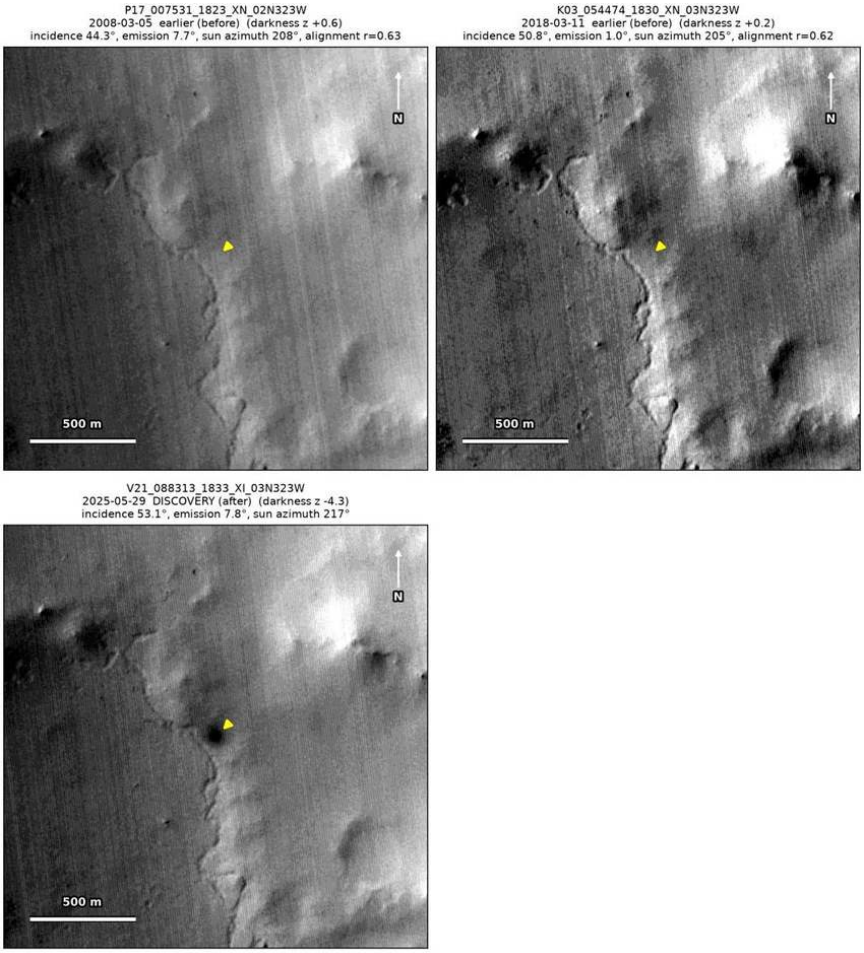

Figure 156: Candidate 61: three CTX images of the same 2×2 km patch, 2008 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 62: Knobby plains of Tharsis, 315 km south of Pavonis Mons

Location	3.7684° S, 247.8767° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	26 March 2009 and 6 February 2025
Images	1 without the spot (absent: 26 March 2009); 2 with it (present: 6 February 2025 and 11 February 2025)
Size	Dark core about 40 to 50 m across, with faint rays and a diffuse dark zone about 400 m across
HiRISE	none covers the spot (nearest footprint edge 23.9 km)
Nearest catalogued new impact	73 km (Daubar et al., 2022)

What changed. A compact dark spot with a core about 40 to 50 m across, faint dark rays and a diffuse dark zone appeared among the knobs of the Tharsis plains, 315 km south of Pavonis Mons, between March 2009 and February 2025. Two CTX images from two orbits, five days apart, show it.

Why it may be a new impact. The March 2009 image shows only a faint mark there, while a knob 300 m to the east shows in it. The two February 2025 images, taken five days apart under the same sun, both show the spot, and a fresh impact after the February 2023 image fits its faint rays best.

Caveats. A poor February 2023 image, oblique and heavily compressed, shows only a faint dark blob about 25 m across at the same place, so the spot may have formed before 2023 and appear only faintly in it, or wind may have darkened and enlarged a small older dark feature. The March 2009 image lines up poorly and has the sun from a different direction. No image from the high-resolution HiRISE camera covers it; the nearest ends 23.9 km away.

Figure note. This figure was made for this paper (Section 3.9) from the map-projected versions of the three images. It includes the poor February 2023 image, in which a faint, smaller dark mark is visible at the place, because a two-panel "before" and "after" comparison would overstate the case. In all three map-projected images, which agree with one another to within about 100 m, the spot lies 1.47 km from the catalogued position, at 3.7733° S, 247.8524° E; the catalogued coordinate is therefore probably in error by about 1.5 km, and follow-up imaging should cover both positions.

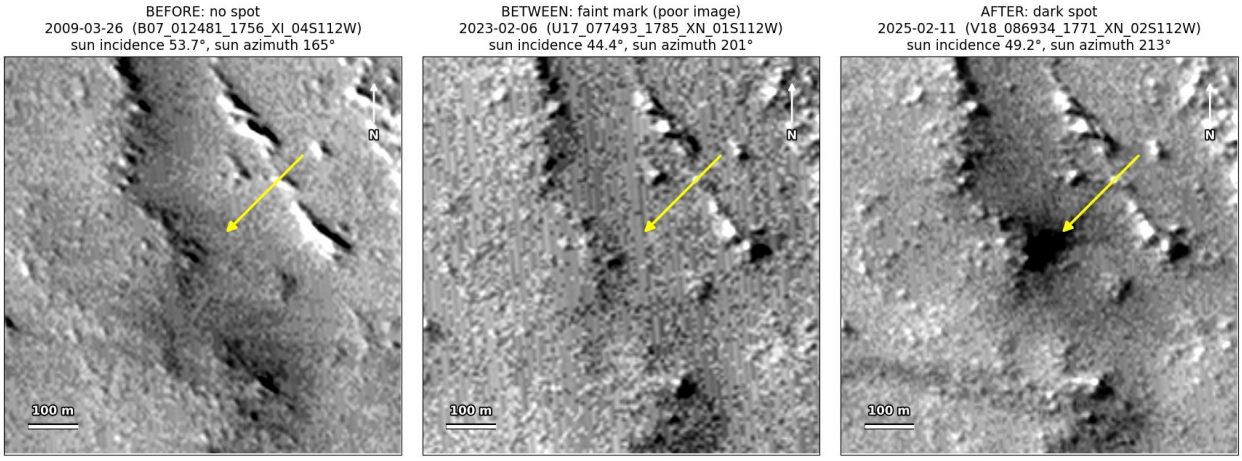

Figure 157: Candidate 62, before and after: the same 800 × 800 m of ground in 26 March 2009 (`B07_012481_1756_XI_04S112W`), 6 February 2023 (`U17_077493_1785_XN_01S112W`) and 11 February 2025 (`V18_086934_1771_XN_02S112W`), under the suns given above the panels; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS. Made for this paper from map-projected images distributed by Arizona State University.

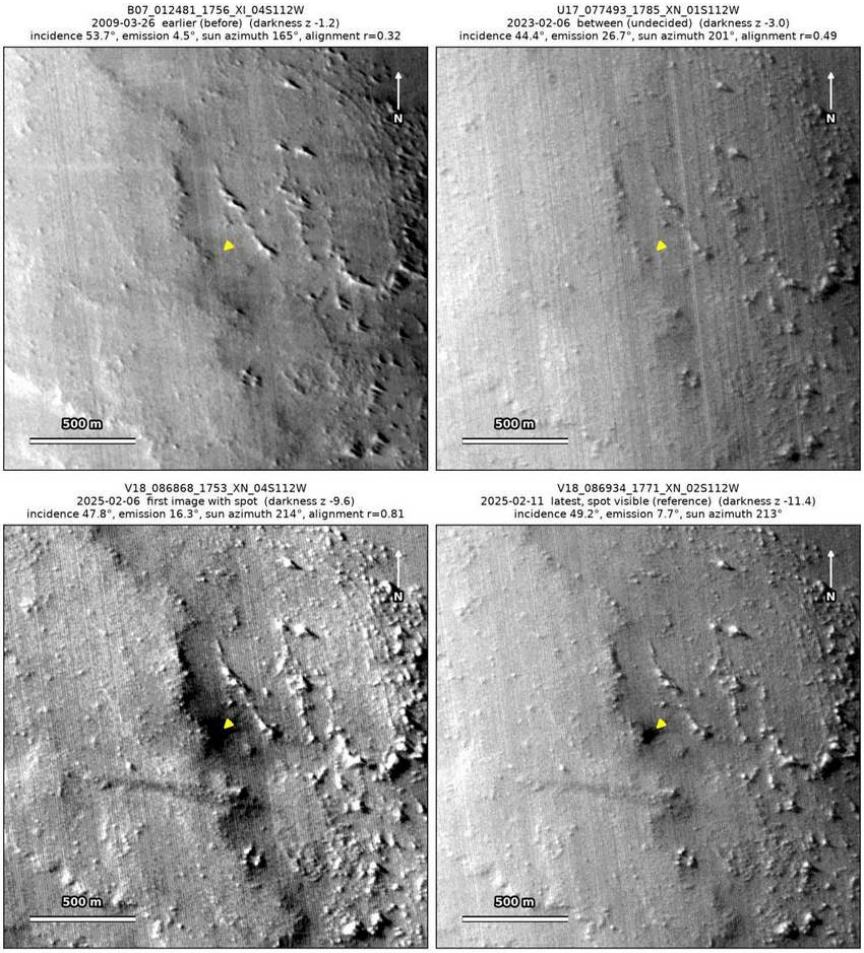

Figure 158: Candidate 62: four CTX images of the same 2×2 km patch, 2009 to 2025. The poor February 2023 image, which shows only a faint, smaller dark blob at the same place, is shown but not used to date the spot. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 63: Smooth plains at the western rim of Echus Chasma

Location	2.5252° N, 278.7758° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	25 July 2016 and 29 March 2023
Images	3 without the spot (absent: 22 December 2006, 12 October 2012 and 25 July 2016); 2 with it (present: 29 March 2023 and 12 February 2025)
Size	Compact, very dark spot about 47 m across (60 × 50 m in the February 2025 image; larger in the March 2023 one)
HiRISE	none covers the spot (nearest footprint edge 3.3 km)
Nearest catalogued new impact	63 km (Daubar et al., 2022)
Nearby	Candidate 75, 16.5 km to the north, appears in the same pair of images and first shows in the same March 2023 image.

What changed. A compact, very dark spot about 47 m across appeared on the smooth plains 2 km west of the rim of Echus Chasma between July 2016 and March 2023, and is still there, smaller and fainter, in February 2025; two CTX images from two orbits show it.

Why it may be a new impact. Images from 2006, 2012 and 2016, all taken under a lower sun, show plain ground there. The March 2023 and February 2025 images, from two orbits, show the spot fading, and a compact spot fading over two years is the usual look of a fresh blast zone. The wall of Echus Chasma has slope streaks, dark streaks left by dust sliding down a slope, but they are long and thin, and this spot lies on level ground 2 km from the rim.

Caveats. No comparable dark feature lies nearby to confirm that the earlier images would have shown a spot like this one. No image from the high-resolution HiRISE camera covers it; the nearest ends 3.3 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 90% that the spot is a new impact. It placed the spot at 2.5240° N, 278.7785° E, 174 m from the position given above.

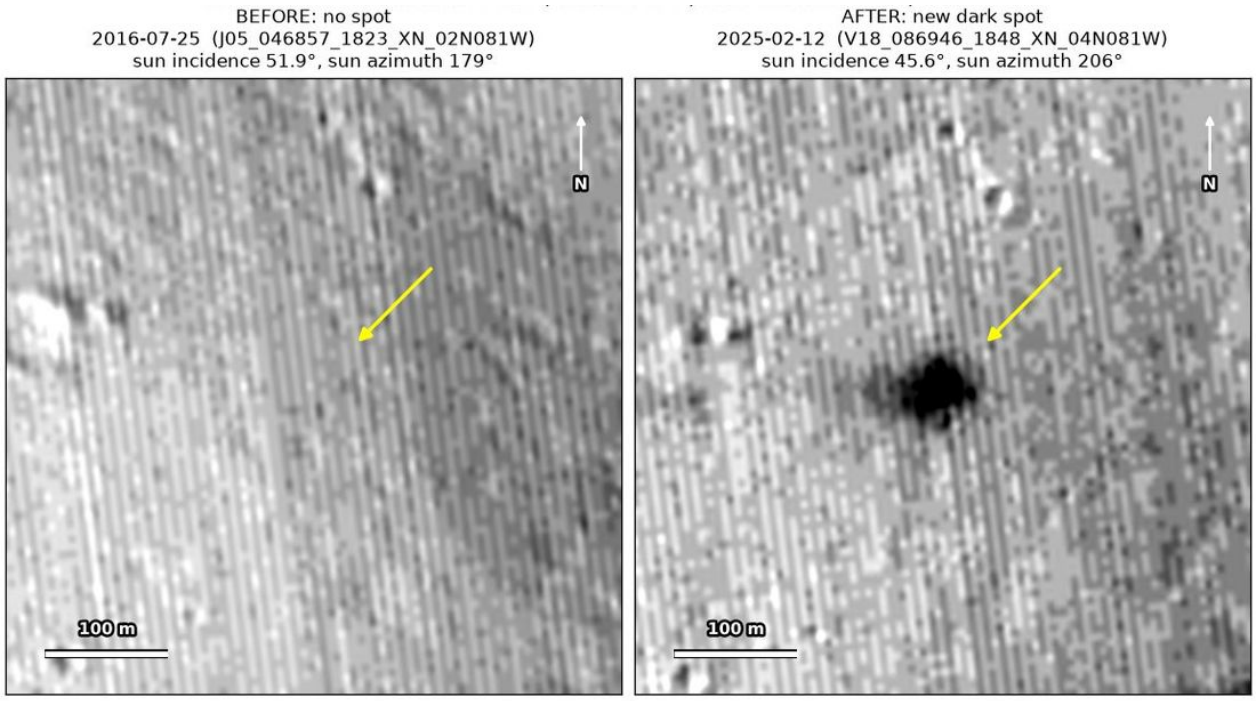

Figure 159: Candidate 63, before and after: the same 500 × 500 m of ground in 25 July 2016 (`J05_046857_1823_XN_02N081W`) and 12 February 2025 (`V18_086946_1848_XN_04N081W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

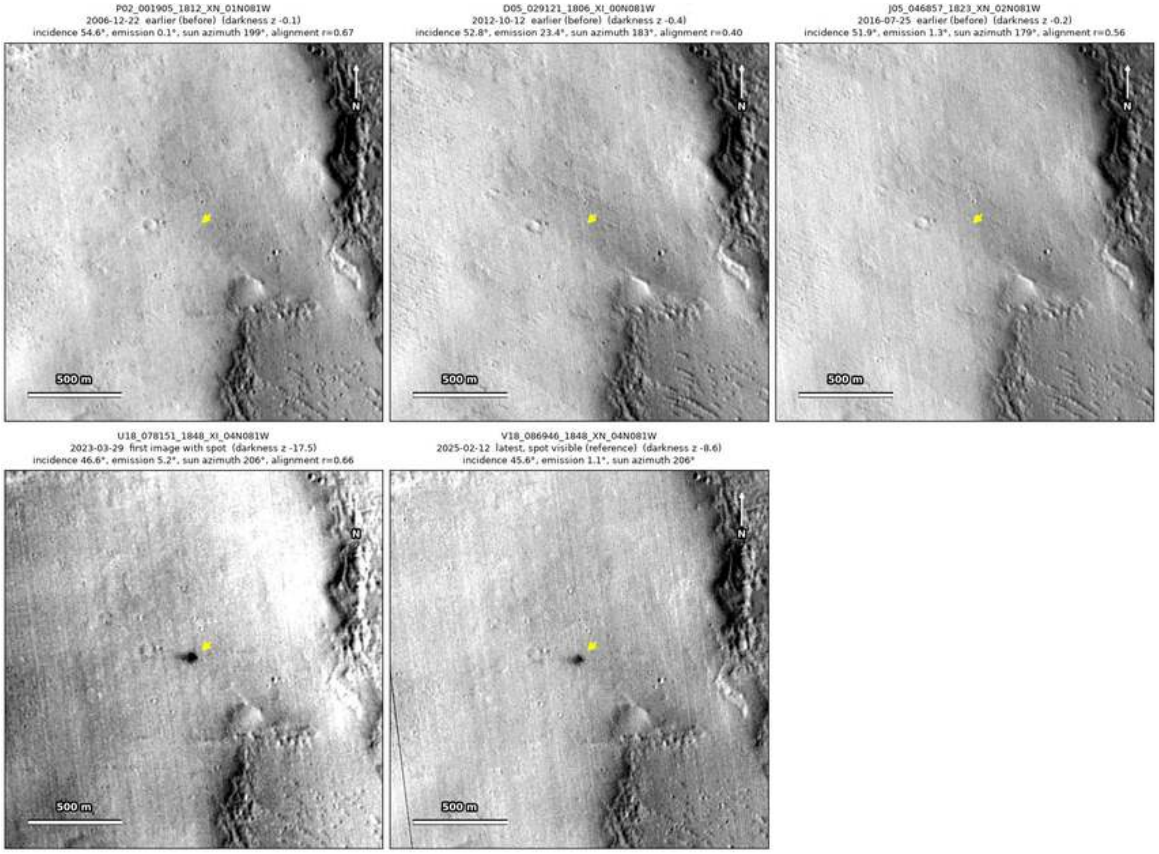

Figure 160: Candidate 63: five CTX images of the same 2×2 km patch, 2006 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

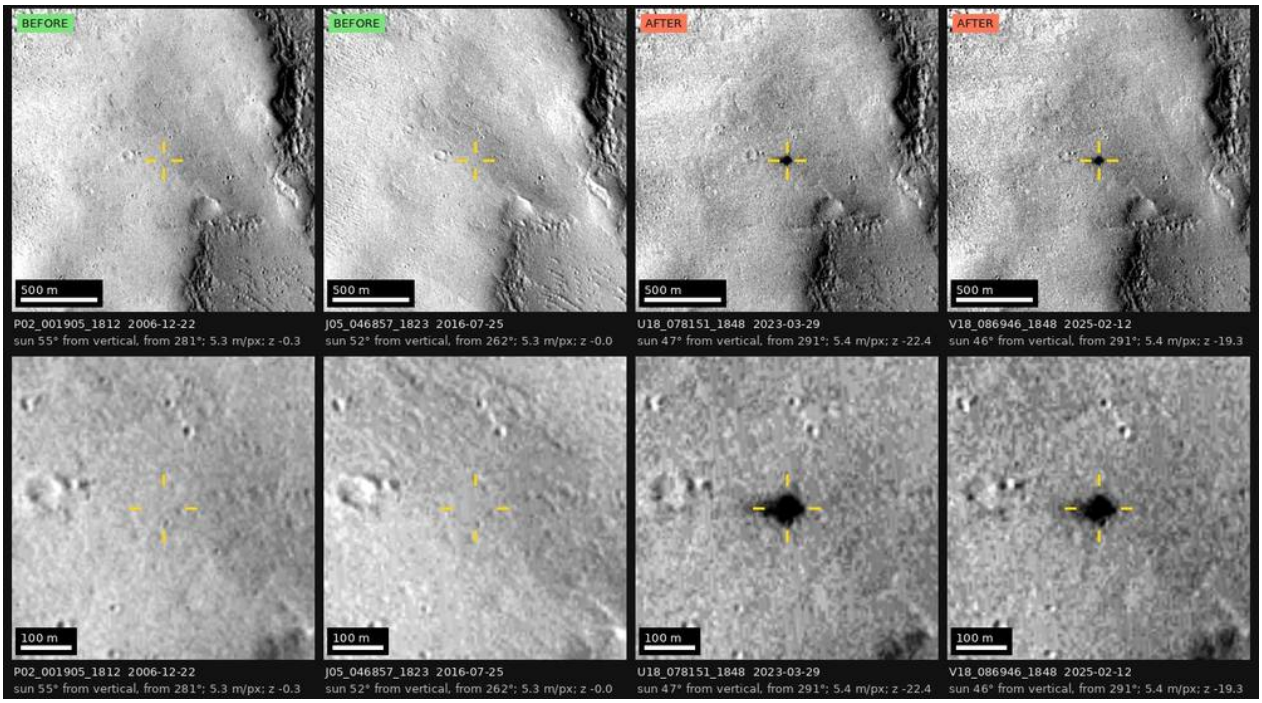

Figure 161: Candidate 63, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 64: Rippled plains 212 km south-west of Biblis Tholus

Location	0.6852° S, 234.0286° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	30 May 2012 and 14 July 2023
Images	1 without the spot (absent: 30 May 2012); 2 with it (present: 14 July 2023 and 21 April 2025)
Size	Compact dark spot about 26 m across (30 × 28 m)
HiRISE	none covers the spot (nearest footprint edge 31.0 km)
Nearest catalogued new impact	72 km (Daubar et al., 2022)

What changed. A compact dark spot about 26 m across appeared on wind-rippled lava plains 212 km south-west of Biblis Tholus between May 2012 and July 2023. It shows in two CTX images from two orbits and is still there in April 2025.

Why it may be a new impact. A May 2012 image, taken under the same sun as the April 2025 one and well aligned with it, shows plain ground at the site, yet a dark mark 220 m to the west does show in it, so a new spot would have shown too. The spot is round, isolated on a plain of uniform ripples and unchanged over 21 months, which is how the blast zone of a small new crater looks.

Caveats. Only one earlier image could have shown the spot, so it may have formed at any time in an 11-year span. Dark sand gathered between the ripples is an unlikely alternative; a dust devil is possible, although no track shows. No image from the high-resolution HiRISE camera covers it; the nearest ends 31.0 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 80% that the spot is a new impact. Because the spot is small and only moderately dark on rippled ground, it kept dark sand gathered in ripple troughs as the main alternative and estimated the probability at about 80%. It placed the spot at 0.6833° S, 234.0278° E, 121 m from the position given above.

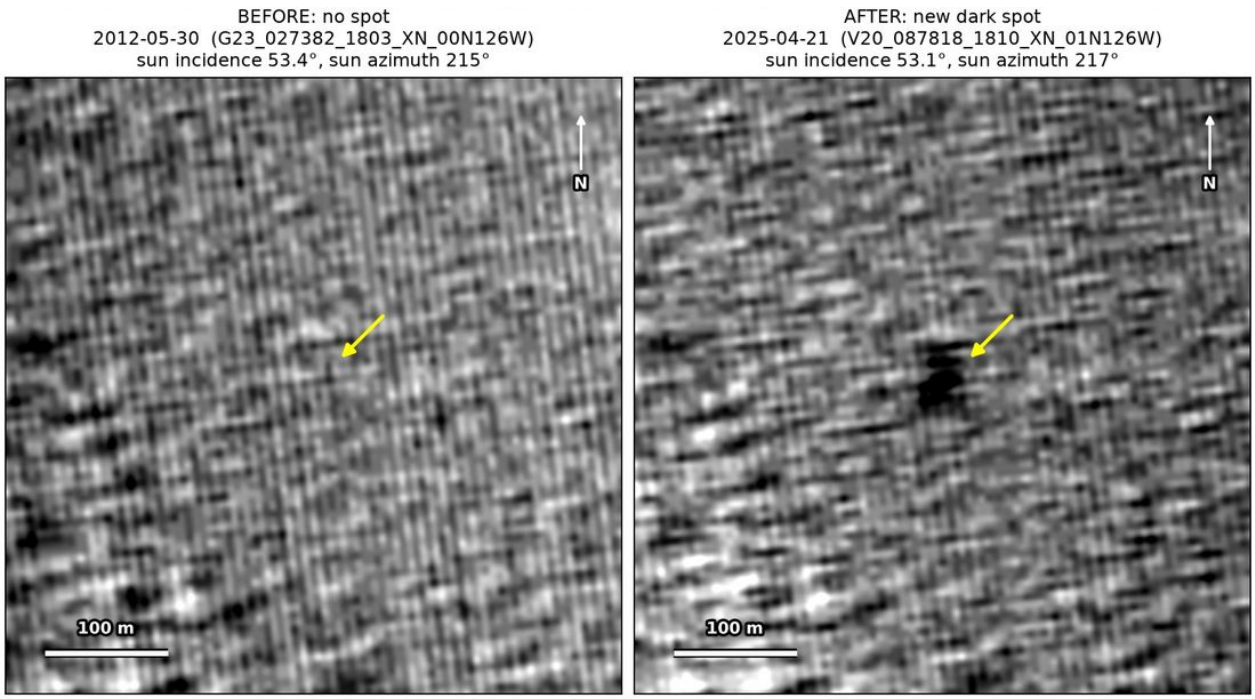

Figure 162: Candidate 64, before and after: the same 500 × 500 m of ground in 30 May 2012 (`G23_027382_1803_XN_00N126W`) and 21 April 2025 (`V20_087818_1810_XN_01N126W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

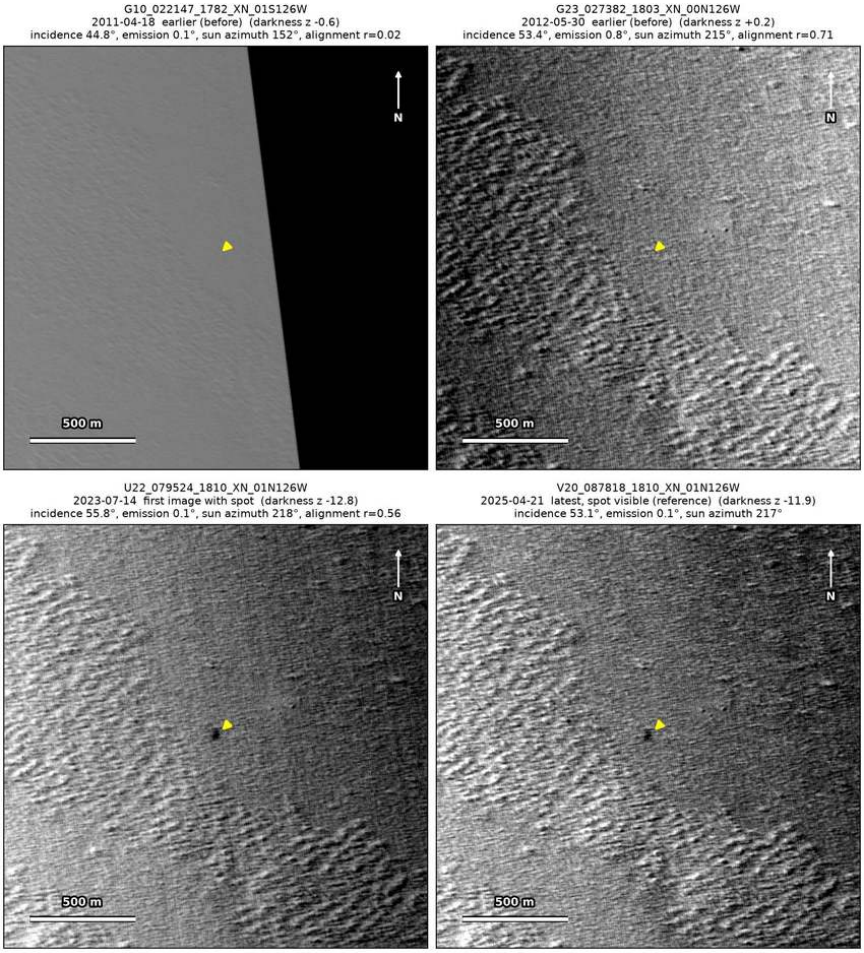

Figure 163: Candidate 64: four CTX images of the same 2×2 km patch, 2011 to 2025. The April 2011 image shows no surface detail at the site and is not used to date the spot. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

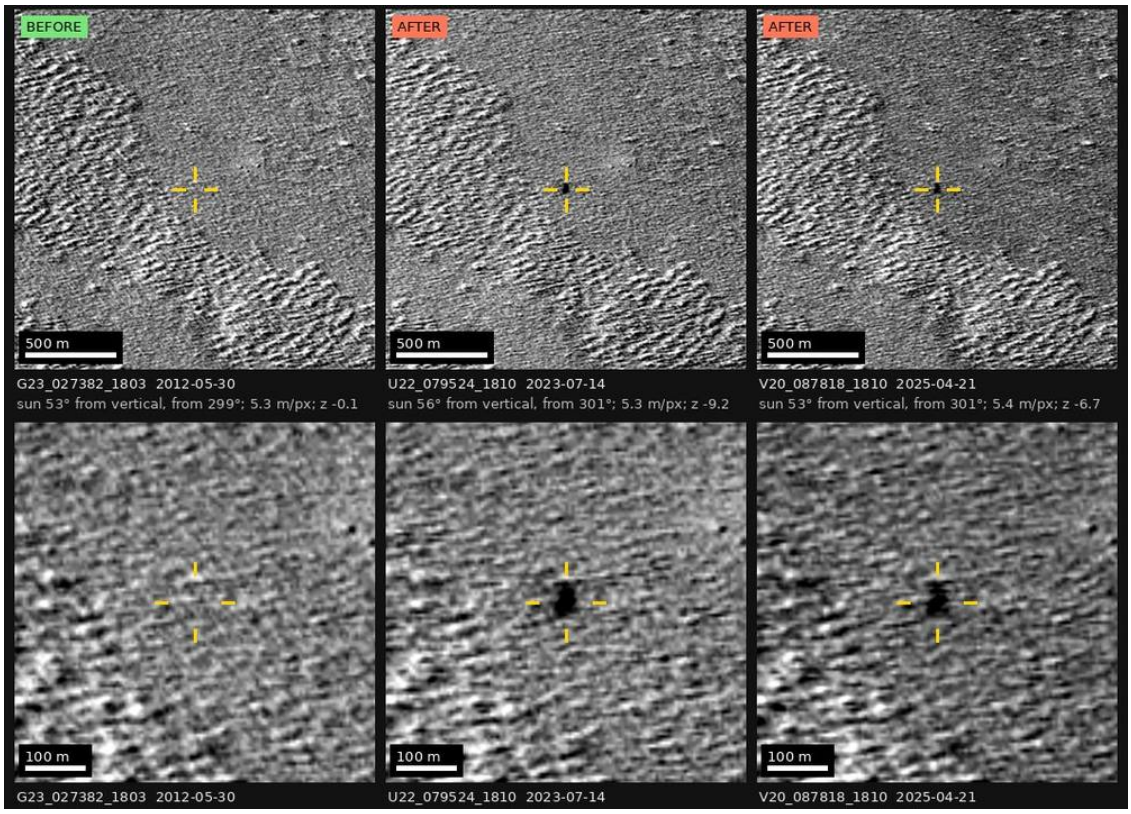

Figure 164: Candidate 64, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 65: Lava plains at the south-east margin of Ceraunius Fossae, 525 km north of Ascraeus Mons

Location	20.4810° N, 253.4915° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 70%
Formed between	13 July 2023 and 20 April 2025
Images	3 without the spot (absent: 22 July 2013, 1 December 2013 and 13 July 2023); 1 with it (present: 20 April 2025)
Size	Compact dark spot about 34 m across (44 × 32 m)
HiRISE	none covers the spot (nearest footprint edge 14.9 km)
Nearest catalogued new impact	75 km (Daubar et al., 2022)

What changed. A compact dark spot about 34 m across appeared beside a small knob on cratered lava plains at the south-east margin of Ceraunius Fossae between July 2023 and April 2025. It shows in one CTX image so far.

Why it may be a new impact. Three earlier images, from 2013 and July 2023, show no spot there. The July 2023 image was taken under the same sun as the April 2025 one and lines up well with it, and a shadowed crater 300 m to the east-north-east shows in both. The spot lies on the knob's sunward side, so it is not the knob's shadow, and a compact dark spot appearing on lava plains is how the blast zone of a small new crater can look.

Caveats. Only one image shows it so far, and only a second image can rule out a short-lived mark: a dust devil caught in the act, or ground freshly scoured by a dust devil, is possible, though there is no track. Dark material exposed by wind on the knob's flank is also possible. No image from the high-resolution HiRISE camera covers it; the nearest ends 14.9 km away.

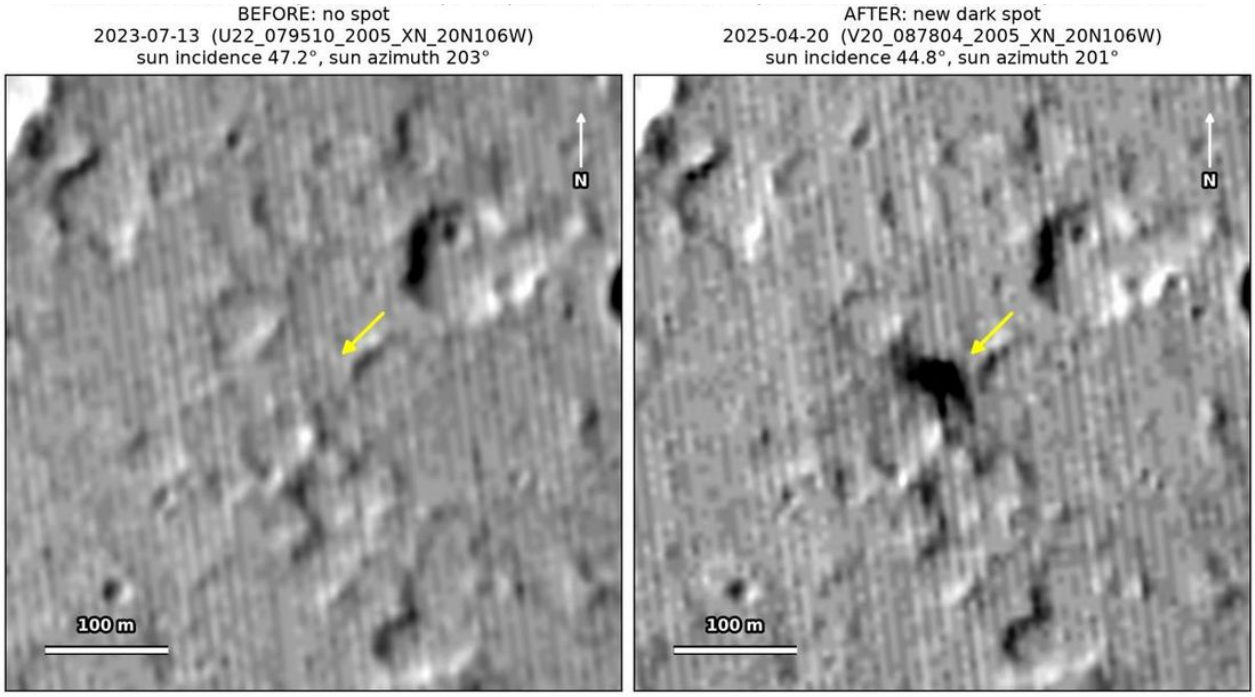

Figure 165: Candidate 65, before and after: the same 500 × 500 m of ground in 13 July 2023 (`U22_079510_2005_XN_20N106W`) and 20 April 2025 (`V20_087804_2005_XN_20N106W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

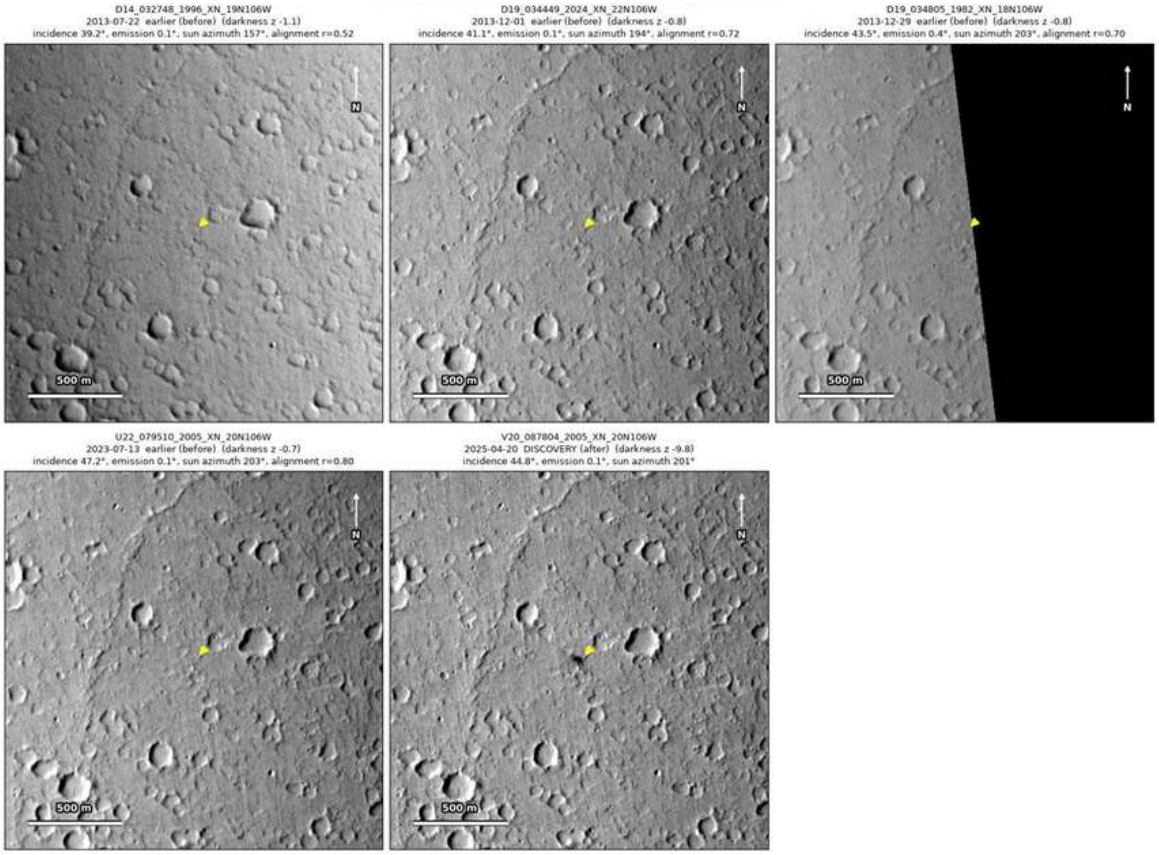

Figure 166: Candidate 65: five CTX images of the same 2×2 km patch, 2013 to 2025. The image of 29 December 2013, whose edge falls at the spot, is shown but not used to date it. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 66: Knob chain in the Fortuna Fossae region, 400 km south-south-west of Tharsis Tholus

Location	6.9479° N, 266.7737° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 70%
Formed between	9 January 2012 and 29 January 2025
Images	2 without the spot (absent: 29 May 2008 and 9 January 2012); 1 with it (present: 29 January 2025)
Size	Very dark spot about 42 m across (46 × 62 m), with a faint halo
HiRISE	none covers the spot (nearest footprint edge 54.4 km)
Nearest catalogued new impact	140 km (Daubar et al., 2022)

What changed. A very dark spot about 42 m across, with a faint halo, appeared beside a small knob at the end of a chain of knobs in the Fortuna Fossae region between January 2012 and January 2025. It shows in one CTX image so far.

Why it may be a new impact. Two earlier images, from 2008 and January 2012, show the knob but no spot, and the 2012 image was taken under the same sun as the January 2025 one. The spot lies on the knob's south-west side, the side facing the sun, so it is not the knob's shadow, and a very dark new spot with a faint halo is how the blast zone of a small new crater can look.

Caveats. Only one image shows it so far, and dark material shed off the knob is possible. A knob shadow 210 m to the east-north-east would show that the earlier images could have caught a new spot, but it is only weakly dark, so on its own it is not conclusive. No image from the high-resolution HiRISE camera covers it; the nearest ends 54.4 km away.

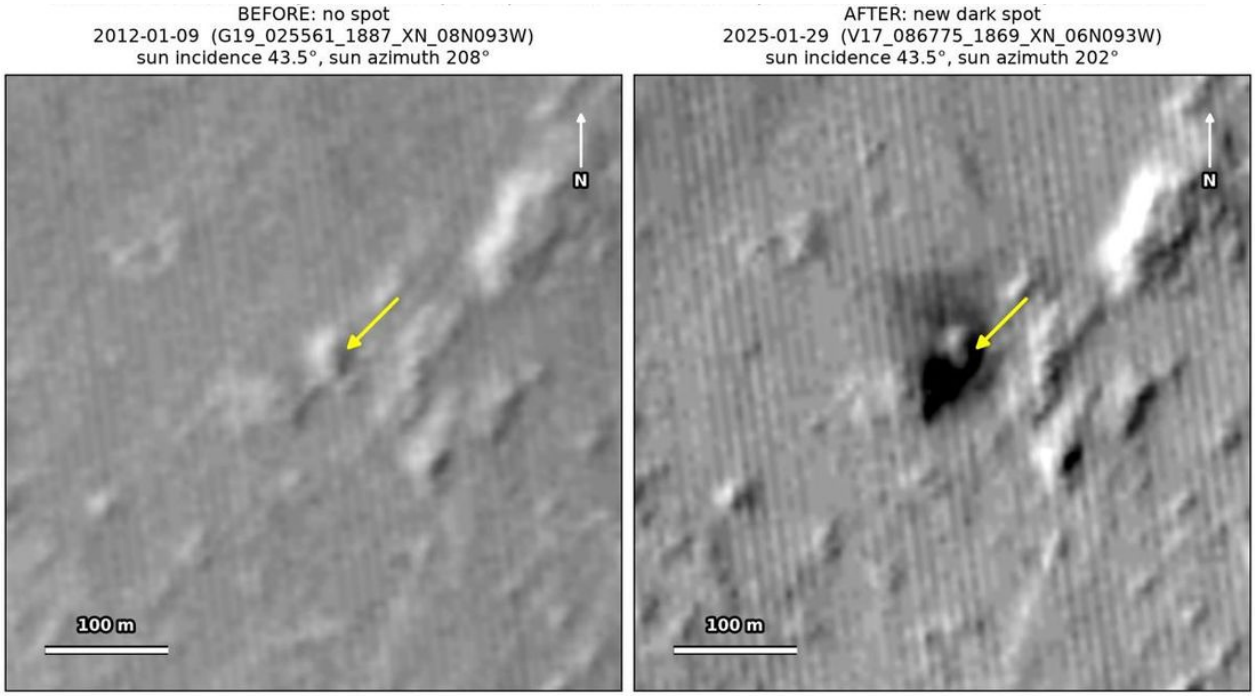

Figure 167: Candidate 66, before and after: the same 500 × 500 m of ground in 9 January 2012 (`G19_025561_1887_XN_08N093W`) and 29 January 2025 (`V17_086775_1869_XN_06N093W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

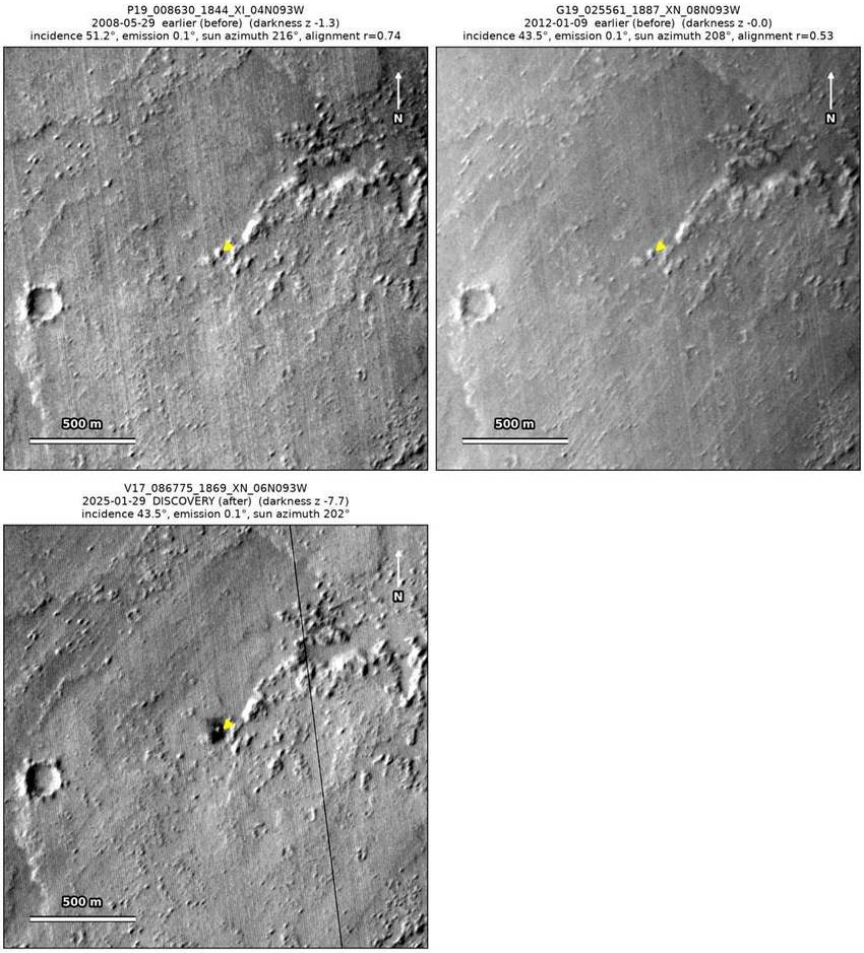

Figure 168: Candidate 66: three CTX images of the same 2×2 km patch, 2008 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 67: Cratered plains in the Memnonia region, 430 km east of Memnonia Sulci

Location	9.2688° S, 191.2269° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	29 January 2014 and 3 April 2023
Images	3 without the spot (absent: 2 March 2010, 27 November 2011 and 29 January 2014); 2 with it (present: 3 April 2023 and 16 February 2025)
Size	Compact, round dark spot about 34 m across (34 × 36 m)
HiRISE	none covers the spot (nearest footprint edge 49.8 km)
Nearest catalogued new impact	56 km (Daubar et al., 2022)

What changed. A compact, round dark spot about 34 m across appeared on cratered plains in the Memnonia region between January 2014 and April 2023. It shows in two CTX images from two orbits and is still there in February 2025.

Why it may be a new impact. Three earlier images, from 2010 to January 2014, show nothing there, although a dark mark 200 m to the west-north-west shows in all three; the 2010 and 2014 images were taken under a sun much like that of the April 2023 and February 2025 ones. A compact, round spot with no track, isolated on the plain and still there 22 months later, is how the blast zone of a small new crater looks.

Caveats. An image from December 2018 lines up too poorly to show whether the spot had formed by then, so the spot could have formed at any time in a nine-year span. Ground scoured by a dust devil, or a patch of dark sand, is unlikely but not ruled out. No image from the high-resolution HiRISE camera covers it; the nearest ends 49.8 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 85% that the spot is a new impact. It found the spot already present on 16 December 2018 (`K12_058069_1715`), which narrows the window to January 2014 to December 2018. It placed the spot at 9.2691° S, 191.2234° E, 203 m from the position given above.

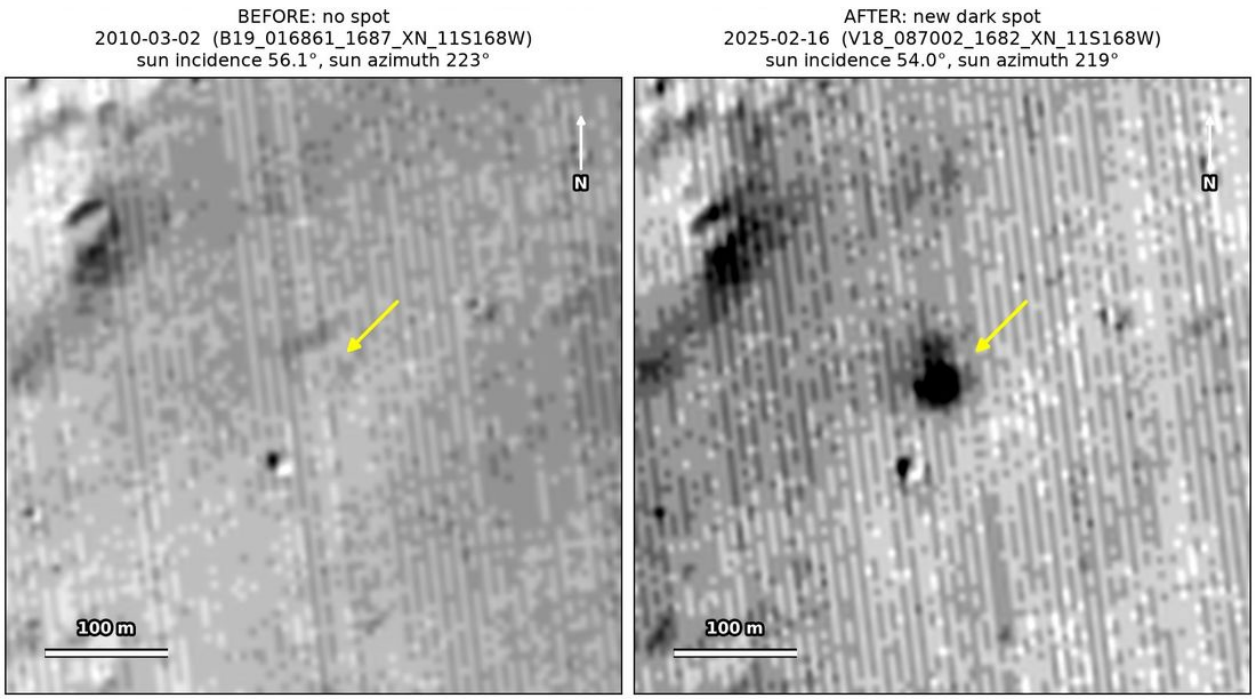

Figure 169: Candidate 67, before and after: the same 500 × 500 m of ground in 2 March 2010 (`B19_016861_1687_XN_11S168W`) and 16 February 2025 (`V18_087002_1682_XN_11S168W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

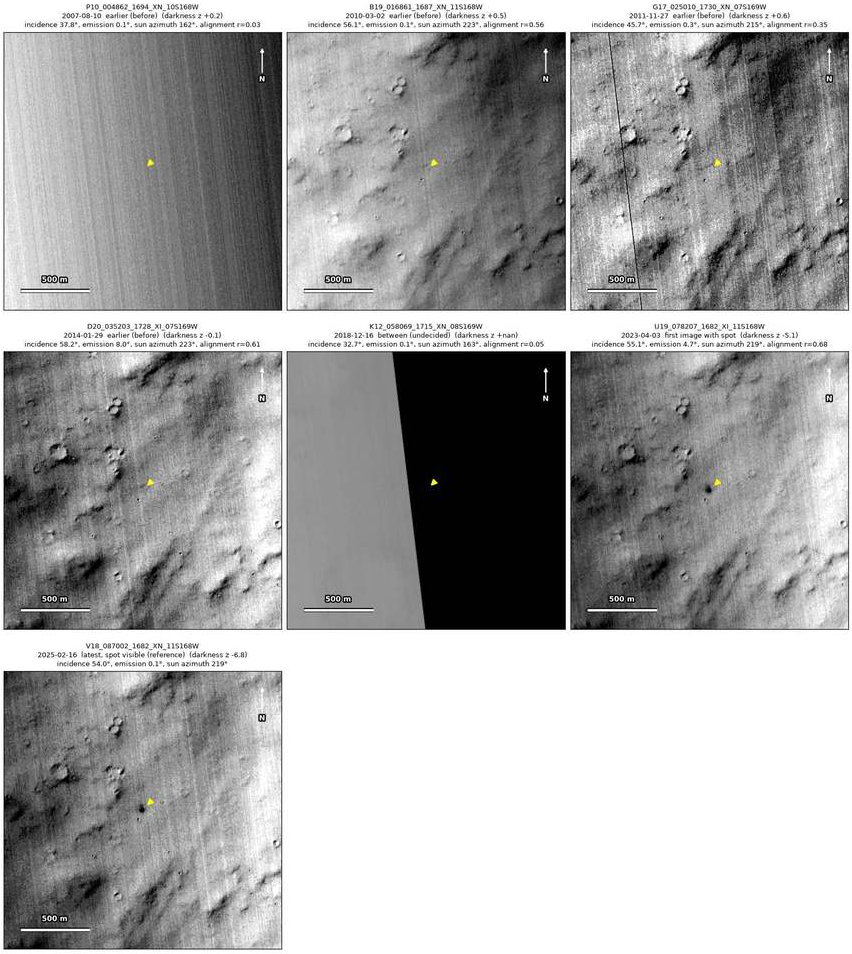

Figure 170: Candidate 67: seven CTX images of the same 2×2 km patch, 2007 to 2025. The August 2007 image, taken during a global dust storm, and the December 2018 image line up too poorly to tell whether the spot is there; they are shown but not used to date it. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

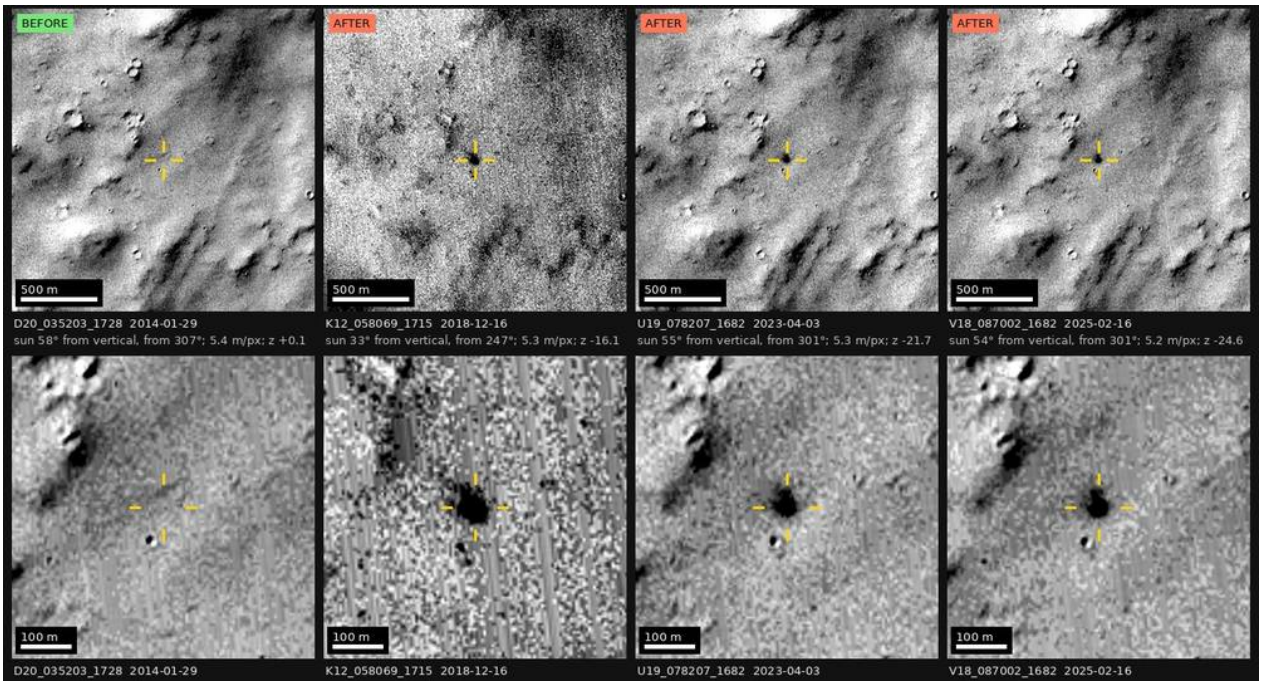

Figure 171: Candidate 67, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 68: Northern edge of Phison Patera, 220 km south of Rudaux crater

Location	34.3303° N, 50.2662° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	25 June 2023 and 9 November 2024
Images	4 without the spot (absent: 14 May 2014, 3 July 2019, 8 September 2021 and 25 June 2023); 2 with it (present: 9 November 2024 and 31 January 2025)
Size	Compact, very dark round spot about 42 m across (42 × 44 m)
HiRISE	none covers the spot (nearest footprint edge 24.0 km)
Nearest catalogued new impact	166 km (Daubar et al., 2022)

What changed. A compact, very dark round spot about 42 m across appeared on the plains at the northern edge of Phison Patera, in Arabia Terra, between June 2023 and November 2024. It shows in two CTX images from two orbits and is still there in January 2025.

Why it may be a new impact. Four earlier images, from 2014 to June 2023, show nothing there, although a dark mark 300 m to the east-north-east shows in them. The June 2023 image was taken under a sun like that of the January 2025 one, so the spot is not a shadow. A compact, round, very dark spot that appeared within 16 months on ground unchanged for the nine years before is how the blast zone of a small new crater looks.

Caveats. Only two images show it, less than three months apart, and the first, of November 2024, was taken with the sun from another direction and lines up poorly. Ground scoured by a dust devil, or dark material exposed by wind, is unlikely but not excluded. No image from the high-resolution HiRISE camera covers it; the nearest ends 24.0 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 80% that the spot is a new impact. It noted that the spot had been followed for only 2.7 months, so a short-lived wind-made patch is not excluded, and estimated the probability at about 80%. It placed the spot at 34.3352° N, 50.2677° E, 302 m from the position given above.

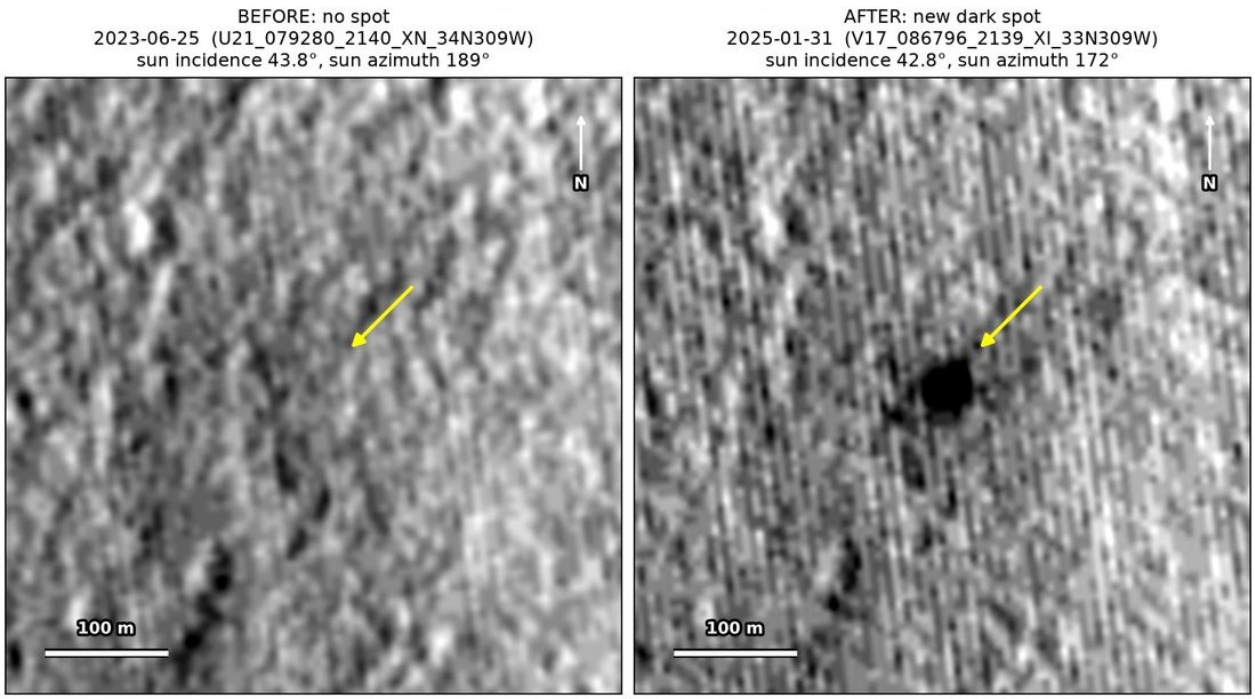

Figure 172: Candidate 68, before and after: the same 500 × 500 m of ground in 25 June 2023 (`U21_079280_2140_XN_34N309W`) and 31 January 2025 (`V17_086796_2139_XI_33N309W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

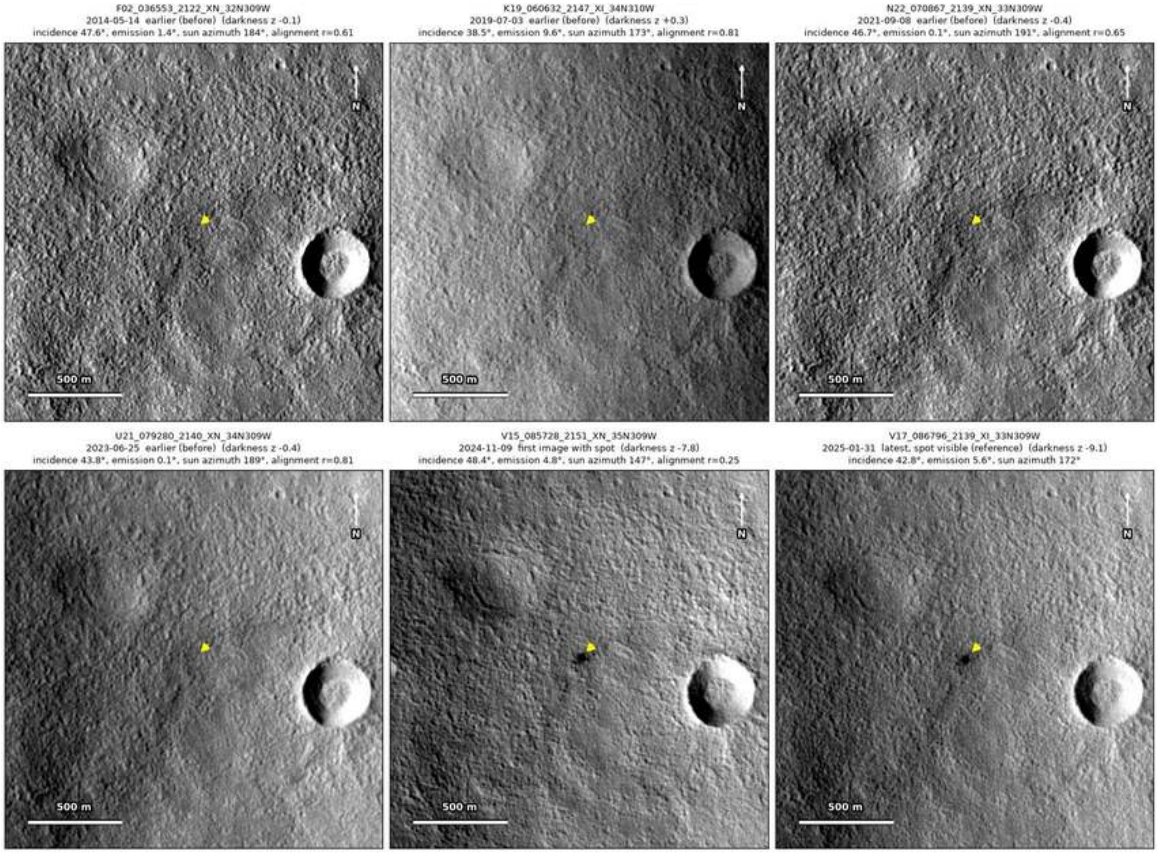

Figure 173: Candidate 68: six CTX images of the same 2×2 km patch, 2014 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

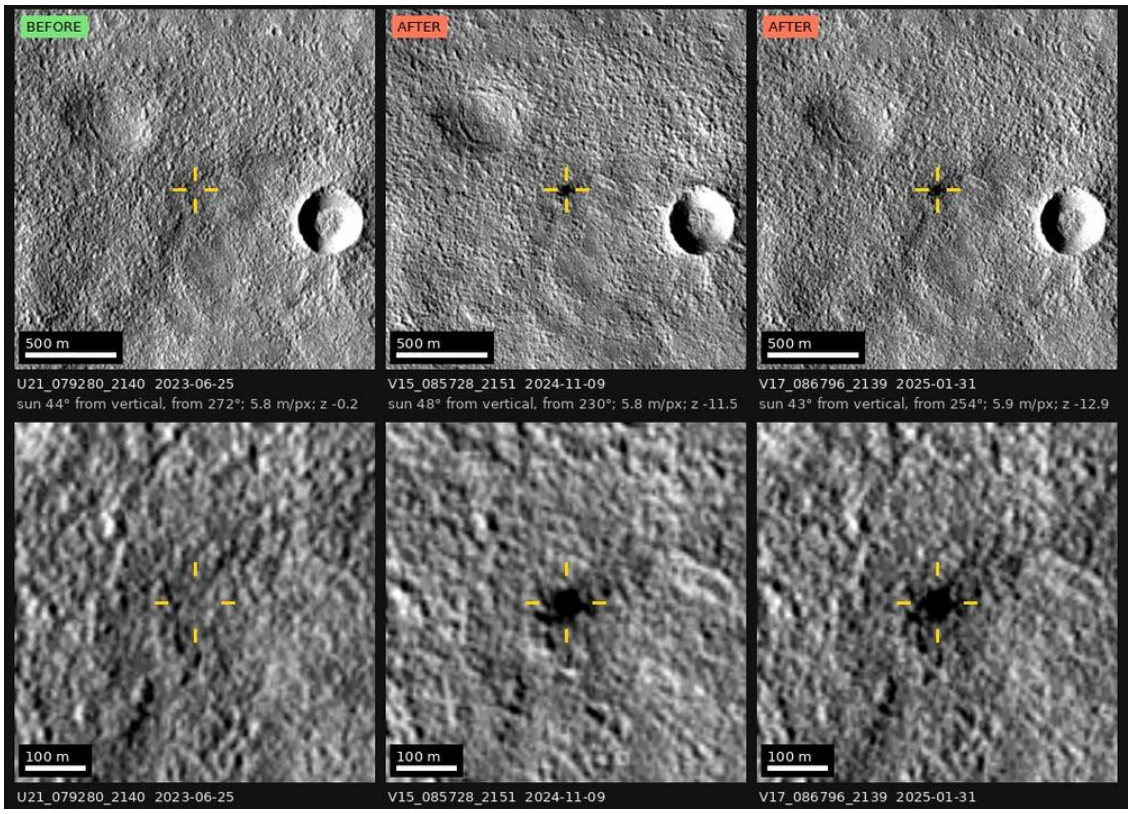

Figure 174: Candidate 68, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 69: Plains among the Erebus Montes

Location	31.0754° N, 181.4958° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	13 April 2023 and 3 March 2025
Images	2 without the spot (absent: 5 September 2010 and 13 April 2023 (CTX and HiRISE)); 2 with it (present: 3 March 2025 (CTX and HiRISE) and 17 April 2025)
Size	Diffuse dark spot about 70 m across (74 × 130 m in CTX, 72 × 90 m in HiRISE) around a pit about 2 m across
HiRISE	ESP_078339_2115 (2023-04-13, before) and ESP_087200_2115 (2025-03-03, after); same orbits as U19 and V19, not impact-targeted
Nearest catalogued new impact	60 km (Daubar et al., 2022)

What changed. A diffuse dark spot about 70 m across appeared on the plains among the Erebus Montes between April 2023 and March 2025, and shows in two CTX images from two orbits. Much sharper images from the HiRISE camera show a new pit about 2 m across at its centre.

Why it may be a new impact. The two earlier CTX images, from 2010 and April 2023, show nothing there, although a dark mark 280 m to the west-south-west shows in both; the 2023 image was taken under the same sun as the March and April 2025 images. A dust devil or wind could darken the ground but would not make a new pit at its centre, and darkened ground around a small new pit is what a fresh impact would leave.

Caveats. By chance, NASA's high-resolution HiRISE camera photographed the site on 13 April 2023, before the spot formed, and on 3 March 2025, after it formed, on the same orbits as two of the CTX images. In the 2025 image the ground within 40 m of the spot is 11% darker than in 2023, and at its darkest point is a new feature about 2 m across, dark on its west side and bright on its east side under a western sun, as a small pit would be, while a boulder 5 m to the north-east shows in both images. The spot has been followed for only 45 days in CTX images, and the pit shows in only one HiRISE image. Dust removed by a dust devil or wind could account for the diffuse darkening, though not for the pit.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 85% that the spot is a new impact. Registering the two HiRISE images to each other, it found the ground within 20 m of the centre 5.2% darker in 2025 than in 2023, over a zone 78 × 90 m, with a new compact dark feature about 2.5 m across inside it. It saw no bowl-like shading in that feature, so a crater is not proven (Figure 179). It placed the spot at 31.0735° N, 181.4839° E, 614 m from the position given above.

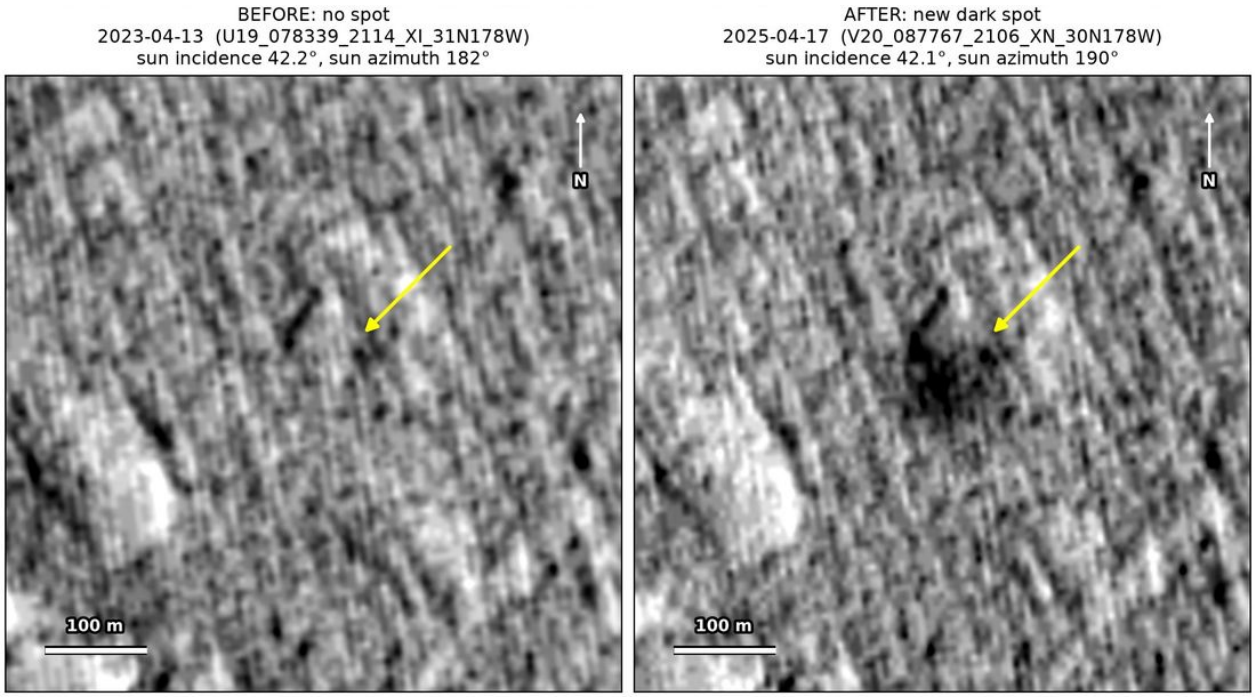

Figure 175: Candidate 69, before and after: the same 600 × 600 m of ground in 13 April 2023 (`U19_078339_2114_XI_31N178W`) and 17 April 2025 (`V20_087767_2106_XN_30N178W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

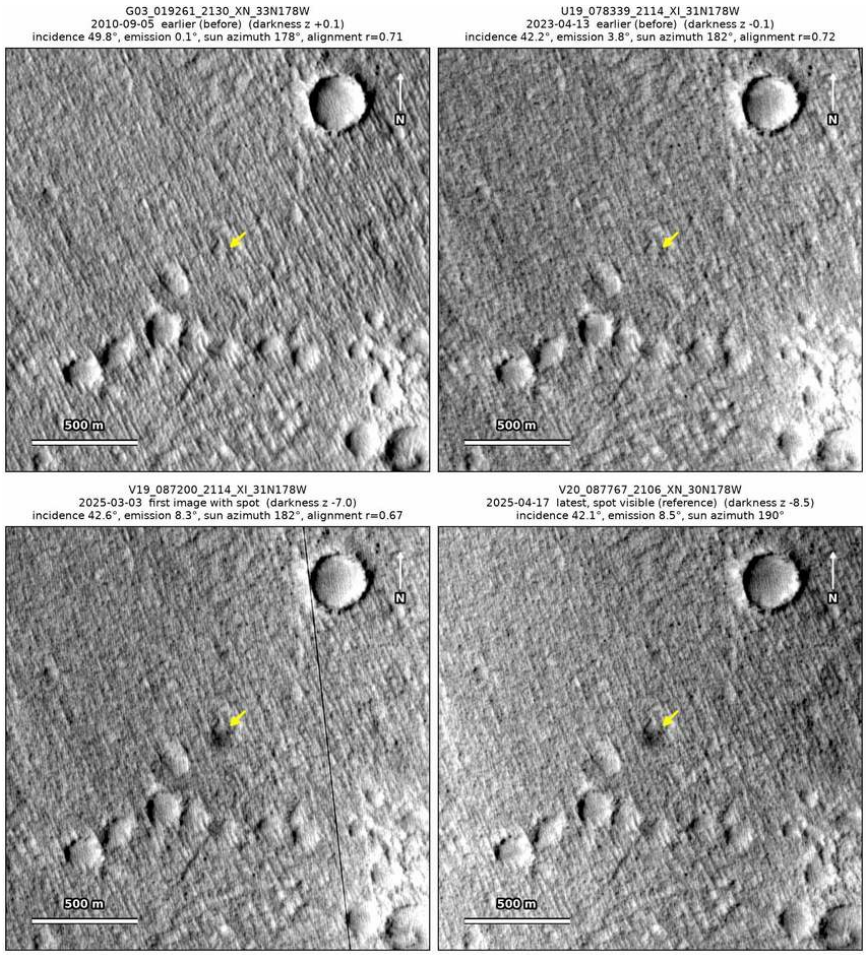

Figure 176: Candidate 69: four CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

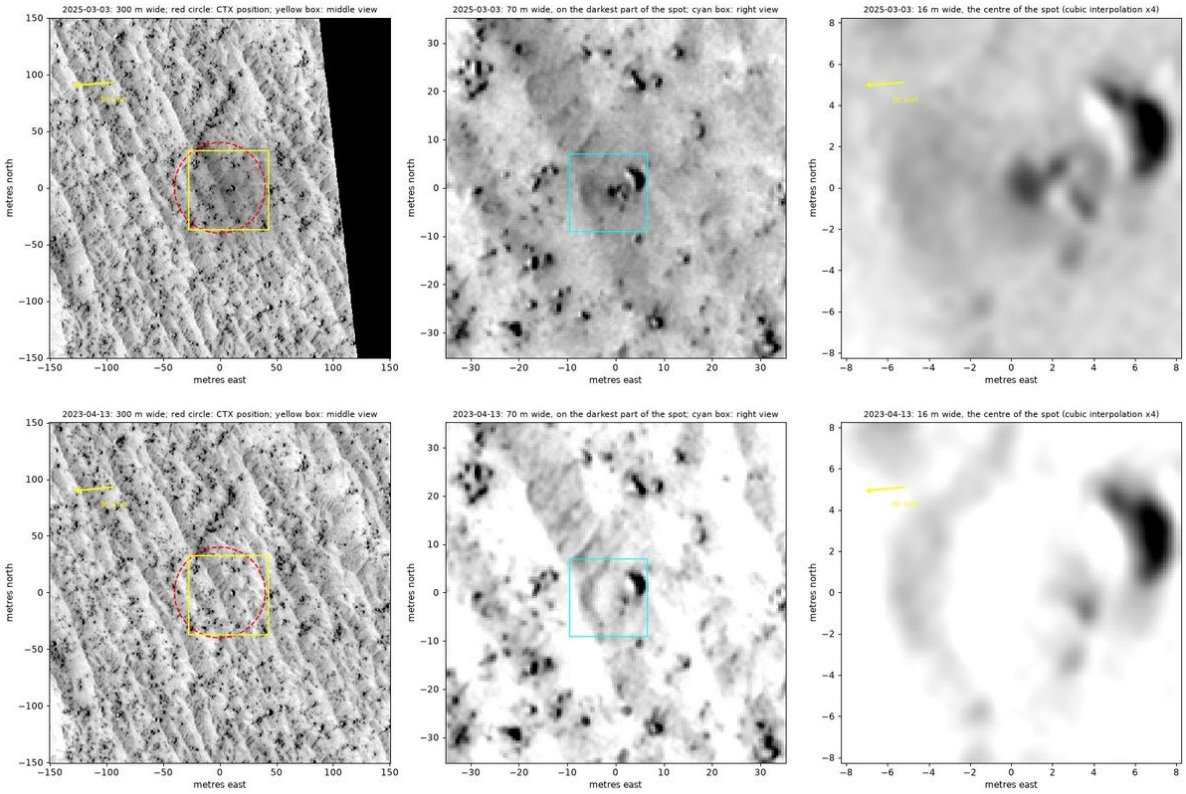

Figure 177: Candidate 69, HiRISE images `ESP_087200_2115` (top) and `ESP_078339_2115` (bottom): HiRISE views of the spot from 3 March 2025 (top) and 13 April 2023 (bottom), 0.5 m per pixel, taken by chance on the same orbits as two of the CTX images and under the same light: 300 m wide, then a 70 m enlargement of the darkest part, then the central 16 m. Both rows use the same contrast. The ground around the centre is darker in 2025, and a small dark feature near the centre is new; a boulder 5 m to its north-east is in both images. The red dashed circle marks the CTX position. NASA/JPL-Caltech/UArizona.

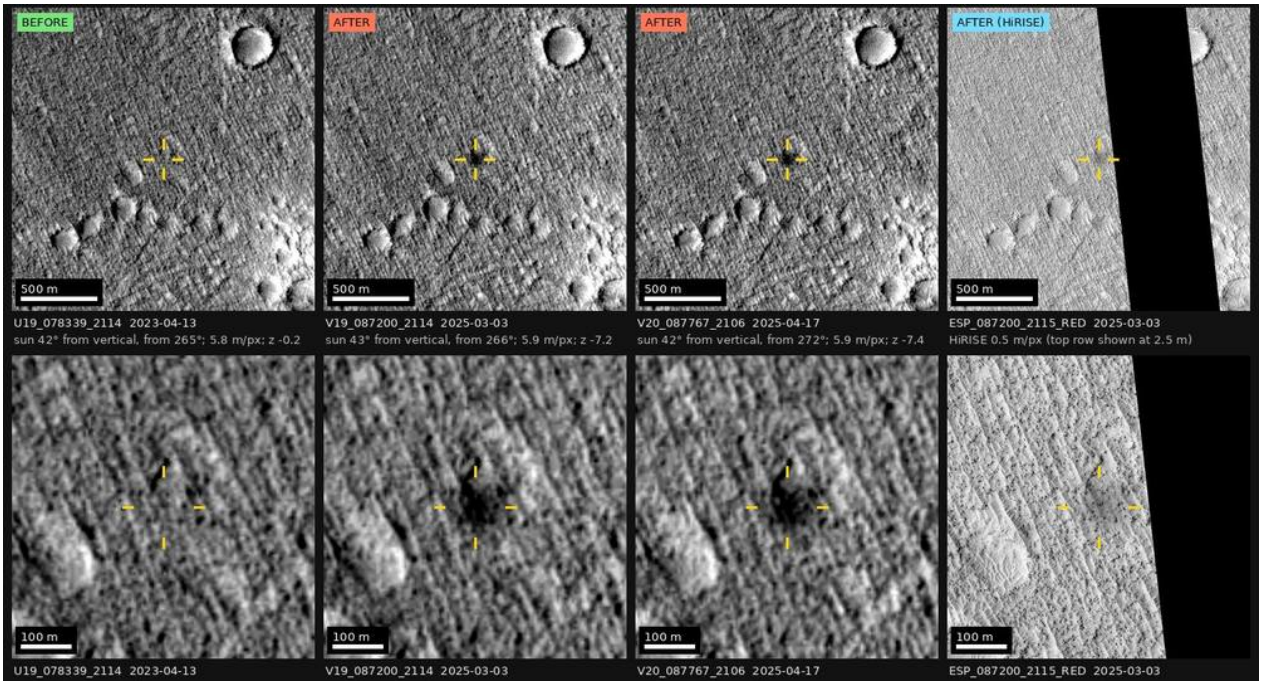

Figure 178: Candidate 69, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

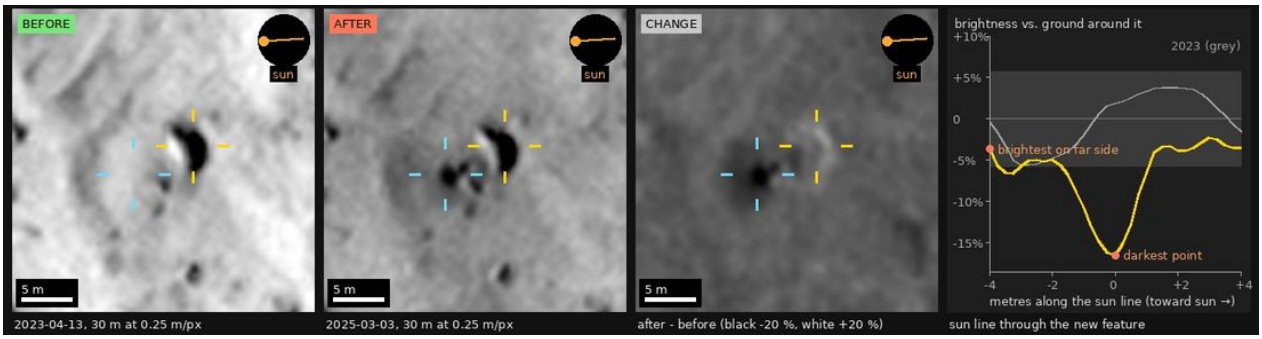

Figure 179: Candidate 69, HiRISE images `ESP_087200_2115` (after) and `ESP_078339_2115` (before). The independent check: HiRISE comparison. Top: the 2023 and 2025 images, aligned and matched in brightness, and their ratio, which shows the darkened zone, 78×90 m. Bottom: the central 30 m of both and their difference. Cyan ticks mark the new dark feature, about 2.5 m across; yellow ticks mark a dark dot that is in both images. The graph follows the brightness along the sun line through the new feature: its far side never becomes brighter than the ground around it, so on this measure it shows no clear bowl shading and the check could not show it to be a crater, although in the HiRISE images the feature, about 2 m across, is dark on its west side and bright on its east side, as a small pit would be. NASA/JPL-Caltech/UArizona.

Candidate 70: Tharsis plains 490 km south of Ascraeus Mons

Location	3.8713° N, 257.6935° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	8 April 2024 and 29 March 2025
Images	4 without the spot (absent: 20 February 2012, 15 September 2012, 19 June 2022 and 8 April 2024); 1 with it (present: 29 March 2025)
Size	Compact, very dark spot about 19 m across (16 × 22 m)
HiRISE	none covers the spot (nearest footprint edge 6.9 km)
Nearest catalogued new impact	22 km (Daubar et al., 2022)

What changed. A compact, very dark spot about 19 m across appeared on the Tharsis lava plains 490 km south of Ascraeus Mons between April 2024 and March 2025. It shows in one CTX image so far.

Why it may be a new impact. Four earlier images, from 2012 to April 2024, show plain ground there; one of them, from February 2012, was taken under the same sun as the March 2025 image, and a dark mark 280 m to the east shows in the first three. The spot is a smooth blob about four pixels across rather than a blocky flaw from image compression, and a compact, very dark new spot is how the blast zone of a small new crater can look.

Caveats. Only one image shows it so far, and that image is heavily compressed, so a compression flaw or a dust devil caught in the act is possible; only a second image can rule out a short-lived mark. The April 2024 image lines up poorly, and without it the spot could have formed as early as June 2022. No image from the high-resolution HiRISE camera covers it; the nearest ends 6.9 km away.

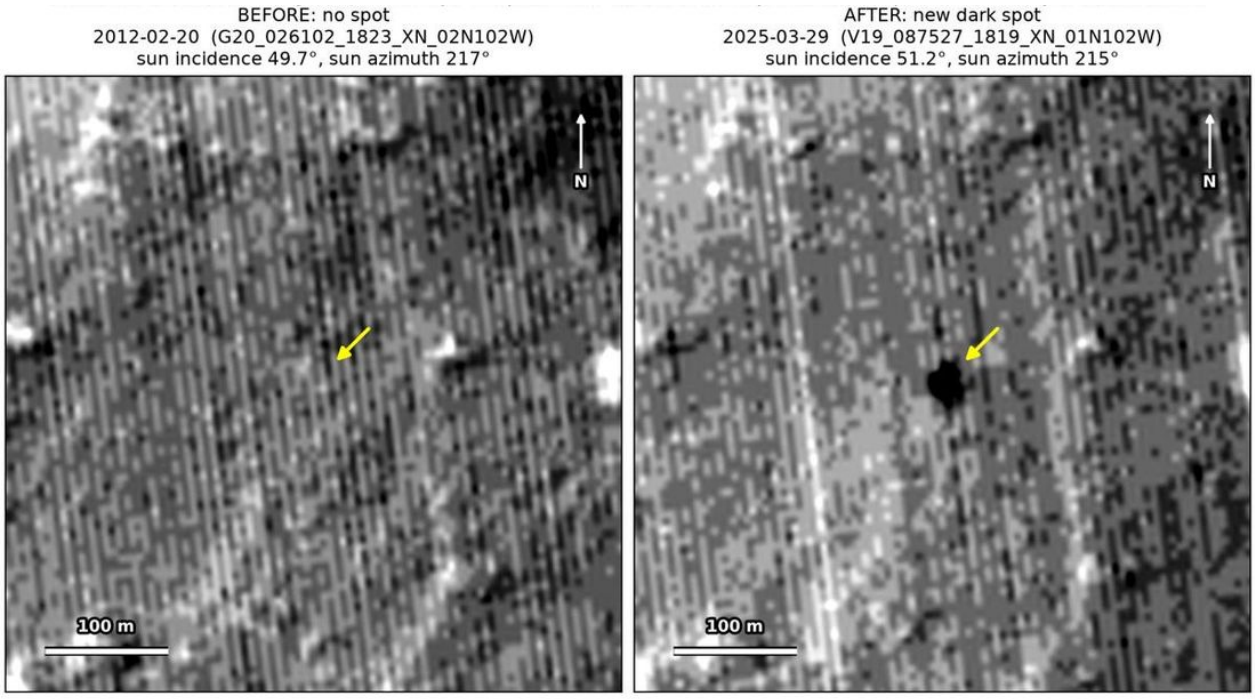

Figure 180: Candidate 70, before and after: the same 500 × 500 m of ground in 20 February 2012 (`G20_026102_1823_XN_02N102W`) and 29 March 2025 (`V19_087527_1819_XN_01N102W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

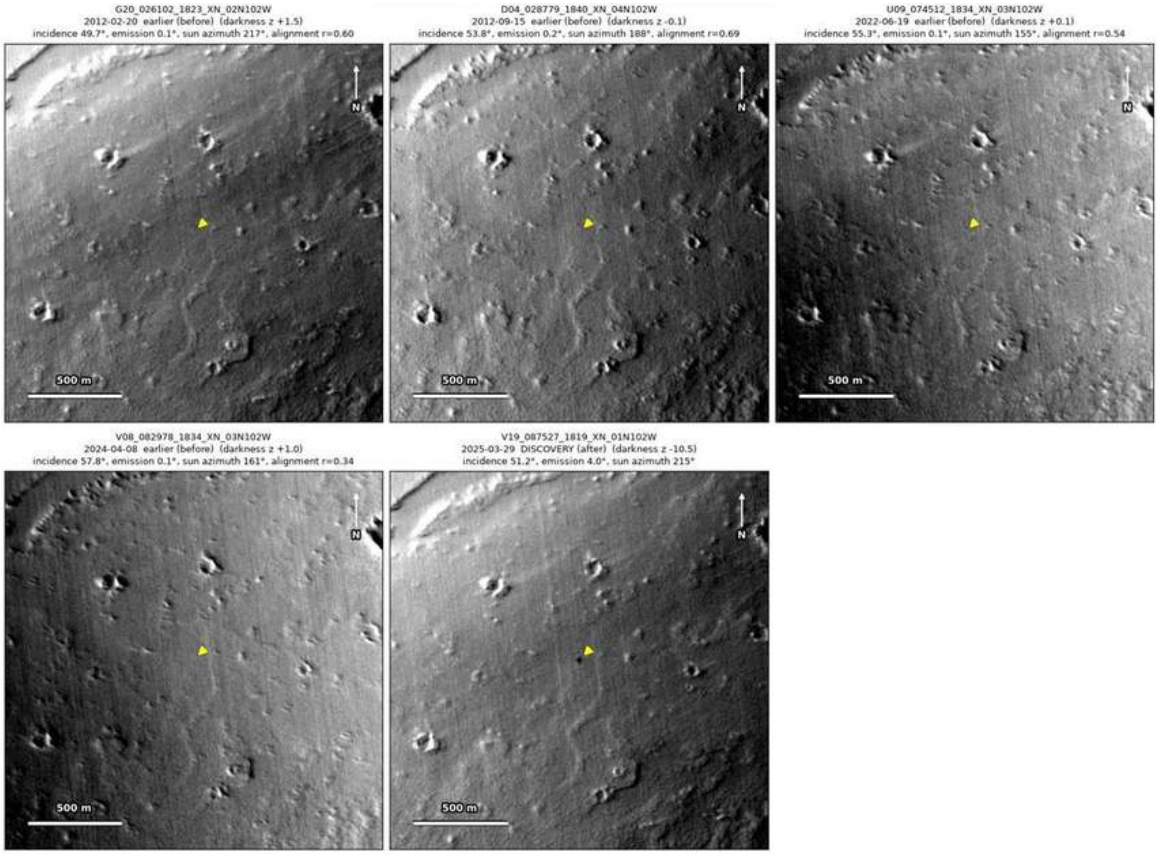

Figure 181: Candidate 70: five CTX images of the same 2×2 km patch, 2012 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 71: Tharsis lava plains 350 km west of Echus Chasma

Location	2.4389° N, 274.0536° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	23 January 2010 and 27 January 2023
Images	1 without the spot (absent: 23 January 2010); 3 with it (present: 27 January 2023, 12 April 2025 and 5 July 2025)
Size	Compact, very dark spot about 33 m across (34 × 42 m), with a smaller dark dot about 70 m to its north-west
HiRISE	none covers the spot (nearest footprint edge 11.0 km)
Nearest catalogued new impact	107 km (Daubar et al., 2022)

What changed. A compact, very dark spot about 33 m across, with a smaller dark dot about 70 m to its north-west, appeared on smooth Tharsis lava plains between January 2010 and January 2023. It shows in three CTX images from three orbits and is still there in July 2025.

Why it may be a new impact. The only earlier image, from January 2010, was taken under a sun much like that of the January 2023 image and shows nothing there, yet a dark mark 225 m to the south-west shows clearly in it, more clearly than in the later images. A round spot with no track, seen in the same place on three orbits and unchanged over two and a half years, is how the blast zone of a small new crater looks.

Caveats. Only one earlier image exists, so the spot could have formed at any time in a 13-year span, and that image is noisy, although the dark mark nearby shows clearly in it. No image from the high-resolution HiRISE camera covers it; the nearest ends 11.0 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 85% that the spot is a new impact. It placed the spot at 2.4370° N, 274.0570° E, 229 m from the position given above.

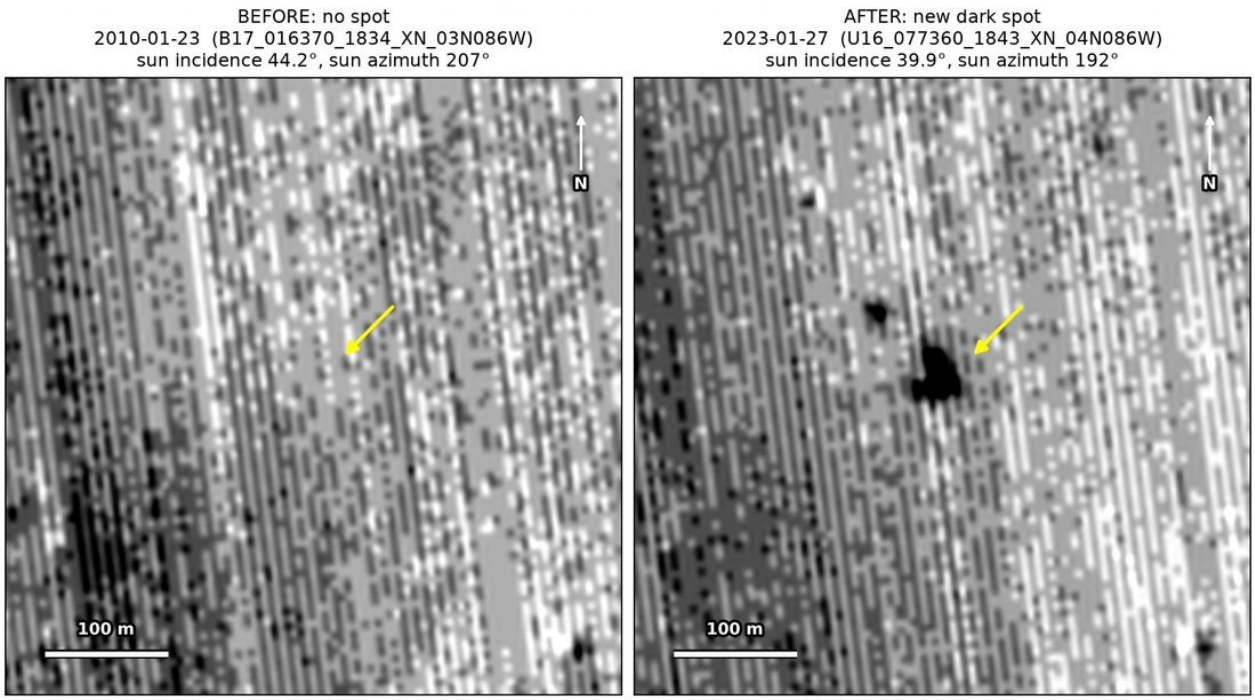

Figure 182: Candidate 71, before and after: the same 500 × 500 m of ground in 23 January 2010 (`B17_016370_1834_XN_03N086W`) and 27 January 2023 (`U16_077360_1843_XN_04N086W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

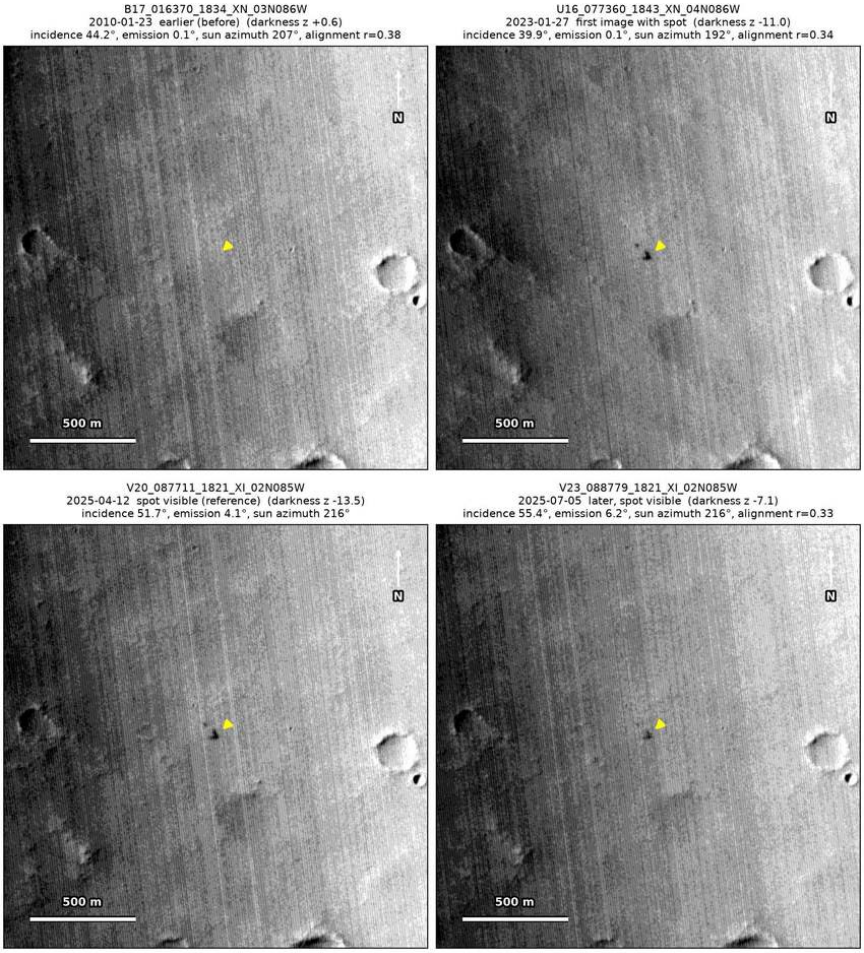

Figure 183: Candidate 71: four CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

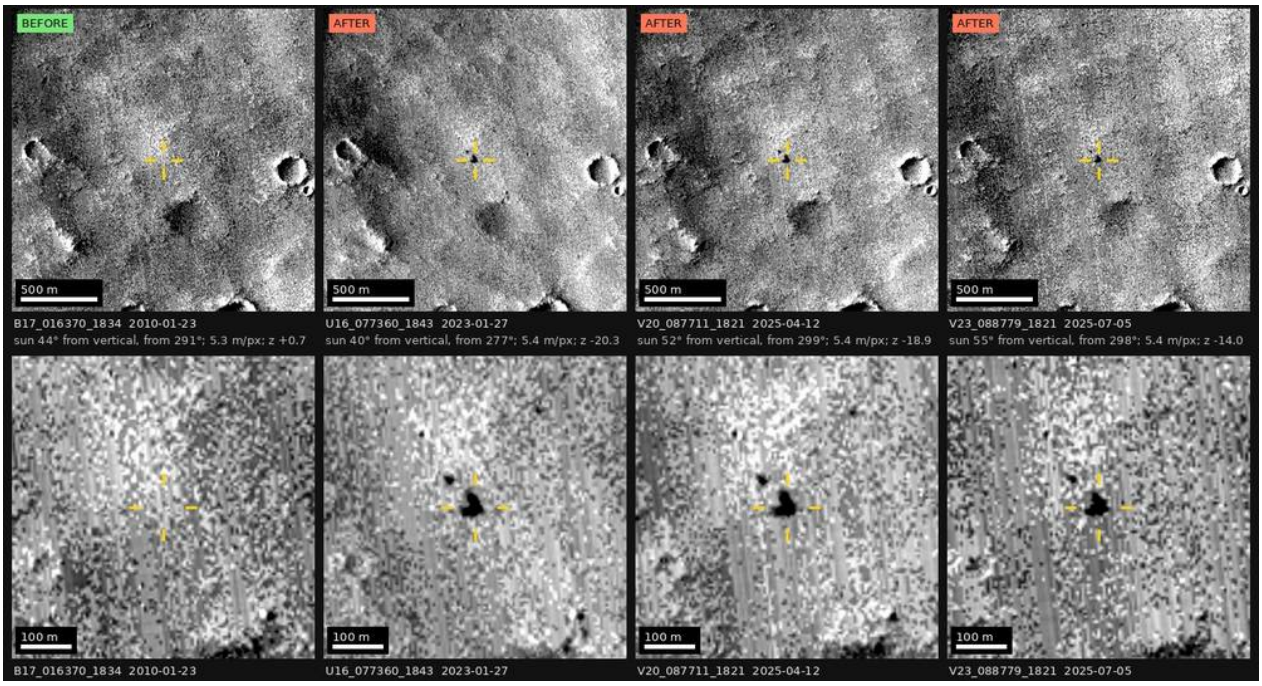

Figure 184: Candidate 71, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 72: Eroded plains in western Arabia Terra, 5 km north-north-west of candidate 57

Location	19.2191° N, 12.1485° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	18 March 2014 and 4 May 2023
Images	2 without the spot (absent: 17 February 2010 and 18 March 2014); 4 with it (present: 4 May 2023, 6 January 2025, 3 February 2025 and 25 March 2025)
Size	Main spot about 55 m across (48 × 74 m) in an irregular dark cluster about 80 × 140 m
HiRISE	none covers the spot (nearest footprint edge 4.0 km)
Nearest catalogued new impact	189 km (Daubar et al., 2022)
Nearby	Candidate 57, 5.2 km to the south-south-east and covered by a HiRISE image, formed between the same two images, of March 2014 and May 2023. CTX images cannot show whether the two come from one body that broke up, although 5 km is far wider than typical impact clusters.

What changed. An irregular cluster of dark spots about 80 × 140 m, with a main spot about 55 m across, appeared on eroded plains in western Arabia Terra between March 2014 and May 2023. It shows in four CTX images from four orbits.

Why it may be a new impact. Two earlier images, from 2010 and March 2014, show nothing there, and the 2014 one was taken under the same sun as the May 2023 image. No steep slope is visible at the spot and the dark marks are blobs, not streaks running downslope, so they are unlikely to be slope streaks, the dark streaks left by small dust avalanches. A new cluster of dark spots that has lasted two years with no track fits a small new impact.

Caveats. A dark mark 300 m to the north-east would show that the two earlier images could have caught a new spot, but it is only faintly dark in them, so on its own it is weak evidence. No image from the high-resolution HiRISE camera covers it; the nearest ends 4.0 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 85% that the spot is a new impact. It noted that candidates 57 and 72, 5.2 km apart, are bracketed by the same two CTX images and may be fragments of one meteoroid; either way the site is most probably an impact. It placed the spot at 19.2186° N, 12.1405° E, 449 m from the position given above.

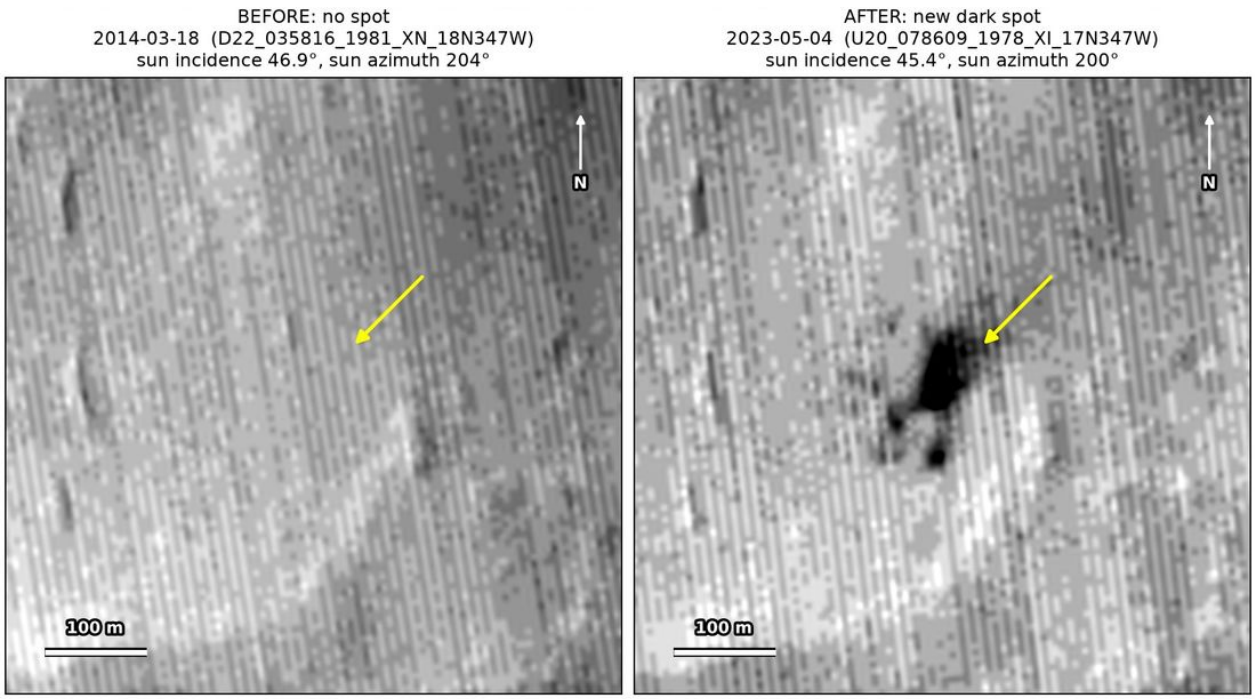

Figure 185: Candidate 72, before and after: the same 600 × 600 m of ground in 18 March 2014 (`D22_035816_1981_XN_18N347W`) and 4 May 2023 (`U20_078609_1978_XI_17N347W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

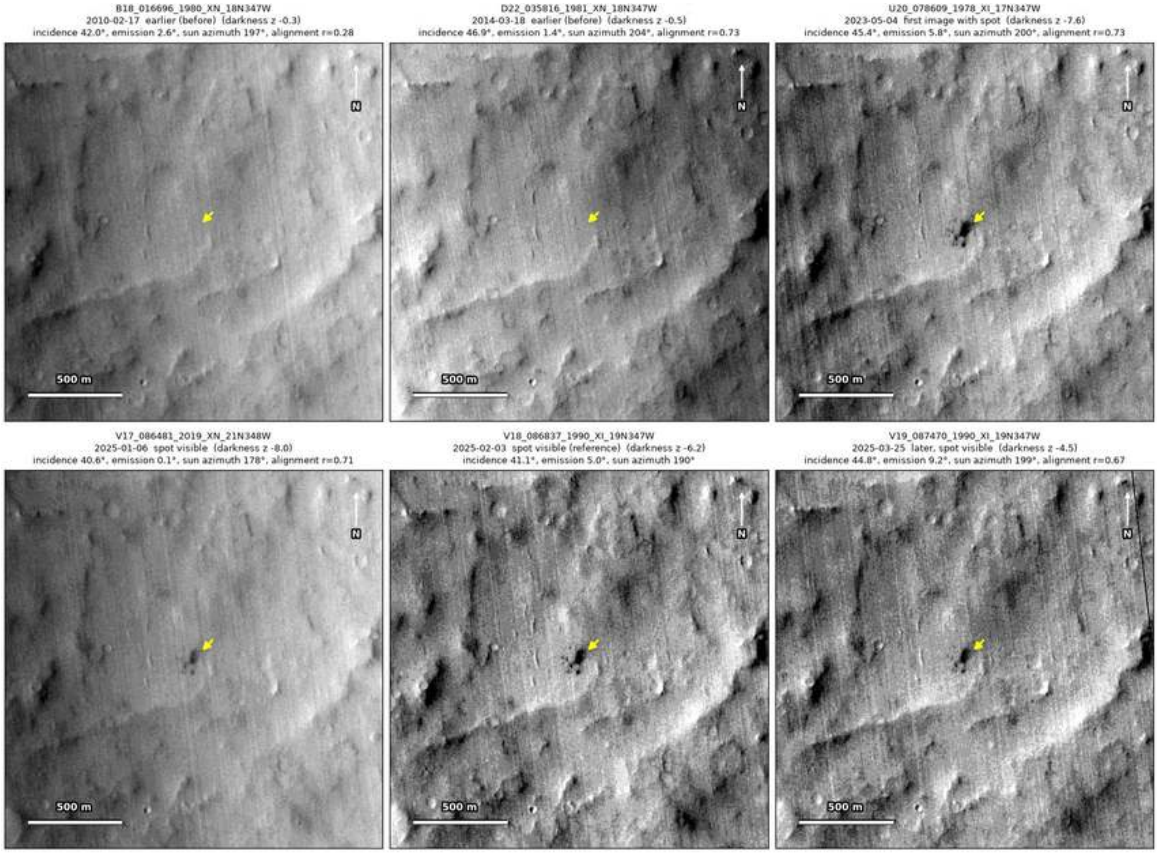

Figure 186: Candidate 72: six CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

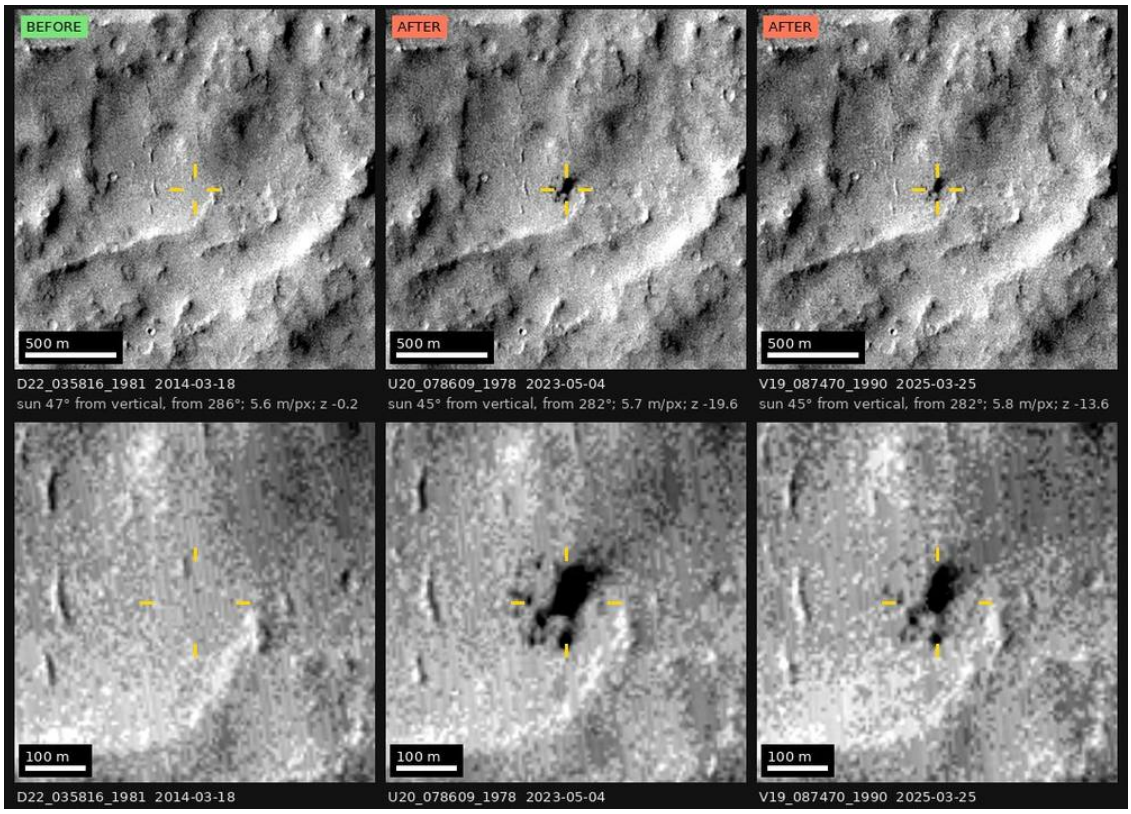

Figure 187: Candidate 72, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 73: Flat ground in northern Amazonis Planitia

Location	39.4963° N, 209.8523° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 75%
Formed between	14 July 2022 and 30 December 2022
Images	4 without the spot (absent: 31 August 2010, 27 October 2021, 3 March 2022 and 14 July 2022); 9 with it (present: 30 December 2022, 8 April 2023, 23 August 2023, 23 December 2023, 25 May 2024, 28 December 2024, 28 April 2025, 28 August 2025 and 30 January 2026)
Size	Diffuse dark patch about 210 m across (172 × 292 m), in two lobes in a north–south line
HiRISE	none covers the spot (nearest footprint edge 25.1 km)
Nearest catalogued new impact	54 km (Daubar et al., 2022)

What changed. A diffuse dark patch about 210 m across, in two lobes lined up north to south, appeared on flat ground among small knobs at a site in northern Amazonis Planitia that CTX photographs regularly, between July and December 2022. Nine CTX images from nine orbits, up to January 2026, show it.

Why it may be a new impact. Four earlier images, from 2010 to July 2022, show no patch there. The October 2021 image was taken under the same sun as the April 2023 one, and the July 2022 image under the same sun as the May 2024 one. Much of the darkening, including the whole northern lobe, lies on flat ground between the knobs, and a patch that appeared suddenly, with no track, and stayed for three years fits a small new impact.

Caveats. No crater can be made out in these images, so the case for an impact rests on the patch's shape and its sudden appearance. Dust avalanches on the knobs' flanks, perhaps set off together, remain possible. No image from the high-resolution HiRISE camera covers it; the nearest ends 25.1 km away.

Independent check. The independent re-examination (Section 3.8) would rate this site promising rather than strong, with its own estimate of about 60% that the spot is a new impact. It found the change real and well dated (29 earlier images without it) but judged that a diffuse two-lobed zone without a compact core does not meet the criteria for a strong rating, and that dust stripped by wind during the dusty season explains it nearly as well. It would rate the site promising, at about 60%. It placed the spot at 39.4960° N, 209.8607° E, 383 m from the position given above.

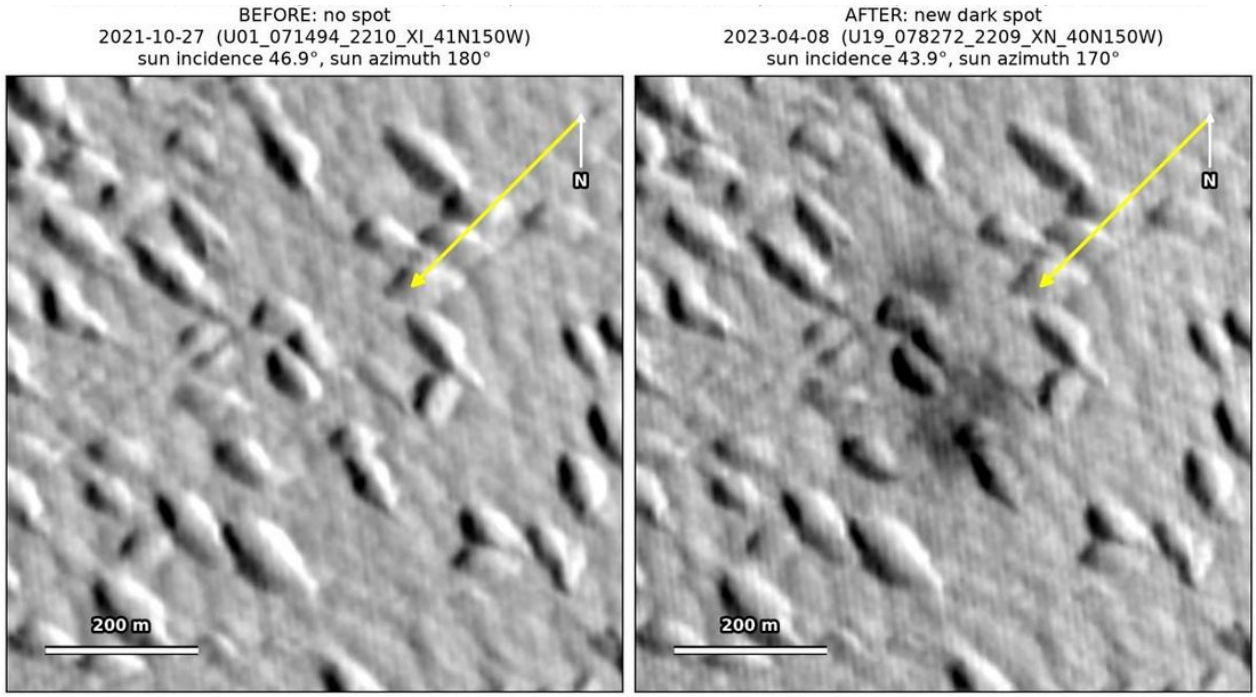

Figure 188: Candidate 73, before and after: the same 800 × 800 m of ground in 27 October 2021 (`U01_071494_2210_XI_41N150W`) and 8 April 2023 (`U19_078272_2209_XN_40N150W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

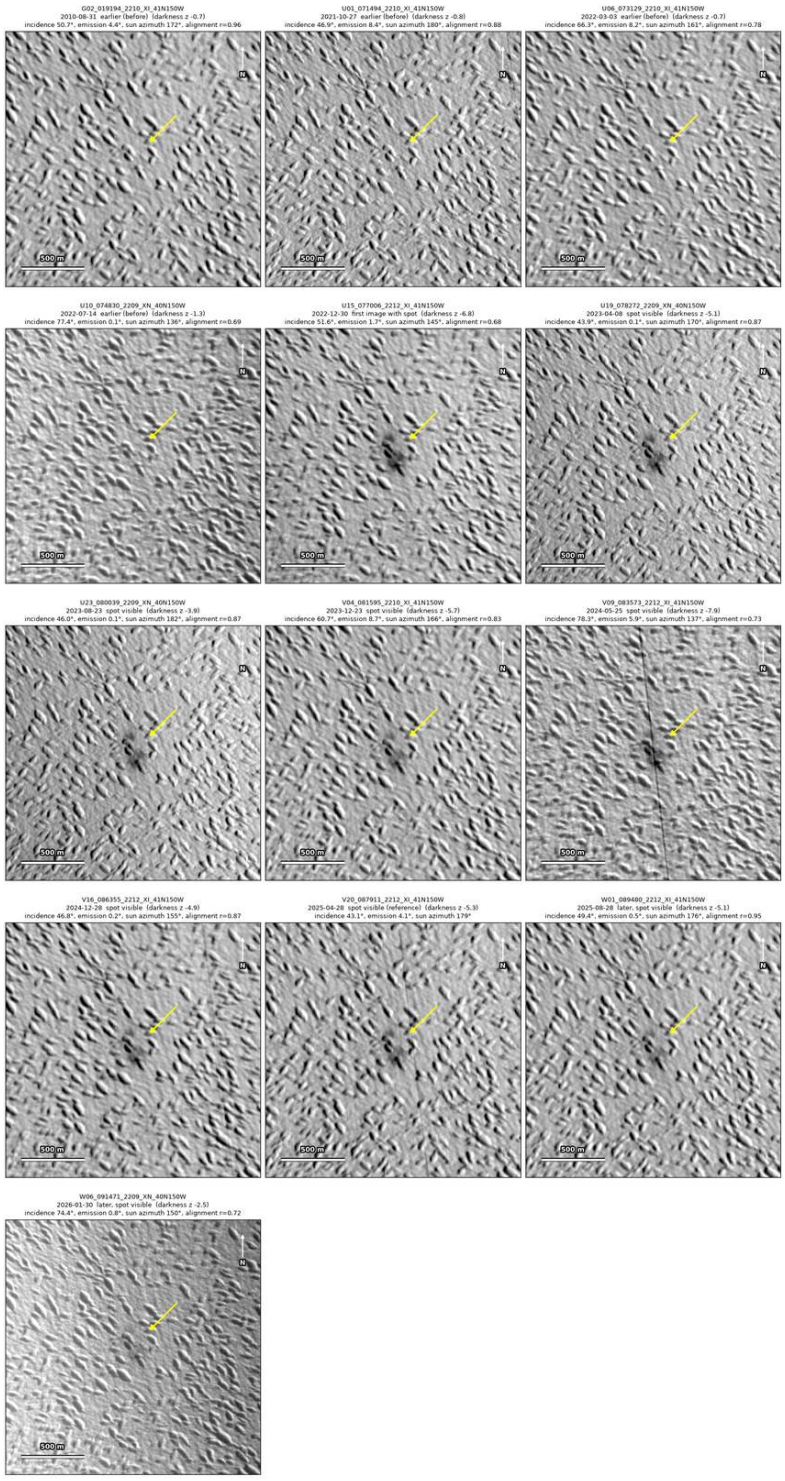

Figure 189: Candidate 73: thirteen CTX images of the same 2×2 km patch, 2010 to 2026. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 74: Knobby floor of northern Orcus Patera

Location	16.0736° N, 178.9461° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 70%
Formed between	9 June 2014 and 30 January 2023
Images	2 without the spot (absent: 18 August 2010 and 9 June 2014); 3 with it (present: 30 January 2023, 30 March 2025 and 4 October 2025)
Size	Compact dark core about 55 m across (48 × 114 m in the March 2025 image), with fainter darkening around it
HiRISE	none covers the spot (nearest footprint edge 10.8 km)
Nearest catalogued new impact	156 km (Daubar et al., 2022)

What changed. A compact dark spot about 55 m across, with fainter darkening around it, appeared on knobby ground on the floor of northern Orcus Patera between June 2014 and January 2023. It shows in three CTX images from three orbits and is still there in October 2025.

Why it may be a new impact. Two earlier images, from 2010 and June 2014, taken under the same sun as the March and October 2025 images, show no spot there. The core is compact rather than a slope streak (a dark streak left by dust sliding down a slope), the knobs around it are small, and no streak or track of wind-blown material is visible, so a compact new dark spot like this fits the blast zone of a small new crater.

Caveats. A dark mark nearby, which would show that the earlier images could have caught a new spot, is faint in both of them, and the January 2023 image, under a higher sun, shows the spot less dark, so it may have darkened later. Slope streaks are common on Orcus Patera's rim and ridges, and wind makes dark spots and streaks elsewhere in Orcus Patera, though the knobs here are small and no streak or track is visible. The spot lies within the area of a catalogue of new impacts around the InSight lander (Bickel et al. 2025) and may have formed within its period, but all 123 impacts in it have HiRISE images, and this spot has none. No image from the high-resolution HiRISE camera covers it; the nearest ends 10.8 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 65% that the spot is a new impact. It found a faint dark patch already at the site in 2010 and 2014, about a quarter as deep as the later spot. Such faint patches are common nearby and none of the others darkened, but dark material gathering where some already lay, or a short slope streak, remains possible; it estimated about 65%. It placed the spot at 16.0743° N, 178.9482° E, 124 m from the position given above.

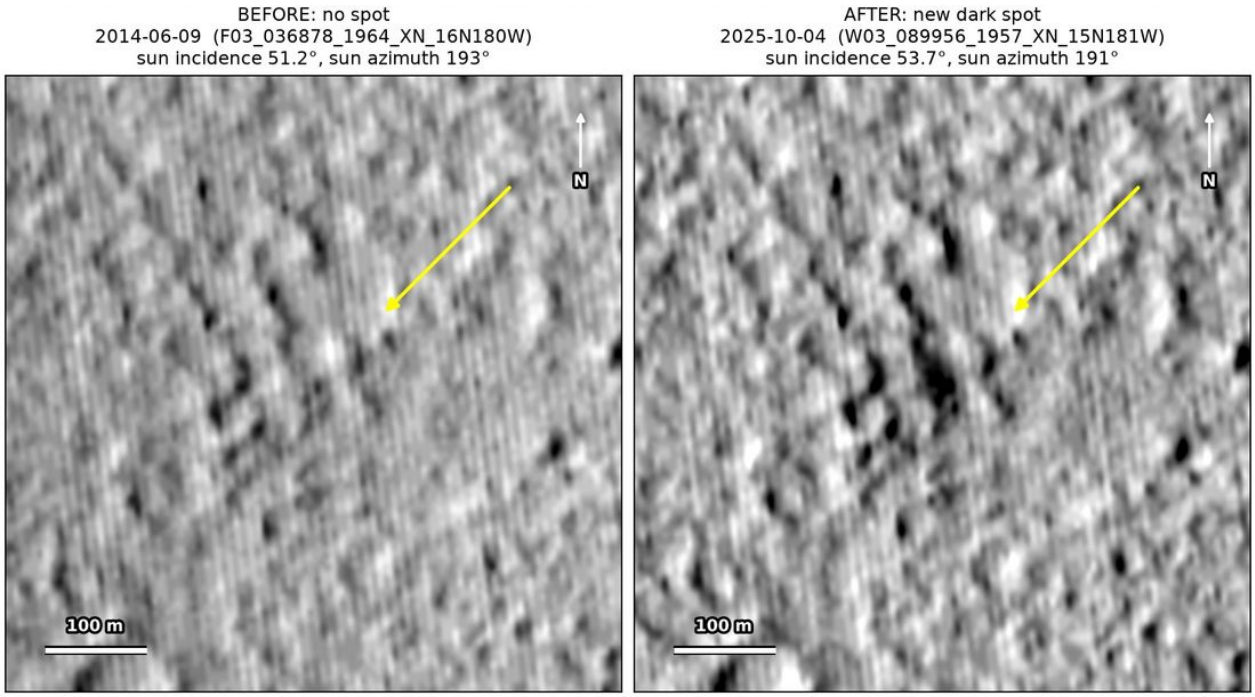

Figure 190: Candidate 74, before and after: the same 600 × 600 m of ground in 9 June 2014 (`F03_036878_1964_XN_16N180W`) and 4 October 2025 (`W03_089956_1957_XN_15N181W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

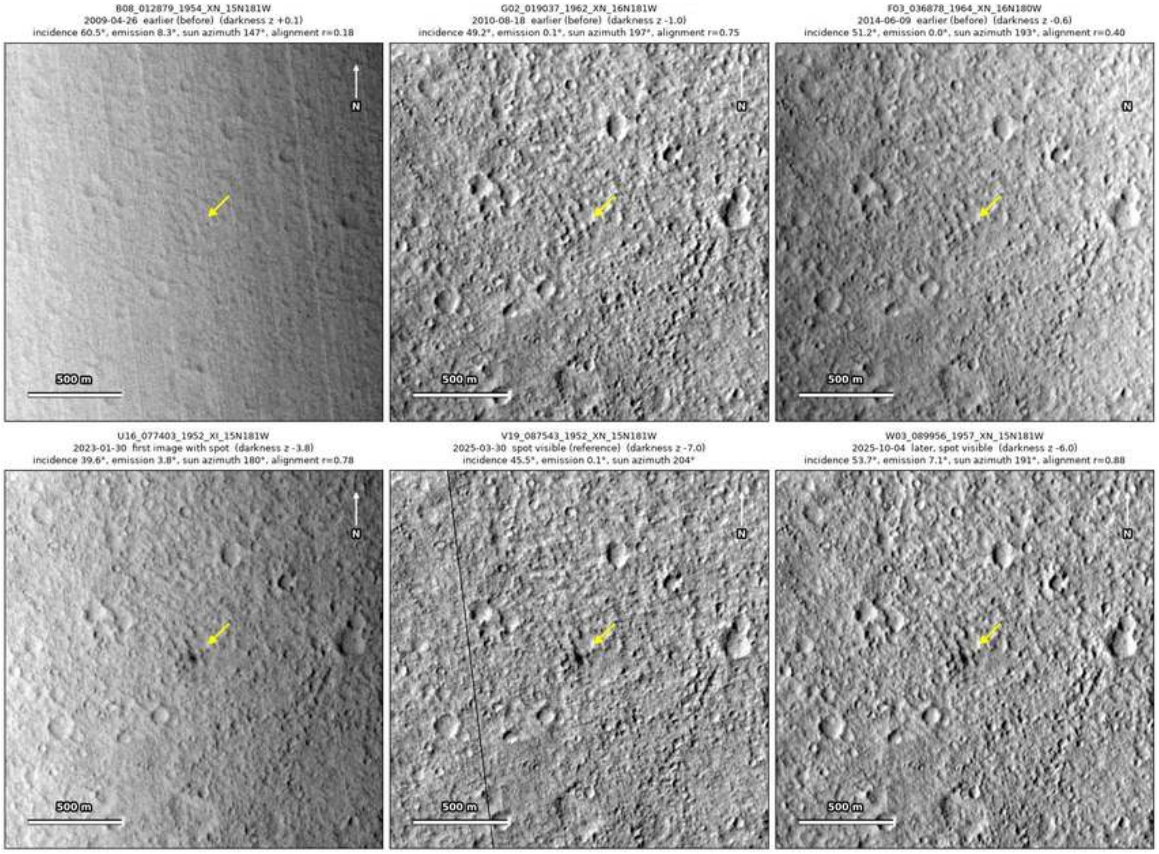

Figure 191: Candidate 74: six CTX images of the same 2×2 km patch, 2009 to 2025. The hazy April 2009 image is shown but not used to date the spot. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

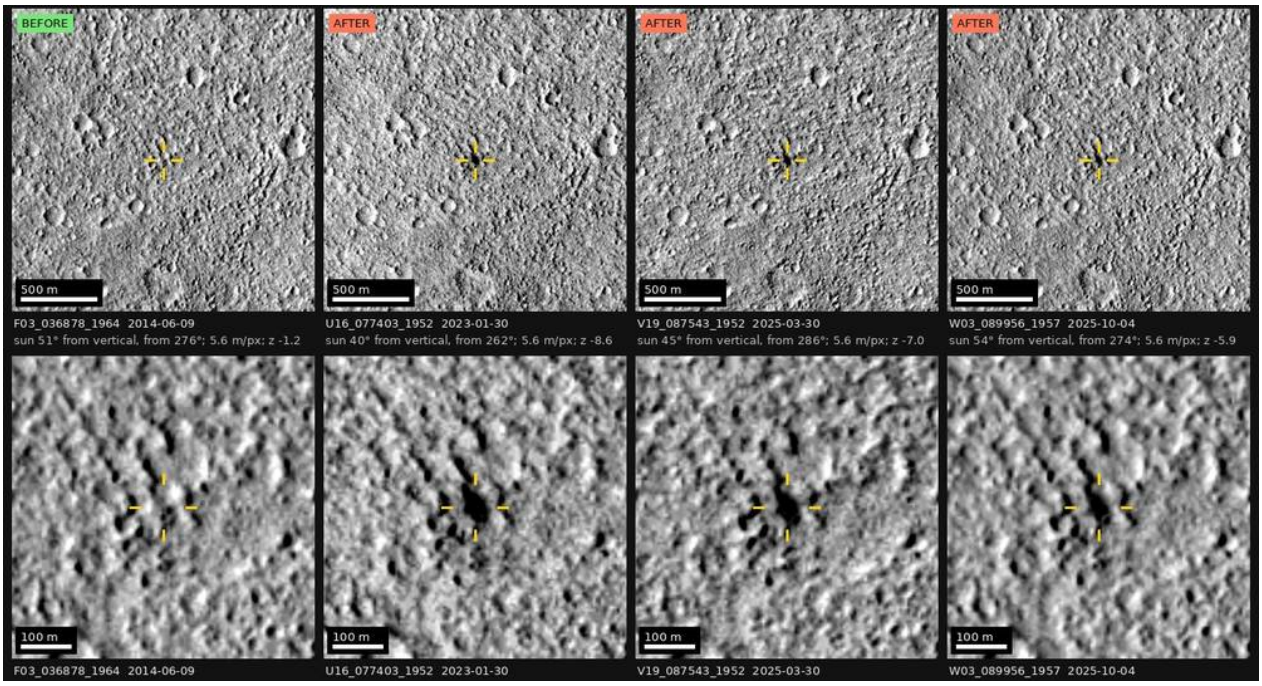

Figure 192: Candidate 74, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 75: Plains west of Echus Chasma, 16 km north of candidate 63

Location	2.8034° N, 278.7676° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 50%
Formed between	12 October 2012 and 29 March 2023
Images	2 without the spot (absent: 22 December 2006 and 12 October 2012); 2 with it (present: 29 March 2023 and 12 February 2025)
Size	Compact dark spot about 45 m across (48 × 50 m in the February 2025 image)
HiRISE	none covers the spot (nearest footprint edge 12.5 km)
Nearest catalogued new impact	62 km (Daubar et al., 2022)
Nearby	Candidate 63, 16.5 km to the south, appears in the same pair of images and formed in an overlapping time span, but at that distance this spot is unlikely to be a fragment of the same event.

What changed. A compact dark spot about 45 m across, with a lobe to the north-west, appeared on wind-textured plains west of Echus Chasma between October 2012 and March 2023. It shows in two CTX images from two orbits.

Why it may be a new impact. Two earlier images, from December 2006 and October 2012, show no spot, even under a lower sun than in the March 2023 and February 2025 images, which would have lengthened any shadow. Allowing for a gradual change in brightness across the area, the spot is the darkest point within 200 m in both later images, and the aligned before-and-after difference changes more at the spot than anywhere else in the surrounding 2 × 2 km, so it could be the blast zone of a small new crater.

Caveats. The spot is faint, and the hazy March 2023 image shows only a smudge, so the case rests mostly on the blocky, compressed February 2025 image. On these wind-textured plains, dust stripped by wind from a small patch, for example in the lee of a knob, can make a faint change like this. A dark mark nearby would show that the earlier images could have caught a new spot, but it is only weakly to moderately dark in them. No image from the high-resolution HiRISE camera covers it; the nearest ends 12.5 km away.

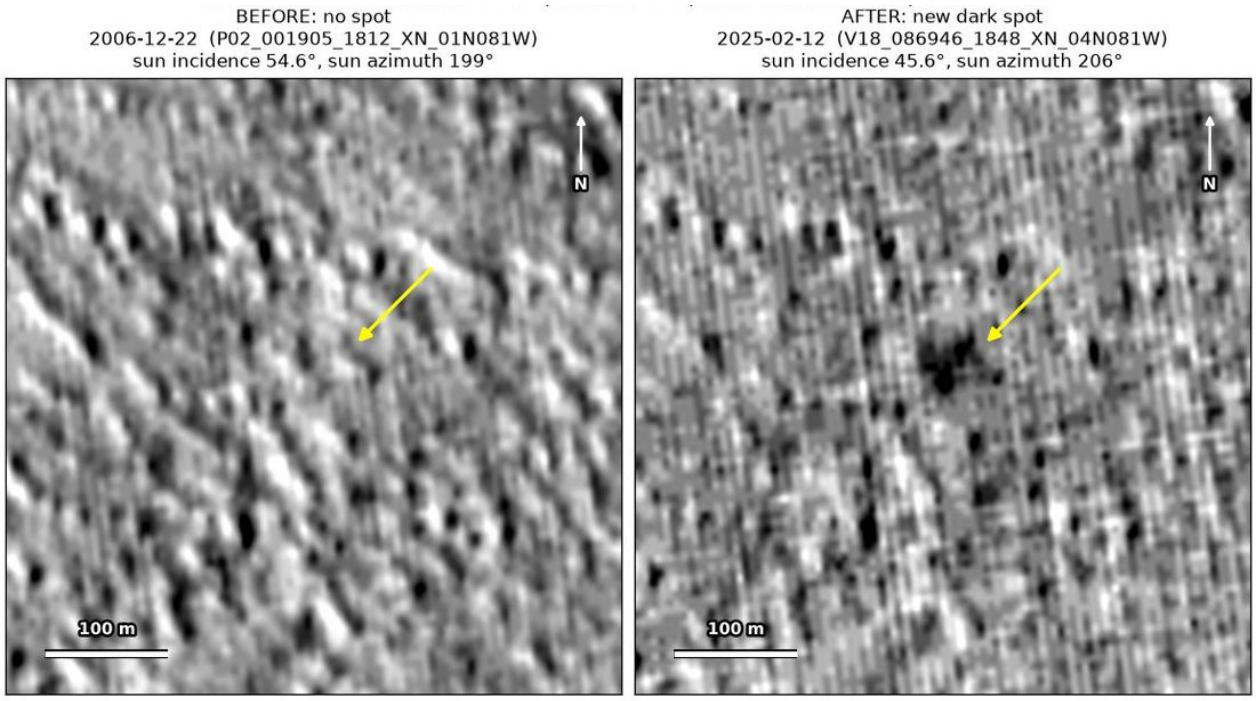

Figure 193: Candidate 75, before and after: the same 500 × 500 m of ground in 22 December 2006 (`P02_001905_1812_XN_01N081W`) and 12 February 2025 (`V18_086946_1848_XN_04N081W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

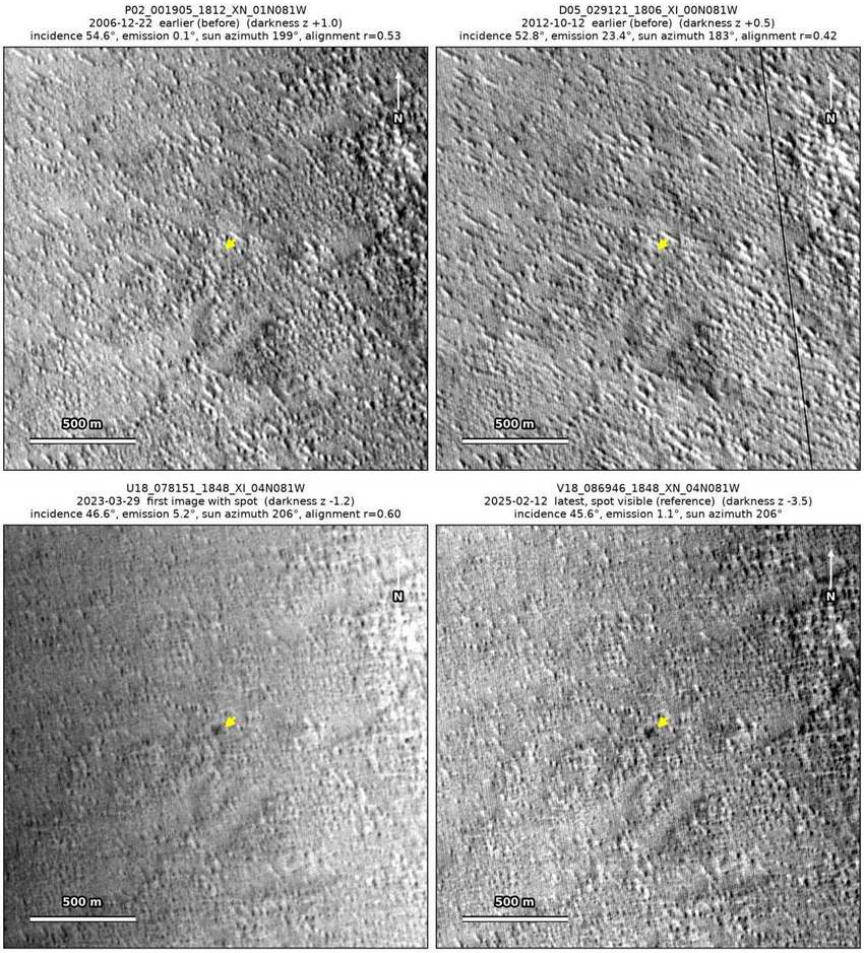

Figure 194: Candidate 75: four CTX images of the same 2×2 km patch, 2006 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 76: Western Terra Sabaea, 210 km south of Tuscaloosa crater

Location	3.4926° S, 27.7793° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	27 June 2016 and 27 May 2025
Images	2 without the spot (absent: 8 July 2010 and 27 June 2016); 1 with it (present: 27 May 2025)
Size	Compact, round dark spot about 55 m across (58 × 52 m), with a darker core about 20 m wide
HiRISE	none covers the spot (nearest footprint edge 24.7 km)
Nearest catalogued new impact	97 km (Daubar et al., 2022)

What changed. A compact, round dark spot about 55 m across, with a darker core about 20 m wide, appeared on flat ground 100 m south-west of a dark knobby ridge in western Terra Sabaea between June 2016 and May 2025. It shows in one CTX image so far.

Why it may be a new impact. Two earlier images, from July 2010 and June 2016, show no spot there. The 2010 image, taken under the same sun as the May 2025 one, lines up well and shows a dark knob on the ridge 260 m to the north-east as clearly as in 2025. The aligned before-and-after difference changes more at the spot than anywhere else in the surrounding 2 × 2 km, and a round spot with a sharp darker core could be the blast zone of a small new crater.

Caveats. Only one image shows it so far, so a short-lived surface change cannot be excluded, and the June 2016 image is striped. Dust stripped by wind from a small patch at the foot of the ridge, where bedforms (ripples or dunes built by the wind) lie just to the east, or a dark boulder or outcrop exposed by dust removal could also look like this. No image from the high-resolution HiRISE camera covers it; the nearest ends 24.7 km away.

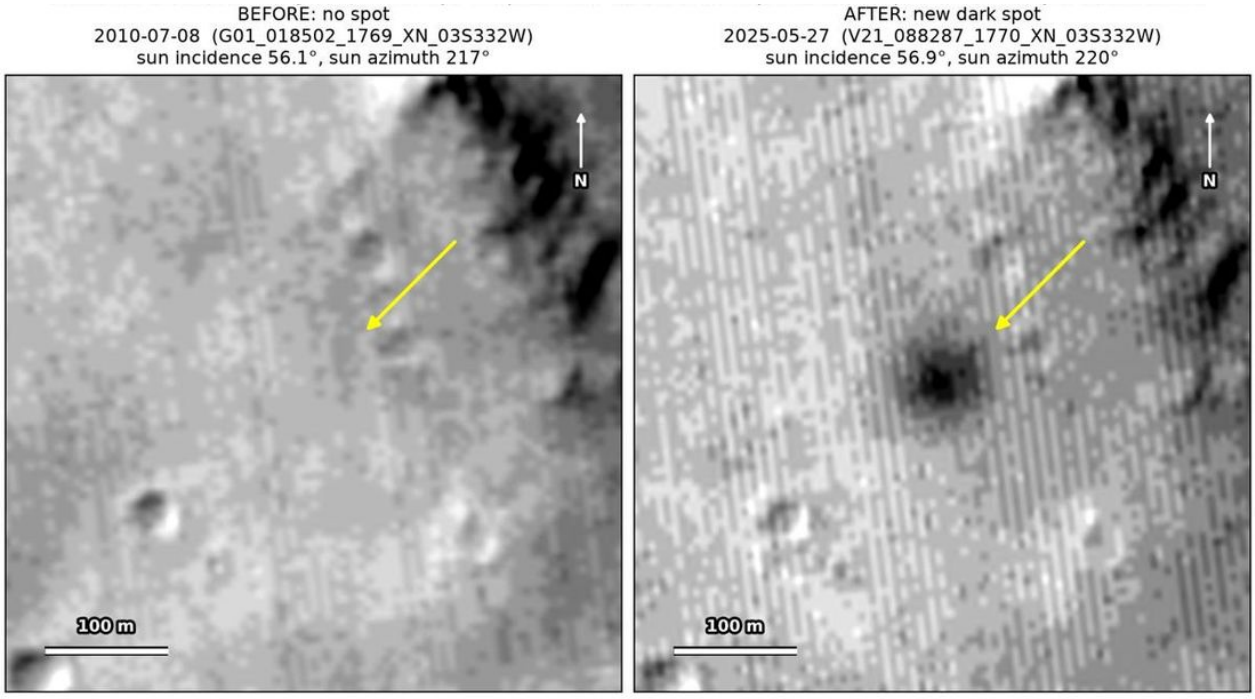

Figure 195: Candidate 76, before and after: the same 500 × 500 m of ground in 8 July 2010 (`G01_018502_1769_XN_03S332W`) and 27 May 2025 (`V21_088287_1770_XN_03S332W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

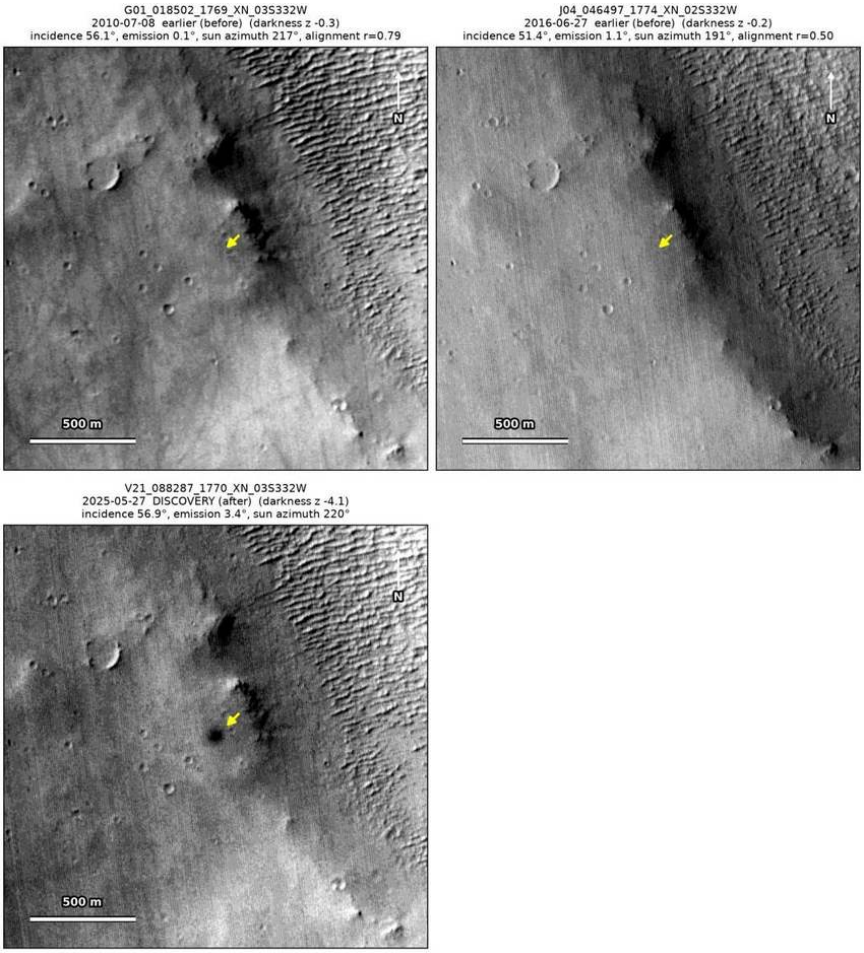

Figure 196: Candidate 76: three CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 77: Lava plains of Echus Palus

Location	13.3505° N, 277.9001° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	18 July 2020 and 13 October 2024
Images	3 without the spot (absent: 20 May 2007, 8 January 2008 and 18 July 2020); 2 with it (present: 13 October 2024 and 30 December 2024)
Size	Dark spot about 120 to 140 m across (134 × 162 m in the October 2024 image), with a darker core about 60 m wide
HiRISE	none covers the spot (nearest footprint edge 22.7 km)
Nearest catalogued new impact	130 km (Daubar et al., 2022)

What changed. A dark spot about 120 to 140 m across, with a darker core about 60 m wide, appeared on ridged lava plains of Echus Palus between July 2020 and October 2024. It shows in two CTX images from two orbits and is still there in December 2024.

Why it may be a new impact. Three earlier images, from 2007, 2008 and July 2020, show the nearby ridges clearly but no spot. The spot is round, sits on no landform and has a sharp darker core, with no source of dark sand nearby, and a new dark spot like this fits the blast zone of a small new crater.

Caveats. Only two images show it, two and a half months apart, and a dark mark nearby, which would show that the earlier images could have caught a new spot, is only weakly dark in them, far fainter than the spot. The spot could instead be a patch of the plain stripped of dust by wind, though it is round, sits on no landform and has a sharp darker core. It is most likely a separate event from a catalogued candidate impact site 27 km away, at which the nearest HiRISE image was aimed in April 2026. No image from the high-resolution HiRISE camera covers it; the nearest ends 22.7 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 85% that the spot is a new impact. It noted that the spot lies 58 km west of the mapped western edge of Echus Palus and that it had been followed for only 2.6 months. It placed the spot at 13.3537° N, 277.8947° E, 367 m from the position given above.

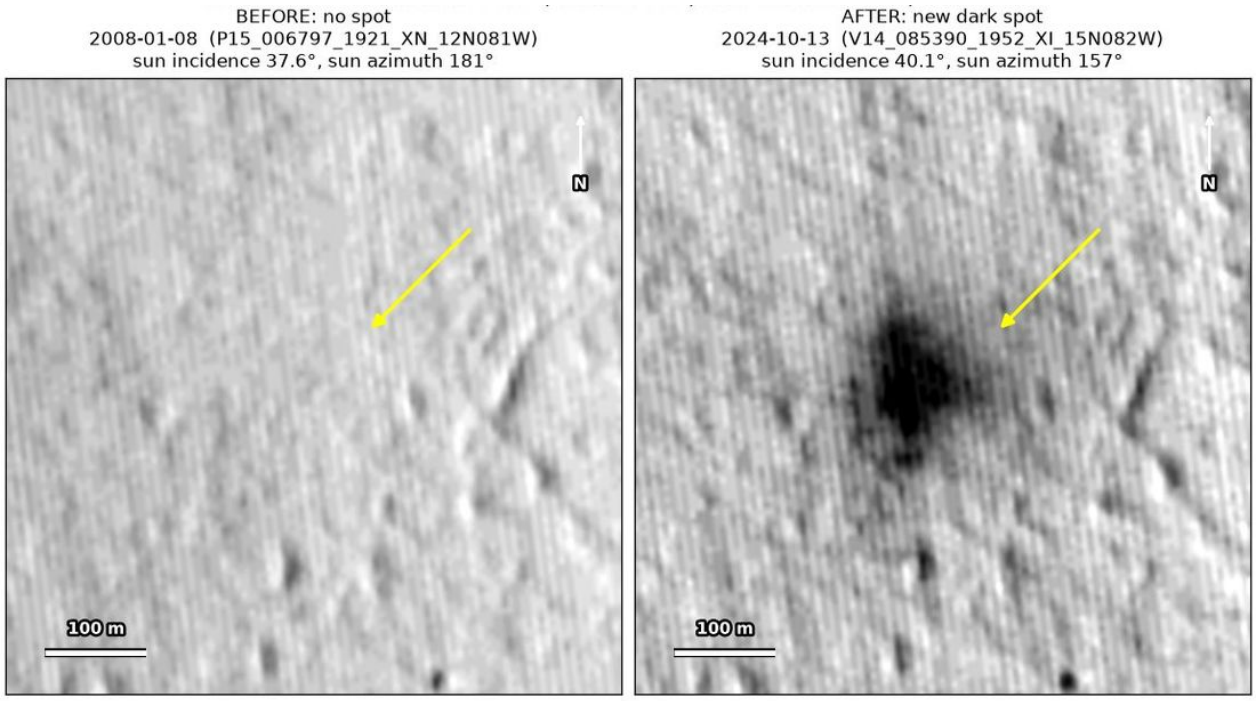

Figure 197: Candidate 77, before and after: the same 600 × 600 m of ground in 8 January 2008 (`P15_006797_1921_XN_12N081W`) and 13 October 2024 (`V14_085390_1952_XI_15N082W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

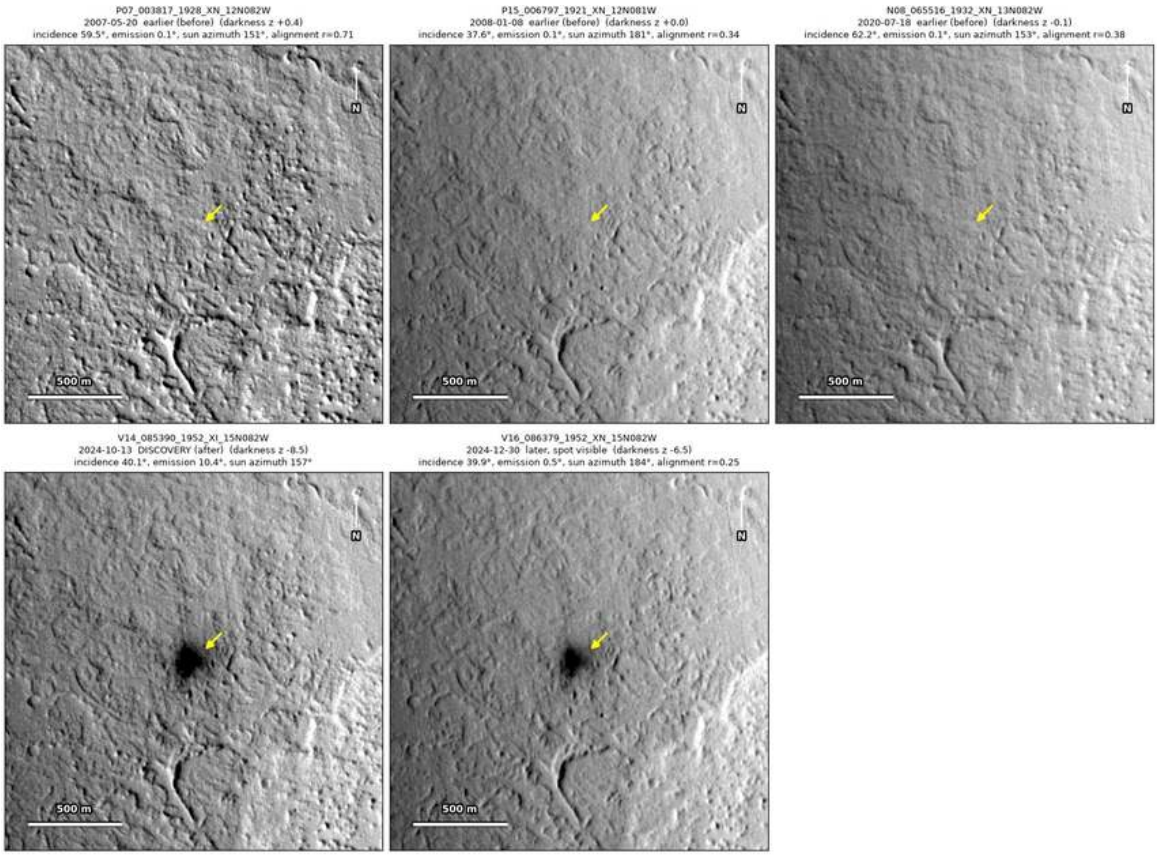

Figure 198: Candidate 77: five CTX images of the same 2×2 km patch, 2007 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

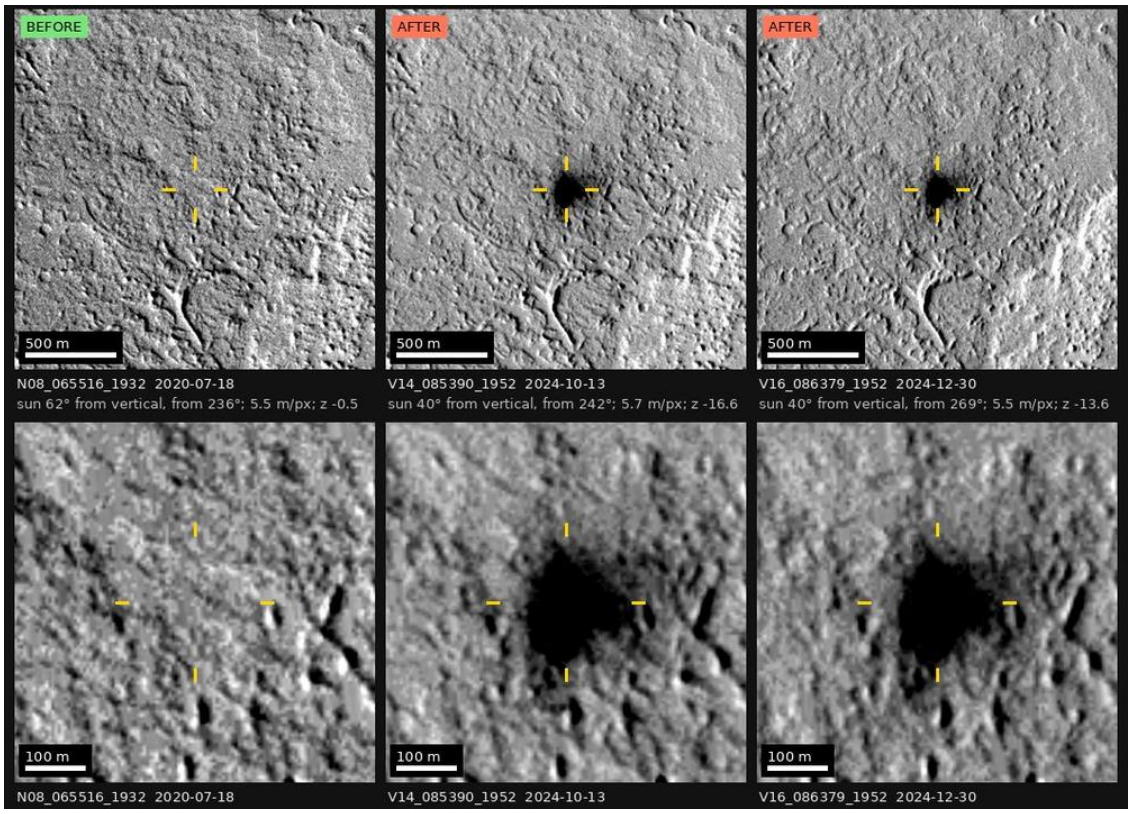

Figure 199: Candidate 77, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 78: Lava plains north of Ascraeus Mons

Location	16.0710° N, 256.4611° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	28 February 2008 and 12 February 2021
Images	1 without the spot (absent: 28 February 2008); 2 with it (present: 12 February 2021 and 11 January 2025)
Size	Compact dark spot about 70 m across (88 × 76 m in the February 2021 image), fading
HiRISE	none covers the spot (nearest footprint edge 28.9 km)
Nearest catalogued new impact	158 km (Daubar et al., 2022)

What changed. A compact dark spot about 70 m across appeared on knobby lava plains 250 km north of the summit of Ascraeus Mons between February 2008 and February 2021, and shows in two CTX images from two orbits. In 2021 a faint dark streak ran about 300 m north-west from it; by January 2025 the streak was gone and the spot was fading.

Why it may be a new impact. The only earlier image, from February 2008, was taken under the same sun as the January 2025 one; it shows knob shadows near the site but nothing like the spot, and a dark mark 250 m to the west does show in it. Four years later the streak had gone, as a ray of a fresh blast zone would, but the spot was still clear, longer than dust-devil marks usually last.

Caveats. Only one earlier image exists, so the spot could have formed at any time in a 13-year span. The 2021 streak could be the track of a dust devil that stalled, and dust stripped from a patch by wind is also possible. No image from the high-resolution HiRISE camera covers it; the nearest ends 28.9 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 85% that the spot is a new impact. It confirmed a dark tail about 80 m wide extending up to about 700 m north-west of the spot in 2021, which had faded by 2025 while the spot persisted; it reads this as a ray or as material thrown out by the impact. It placed the spot at 16.0735° N, 256.4609° E, 149 m from the position given above.

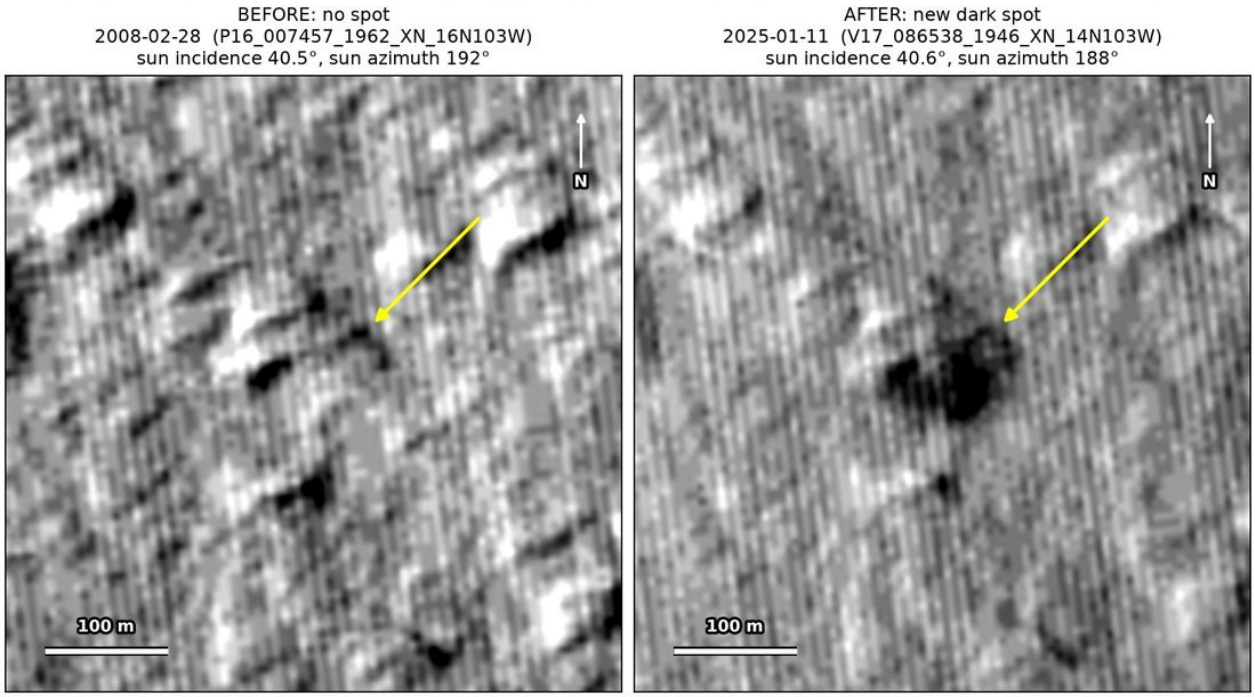

Figure 200: Candidate 78, before and after: the same 500 × 500 m of ground in 28 February 2008 (`P16_007457_1962_XN_16N103W`) and 11 January 2025 (`V17_086538_1946_XN_14N103W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

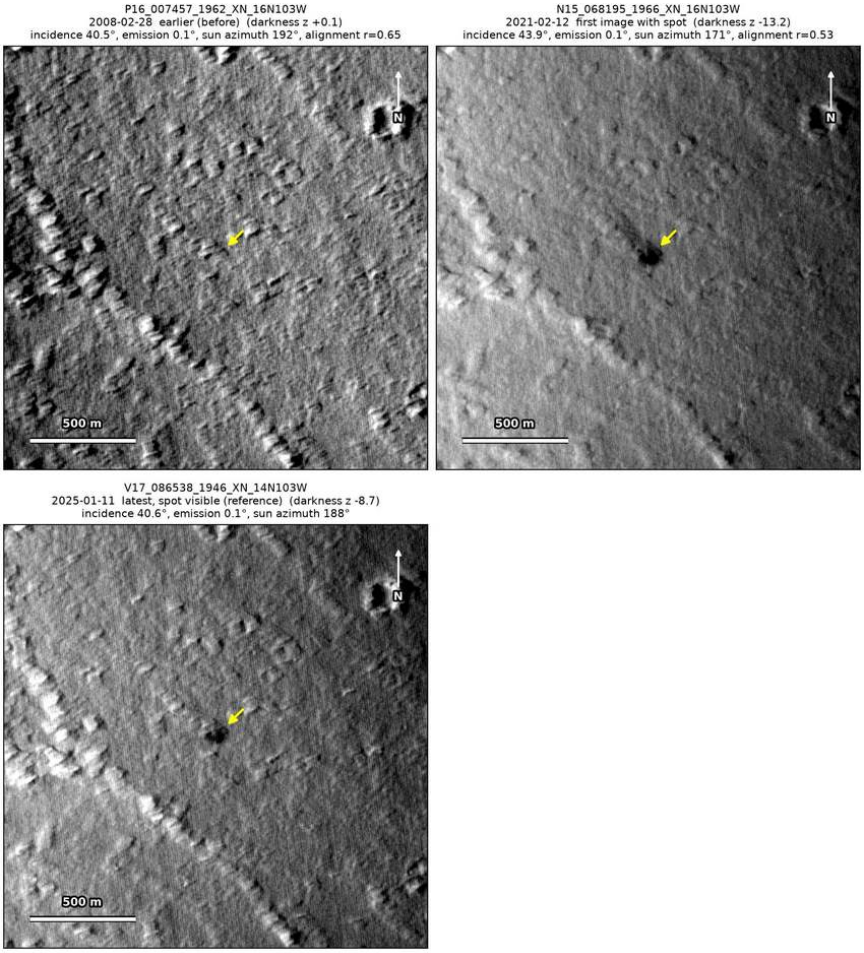

Figure 201: Candidate 78: three CTX images of the same 2×2 km patch, 2008 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

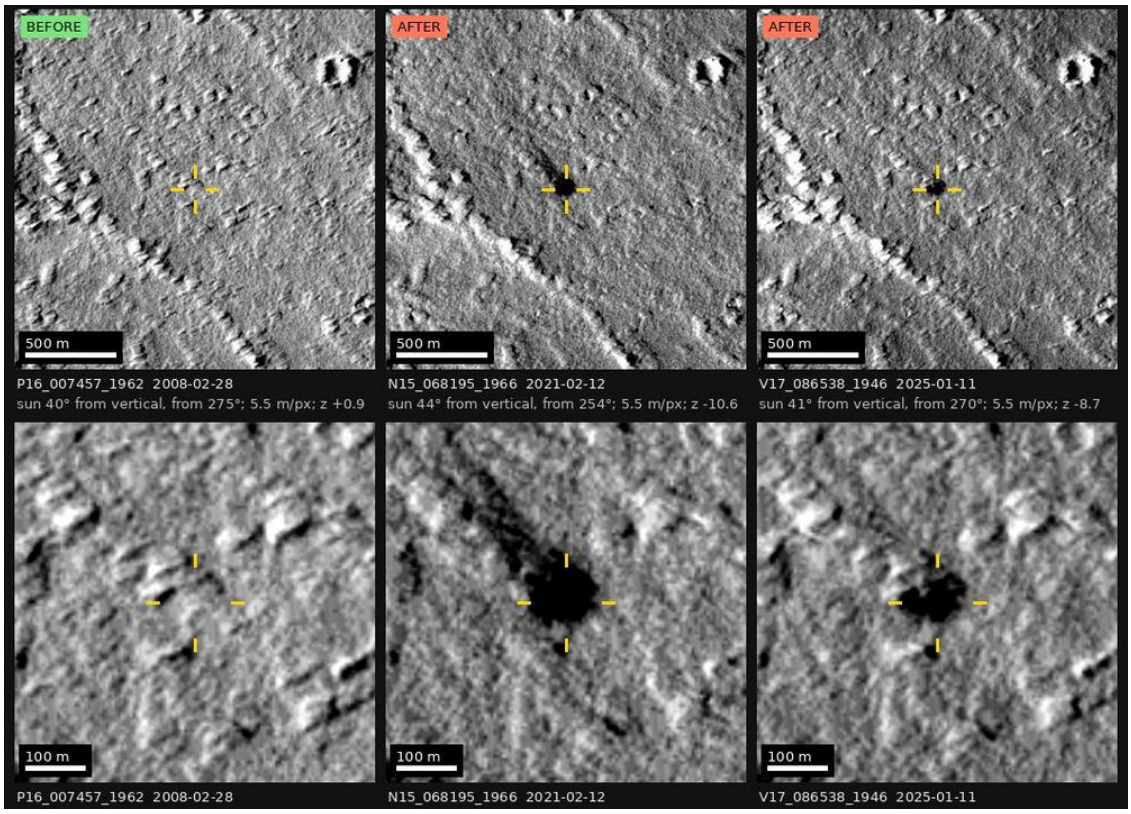

Figure 202: Candidate 78, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

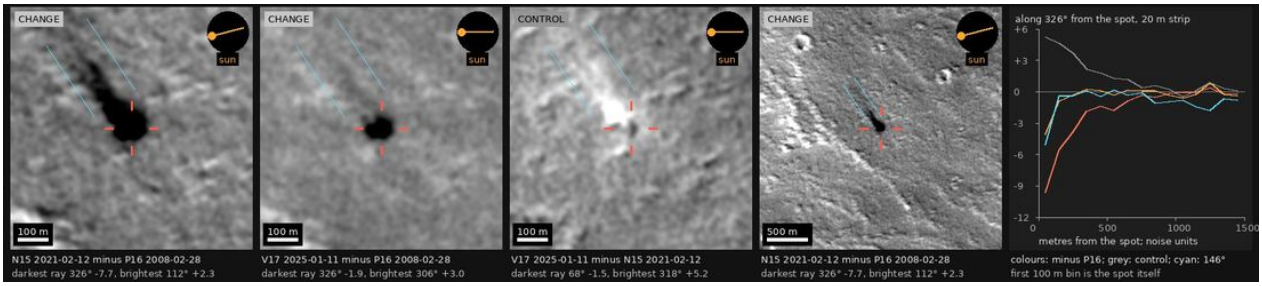

Figure 203: Candidate 78. The independent check: test of the tail. Each panel is a later image minus an earlier one, matched in brightness, so that ground that darkened is black and ground that brightened is white. In February 2021 a dark tail extends north-west from the spot along the cyan lines (first panel, and over a wider area in the fourth); by January 2025 only the spot is left (second panel), and between the two the ground along the tail brightened again (third panel). The graph traces the change along the tail's direction. NASA/JPL-Caltech/MSSS.

Candidate 79: Smooth plains in Arabia Terra, 97 km west of Flammarion crater

Location	24.8734° N, 46.5175° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	21 June 2022 and 6 December 2024
Images	2 without the spot (absent: 3 October 2010 and 21 June 2022); 2 with it (present: 6 December 2024 and 17 October 2025)
Size	Compact, very dark spot about 55 to 60 m across (58 × 58 m in the December 2024 image, 56 × 54 m in the October 2025 one)
HiRISE	none covers the spot (nearest footprint edge 18.6 km)
Nearest catalogued new impact	217 km (Daubar et al., 2022)

What changed. A compact, very dark spot about 55 to 60 m across appeared on smooth plains of Arabia Terra, 97 km west of Flammarion crater, between June 2022 and December 2024. It shows in two CTX images from two orbits and is still as dark 10 months later, in October 2025.

Why it may be a new impact. An October 2010 image, taken under the same sun as the October 2025 one, and a June 2022 image show no spot there, although a small fresh crater 300 m to the east and the pits around the site show in all four images. The aligned before-and-after difference picks out the spot and nothing else in the surrounding 2 × 2 km, no dust-devil tracks are visible, and a round, sharp, very dark spot fits the blast zone of a small new crater.

Caveats. The June 2022 image has a low sun and lines up only moderately well, so the spot could have formed as early as October 2010. A dust devil that stalled is one alternative, though no dust-devil tracks are visible and the spot was still as dark 10 months later; dust stripped by wind from a patch is another, though wind seldom leaves so round and sharp a spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 18.6 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 85% that the spot is a new impact. It measured the spot as 95–105 m across at half depth, larger than the size given above. It placed the spot at 24.8751° N, 46.5177° E, 103 m from the position given above.

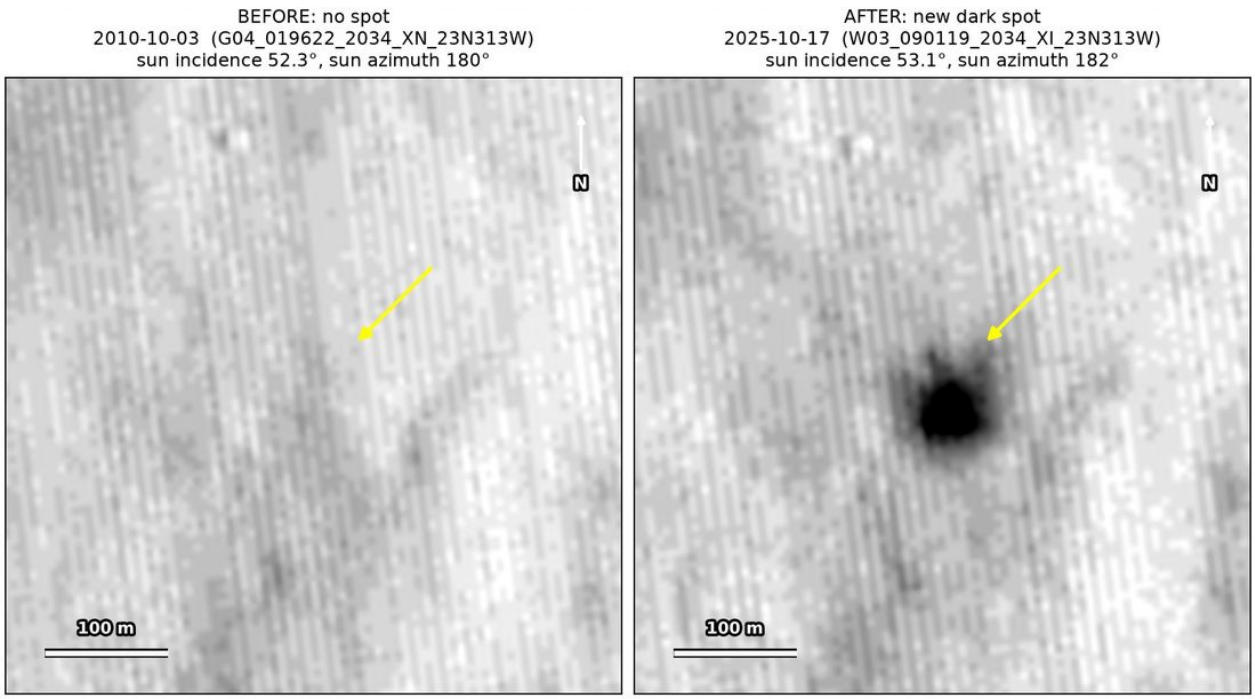

Figure 204: Candidate 79, before and after: the same 500 × 500 m of ground in 3 October 2010 (`G04_019622_2034_XN_23N313W`) and 17 October 2025 (`W03_090119_2034_XI_23N313W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

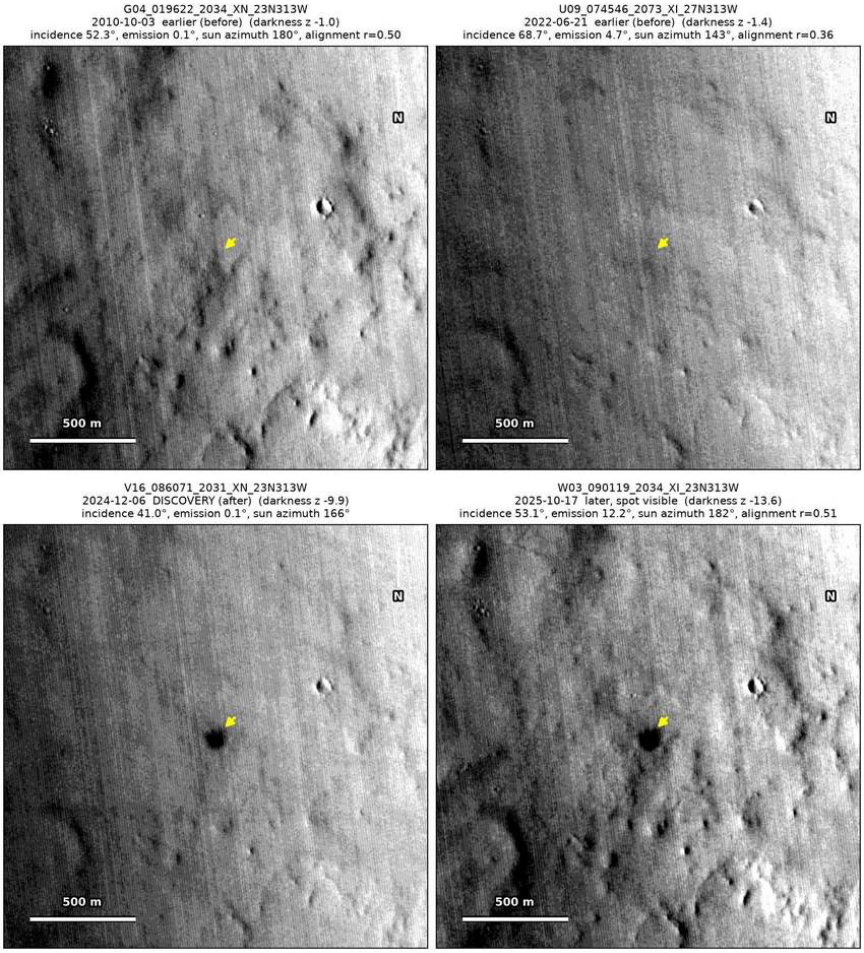

Figure 205: Candidate 79: four CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

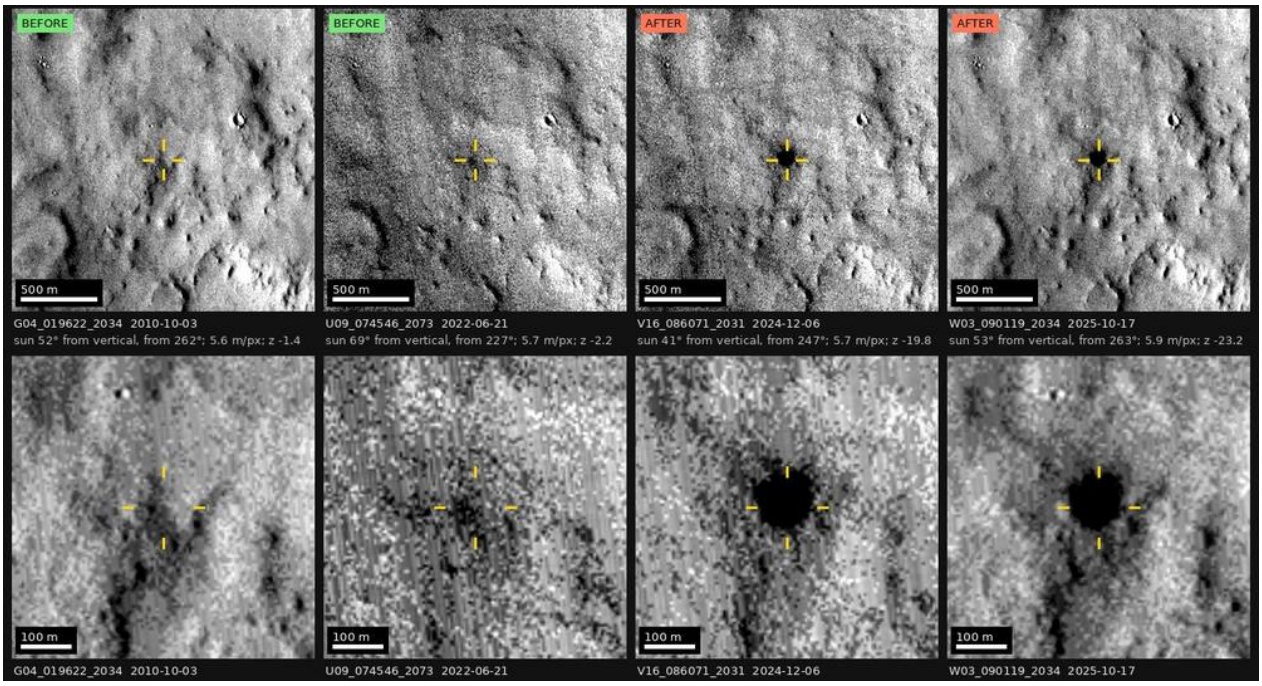

Figure 206: Candidate 79, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 80: Arabia Terra, 116 km north of Barth crater

Location	9.3943° N, 25.7511° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	11 January 2023 and 14 November 2024
Images	5 without the spot (absent: 28 January 2010, 18 June 2012, 1 October 2012, 22 October 2012 and 11 January 2023); 2 with it (present: 14 November 2024 and 3 January 2025)
Size	Compact dark spot about 25 m across (22×28 m in the November 2024 image, 20×28 m in the January 2025 one)
HiRISE	none covers the spot (nearest footprint edge 17.9 km)
Nearest catalogued new impact	141 km (Daubar et al., 2022)

What changed. A small, compact dark spot about 25 m across appeared on smooth ground in Arabia Terra, 116 km north of Barth crater, between January 2023 and November 2024. It shows in two CTX images from two orbits, taken seven weeks apart.

Why it may be a new impact. Five earlier images, from 2010 to January 2023, show no spot there. The January 2023 image was taken under the same sun as the November 2024 one, and the January 2010 image under nearly the same sun as the January 2025 one. A dark blob 100 m to the south was there all along, but the aligned before-and-after difference changes most at the spot, and a compact new dark spot is how the blast zone of a small new crater looks.

Caveats. At about 25 m the spot is only four or five pixels across, so its shape is poorly resolved. It has been followed for only seven weeks, so a dust devil that stalled is not excluded; dark material moved by wind into a small round patch on smooth ground would be unusual and would have had to stay unchanged for those seven weeks. The earlier images line up only moderately well, and no dark mark like the spot lies within 300 m to show that they could have caught a new one. No image from the high-resolution HiRISE camera covers it; the nearest ends 17.9 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 80% that the spot is a new impact. It placed the spot at 9.3959° N, 25.7512° E, 93 m from the position given above.

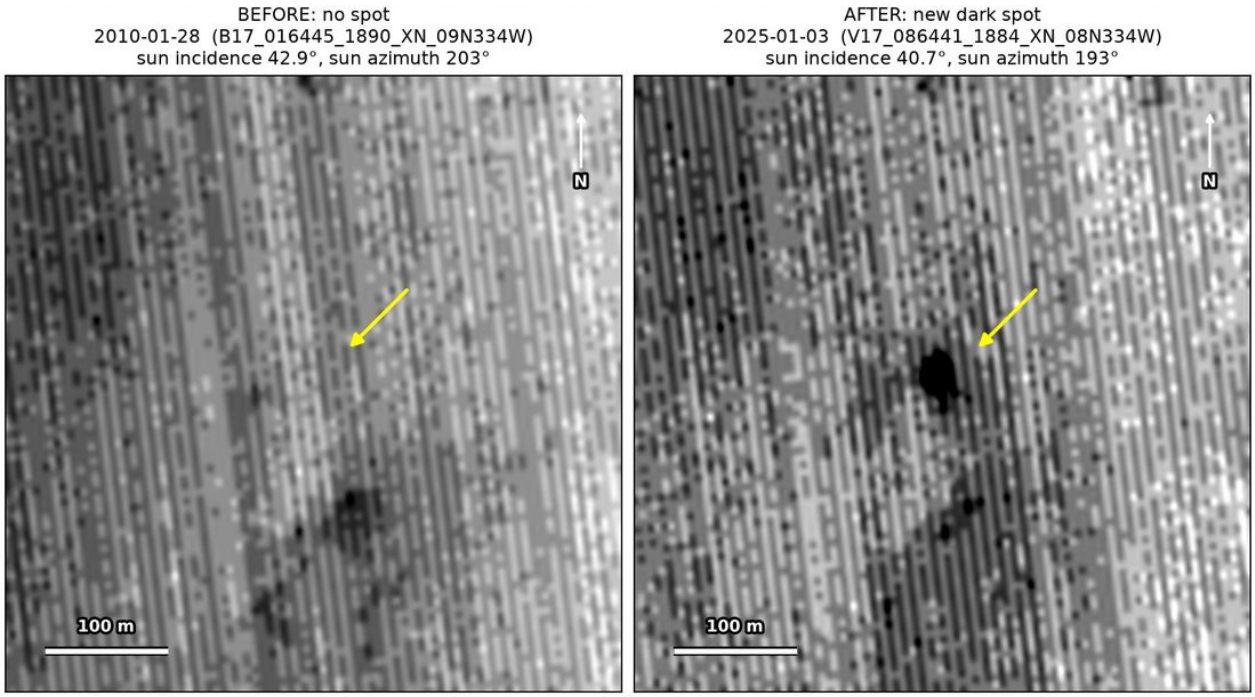

Figure 207: Candidate 80, before and after: the same 500 × 500 m of ground in 28 January 2010 (`B17_016445_1890_XN_09N334W`) and 3 January 2025 (`V17_086441_1884_XN_08N334W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

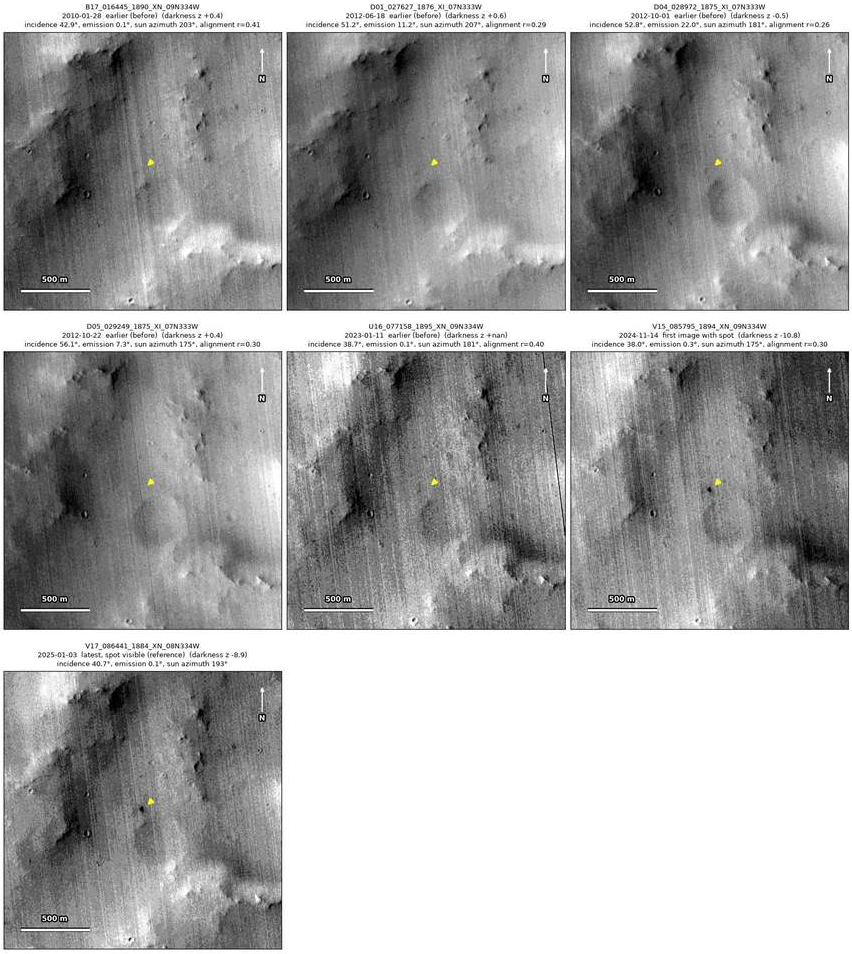

Figure 208: Candidate 80: seven CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

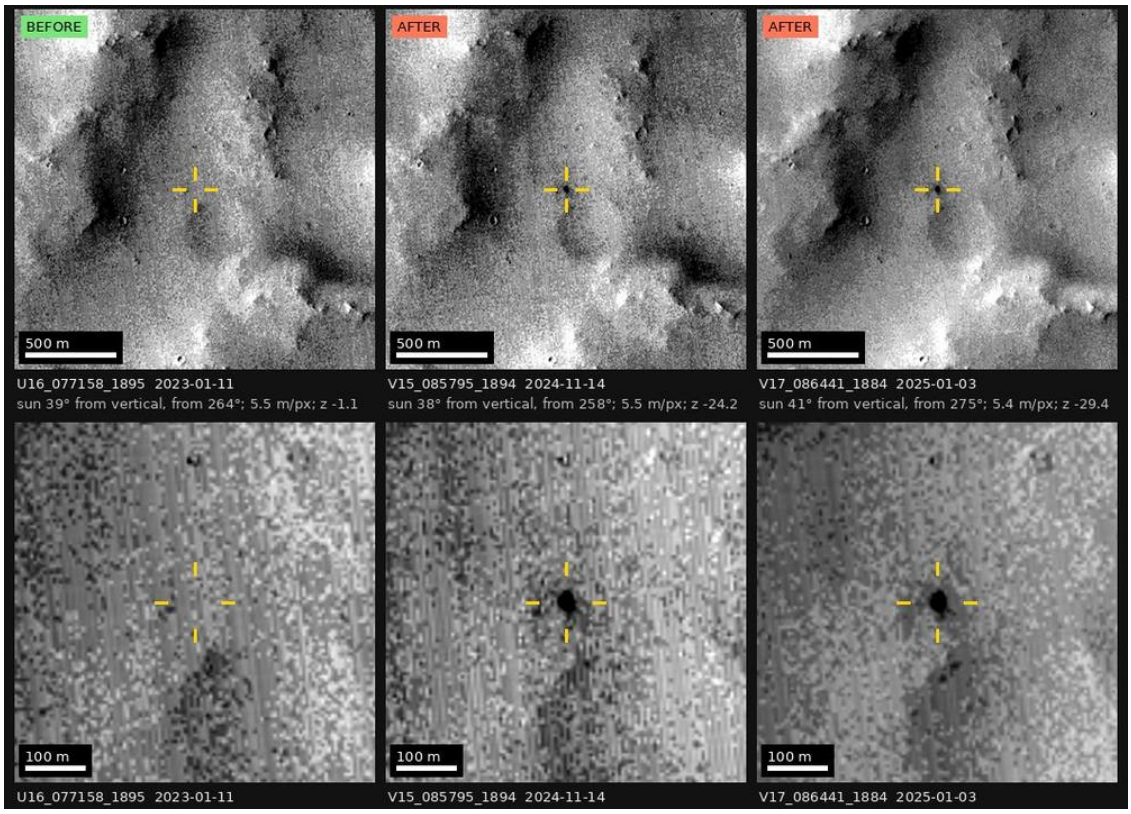

Figure 209: Candidate 80, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 81: Plains west of Ascraeus Mons

Location	11.8779° N, 251.7127° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	16 March 2022 and 2 December 2024
Images	7 without the spot (absent: 7 June 2008, 18 June 2008, 16 June 2010, 12 October 2013, 1 May 2015, 11 May 2015 and 16 March 2022); 1 with it (present: 2 December 2024)
Size	Compact, very dark spot about 60 m across (60 × 64 m), with a diffuse halo
HiRISE	none covers the spot (nearest footprint edge 6.3 km)
Nearest catalogued new impact	54 km (Daubar et al., 2022)

What changed. A compact, very dark spot about 60 m across, inside a diffuse dark halo, appeared on knobby lava plains west of Ascraeus Mons, 25 km west of Ascraeus Mensa, between March 2022 and December 2024. It shows in one CTX image so far.

Why it may be a new impact. Seven earlier images, from 2008 to March 2022, show nothing as dark there, among them an October 2013 image taken under nearly the same sun as the December 2024 one. The aligned before-and-after difference picks out the spot and nothing else in the surrounding 2 × 2 km, and a compact, very dark spot inside a diffuse halo that fades into the surrounding ground could be the blast zone of a small new crater.

Caveats. Only one image shows it so far, so an image defect or a dust devil that stalled cannot be fully excluded. The ground is almost featureless in CTX images: the earlier images line up poorly, two of them, of 7 June 2008 and 16 June 2010, show no texture at all, and a dark mark that would show they could have caught a new spot does not appear in any of them, so a faint dark patch already there cannot be ruled out. No image from the high-resolution HiRISE camera covers it; the nearest ends 6.3 km away.

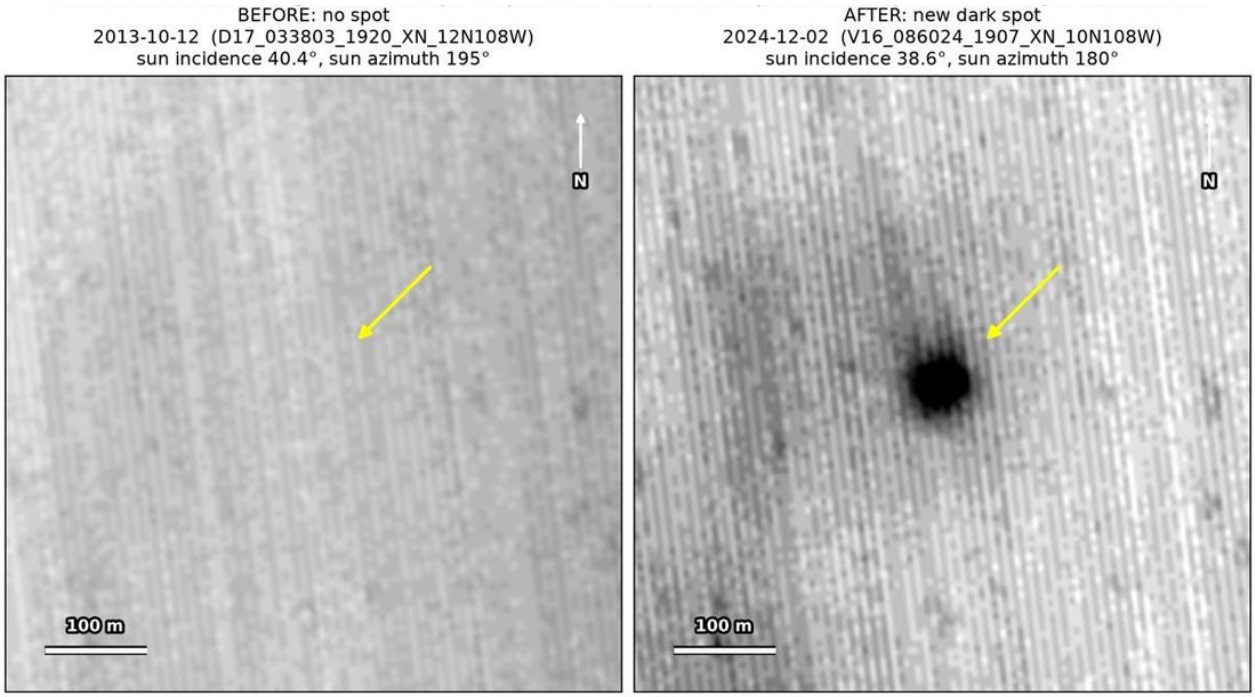

Figure 210: Candidate 81, before and after: the same 600 × 600 m of ground in 12 October 2013 (`D17_033803_1920_XN_12N108W`) and 2 December 2024 (`V16_086024_1907_XN_10N108W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

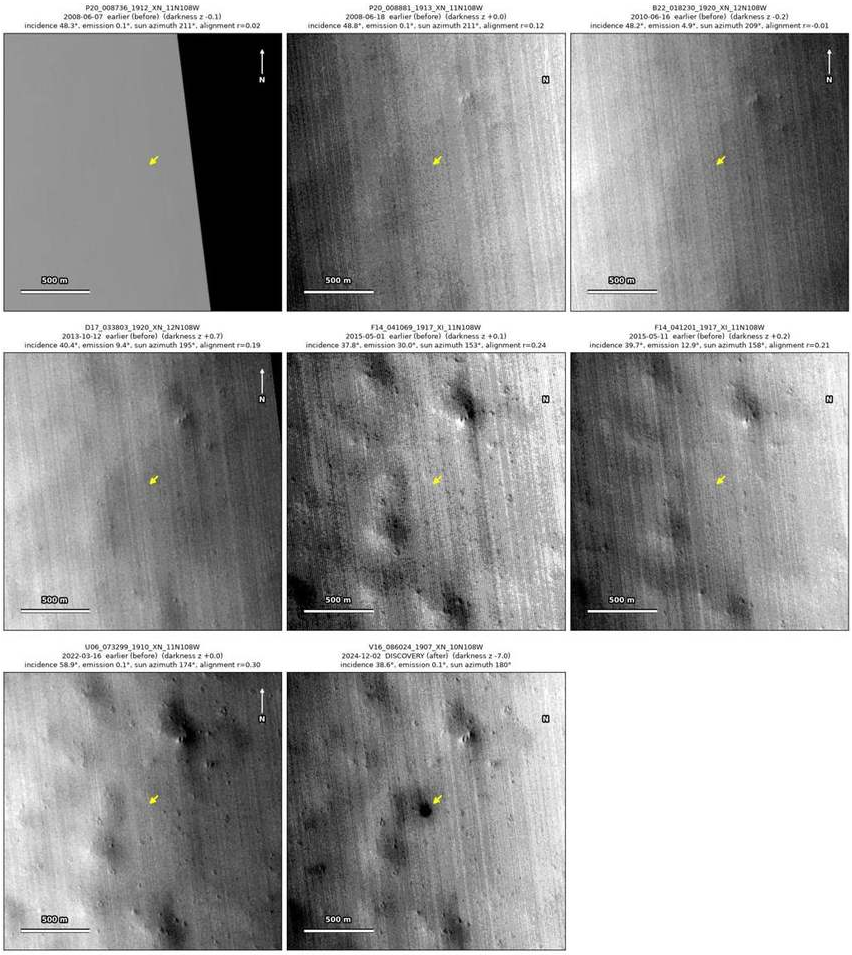

Figure 211: Candidate 81: eight CTX images of the same 2×2 km patch, 2008 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 82: Lava plains in south-eastern Ulysses Fossae

Location	8.3239° N, 239.2944° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	13 May 2024 and 21 December 2024
Images	3 without the spot (absent: 12 March 2010, 19 July 2022 and 13 May 2024); 1 with it (present: 21 December 2024)
Size	Compact dark spot about 40 m across (34 × 44 m)
HiRISE	none covers the spot (nearest footprint edge 14.4 km)
Nearest catalogued new impact	20 km (Daubar et al., 2022)

What changed. A compact dark spot about 40 m across appeared on flat lava plains about 150 m west of the foot of a ridge in south-eastern Ulysses Fossae between May and December 2024. It shows in one CTX image so far.

Why it may be a new impact. Three earlier images, from 2010, 2022 and May 2024, line up well and show no spot there, while knobs and small dark marks around the site show in every image; the last was taken only 222 days before the image with the spot. The spot has a diffuse margin and sits on the ground texture, which makes a flaw in the image less likely, and a compact new dark spot like this could be the blast zone of a small new crater.

Caveats. Only one image shows it so far, so an image defect or a dust devil that stalled cannot be fully excluded. Dark material shed from the ridge 150 m away is also possible, though no streak links the two. No image from the high-resolution HiRISE camera covers it; the nearest ends 14.4 km away.

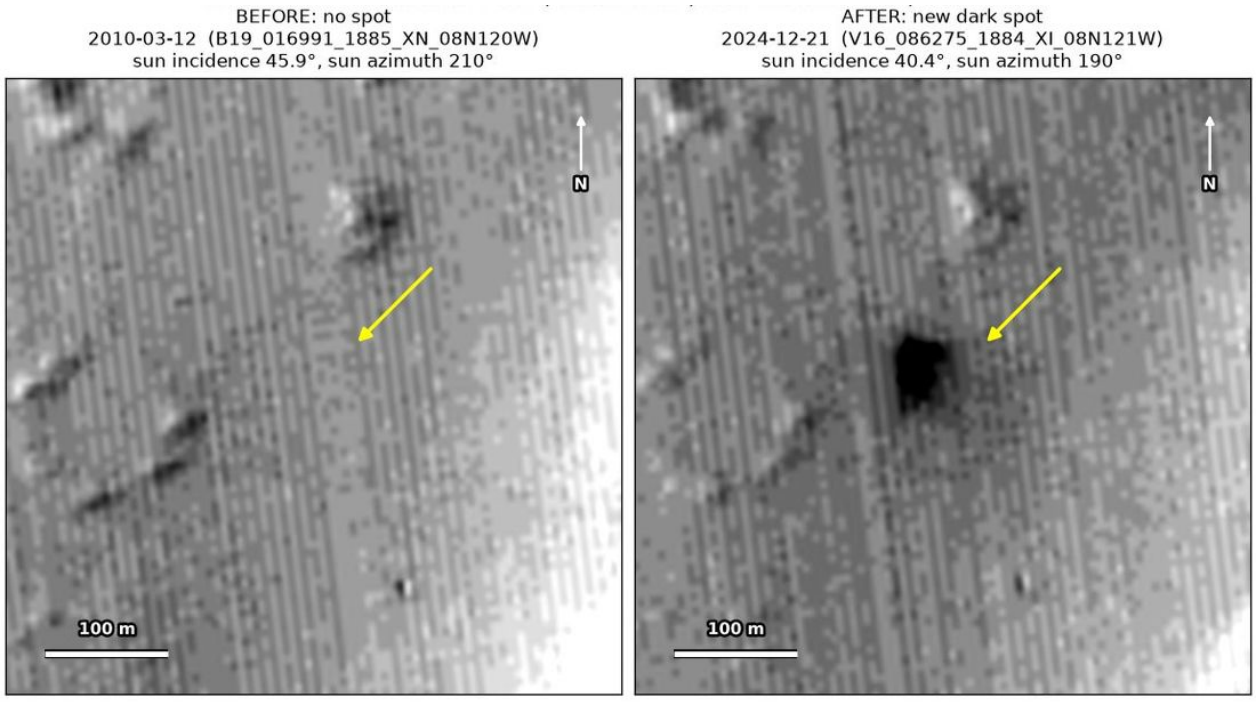

Figure 212: Candidate 82, before and after: the same 500 × 500 m of ground in 12 March 2010 (`B19_016991_1885_XN_08N120W`) and 21 December 2024 (`V16_086275_1884_XI_08N121W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

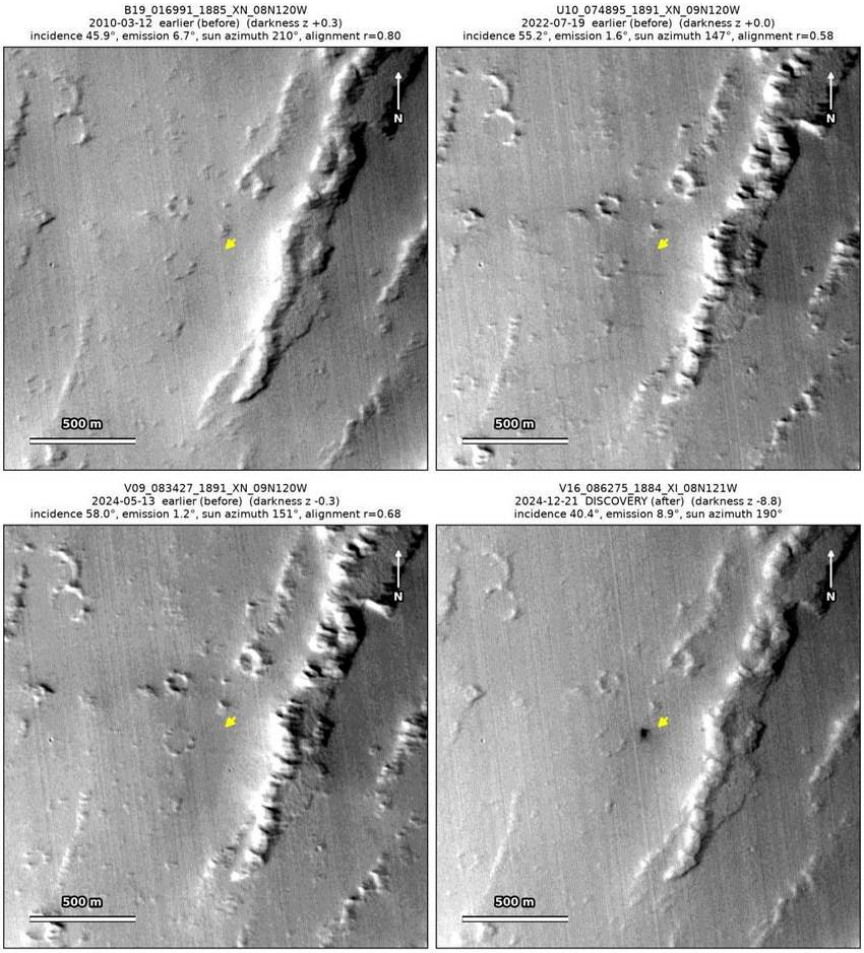

Figure 213: Candidate 82: four CTX images of the same 2×2 km patch, 2010 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 83: Lucus Planum, in the Medusae Fossae Formation

Location	4.6753° S, 179.5549° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 50%
Formed between	16 March 2010 and 9 December 2024
Images	1 without the spot (absent: 16 March 2010); 3 with it (present: 9 December 2024, 19 February 2025 and 13 May 2025)
Size	Diffuse dark patch about 80 to 100 m across (84 × 64 m in the aligned before-and-after difference), with no distinct core
HiRISE	none covers the spot (nearest footprint edge 8.6 km)
Nearest catalogued new impact	289 km (Daubar et al., 2022)

What changed. A diffuse dark patch about 80 to 100 m across appeared among the wind-eroded ridges of Lucus Planum, 139 km east of Tartarus Scopulus, between March 2010 and December 2024. It shows in three CTX images from three orbits, faintly in the first and clearly in the two from 2025.

Why it may be a new impact. The only earlier image, from March 2010, shows no patch there. It was taken under the same sun as the February 2025 image, its ridges line up with those of the later images, and a dark mark nearby shows in it. Unlike the floor-wide darkenings with no centre seen elsewhere on the Medusae Fossae Formation, which are not among the candidates here, the change is local and centred, darkening both the ground and the shaded troughs, so it could mark a small new impact.

Caveats. Wind removing dust from a patch of the ridged surface fits the diffuse outline and the lack of a distinct core about as well as an impact does, and the single earlier image leaves a 15-year span in which the patch could have formed. It lies within the area and period of a catalogue of new impacts around the InSight lander (Bickel et al. 2025), but every entry in it has a HiRISE image and this site has none. No image from the high-resolution HiRISE camera covers it; the nearest ends 8.6 km away.

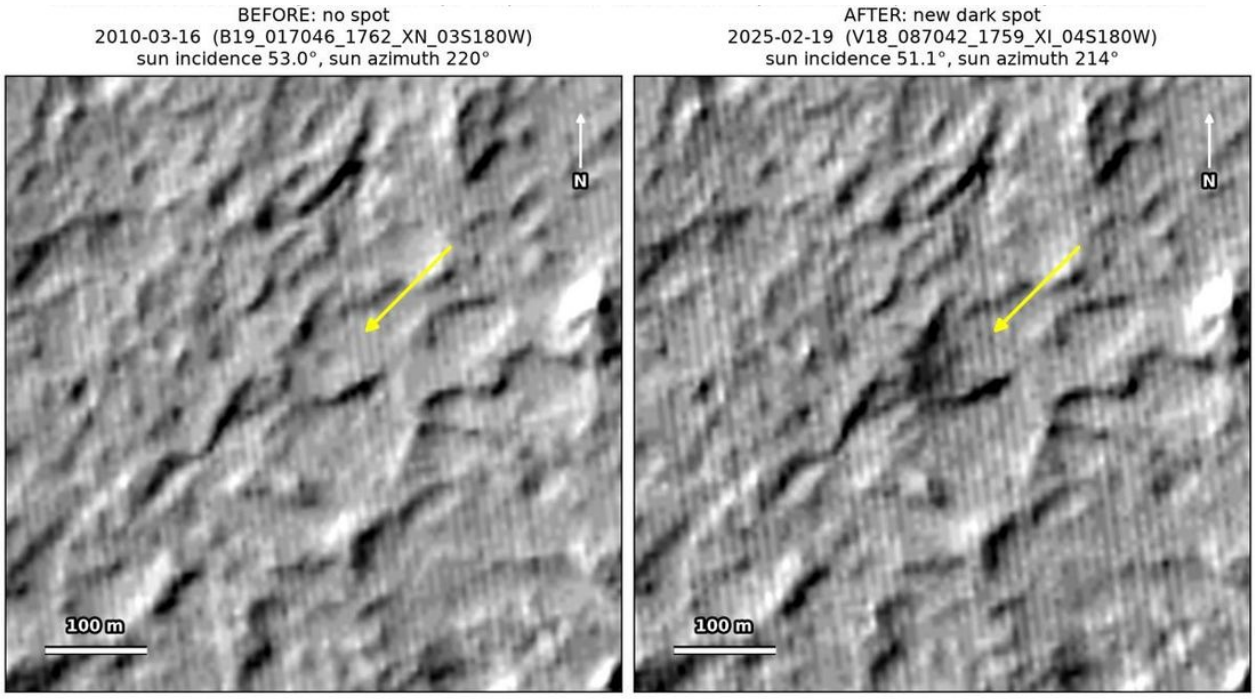

Figure 214: Candidate 83, before and after: the same 600 × 600 m of ground in 16 March 2010 (`B19_017046_1762_XN_03S180W`) and 19 February 2025 (`V18_087042_1759_XI_04S180W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

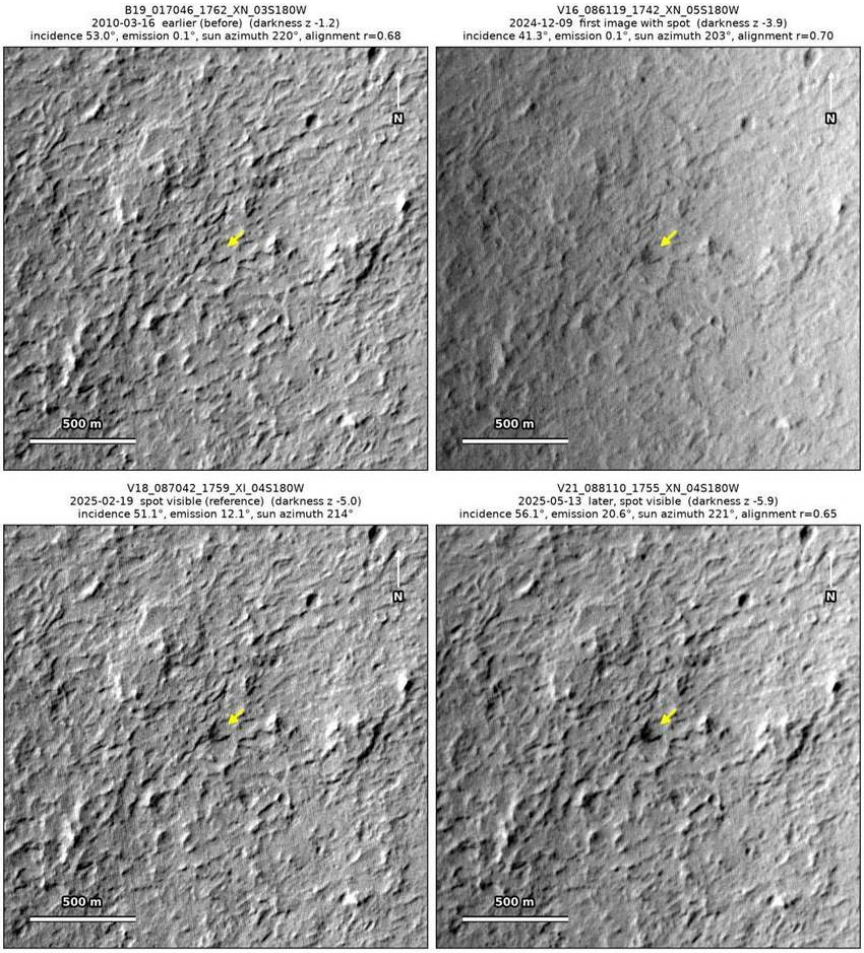

Figure 215: Candidate 83: four CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 84: Knobby plains in northern Cyane Fossae, 250 km north-east of candidate 21

Location	29.3380° N, 239.8719° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	23 August 2020 and 25 December 2022
Images	3 without the spot (absent: 20 December 2007, 30 December 2011 and 23 August 2020); 4 with it (present: 25 December 2022, 12 August 2024, 11 October 2024 and 28 October 2024)
Size	Compact dark spot about 80 m across (94 × 82 m in the image of 28 October 2024, 104 × 92 m in that of December 2022)
HiRISE	none covers the spot (nearest footprint edge 22.0 km)
Nearest catalogued new impact	89 km (Daubar et al., 2022)

What changed. A compact dark spot about 80 m across appeared on knobby plains in northern Cyane Fossae, 4 km west of a chain of pits, between August 2020 and December 2022. It shows in four CTX images from four orbits and is still there in October 2024.

Why it may be a new impact. Three well-aligned earlier images, from 2007, 2011 and August 2020, show no spot there; the December 2007 one was taken under the same sun as the December 2022 image, and a dark mark 250 m to the east-north-east shows in the 2007 and 2020 images. The aligned before-and-after difference picks out the spot and nothing else in the surrounding 2 × 2 km, and a compact dark spot still there 22 months later is how the blast zone of a small new crater looks.

Caveats. A faint, thin dark line extends south-west from the spot in the 2024 images and may show faintly in the August 2020 one, so the spot could mark where a dust devil stalled at the end of its track, although it is far darker and wider than the line. No image from the high-resolution HiRISE camera covers it; the nearest ends 22.0 km away.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 75% that the spot is a new impact. It found no support for a dust devil stalling at the end of a track: no ray or track from the spot stands out in any image. It placed the site in central rather than northern Cyane Fossae and estimated about 75%. It placed the spot at 29.3396° N, 239.8731° E, 113 m from the position given above.

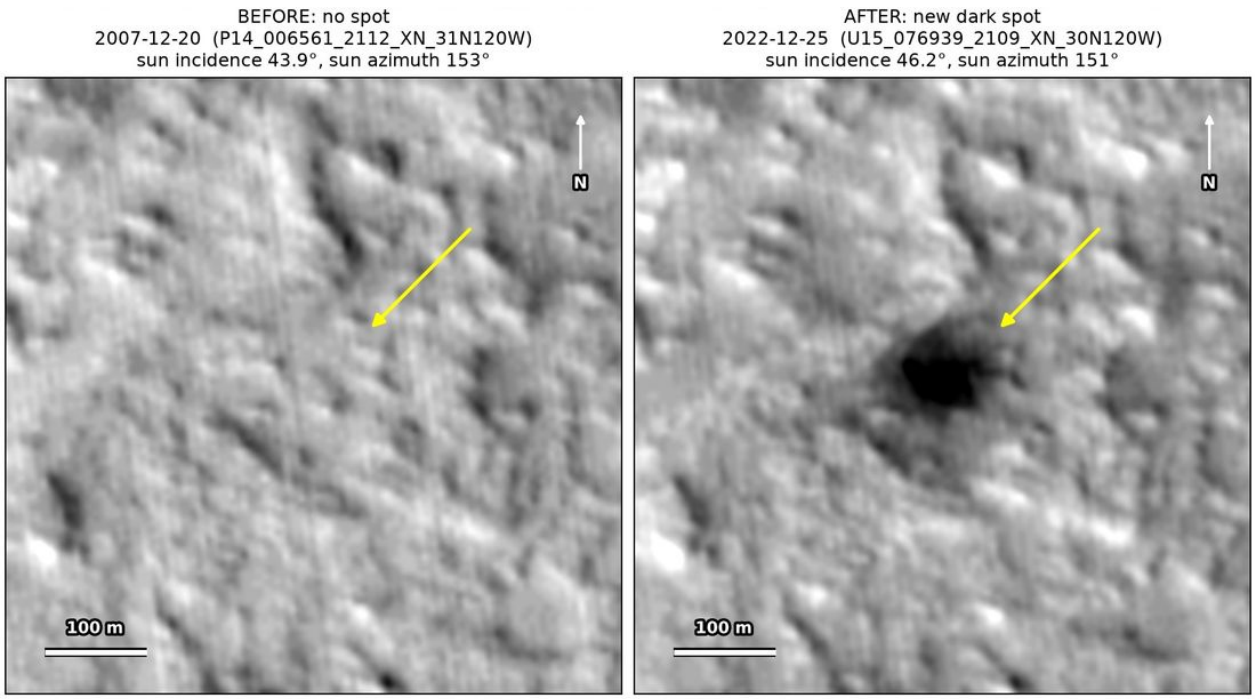

Figure 216: Candidate 84, before and after: the same 600 × 600 m of ground in 20 December 2007 (`P14_006561_2112_XN_31N120W`) and 25 December 2022 (`U15_076939_2109_XN_30N120W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

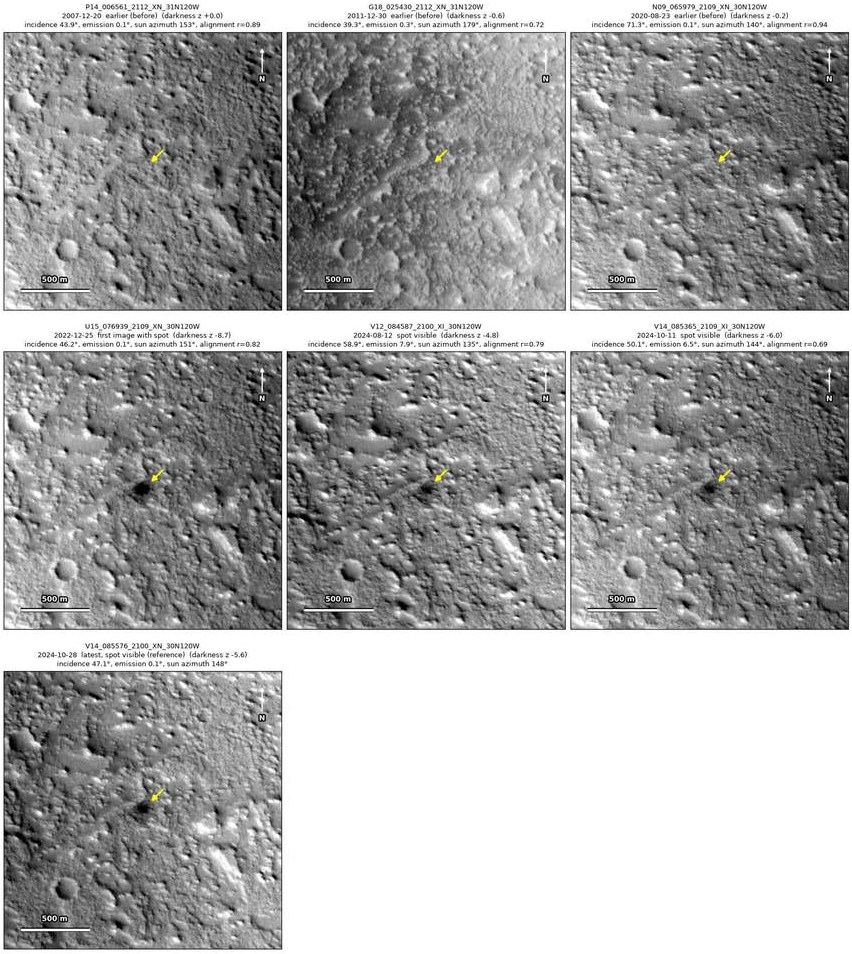

Figure 217: Candidate 84: seven CTX images of the same 2×2 km patch, 2007 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

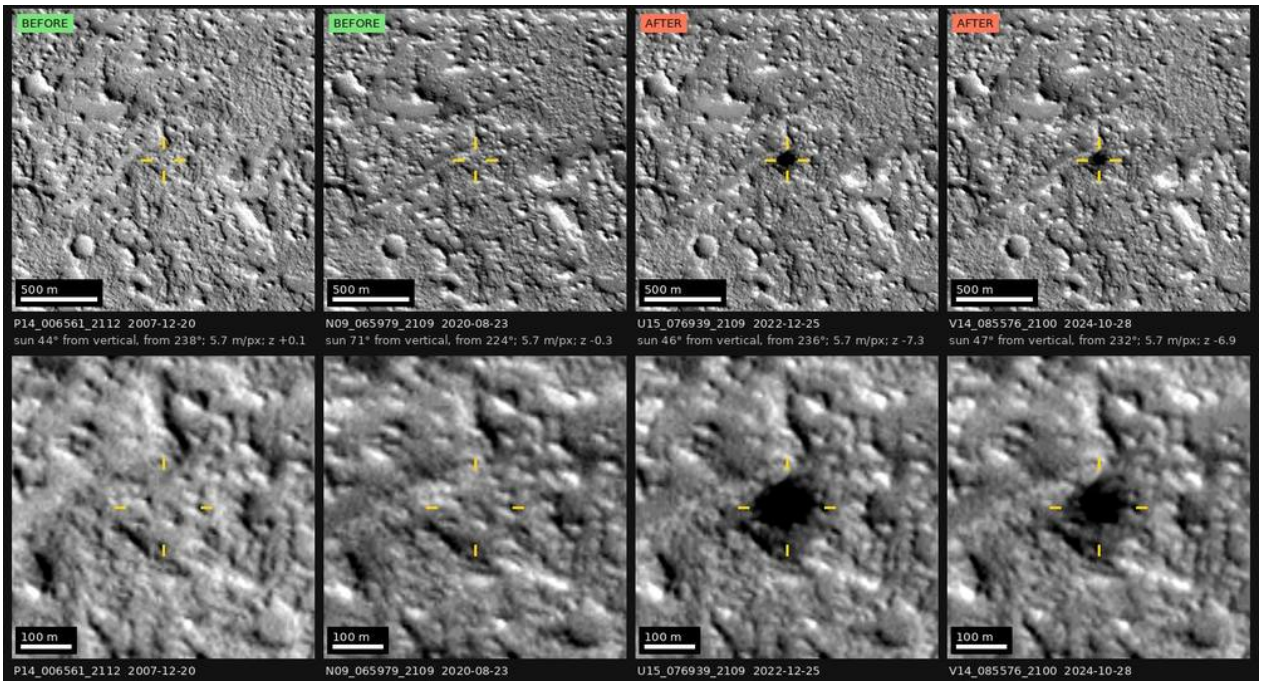

Figure 218: Candidate 84, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

Candidate 85: Wind-ridged ground in the Medusae Fossae Formation, 295 km west of Gordii Dorsum

Location	4.2241° N, 210.8642° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	12 March 2020 and 21 December 2024
Images	2 without the spot (absent: 18 May 2009 and 12 March 2020); 3 with it (present: 21 December 2024, 30 January 2025 (CTX and HiRISE) and 25 July 2025)
Size	Dark spot about 60 m across (66 × 76 m), with ragged, ray-like margins; in the HiRISE image, a dark zone 76 × 81 m with a pit about 2 m across
HiRISE	ESP_086777_1840 (2025-01-30, same orbit as V17, terrain sample, not impact-targeted)
Nearest catalogued new impact	86 km (Daubar et al., 2022)

What changed. A compact dark spot about 60 m across, with ragged, ray-like edges, appeared on wind-ridged ground in the Medusae Fossae Formation, 295 km west of Gordii Dorsum, between March 2020 and December 2024. Three CTX images from three orbits show it, the last in July 2025.

Why it may be a new impact. Images from May 2009 and March 2020 show nothing there, yet both show a dark mark 290 m to the north, so they would have shown the spot too. When an earlier image is subtracted from a later image, the strongest change in the surrounding 2 × 2 km is right at the spot and nowhere else. A compact dark spot with ray-like edges, seen on three orbits, is how the blast zone of a small new crater looks.

Caveats. By chance, NASA's HiRISE camera photographed the spot on 30 January 2025, on the same orbit as a CTX image, while taking a sample of the terrain rather than aiming at the spot. It shows a dark zone 76 × 81 m with dark streaks and, inside it, a dark pit about 2 m across with a brighter east rim, as a small crater lit from the west-north-west would look. The wind ridges cast shadows, and the pit is only about four HiRISE pixels across, so it may be a shadowed hollow rather than a crater; with no earlier HiRISE image, its age is unknown.

Independent check. The independent re-examination (Section 3.8) kept this site as a strong candidate, with its own estimate of about 75% that the spot is a new impact. It found the dark zone in the HiRISE image but judged the 2 m dark dot inside it no evidence of a crater, since dots like it are as common on the surrounding ground; wind-driven albedo change on the Medusae Fossae Formation remains its main alternative, and it estimated about 75%. It placed the spot at 4.2268° N, 210.8519° E, 745 m from the position given above.

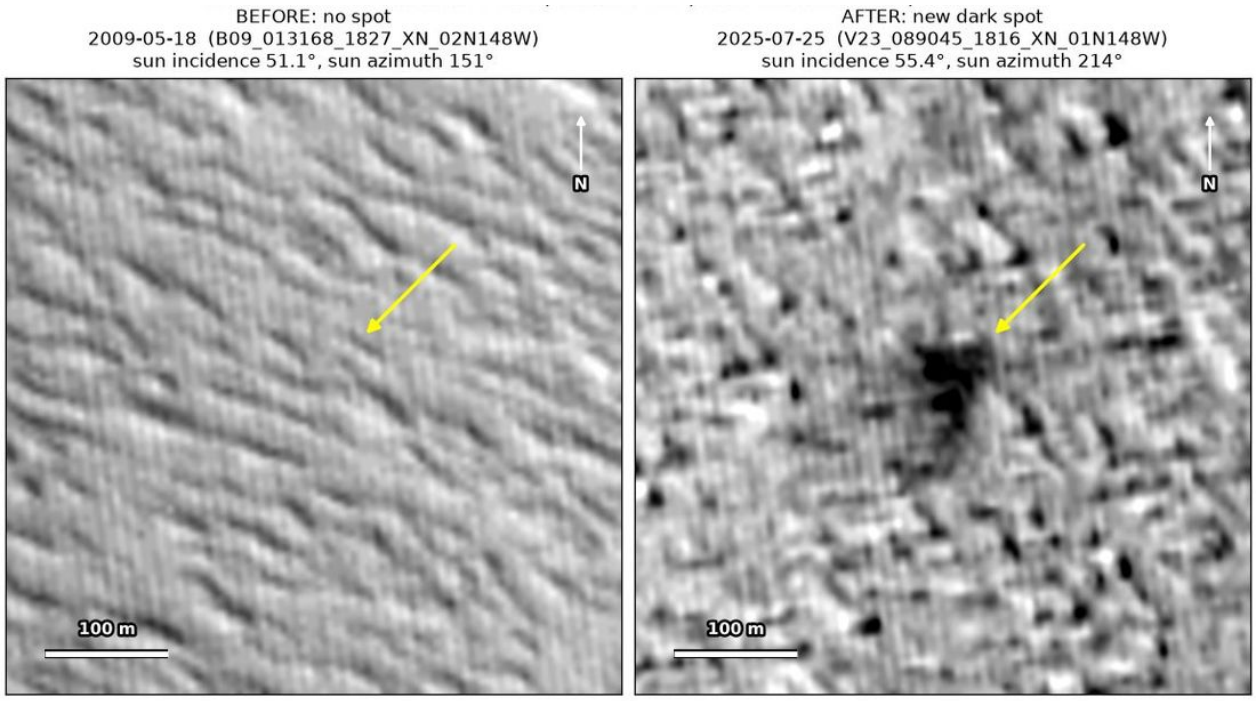

Figure 219: Candidate 85, before and after: the same 500 × 500 m of ground in 18 May 2009 (`B09_013168_1827_XN_02N148W`) and 25 July 2025 (`V23_089045_1816_XN_01N148W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

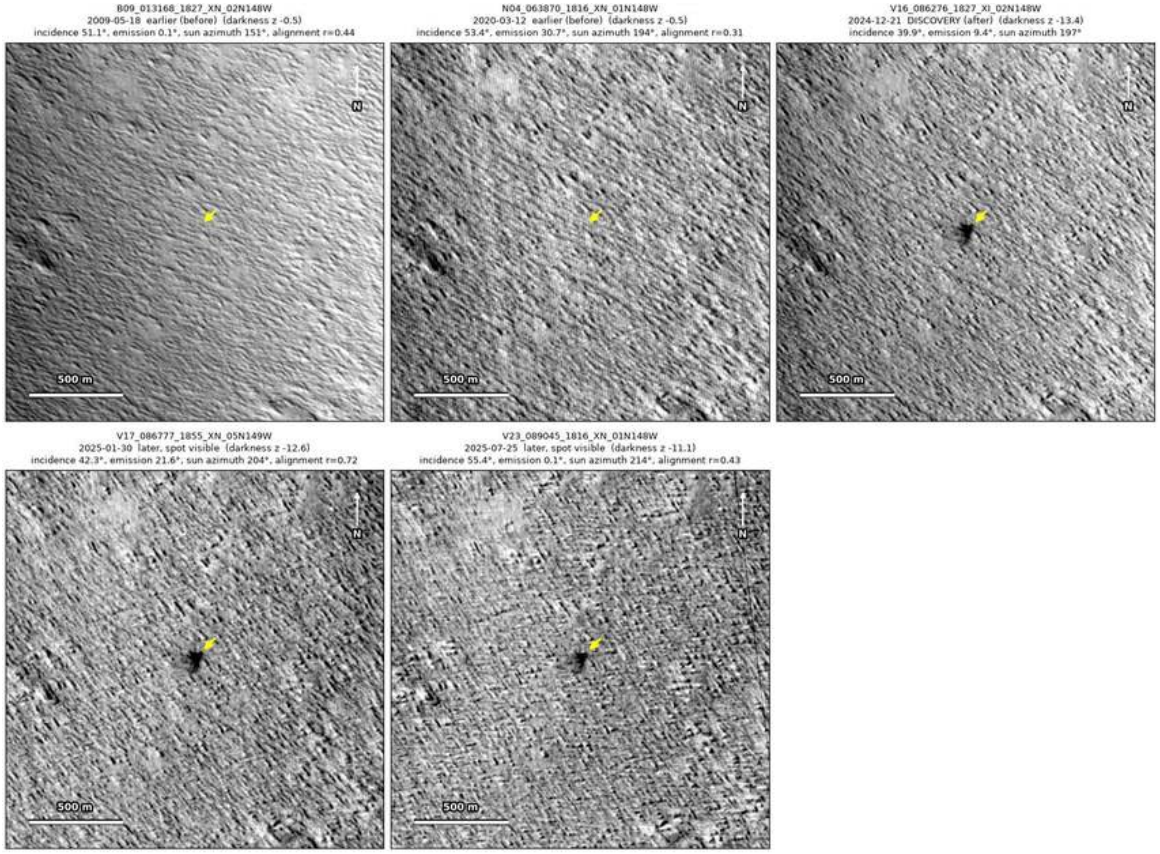

Figure 220: Candidate 85: five CTX images of the same 2×2 km patch, 2009 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

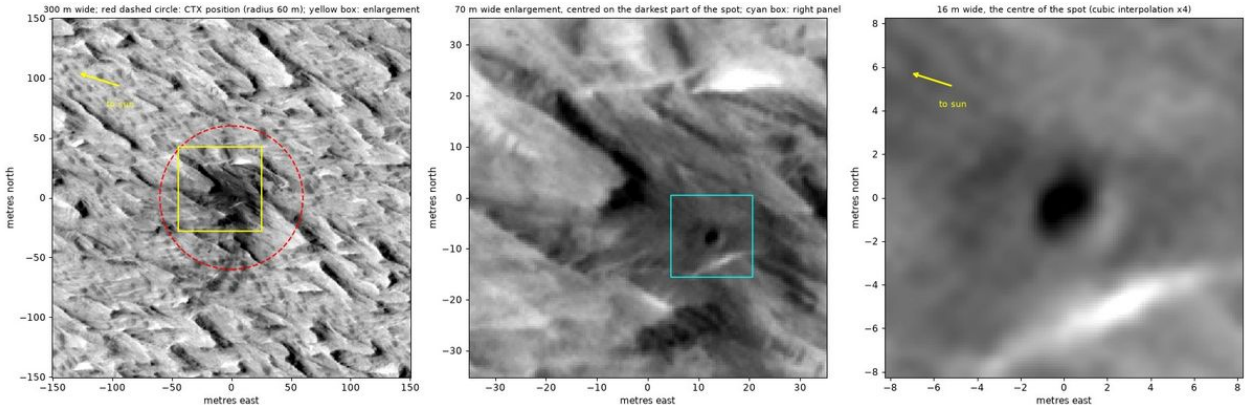

Figure 221: Candidate 85, HiRISE image `ESP_086777_1840`: HiRISE view of the spot (30 January 2025, 0.5 m per pixel), taken by chance on the same orbit as a CTX image: 300 m wide, then a 70 m enlargement of its darkest part, then the central 16 m. The red dashed circle marks the CTX position. HiRISE shows a dark zone about 76×81 m, and inside it a dark pit about 2 m across with a brighter east rim, as a small crater lit from the west-north-west would look; the pit is only about four pixels across, so it may be a shadowed hollow instead. NASA/JPL-Caltech/UArizona.

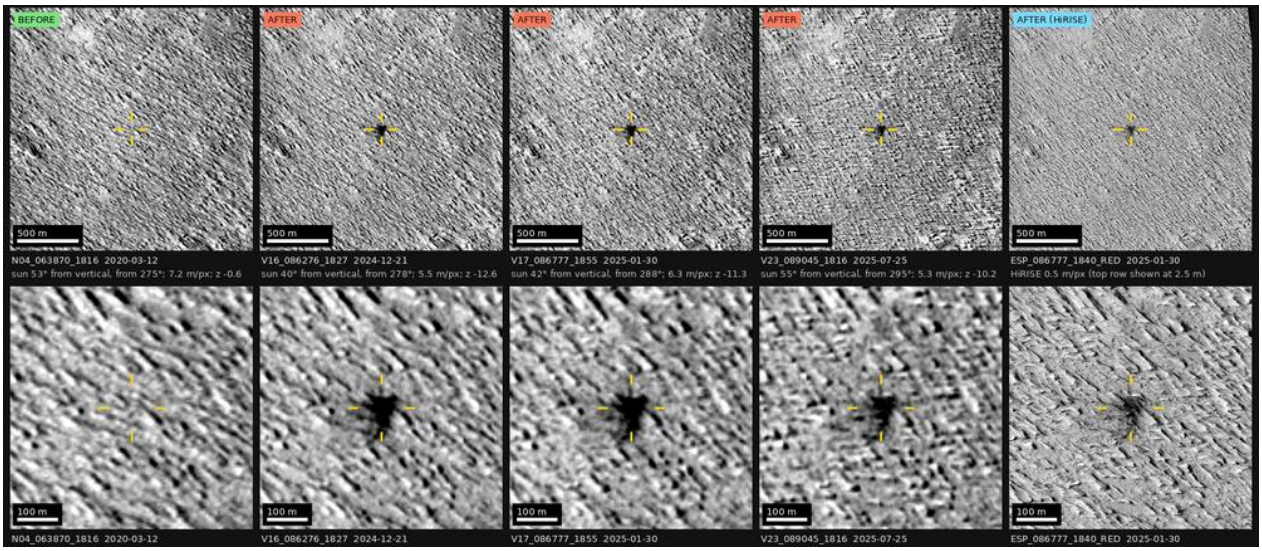

Figure 222: Candidate 85, independent check. Each column is the same co-registered ground in one image, north up, 2 km wide (top row) and 600 m wide (bottom row); yellow ticks bracket the spot. Columns marked BEFORE are earlier images in which, by the check's detectability test, the spot would have been visible had it been there; columns marked AFTER show it (the last is HiRISE where so marked). Each image keeps one contrast stretch in both rows. Below each panel: product, date, sun direction and height, pixel scale and the darkness z measured by the check. CTX (map-projected by Arizona State University): NASA/JPL-Caltech/MSSS; HiRISE: NASA/JPL-Caltech/UArizona.

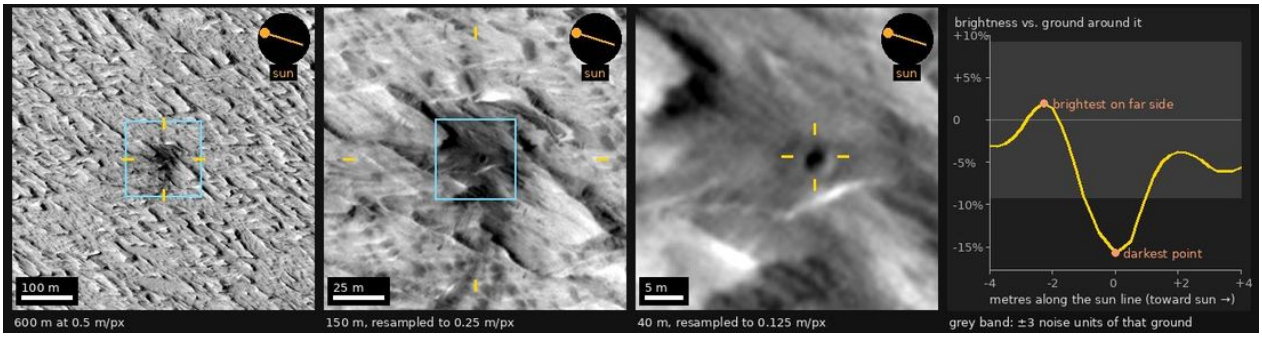

Figure 223: Candidate 85, HiRISE image `ESP_086777_1840`. The independent check: HiRISE close-ups, 600 m, 150 m and 40 m wide, and the brightness along the sun line through the small dark dot. A small pit would be dark on the side towards the sun and bright on the far side; here the far side is no brighter than the noise of the ground around it (grey band), so on this measure the dot shows no sign of being a pit, although in the HiRISE view above it looks like a pit with a brighter east rim, which could also be a shadowed hollow. NASA/JPL-Caltech/UArizona.

Candidate 86: Plains 273 km north-west of Reuyl crater

Location	6.0430° S, 164.0042° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 55%
Formed between	20 July 2022 and 3 January 2025
Images	4 without the spot (absent: 27 June 2014, 17 September 2014, 1 May 2020 and 20 July 2022); 2 with it (present: 3 January 2025 and 4 May 2025)
Size	Diffuse dark patch about 110 m across with a darker core, elongated north–south (106 × 170 m in the January 2025 image; 112 × 220 m in the difference between the earlier and later images)
HiRISE	none covers the spot (nearest footprint edge 47.2 km)
Nearest catalogued new impact	210 km (Daubar et al., 2022)

What changed. A diffuse dark patch about 110 m across, with a darker core, appeared on mantled plains 273 km north-west of Reuyl crater between July 2022 and January 2025. Two CTX images from two orbits show it, the second, from May 2025, more faintly.

Why it may be a new impact. Four earlier images, from 2014 to 2022, show the site slightly brighter than its surroundings, and three of them show a dark mark 195 m to the north. When an earlier image is subtracted from a later image, the strongest change in the surrounding 2 × 2 km is right at the patch and nowhere else. A new dark patch with a darker core can be the blast zone of a small new crater.

Caveats. It may instead be a patch where wind or a dust devil removed dust from the mantle covering the plains: its soft outline, its north–south elongation and its weaker darkening in May 2025 fit that about as well as an impact. Since the site was slightly bright before, the loss of a bright patch of dust would also darken it. No image from the high-resolution HiRISE camera covers it; the nearest ends 47.2 km away.

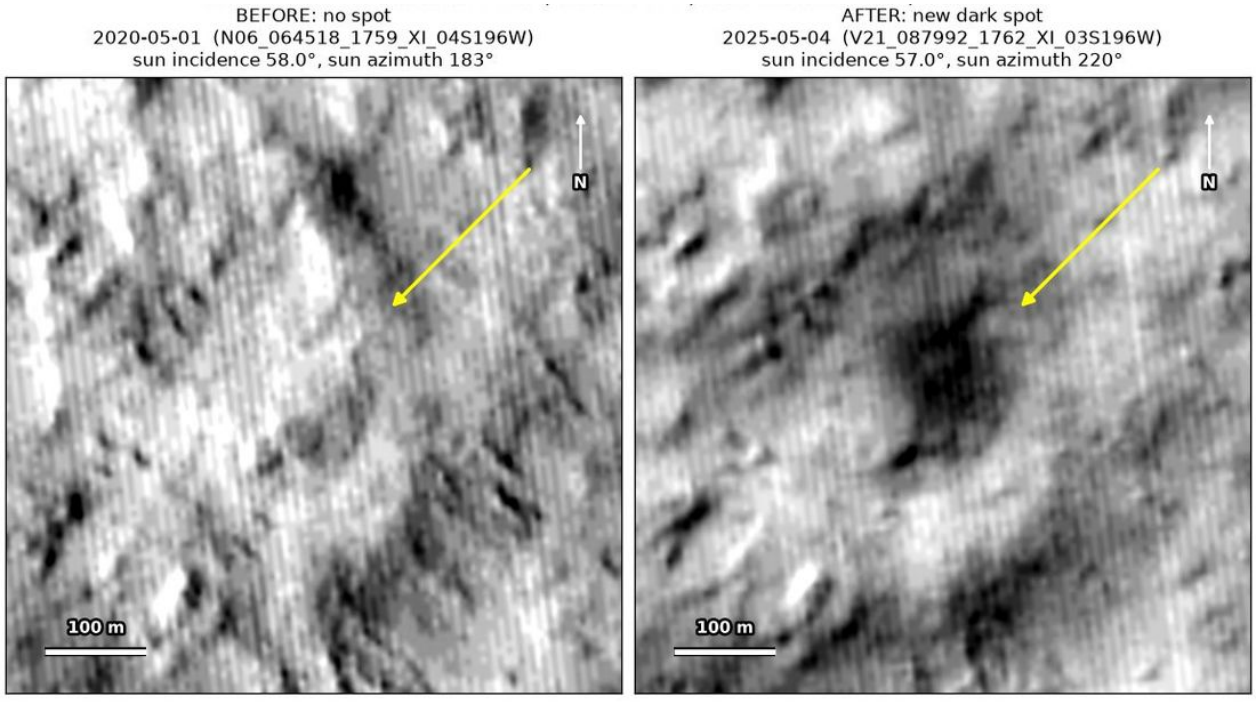

Figure 224: Candidate 86, before and after: the same 600 × 600 m of ground in 1 May 2020 (`N06_064518_1759_XI_04S196W`) and 4 May 2025 (`V21_087992_1762_XI_03S196W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

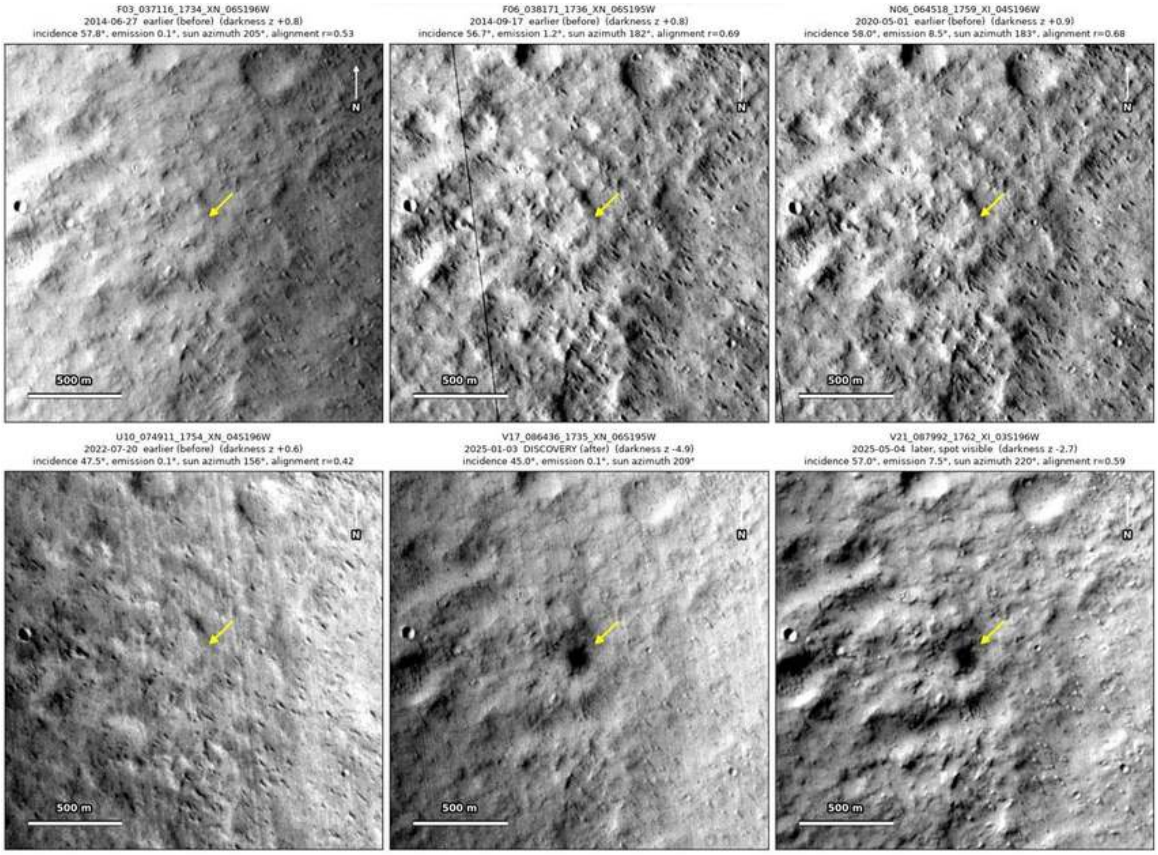

Figure 225: Candidate 86: six CTX images of the same 2×2 km patch, 2014 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 87: Smooth plains in Arabia Terra, 165 km north-east of Schoner crater

Location	22.3387° N, 52.2009° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	4 September 2020 and 7 December 2024
Images	5 without the spot (absent: 26 August 2011, 1 August 2012, 19 November 2012, 30 November 2012 and 4 September 2020); 1 with it (present: 7 December 2024)
Size	Rayed dark spot about 150 to 200 m across (188 × 222 m), with a bright central dot
HiRISE	none covers the spot (nearest footprint edge 35.0 km)
Nearest catalogued new impact	58 km (Daubar et al., 2022)

What changed. A dark spot about 150 to 200 m across, with rays and a small bright dot near its centre, appeared on smooth plains in Arabia Terra, 165 km north-east of Schoner crater, between September 2020 and December 2024. It is visible in one CTX image so far.

Why it may be a new impact. Five earlier images from 2011 to 2020 show no spot there, including one of August 2011 taken under the same sun as the December 2024 image. When an earlier image is subtracted from the later one, the strongest change in the surrounding 2 × 2 km is right at the spot and nowhere else. Dark rays around a small bright centre, hard to explain as an image defect or a dust devil, are how the blast zone of a small new crater looks.

Caveats. Only one image shows it, so an image defect cannot be fully ruled out, though rays and a bright centre are hard to produce that way and the image's striping is even across the area. No nearby feature can confirm that the earlier images would have shown such a spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 35.0 km away.

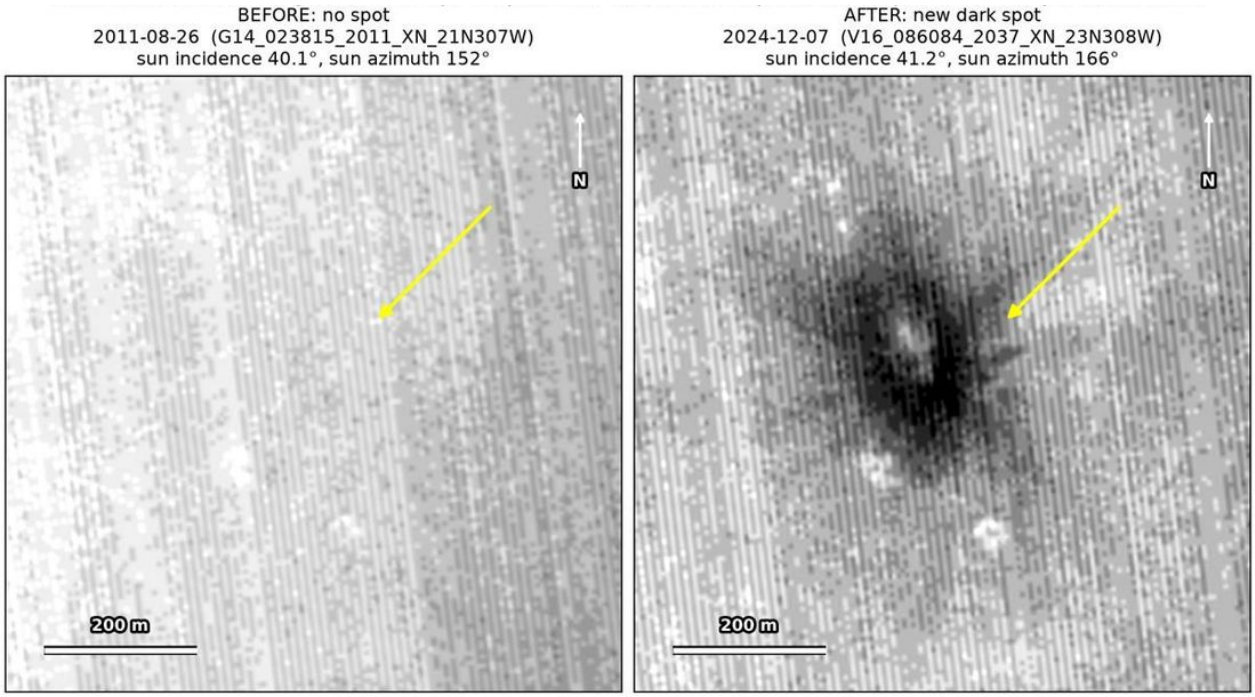

Figure 226: Candidate 87, before and after: the same 800 × 800 m of ground in 26 August 2011 (`G14_023815_2011_XN_21N307W`) and 7 December 2024 (`V16_086084_2037_XN_23N308W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

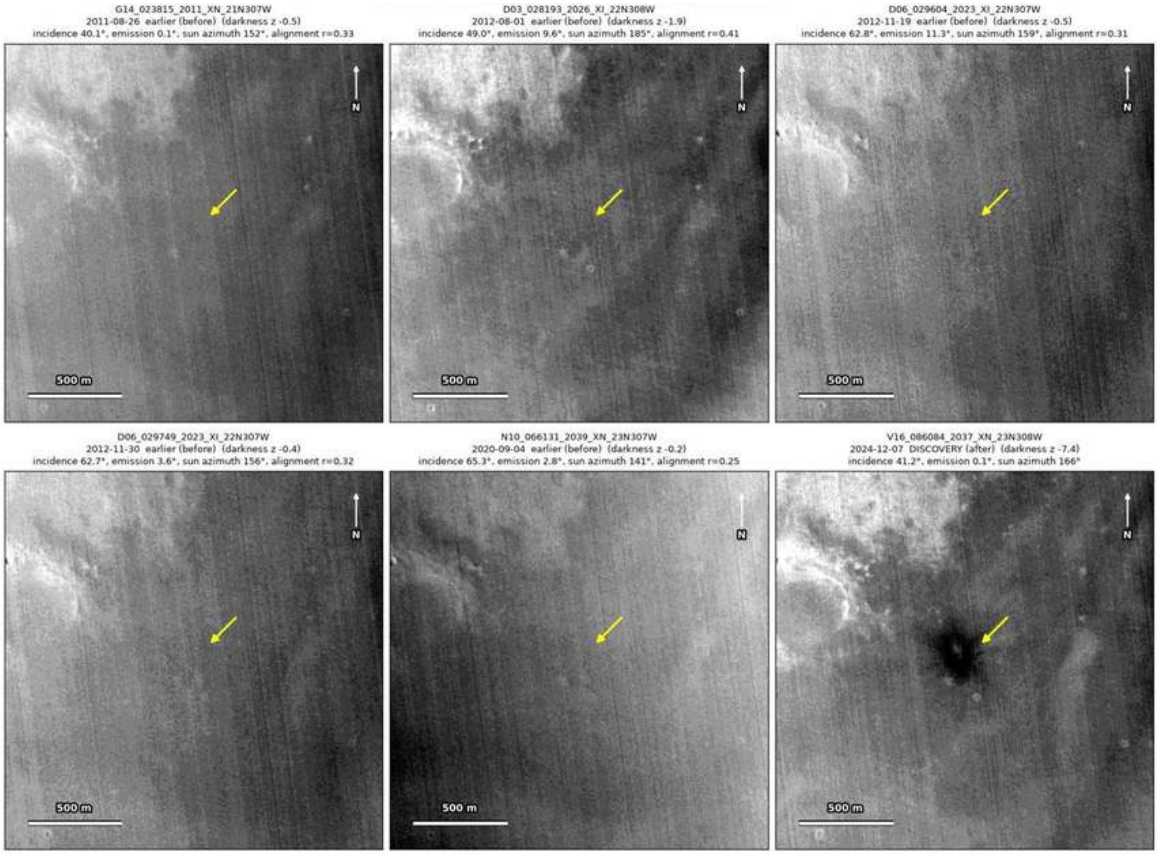

Figure 227: Candidate 87: six CTX images of the same 2×2 km patch, 2011 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 88: Plains north of western Tithonium Chasma, 72 km south of candidate 47

Location	3.9974° S, 268.6457° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	13 February 2020 and 5 December 2024
Images	3 without the spot (absent: 31 October 2007, 17 September 2008 and 13 February 2020); 1 with it (present: 5 December 2024)
Size	Compact black spot about 35 m across (36 × 46 m)
HiRISE	none covers the spot (nearest footprint edge 20.7 km)
Nearest catalogued new impact	106 km (Daubar et al., 2022)

What changed. A compact black spot about 35 m across, with a faint halo, appeared on rippled plains north of western Tithonium Chasma, 72 km south of candidate 47, between February 2020 and December 2024. It is visible in one CTX image so far.

Why it may be a new impact. Three earlier images, from 2007, 2008 and 2020, show no spot there, and when an earlier image is subtracted from the later one, the strongest change is right at the spot. The spot sits on the ripple texture and has a faint halo, unlike an image defect, and a compact black spot with a halo is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, as a later image, from April 2025, has the spot in the dark strip along its edge, so an image defect cannot be fully ruled out. The earlier images were taken under other suns, though none shows anything as dark. In the October 2007 image, a nearby feature that was already there looks no darker than the spot's site, so it cannot confirm that the earlier images would have shown the spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 20.7 km away.

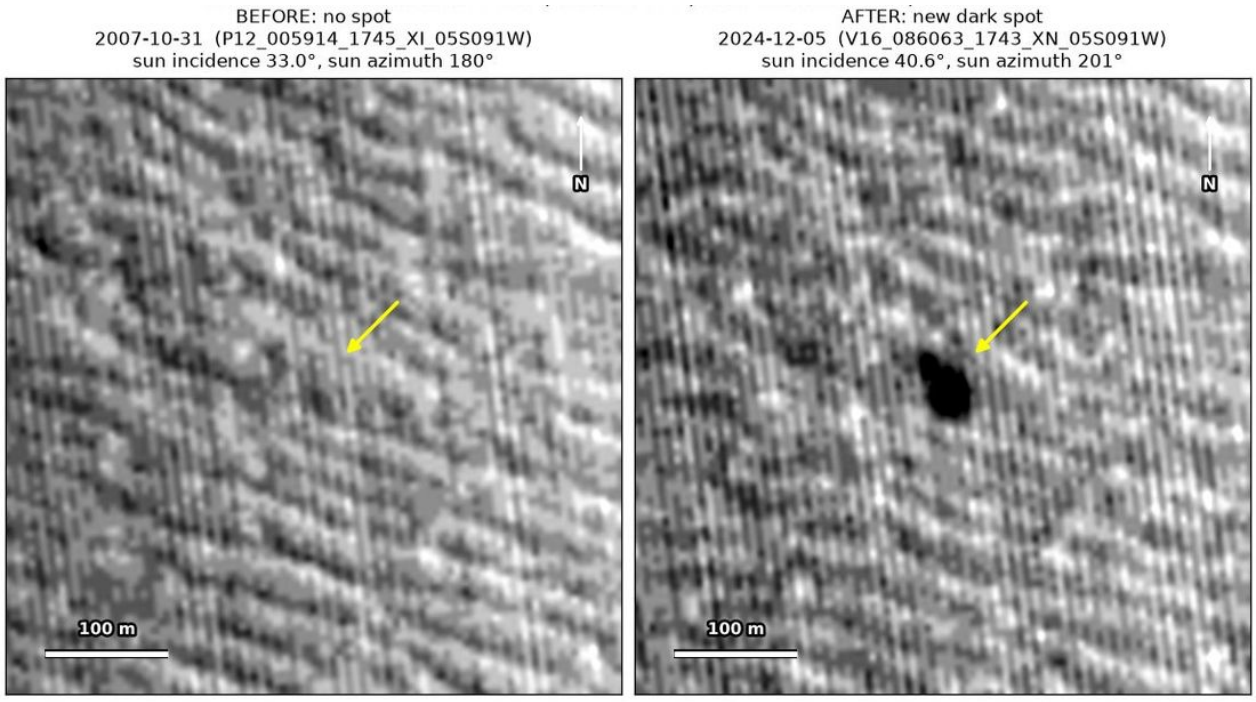

Figure 228: Candidate 88, before and after: the same 500 × 500 m of ground in 31 October 2007 (`P12_005914_1745_XI_05S091W`) and 5 December 2024 (`V16_086063_1743_XN_05S091W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

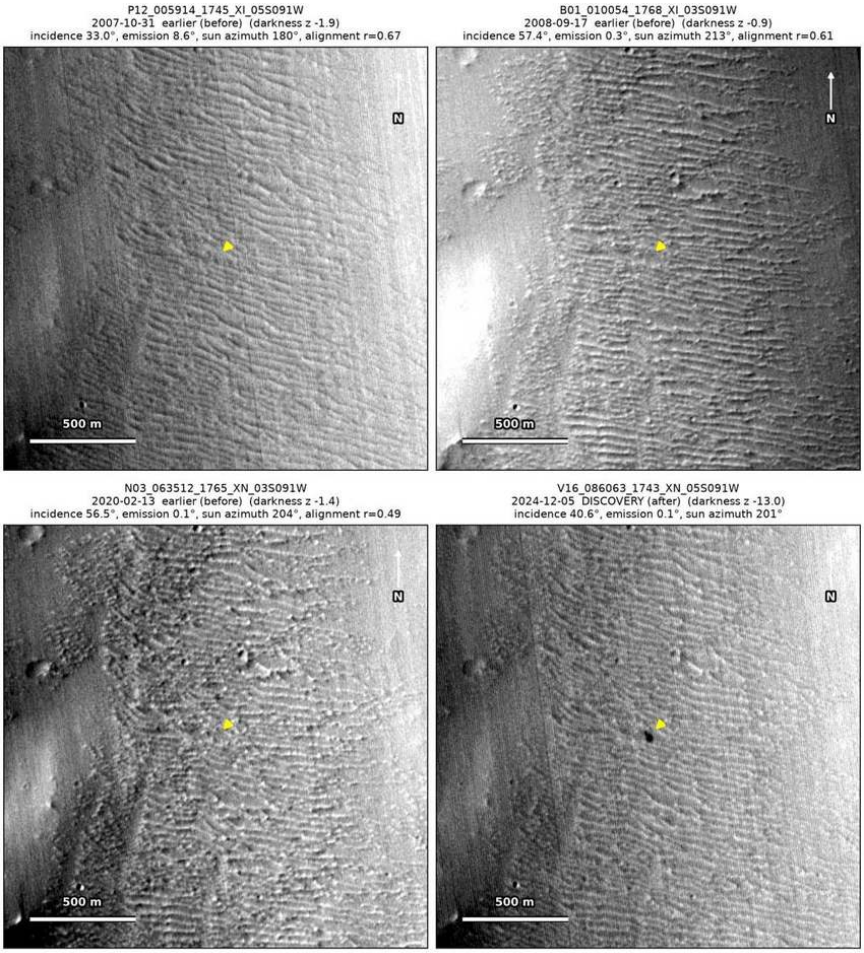

Figure 229: Candidate 88: four CTX images of the same 2×2 km patch, 2007 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 89: Plains among the Aeolis Mensae, 325 km north of Gale crater

Location	0.0674° N, 138.5886° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 55%
Formed between	12 August 2022 and 20 November 2024
Images	6 without the spot (absent: 20 July 2009, 1 April 2014, 6 June 2014, 22 February 2019, 17 June 2022 and 12 August 2022); 1 with it (present: 20 November 2024)
Size	Compact black spot about 30 m across (24 × 34 m), with a faint halo
HiRISE	none covers the spot (nearest footprint edge 28.4 km)
Nearest catalogued new impact	195 km (Daubar et al., 2022)

What changed. A compact black spot about 30 m across, with a faint halo, appeared on plains among the Aeolis Mensae, 325 km north of Gale crater, between August 2022 and November 2024. It is visible in one CTX image so far.

Why it may be a new impact. Six earlier images, from 2009 to 2022, show no spot there, while a small crater 160 m to the south lines up and shows in all of them. When an earlier image is subtracted from the later one, the strongest change in the surrounding 2 × 2 km is right at the spot and nowhere else. A round black spot with a halo that fades gradually is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, and that image has a thin vertical line defect 300 m to the east, though the spot is round with a gradually fading halo rather than a line or a block. Because the plains are smooth at this scale, the earlier images line up only moderately, though the small crater nearby lines up in all of them. No image from the high-resolution HiRISE camera covers it; the nearest ends 28.4 km away.

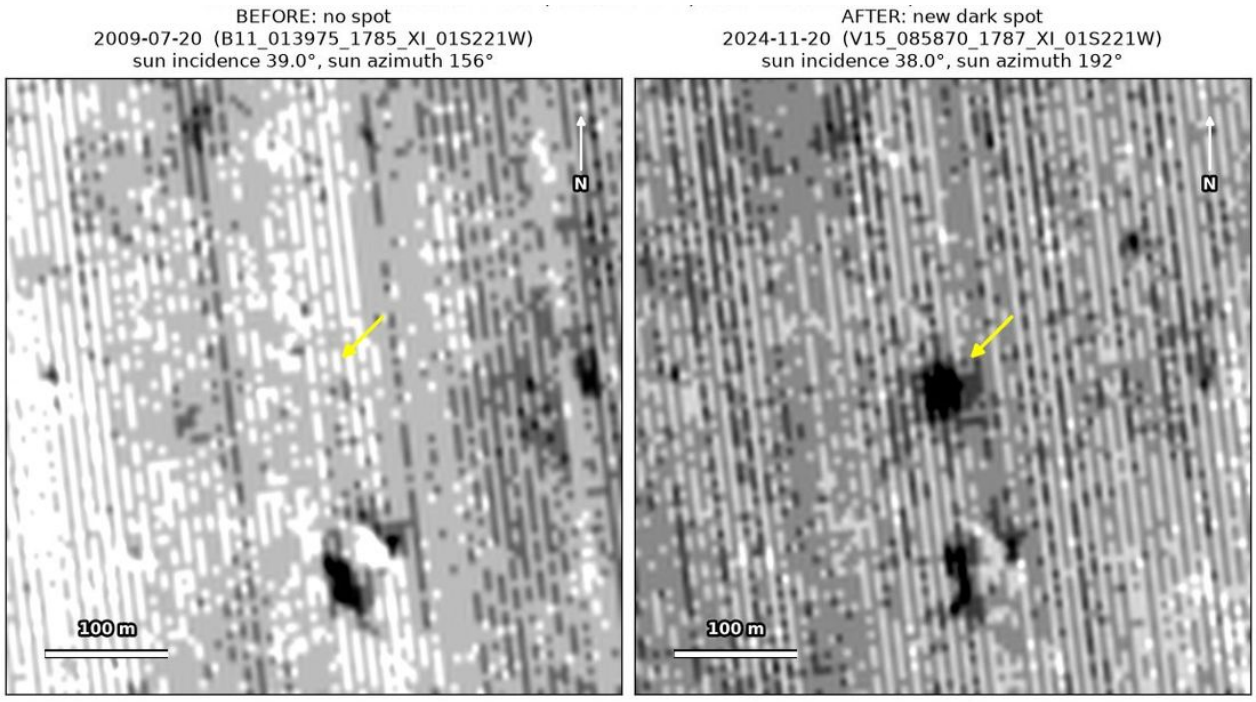

Figure 230: Candidate 89, before and after: the same 500 × 500 m of ground in 20 July 2009 (`B11_013975_1785_XI_01S221W`) and 20 November 2024 (`V15_085870_1787_XI_01S221W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

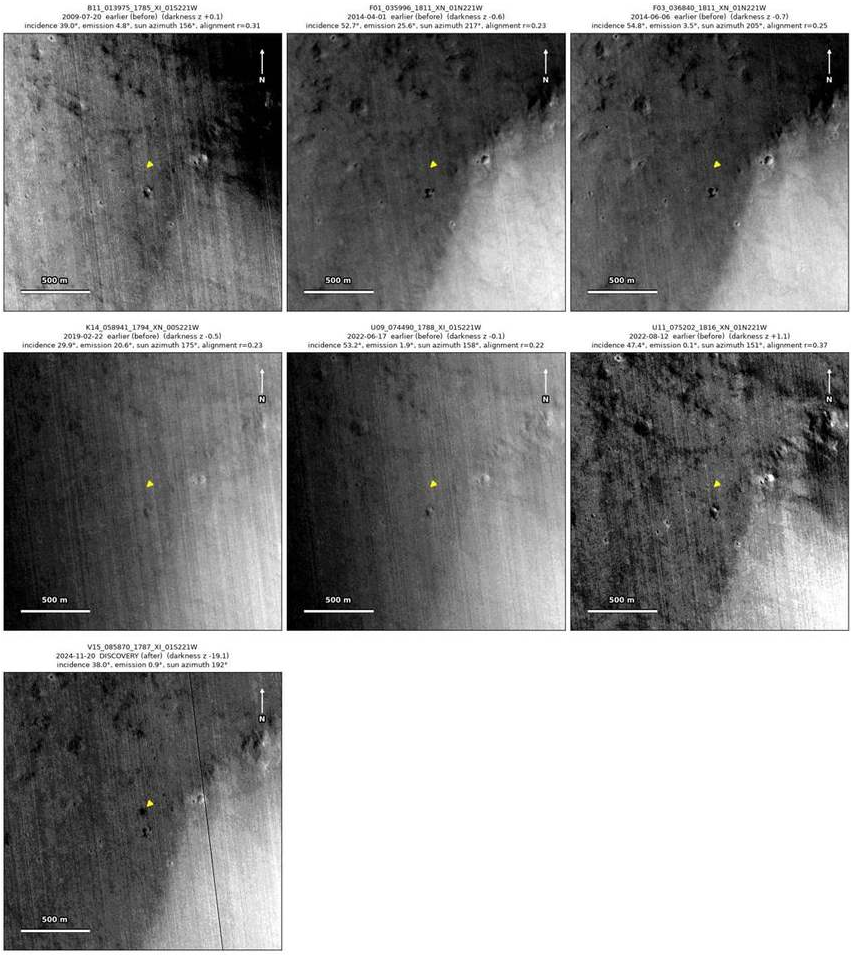

Figure 231: Candidate 89: seven CTX images of the same 2×2 km patch, 2009 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 90: Edge of hummocky terrain in northern Acheron Fossae, 275 km north of candidate 13

Location	39.9957° N, 228.8713° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	8 April 2018 and 21 October 2024
Images	4 without the spot (absent: 11 May 2010, 26 February 2012, 16 July 2016 and 8 April 2018); 1 with it (present: 21 October 2024)
Size	Compact dark spot about 50 m across (82 × 48 m), with a darker core about 20 m wide in a diffuse halo
HiRISE	none covers the spot (nearest footprint edge 65.7 km)
Nearest catalogued new impact	201 km (Daubar et al., 2022)

What changed. A compact dark spot about 50 m across, with a darker core in a diffuse halo, appeared at the southern edge of hummocky terrain in northern Acheron Fossae between April 2018 and October 2024. It is visible in one CTX image so far.

Why it may be a new impact. Four earlier images, from 2010 to 2018, all line up with the later one and show no spot there, while a dark mark 276 m to the east shows in all of them. When an earlier image is subtracted from the later one, the strongest change in the surrounding 2 × 2 km is right at the spot and nowhere else. A dark core in a gradually fading halo is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, and that image has a thin vertical line defect about 220 m to the east, though the spot is round with a gradually fading halo. Wind could also have changed the brightness of the ground at the edge of the hummocky terrain. No image from the high-resolution HiRISE camera covers it; the nearest ends 65.7 km away.

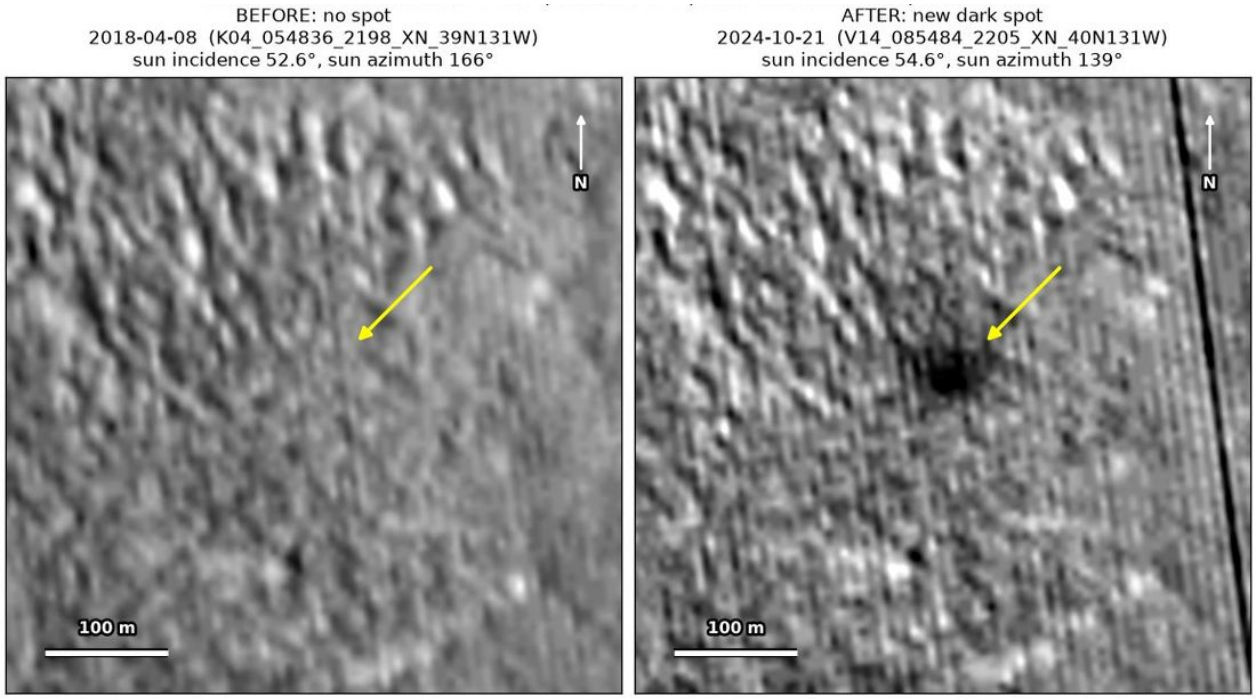

Figure 232: Candidate 90, before and after: the same 500 × 500 m of ground in 8 April 2018 (`K04_054836_2198_XN_39N131W`) and 21 October 2024 (`V14_085484_2205_XN_40N131W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

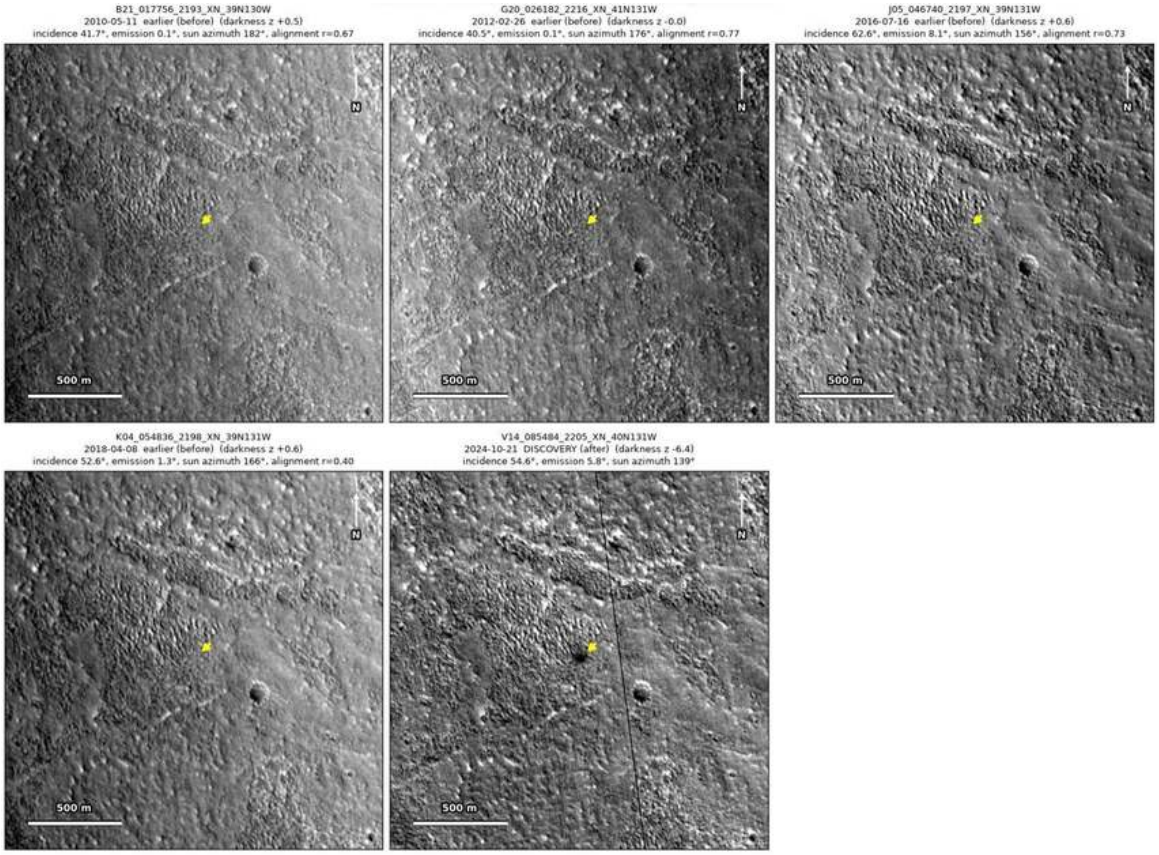

Figure 233: Candidate 90: five CTX images of the same 2×2 km patch, 2010 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 91: Knobby plains in northern Amazonis Planitia, 427 km south-west of candidate 73

Location	33.6203° N, 204.6124° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 45%
Formed between	15 November 2010 and 27 December 2024
Images	2 without the spot (absent: 21 January 2008 and 15 November 2010); 1 with it (present: 27 December 2024)
Size	Compact dark spot about 50 m across (62 × 66 m), beside a small knob that was already there
HiRISE	none covers the spot (nearest footprint edge 25.5 km)
Nearest catalogued new impact	188 km (Daubar et al., 2022)

What changed. A compact dark spot about 50 m across appeared on knobby plains in northern Amazonis Planitia, beside a small knob that was already there, between November 2010 and December 2024. It is visible in one CTX image so far.

Why it may be a new impact. An image from January 2008, taken under the same sun as the December 2024 image, shows the knob as a small dark mark but no spot, so the spot is not a shadow, and an image from November 2010 shows no spot either. When the January 2008 image is subtracted from the later one, the spot is the only compact dark change in the surrounding 2 × 2 km. A compact new dark spot can be the blast zone of a small new crater.

Caveats. Only one image shows it, and dust devils are frequent in northern Amazonis: wind or a dust devil could darken a patch this size, though the spot is compact with no tracks near it. It could also be a small slope streak, left by dust sliding down the knob, though it is round rather than a tapering fan. The November 2010 image, under a low sun, lines up poorly, and no nearby feature settles whether the earlier images would have shown the spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 25.5 km away.

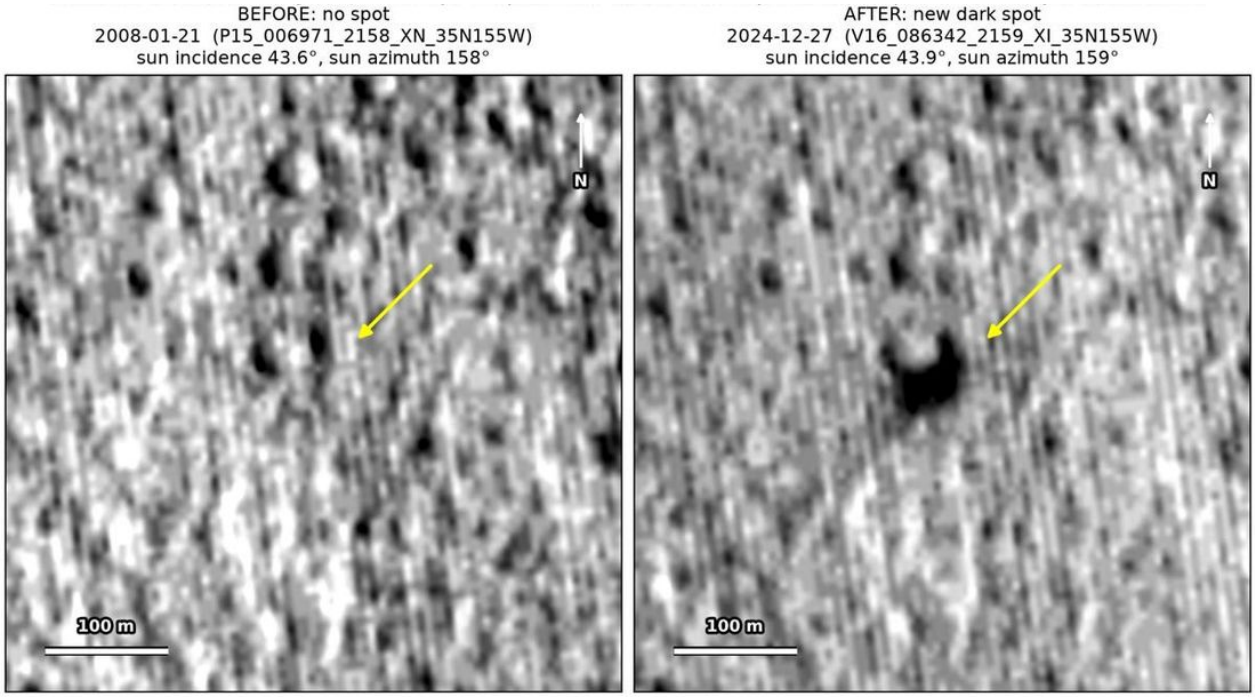

Figure 234: Candidate 91, before and after: the same 500 × 500 m of ground in 21 January 2008 (`P15_006971_2158_XN_35N155W`) and 27 December 2024 (`V16_086342_2159_XI_35N155W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

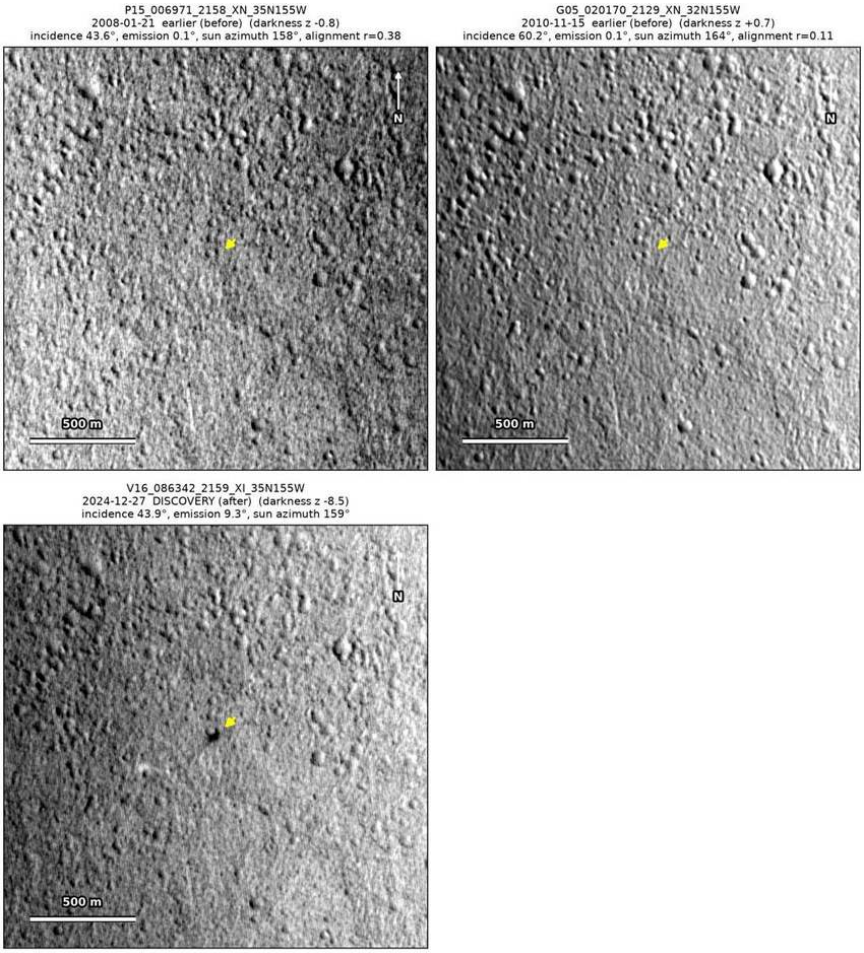

Figure 235: Candidate 91: three CTX images of the same 2×2 km patch, 2008 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 92: Western Acheron Fossae

Location	38.5305° N, 220.8085° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	13 April 2025 and 3 June 2025
Images	7 without the spot (absent: 18 June 2008, 4 July 2016, 2 September 2016, 1 April 2018, 17 April 2018, 12 March 2025 and 13 April 2025); 1 with it (present: 3 June 2025)
Size	Compact dark spot about 65 m across (76 × 66 m), a darker core with a lighter patch on its west side
HiRISE	none covers the spot (nearest footprint edge 1.7 km)
Nearest catalogued new impact	92 km (Daubar et al., 2022)

What changed. A compact dark spot about 65 m across, with a darker core, appeared on smooth ground on the floor of a trough in western Acheron Fossae between 13 April and 3 June 2025, a window of only 51 days. It is visible in one CTX image so far.

Why it may be a new impact. Seven earlier images, from 2008 to 2025, show no spot there, including two of 12 March and 13 April 2025 taken under the same sun as the June 2025 image. When an earlier image is subtracted from the later one, the strongest change in the surrounding 2 × 2 km is right at the spot and nowhere else. A round dark spot with a darker core and a gradually fading edge, appearing within about seven weeks, is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, though the spot is round with a gradually fading edge and that image has no line defect near it. No comparably dark feature nearby can confirm that the earlier images would have shown it. No image from the high-resolution HiRISE camera covers it; the nearest ends 1.7 km away.

Independent check. The independent re-examination (Section 3.8) judged this site unlikely to be a new impact. It confirmed the 51-day window, but in the June 2025 image it found a new bright patch 91 m from the spot on its sunward side and a new dark band 60–70 m wide extending east-south-east from it. A bright patch on the sunward side of a dark one is how a sunlit dust column and its shadow look, so it judged the spot most probably a dust devil and its track, and a new impact unlikely (about 5%). We keep the probability assigned on the public pages but regard this candidate as doubtful; any later image of the site would settle the question. It placed the spot at 38.5289° N, 220.8006° E, 377 m from the position given above.

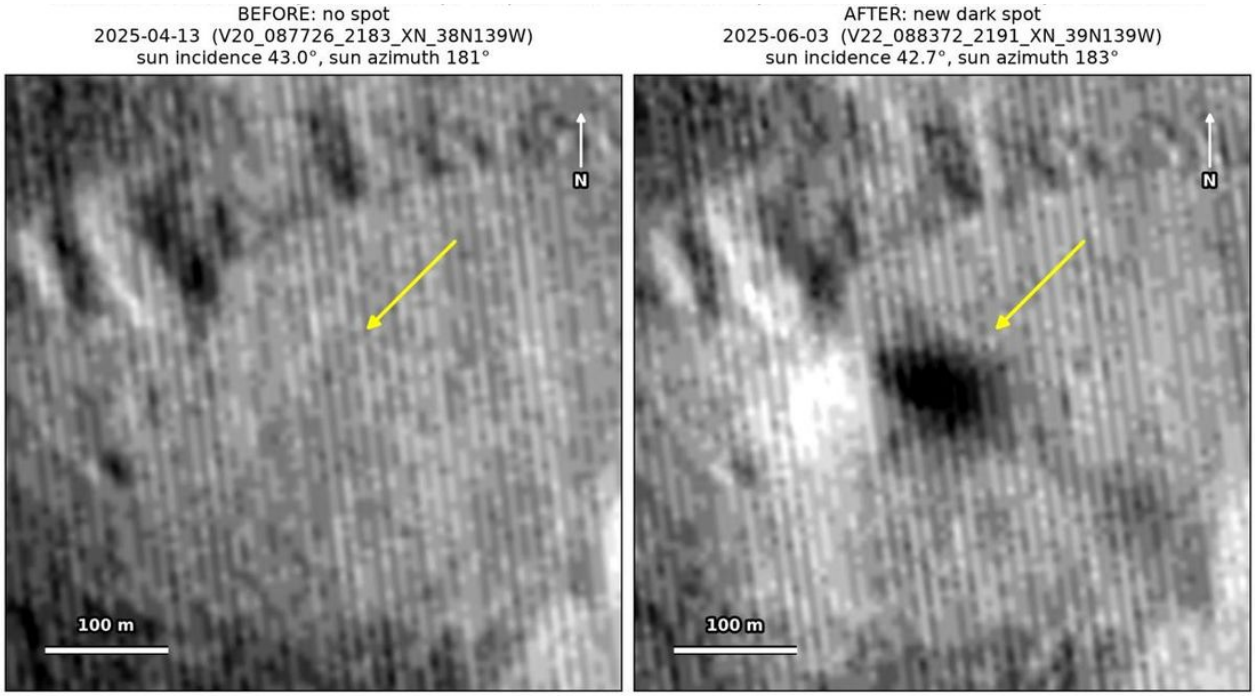

Figure 236: Candidate 92, before and after: the same 500 × 500 m of ground in 13 April 2025 (`V20_087726_2183_XN_38N139W`) and 3 June 2025 (`V22_088372_2191_XN_39N139W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

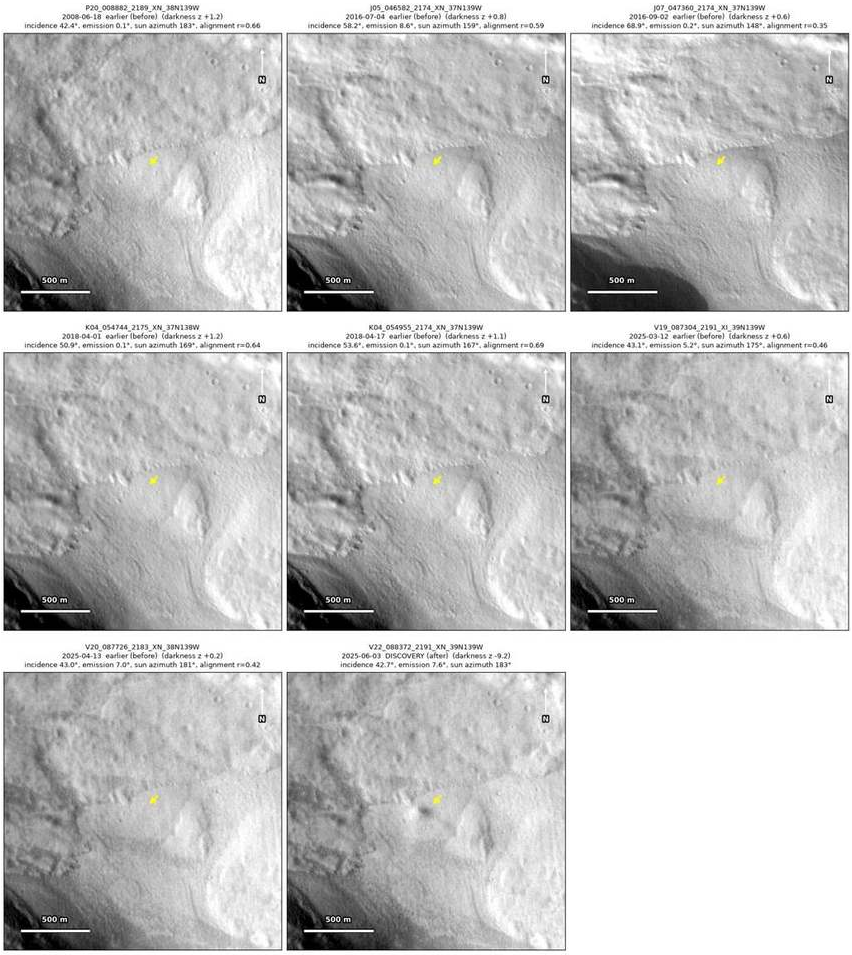

Figure 237: Candidate 92: eight CTX images of the same 2×2 km patch, 2008 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 93: Lava plains in eastern Daedalia Planum, 555 km south of candidate 18

Location	17.3508° S, 244.9885° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	31 December 2017 and 19 November 2024
Images	3 without the spot (absent: 15 August 2007, 4 April 2010 and 31 December 2017); 1 with it (present: 19 November 2024)
Size	Compact dark spot about 75 m across (86 × 76 m), a dark core with small dark specks around it
HiRISE	none covers the spot (nearest footprint edge 35.4 km)
Nearest catalogued new impact	256 km (Daubar et al., 2022)

What changed. A compact dark spot about 75 m across, a dark core with small dark specks around it, appeared on lava plains in eastern Daedalia Planum between December 2017 and November 2024. It is visible in one CTX image so far.

Why it may be a new impact. Three earlier images, from 2007 to 2017 and all aligned with the later one, show no spot there, while the bright patches around the site line up in all four images. When an earlier image is subtracted from the later one, the strongest change in the surrounding 2 × 2 km is right at the spot and nowhere else. A compact dark core with small dark specks around it fits the blast zone of a small new crater.

Caveats. Only one image shows it, so an image defect cannot be fully ruled out, though the spot spans many image lines and has small specks around it. The earlier images were taken under other suns, the August 2007 one hazy from the 2007 global dust storm and the other two with a lower sun, though none shows anything as dark. No comparably dark feature nearby can confirm that they would have shown it. No image from the high-resolution HiRISE camera covers it; the nearest ends 35.4 km away.

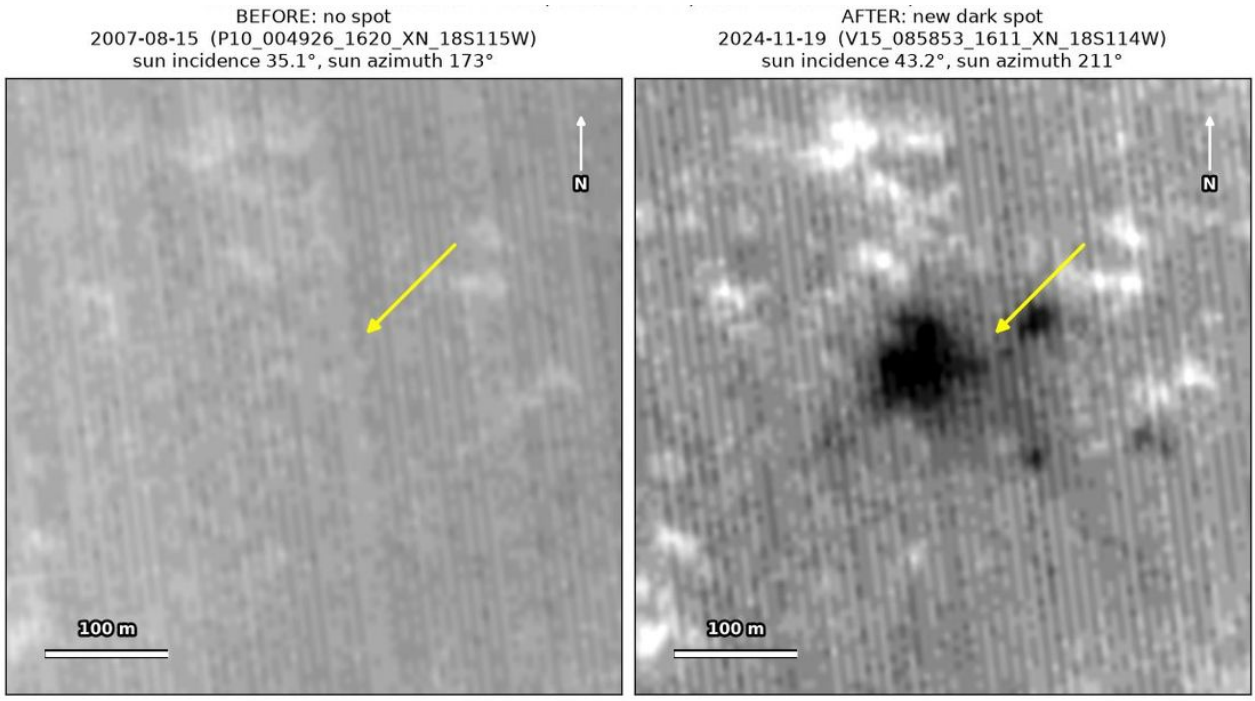

Figure 238: Candidate 93, before and after: the same 500 × 500 m of ground in 15 August 2007 (`P10_004926_1620_XN_18S115W`) and 19 November 2024 (`V15_085853_1611_XN_18S114W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

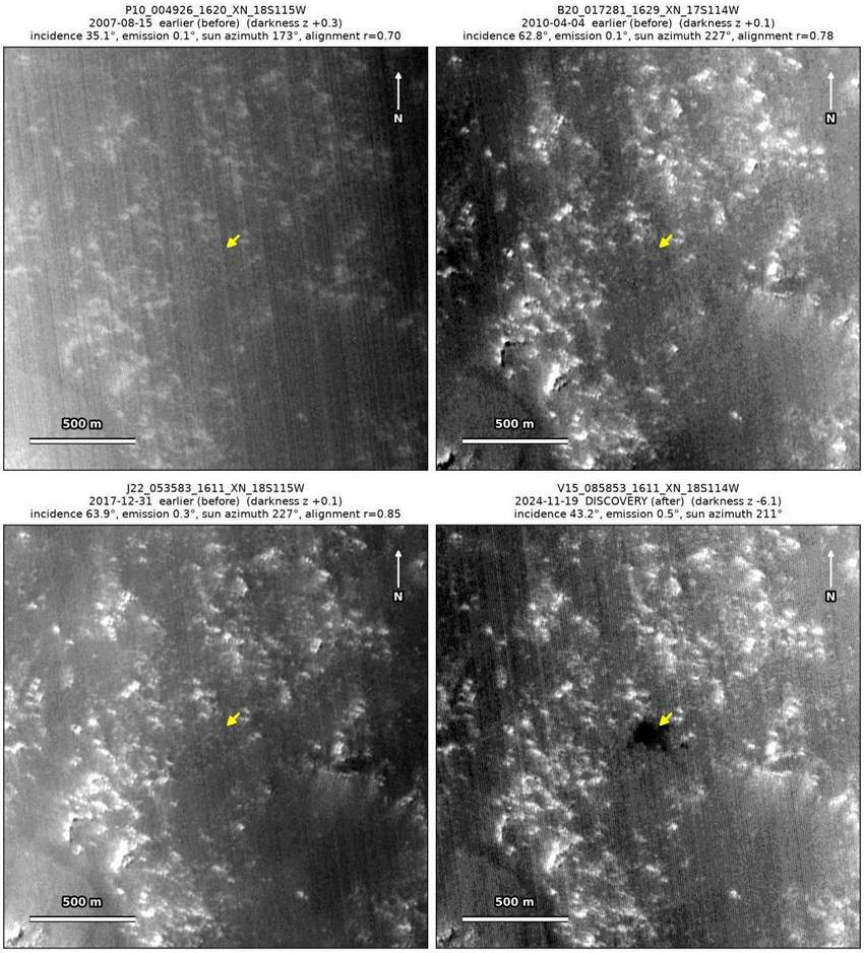

Figure 239: Candidate 93: four CTX images of the same 2×2 km patch, 2007 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 94: Plains between Aeolis Dorsa and Zephyria Planum, 50 km west of Kalba crater

Location	6.0510° S, 154.0089° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 70%
Formed between	29 December 2010 and 9 August 2021
Images	6 without the spot (absent: 19 October 2007, 28 March 2008, 24 April 2010, 24 August 2010, 18 December 2010 and 29 December 2010); 7 with it (present: 9 August 2021 (CTX and HiRISE), 22 September 2021 (CTX and HiRISE), 23 June 2023, 6 August 2023, 13 September 2024, 12 November 2024 and 29 January 2025)
Size	Dark patch about 100 m across (58 × 112 m in the September 2024 image), a darker core in a halo about 200 m across; in the HiRISE images, a dark zone 89 × 107 m with no definite crater
HiRISE	ESP_070481_1740 and ESP_071048_1740 (2021-08-09 and 2021-09-22, same orbits as N21 and N22, 'Sinuous channels near Zephyria Planum', not impact-targeted)
Nearest catalogued new impact	148 km (Daubar et al., 2022)

What changed. A dark patch about 100 m across, with a darker core in a fainter halo about 200 m across, appeared on rough, wind-eroded plains 50 km west of Kalba crater between December 2010 and August 2021. Seven CTX images from seven orbits show it, fading in the last two, from November 2024 and January 2025.

Why it may be a new impact. Six earlier images, from 2007 to 2010, show no patch there, including one of August 2010 taken under the same sun as the June 2023 image. When an earlier image is subtracted from a later one, the strongest change in the surrounding 2 × 2 km is right at the patch and nowhere else. A round dark zone with a darker centre and no streaks, seen on seven orbits from 2021 to 2025, is how the blast zone of a small new crater looks.

Caveats. By chance, NASA's HiRISE camera photographed the patch on 9 August 2021 and again on 22 September 2021, on the same orbits as two CTX images, while aimed at sinuous channels near Zephyria Planum. Both show the same dark zone, 89 × 107 m and darker towards the centre, but no definite crater, only a dark blob about 4 m across at its centre. HiRISE shows no definite crater, and the patch faded after September 2024, so a wind-driven change in the dust, without a crater, cannot be ruled out. With no earlier HiRISE image and no CTX image between December 2010 and August 2021, its date of formation is poorly known. It lies in the area of a catalogue of new impacts near the InSight lander (Bickel et al. 2025) whose period overlaps its window, so it may already be listed there, which could not be checked.

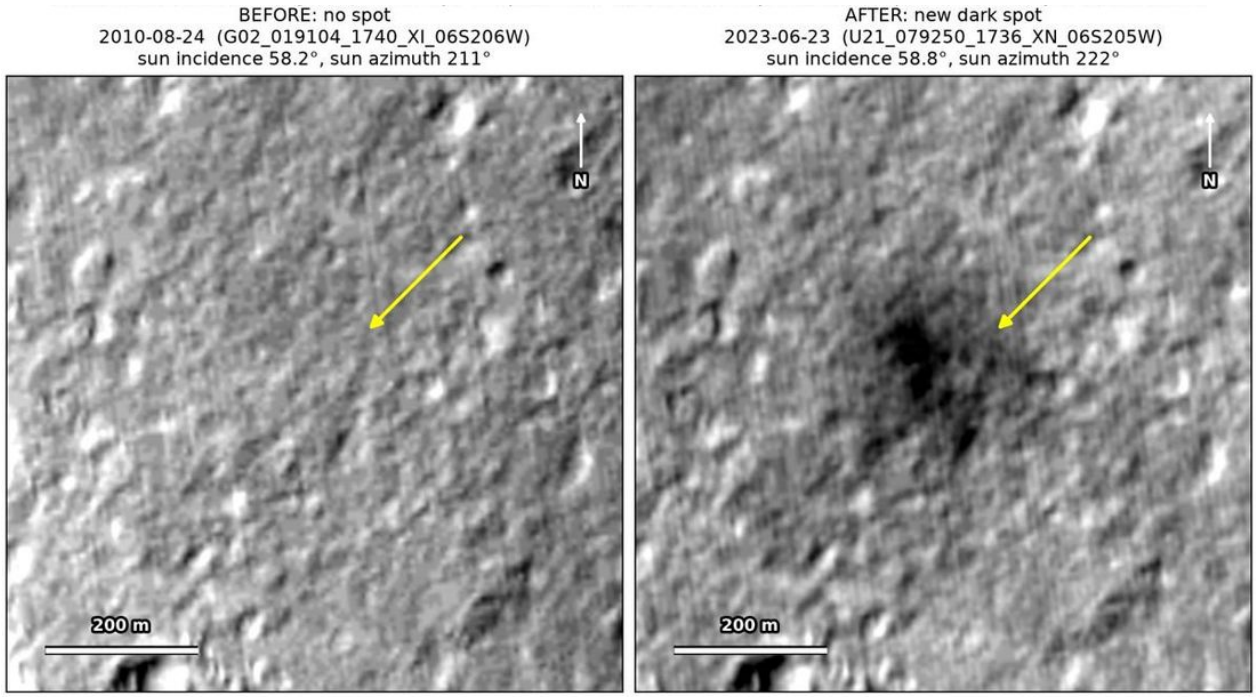

Figure 240: Candidate 94, before and after: the same 800 × 800 m of ground in 24 August 2010 (`G02_019104_1740_XI_06S206W`) and 23 June 2023 (`U21_079250_1736_XN_06S205W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

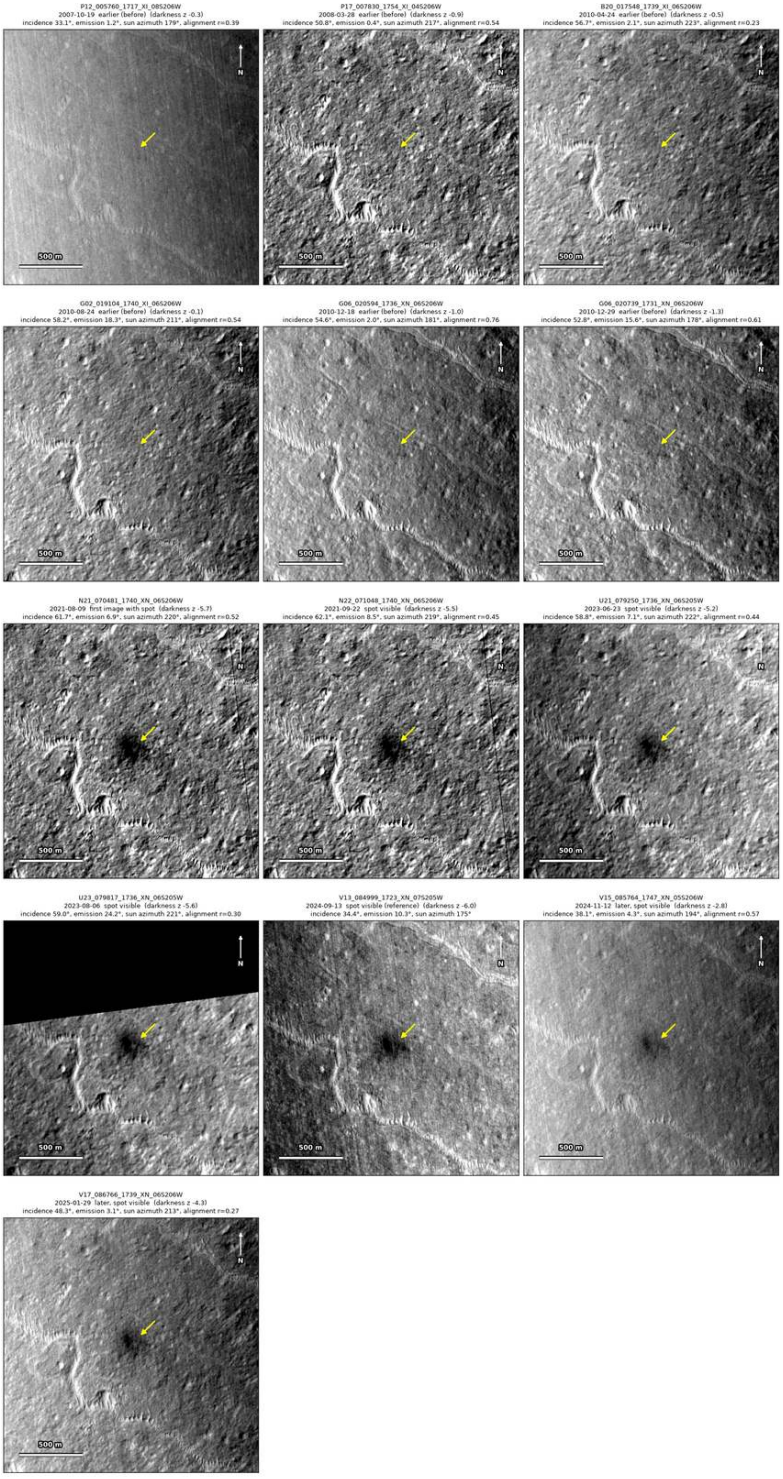

Figure 241: Candidate 94: thirteen CTX images of the same 2×2 km patch, 2007 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

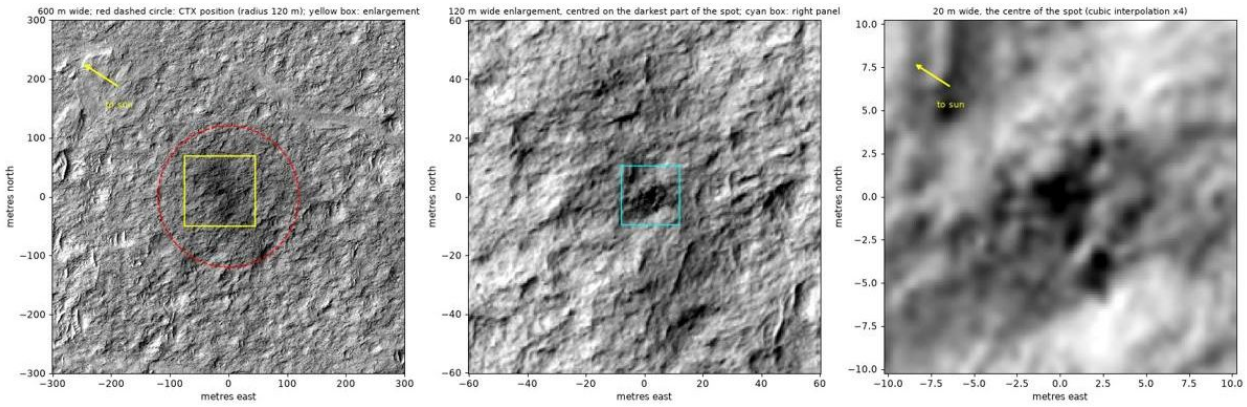

Figure 242: Candidate 94, HiRISE image `ESP_070481_1740`; the second image is `ESP_071048_1740`: HiRISE view of the spot (9 August 2021, 0.5 m per pixel), taken by chance on the same orbit as a CTX image: 600 m wide, then a 120 m enlargement centred on its darkest part, then the central 20 m. The red dashed circle marks the CTX position. HiRISE shows a dark zone about 89×107 m, darker towards the centre, but no definite crater; a second HiRISE image, of 22 September 2021, shows it the same. NASA/JPL-Caltech/UArizona.

Candidate 95: Plains north of Tithonium Chasma, 58 km north-east of candidate 47

Location	1.9174° S, 268.7421° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	10 July 2023 and 25 May 2025
Images	4 without the spot (absent: 6 April 2012, 23 April 2012, 27 March 2023 and 10 July 2023); 1 with it (present: 25 May 2025)
Size	Compact dark spot about 25 m across (26 × 24 m), with a dark core
HiRISE	none covers the spot (nearest footprint edge 15.3 km)
Nearest catalogued new impact	18 km (Daubar et al., 2022)

What changed. A compact dark spot about 25 m across, with a dark core, appeared on cratered plains north of Tithonium Chasma, 58 km north-east of candidate 47, between July 2023 and May 2025. It is visible in one CTX image so far.

Why it may be a new impact. Four earlier images, from 2012 and 2023, all line up with the later one and show no spot there, while a small crater 276 m to the south shows clearly in every one; the last, from July 2023, was taken under the same sun as the May 2025 image. The spot is the strongest small-scale change between the earlier and later images, and a compact dark spot with a dark core is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, so an image defect cannot be fully ruled out, though the spot spans several of the blocks in which that image was compressed. A dust devil that stalled is also possible, though there are no tracks near it. No image from the high-resolution HiRISE camera covers it; the nearest ends 15.3 km away.

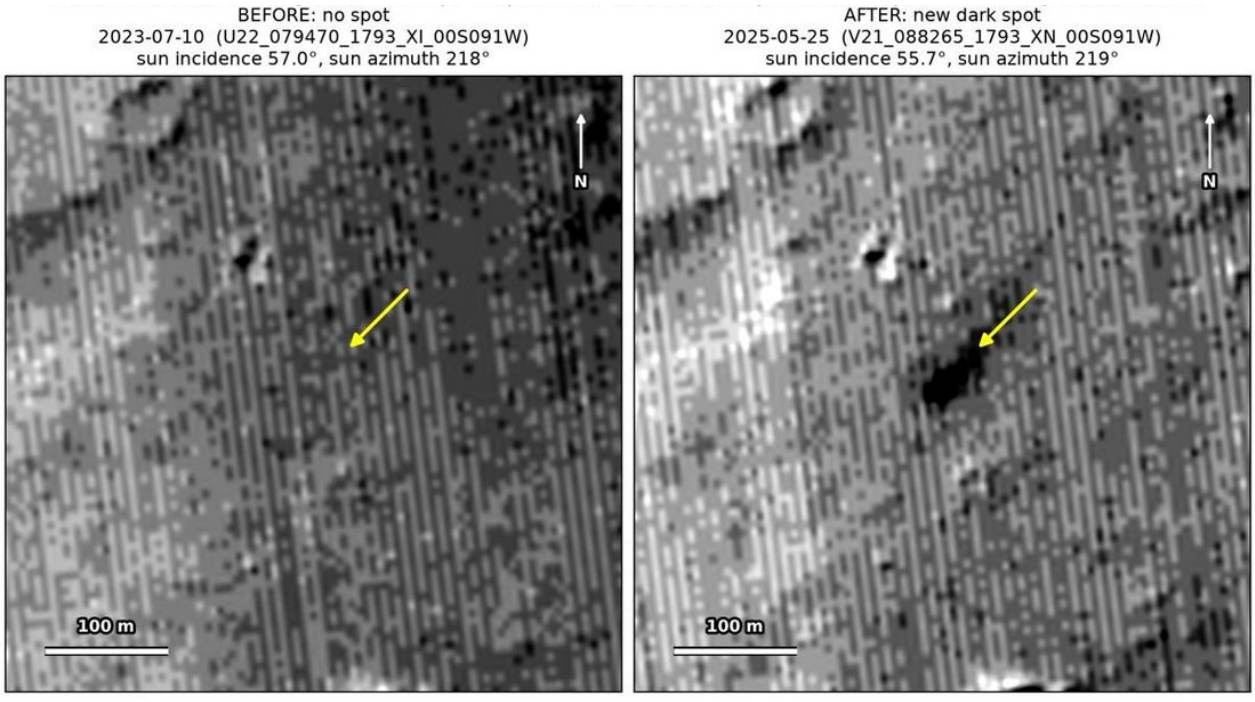

Figure 243: Candidate 95, before and after: the same 500 × 500 m of ground in 10 July 2023 (`U22_079470_1793_XI_00S091W`) and 25 May 2025 (`V21_088265_1793_XN_00S091W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

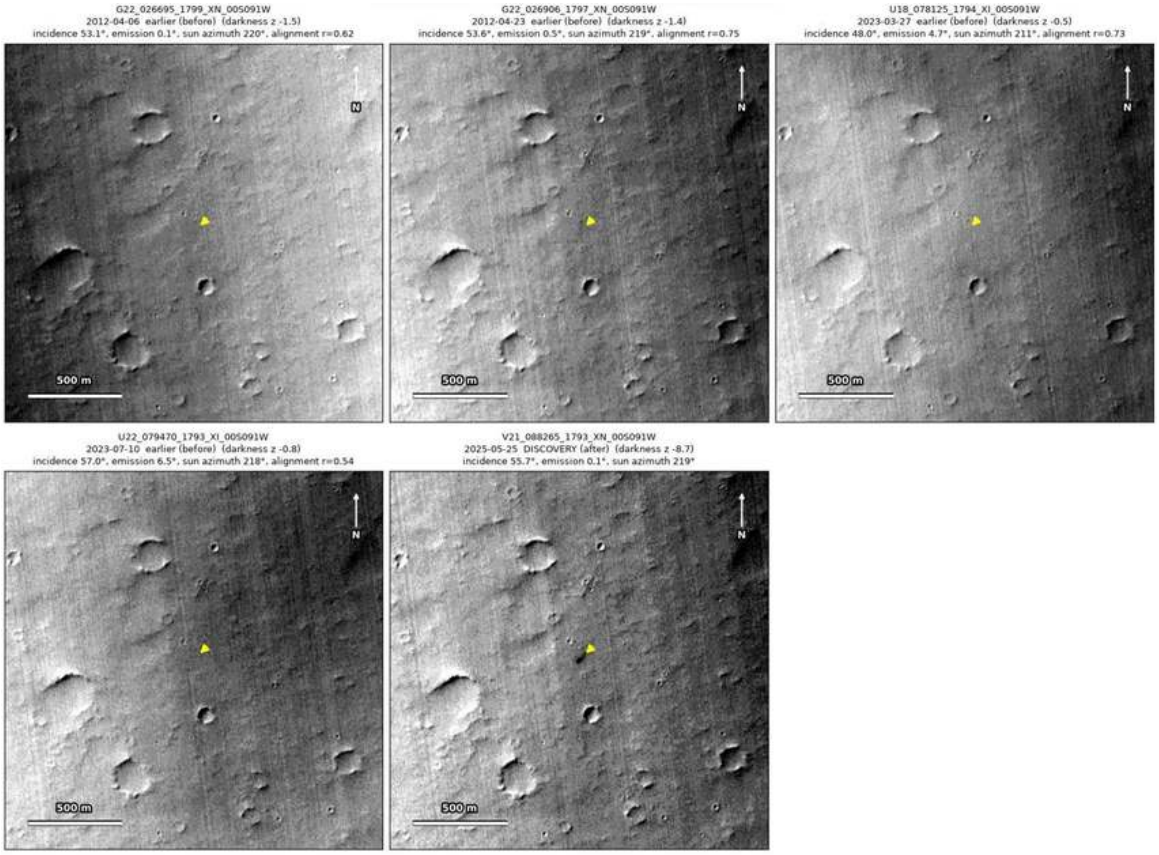

Figure 244: Candidate 95: five CTX images of the same 2×2 km patch, 2012 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 96: Floor of Tikhonravov crater, Arabia Terra

Location	12.8483° N, 36.1185° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	25 July 2020 and 4 April 2025
Images	3 without the spot (absent: 14 January 2010, 31 May 2020 and 25 July 2020); 1 with it (present: 4 April 2025)
Size	Compact dark spot about 40 m across (38 × 36 m), a dark core in a diffuse halo about 150 m across, mostly on its south-west side
HiRISE	none covers the spot (nearest footprint edge 22.1 km)
Nearest catalogued new impact	74 km (Daubar et al., 2022)

What changed. A compact dark spot about 40 m across, a dark core in a diffuse halo, appeared on the floor of Tikhonravov crater in Arabia Terra between July 2020 and April 2025. It is visible in one CTX image so far.

Why it may be a new impact. Three earlier images, from 2010 and 2020, all line up with the later one and show no spot there, while a small crater about 120 m to the north-north-east shows in all four images. A compact dark core in a diffuse halo is how the blast zone of a small new crater can look.

Caveats. Only one image shows it, and that image is heavily compressed, though the core and halo span many of the blocks in which it was compressed. The 2020 images were taken under a lower sun, though a spot this dark would show under it, and a nearby feature that was already there shows only faintly in the earlier images. When an earlier image is subtracted from the later one, a broad change at the eastern edge of the surrounding 2 × 2 km is slightly stronger than the spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 22.1 km away.

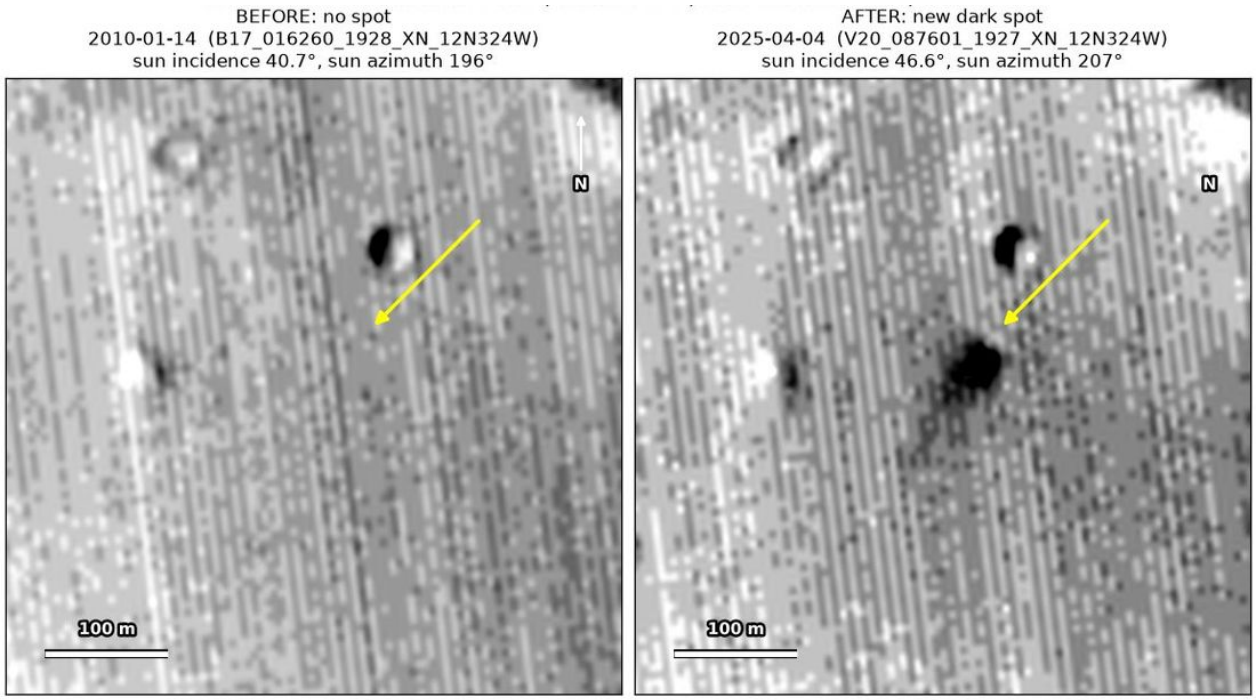

Figure 245: Candidate 96, before and after: the same 500 × 500 m of ground in 14 January 2010 (`B17_016260_1928_XN_12N324W`) and 4 April 2025 (`V20_087601_1927_XN_12N324W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

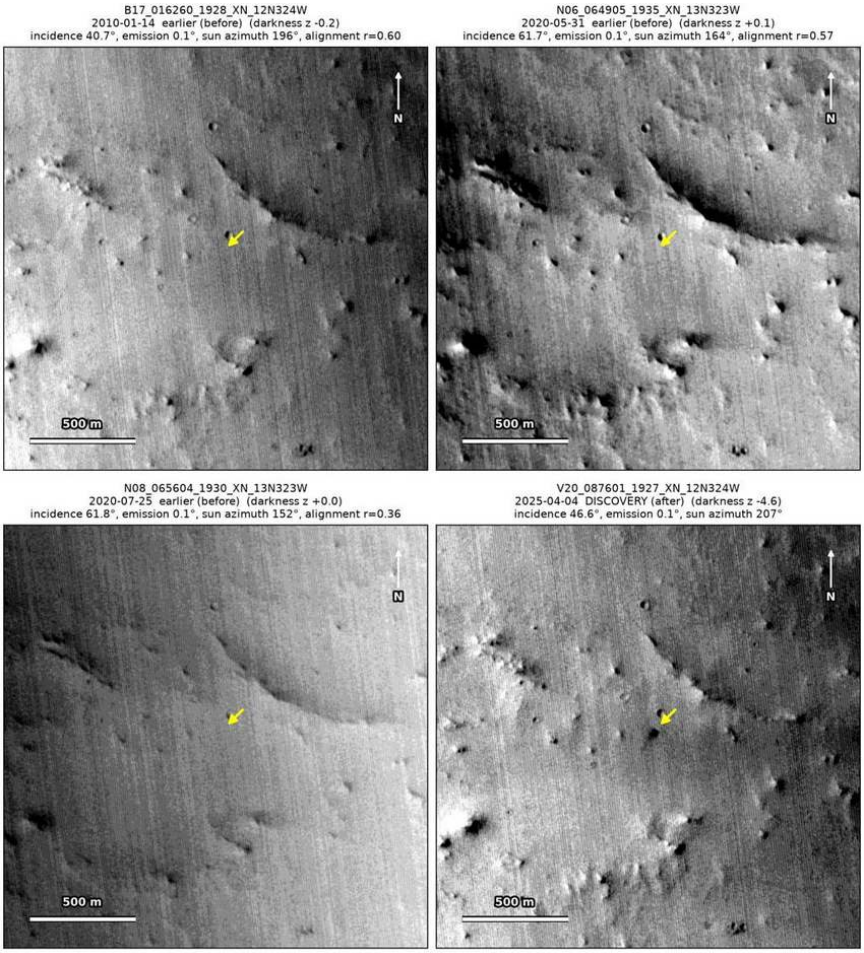

Figure 246: Candidate 96: four CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 97: Mangala Valles

Location	9.3950° S, 208.6036° E (planetocentric)
Assessment	weak candidate; probability of a new impact about 20%
Formed between	18 September 2023 and 13 May 2025
Images	6 without the spot (absent: 23 November 2006, 12 January 2015, 13 June 2016, 11 May 2018, 28 April 2023 and 18 September 2023); 1 with it (present: 13 May 2025)
Size	Teardrop-shaped dark streak about 90 × 30 m (within a box 94 × 54 m), with its broad end to the west-north-west
HiRISE	none covers the spot (nearest footprint edge 9.7 km)
Nearest catalogued new impact	155 km (Daubar et al., 2022)

What changed. An elongated dark streak about 90 × 30 m, shaped like a teardrop with its broad end to the west-north-west, appeared on smooth ground north-west of a rough slope in Mangala Valles between September 2023 and May 2025. It is visible in one CTX image so far.

Why it may be a new impact. Six earlier CTX images, from 2006 to 2023, each with its position checked against the ground in four parts of the scene, show nothing there; two of them, from April and September 2023, were taken under nearly the same sun as the May 2025 image. An image of 15 May 2002 from another camera, the Mars Orbiter Camera (MOC), shows nothing at the spot either, and an impact that strikes at a slant can make an elongated dark zone like this.

Caveats. It may well be a wind streak or part of a dust-devil track: the 2002 image shows a dark streak of the same orientation and size about 130 m to the north-east, gone by 2006, so streaks like this form and fade here. It is also seen in only one image and is a teardrop rather than a round spot. A fainter streak 290 m to the west-north-west shows clearly in only two of the CTX images, from April 2023 and May 2025, so it cannot show that the others would have caught a new streak. No image from the high-resolution HiRISE camera covers it; the nearest ends 9.7 km away.

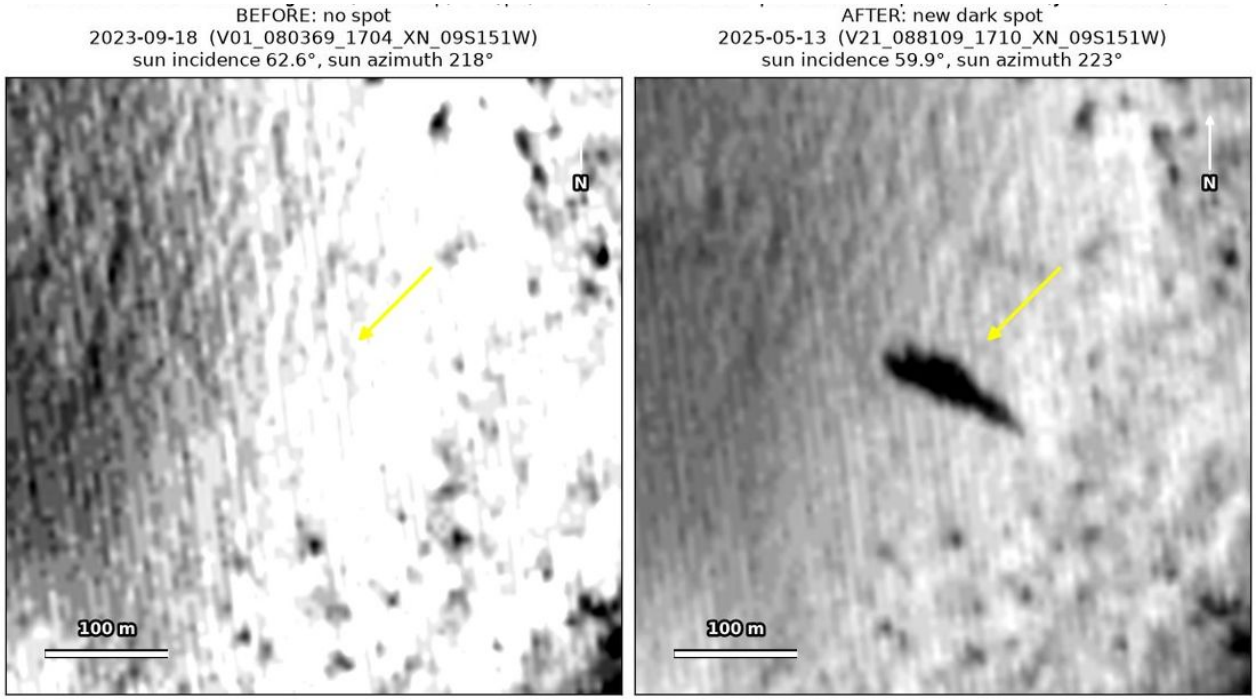

Figure 247: Candidate 97, before and after: the same 500 × 500 m of ground in 18 September 2023 (`V01_080369_1704_XN_09S151W`) and 13 May 2025 (`V21_088109_1710_XN_09S151W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

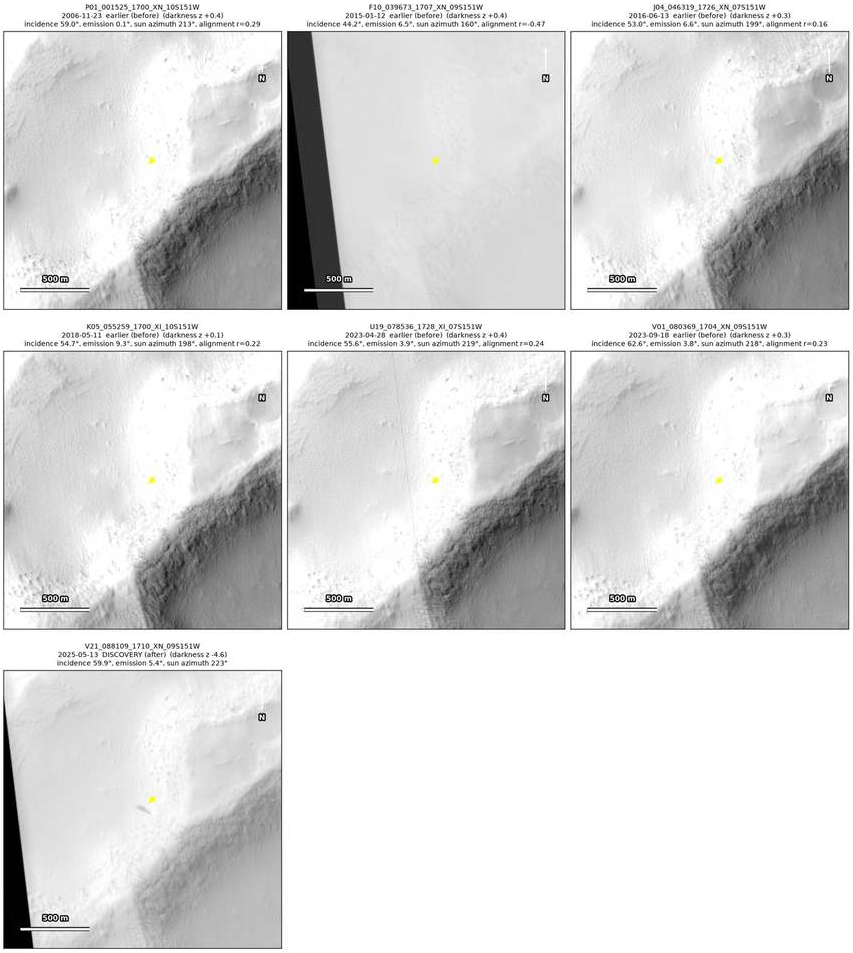

Figure 248: Candidate 97: seven CTX images of the same 2×2 km patch, 2006 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

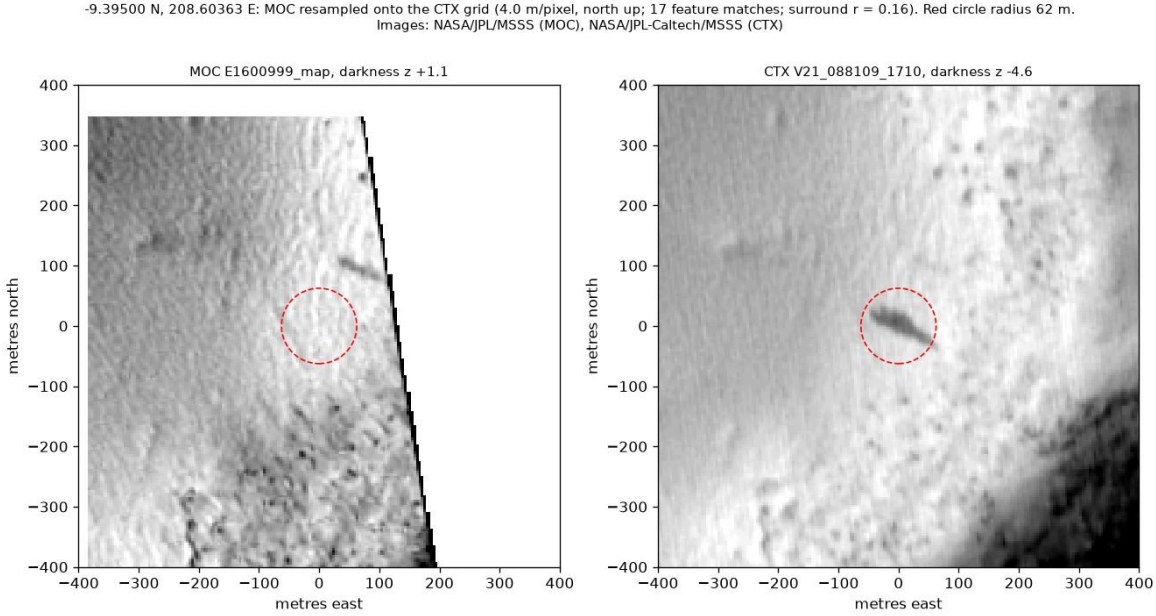

Figure 249: Candidate 97: An image from the older Mars Orbiter Camera (MOC) of 15 May 2002 (left), matched to the CTX image of 13 May 2025 (right); the red circle marks the spot. In 2002 there is nothing at the spot, but a similar dark streak lies about 130 m to the north-east, and no CTX image from 2006 on shows that streak; a fainter streak about 290 m to the west-north-west shows in both. MOC: NASA/JPL/MSSS; CTX: NASA/JPL-Caltech/MSSS.

Candidate 98: Lightly ridged plains in Sulci Gordii, east of Olympus Mons

Location	17.9059° N, 231.7729° E (planetocentric)
Assessment	weak candidate; probability of a new impact about 35%
Formed between	25 October 2020 and 4 April 2025
Images	2 without the spot (absent: 25 July 2020 and 25 October 2020); 1 with it (present: 4 April 2025)
Size	Compact dark spot about 35 m across (28 × 40 m), a dark core with a short faint tail to the south
HiRISE	none covers the spot (nearest footprint edge 9.5 km)
Nearest catalogued new impact	53 km (Daubar et al., 2022)

What changed. A compact dark spot about 35 m across, with a short faint tail to the south, appeared on lightly ridged plains in Sulci Gordii, 317 km east of the summit of Olympus Mons, between October 2020 and April 2025. It is visible in one CTX image so far.

Why it may be a new impact. Two earlier images, from July and October 2020, show no dark spot there, and when an earlier image is subtracted from the later one, the spot shows up as a change. A compact spot with a dark core that was not there before is how the blast zone of a small new crater can look.

Caveats. It is seen in only one heavily compressed image and spans only about 5 × 7 CTX pixels, so it may be noise or a compression artefact. Both earlier images were taken under a lower sun from another direction, so it may be shading or a brightness difference seen only under the April 2025 sun. Those images are also strongly striped and poorly aligned, a dark mark nearby is too faint to show that they would have caught the spot, and a dust devil that stalled is possible too. No image from the high-resolution HiRISE camera covers it; the nearest ends 9.5 km away.

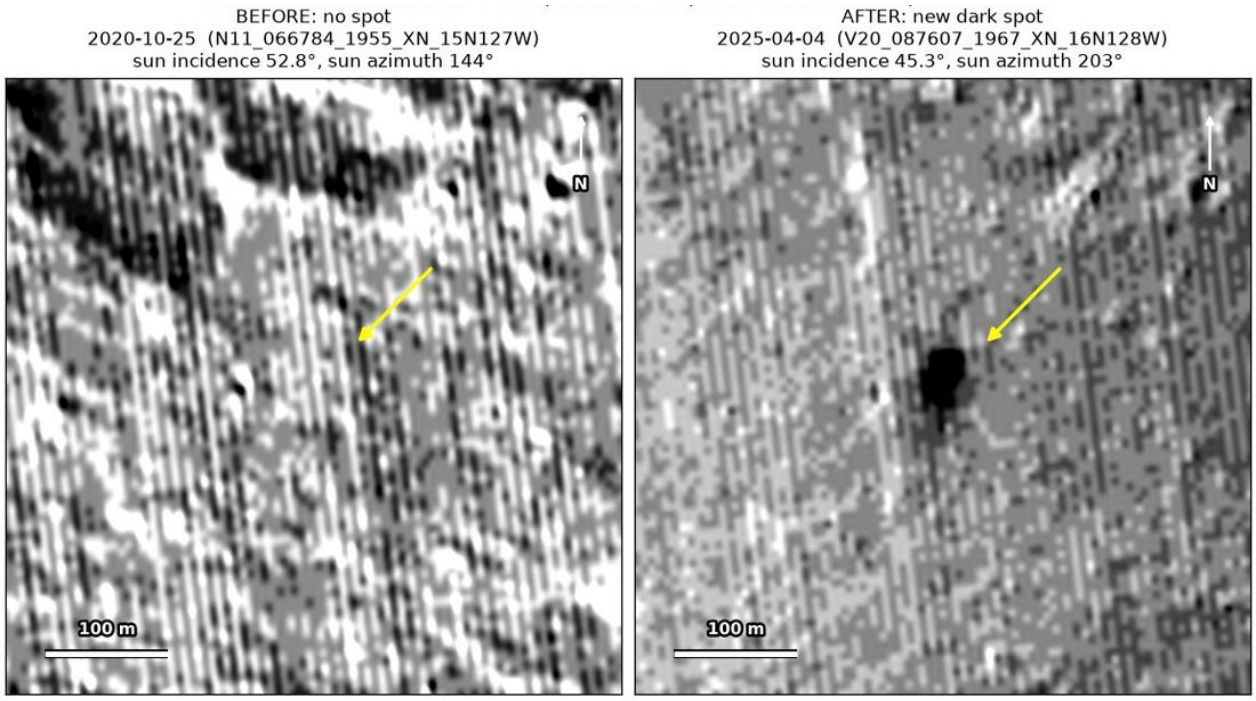

Figure 250: Candidate 98, before and after: the same 500 × 500 m of ground in 25 October 2020 (`N11_066784_1955_XN_15N127W`) and 4 April 2025 (`V20_087607_1967_XN_16N128W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

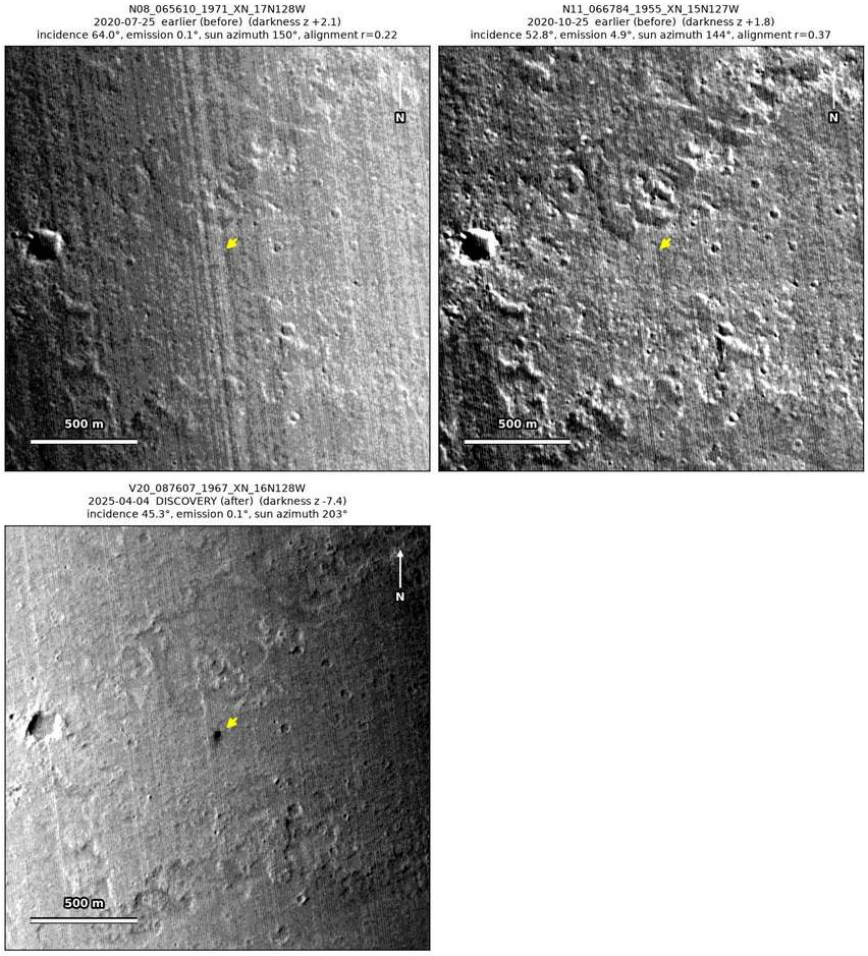

Figure 251: Candidate 98: three CTX images of the same 2×2 km patch, 2020 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 99: Cratered plains of Nepenthes Planum, 148 km south-west of Phedra crater

Location	12.0131° N, 122.1235° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 70%
Formed between	13 March 2022 and 7 May 2025
Images	4 without the spot (absent: 15 February 2012, 23 November 2015, 31 July 2020 and 13 March 2022); 2 with it (present: 7 May 2025 and 24 July 2025)
Size	Compact dark spot about 25 m across (22 × 26 m in the May 2025 image), a dark core in a faint halo
HiRISE	none covers the spot (nearest footprint edge 5.8 km)
Nearest catalogued new impact	95 km (Daubar et al., 2022)

What changed. A compact dark spot about 25 m across, with a faint halo, appeared on cratered plains of Nepenthes Planum, 148 km south-west of Phedra crater, between March 2022 and May 2025. Two CTX images, taken on two orbits 78 days apart, show it.

Why it may be a new impact. Four earlier images, from 2012 to 2022, all line up with the later ones and show no spot there, while a fainter dark mark 288 m to the west shows in every one. The second image with the spot, taken on another orbit, rules out an image defect, and a compact dark spot with a faint halo is how the blast zone of a small new crater looks.

Caveats. The last two images without the spot, from July 2020 and March 2022, were taken under a lower sun, and the March 2022 one has coarser pixels; if it is set aside, the spot may have formed as early as July 2020. A dust devil that stalled is also possible, though there are no tracks near it. No image from the high-resolution HiRISE camera covers it; the nearest ends 5.8 km away.

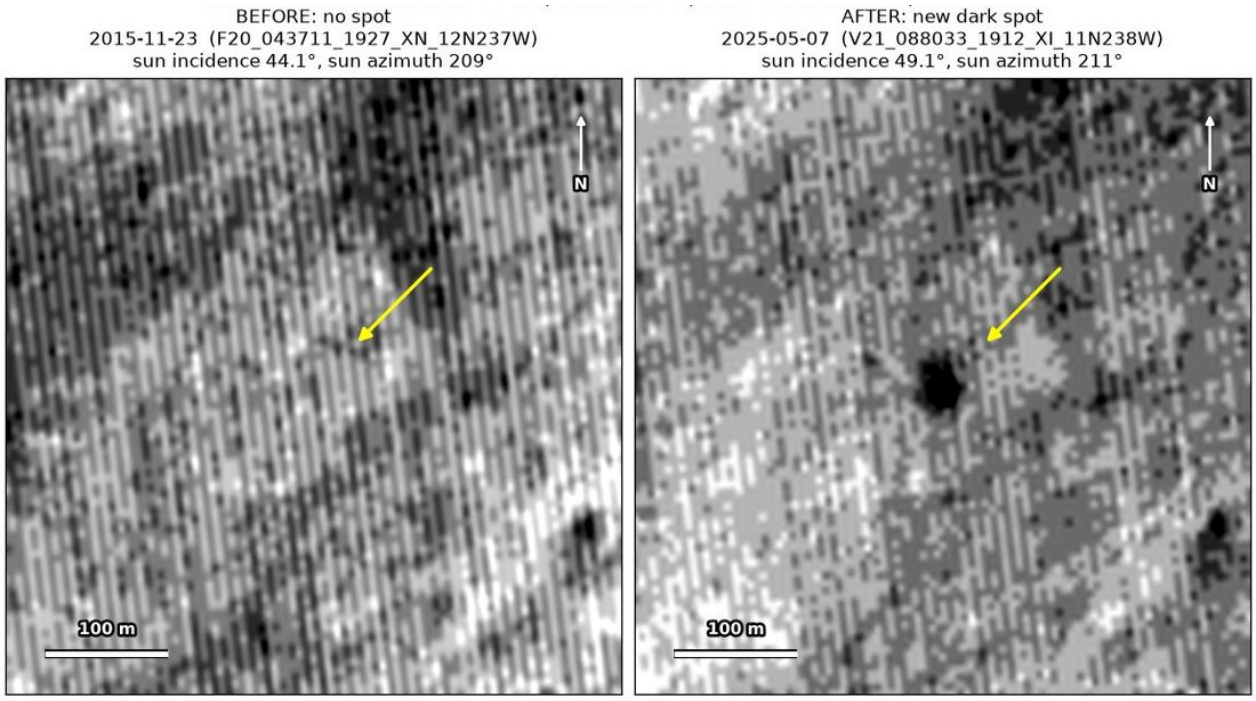

Figure 252: Candidate 99, before and after: the same 500 × 500 m of ground in 23 November 2015 (`F20_043711_1927_XN_12N237W`) and 7 May 2025 (`V21_088033_1912_XI_11N238W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

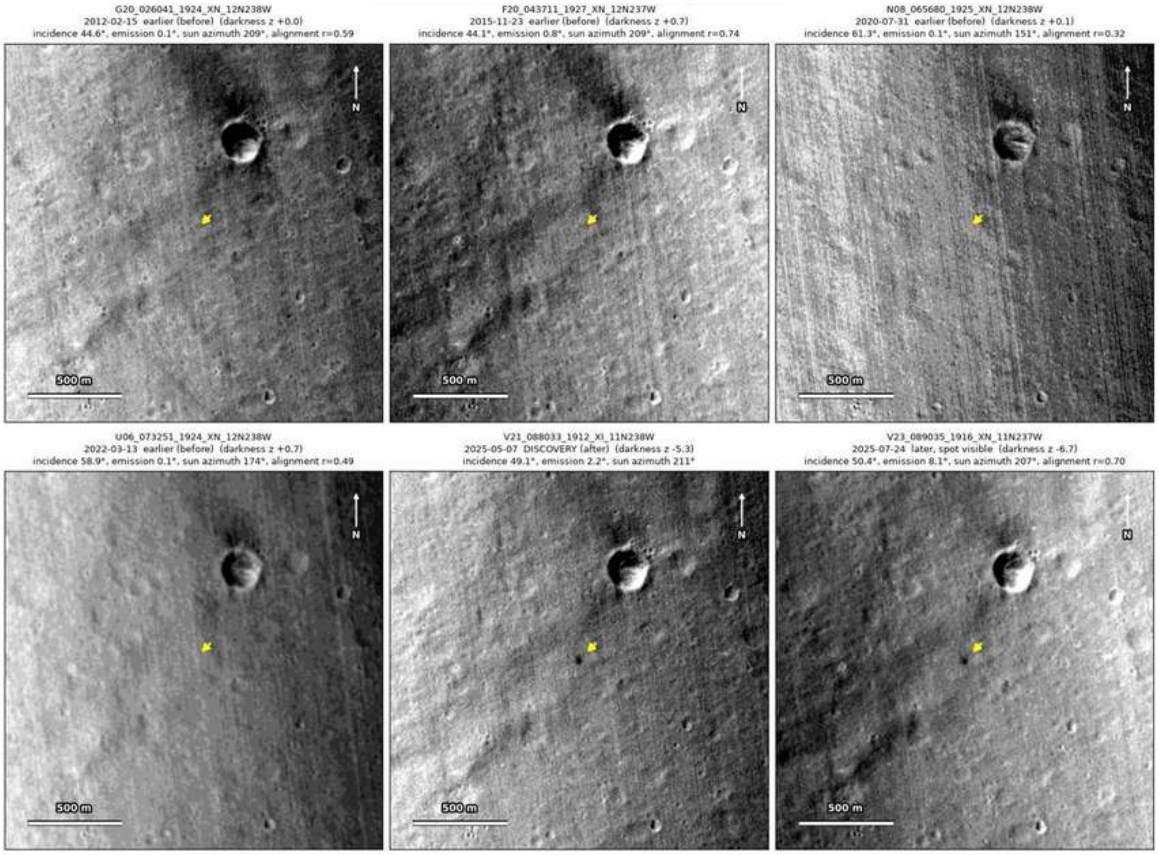

Figure 253: Candidate 99: six CTX images of the same 2×2 km patch, 2012 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 100: Plains 123 km north of Labeatis Mensa

Location	27.5278° N, 285.0236° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	27 May 2017 and 16 August 2020
Images	4 without the spot (absent: 24 October 2007, 27 January 2014, 16 July 2016 and 27 May 2017); 3 with it (present: 16 August 2020, 23 July 2022 and 20 December 2024)
Size	Compact dark spot about 55 m across (50 × 64 m in the December 2024 image), a dark core with fainter darkening out to about 150 m
HiRISE	ESP_026180_2080 (2012-02-26, before the spot formed; not impact-targeted)
Nearest catalogued new impact	145 km (Daubar et al., 2022)

What changed. A compact dark spot about 55 m across, with fainter darkening around it, appeared on cratered plains 123 km north of Labeatis Mensa between May 2017 and August 2020. Three CTX images from three orbits show it, the last in December 2024.

Why it may be a new impact. Four earlier images, from 2007 to 2017, show plain ground at the spot, while a small crater about 150 m to the east and a dark mark 300 m away show in every image. The last of the four, from May 2017, was taken under nearly the same sun as the December 2024 image. A compact spot with a dark core and fainter darkening around it, lasting more than four years, is how the blast zone of a small new crater looks.

Caveats. An image from NASA's high-resolution HiRISE camera covers the site, but it was taken on 26 February 2012, before the spot formed. The first two images with the spot, from August 2020 and July 2022, were taken under a low sun, though the December 2024 image, under the May 2017 sun, shows it clearly. When an earlier image is subtracted from a later image, a change at the edge of the surrounding 2 × 2 km is about as strong as the spot.

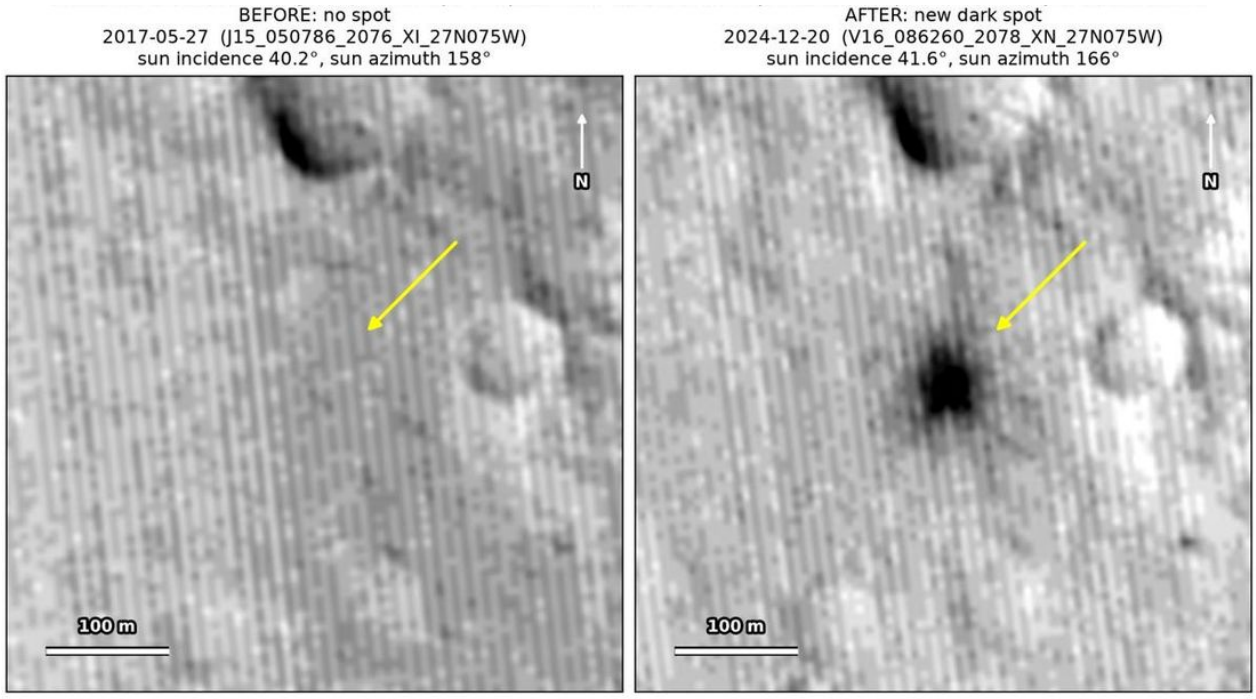

Figure 254: Candidate 100, before and after: the same 500 × 500 m of ground in 27 May 2017 (`J15_050786_2076_XI_27N075W`) and 20 December 2024 (`V16_086260_2078_XN_27N075W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

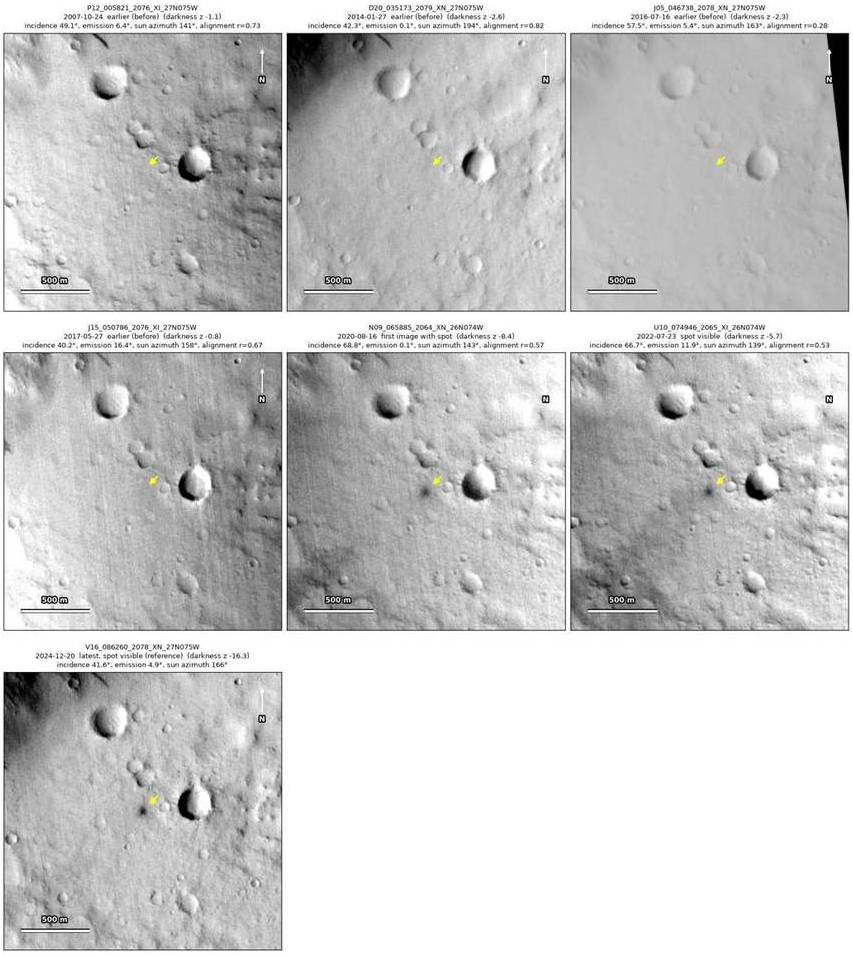

Figure 255: Candidate 100: seven CTX images of the same 2×2 km patch, 2007 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 101: Gordii Fossae

Location	16.4977° N, 233.9210° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	30 January 2023 and 4 December 2024
Images	5 without the spot (absent: 21 January 2008, 19 February 2016, 24 February 2016, 14 October 2022 and 30 January 2023); 1 with it (present: 4 December 2024)
Size	Compact dark spot about 25 m across (28 × 24 m)
HiRISE	none covers the spot (nearest footprint edge 10.0 km)
Nearest catalogued new impact	37 km (Daubar et al., 2022)

What changed. A compact dark spot about 25 m across appeared on ridged plains in Gordii Fossae between January 2023 and December 2024. It is visible in one CTX image so far.

Why it may be a new impact. Five earlier images, from 2008 to 2023, all line up with the later one and show no spot there; two of them, from January 2008 and January 2023, were taken under the same sun as the December 2024 image. A dark mark 216 m to the west shows clearly in the January 2008 image, and the spot is far darker than the striping of the image that shows it. A compact new dark spot is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, so an image defect or a cluster of noise cannot be fully ruled out, though the spot spans about 5 × 4 CTX pixels. When an earlier image is subtracted from the later one, a change at a corner of the surrounding 2 × 2 km is stronger than the spot, and the dark mark to the west shows clearly only in the January 2008 image. No image from the high-resolution HiRISE camera covers it; the nearest ends 10.0 km away.

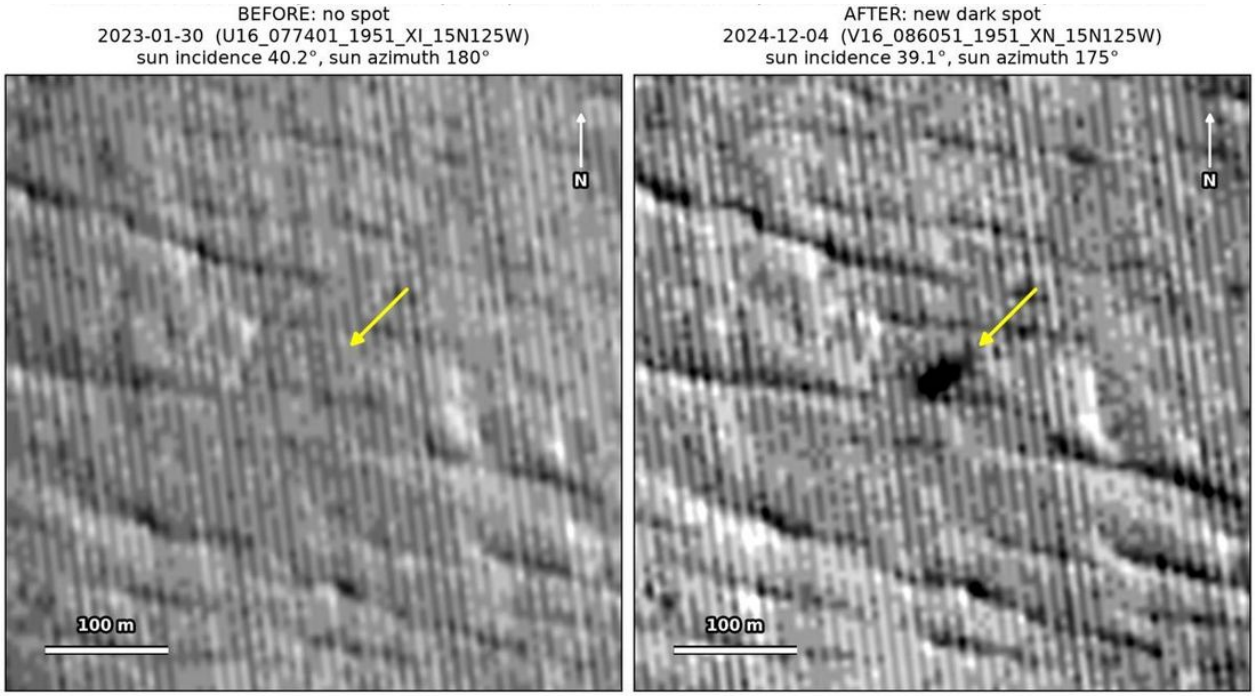

Figure 256: Candidate 101, before and after: the same 500 × 500 m of ground in 30 January 2023 (`U16_077401_1951_XI_15N125W`) and 4 December 2024 (`V16_086051_1951_XN_15N125W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

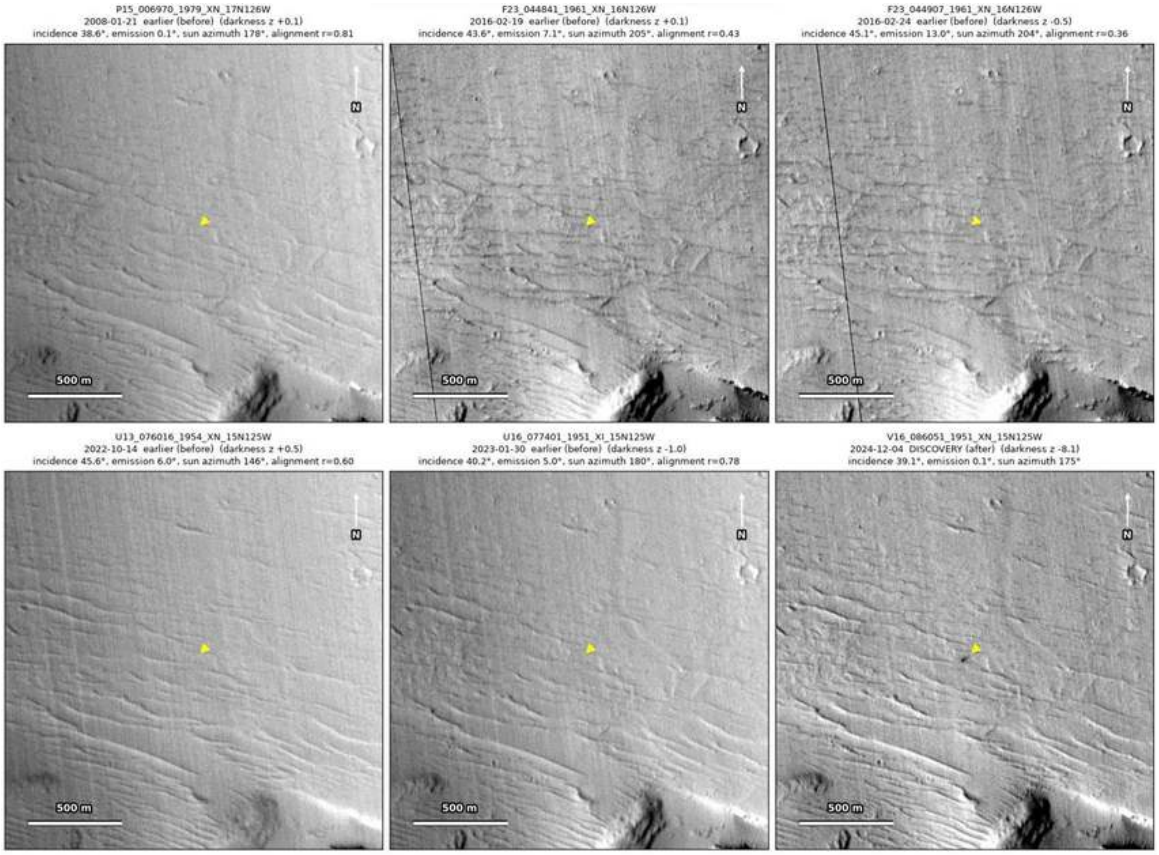

Figure 257: Candidate 101: six CTX images of the same 2×2 km patch, 2008 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 102: Plains 272 km south-east of Apollinaris Sulci

Location	13.3031° S, 181.5627° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 70%
Formed between	26 May 2023 and 10 June 2025
Images	2 without the spot (absent: 11 February 2010 and 26 May 2023); 2 with it (present: 10 June 2025 and 21 August 2025)
Size	Dark head about 40 × 25 m (42 × 26 m in the June 2025 image) at the west-north-west end of a fainter tail 73 m long
HiRISE	none covers the spot (nearest footprint edge 33.5 km)
Nearest catalogued new impact	141 km (Daubar et al., 2022)

What changed. An elongated dark spot, a dark head about 40 × 25 m with a fainter tail, appeared on smooth plains 272 km south-east of Apollinaris Sulci, about 200 m west of a small crater, between May 2023 and June 2025. Two CTX images, taken on two orbits 72 days apart, show it.

Why it may be a new impact. Two earlier images, from February 2010 and May 2023, show no spot there, the second taken under the same sun as the June 2025 image, while the small crater shows in all four images. When an earlier image is subtracted from a later one, the strongest change is right at the spot, and the second image with it, taken on another orbit, rules out an image defect. An impact that strikes at a slant can make an elongated dark zone like this one.

Caveats. With its elongated shape, it could also be a dark deposit left by wind or by a dust devil that stalled, though there are no tracks near it, and HiRISE was aimed at a 'new dark deposit' 100 km away in 2024. No nearby feature is dark enough to show that the earlier images would have caught it. No image from the high-resolution HiRISE camera covers it; the nearest ends 33.5 km away.

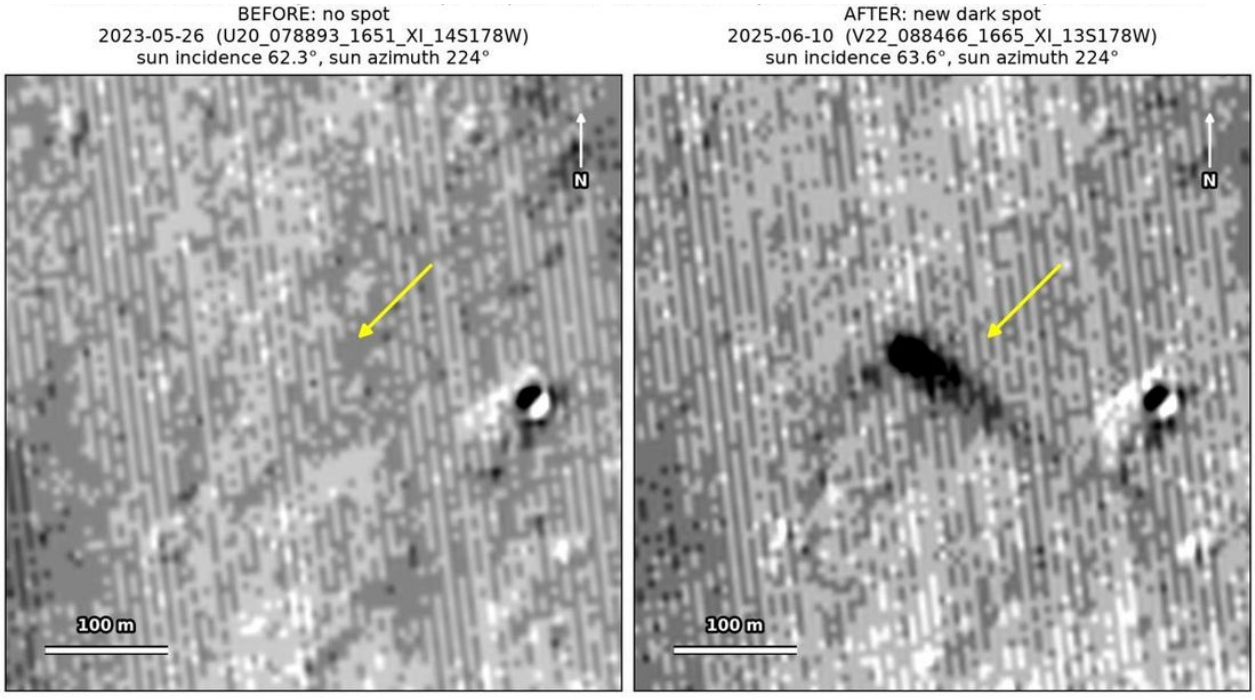

Figure 258: Candidate 102, before and after: the same 500 × 500 m of ground in 26 May 2023 (`U20_078893_1651_XI_14S178W`) and 10 June 2025 (`V22_088466_1665_XI_13S178W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

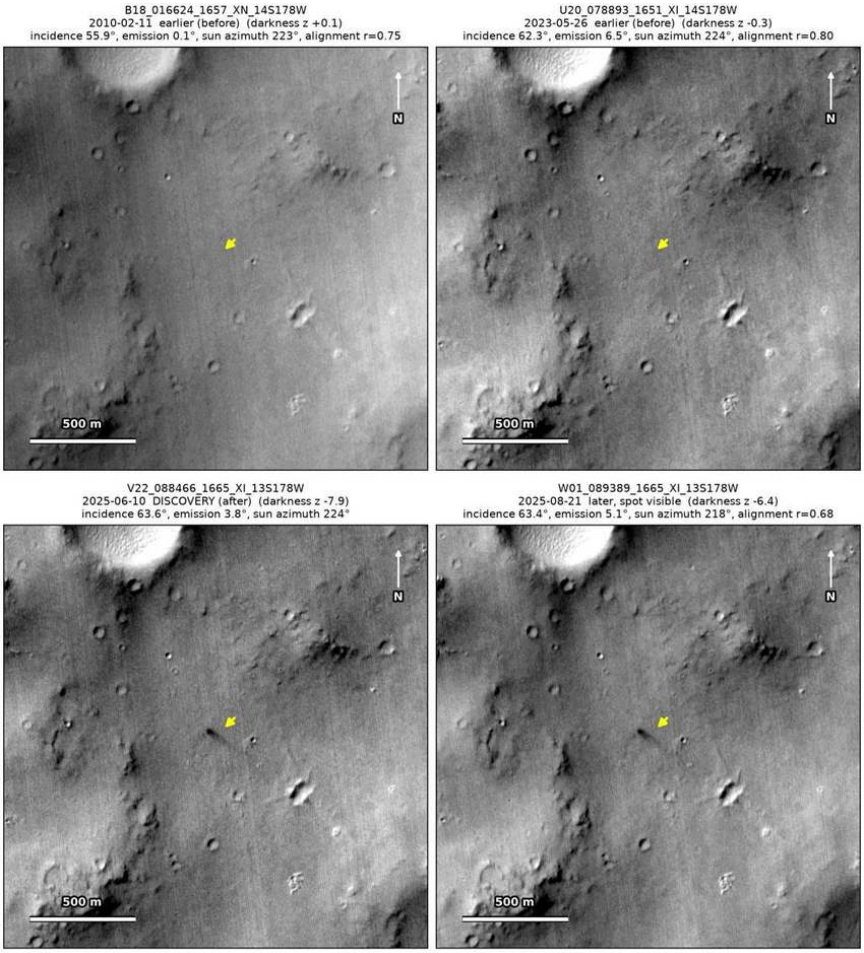

Figure 259: Candidate 102: four CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 103: Cratered plains in Nilus Chaos, 178 km east of candidate 45

Location	24.8247° N, 282.1434° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 70%
Formed between	21 May 2018 and 21 November 2024
Images	3 without the spot (absent: 15 September 2008, 10 March 2010 and 21 May 2018); 1 with it (present: 21 November 2024)
Size	Compact dark spot about 55 m across (44 × 76 m), a very dark core about 30 m across in a patchy dark zone
HiRISE	none covers the spot (nearest footprint edge 29.7 km)
Nearest catalogued new impact	43 km (Daubar et al., 2022)

What changed. A compact dark spot about 55 m across, a very dark core with a few dark specks around it, appeared on pitted and cratered plains in Nilus Chaos between May 2018 and November 2024. It is visible in one CTX image so far.

Why it may be a new impact. Three earlier images, from 2008 to 2018, all line up with the later one and show no spot there, while a dark mark 240 m to the west-south-west shows in every one. When an earlier image is subtracted from the later one, the strongest change anywhere in the surrounding 2 × 2 km is right at the spot. A very dark core with specks around it is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, so an image defect or a short-lived change of the surface cannot be fully ruled out, though the spot spans about 8 × 13 CTX pixels and the image's striping is far fainter. It could also be dark material exposed in one of the pits on these plains, though no pit shows there in March 2010. No image from the high-resolution HiRISE camera covers it; the nearest ends 29.7 km away.

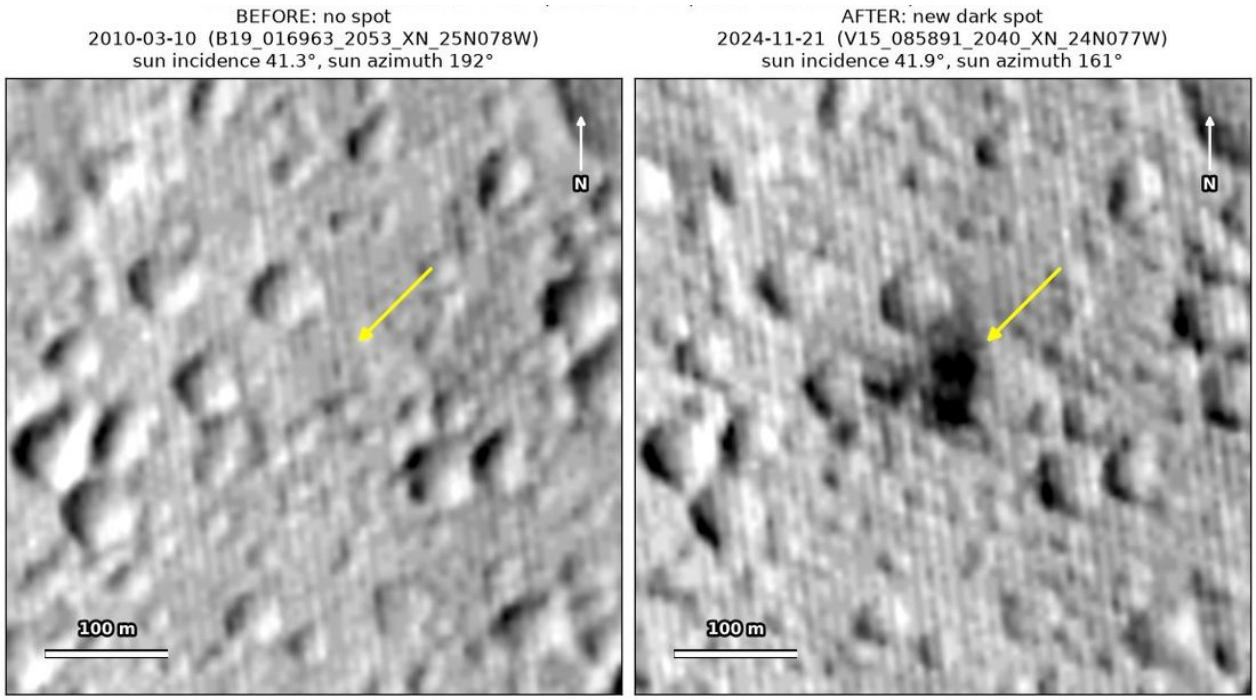

Figure 260: Candidate 103, before and after: the same 500 × 500 m of ground in 10 March 2010 (`B19_016963_2053_XN_25N078W`) and 21 November 2024 (`V15_085891_2040_XN_24N077W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

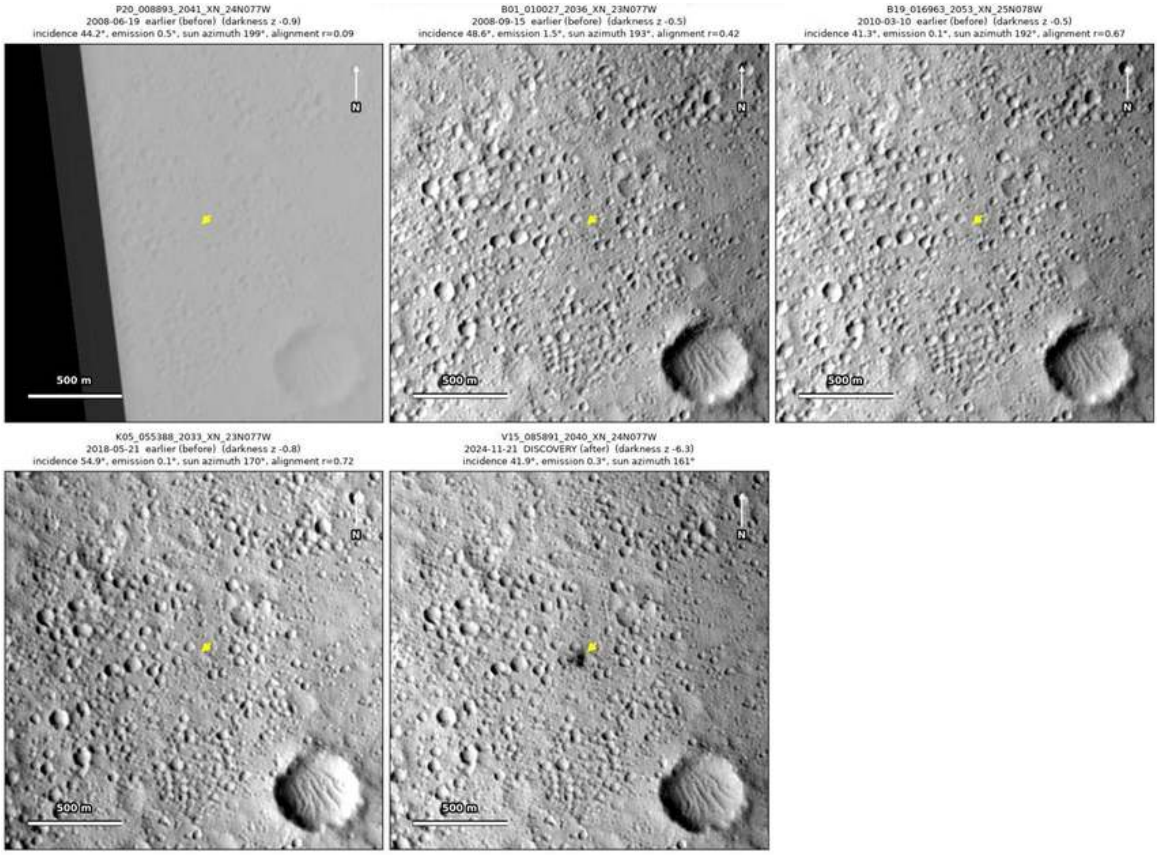

Figure 261: Candidate 103: five CTX images of the same 2×2 km patch, 2008 to 2024. The June 2008 image, washed out at the site, is shown but not used to date the spot. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 104: Smooth plains in Amazonis Planitia, 272 km south-west of Qara crater

Location	14.3619° N, 205.4454° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	21 January 2018 and 8 January 2023
Images	3 without the spot (absent: 27 March 2010, 10 July 2010 and 21 January 2018); 2 with it (present: 8 January 2023 and 11 November 2024)
Size	Compact dark spot about 36 m across (28 × 50 m)
HiRISE	none covers the spot (nearest footprint edge 43.1 km)
Nearest catalogued new impact	183 km (Daubar et al., 2022)

What changed. A compact dark spot about 36 m across appeared on smooth plains in Amazonis Planitia, about 400 m west of the edge of rough, knobby ground, between January 2018 and January 2023. Two CTX images from two orbits show it, the later in November 2024.

Why it may be a new impact. Three earlier images, from 2010 and 2018 and all aligned with the later ones, show no spot there, while small craters 40 m to the west show in all five images. When an earlier image is subtracted from a later one, the strongest change in the surrounding 2 × 2 km is right at the spot, except at the very edge of that area. A compact dark spot, unchanged after 22 months and with no dust-devil tracks near it, is how the blast zone of a small new crater looks.

Caveats. The earlier images were all taken under a lower sun than the later ones, and a dark mark nearby is only faint in them, so it cannot show that they would have caught the spot. A small hollow that was already there is unlikely, as none shows in the earlier images, whose lower sun would show one better. No image from the high-resolution HiRISE camera covers it; the nearest ends 43.1 km away.

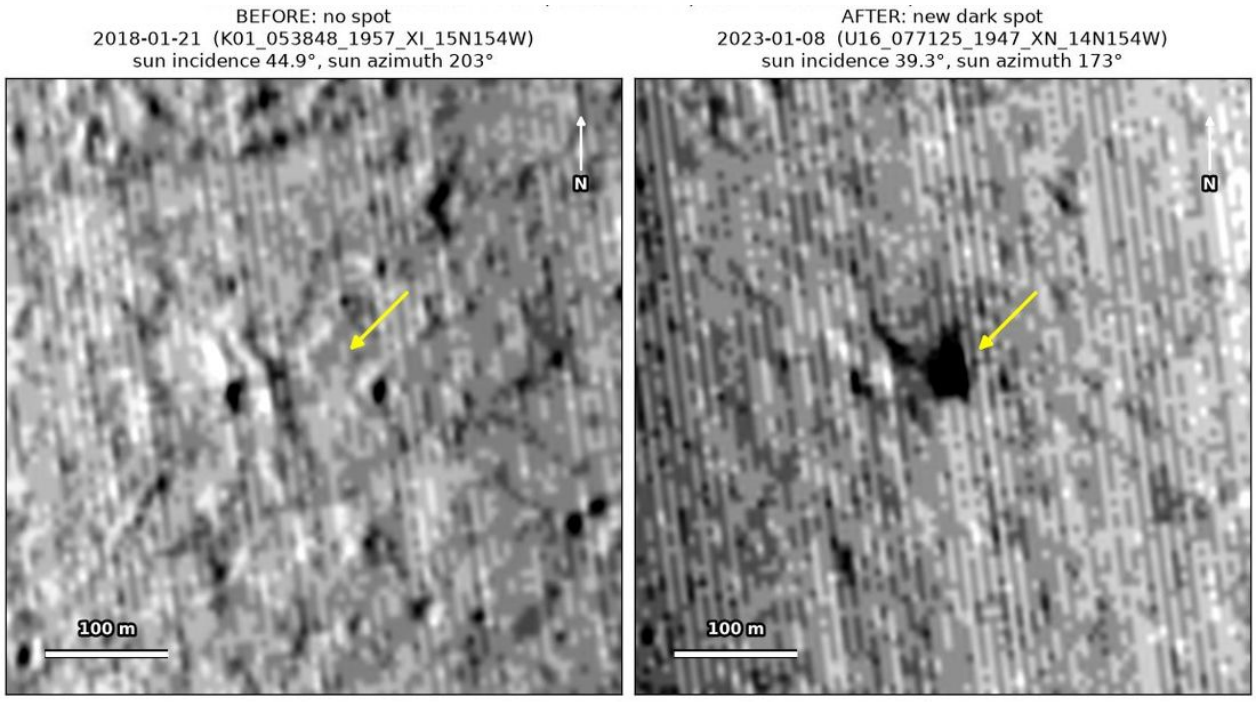

Figure 262: Candidate 104, before and after: the same 500 × 500 m of ground in 21 January 2018 (`K01_053848_1957_XI_15N154W`) and 8 January 2023 (`U16_077125_1947_XN_14N154W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

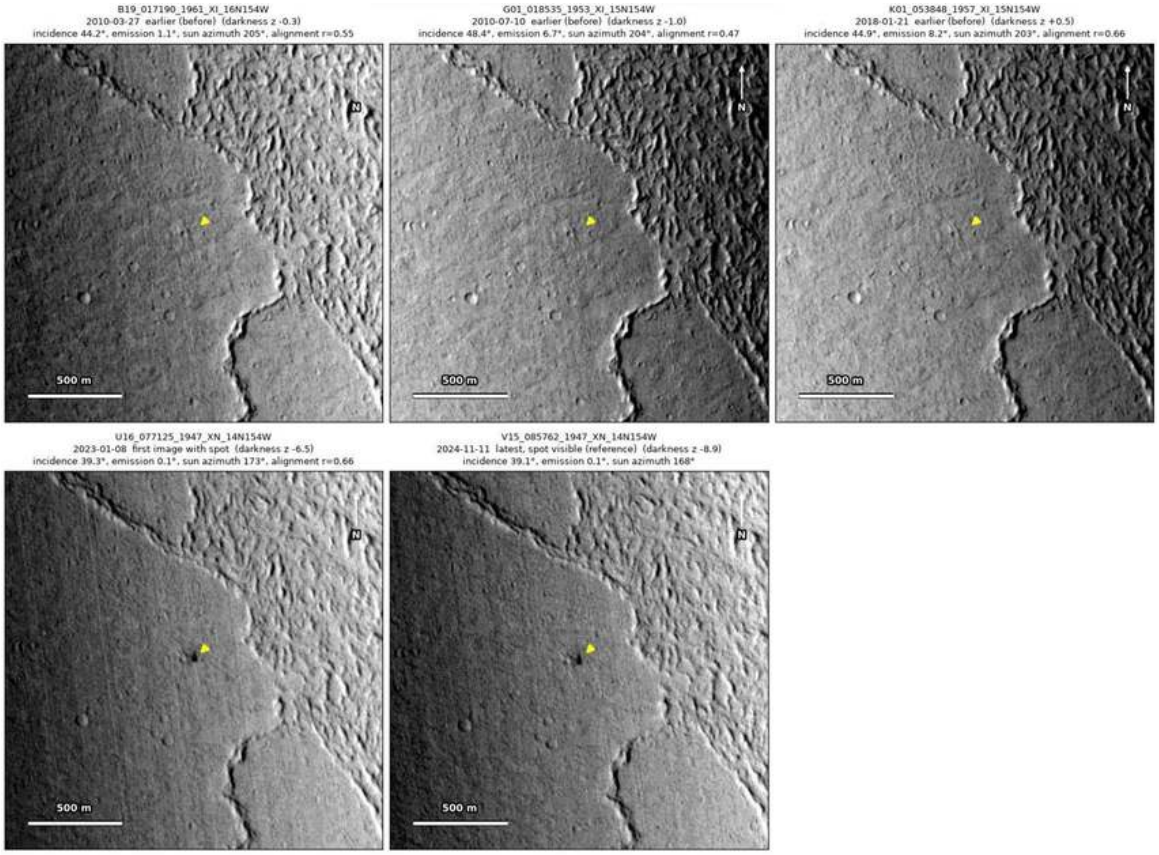

Figure 263: Candidate 104: five CTX images of the same 2×2 km patch, 2010 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 105: Plains 276 km north-east of Tharsis Tholus

Location	15.5441° N, 273.4954° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 70%
Formed between	19 February 2010 and 25 June 2020
Images	1 without the spot (absent: 19 February 2010); 2 with it (present: 25 June 2020 and 22 February 2025)
Size	Diffuse dark spot about 69 m across (58 × 114 m), a darker core in a diffuse zone elongated north–south
HiRISE	none covers the spot (nearest footprint edge 40.8 km)
Nearest catalogued new impact	122 km (Daubar et al., 2022)

What changed. A diffuse dark spot about 69 m across, elongated north–south and with a darker core, appeared on lava plains 276 km north-east of Tharsis Tholus between February 2010 and June 2020. Two CTX images from two orbits show it, the second, from February 2025, still unfaded.

Why it may be a new impact. An image from February 2010, taken under almost the same sun as the February 2025 image, shows no spot there but does show a dark mark 260 m to the west-south-west. When it is subtracted from a later image, the strongest change in the surrounding 2 × 2 km is right at the spot, except at the edge of that area. A dark spot with a darker core that has not faded over the 4.7 years between its two images is how the blast zone of a small new crater looks.

Caveats. Only one usable earlier image exists, and a dark band that was already there, 100 to 150 m to the south-west, lies parallel to the spot's long axis, so the spot may be a newly darkened stretch of it, caused by dust sliding in a shallow fissure or by wind scouring along it. No image from the high-resolution HiRISE camera covers it; the nearest ends 40.8 km away.

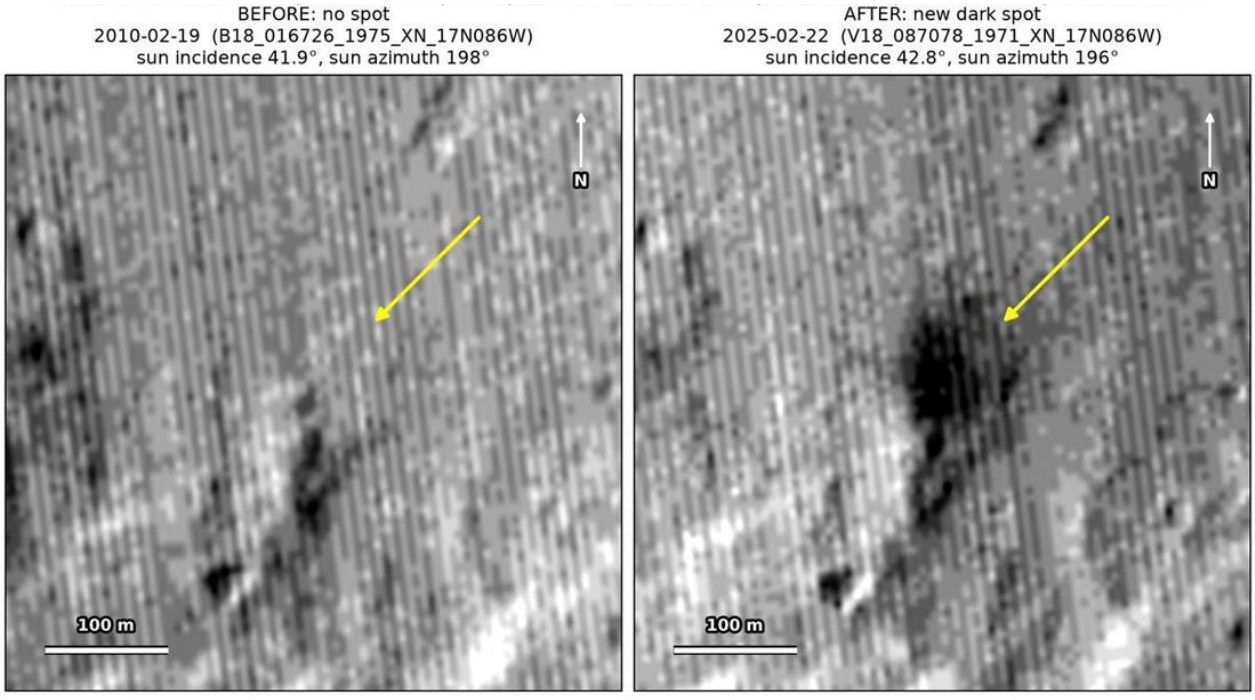

Figure 264: Candidate 105, before and after: the same 500 × 500 m of ground in 19 February 2010 (`B18_016726_1975_XN_17N086W`) and 22 February 2025 (`V18_087078_1971_XN_17N086W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

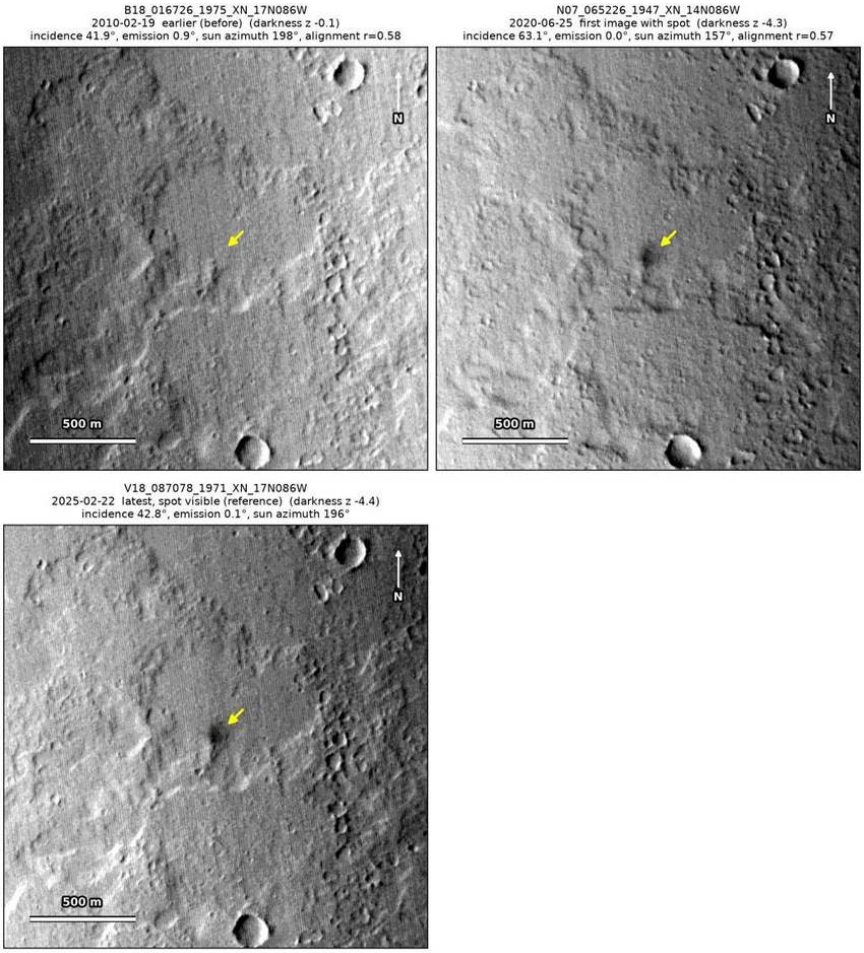

Figure 265: Candidate 105: three CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 106: Plains near Phlegra Montes, 269 km east of Aesacus Dorsum

Location	37.2860° N, 158.8058° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	6 December 2023 and 3 January 2025
Images	4 without the spot (absent: 12 January 2010, 26 April 2010, 13 May 2016 and 6 December 2023); 1 with it (present: 3 January 2025)
Size	Cluster of dark spots: a dark halo about 130 m across (154 × 170 m) with a darker core about 60 m across, and three smaller dark spots 110 to 280 m away
HiRISE	ESP_056685_2180 (2018-08-30, before the spot formed; not impact-targeted)
Nearest catalogued new impact	173 km (Daubar et al., 2022)

What changed. A cluster of dark spots appeared on plains near Phlegra Montes between December 2023 and January 2025: a dark halo about 130 m across with a darker core, and three smaller spots within 280 m of it. So far only one CTX image shows it.

Why it may be a new impact. Four earlier images, from 2010 to 2023, show the site no darker than its surroundings, including one of January 2010 taken under nearly the same sun as the January 2025 image. Subtracting an earlier image from the later one shows the strongest change in the surrounding 2 × 2 km right at the halo. Separate round dark spots with a halo and no tracks are not what dust devils or wind leave, but fit a small new impact.

Caveats. A HiRISE image of 30 August 2018 covers the site, but it was taken years before the spots formed. Only one CTX image shows the cluster, so an image defect cannot be fully ruled out, although its separate parts span about 80 image pixels. The spots' absence from the two 2023 images falls just short of being confirmed, so they may have formed as early as May 2016. A dark mark nearby, which would show what the earlier images could reveal, lies too close to the cluster to be told apart from it cleanly.

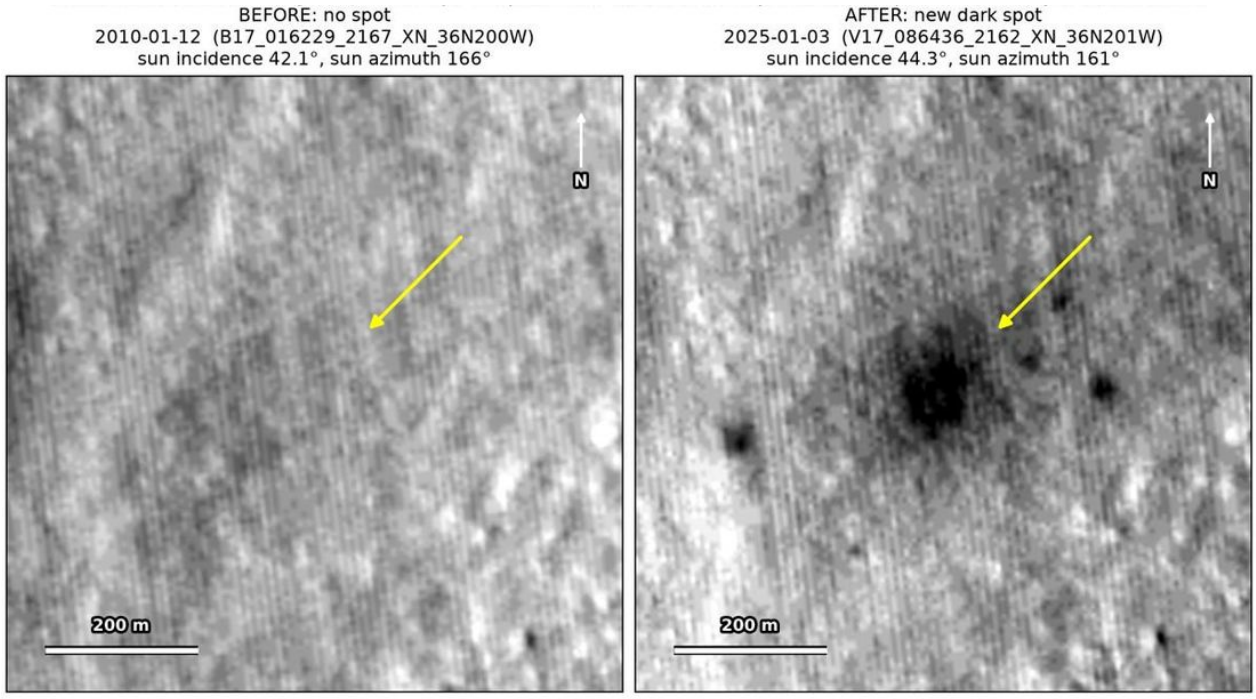

Figure 266: Candidate 106, before and after: the same 800 × 800 m of ground in 12 January 2010 (`B17_016229_2167_XN_36N200W`) and 3 January 2025 (`V17_086436_2162_XN_36N201W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

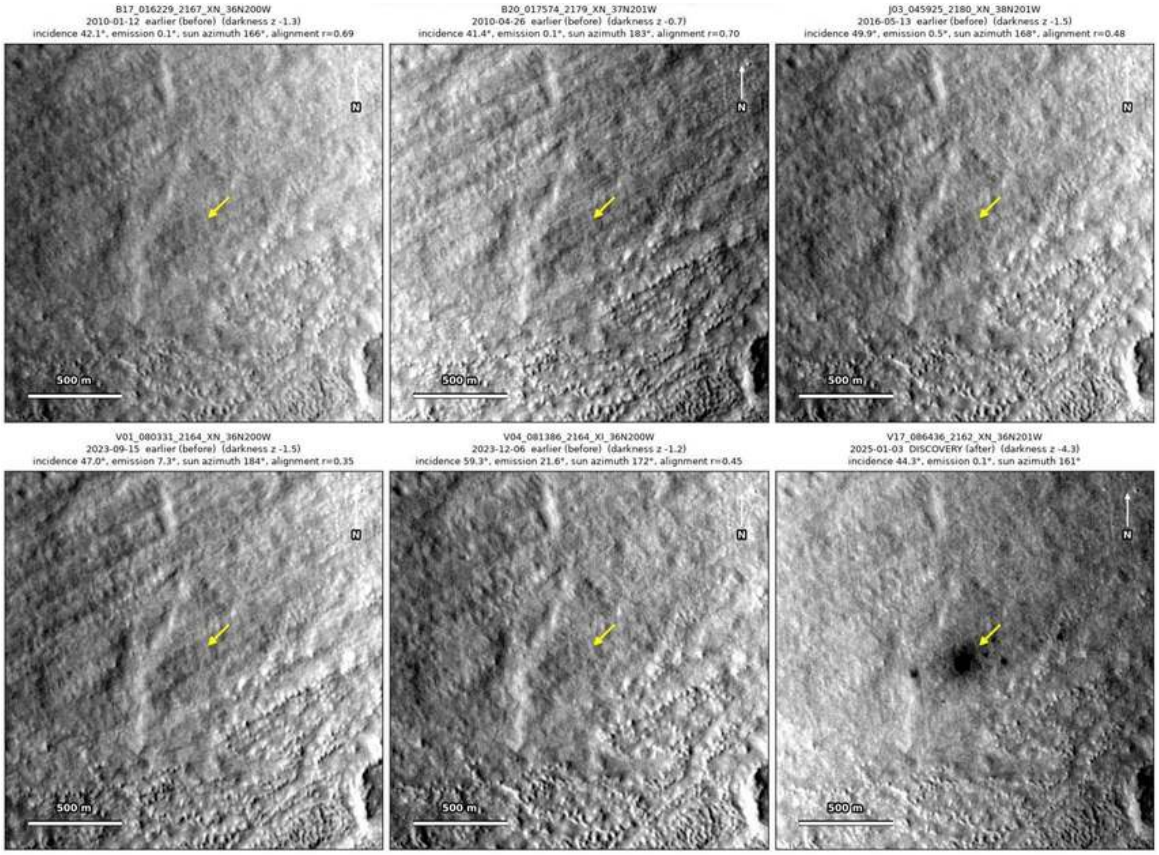

Figure 267: Candidate 106: six CTX images of the same 2×2 km patch, 2010 to 2025. The September 2023 image, which shows nothing there but lines up only roughly with the others, is shown but not used to date the spots. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 107: Plains in Arabia Terra, 147 km north-west of Ramanathan crater

Location	29.2900° N, 25.4220° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 55%
Formed between	21 March 2014 and 18 December 2024
Images	1 without the spot (absent: 21 March 2014); 1 with it (present: 18 December 2024)
Size	Compact dark spot about 35 m across (32 × 40 m), with a smaller dark speck about 100 m to the east
HiRISE	none covers the spot (nearest footprint edge 7.7 km)
Nearest catalogued new impact	51 km (Daubar et al., 2022)

What changed. A compact, very dark spot about 35 m across, with a smaller dark speck 100 m to the east, appeared on plains in northern Arabia Terra between March 2014 and December 2024. So far only one CTX image shows it.

Why it may be a new impact. The March 2014 image, the only earlier one that could be measured, shows no spot there, and the February 2010 image shows none by eye. Both earlier images are noisy, but a spot this dark could not hide in them, and a compact, very dark spot with no tracks around it is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, so an image defect cannot be fully ruled out, although the spot spans about 7 × 8 image pixels. The earlier images are noisy and striped: a dark mark nearby, which would show what they could reveal, does not appear in the March 2014 image, and the difference between the images is noisy, with a stronger change 780 m away. No image from the high-resolution HiRISE camera covers it; the nearest ends 7.7 km away.

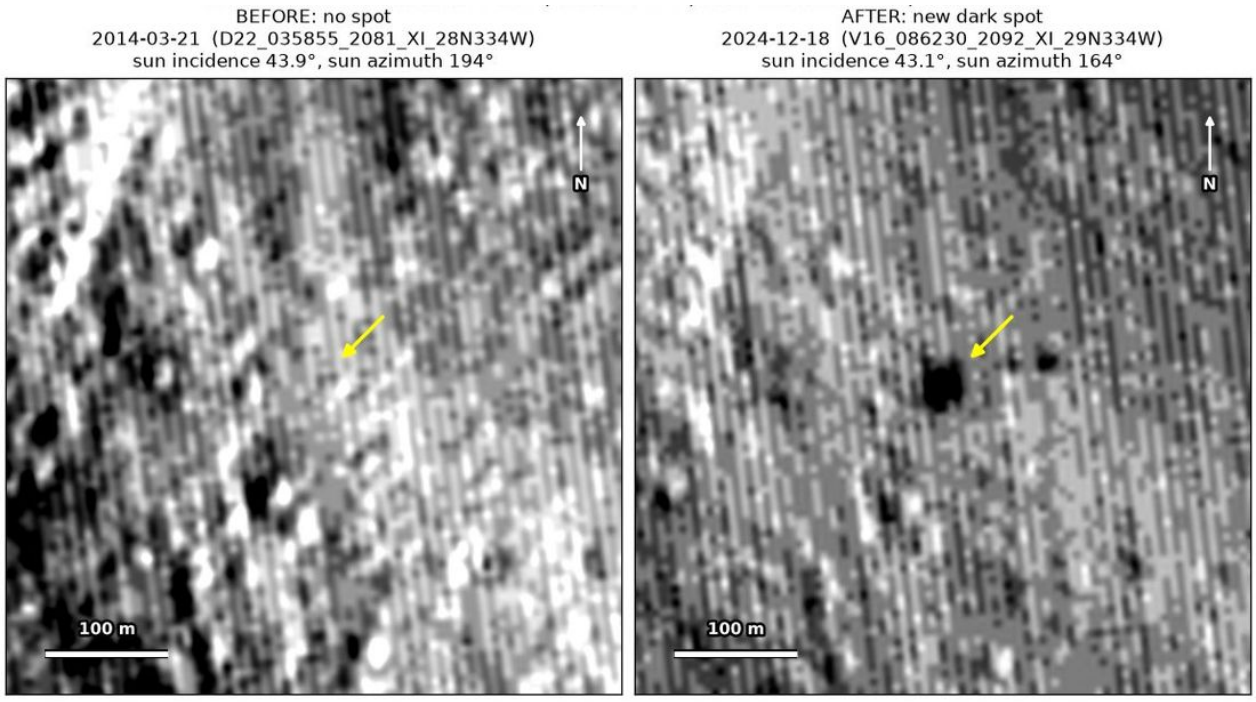

Figure 268: Candidate 107, before and after: the same 500 × 500 m of ground in 21 March 2014 (`D22_035855_2081_XI_28N334W`) and 18 December 2024 (`V16_086230_2092_XI_29N334W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

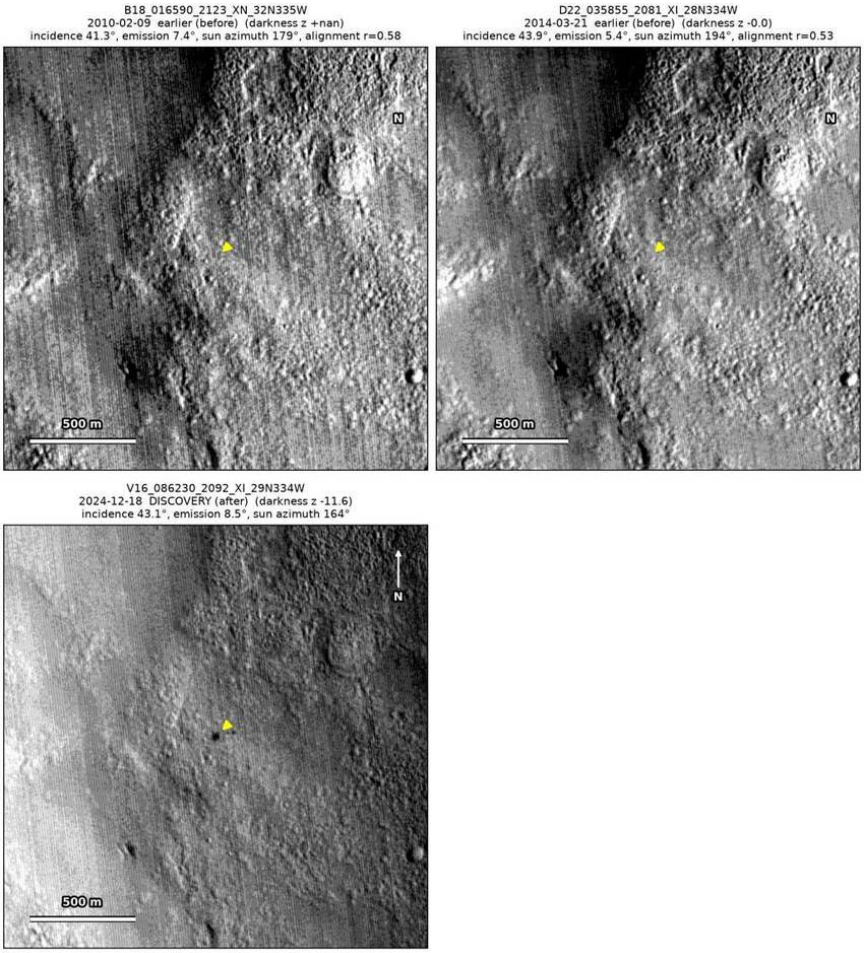

Figure 269: Candidate 107: three CTX images of the same 2×2 km patch, 2010 to 2024. The February 2010 image shows no spot by eye but could not be measured, so it is shown but not used to date the spot. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 108: Lava plains of Elysium Planitia, among the Cerberus Tholi

Location	4.7761° N, 169.2420° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 70%
Formed between	19 December 2015 and 19 December 2024
Images	1 without the spot (absent: 19 December 2015); 1 with it (present: 19 December 2024)
Size	Dark spot about 110 m across (140 × 128 m), a ragged dark core with a short ray to the east and a fainter tail reaching about 290 m to the west-south-west
HiRISE	none covers the spot (nearest footprint edge 6.4 km)
Nearest catalogued new impact	228 km (Daubar et al., 2022)

What changed. A dark spot about 110 m across, with ragged edges, a short ray to the east and a fainter tail to the west-south-west, appeared on smooth lava plains of Elysium Planitia among the Cerberus Tholi between December 2015 and December 2024. So far only one CTX image shows it.

Why it may be a new impact. The earlier image of 19 December 2015 shows no spot there, and its sun, lower than in the December 2024 image, would show a hollow better, not worse; two more earlier images show nothing by eye. Subtracting the earlier image from the later one shows the strongest change at the spot and no other new dark mark in the surrounding 5 × 5 km. A dark core with a ray and no tracks fits the blast zone of a small new crater better than wind or a dust devil.

Caveats. Only one image shows it, so an image defect cannot be fully ruled out, although the spot spans about 25 × 23 image pixels and has a ray and a tail. Only one earlier image is used to date it, and a dark mark nearby is too faint in the earlier images to show that they would have caught a spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 6.4 km away.

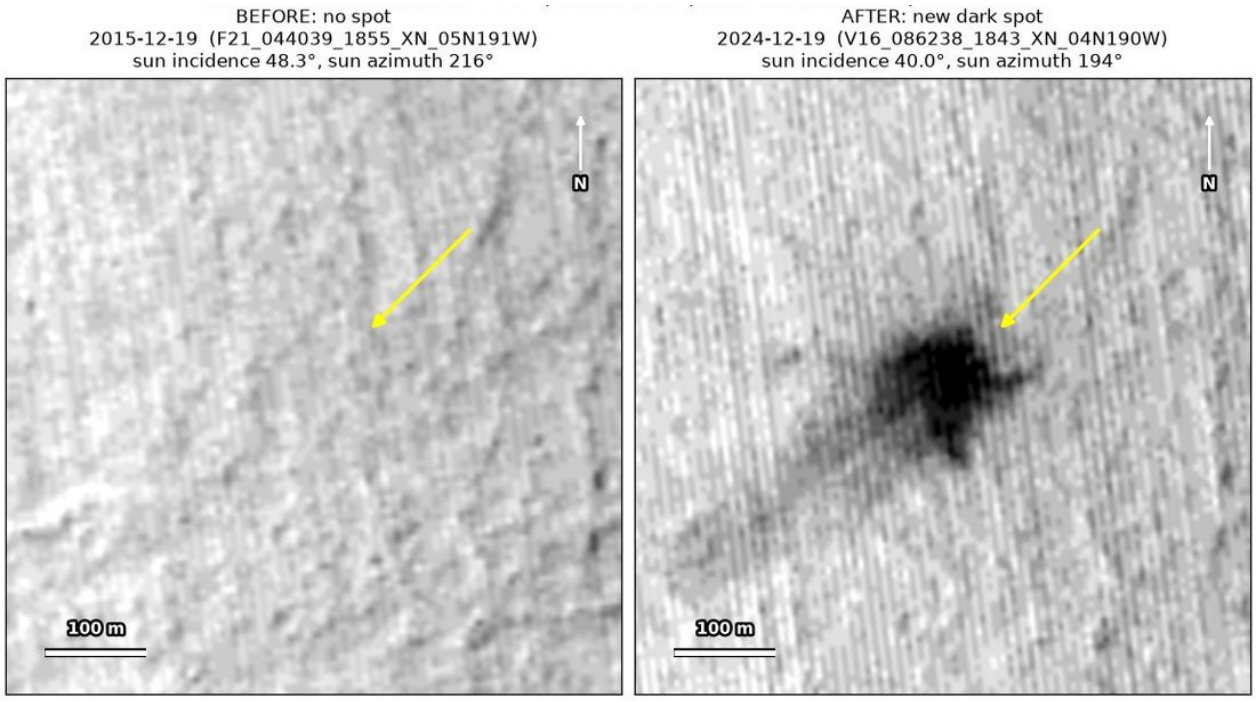

Figure 270: Candidate 108, before and after: the same 600 × 600 m of ground in 19 December 2015 (`F21_044039_1855_XN_05N191W`) and 19 December 2024 (`V16_086238_1843_XN_04N190W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

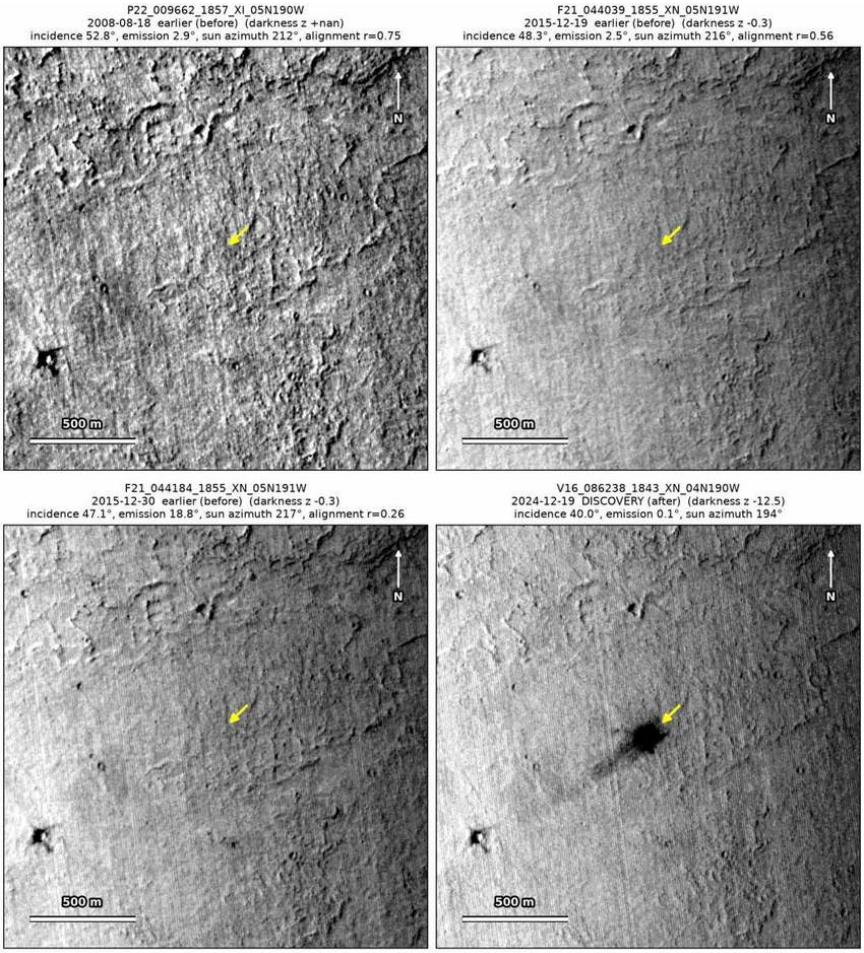

Figure 271: Candidate 108: four CTX images of the same 2×2 km patch, 2008 to 2024. The images of August 2008 and 30 December 2015, which show nothing there by eye, are shown but not used to date the spot: the first could not be measured at the spot and the second lines up poorly. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 109: Plains 272 km south of Medusae Fossae

Location	6.4592° S, 194.1982° E (planetocentric)
Assessment	weak candidate; probability of a new impact about 40%
Formed between	6 April 2014 and 10 May 2025
Images	1 without the spot (absent: 6 April 2014); 1 with it (present: 10 May 2025)
Size	Irregular dark patch about 190 m across (196 × 270 m), darkest in its north-western part, with no compact core
HiRISE	none covers the spot (nearest footprint edge 20.5 km)
Nearest catalogued new impact	87 km (Daubar et al., 2022)

What changed. An irregular dark patch about 190 m across, darkest in its north-western part, appeared on rolling plains 272 km south of Medusae Fossae between April 2014 and May 2025. So far only one CTX image shows it.

Why it may be a new impact. The only other CTX image of the site, taken in April 2014 under the same sun as the May 2025 image, shows no patch there. Subtracting it from the later image shows the strongest change at the patch and no other new dark mark in the surrounding 5 × 5 km. A new dark mark with no tracks, far too large to be an image defect, could be the blast zone of a small new impact.

Caveats. The patch is irregular, has no compact core and lies on dusty ground, so it may be where the wind stripped away a thin layer of bright dust; this is the main reason it is rated weak. Only one image shows it, only two CTX images cover the site, and a dark mark nearby is only faintly dark in the April 2014 image, a weak sign that this image would have shown the patch. No image from the high-resolution HiRISE camera covers it; the nearest ends 20.5 km away.

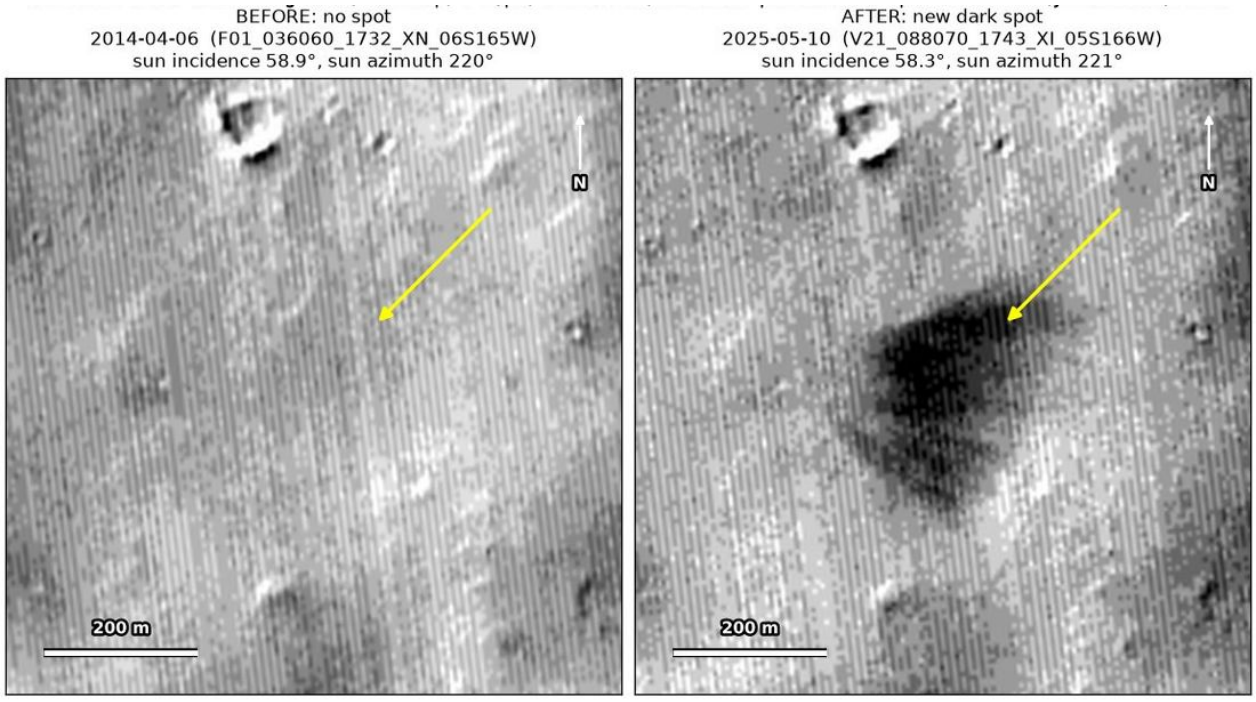

Figure 272: Candidate 109, before and after: the same 800 × 800 m of ground in 6 April 2014 (`F01_036060_1732_XN_06S165W`) and 10 May 2025 (`V21_088070_1743_XI_05S166W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

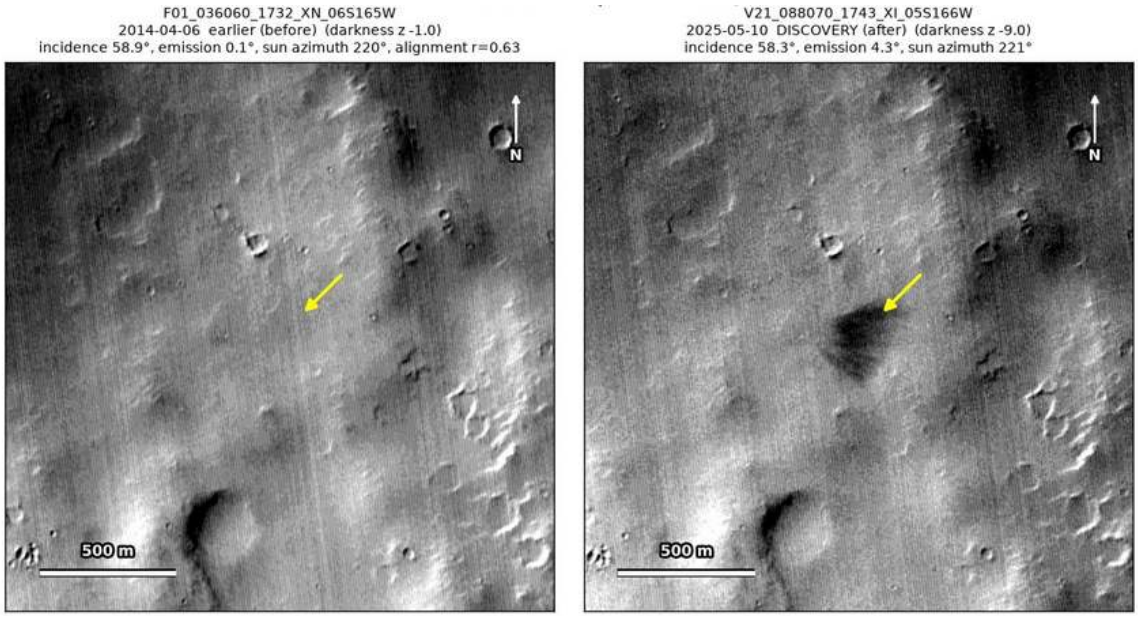

Figure 273: Candidate 109: both CTX images of the same 2×2 km patch, April 2014 and May 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 110: Plains of Arabia Terra near Scamander Vallis

Location	16.1878° N, 29.7458° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	28 February 2016 and 26 November 2024
Images	3 without the spot (absent: 21 February 2008, 10 January 2016 and 28 February 2016); 1 with it (present: 26 November 2024)
Size	Compact dark spot about 27 m across (24 × 30 m), with a darker centre and a soft edge
HiRISE	none covers the spot (nearest footprint edge 1.85 km)
Nearest catalogued new impact	58 km (Daubar et al., 2022)

What changed. A compact dark spot about 27 m across, with a darker centre and a soft edge, appeared on smooth, light-toned plains of Arabia Terra near Scamander Vallis between February 2016 and November 2024. So far only one CTX image shows it.

Why it may be a new impact. Three earlier images, from 2008 and 2016, show no spot there, including one of February 2008 taken under almost the same sun as the November 2024 image. A small pit 530 m away shows in all three, so they would have shown the spot. It is real image data that fades smoothly from its centre, and a compact new dark spot with no tracks around it is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, so an image defect cannot be fully ruled out, although the spot is a smooth dip in brightness over about 5 × 5 image pixels and a straight dark defect line 80 m to the east is separate from it. No image from the high-resolution HiRISE camera covers it; the nearest ends 1.85 km away.

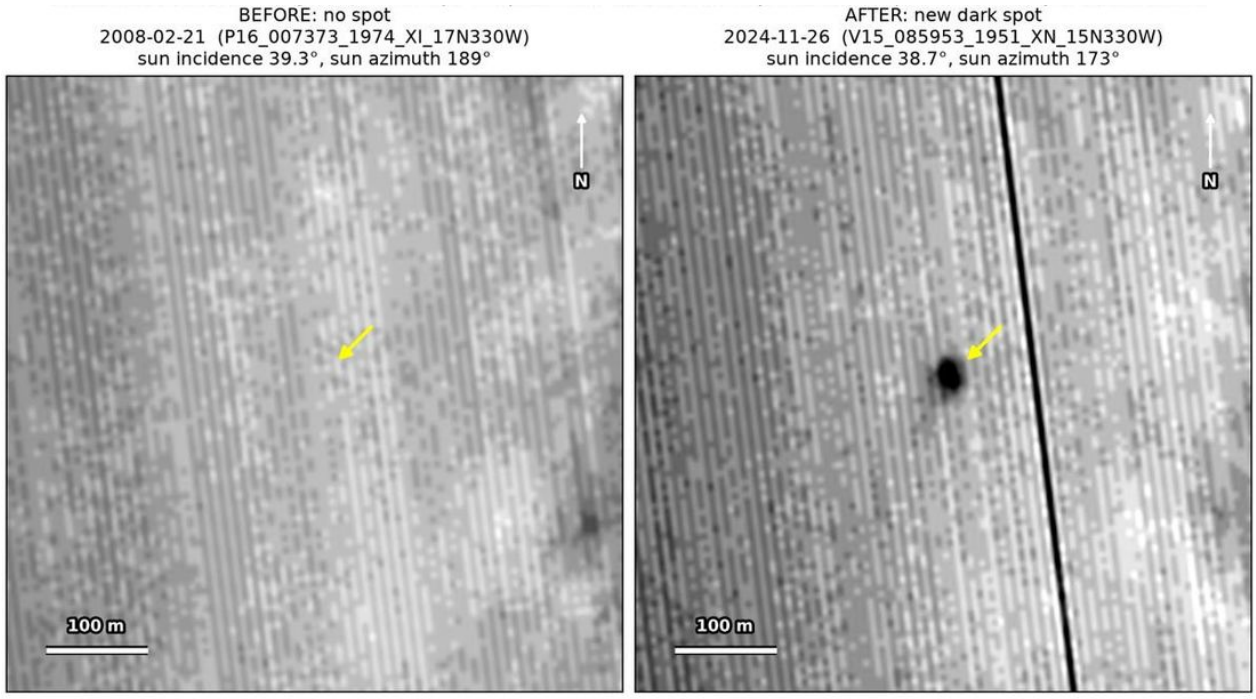

Figure 274: Candidate 110, before and after: the same 600 × 600 m of ground in 21 February 2008 (`P16_007373_1974_XI_17N330W`) and 26 November 2024 (`V15_085953_1951_XN_15N330W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

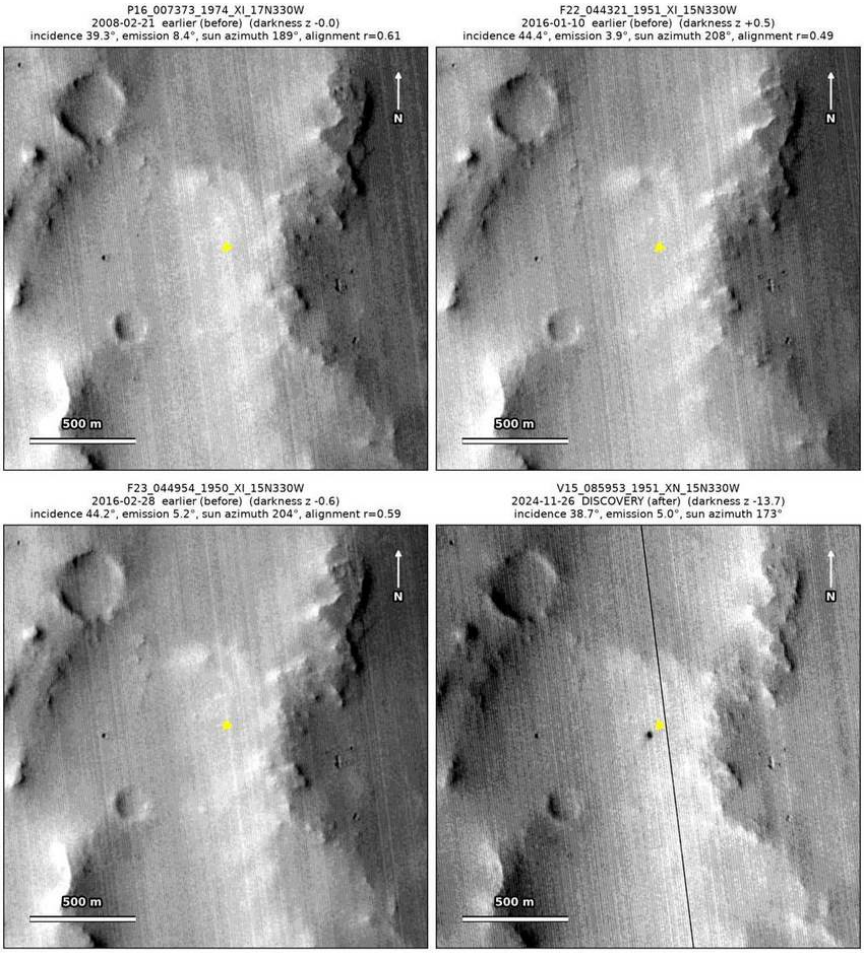

Figure 275: Candidate 110: four CTX images of the same 2×2 km patch, 2008 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 111: Knobby plains near Mangala Fossa

Location	17.4903° S, 212.9255° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	3 August 2014 and 24 October 2019
Images	2 without the spot (absent: 30 April 2010 and 3 August 2014); 2 with it (present: 24 October 2019 and 7 January 2025)
Size	Dark spot about 40 m across (44 × 42 m in the October 2019 image), a darker core in a soft halo
HiRISE	none covers the spot (nearest footprint edge 28.8 km)
Nearest catalogued new impact	57 km (Daubar et al., 2022)

What changed. A dark spot about 40 m across, a darker core in a soft halo, appeared on a smooth patch of ground among low knobs near Mangala Fossa between August 2014 and October 2019. Two CTX images from two orbits, more than five years apart, show it.

Why it may be a new impact. Two earlier images, from 2010 and 2014, show no spot there. The April 2010 one, taken under the same sun as the October 2019 image, shows the same knob shadows around the spot, and a dark mark 280 m away shows in both, so they would have shown the spot. A dark spot that has lasted more than five years, with no tracks around it, is how the blast zone of a small new crater looks.

Caveats. Only two images show it, and a straight defect line in the April 2010 image passes 70 m west of the spot, although the August 2014 image shows nothing there either. No image from the high-resolution HiRISE camera covers it; the nearest ends 28.8 km away.

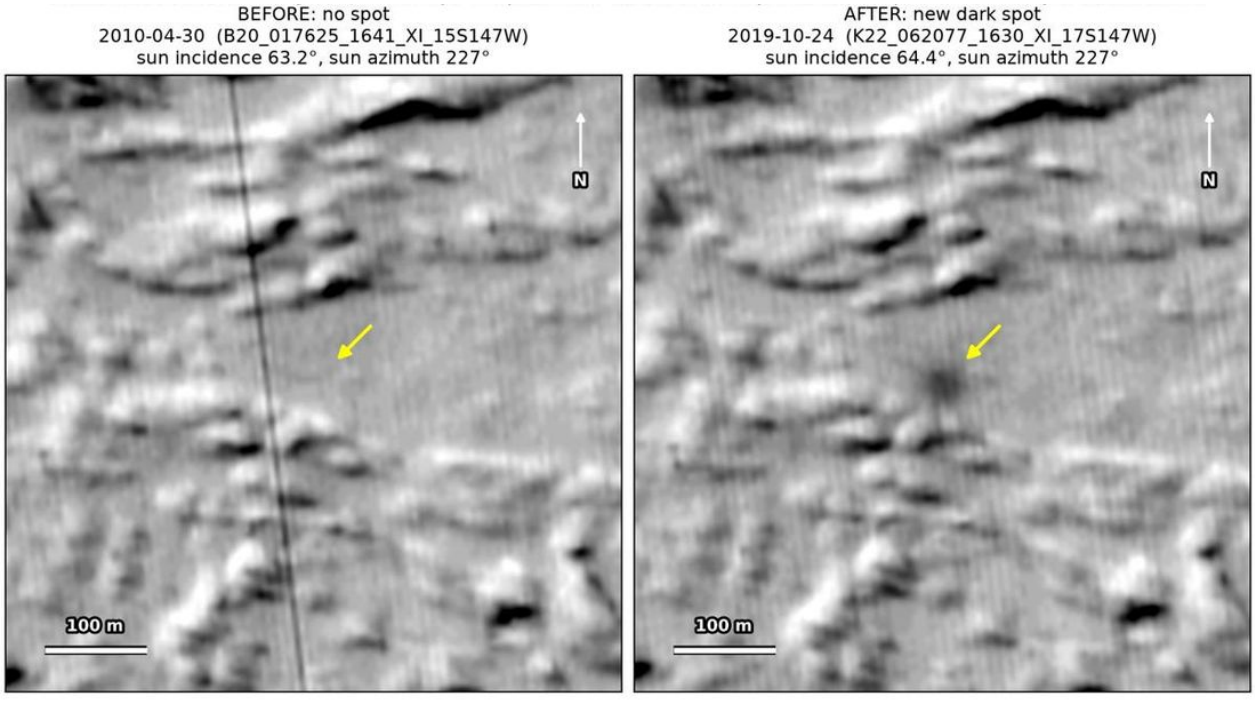

Figure 276: Candidate 111, before and after: the same 600 × 600 m of ground in 30 April 2010 (`B20_017625_1641_XI_15S147W`) and 24 October 2019 (`K22_062077_1630_XI_17S147W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

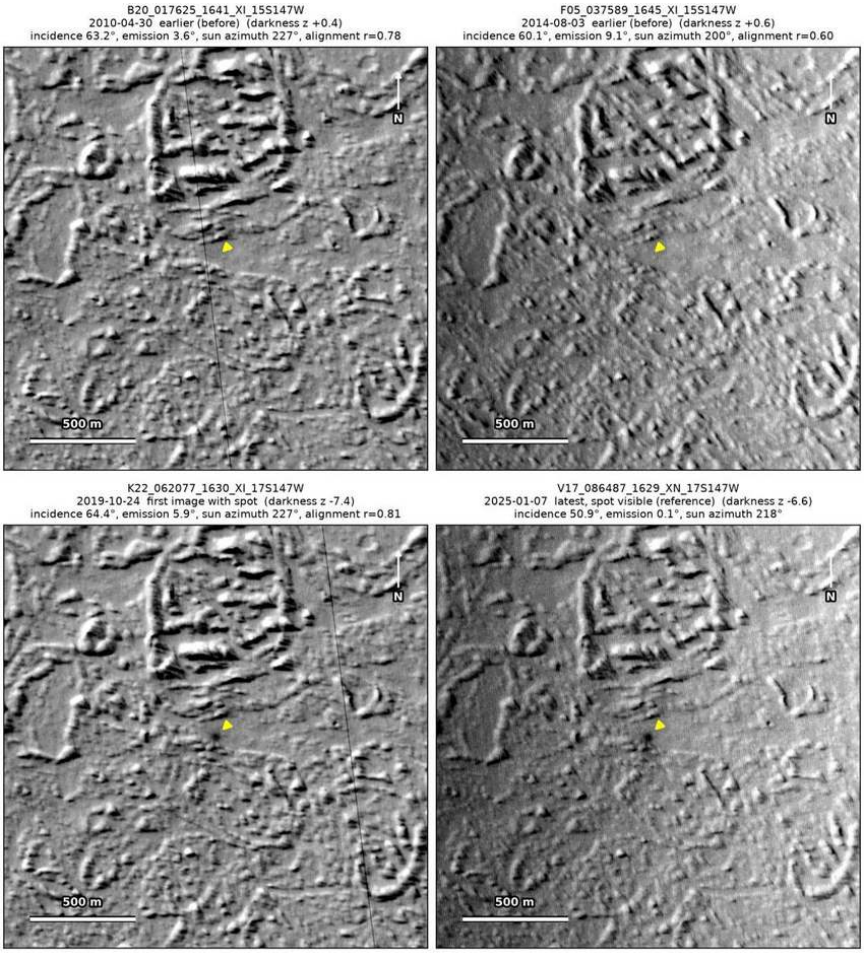

Figure 277: Candidate 111: four CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 112: Ridged plateau in eastern Noctis Labyrinthus

Location	8.8793° S, 265.2273° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	16 September 2016 and 24 September 2024
Images	5 without the spot (absent: 1 May 2007, 30 August 2007, 22 May 2016, 6 July 2016 and 16 September 2016); 1 with it (present: 24 September 2024)
Size	Diffuse dark spot about 160 m across (212 × 158 m), with a darker core
HiRISE	none covers the spot (nearest footprint edge 5.2 km)
Nearest catalogued new impact	48 km (Daubar et al., 2022)

What changed. A diffuse dark spot about 160 m across, with a darker core, appeared on a ridged plateau in eastern Noctis Labyrinthus between September 2016 and September 2024. So far only one CTX image shows it.

Why it may be a new impact. Five earlier images, from 2007 and 2016, show plain ground there, including one of August 2007 taken under almost the same sun as the September 2024 image. Subtracting an earlier image from the later one shows the strongest change at the spot, with no other new dark mark and no tracks in the surrounding 5 × 5 km, and a roundish new dark spot with a darker core, not drawn out downslope, fits the blast zone of a small new crater.

Caveats. Only one image shows it, and a short one, so an image defect cannot be fully ruled out, although the spot is a smooth dip in brightness spread over about 40 × 30 image pixels. A dark mark nearby is faint in the earlier images, so it does not show that they would have caught a spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 5.2 km away.

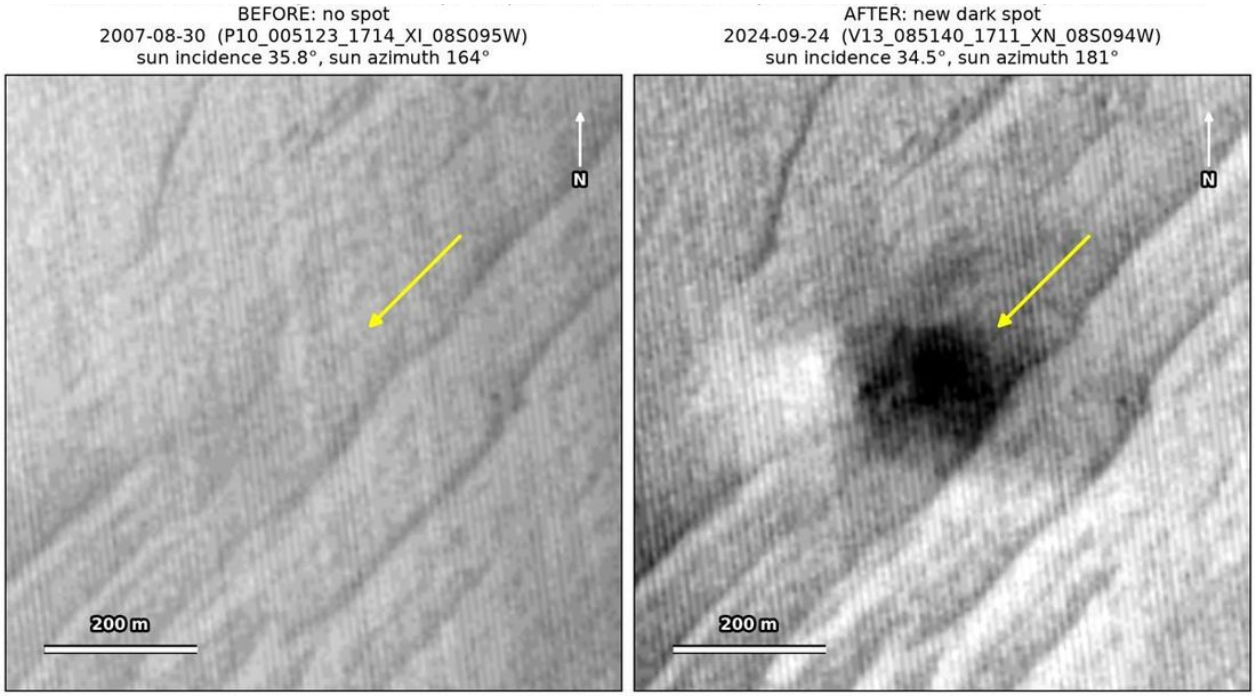

Figure 278: Candidate 112, before and after: the same 800 × 800 m of ground in 30 August 2007 (`P10_005123_1714_XI_08S095W`) and 24 September 2024 (`V13_085140_1711_XN_08S094W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

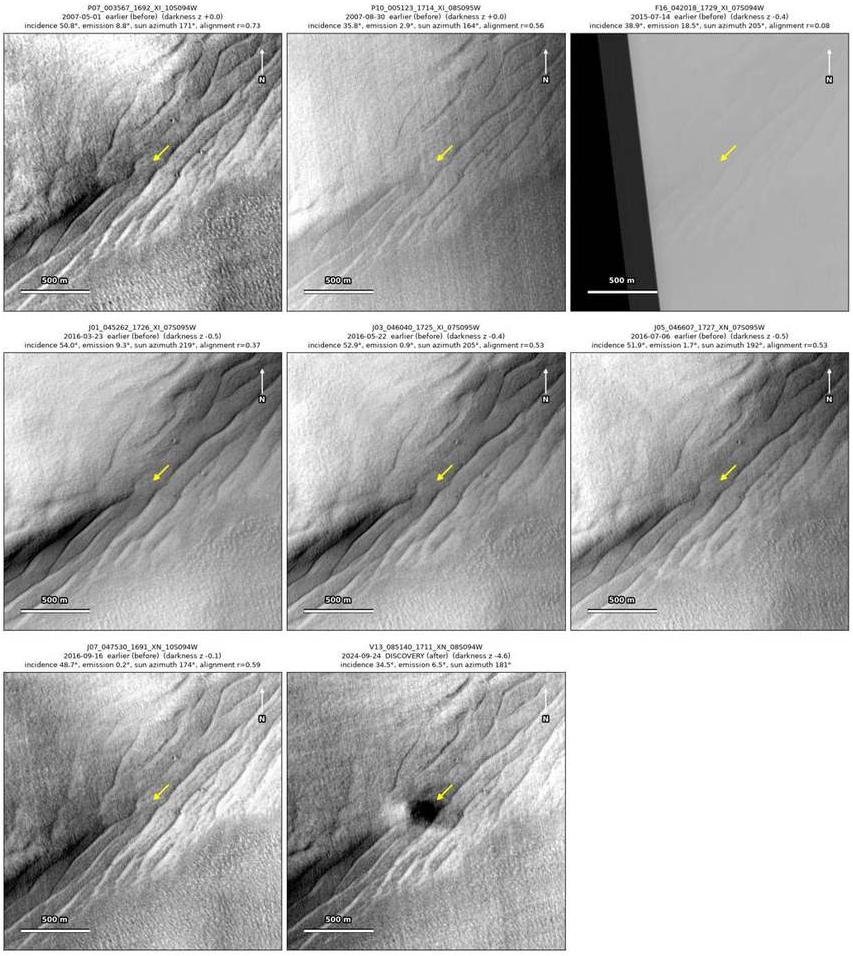

Figure 279: Candidate 112: eight CTX images of the same 2×2 km patch, 2007 to 2024. The July 2015 image, washed out at the site, and the March 2016 image, which shows plain ground but lines up only roughly, are shown but not used to date the spot. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 113: Plains of Arabia Terra near Indus Vallis

Location	19.9814° N, 39.4433° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 50%
Formed between	24 March 2022 and 6 January 2025
Images	4 without the spot (absent: 24 February 2008, 15 September 2019, 25 October 2021 and 24 March 2022); 1 with it (present: 6 January 2025)
Size	Round dark spot about 230 m across (234 × 236 m), with a mottled interior, a soft edge and no compact core
HiRISE	none covers the spot (nearest footprint edge 26 km)
Nearest catalogued new impact	128 km (Daubar et al., 2022)

What changed. A round dark spot about 230 m across, with a mottled interior and a soft edge, appeared at the foot of a shaded scarp on the plains of Arabia Terra near Indus Vallis between March 2022 and January 2025. So far only one CTX image shows it.

Why it may be a new impact. Four earlier images, from 2008 to 2022, show plain ground there and the same scarp, including one of February 2008 taken under almost the same sun as the January 2025 image. Subtracting an earlier image from the later one shows the strongest change at the spot, with no other new dark mark and no tracks in the surrounding 5 × 5 km, and a round dark spot that appeared within three years could be the blast zone of a small new crater.

Caveats. It has no compact core, so it could be a patch where the wind stripped away bright dust, and only one image shows it, although at about 42 × 42 image pixels it is far larger than an image defect. A dark mark nearby is only just dark enough in the February 2008 and October 2021 images to show that they would have caught a spot, and fainter in the other two. No image from the high-resolution HiRISE camera covers it; the nearest ends 26 km away.

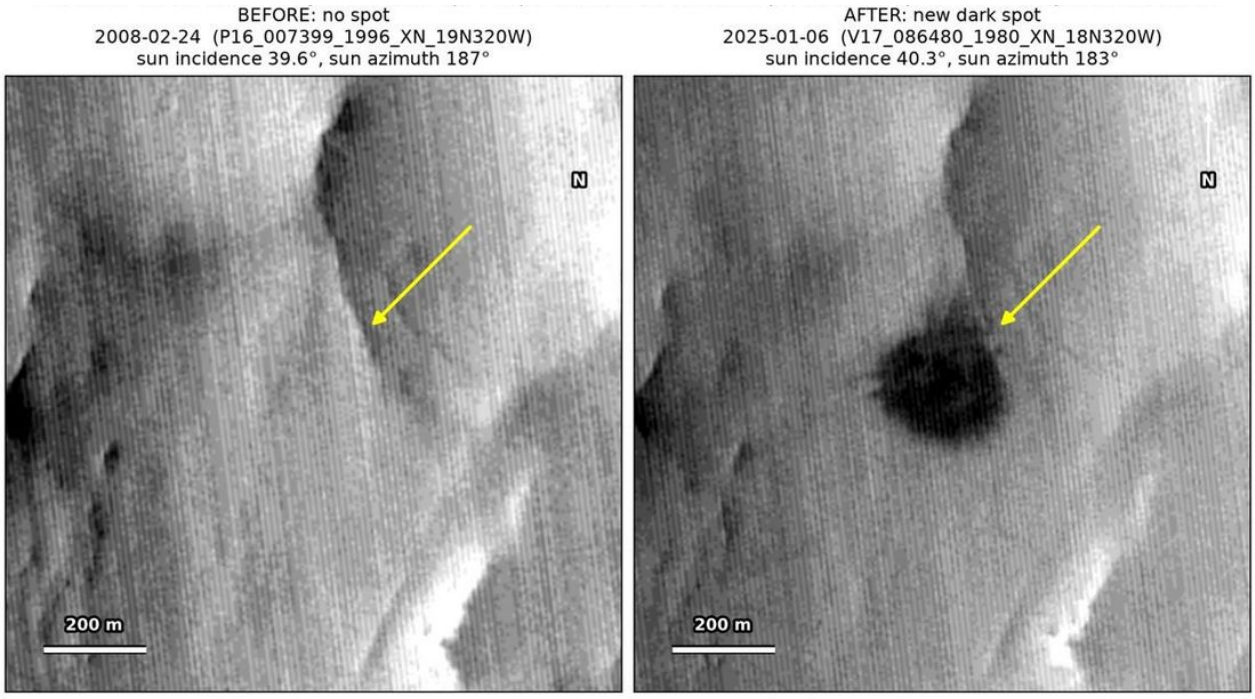

Figure 280: Candidate 113, before and after: the same 1200 × 1200 m of ground in 24 February 2008 (`P16_007399_1996_XN_19N320W`) and 6 January 2025 (`V17_086480_1980_XN_18N320W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

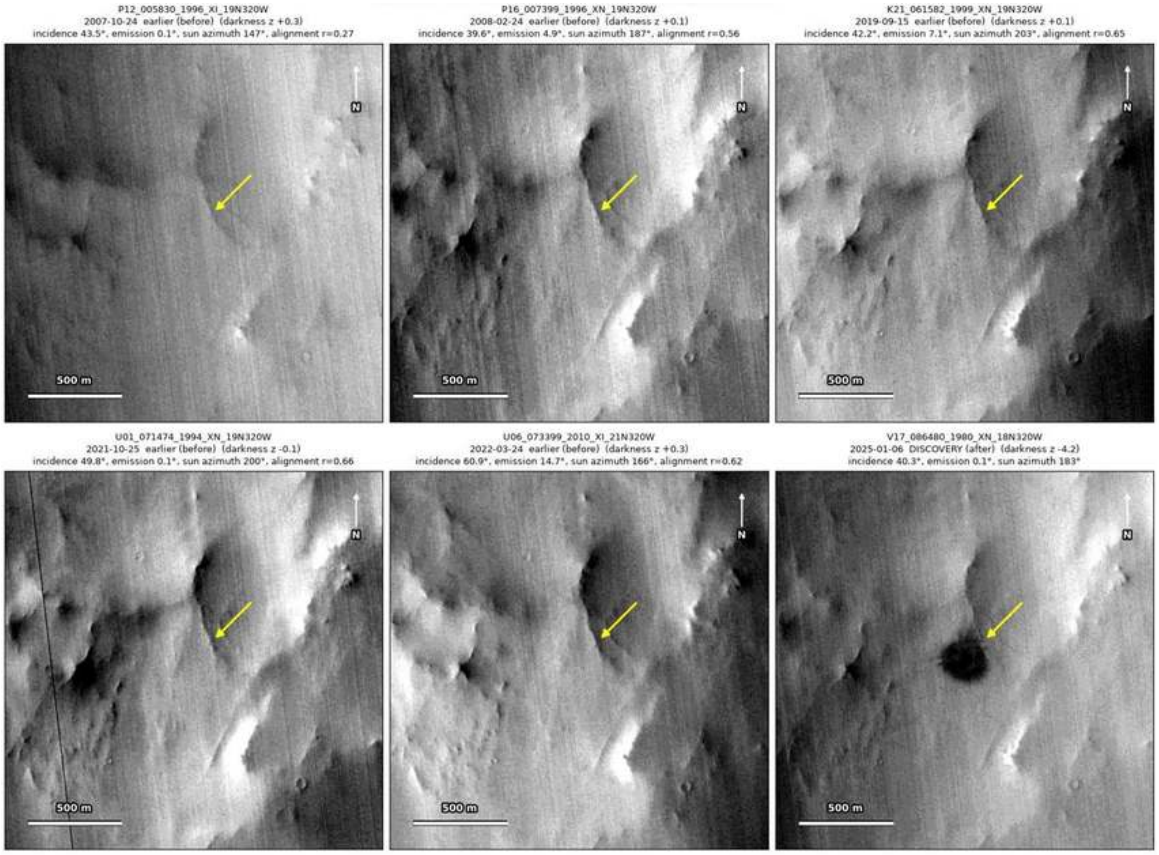

Figure 281: Candidate 113: six CTX images of the same 2×2 km patch, 2007 to 2025. The October 2007 image, which shows plain ground but lines up poorly with the others, is shown but not used to date the spot. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 114: Knobby plains 202 km west of Rudaux crater

Location	37.8280° N, 46.6507° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	2 February 2010 and 12 October 2023
Images	1 without the spot (absent: 2 February 2010); 3 with it (present: 12 October 2023, 10 February 2025 and 10 May 2025)
Size	Compact dark spot about 35 m across (36 × 46 m in the February 2025 image), on the side of a bright knob
HiRISE	none covers the spot (nearest footprint edge 7.9 km)
Nearest catalogued new impact	77 km (Daubar et al., 2022)

What changed. A compact dark spot about 35 m across appeared on the side of a bright knob, on knobby plains in northern Arabia Terra, between February 2010 and October 2023. It is still there in May 2025, seen in three CTX images from three orbits.

Why it may be a new impact. The only earlier image, of February 2010, was taken under almost the same sun and in the same season as the February 2025 image and shows the side of the knob bright there, while a dark mark 110 m away shows in it, so a spot this dark would have shown. A compact dark spot that stays in the same place in three images from three orbits fits the blast zone of a small new crater.

Caveats. The spot is faint, no darker than the shadows of the knobs nearby, and only one earlier image covers the site. The difference between the images singles the spot out only in fine detail, as other places change as much at a coarser scale, and the February 2025 image is heavily compressed, although the October 2023 and May 2025 images show the spot in the same place. No image from the high-resolution HiRISE camera covers it; the nearest ends 7.9 km away.

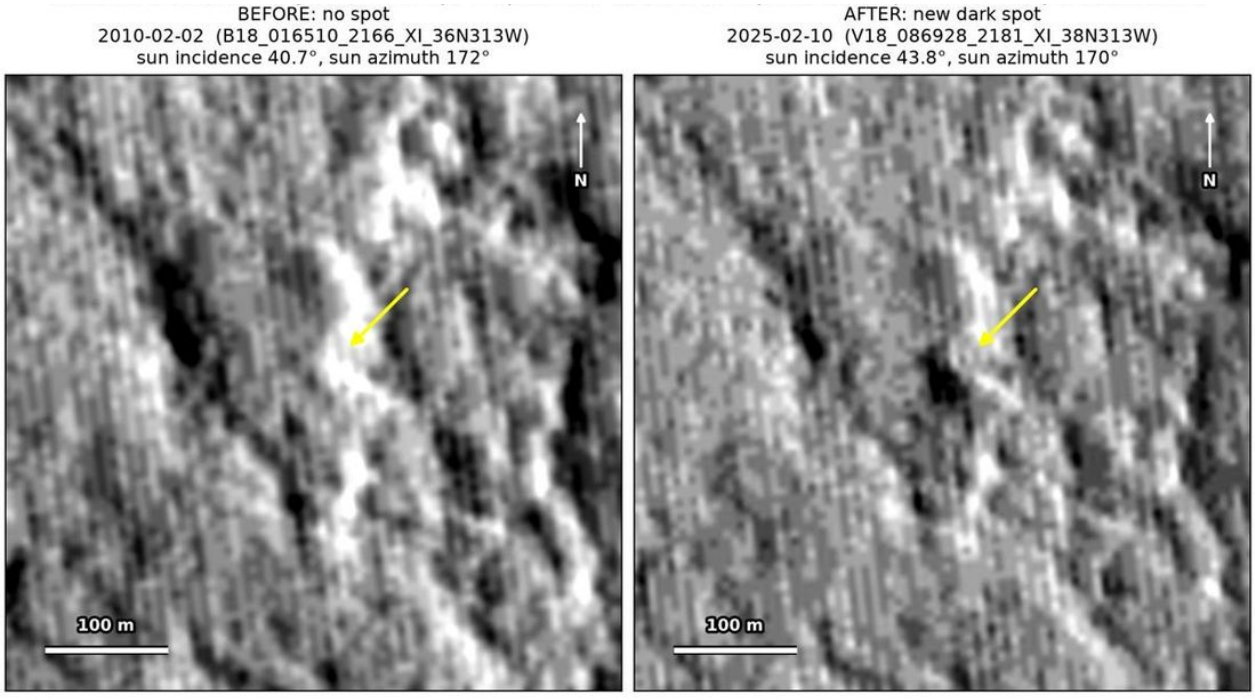

Figure 282: Candidate 114, before and after: the same 500 × 500 m of ground in 2 February 2010 (`B18_016510_2166_XI_36N313W`) and 10 February 2025 (`V18_086928_2181_XI_38N313W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

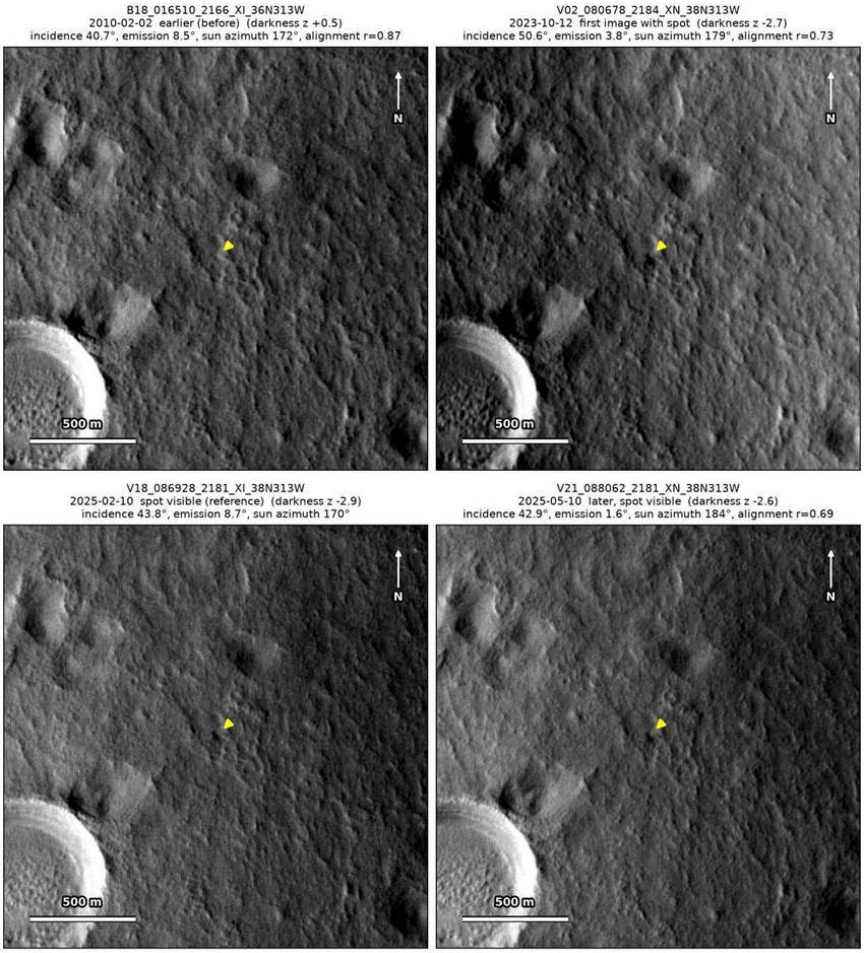

Figure 283: Candidate 114: four CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 115: Plains of the Nepenthes Mensae region, 408 km west of the InSight lander

Location	5.6256° N, 128.7919° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	16 September 2019 and 18 April 2023
Images	2 without the spot (absent: 14 August 2016 and 16 September 2019); 2 with it (present: 18 April 2023 and 6 October 2024)
Size	Dark core about 40 m across in a halo about 120 m wide (112 × 128 m in the October 2024 image)
HiRISE	none covers the spot (nearest footprint edge 30 km)
Nearest catalogued new impact	110 km (Daubar et al., 2022)

What changed. A dark spot, a dark core about 40 m across in a halo about 120 m wide, appeared on plains of the Nepenthes Mensae region, 408 km west of the InSight lander, between September 2019 and April 2023. Two CTX images from two orbits show it.

Why it may be a new impact. Two earlier images, from 2016 and 2019, show no spot there; the September 2019 one was taken under almost the same sun as the April 2023 image. A dark mark 190 m away shows in both, so they would have shown a spot this dark. A dark core in a halo, with no tracks and no other new dark mark in the surrounding 5 × 5 km, is how the blast zone of a small new crater looks.

Caveats. Only two images show it, and the two earlier images line up only loosely with the striped October 2024 image, although the dark mark 190 m away and the craters around the spot line up in both. No image from the high-resolution HiRISE camera covers it; the nearest ends 30 km away.

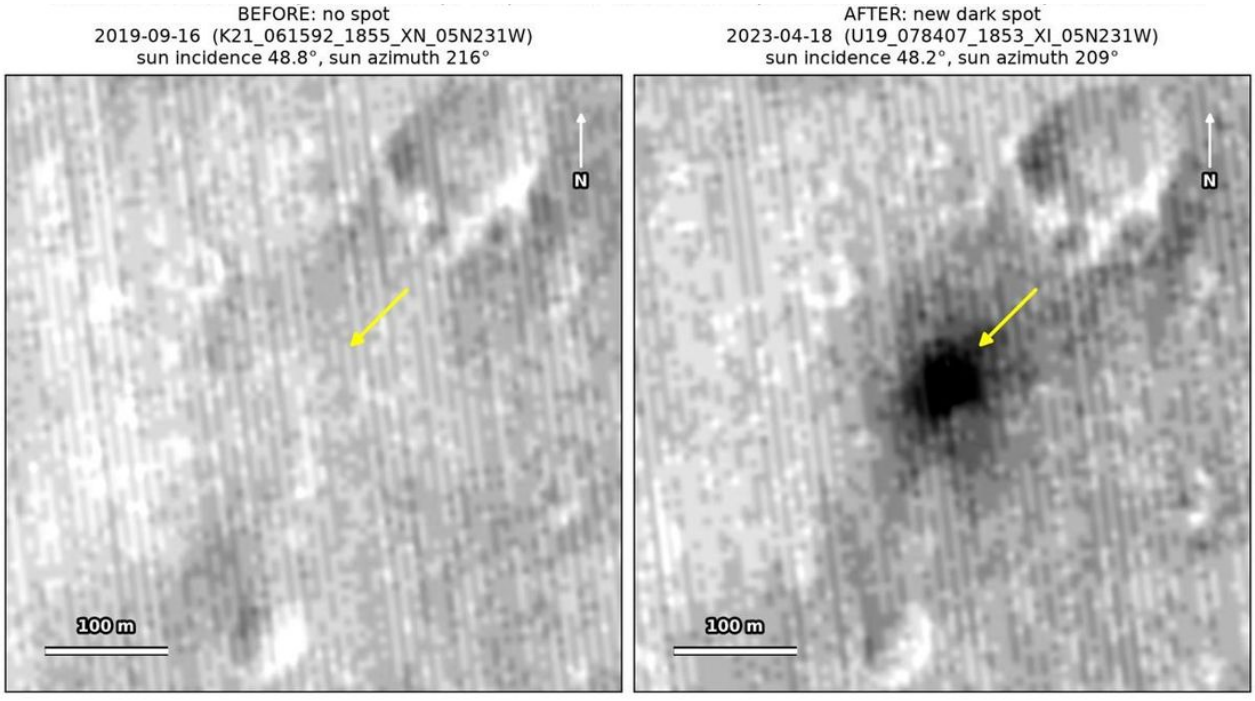

Figure 284: Candidate 115, before and after: the same 500 × 500 m of ground in 16 September 2019 (`K21_061592_1855_XN_05N231W`) and 18 April 2023 (`U19_078407_1853_XI_05N231W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

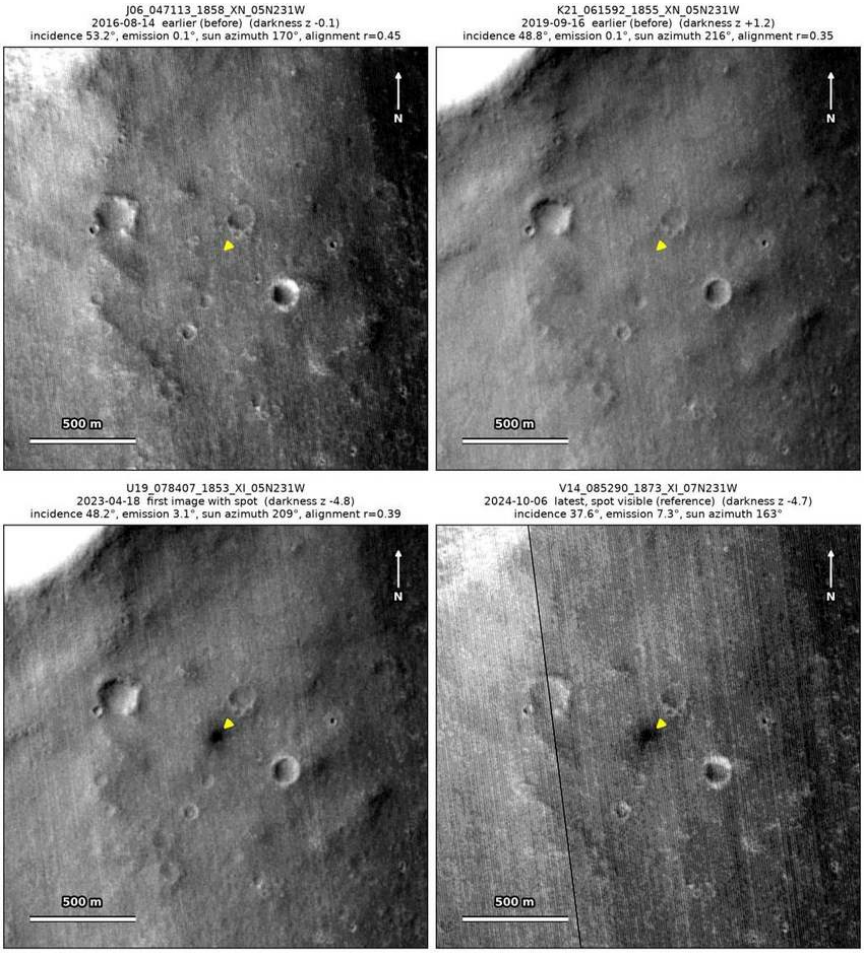

Figure 285: Candidate 115: four CTX images of the same 2×2 km patch, 2016 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 116: Etched plains of Arabia Terra near Indus Vallis

Location	17.9591° N, 39.0062° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	26 February 2023 and 19 July 2024
Images	5 without the spot (absent: 4 January 2008, 3 April 2008, 23 July 2022, 16 September 2022 and 26 February 2023); 2 with it (present: 19 July 2024 and 26 December 2024)
Size	Dark spot about 75 m across (110 × 86 m in the December 2024 image), a dark core with ragged edges
HiRISE	none covers the spot (nearest footprint edge 30 km)
Nearest catalogued new impact	119 km (Daubar et al., 2022)

What changed. A dark spot about 75 m across, with a darker core and ragged edges, appeared on etched plains of Arabia Terra near Indus Vallis between February 2023 and July 2024. Two CTX images from two orbits show it.

Why it may be a new impact. Five earlier images, from 2008 to 2023, show no spot there; the last, of February 2023, was taken under almost the same sun as the December 2024 image. A dark mark 300 m away shows in all five, so each would have shown the spot. A dark spot with a darker core that appeared within 17 months, with nothing else changing in the surrounding 5 × 5 km and no tracks, is how the blast zone of a small new crater looks.

Caveats. Only two images show it, and a thin defect line in the July 2024 image crosses its western edge, although the December 2024 image has none and shows the spot as dark. No image from the high-resolution HiRISE camera covers it; the nearest ends 30 km away.

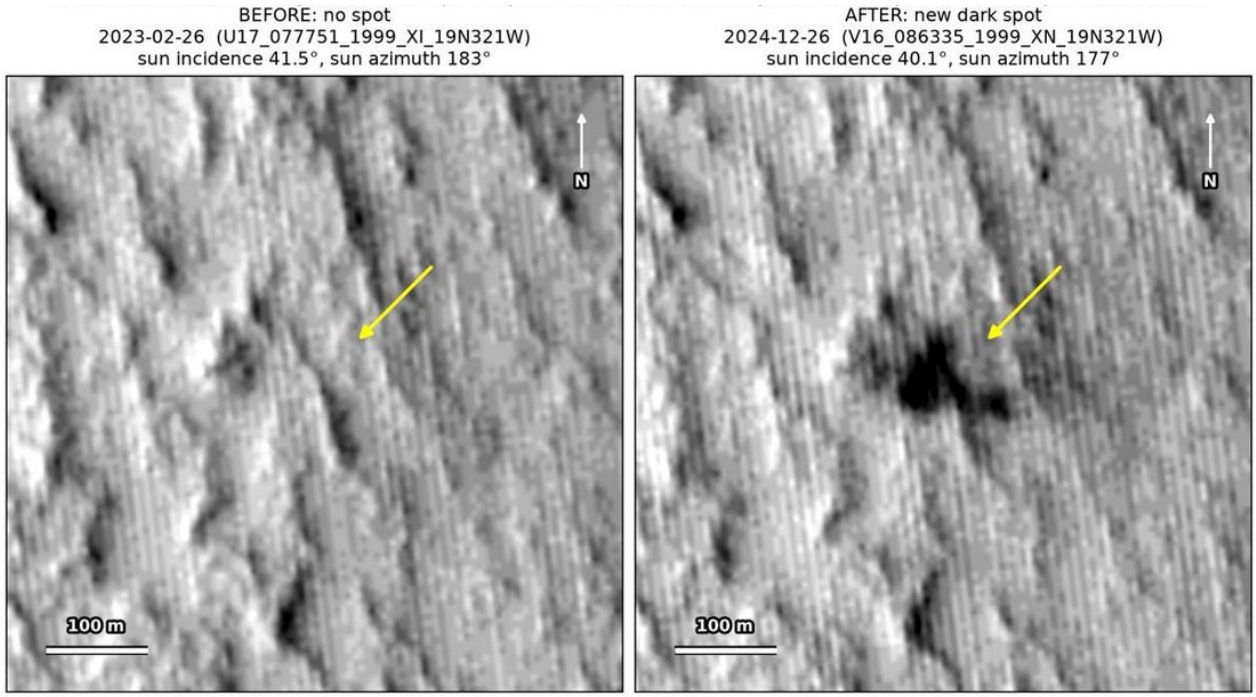

Figure 286: Candidate 116, before and after: the same 600 × 600 m of ground in 26 February 2023 (`U17_077751_1999_XI_19N321W`) and 26 December 2024 (`V16_086335_1999_XN_19N321W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

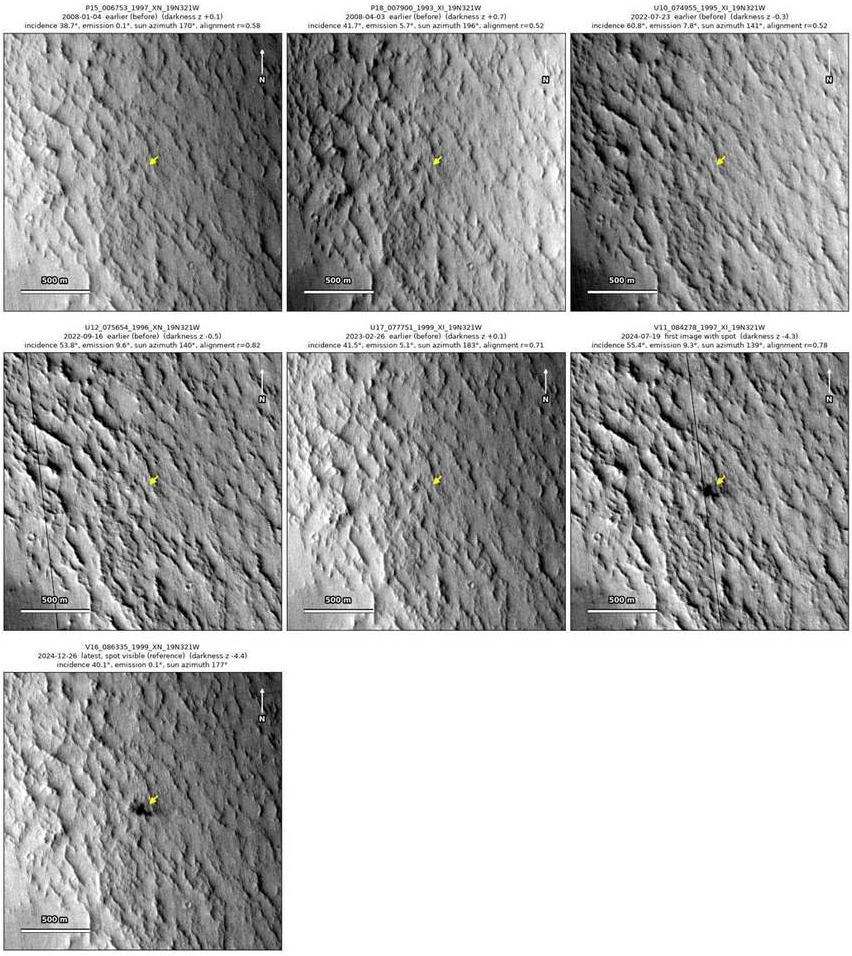

Figure 287: Candidate 116: seven CTX images of the same 2×2 km patch, 2008 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 117: Crater ejecta in western Phlegra Montes

Location	40.4237° N, 160.1682° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	23 January 2010 and 20 February 2021
Images	1 without the spot (absent: 23 January 2010); 3 with it (present: 20 February 2021, 7 May 2023 and 18 December 2024)
Size	Compact dark spot about 50 m across (58 × 78 m in the December 2024 image), a cluster of dark blotches with a faint halo
HiRISE	none covers the spot (nearest footprint edge 27 km)
Nearest catalogued new impact	103 km (Daubar et al., 2022)

What changed. A compact dark spot about 50 m across, a cluster of dark blotches with a faint halo, appeared on pitted ejecta, debris thrown out of a crater, in western Phlegra Montes between January 2010 and February 2021. It is still there in December 2024, seen in three CTX images from three orbits.

Why it may be a new impact. The only usable earlier image, of January 2010, taken under almost the same sun as the May 2023 image, shows no spot there. The spot is darker than any pit shadow around it, so it is unlikely to be pit shadows merging under a change of light. It stays in the same place in three images from three orbits, with no tracks and no other new dark mark in the surrounding 5 × 5 km, as the blast zone of a small new crater would.

Caveats. Only one earlier image is usable, and in it the spot's place already comes out slightly dark because of the shadows of the pitted surface, while a dark mark nearby is too faint in that image to show that it would have caught a spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 27 km away.

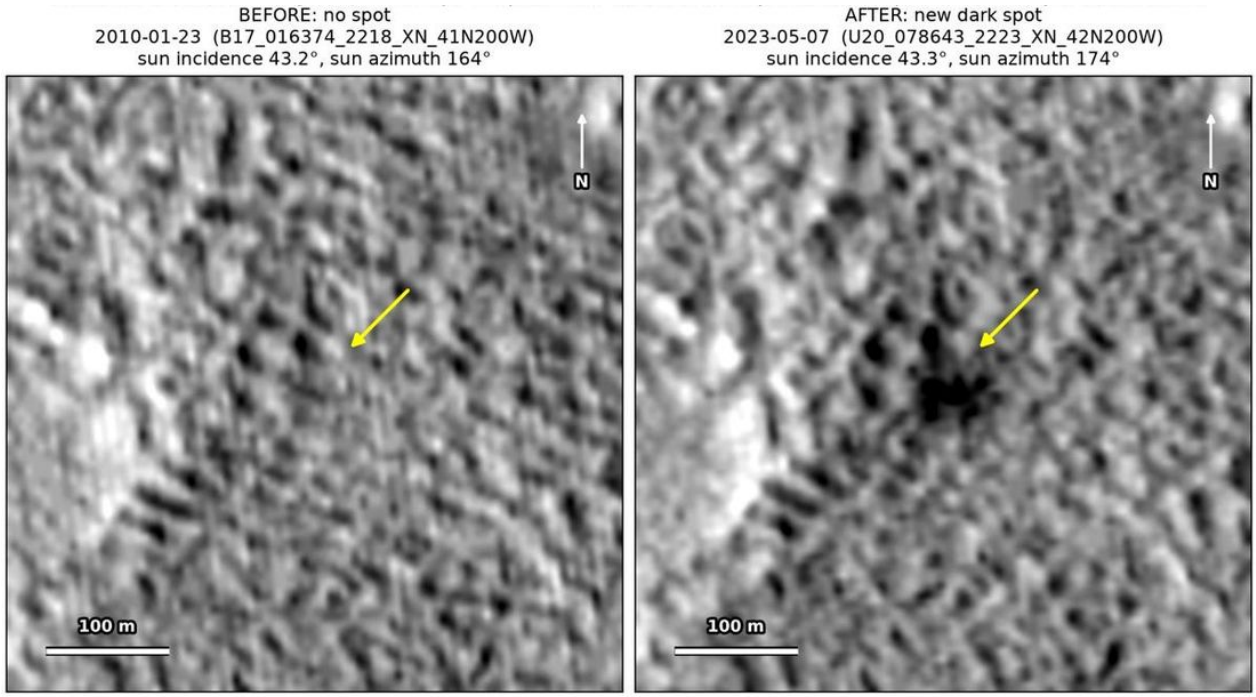

Figure 288: Candidate 117, before and after: the same 500 × 500 m of ground in 23 January 2010 (`B17_016374_2218_XN_41N200W`) and 7 May 2023 (`U20_078643_2223_XN_42N200W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

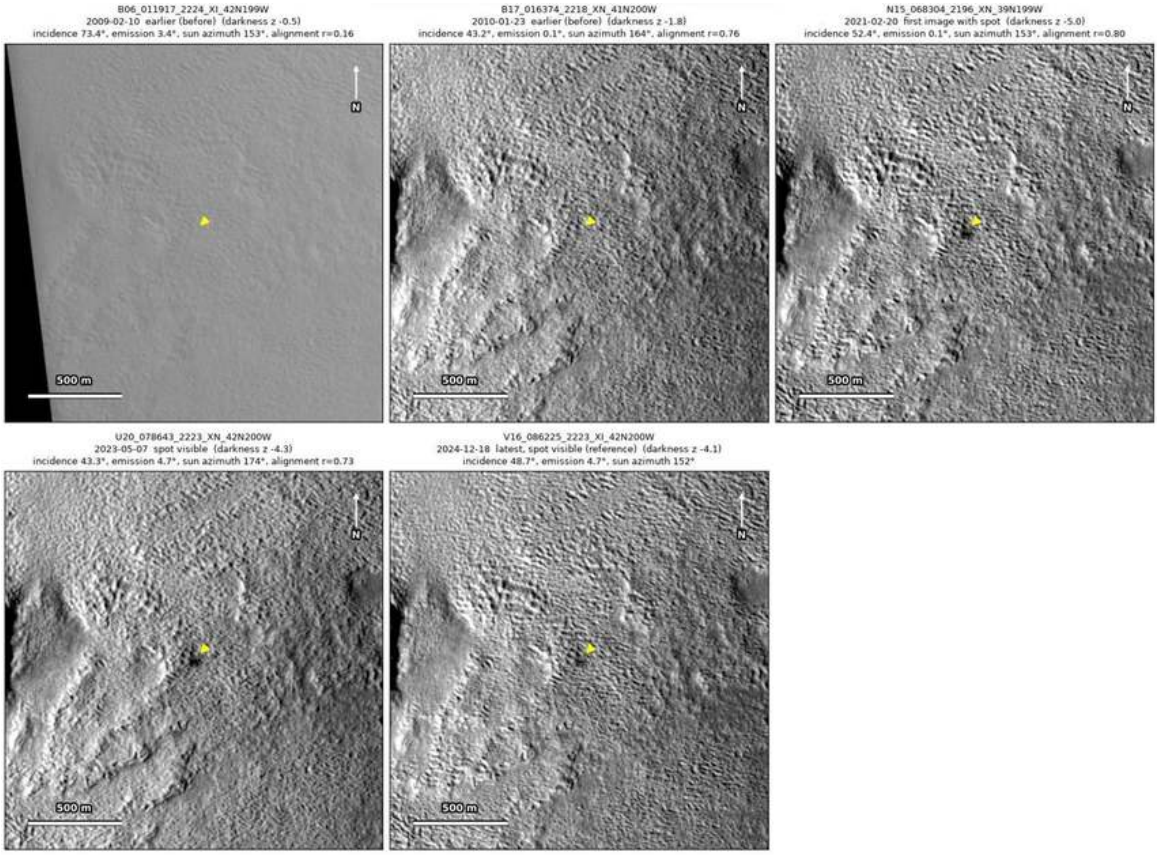

Figure 289: Candidate 117: five CTX images of the same 2×2 km patch, 2009 to 2024. The February 2009 image, taken under a low sun, is washed out and lines up poorly, so it is shown but not used to date the spot. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 118: Plains of west-central Arabia Terra

Location	21.2241° N, 14.9784° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	26 April 2014 and 27 March 2023
Images	1 without the spot (absent: 26 April 2014); 2 with it (present: 27 March 2023 and 10 July 2023)
Size	Cluster of at least six dark spots within about 350 m, the largest about 40 m across (40 × 44 m in the July 2023 image)
HiRISE	none covers the spot (nearest footprint edge 41 km)
Nearest catalogued new impact	48 km (Daubar et al., 2022)

What changed. A cluster of at least six dark spots within about 350 m, the largest about 40 m across, appeared on the plains of west-central Arabia Terra between April 2014 and March 2023. Two CTX images from two orbits show it.

Why it may be a new impact. The only earlier image, of April 2014, was taken under almost the same sun as the July 2023 image and shows the small ridges around the cluster sharply, but no spots. Subtracting it from the later image shows the strongest change at the largest spot, with no other new dark mark and no tracks in the surrounding 5 × 5 km, and a cluster of separate round dark spots is not a dust-devil pattern but fits a small new impact.

Caveats. Only one earlier image covers the site. A mark nearby is dark in that image but not in the 2023 images, so it may be a shading detail, which makes it a weaker sign than usual that the earlier image could have caught the spots. No image from the high-resolution HiRISE camera covers it; the nearest ends 41 km away.

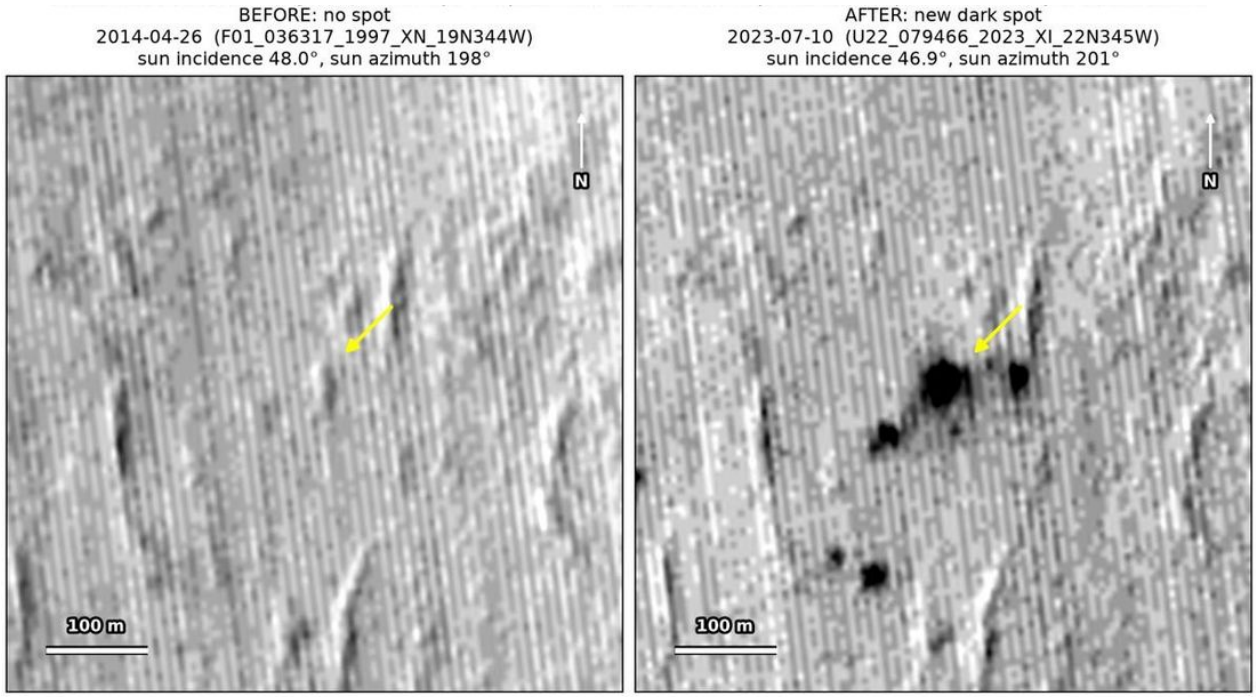

Figure 290: Candidate 118, before and after: the same 600 × 600 m of ground in 26 April 2014 (`F01_036317_1997_XN_19N344W`) and 10 July 2023 (`U22_079466_2023_XI_22N345W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

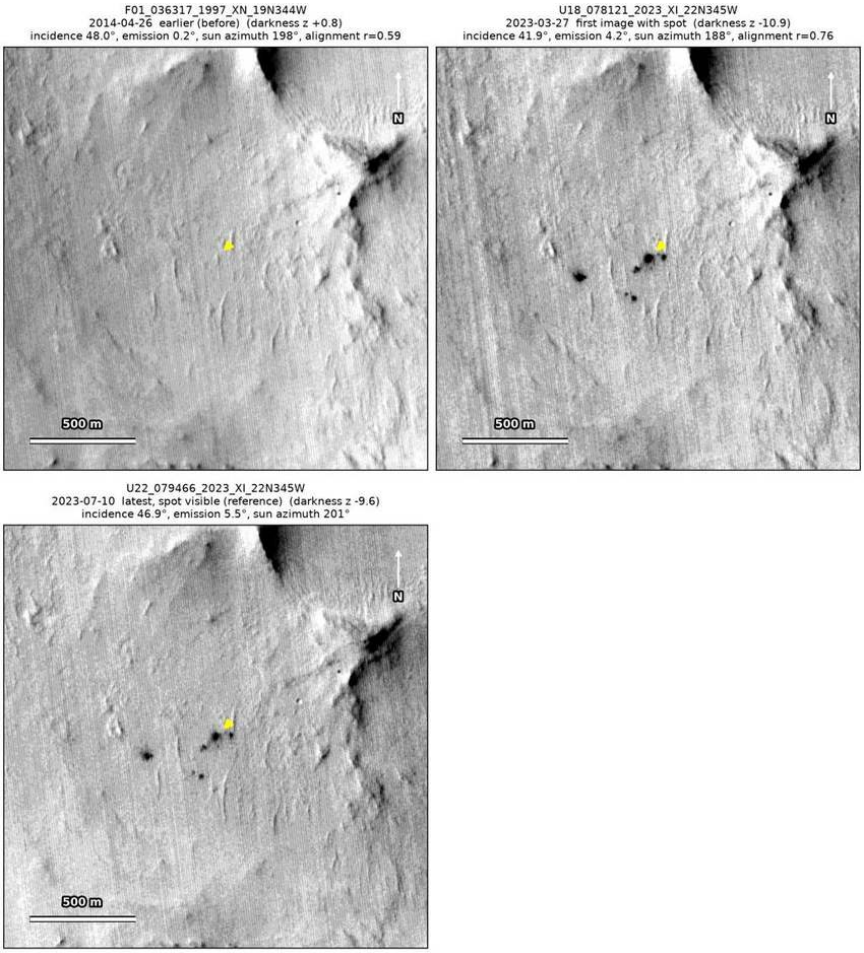

Figure 291: Candidate 118: three CTX images of the same 2×2 km patch, 2014 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 119: Plains 245 km south of Uranius Mons

Location	22.8028° N, 268.4438° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 55%
Formed between	29 May 2010 and 16 July 2023
Images	1 without the spot (absent: 29 May 2010); 1 with it (present: 16 July 2023)
Size	Compact dark spot about 35 m across (32 × 38 m)
HiRISE	none covers the spot (nearest footprint edge 38 km)
Nearest catalogued new impact	64 km (Daubar et al., 2022)

What changed. A compact dark spot about 35 m across appeared on plains 245 km south of Uranius Mons between May 2010 and July 2023. So far only one CTX image shows it.

Why it may be a new impact. The only earlier image, of May 2010, was taken under almost the same sun as the July 2023 image and shows no spot there. Subtracting it from the later image shows the strongest change at the spot, with no other new dark mark and no tracks in the surrounding 5 × 5 km, and a compact new dark spot like this is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, so an image defect cannot be fully ruled out, although the spot is a smooth dip in brightness about 6 × 7 image pixels across; a later image, of August 2023, ends a few hundred metres east of it and does not cover it. The only earlier image is softer, and a dark mark nearby is faint in it, so the mark does not show that this image would have caught a spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 38 km away.

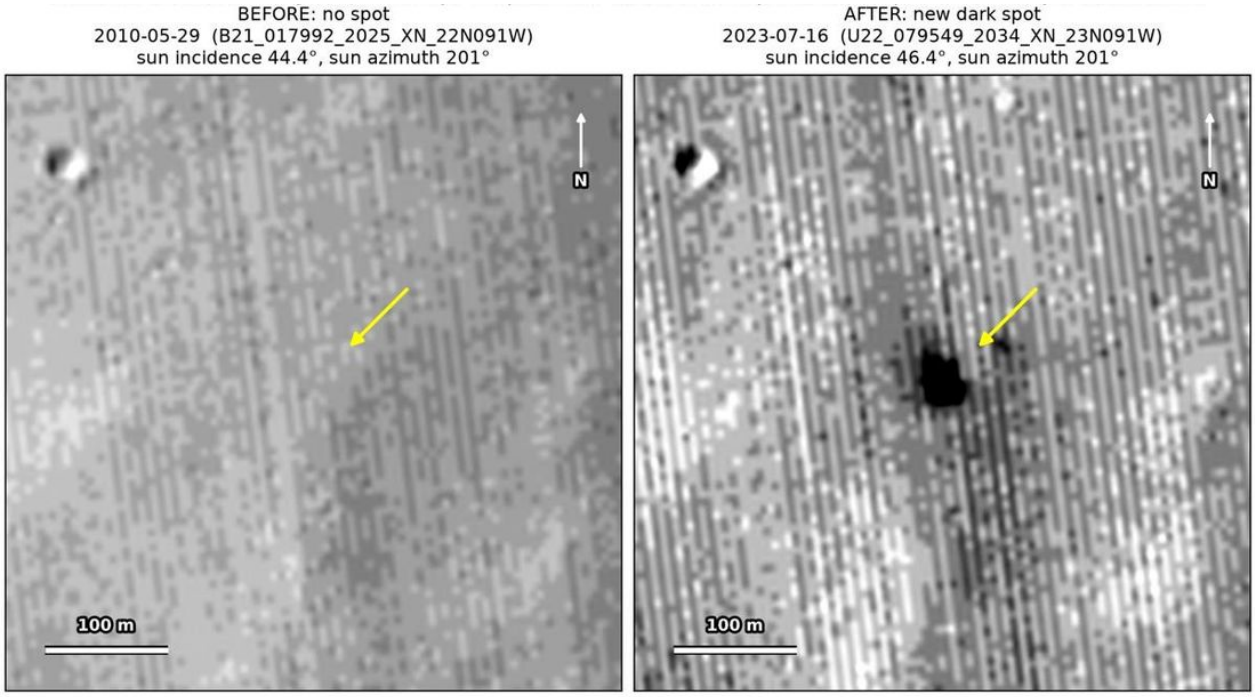

Figure 292: Candidate 119, before and after: the same 500 × 500 m of ground in 29 May 2010 (`B21_017992_2025_XN_22N091W`) and 16 July 2023 (`U22_079549_2034_XN_23N091W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

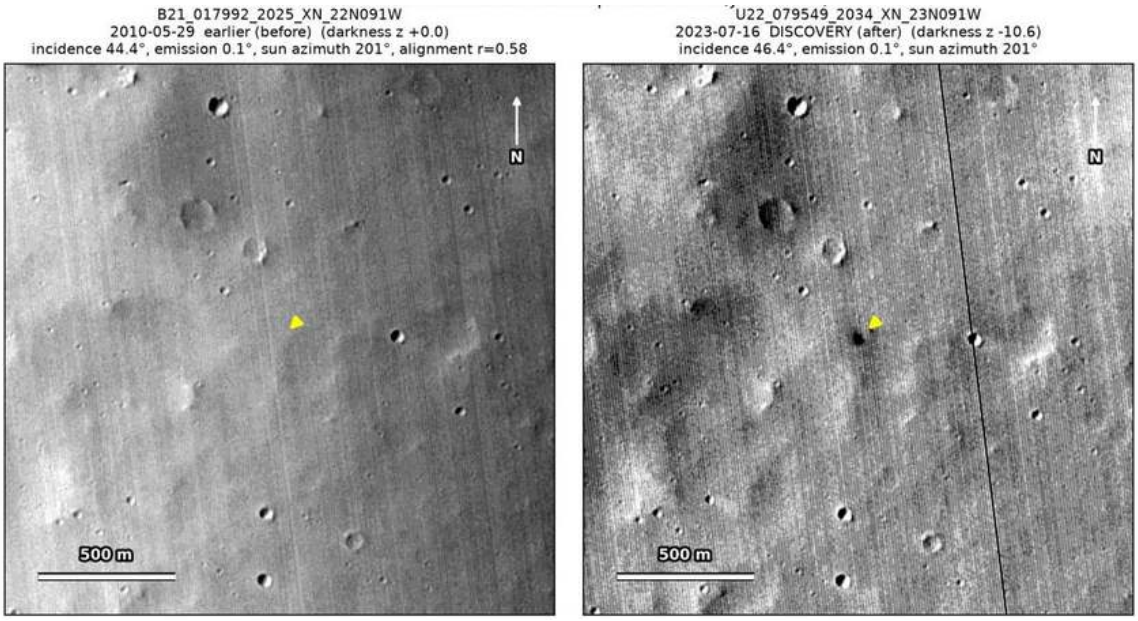

Figure 293: Candidate 119: both CTX images of the same 2×2 km patch, May 2010 and July 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 120: Lava plains of Elysium Planitia, north of Hibes Montes

Location	5.1641° N, 170.6673° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	10 May 2014 and 25 March 2023
Images	1 without the spot (absent: 10 May 2014); 2 with it (present: 25 March 2023 and 21 June 2023)
Size	Compact dark spot about 50 m across (44 × 60 m in the June 2023 image), with a faint halo
HiRISE	none covers the spot (nearest footprint edge 20 km)
Nearest catalogued new impact	150 km (Daubar et al., 2022)

What changed. A compact dark spot about 50 m across, with a faint halo, appeared on the lava plains of Elysium Planitia north of Hibes Montes between May 2014 and March 2023. Two CTX images from two orbits show it.

Why it may be a new impact. The only earlier image, of May 2014, taken under almost the same sun as the June 2023 image, shows no spot there, while a dark mark 135 m away shows clearly in it, so it would have shown a spot this dark. Two older dark-halo craters, 1.5 and 2.2 km away, look the same in all three images, and a compact new dark spot with a faint halo and no tracks is how the blast zone of a small new crater looks.

Caveats. Only one earlier image covers the site, and only two images, from March and June 2023, show the spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 20 km away.

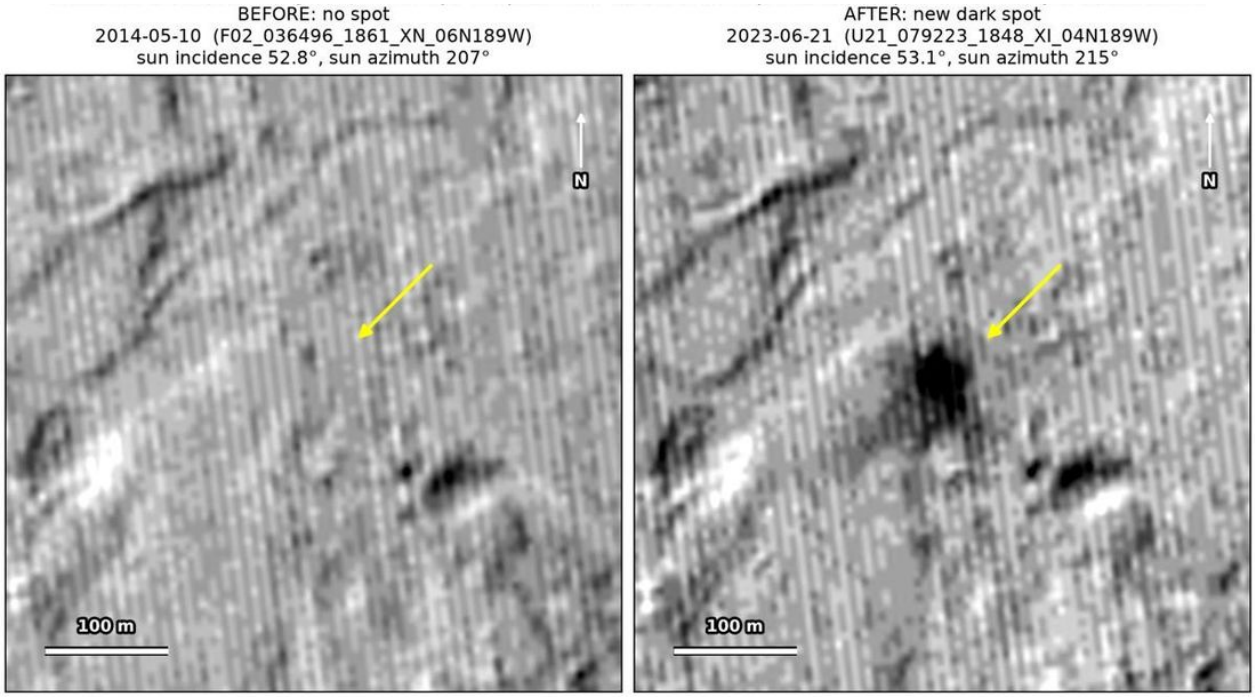

Figure 294: Candidate 120, before and after: the same 500 × 500 m of ground in 10 May 2014 (`F02_036496_1861_XN_06N189W`) and 21 June 2023 (`U21_079223_1848_XI_04N189W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

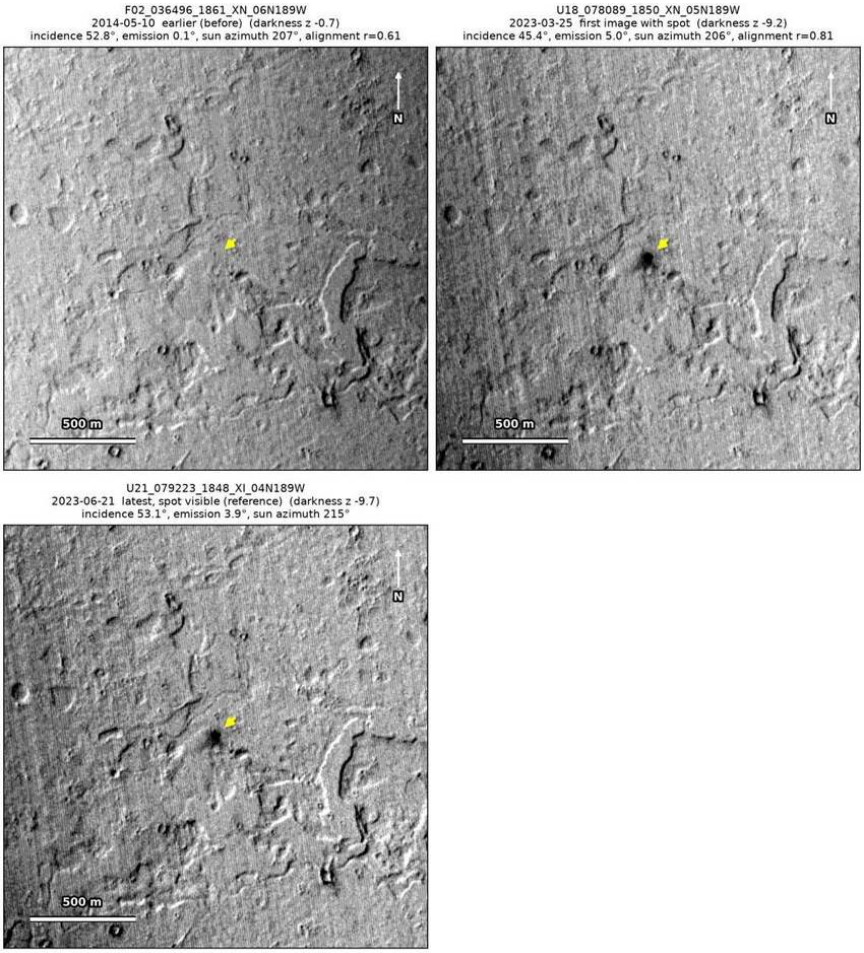

Figure 295: Candidate 120: three CTX images of the same 2×2 km patch, 2014 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 121: Western edge of Becquerel crater, Arabia Terra

Location	21.5593° N, 350.3671° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 45%
Formed between	31 May 2012 and 13 September 2020
Images	2 without the spot (absent: 8 December 2010 and 31 May 2012); 2 with it (present: 13 September 2020 and 11 December 2024)
Size	Irregular dark patch about 100 to 150 m across (86 × 142 m in the December 2024 image) around an older crater about 30 m across
HiRISE	none covers the spot (nearest footprint edge 61 km)
Nearest catalogued new impact	263 km (Daubar et al., 2022)

What changed. An irregular dark patch about 100 to 150 m across appeared around and just north-east of an older crater about 30 m across, on plains at the western edge of Becquerel crater, between May 2012 and September 2020. It is still there in December 2024, seen in two CTX images from two orbits.

Why it may be a new impact. Two earlier images, of December 2010 and May 2012, show only the crater there. Subtracting an earlier image from a later one shows the strongest change at the patch, with no other new dark mark and no tracks in the surrounding 5 × 5 km, and a new dark patch like this could be the blast zone of a small new impact beside the older crater.

Caveats. The patch has an irregular edge and surrounds an older crater, so the wind may have exposed dark material or blown it out of that crater; a new impact right beside an older one would be a coincidence, though small craters are common here. The earlier images line up only modestly, though the crater lines up in all four, and a dark mark nearby, which would show what they could reveal, is faint in the May 2012 image. No image from the high-resolution HiRISE camera covers it; the nearest ends 61 km away.

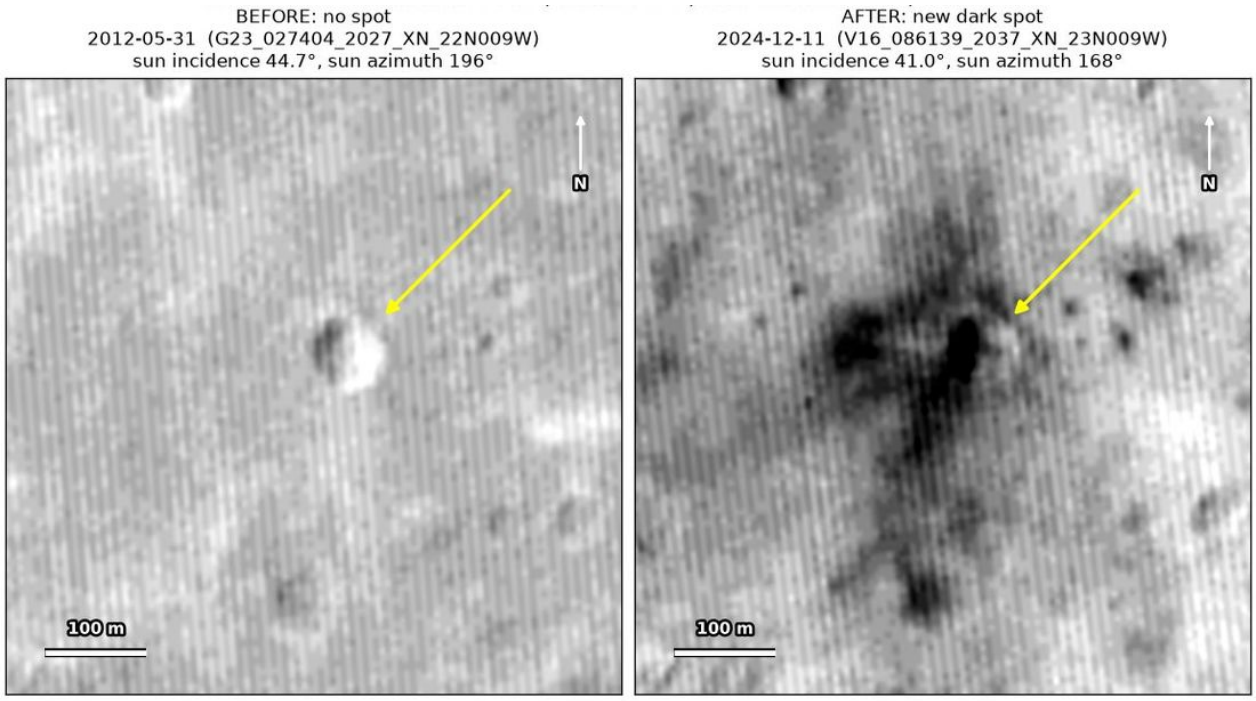

Figure 296: Candidate 121, before and after: the same 600 × 600 m of ground in 31 May 2012 (`G23_027404_2027_XN_22N009W`) and 11 December 2024 (`V16_086139_2037_XN_23N009W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

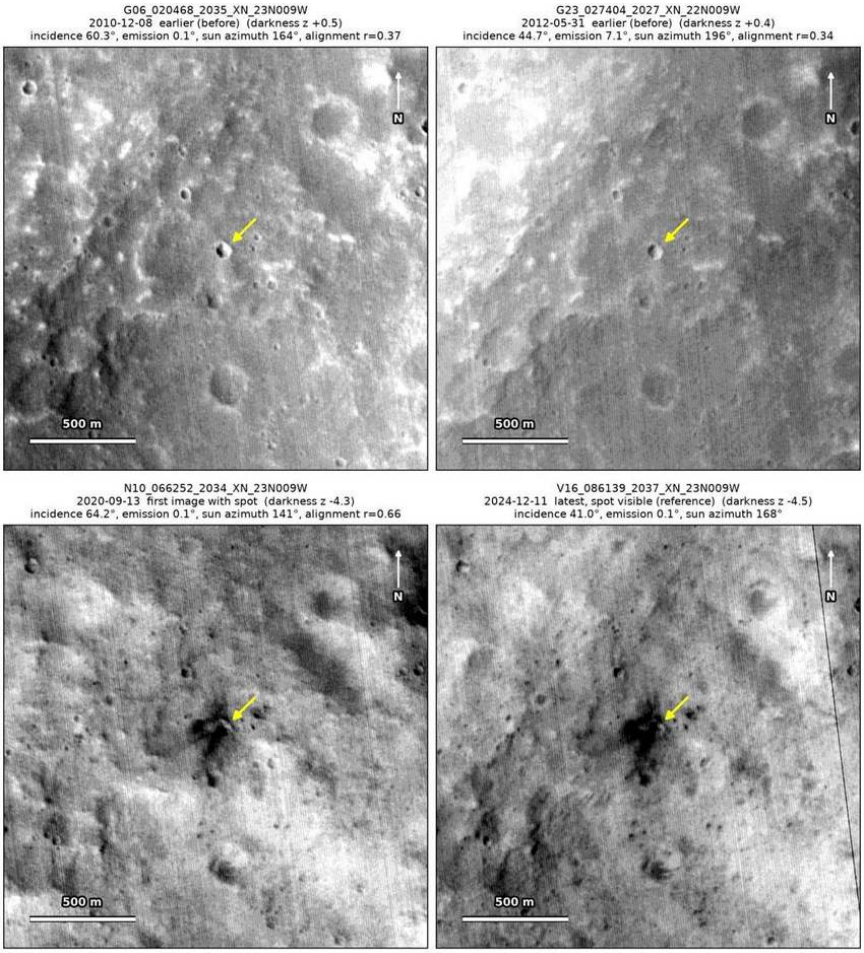

Figure 297: Candidate 121: four CTX images of the same 2×2 km patch, 2010 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 122: Lava plains 222 km east of Tharsis Tholus

Location	12.5218° N, 272.9512° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 70%
Formed between	21 April 2010 and 18 October 2024
Images	1 without the spot (absent: 21 April 2010); 2 with it (present: 18 October 2024 and 28 December 2024)
Size	Compact dark spot about 50 m across (50 × 50 m in the December 2024 image)
HiRISE	none covers the spot (nearest footprint edge 18 km)
Nearest catalogued new impact	87 km (Daubar et al., 2022)

What changed. A compact dark spot about 50 m across appeared on lava plains 222 km east of Tharsis Tholus between April 2010 and October 2024. Two CTX images from two orbits show it.

Why it may be a new impact. The only earlier image, of April 2010, shows plain ground there, and a dark mark 150 m away shows in it, so it would have shown a spot. Its sun, lower than in the December 2024 image, would bring out a hollow or knob more strongly, not less. The spot sits in the same place in the October and December 2024 images, lit from directions 25° apart, and a compact new dark spot is how the blast zone of a small new crater looks.

Caveats. The only earlier image was taken under a lower sun than the later ones, by more than is accepted for a strong candidate, and the change at the spot between the aligned images is weak, because both later images are noisy and striped. No image from the high-resolution HiRISE camera covers it; the nearest ends 18 km away.

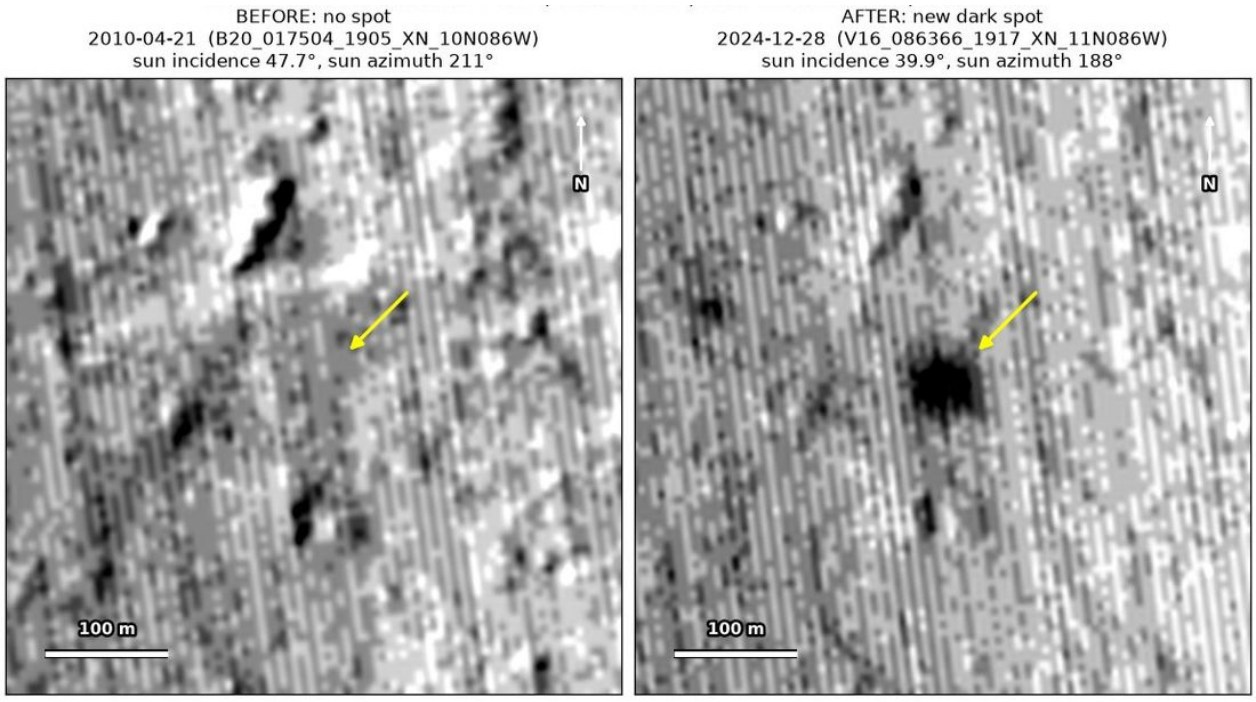

Figure 298: Candidate 122, before and after: the same 500 × 500 m of ground in 21 April 2010 (`B20_017504_1905_XN_10N086W`) and 28 December 2024 (`V16_086366_1917_XN_11N086W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

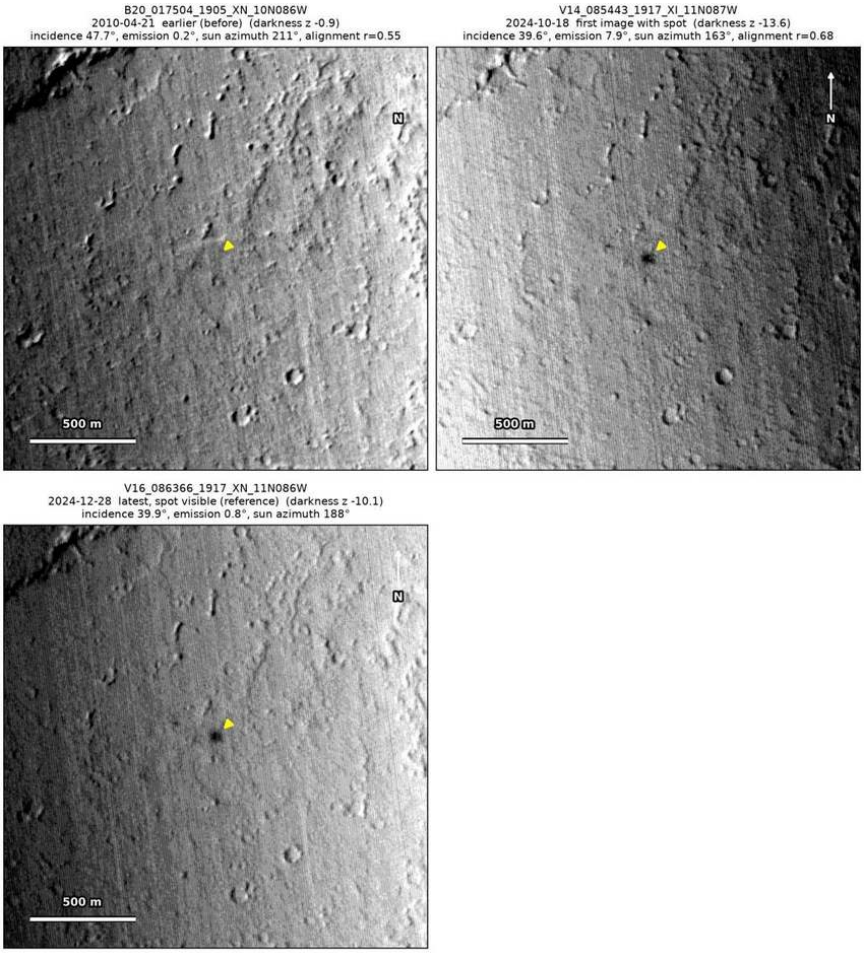

Figure 299: Candidate 122: three CTX images of the same 2×2 km patch, 2010 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 123: Yardang terrain of Zephyria Planum

Location	3.2605° N, 149.7742° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 75%
Formed between	10 November 2021 and 10 October 2023
Images	4 without the spot (absent: 18 April 2010, 2 January 2012, 22 September 2013 and 10 November 2021); 1 with it (present: 10 October 2023)
Size	Rayed dark blast zone about 1300 m across, its dark core about 500 m wide (416 × 616 m)
HiRISE	none covers the spot (nearest footprint edge 29 km)
Nearest catalogued new impact	58 km (Daubar et al., 2022)

What changed. A dark blast zone about 1.3 km across, with radial rays around a dark core about 500 m wide, appeared on the yardangs, or wind-carved ridges, of Zephyria Planum between November 2021 and October 2023. So far only one CTX image shows it.

Why it may be a new impact. Four earlier images, from 2010 to 2021, show the same yardangs there without the dark zone; the last, of November 2021, was taken under almost the same sun as the October 2023 image, and a zone this large would show in any of them. Radial rays around a dark core, with no tracks and no other new dark mark in the surrounding 5 × 5 km, are the pattern of an impact blast.

Caveats. Only one image shows it, although the zone spans about 240 image pixels and the yardang texture shows through it, so an image defect is very unlikely. No dark mark of its size lies nearby to confirm that the earlier images would have shown it. No image from the high-resolution HiRISE camera covers it; the nearest ends 29 km away.

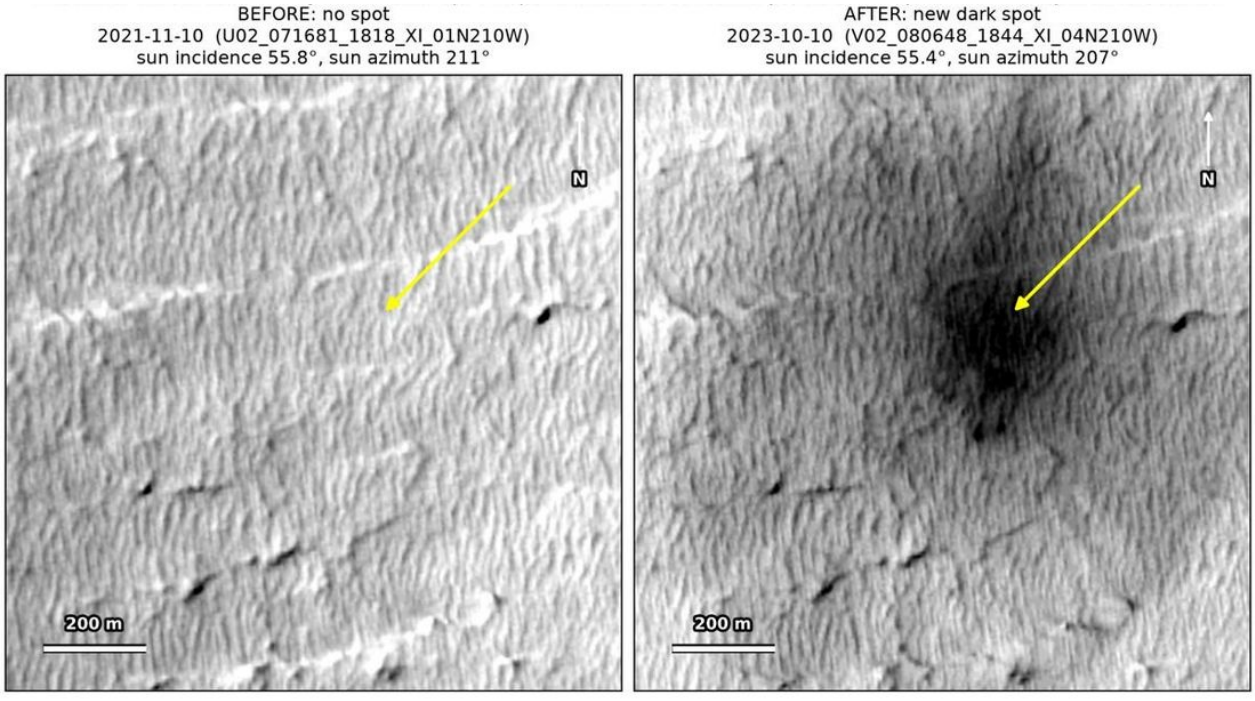

Figure 300: Candidate 123, before and after: the same 1200 × 1200 m of ground in 10 November 2021 (`U02_071681_1818_XI_01N210W`) and 10 October 2023 (`V02_080648_1844_XI_04N210W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

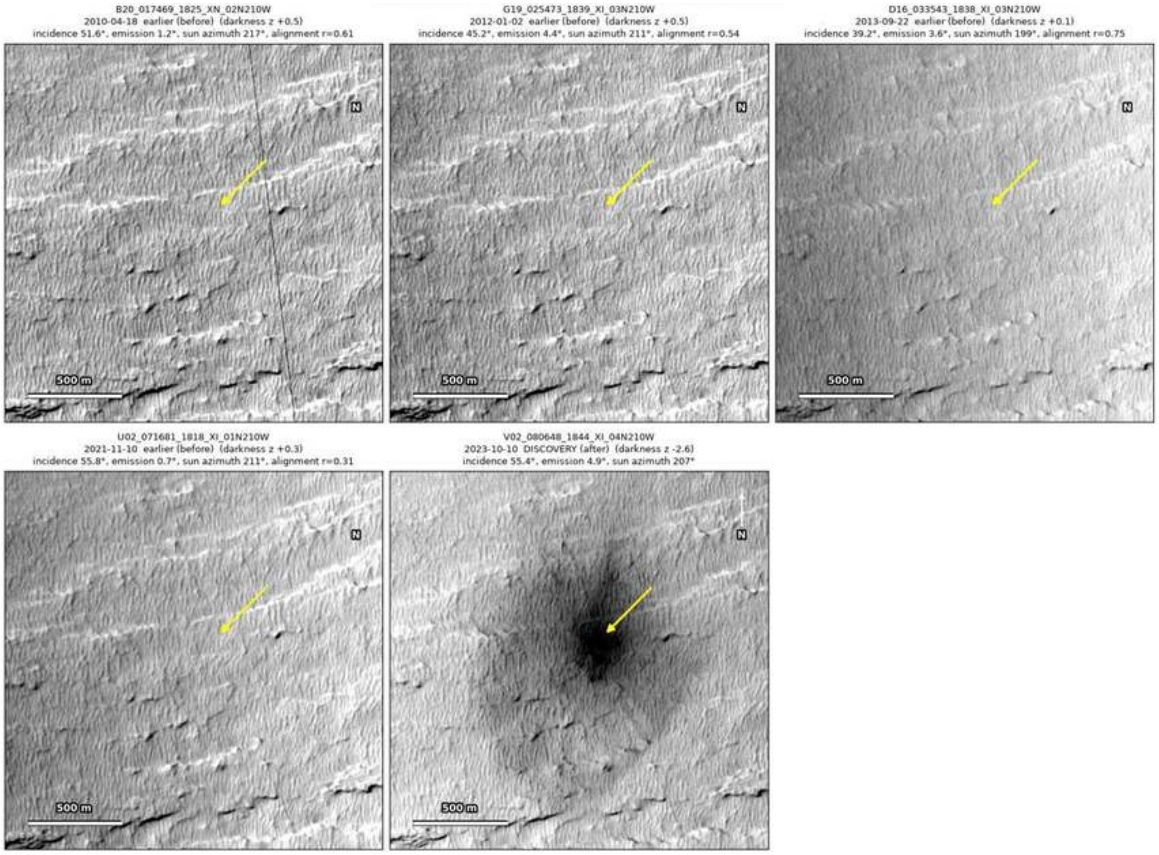

Figure 301: Candidate 123: five CTX images of the same 2×2 km patch, 2010 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 124: Cratered plains of northern Arabia Terra, south-east of Quenisset crater

Location	33.5460° N, 42.6671° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	1 May 2018 and 22 March 2023
Images	2 without the spot (absent: 19 December 2009 and 1 May 2018); 3 with it (present: 22 March 2023, 18 June 2023 and 7 January 2025)
Size	Compact dark spot about 40 m across (38 × 54 m in the June 2023 image), with a smaller new spot 154 m to the south-south-west
HiRISE	none covers the spot (nearest footprint edge 25 km)
Nearest catalogued new impact	29 km (Daubar et al., 2022)

What changed. A compact dark spot about 40 m across, with a smaller new spot 154 m away and possibly a fainter third, appeared on cratered plains of northern Arabia Terra between May 2018 and March 2023. It is still there in January 2025, seen in three CTX images from three orbits.

Why it may be a new impact. Two earlier images, from 2009 and 2018, show no spot there, the December 2009 one under almost the same sun as the March 2023 and January 2025 images, and a dark mark 253 m away shows in both, so they could have shown a spot. The change between the images is strongest at the spot and next strongest at the smaller one. Compact, separate dark spots that stay in place in three images from three orbits, with no tracks, fit a small new impact.

Caveats. The dark mark 253 m away is only just dark enough in the December 2009 image to show that this image could have caught a spot, although the spot is far darker than that mark in every later image. No image from the high-resolution HiRISE camera covers it; the nearest ends 25 km away.

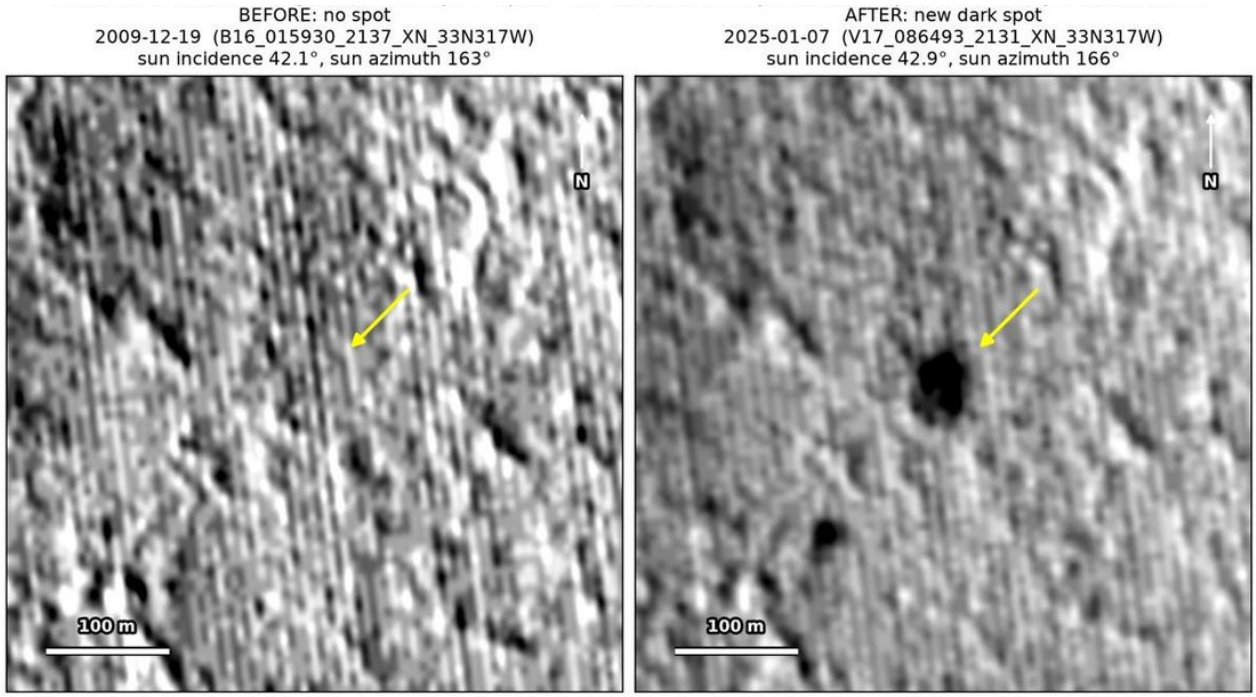

Figure 302: Candidate 124, before and after: the same 500 × 500 m of ground in 19 December 2009 (`B16_015930_2137_XN_33N317W`) and 7 January 2025 (`V17_086493_2131_XN_33N317W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

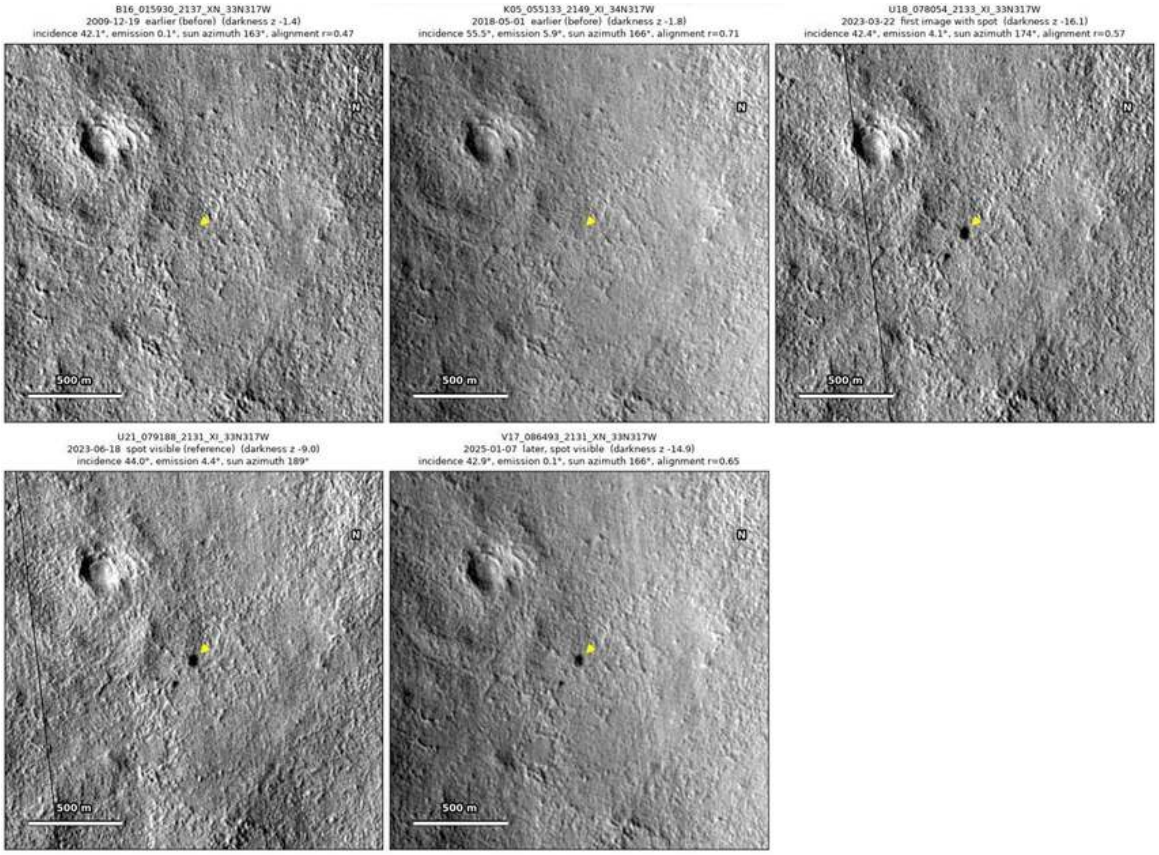

Figure 303: Candidate 124: five CTX images of the same 2×2 km patch, 2009 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 125: Pitted plains 220 km south of Ceraunius Tholus

Location	20.3980° N, 263.7310° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	2 June 2020 and 11 August 2024
Images	2 without the spot (absent: 27 February 2012 and 2 June 2020); 3 with it (present: 11 August 2024, 27 October 2024 and 26 May 2026)
Size	Compact dark spot about 45 m across (48 × 52 m in the August 2024 image), with a faint dark smear to the north
HiRISE	none covers the spot (nearest footprint edge 8.3 km)
Nearest catalogued new impact	69 km (Daubar et al., 2022)

What changed. A compact dark spot about 45 m across, with a faint dark smear to the north, appeared on pitted plains 220 km south of Ceraunius Tholus between June 2020 and August 2024. It is still there in May 2026, seen in three CTX images from three orbits.

Why it may be a new impact. Two earlier images, of February 2012 and June 2020, show no spot there; the June 2020 one was taken under a lower sun, which would bring out a pit more strongly, not less. A dark mark 284 m away shows in both, so they would have shown the spot, and a compact new dark spot with no tracks is how the blast zone of a small new crater looks.

Caveats. The main alternatives are a dust devil that stalled and a pit brought out by the sun, but there are no tracks and neither earlier image shows a pit. A December 2024 image is washed out at the site and shows the spot only faintly, so it is not used. No image from the high-resolution HiRISE camera covers it; the nearest ends 8.3 km away.

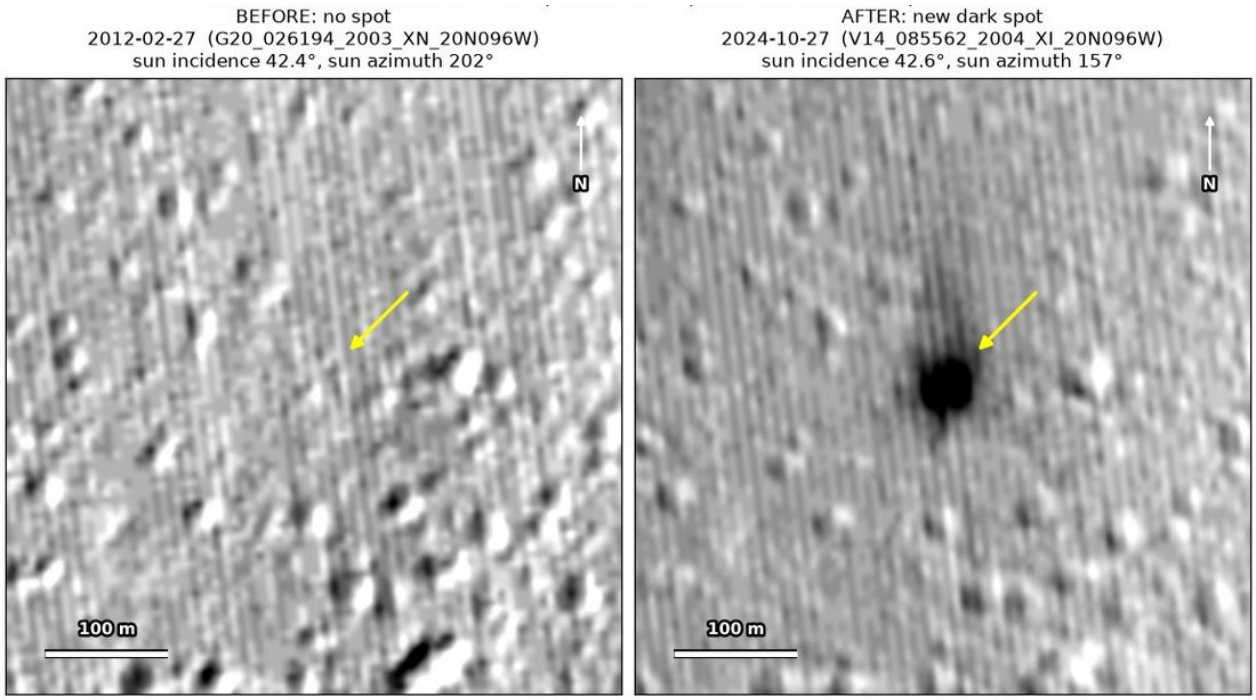

Figure 304: Candidate 125, before and after: the same 500 × 500 m of ground in 27 February 2012 (`G20_026194_2003_XN_20N096W`) and 27 October 2024 (`V14_085562_2004_XI_20N096W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

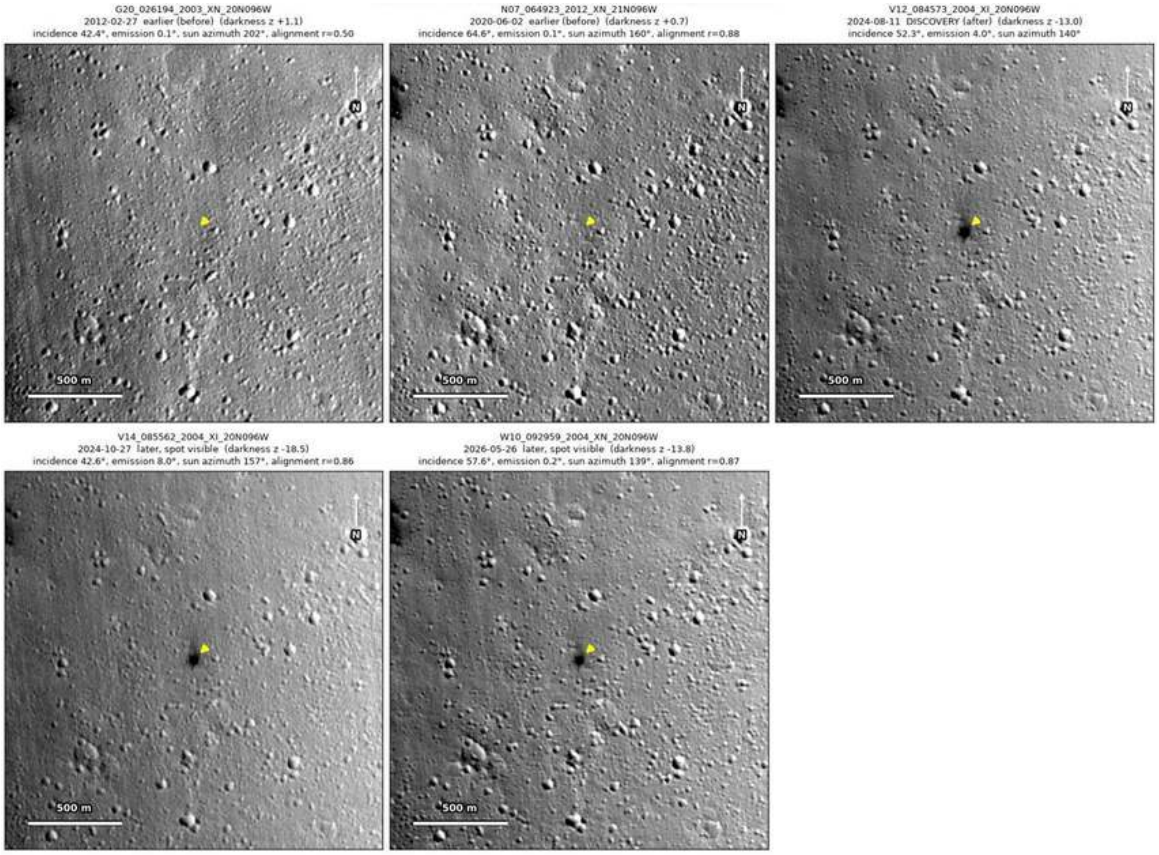

Figure 305: Candidate 125: five CTX images of the same 2×2 km patch, 2012 to 2026. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 126: Plains in south-eastern Gordii Fossae

Location	13.6357° N, 234.0298° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 75%
Formed between	6 May 2007 and 8 June 2016
Images	1 without the spot (absent: 6 May 2007); 3 with it (present: 8 June 2016, 29 March 2019 and 29 March 2024)
Size	Dark spot with a head about 60 m across and a tail about 140 m long to the east (196 × 96 m in the June 2016 image), fading to a core of 38 × 46 m in the March 2024 image
HiRISE	none covers the spot (nearest footprint edge 17.2 km)
Nearest catalogued new impact	22 km (Daubar et al., 2022)

What changed. A dark spot with a head about 60 m across and a tail about 140 m long to the east appeared on plains in south-eastern Gordii Fossae between May 2007 and June 2016. It is still there, fading, in March 2024, seen in three CTX images from three orbits.

Why it may be a new impact. The only earlier image, from 2007, was taken under almost the same sun as the March 2024 image and shows plain ground there. Subtracting it from a later image and removing the broad background leaves the spot as the only compact darkening on open ground. A dark spot with a short tail appearing on open plain, away from a knob and a ridge nearby, fits the blast zone of a small new impact.

Caveats. The spot is faint and fading, no dark mark nearby could show whether the 2007 image would have caught one, and the change at the spot between the aligned images is weak, so it is rated about 75% rather than 90%. No image from the high-resolution HiRISE camera covers it; the nearest ends 17.2 km away.

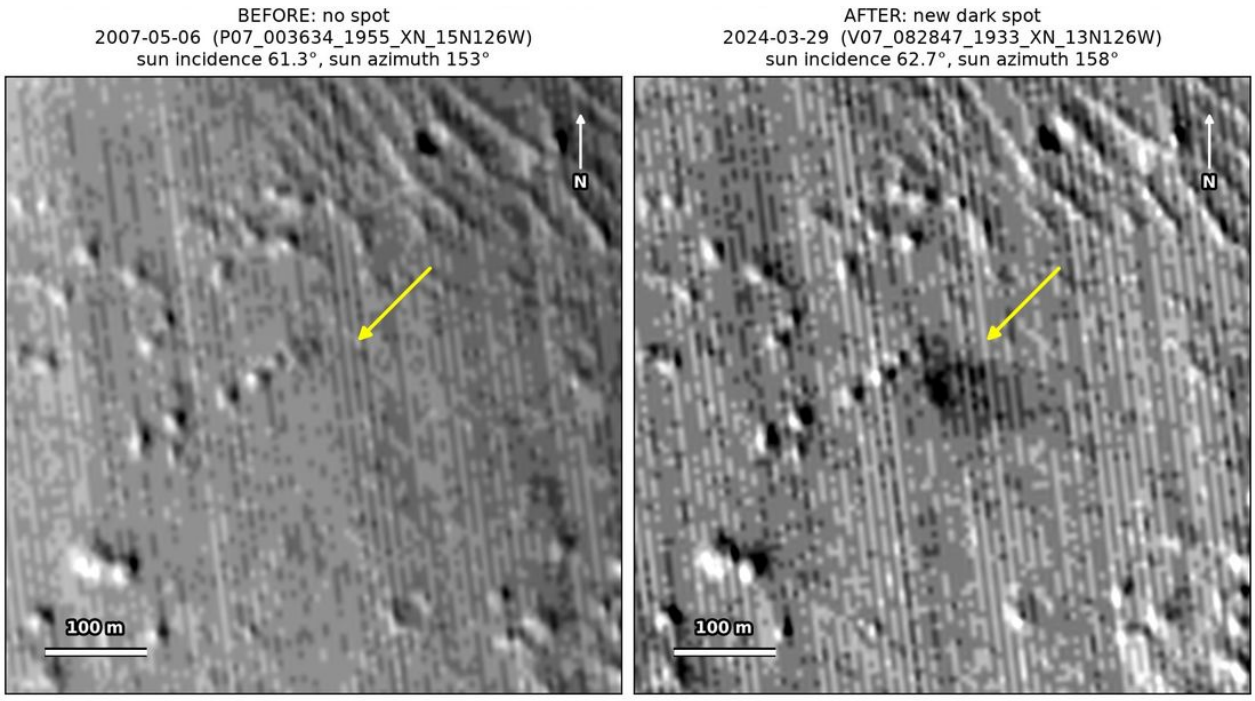

Figure 306: Candidate 126, before and after: the same 600 × 600 m of ground in 6 May 2007 (`P07_003634_1955_XN_15N126W`) and 29 March 2024 (`V07_082847_1933_XN_13N126W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

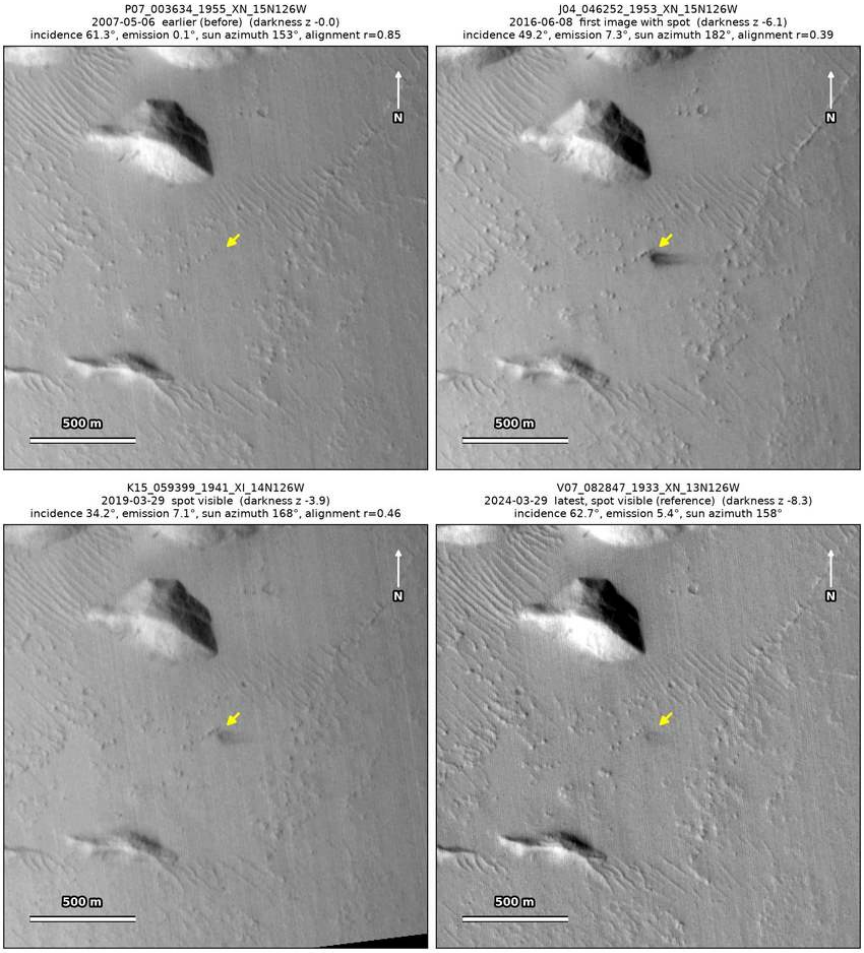

Figure 307: Candidate 126: four CTX images of the same 2×2 km patch, 2007 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 127: Wind-ridged plains of southern Amazonis Planitia, north-east of Nicholson crater

Location	3.8206° N, 197.5165° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	3 November 2021 and 9 August 2023
Images	2 without the spot (absent: 27 May 2008 and 3 November 2021); 1 with it (present: 9 August 2023)
Size	Three dark spots within about 180 m: the main one about 40 m across (48 × 36 m in the August 2023 image), and two more new spots, 42 × 30 m and 50 × 52 m, 119 m to the west and 177 m to the west-north-west of it
HiRISE	none covers the spot (nearest footprint edge 39.7 km)
Nearest catalogued new impact	121 km (Daubar et al., 2022)

What changed. A cluster of three dark spots, each about 30 to 50 m across and all within about 180 m, appeared on wind-ridged plains of southern Amazonis Planitia between November 2021 and August 2023. So far only one CTX image shows it.

Why it may be a new impact. Two earlier images, from 2008 and 2021, both taken under a sun like that of the August 2023 image, show plain ground there. A dark mark 156 m away shows in both, so they would have shown new spots too, and the change between the aligned before and after images is greatest at the three spots. A cluster of small, compact dark spots is the usual pattern of an impactor that broke up in the atmosphere.

Caveats. Only one image shows it, so it is rated promising rather than strong. An image defect is unlikely, since these are three separate spots of different shapes, not a stripe or speckle. No image from the high-resolution HiRISE camera covers it; the nearest ends 39.7 km away.

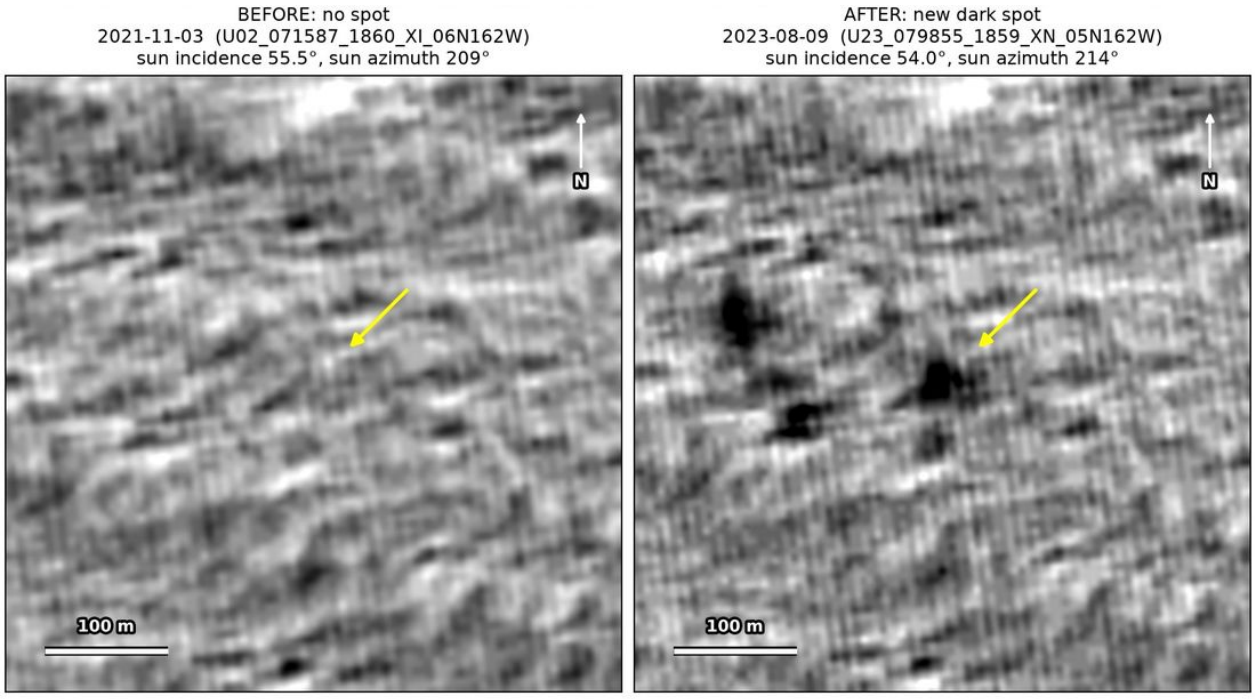

Figure 308: Candidate 127, before and after: the same 500 × 500 m of ground in 3 November 2021 (`U02_071587_1860_XI_06N162W`) and 9 August 2023 (`U23_079855_1859_XN_05N162W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

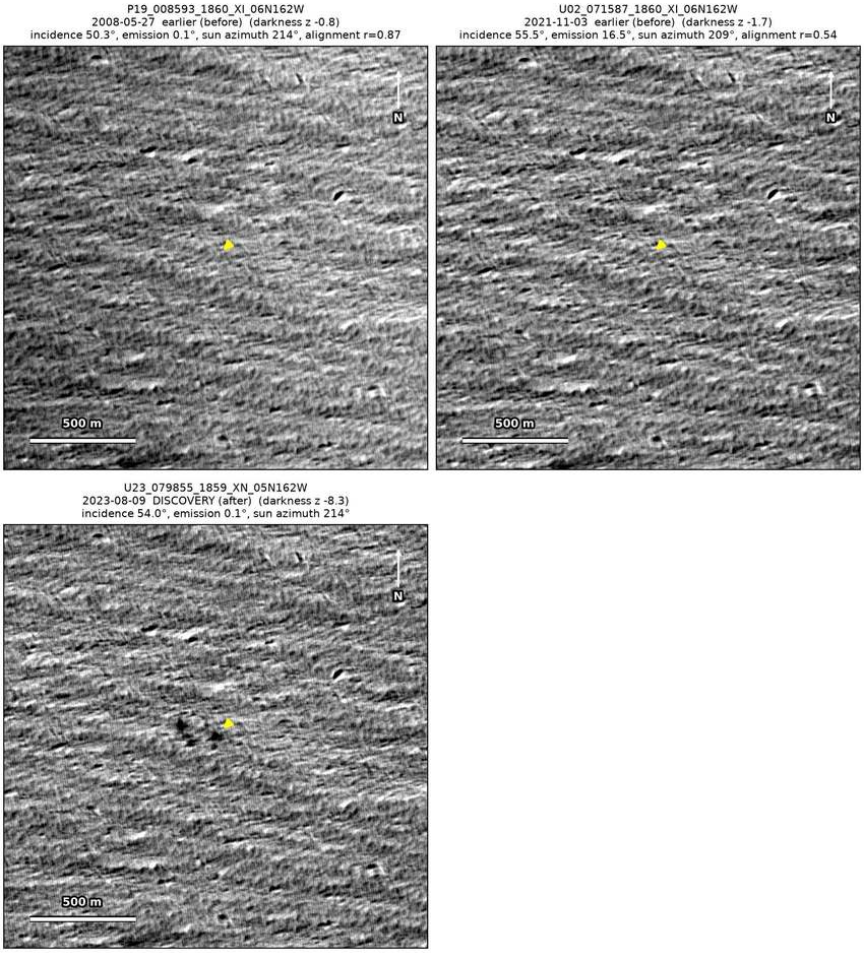

Figure 309: Candidate 127: three CTX images of the same 2×2 km patch, 2008 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 128: Textured plains in north-western Phlegra Montes

Location	43.4682° N, 155.5156° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 65%
Formed between	19 February 2010 and 2 June 2023
Images	1 without the spot (absent: 19 February 2010); 1 with it (present: 2 June 2023)
Size	Compact dark spot about 40 m across (32 × 50 m in the June 2023 image)
HiRISE	none covers the spot (nearest footprint edge 9.9 km)
Nearest catalogued new impact	122 km (Daubar et al., 2022)

What changed. A compact dark spot about 40 m across appeared on textured plains in north-western Phlegra Montes between February 2010 and June 2023. So far it is seen in one CTX image; the interval is so long because only two CTX images cover the site.

Why it may be a new impact. The only earlier image, from 2010, taken under almost the same sun as the June 2023 image, shows plain ground there, and a dark mark 234 m away shows in it, so that image would have shown a new spot too. Subtracting the aligned earlier image from the later one shows the strongest change in the surrounding 2 × 2 km right at the spot. A compact dark spot appearing on plain ground fits the blast zone of a small new impact.

Caveats. Only one image shows it, so it is rated promising rather than strong, and an image defect cannot be ruled out with one image. It lies inside the area, and possibly the period, covered by the published catalogue of impacts by Bickel et al. (2025), but every entry in that catalogue has a HiRISE image, so it is probably not among them. No image from the high-resolution HiRISE camera covers it; the nearest ends 9.9 km away.

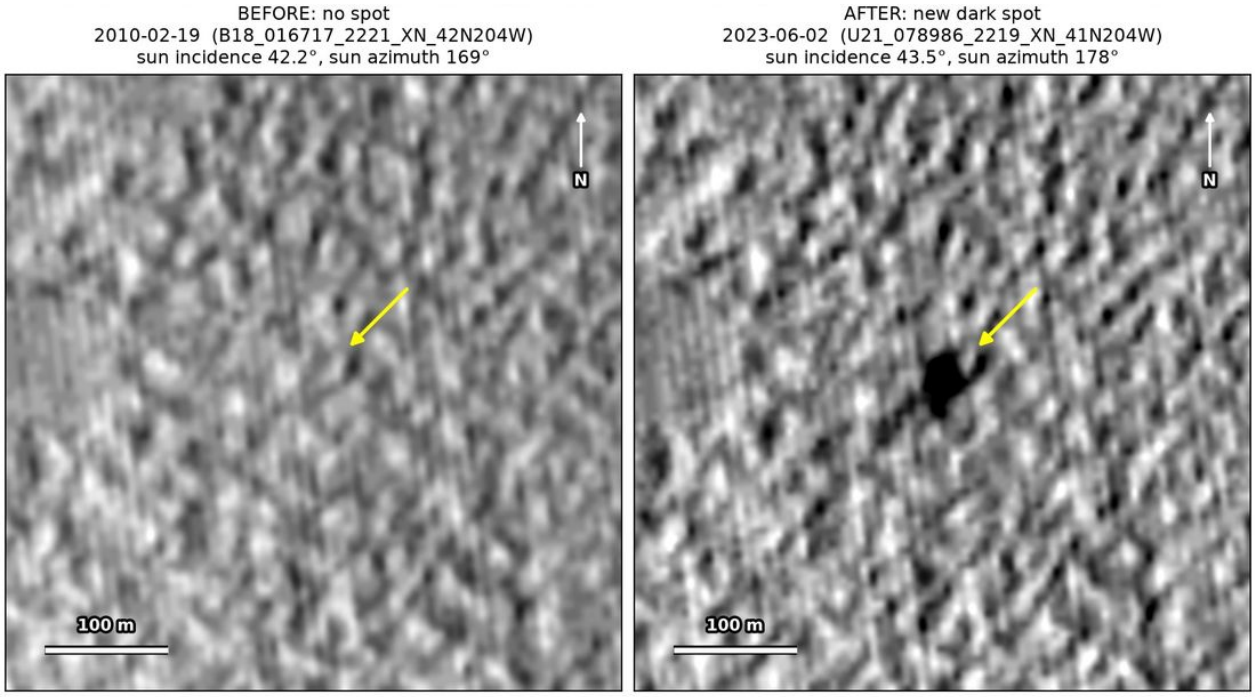

Figure 310: Candidate 128, before and after: the same 500 × 500 m of ground in 19 February 2010 (`B18_016717_2221_XN_42N204W`) and 2 June 2023 (`U21_078986_2219_XN_41N204W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

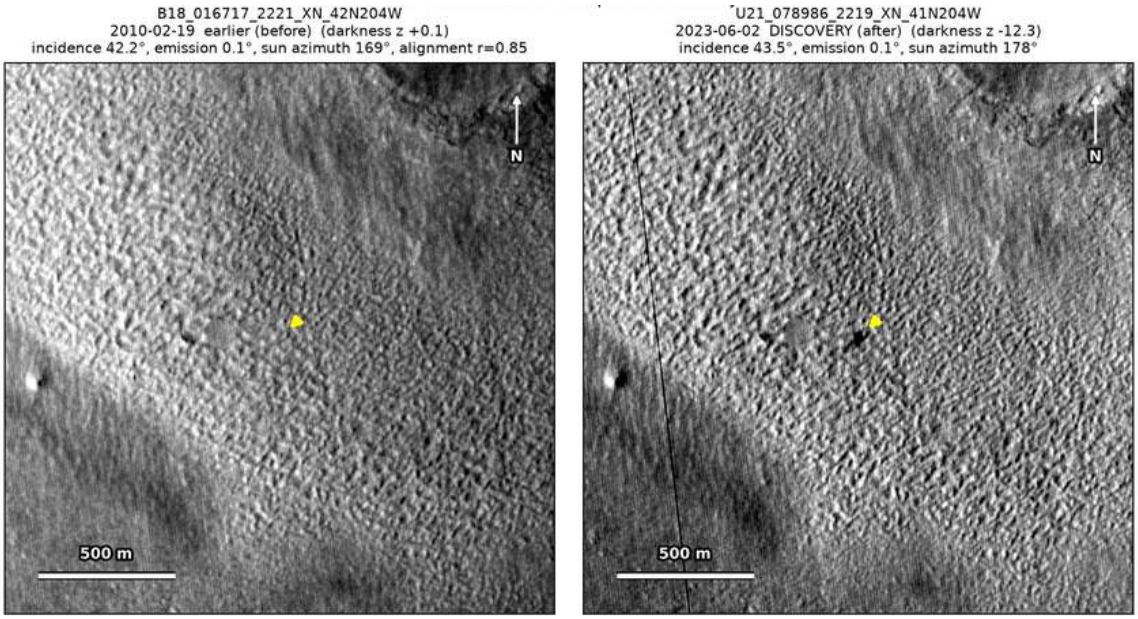

Figure 311: Candidate 128: both CTX images of the same 2×2 km patch, February 2010 and June 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 129: Wind-ridged plains south of Medusae Fossae, 183 km west of Medusae Sulci

Location	6.1020° S, 197.3791° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	14 April 2015 and 26 April 2023
Images	2 without the spot (absent: 11 May 2012 and 14 April 2015); 2 with it (present: 26 April 2023 and 12 July 2023)
Size	Compact dark spot about 45 m across with soft edges (70 × 62 m in the July 2023 image)
HiRISE	none covers the spot (nearest footprint edge 36.5 km)
Nearest catalogued new impact	85 km (Daubar et al., 2022)

What changed. A compact dark spot about 45 m across appeared over a small wind ridge on the plains south of Medusae Fossae between April 2015 and April 2023, and was still there in July 2023. Two CTX images from separate orbits show it.

Why it may be a new impact. Two earlier images, from 2012 and 2015, show only the ridge's faint shadow there, the first under almost the same sun as the April 2023 image. A dark mark 296 m away shows in the 2012 image, so that image would have shown a new spot too. A compact dark spot appearing there, with no tracks around it, fits the blast zone of a small new impact.

Caveats. The two images with the spot are only 77 days apart. It also sits over a small wind ridge whose shading could mimic a spot, although the 2012 image, taken under almost the same sun as the April 2023 one, shows only the ridge's faint shadow there. No image from the high-resolution HiRISE camera covers it; the nearest ends 36.5 km away.

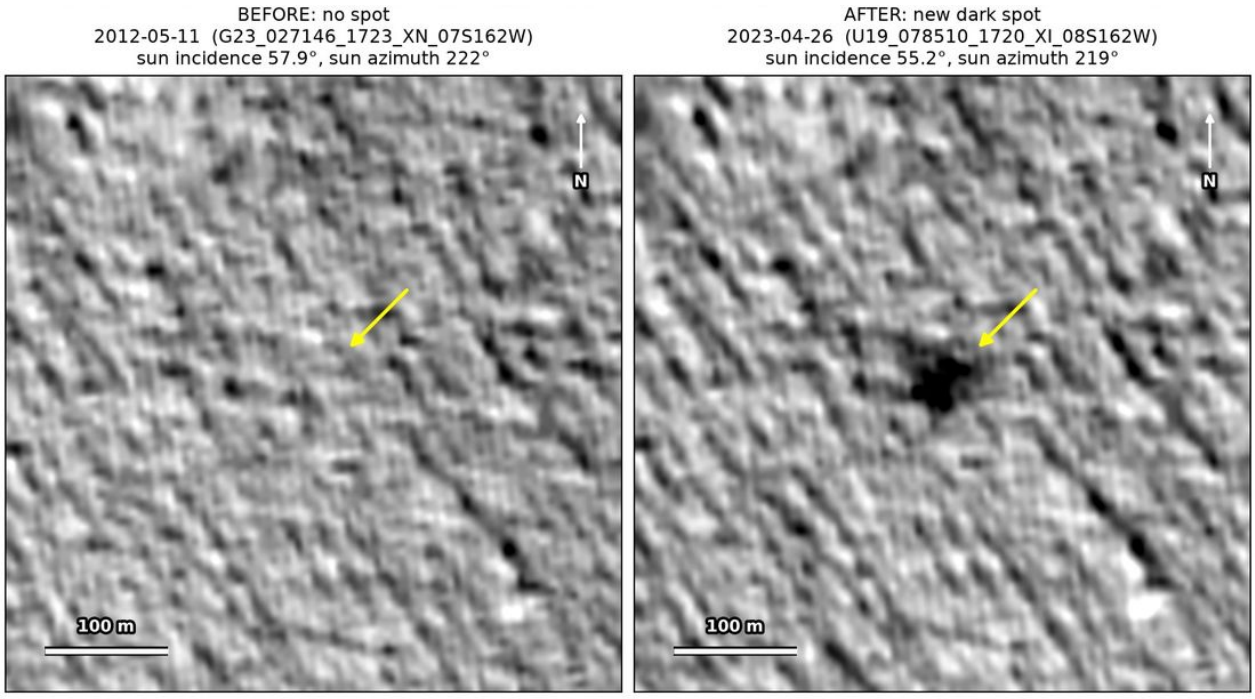

Figure 312: Candidate 129, before and after: the same 500 × 500 m of ground in 11 May 2012 (`G23_027146_1723_XN_07S162W`) and 26 April 2023 (`U19_078510_1720_XI_08S162W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

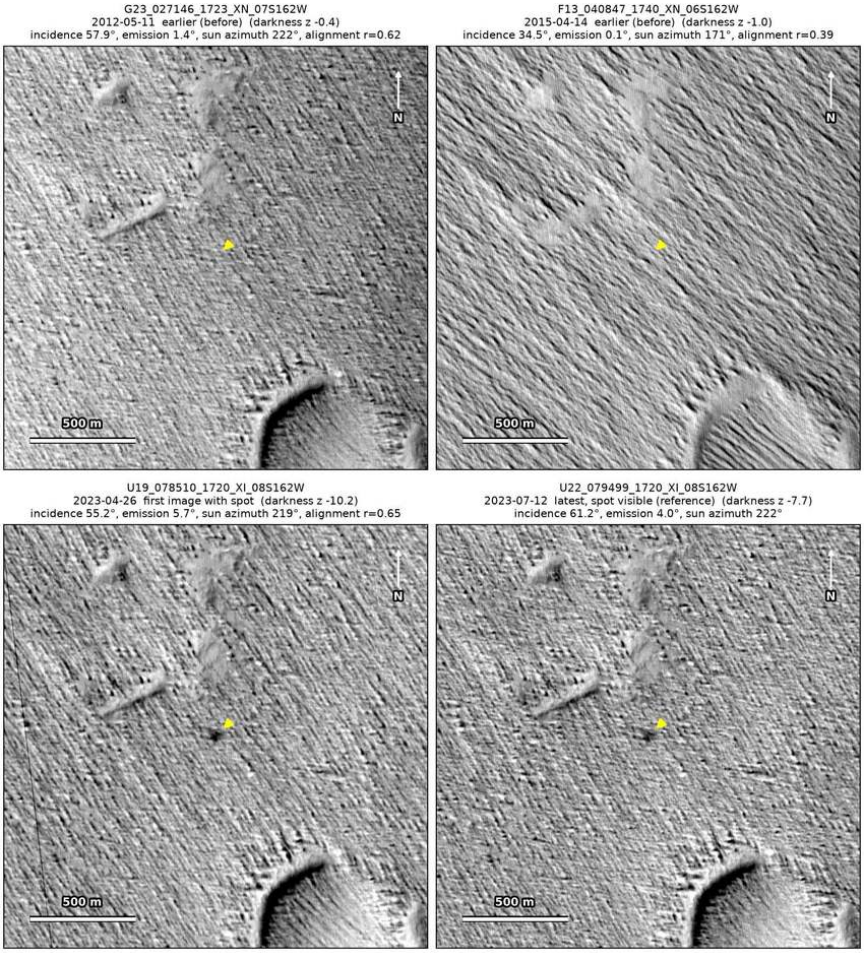

Figure 313: Candidate 129: four CTX images of the same 2×2 km patch, 2012 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 130: Plains among the eastern Erebus Montes, northern Amazonis Planitia

Location	36.5465° N, 192.8396° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	23 March 2012 and 10 April 2023
Images	2 without the spot (absent: 1 November 2010 and 23 March 2012); 3 with it (present: 10 April 2023, 10 June 2023 and 30 December 2024)
Size	Compact dark spot about 30 m across (26 × 38 m in the June 2023 image)
HiRISE	none covers the spot (nearest footprint edge 28.1 km)
Nearest catalogued new impact	109 km (Daubar et al., 2022)

What changed. A compact dark spot about 30 m across appeared on plains among the eastern Erebus Montes between March 2012 and April 2023, and was still there in June 2023 and December 2024. Three CTX images from separate orbits show it.

Why it may be a new impact. Two earlier images, from 2010 and 2012, show plain ground there, the second under almost the same sun as the April and June 2023 images. A faint dark mark 100 m away shows in the 2012 image just as it does in the later ones, so that image would have shown a new spot too. A compact dark spot appearing on plain ground, with no tracks around it, fits the blast zone of a small new impact.

Caveats. The faint dark mark nearby is only just dark enough to show that the earlier images could have caught the spot, and the aligned images show only a weak change at the spot, so it is rated about 80% rather than 90%. The spot could also be a distant secondary crater, made by debris from the S1094b impact of December 2021, but that is a remote possibility: S1094b lies 168 km away, far beyond its secondary craters imaged within about 30 km of it, and the spot would still be a new crater. No image from the high-resolution HiRISE camera covers it; the nearest ends 28.1 km away.

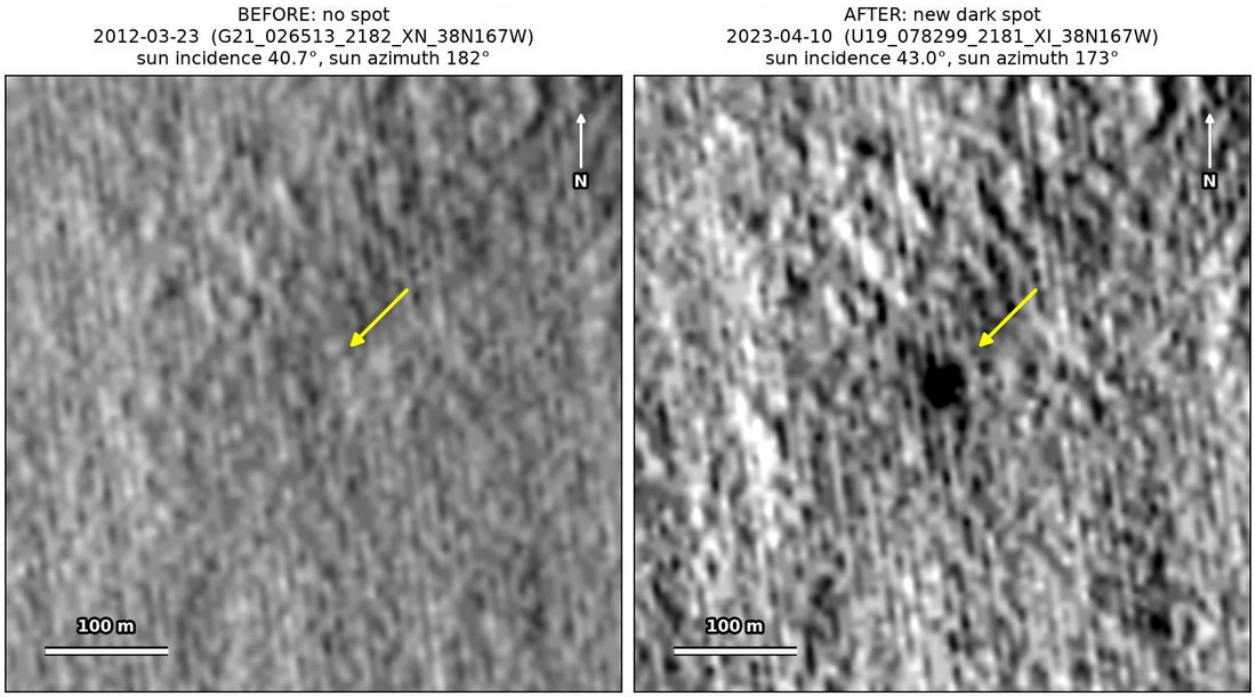

Figure 314: Candidate 130, before and after: the same 500 × 500 m of ground in 23 March 2012 (`G21_026513_2182_XN_38N167W`) and 10 April 2023 (`U19_078299_2181_XI_38N167W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

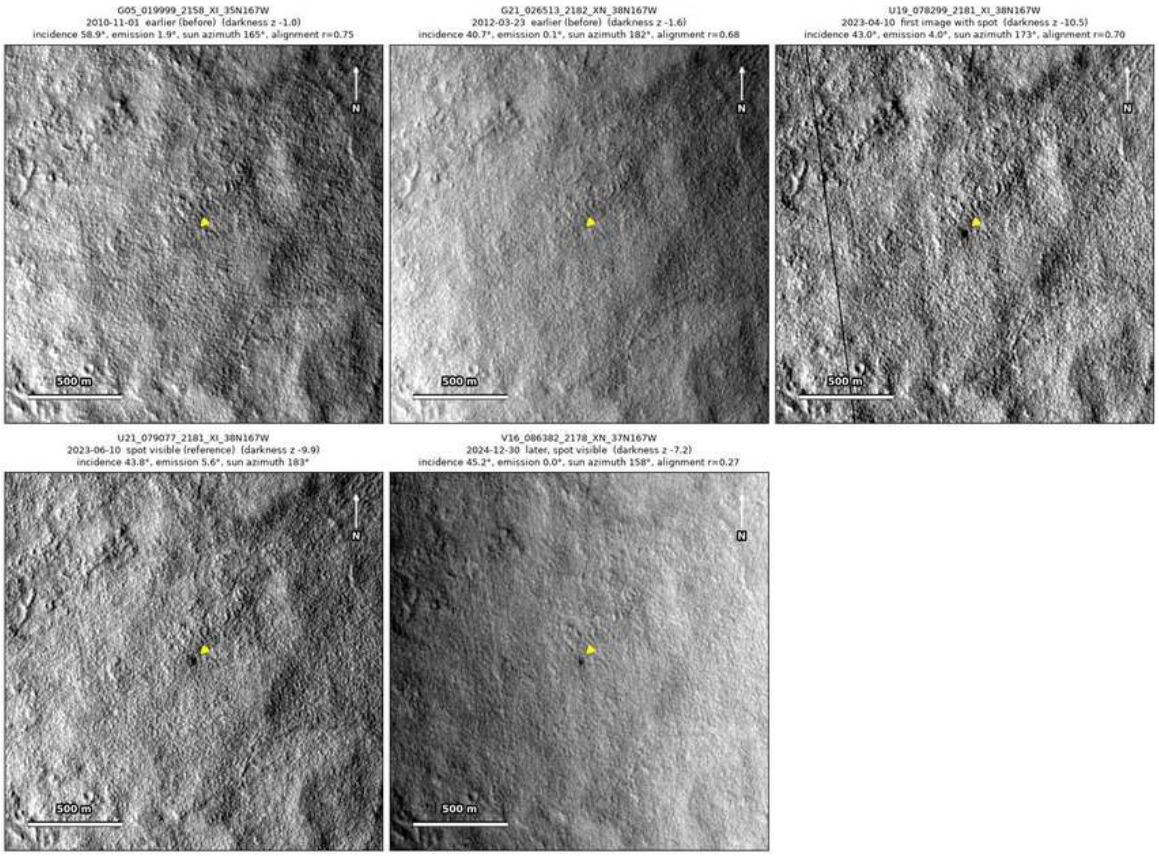

Figure 315: Candidate 130: five CTX images of the same 2×2 km patch, 2010 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 131: Terrain between Lycus Sulci and Gordii Dorsum, 63 km south of Lycus Sulci

Location	15.8187° N, 213.5247° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	18 May 2010 and 2 June 2023
Images	1 without the spot (absent: 18 May 2010); 2 with it (present: 2 June 2023 and 10 March 2025)
Size	Elongated, wedge-shaped dark mark about 90 × 25 m (94 × 38 m in the June 2023 image), blunt at its west end and tapering to the east-north-east
HiRISE	none covers the spot (nearest footprint edge 54.9 km)
Nearest catalogued new impact	121 km (Daubar et al., 2022)

What changed. A pitch-black, wedge-shaped mark about 90 × 25 m appeared on smooth ground 63 km south of Lycus Sulci between May 2010 and June 2023, and was little changed in March 2025. Two CTX images from separate orbits show it.

Why it may be a new impact. The only earlier image, from 2010, taken under almost the same sun as the June 2023 image, shows plain ground there, and comparing the aligned before and after images shows the strongest change in the surrounding 2 × 2 km right at the mark. A sharp-edged dark mark appearing on smooth ground fits the blast zone of a small new impact, and an impact at a low angle can stretch it into this shape.

Caveats. The mark is elongated, and older slope streaks (dark streaks left by dust sliding down a slope) lie on a scarp 1.5 to 2.5 km away, so a fresh slope streak is the main alternative, although the mark lies on smooth ground, not on a steep slope. With only one earlier image, and no dark mark nearby to show that it would have caught a new spot, it is rated promising rather than strong. No image from the high-resolution HiRISE camera covers it; the nearest ends 54.9 km away.

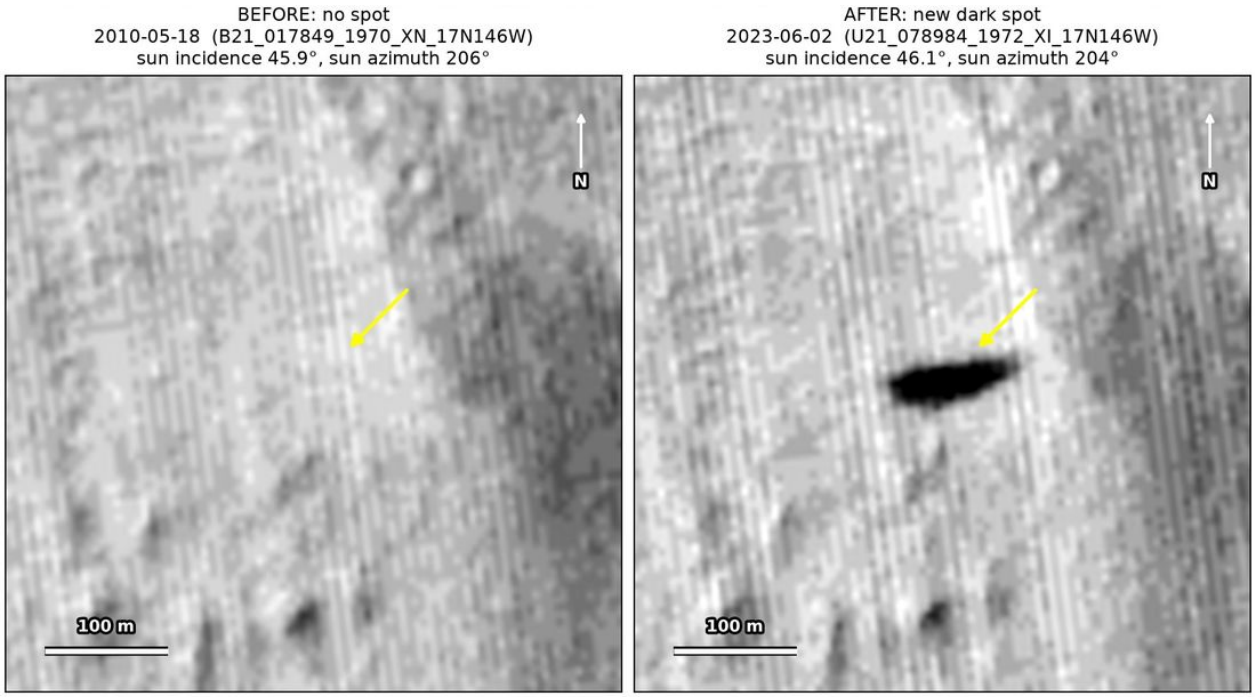

Figure 316: Candidate 131, before and after: the same 500 × 500 m of ground in 18 May 2010 (`B21_017849_1970_XN_17N146W`) and 2 June 2023 (`U21_078984_1972_XI_17N146W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

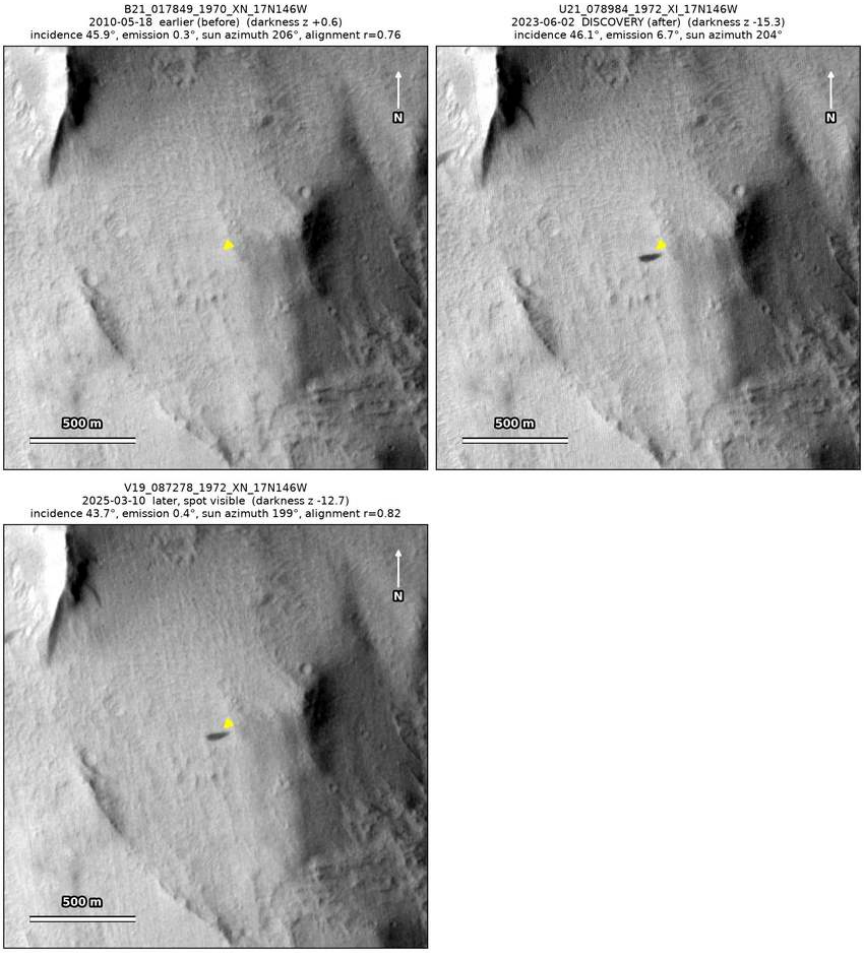

Figure 317: Candidate 131: three CTX images of the same 2×2 km patch, 2010 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 132: Plains in Arabia Terra, 233 km north-east of Janssen crater

Location	6.1215° N, 39.5546° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 50%
Formed between	27 September 2012 and 27 October 2023
Images	2 without the spot (absent: 26 November 2007 and 27 September 2012); 1 with it (present: 27 October 2023)
Size	Two dark spots, about 30 m and about 20 m across, 40 m apart (50 × 44 m in the October 2023 image)
HiRISE	none covers the spot (nearest footprint edge 0.5 km)
Nearest catalogued new impact	88 km (Daubar et al., 2022)

What changed. Two compact dark spots, about 30 and 20 m across and 40 m apart, with a faint trail about 60 m long to the south, appeared at the southern end of a broad dark band on plains in Arabia Terra, north-east of Janssen crater, between September 2012 and October 2023. So far only one CTX image shows them.

Why it may be a new impact. Two earlier images, from 2007 and 2012, show plain ground there, the second taken under a sun like that of the October 2023 image. Two compact dark spots side by side fit the blast zones of a small impactor that broke up in the atmosphere.

Caveats. Only one image shows them, so it is rated promising rather than strong. Dust devils are active there, and one track passes just south of the spots, though their tracks are long lines, not compact spots. The faint trail could also mean a small slope streak (a dark streak left by dust sliding down a slope) at the end of the dark band, and no mark nearby is dark enough in the earlier images to show whether they would have caught the spots. No image from the high-resolution HiRISE camera covers it; the nearest ends 0.5 km away.

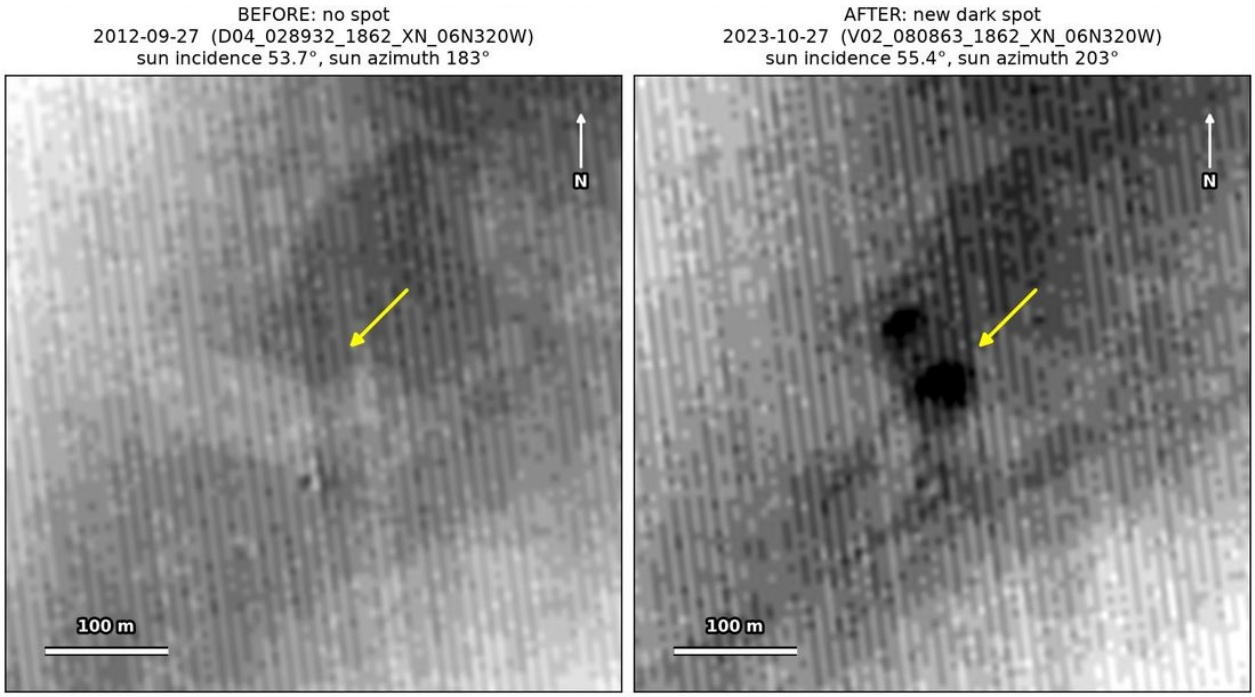

Figure 318: Candidate 132, before and after: the same 500 × 500 m of ground in 27 September 2012 (`D04_028932_1862_XN_06N320W`) and 27 October 2023 (`V02_080863_1862_XN_06N320W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

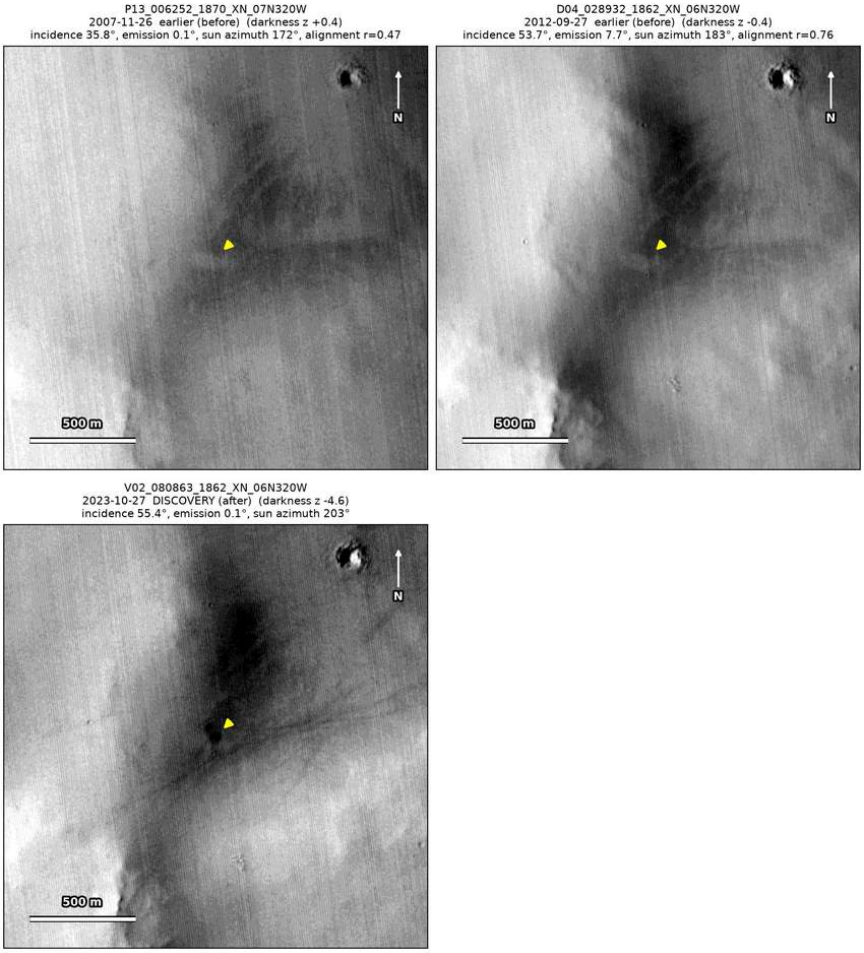

Figure 319: Candidate 132: three CTX images of the same 2×2 km patch, 2007 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 133: Plains in Arabia Terra, 291 km west of Henry crater

Location	9.6103° N, 18.5988° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	27 July 2020 and 6 December 2023
Images	2 without the spot (absent: 3 August 2016 and 27 July 2020); 2 with it (present: 6 December 2023 and 7 January 2025)
Size	Cluster of at least eight dark spots within about 280 m, the largest about 45 m across (50 × 50 m in the December 2023 image) and the others 10 to 26 m
HiRISE	none covers the spot (nearest footprint edge 37.4 km)
Nearest catalogued new impact	70 km (Daubar et al., 2022)

What changed. A cluster of at least eight compact dark spots within about 280 m, the largest about 45 m across, appeared on plains in Arabia Terra west of Henry crater between July 2020 and December 2023, and was still there in January 2025. Two CTX images from separate orbits show it.

Why it may be a new impact. Two earlier images, from 2016 and 2020, show plain ground there, while a small crater 190 m away shows in all four images, so the earlier ones would have shown new spots too. The change between the aligned before and after images is greatest at the largest spot, and a new cluster of compact dark spots, of a kind dust devils do not leave, fits the blast zones of a small new impact.

Caveats. In the 2016 image the small crater nearby is only just dark enough to show that the image would have caught new spots. The ground around the spot is also so featureless in the July 2020 and January 2025 images that darkness at the site is harder to measure reliably in them. No image from the high-resolution HiRISE camera covers it; the nearest ends 37.4 km away.

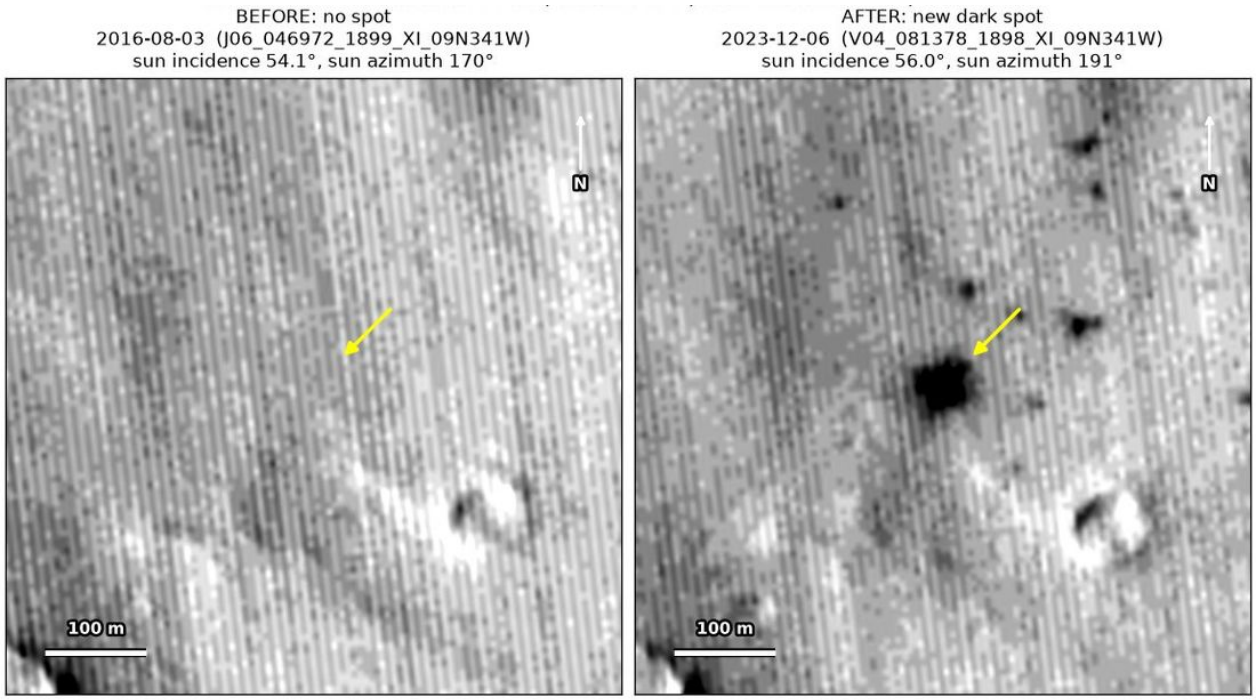

Figure 320: Candidate 133, before and after: the same 600 × 600 m of ground in 3 August 2016 (`J06_046972_1899_XI_09N341W`) and 6 December 2023 (`V04_081378_1898_XI_09N341W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

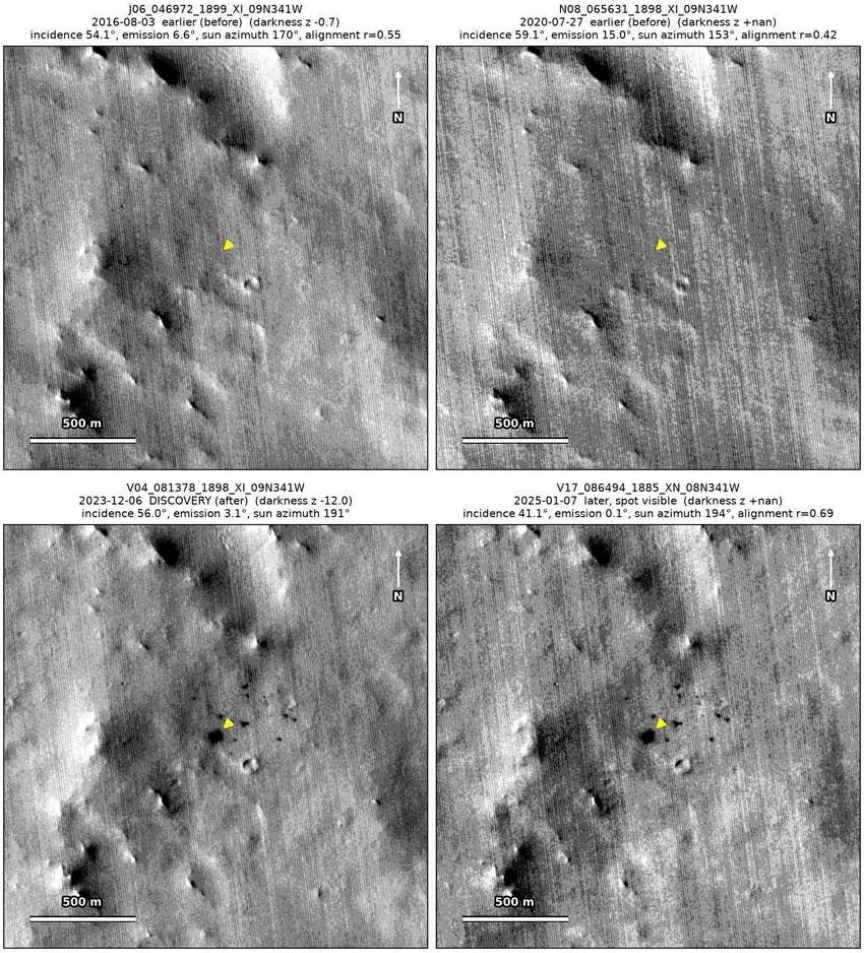

Figure 321: Candidate 133: four CTX images of the same 2×2 km patch, 2016 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 134: Lower flank of a massif in central Phlegra Montes

Location	39.3839° N, 164.7615° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	5 March 2024 and 19 December 2024
Images	4 without the spot (absent: 30 March 2023, 30 October 2023, 30 December 2023 and 5 March 2024); 5 with it (present: 19 December 2024, 17 March 2025, 11 May 2025, 30 August 2025 and 4 November 2025)
Size	Compact dark spot about 40 m across (46 × 62 m in the December 2024 image), roughly round with no tail
HiRISE	none covers the spot (nearest footprint edge 4.3 km)
Nearest catalogued new impact	21 km (Daubar et al., 2022)

What changed. A compact dark spot about 40 m across appeared on the lower flank of a massif in central Phlegra Montes between March and December 2024, and was still there, unchanged, in November 2025. Five CTX images from separate orbits show it.

Why it may be a new impact. Four earlier images from 2023 and 2024 show plain ground there, one of them, from March 2023, under almost the same sun as the December 2024 image. A dark mark 300 m away shows in the March and October 2023 images, so they would have shown a new spot too. A compact dark spot that appears on plain ground and stays unchanged for 11 months fits the blast zone of a small new impact.

Caveats. It lies on a sunlit mountain flank, so a short slope streak (a dark streak left by dust sliding down a slope) is possible, although the spot is compact and has no downslope tail, and no other streaks are seen on these slopes. This setting is why it is rated about 80% rather than 90%. The area is a CTX impact-monitoring site, so the camera team has probably seen the spot already. No image from the high-resolution HiRISE camera covers it; the nearest ends 4.3 km away.

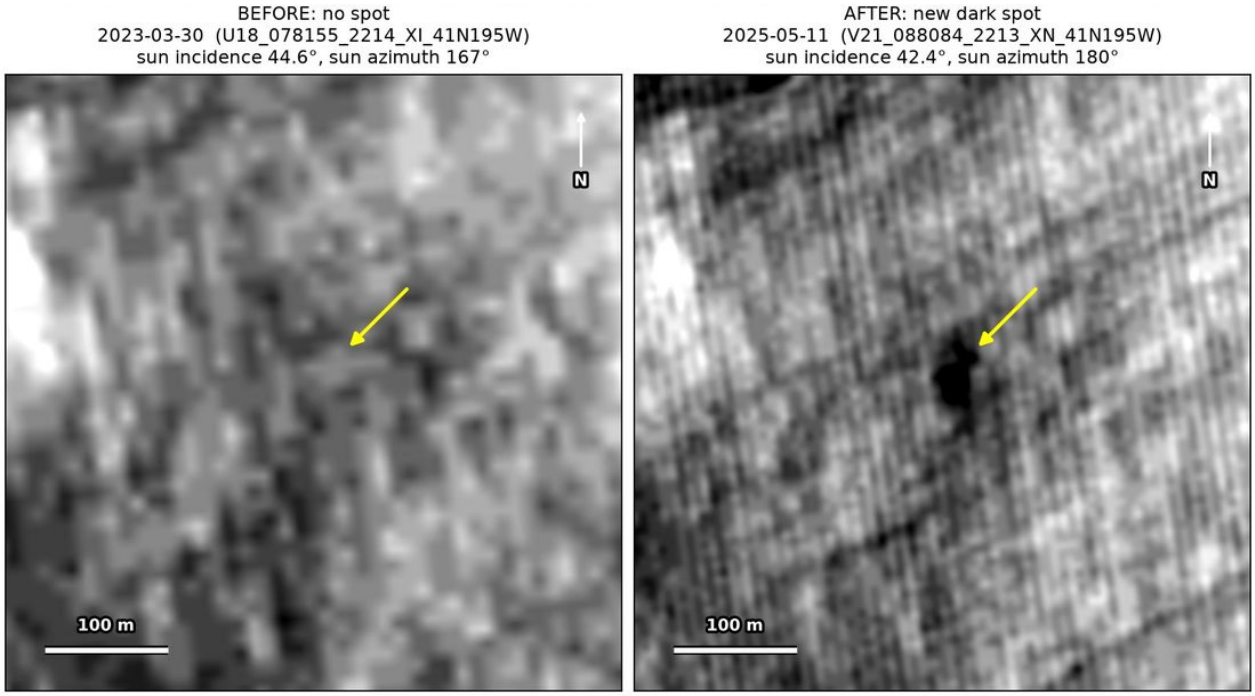

Figure 322: Candidate 134, before and after: the same 500 × 500 m of ground in 30 March 2023 (`U18_078155_2214_XI_41N195W`) and 11 May 2025 (`V21_088084_2213_XN_41N195W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

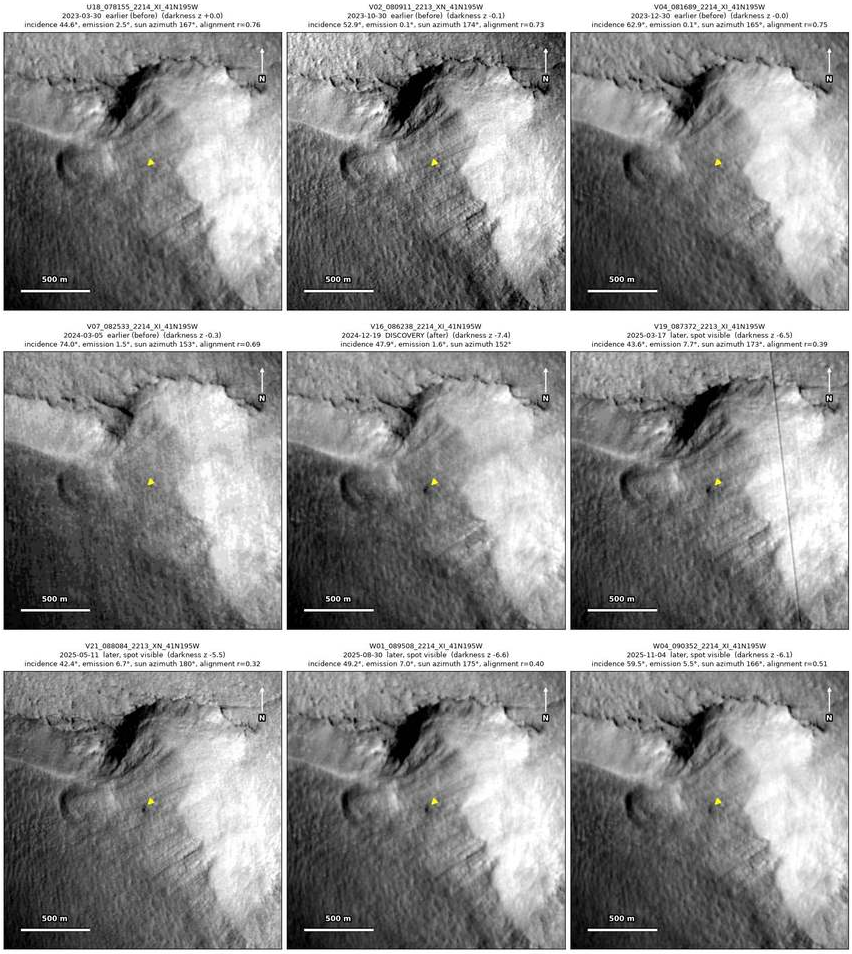

Figure 323: Candidate 134: nine CTX images of the same 2×2 km patch, 2023 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 135: Plains in western Arabia Terra, 328 km south-east of Gill crater

Location	12.4959° N, 10.1536° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	6 October 2014 and 17 January 2024
Images	2 without the spot (absent: 3 January 2011 and 6 October 2014); 3 with it (present: 17 January 2024, 29 March 2024 and 20 October 2025)
Size	Compact dark spot about 50 m across, with two dark cores about 35 m apart (94 × 52 m in the March 2024 image)
HiRISE	none covers the spot (nearest footprint edge 23.3 km)
Nearest catalogued new impact	36 km (Daubar et al., 2022)

What changed. A compact dark spot about 50 m across, with two dark cores, appeared on smooth plains among small knobs in western Arabia Terra between October 2014 and January 2024, and was still there in March 2024 and October 2025. Three CTX images from separate orbits show it.

Why it may be a new impact. Two earlier images, from 2011 and 2014, show plain ground there, the second under almost the same sun as the March 2024 image. A dark mark 300 m away shows in the 2014 image, so that image would have shown a new spot too, and apart from artefacts at the edges of the view, the change between the aligned images is greatest at the spot. A compact dark spot with sharp cores appearing on plain ground fits the blast zone of a small new impact.

Caveats. The images are strongly compressed, and three of them were aimed at a sharp boundary between brighter and darker ground in the region, where the wind moves dust around, although no such boundary is seen within 6 km of the spot. These two points are why it is rated about 80% rather than 90%. No image from the high-resolution HiRISE camera covers it; the nearest ends 23.3 km away.

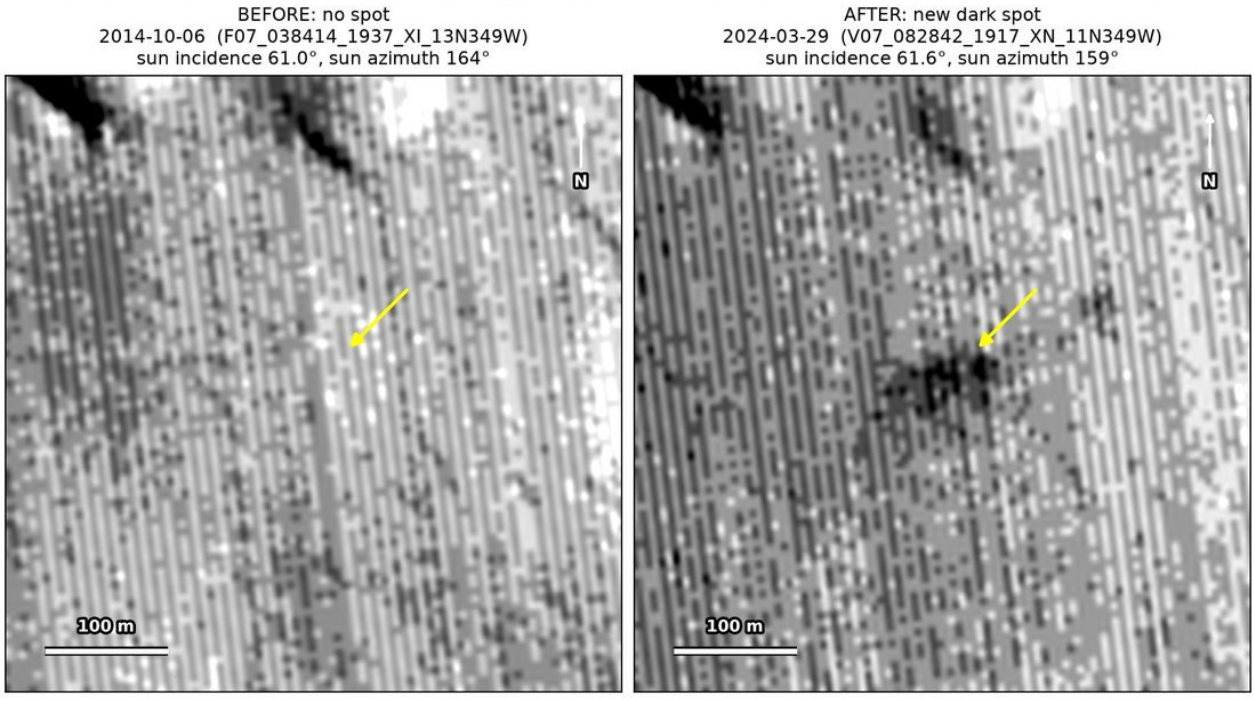

Figure 324: Candidate 135, before and after: the same 500 × 500 m of ground in 6 October 2014 (`F07_038414_1937_XI_13N349W`) and 29 March 2024 (`V07_082842_1917_XN_11N349W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

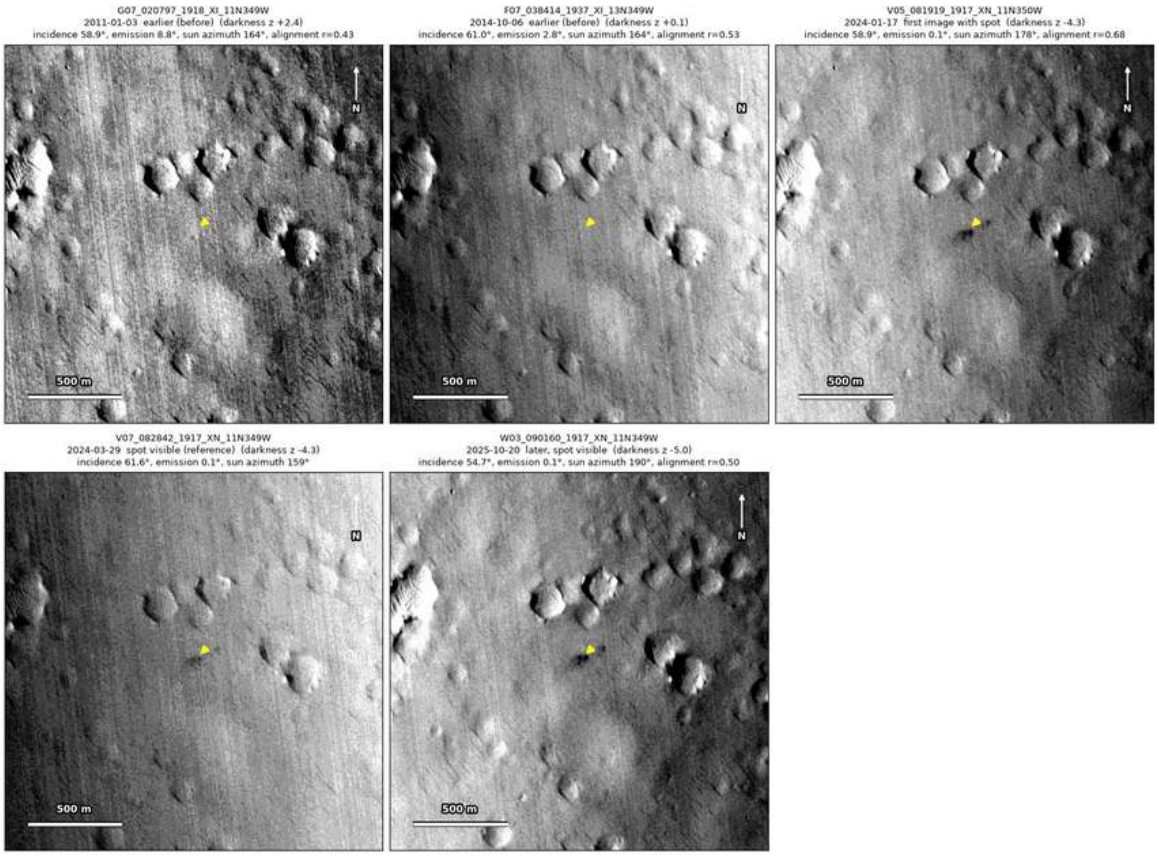

Figure 325: Candidate 135: five CTX images of the same 2×2 km patch, 2011 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 136: Plains in north-western Noctis Labyrinthus

Location	3.7982° S, 256.4596° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	20 January 2019 and 25 November 2020
Images	3 without the spot (absent: 12 June 2014, 12 August 2014 and 20 January 2019); 2 with it (present: 25 November 2020 and 29 December 2021)
Size	Dark blast zone about 170 m across (176 × 182 m in the December 2021 image, 190 × 200 m in the November 2020 one), with rays reaching about 500 m
HiRISE	none covers the spot (nearest footprint edge 15.7 km)
Nearest catalogued new impact	23 km (Daubar et al., 2022)

What changed. A dark blast zone about 170 m across, with dark rays, the longest reaching about 500 m to the west, appeared on smooth plains in the north-west of Noctis Labyrinthus between January 2019 and November 2020, and was still there in December 2021. Two CTX images from separate orbits show it.

Why it may be a new impact. Three earlier images, from 2014 and 2019, show plain ground there, one of them under almost the same sun as the December 2021 image. A small crater 240 m away shows in the 2014 images, so they would have shown a new spot too, and the change between the aligned images is about four times greater at the spot than anywhere else in the 2 × 2 km view. Dark rays radiating from one centre, with no tracks around them, are how the blast zone of a new impact looks.

Caveats. The plains are almost featureless, so in 2 km views the earlier images line up with the later ones only moderately well, although in 5 km views two craters 2 to 2.5 km away line up in all five images. This is why it is rated about 90% and not more. No image from the high-resolution HiRISE camera covers it; the nearest ends 15.7 km away.

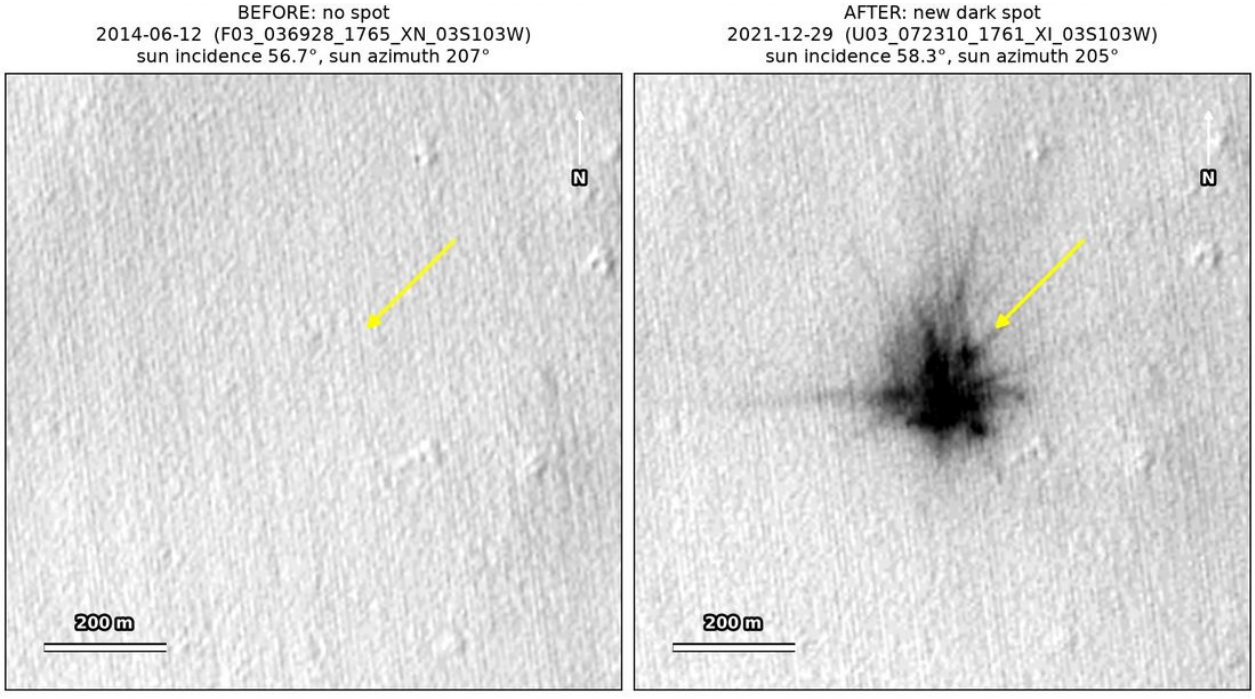

Figure 326: Candidate 136, before and after: the same 1000 × 1000 m of ground in 12 June 2014 (`F03_036928_1765_XN_03S103W`) and 29 December 2021 (`U03_072310_1761_XI_03S103W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

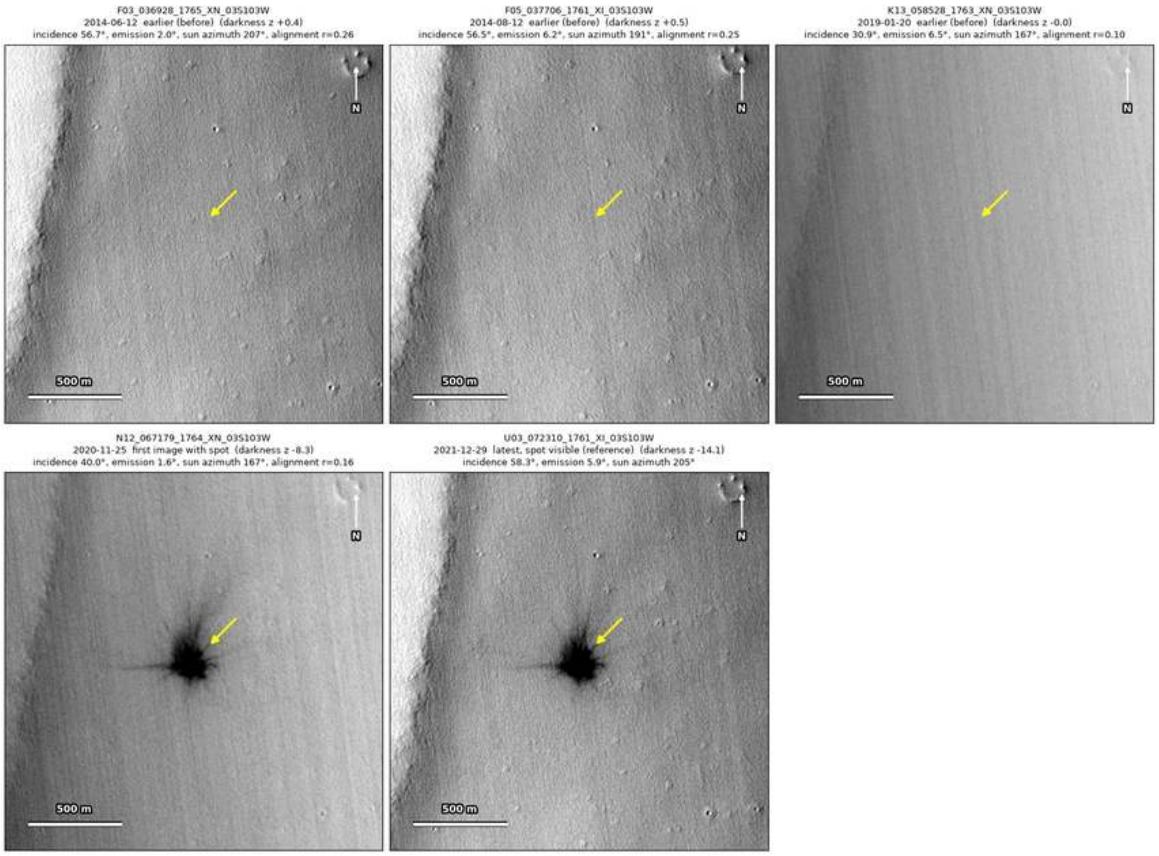

Figure 327: Candidate 136: five CTX images of the same 2×2 km patch, 2014 to 2021. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 137: Plains in western Arcadia Planitia

Location	45.7065° N, 177.1405° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	13 March 2022 and 10 February 2023
Images	4 without the spot (absent: 27 April 2021, 31 August 2021, 11 November 2021 and 13 March 2022); 8 with it (present: 10 February 2023, 16 May 2023, 21 July 2023, 24 September 2023, 22 December 2023, 18 November 2024, 25 March 2025 and 8 September 2025)
Size	Compact dark spot about 30 m across (28 × 28 m in the September 2023 image, at 12 m per pixel), fading
HiRISE	none covers the spot (nearest footprint edge 12.7 km)
Nearest catalogued new impact	40 km (Daubar et al., 2022)

What changed. A compact dark spot about 30 m across appeared on smooth plains in western Arcadia Planitia between March 2022 and February 2023. It is seen in eight CTX images from separate orbits up to September 2025, fading slowly over those two and a half years.

Why it may be a new impact. Four earlier images from 2021 and 2022 show plain ground there, one of them under almost the same sun as the September 2023 image. A dark mark 297 m away shows in three of them, so they would have shown a new spot too, and a compact dark spot that keeps its shape for two and a half years fits the blast zone of a small new impact.

Caveats. The area is a CTX impact-monitoring site, so the camera team has probably already seen the spot. Most of the images were taken at a coarse 12 m per pixel, which blurs it, and it is faint in the 2025 images; these two points are why it is rated about 90% and not more. No image from the high-resolution HiRISE camera covers it; the nearest ends 12.7 km away.

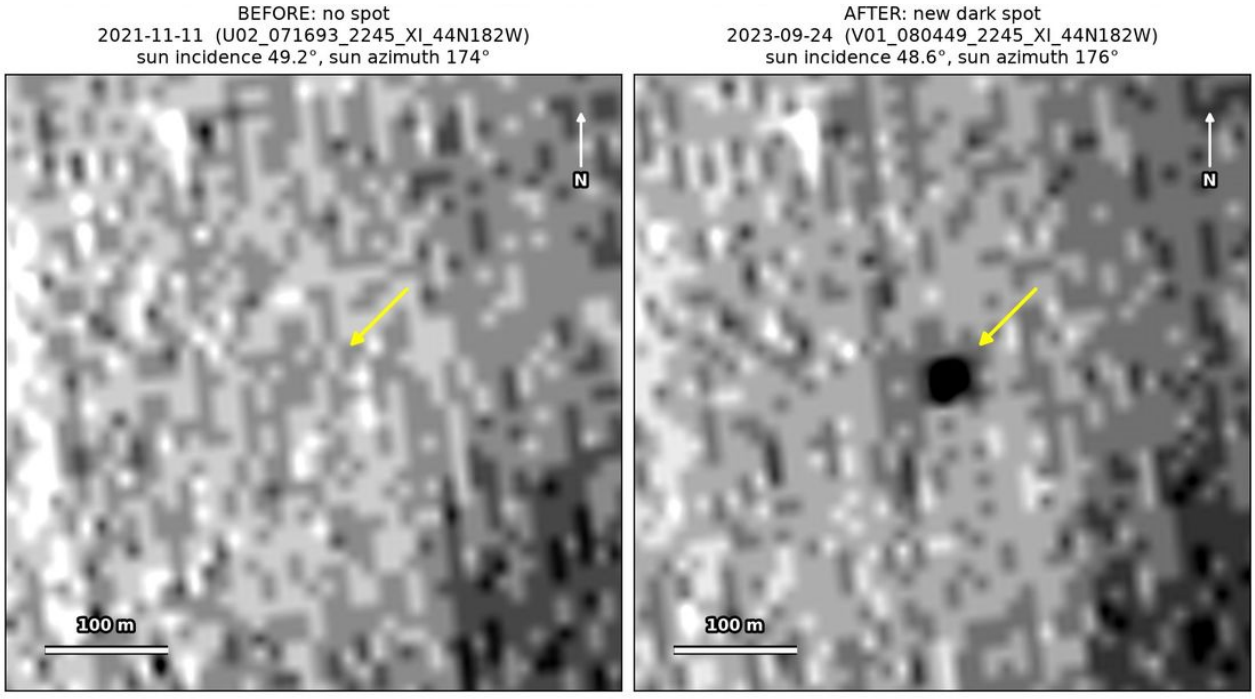

Figure 328: Candidate 137, before and after: the same 500 × 500 m of ground in 11 November 2021 (`U02_071693_2245_XI_44N182W`) and 24 September 2023 (`V01_080449_2245_XI_44N182W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

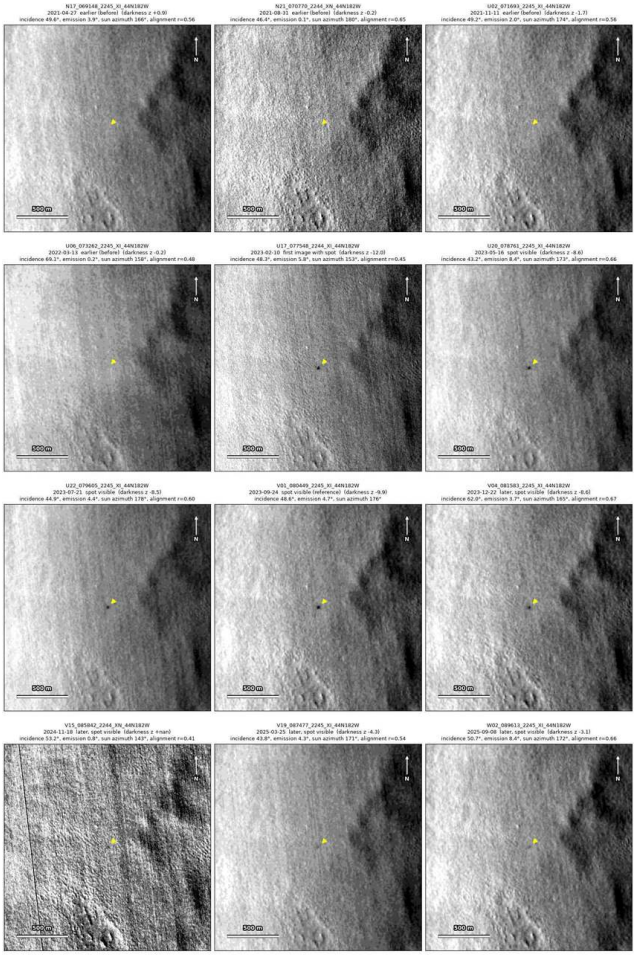

Figure 329: Candidate 137: twelve CTX images of the same 2×2 km patch, 2021 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 138: Lava plains between Pavonis Mons and Arsia Mons

Location	2.4732° S, 243.1012° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 70%
Formed between	13 December 2020 and 10 August 2022
Images	5 without the spot (absent: 19 July 2009, 17 February 2012, 10 October 2020, 6 November 2020 and 13 December 2020); 1 with it (present: 10 August 2022)
Size	Compact dark spot about 50 m across (60 × 42 m in the August 2022 image), with a faint smudge about 80 m long to the north-west
HiRISE	none covers the spot (nearest footprint edge 36.8 km)
Nearest catalogued new impact	182 km (Daubar et al., 2022)

What changed. Between December 2020 and August 2022, a compact dark spot about 50 m across, with a fainter dark smudge on its north-west side, appeared on lava plains between Pavonis Mons and Arsia Mons. So far only one CTX image shows it.

Why it may be a new impact. Five earlier images from 2009 to 2020 show plain ground there, one of them, from October 2020, under almost the same sun as the August 2022 image. A dark mark 269 m away shows in all five, so they would have shown a new spot too. A compact dark core with a fainter smudge on one side is the usual look of a small new impact on dusty ground.

Caveats. Only one image shows it, so it is rated promising rather than strong, and an image defect cannot be ruled out with one image. The shadow of a passing dust devil is unlikely, as the spot is only 7% darker than its surroundings, not black, and no dust column or track is in sight. No image from the high-resolution HiRISE camera covers it; the nearest ends 36.8 km away.

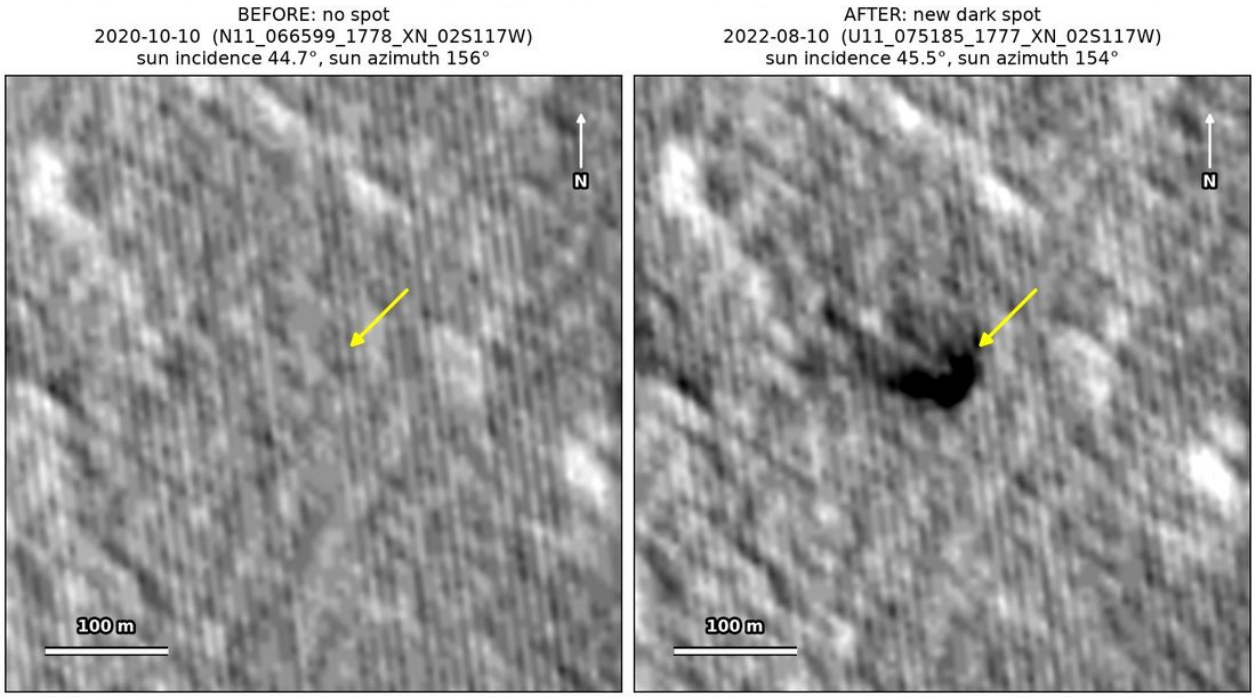

Figure 330: Candidate 138, before and after: the same 500 × 500 m of ground in 10 October 2020 (`N11_066599_1778_XN_02S117W`) and 10 August 2022 (`U11_075185_1777_XN_02S117W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

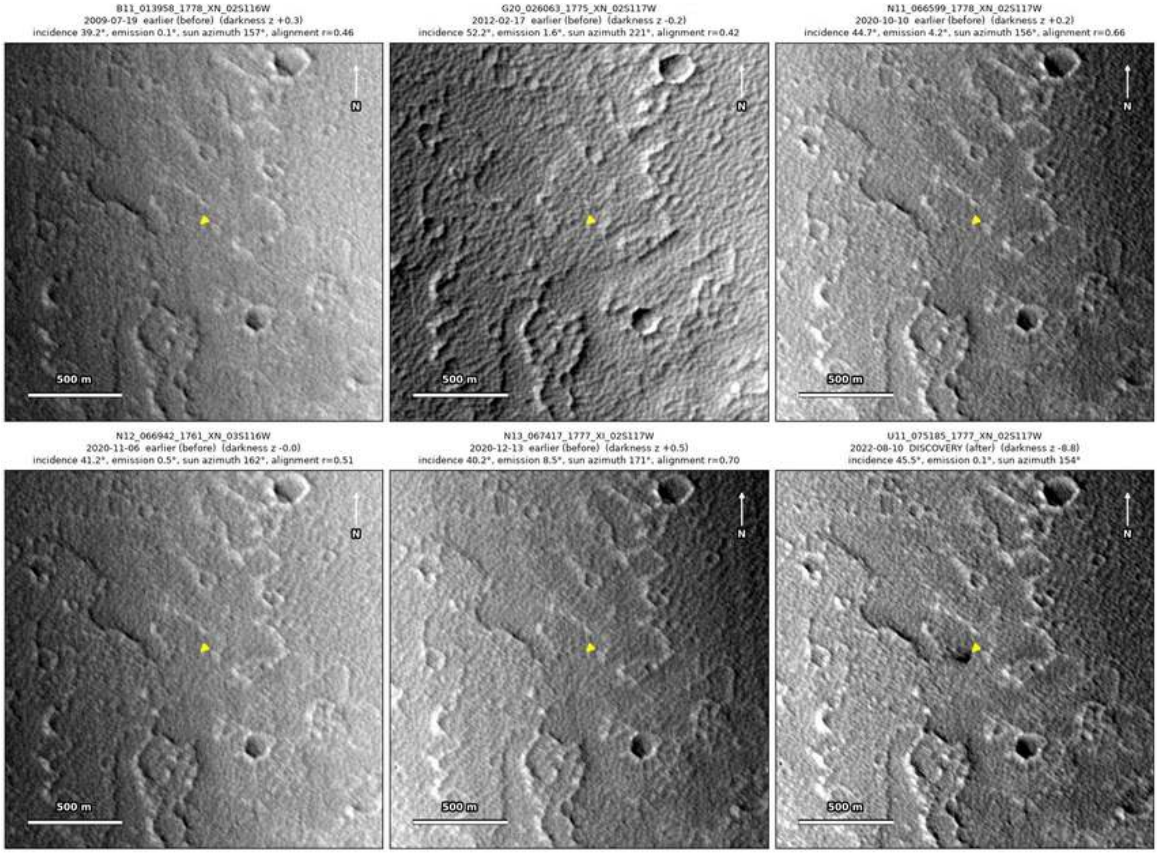

Figure 331: Candidate 138: six CTX images of the same 2×2 km patch, 2009 to 2022. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 139: Plains south-east of Aesacus Dorsum, west of Phlegra Montes

Location	32.9770° N, 158.3317° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 50%
Formed between	20 September 2020 and 5 March 2025
Images	3 without the spot (absent: 31 December 2009, 1 September 2014 and 20 September 2020); 1 with it (present: 5 March 2025)
Size	Chain of five dark patches over about 610 m from west to east; the largest is 82 × 142 m in the March 2025 image, with a darker core about 40 m across
HiRISE	none covers the spot (nearest footprint edge 39.2 km)
Nearest catalogued new impact	242 km (Daubar et al., 2022)

What changed. A chain of five dark patches about 610 m long from west to east, the largest with a darker core about 40 m across, appeared on plains west of Phlegra Montes between September 2020 and March 2025. So far only one CTX image shows it.

Why it may be a new impact. Three earlier images from 2009 to 2020 show plain ground there, one of them, from December 2009, under almost the same sun as the March 2025 image. A dark mark 236 m away shows in all three, so they would have shown new patches too, and a short chain of dark patches with a darker core is the usual look of an impactor that broke up in the air.

Caveats. Only one image shows it, so it is rated promising rather than strong, and an image defect cannot be ruled out with one image. The largest patch is only 3% darker than its surroundings, and it lies near a regional dark band where the wind may also have stripped dust. The latest image without it, from September 2020, is hazy and was taken under a low sun. No image from the high-resolution HiRISE camera covers it; the nearest ends 39.2 km away.

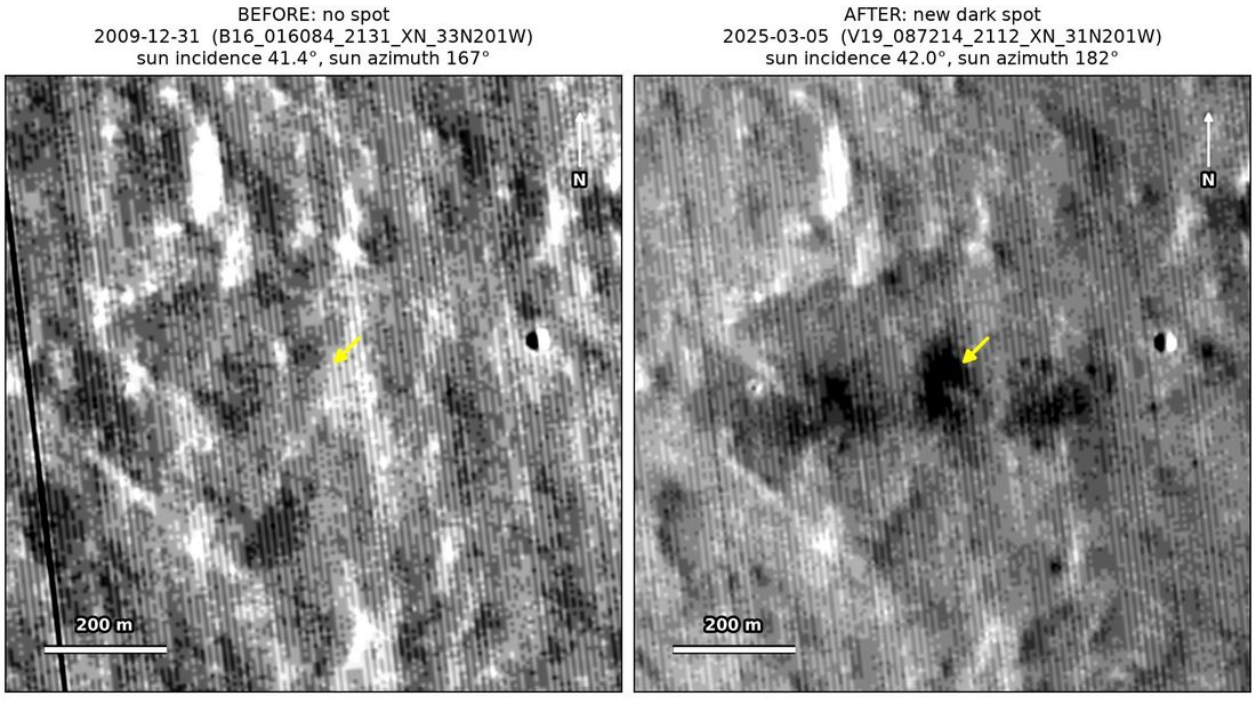

Figure 332: Candidate 139, before and after: the same 1000 × 1000 m of ground in 31 December 2009 (`B16_016084_2131_XN_33N201W`) and 5 March 2025 (`V19_087214_2112_XN_31N201W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

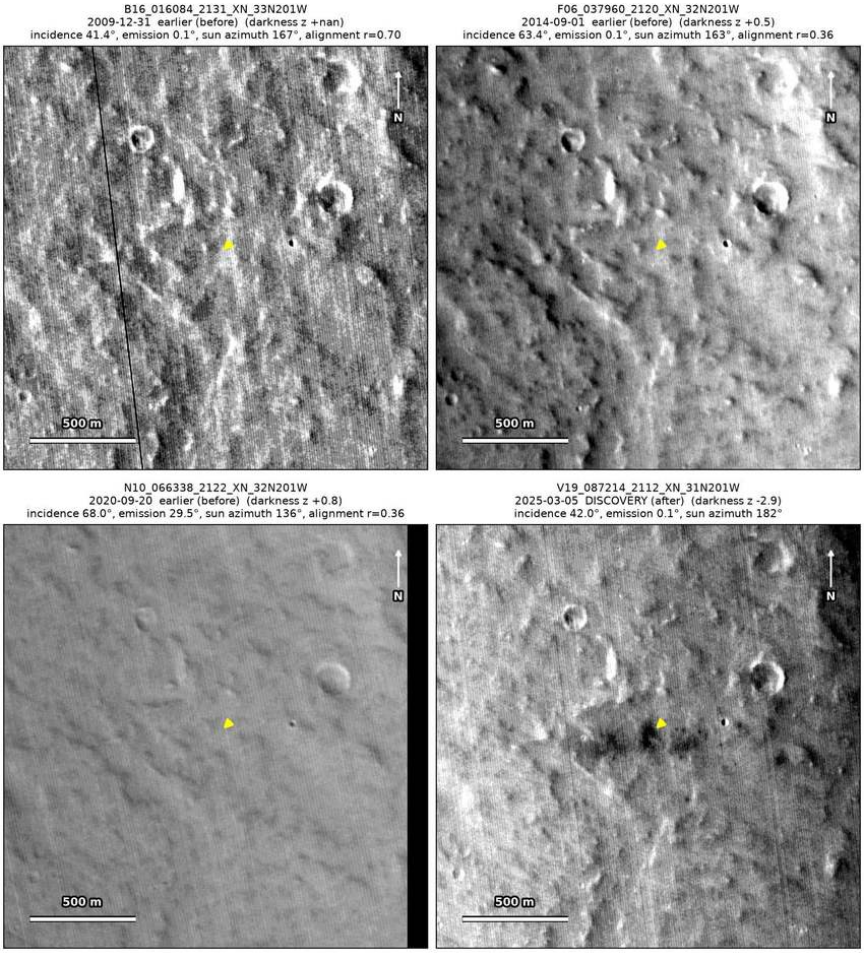

Figure 333: Candidate 139: four CTX images of the same 2×2 km patch, 2009 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 140: South-east flank of Alba Mons, south-west of Tantalus Fluctus

Location	32.5821° N, 255.8855° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	29 October 2022 and 17 April 2023
Images	4 without the spot (absent: 12 February 2008, 1 December 2020, 24 October 2022 and 29 October 2022); 2 with it (present: 17 April 2023 and 17 June 2023)
Size	Compact dark spot about 45 m across (46 × 50 m in the June 2023 image), plus a new companion spot of 42 × 26 m, 175 m to the east-south-east
HiRISE	none covers the spot (nearest footprint edge 23.2 km)
Nearest catalogued new impact	265 km (Daubar et al., 2022)

What changed. A compact dark spot about 45 m across, with a smaller dark spot 175 m to its east-south-east, appeared on rough plains on the south-east flank of Alba Mons between October 2022 and April 2023, and was still there in June 2023. Two CTX images from separate orbits show it.

Why it may be a new impact. Four earlier images from 2008 to 2022 show plain ground there, one of them under almost the same sun as the April 2023 image. A dark mark 291 m away shows in the 2008 image, so that image would have shown new spots too, and the change between the aligned before and after images is greatest at the two spots. A compact pair of new dark spots is the usual look of a small impactor that broke up in the air.

Caveats. The three earlier images closest in time have a lower sun, under which the dark marks nearby look fainter, so it is less certain that those images would have shown the spots. The only later image, from March 2024, is too poorly aligned and too low in contrast to show whether the spots were still there. No image from the high-resolution HiRISE camera covers it; the nearest ends 23.2 km away.

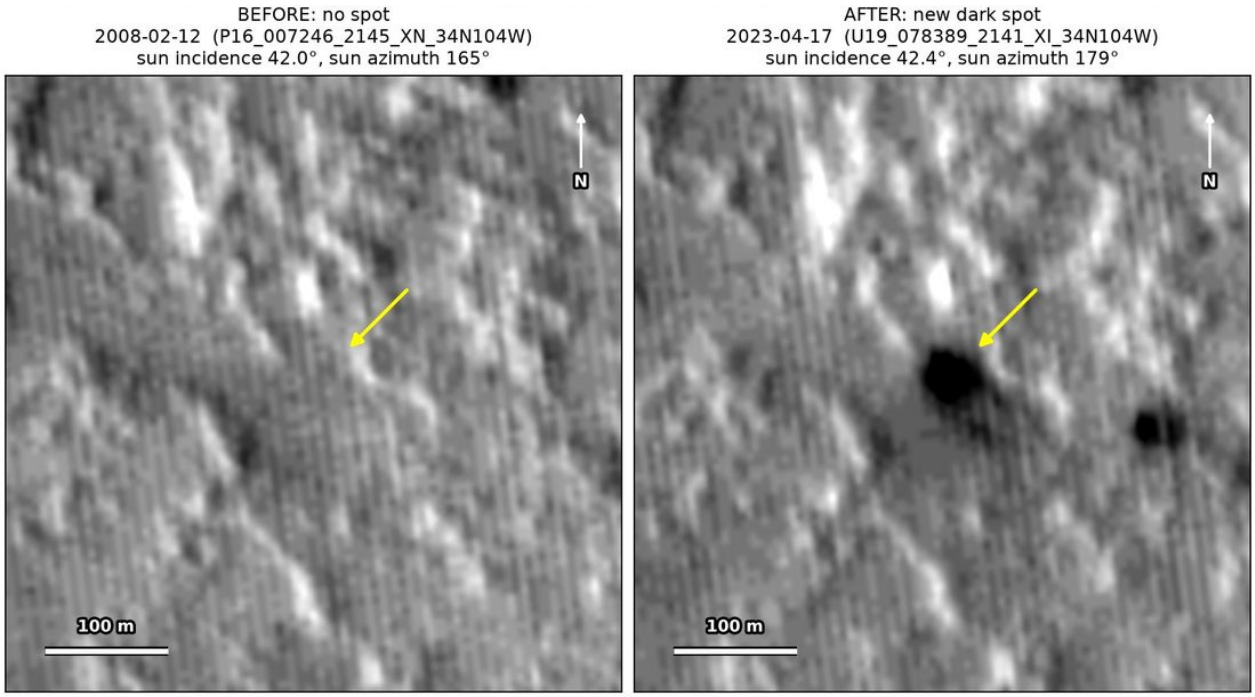

Figure 334: Candidate 140, before and after: the same 500 × 500 m of ground in 12 February 2008 (`P16_007246_2145_XN_34N104W`) and 17 April 2023 (`U19_078389_2141_XI_34N104W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

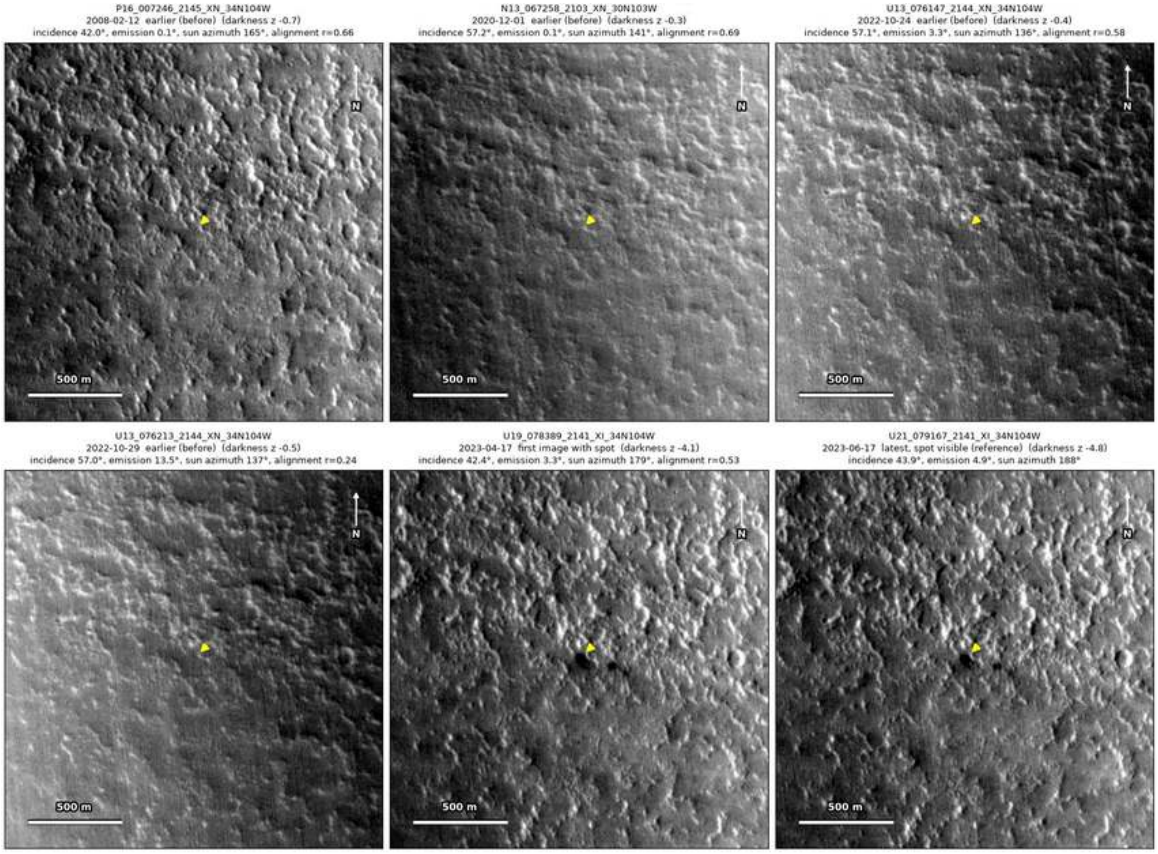

Figure 335: Candidate 140: six CTX images of the same 2×2 km patch, 2008 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 141: Plains in southern Arabia Terra, 249 km west of Tuscaloosa crater

Location	0.1494° N, 24.5323° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	15 March 2022 and 9 January 2024
Images	3 without the spot (absent: 27 October 2008, 23 May 2020 and 15 March 2022); 2 with it (present: 9 January 2024 and 26 March 2024)
Size	Compact dark spot about 30 m across (30 × 26 m in the January 2024 image, 36 × 42 m in the March 2024 one)
HiRISE	none covers the spot (nearest footprint edge 42.8 km)
Nearest catalogued new impact	55 km (Daubar et al., 2022)

What changed. A compact dark spot about 30 m across appeared on plains in southern Arabia Terra between March 2022 and January 2024, and was still there in March 2024. Two CTX images from separate orbits show it.

Why it may be a new impact. Three earlier images from 2008 to 2022 show plain ground there, one of them under almost the same sun as the January 2024 image. A dark mark 270 m away shows in all of them, so they would have shown a new spot too, and a compact new dark spot is how the blast zone of a small new crater looks.

Caveats. The March 2024 image lines up only moderately well with the others, and there is no image after March 2024; this is why it is rated about 90% and not more. No image from the high-resolution HiRISE camera covers it; the nearest ends 42.8 km away.

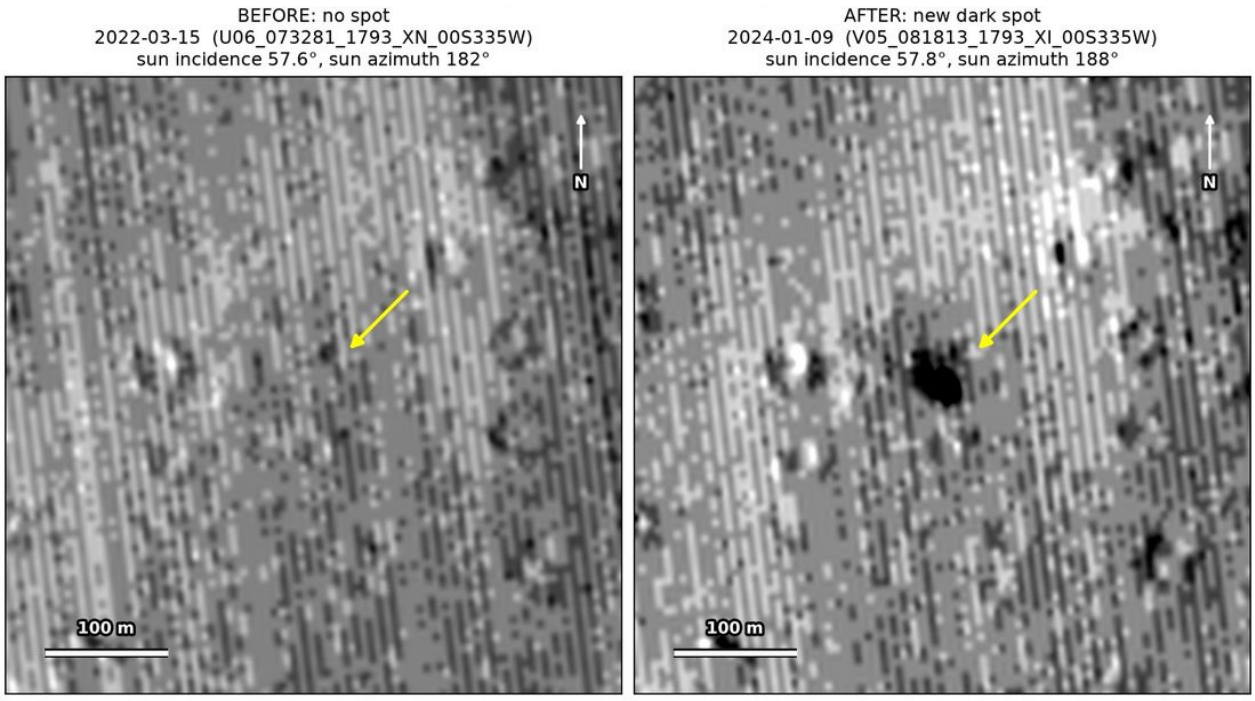

Figure 336: Candidate 141, before and after: the same 500 × 500 m of ground in 15 March 2022 (`U06_073281_1793_XN_00S335W`) and 9 January 2024 (`V05_081813_1793_XI_00S335W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

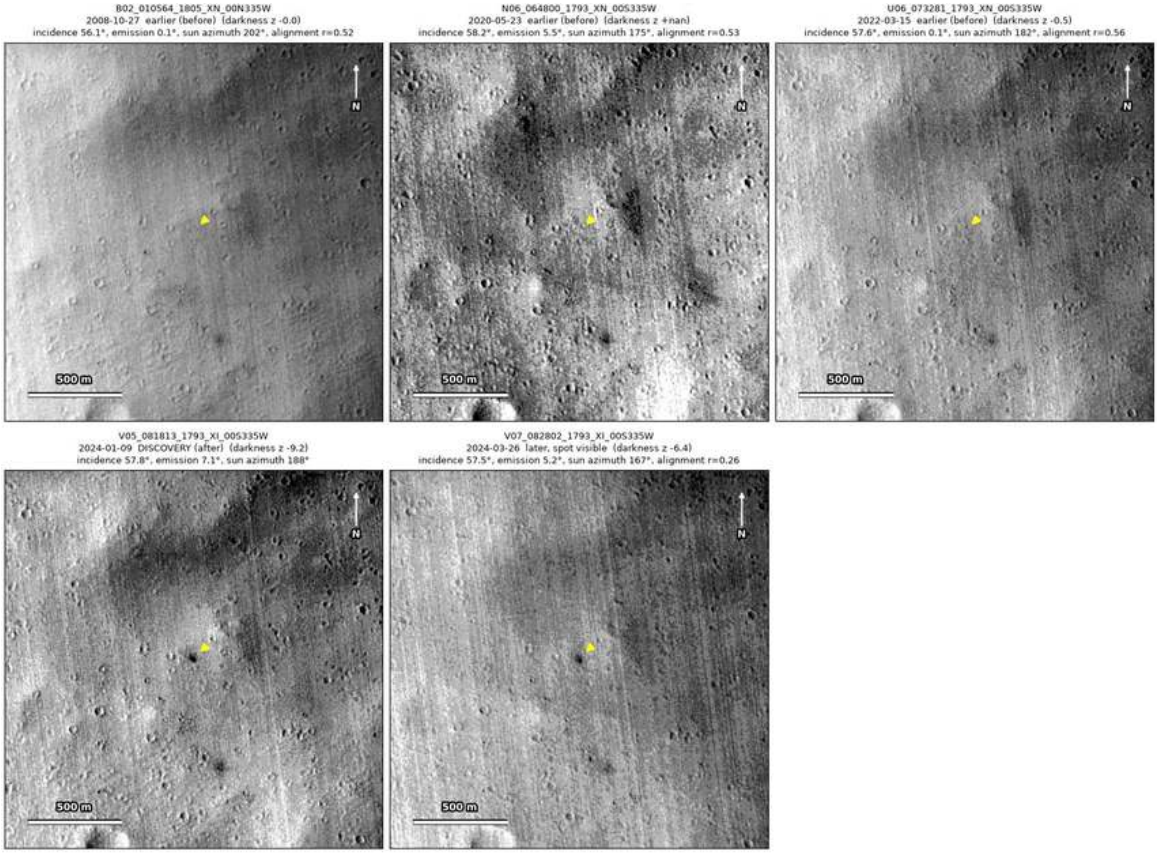

Figure 337: Candidate 141: five CTX images of the same 2×2 km patch, 2008 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 142: Plains in Galaxias Fossae, 20 km north-west of Krasnoye crater

Location	36.0228° N, 143.4838° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	29 January 2023 and 1 September 2024
Images	7 without the spot (absent: 21 January 2007, 10 July 2016, 4 November 2021, 24 December 2021, 18 February 2022, 19 August 2022 and 29 January 2023); 2 with it (present: 1 September 2024 and 2 March 2025)
Size	Compact dark spot about 50 m across (94 × 46 m with its tail to the east in the September 2024 image, 38 × 44 m in the March 2025 one), plus a smaller new spot of 38 × 34 m, 140 m to the south-east
HiRISE	none covers the spot (nearest footprint edge 13.7 km)
Nearest catalogued new impact	171 km (Daubar et al., 2022)

What changed. A compact dark spot about 50 m across, with a smaller one 140 m to its south-east, appeared on plains in Galaxias Fossae, 20 km north-west of Krasnoye crater, between January 2023 and September 2024, and was still there in March 2025. Two CTX images from separate orbits show it.

Why it may be a new impact. Seven earlier images from 2007 to 2023 show plain ground there, while the shaded inner wall of a small crater 102 m away shows in the six of them that can be checked, so they would have shown a new spot too. A compact pair of new dark spots is the usual look of a small impactor that broke up in the air.

Caveats. The site lies at the edge of the March 2025 image, which lines up only moderately well. The main spot lies about 90 m outside the rim of a small crater, but a small dust avalanche off that rim is unlikely, as the spots are blobs on nearly flat ground, not fans. Unless broad brightness differences are removed, the change between the aligned images is slightly greater near a shaded scarp 750 to 1,100 m away than at the spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 13.7 km away.

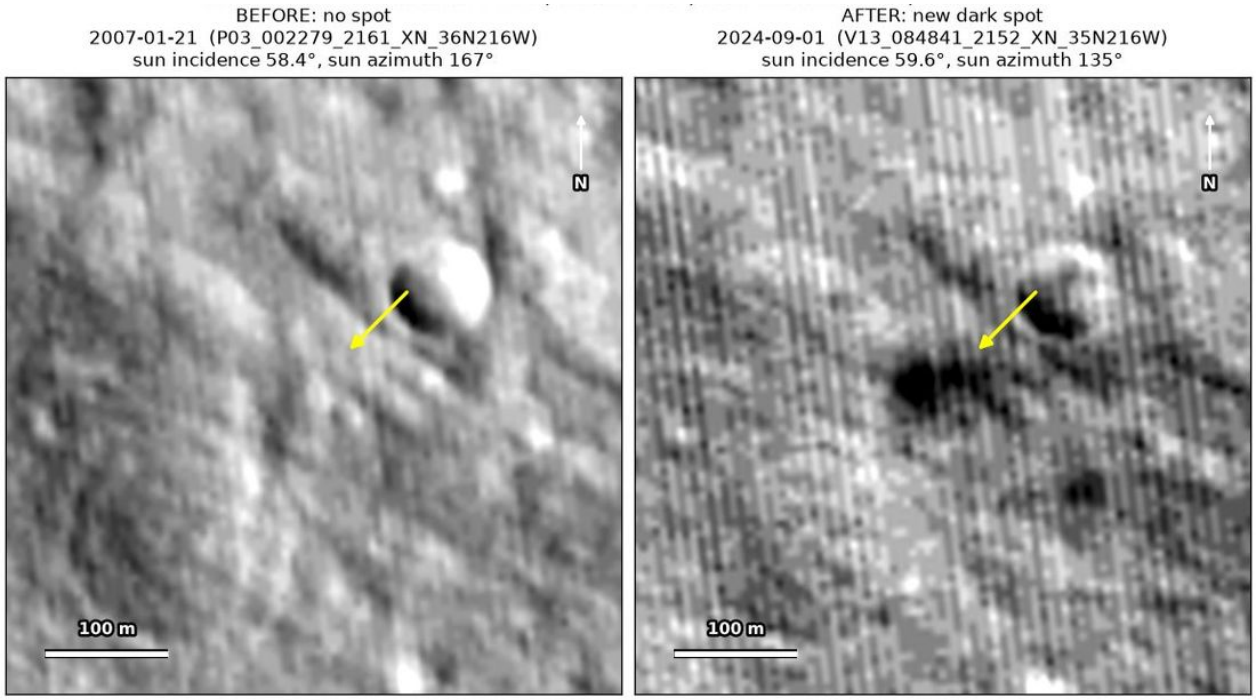

Figure 338: Candidate 142, before and after: the same 500 × 500 m of ground in 21 January 2007 (`P03_002279_2161_XN_36N216W`) and 1 September 2024 (`V13_084841_2152_XN_35N216W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

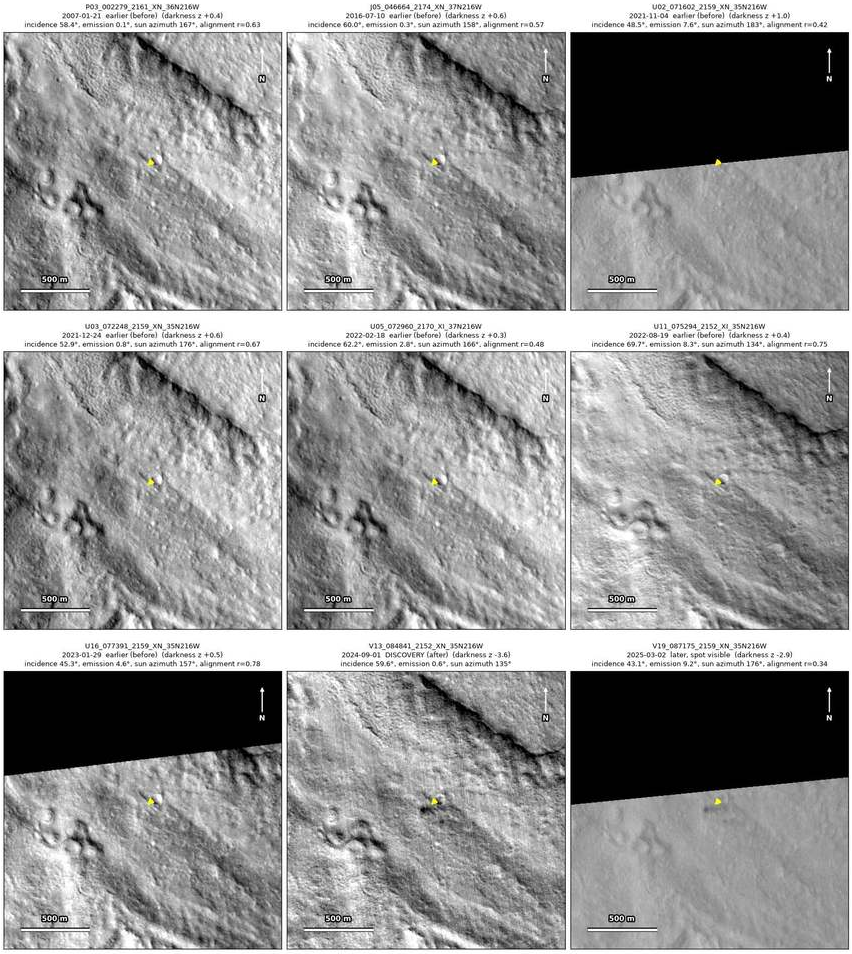

Figure 339: Candidate 142: nine CTX images of the same 2×2 km patch, 2007 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 143: Plains in Jovis Fossae, 163 km north-east of Jovis Tholus

Location	19.7189° N, 245.0063° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	21 June 2016 and 31 December 2022
Images	2 without the spot (absent: 13 September 2008 and 21 June 2016); 1 with it (present: 31 December 2022)
Size	Compact dark spot about 35 m across (36 × 40 m in the December 2022 image), with a fainter tail reaching about 75 m to the south-east
HiRISE	none covers the spot (nearest footprint edge 38.0 km)
Nearest catalogued new impact	43 km (Daubar et al., 2022)

What changed. A compact dark spot about 35 m across, with a fainter tail to the south-east, appeared on plains in Jovis Fossae, 700 m from the rim of a crater 1.1 km across, between June 2016 and December 2022. So far only one CTX image shows it.

Why it may be a new impact. Both earlier images, from 2008 and 2016, show plain ground there. They were taken under a lower sun than the December 2022 image, and a lower sun shows relief more, not less. A dark mark 299 m away shows in both, so they would have shown a new spot too, and a compact dark core with a fainter tail on one side is the usual look of a small new impact on dusty ground.

Caveats. Only one image shows it, so it is rated promising rather than strong, and an image defect cannot be ruled out with one image. A new dust-devil track 2 km to the south shows that dust devils crossed the area in the same years, although no track reaches the spot. The lower sun of the earlier images and the strong compression of the December 2022 image are why it is rated only about 60%. No image from the high-resolution HiRISE camera covers it; the nearest ends 38.0 km away.

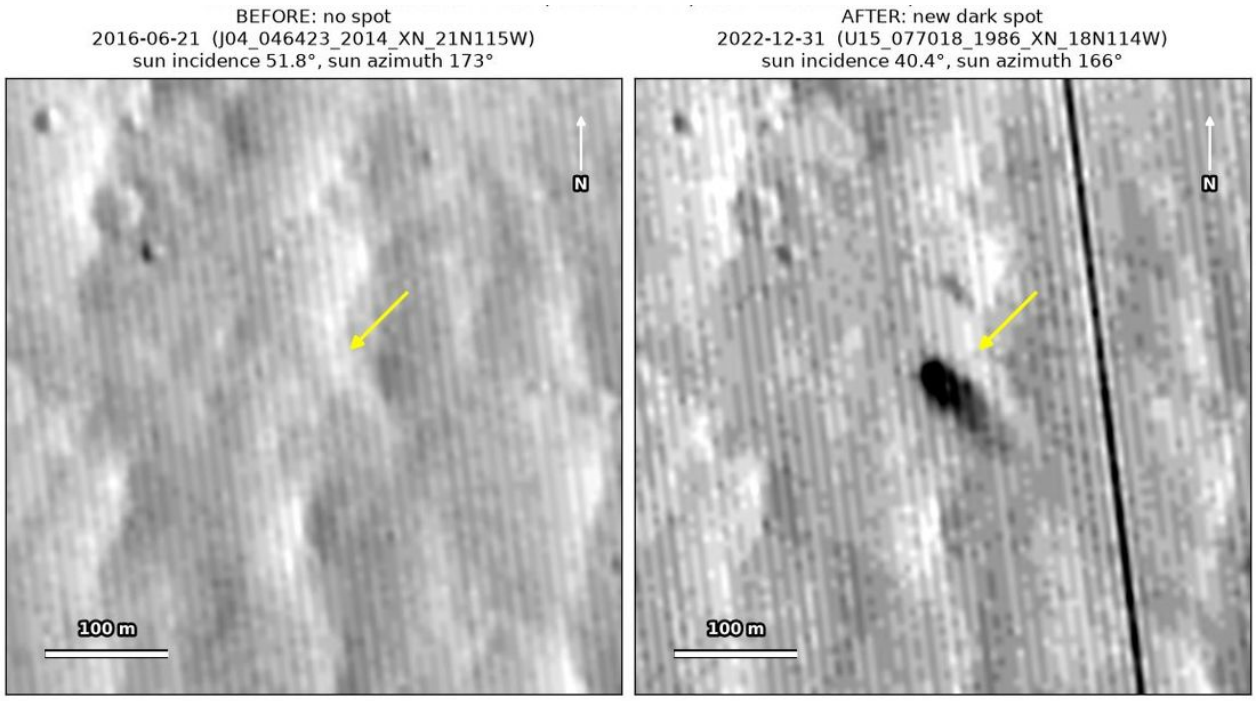

Figure 340: Candidate 143, before and after: the same 500 × 500 m of ground in 21 June 2016 (`J04_046423_2014_XN_21N115W`) and 31 December 2022 (`U15_077018_1986_XN_18N114W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

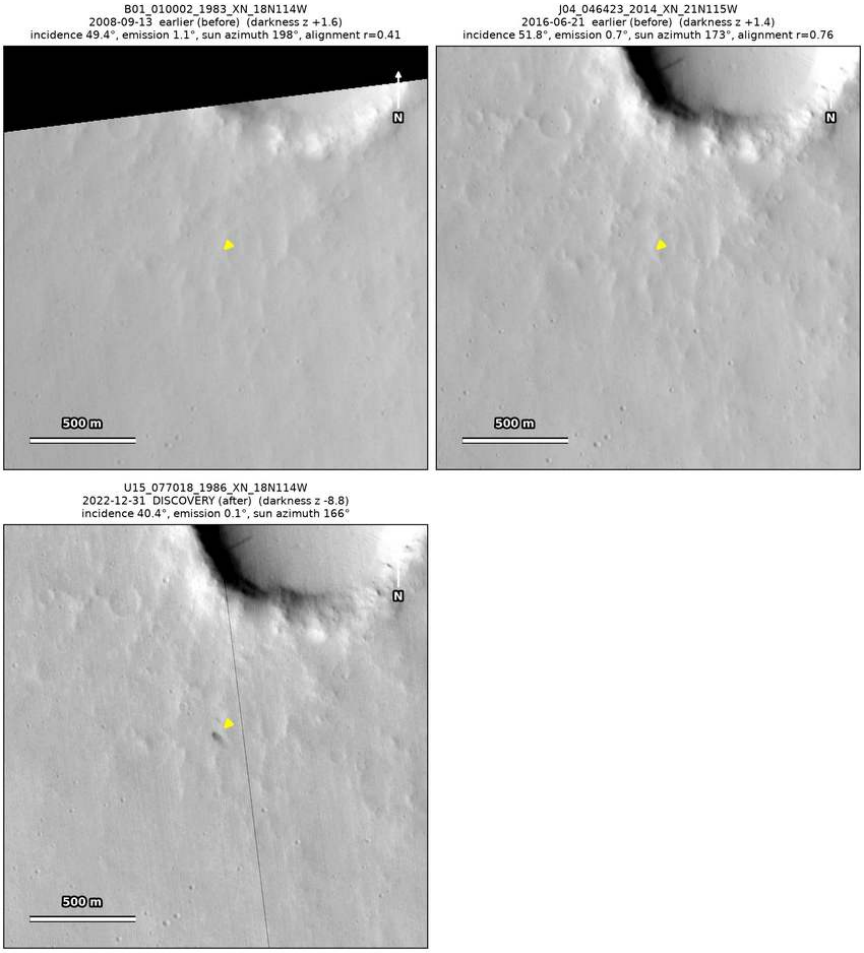

Figure 341: Candidate 143: three CTX images of the same 2×2 km patch, 2008 to 2022. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 144: Lineated terrain in Labeatis Fossae, 115 km north of Fesenkov crater

Location	23.5276° N, 274.0757° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 55%
Formed between	27 April 2018 and 21 December 2022
Images	2 without the spot (absent: 7 November 2007 and 27 April 2018); 1 with it (present: 21 December 2022)
Size	Compact dark spot about 35 m across (42 × 32 m in the December 2022 image), darkest at its west end
HiRISE	none covers the spot (nearest footprint edge 51.2 km)
Nearest catalogued new impact	60 km (Daubar et al., 2022)

What changed. A compact dark spot about 35 m across appeared on lineated terrain in Labeatis Fossae, 115 km north of Fesenkov crater, between April 2018 and December 2022. So far only one CTX image shows it.

Why it may be a new impact. An image from 2007, taken under almost the same sun as the December 2022 image, and one from 2018 show plain ground there. In the new image the spot holds the darkest pixels within 450 m, and once broad differences in brightness are removed, comparing the aligned images reveals only one change in the 2 × 2 km area, a weak one, at the spot.

Caveats. Only one image shows it, so it is rated promising rather than strong, and an image defect cannot be ruled out: that image is strongly compressed, with a narrow range of brightness values, though the dip at the spot is graded and rounded. A small dark dot at its eastern end was already there in 2018, and another old dark dot nearby stands out in the 2007 image but not the 2018 one, so the 2018 image might have missed a new spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 51.2 km away.

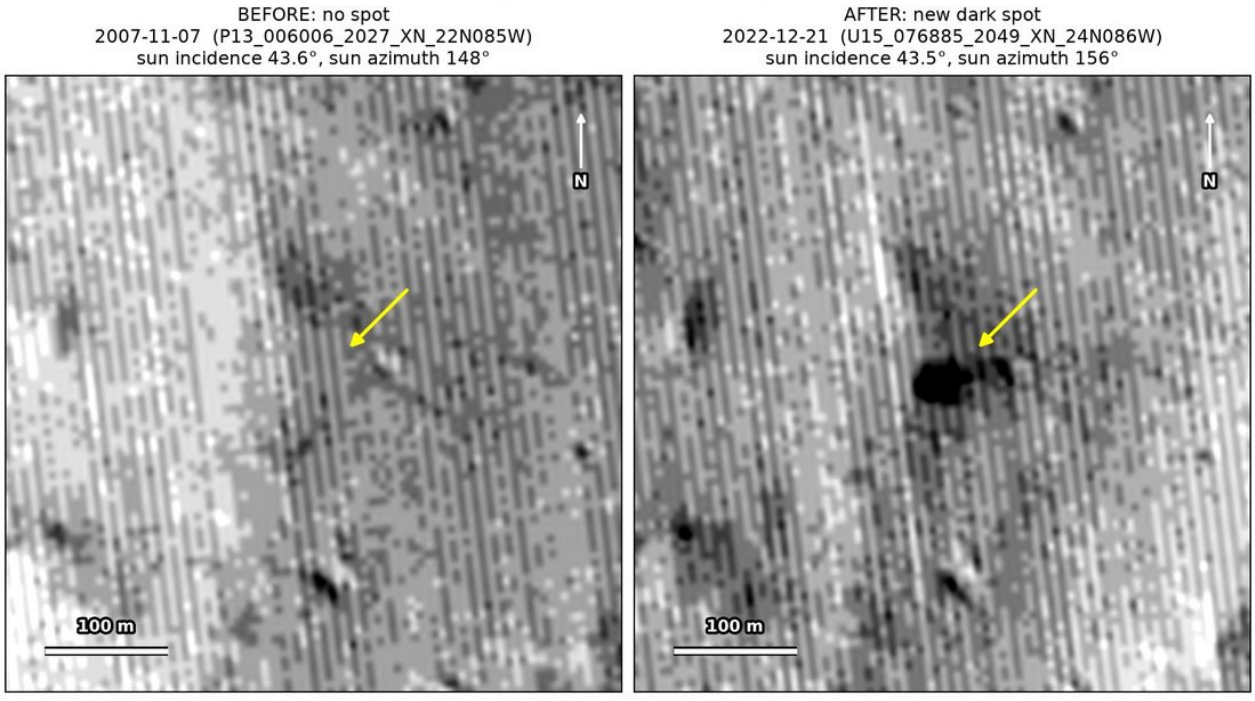

Figure 342: Candidate 144, before and after: the same 500 × 500 m of ground in 7 November 2007 (`P13_006006_2027_XN_22N085W`) and 21 December 2022 (`U15_076885_2049_XN_24N086W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

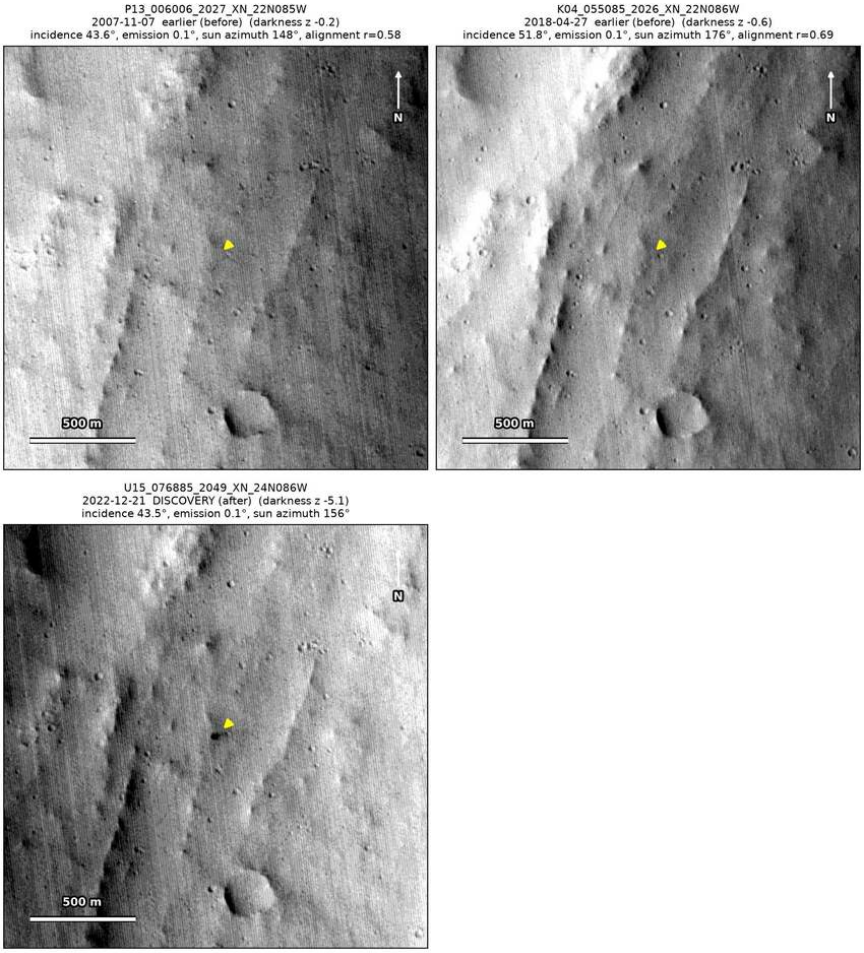

Figure 343: Candidate 144: three CTX images of the same 2×2 km patch, 2007 to 2022. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 145: Lava-flow lobe in Ulysses Fossae, 75 km north-east of candidate 82

Location	9.1389° N, 240.2842° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 45%
Formed between	6 August 2024 and 10 February 2025
Images	2 without the spot (absent: 28 November 2007 and 6 August 2024); 1 with it (present: 10 February 2025)
Size	Three compact dark spots within about 125 m: the largest about 30 m across (28 × 32 m), the others about 25 and 20 m
HiRISE	none covers the spot (nearest footprint edge 14.0 km)
Nearest catalogued new impact	20 km (Daubar et al., 2022)

What changed. Between August 2024 and February 2025, a cluster of three compact dark spots within about 125 m, the largest about 30 m across, appeared on lava plains in Ulysses Fossae, near the edge of a lava-flow lobe. So far only one CTX image shows it.

Why it may be a new impact. Both usable earlier images, from 2007 and 2024, show plain ground there; in the August 2024 one the sun comes from another direction than in the February 2025 image. A dark mark 211 m away shows in both, so they would have shown new spots too, and a tight cluster of compact new dark spots is the usual look of a small impactor that broke up in the air.

Caveats. Only one image shows it, so it is rated promising rather than strong. In that image the sun comes from a direction 35 to 55° away from every earlier image, which also darkens the shaded sides of knobs elsewhere in the area about as much, although the earlier images show no knob or slope where the cluster is. That image also shows broad dark bands, shadows of clouds or very broad dust-devil tracks, the nearest ending about 1 km away, and the earlier images line up only moderately well, the 2007 one with very low contrast. No image from the high-resolution HiRISE camera covers it; the nearest ends 14.0 km away.

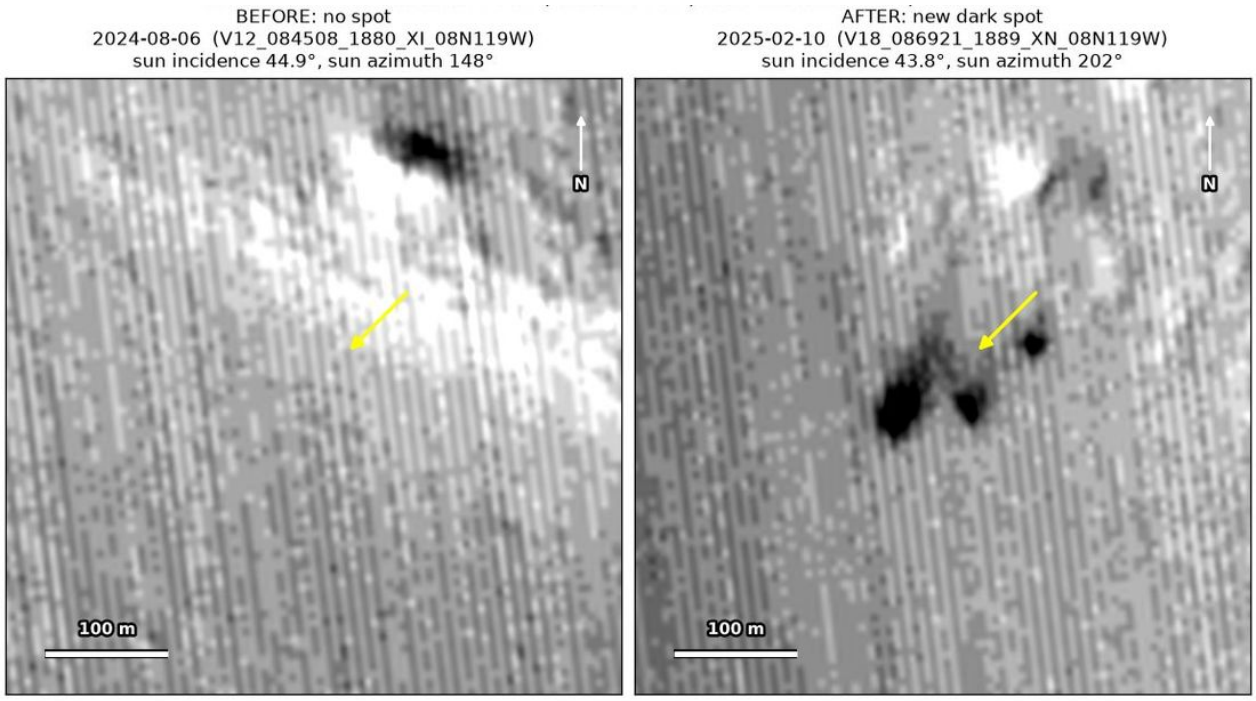

Figure 344: Candidate 145, before and after: the same 500 × 500 m of ground in 6 August 2024 (`V12_084508_1880_XI_08N119W`) and 10 February 2025 (`V18_086921_1889_XN_08N119W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

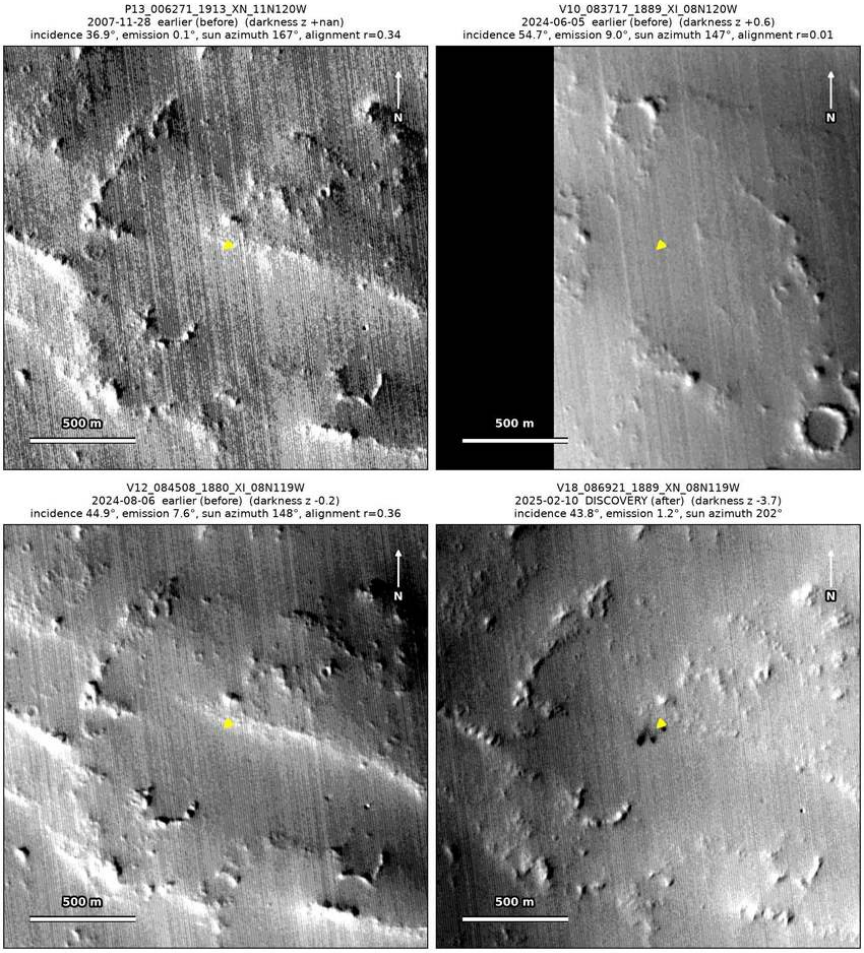

Figure 345: Candidate 145: four CTX images of the same 2×2 km patch, 2007 to 2025. The oblique, hazy June 2024 image, which could not be aligned, is shown but not used to date the spot. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 146: Plains in Arabia Terra, 285 km north-east of Janssen crater

Location	5.7363° N, 41.3369° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 50%
Formed between	30 April 2010 and 4 May 2023
Images	2 without the spot (absent: 17 February 2010 and 30 April 2010); 1 with it (present: 4 May 2023)
Size	Compact dark spot about 40 m across (48 × 40 m in the May 2023 image), with a faint tail reaching 60 to 90 m to the south-east
HiRISE	none covers the spot (nearest footprint edge 15.6 km)
Nearest catalogued new impact	95 km (Daubar et al., 2022)

What changed. A compact dark spot about 40 m across, with a faint tail to the south-east, appeared on plains in Arabia Terra, 285 km north-east of Janssen crater, between April 2010 and May 2023. So far only one CTX image shows it.

Why it may be a new impact. Both earlier images, from 2010, show plain ground there, one of them under almost the same sun as the May 2023 image. In the new image the spot holds the darkest pixels within 220 m, and the change between the images is greatest at the spot. A compact dark core with a fainter tail on one side is the usual look of a small new impact on dusty ground.

Caveats. Only one image shows it, so it is rated promising rather than strong. That image is strongly compressed, there is no dark mark nearby to show whether the earlier images would have caught a spot like it, and the earlier images line up only moderately well. No image from the high-resolution HiRISE camera covers it; the nearest ends 15.6 km away.

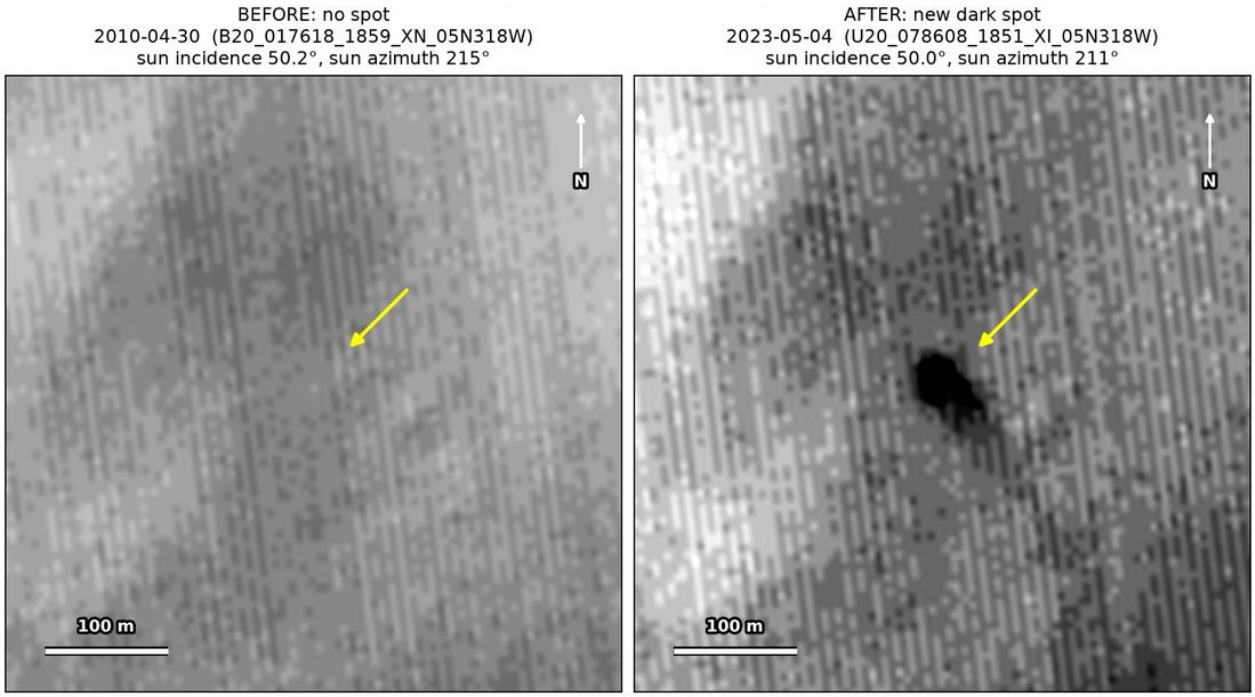

Figure 346: Candidate 146, before and after: the same 500 × 500 m of ground in 30 April 2010 (`B20_017618_1859_XN_05N318W`) and 4 May 2023 (`U20_078608_1851_XI_05N318W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

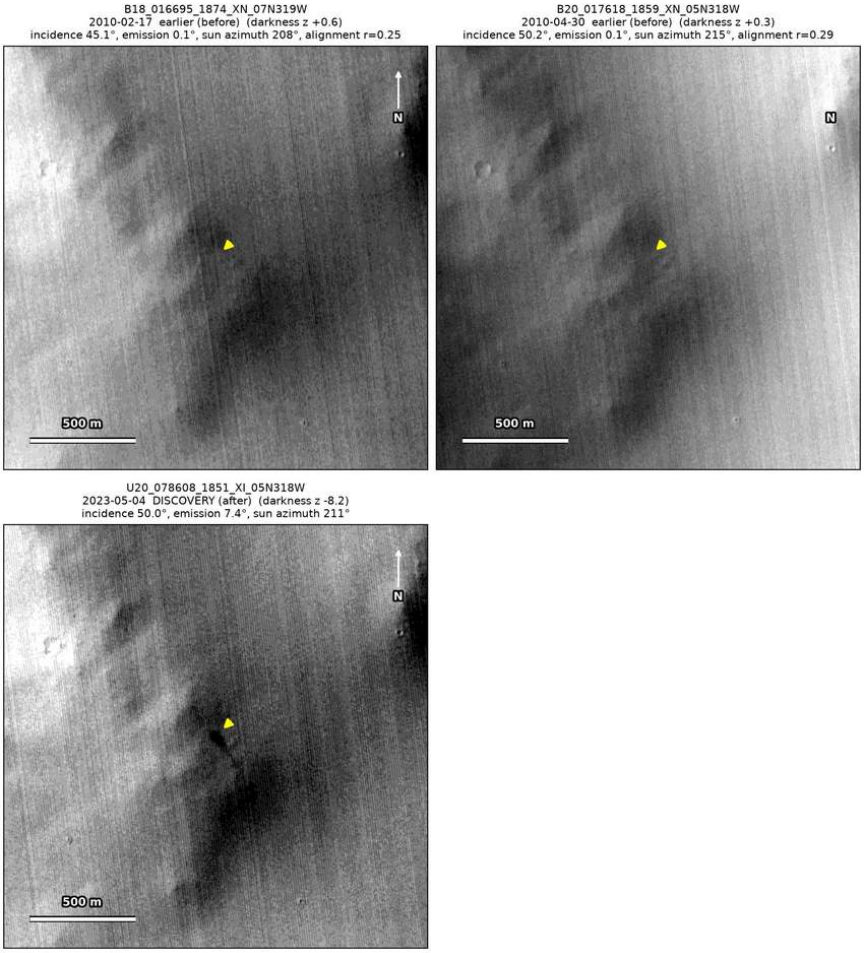

Figure 347: Candidate 146: three CTX images of the same 2×2 km patch, 2010 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 147: Wind-grooved terrain 186 km south-east of Amazonis Sulci

Location	4.5183° S, 218.3412° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	24 November 2018 and 9 September 2020
Images	4 without the spot (absent: 1 January 2009, 14 July 2009, 16 February 2015 and 24 November 2018); 3 with it (present: 9 September 2020, 30 December 2020 and 7 September 2022)
Size	Compact, irregular dark spot about 50 m across (64 × 62 m in the September 2022 image, 66 × 64 m in the first one, of September 2020)
HiRISE	none covers the spot (nearest footprint edge 7.8 km)
Nearest catalogued new impact	13 km (Daubar et al., 2022)

What changed. A compact dark spot about 50 m across appeared on wind-grooved terrain, 186 km south-east of Amazonis Sulci, between November 2018 and September 2020, and was still there in September 2022. Three CTX images from separate orbits show it.

Why it may be a new impact. Four earlier images from 2009 to 2018, one of them under almost the same sun as the 2022 image, show the same grooves there without the spot, while a small pit 1 km away shows in all seven images. A compact new dark spot that fades only a little over two years is how the blast zone of a small new crater looks.

Caveats. A dark patch nearby that was already there looks only moderately dark in the earlier images, so on its own it does not prove that they would have shown a new spot, although the earlier images line up well and the spot is very dark in the later ones. No image from the high-resolution HiRISE camera covers it; the nearest ends 7.8 km away.

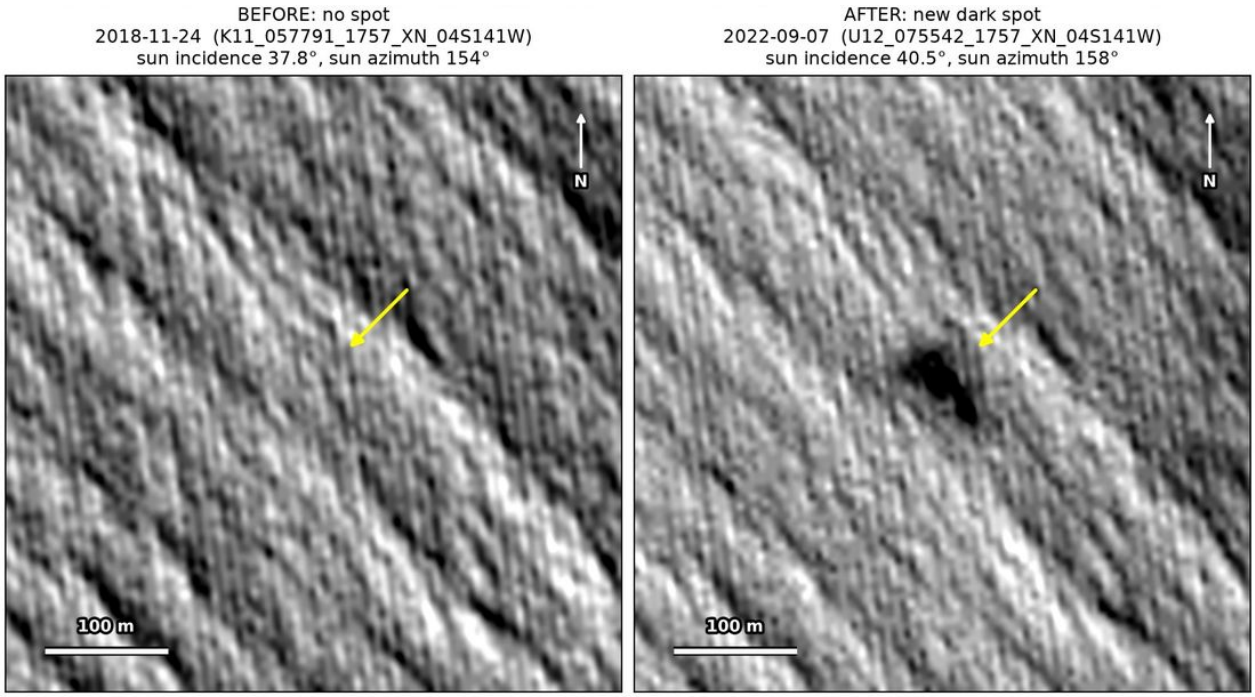

Figure 348: Candidate 147, before and after: the same 500 × 500 m of ground in 24 November 2018 (`K11_057791_1757_XN_04S141W`) and 7 September 2022 (`U12_075542_1757_XN_04S141W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

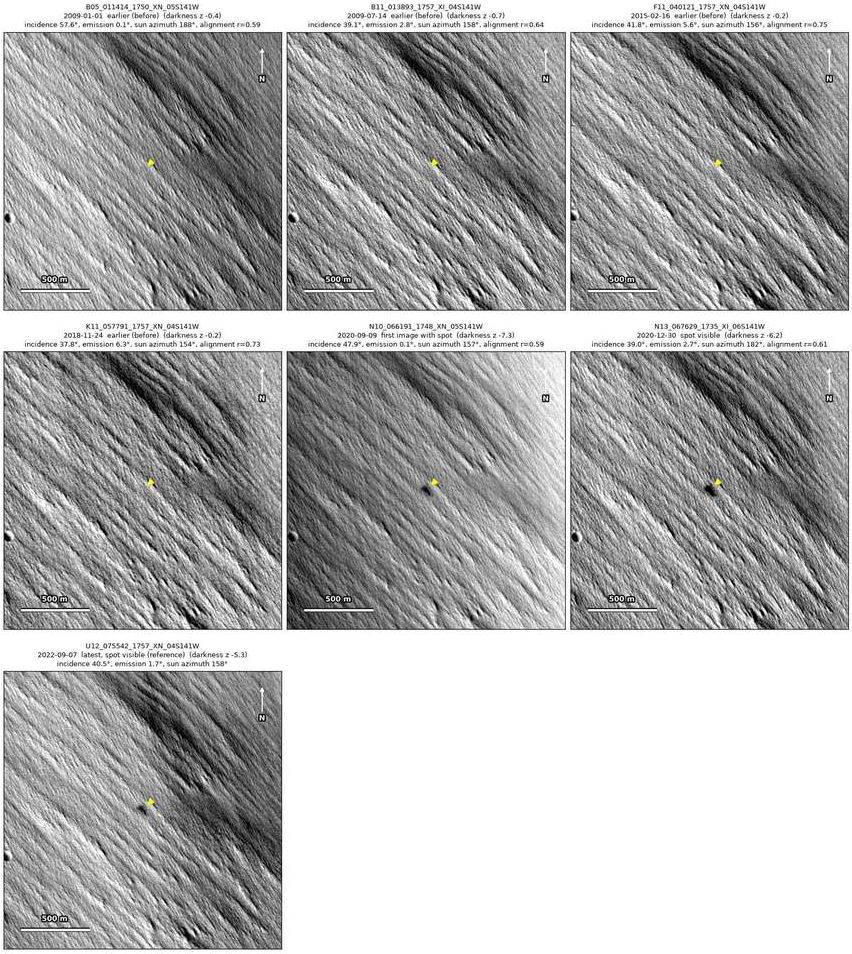

Figure 349: Candidate 147: seven CTX images of the same 2×2 km patch, 2009 to 2022. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 148: Ridged plains, 175 km south of Cobres crater

Location	14.6501° S, 206.4798° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	20 January 2010 and 22 May 2014
Images	1 without the spot (absent: 20 January 2010); 3 with it (present: 22 May 2014, 23 June 2020 and 19 January 2023)
Size	Dark spot about 45 m across (54 × 52 m in the May 2014 image, 38 × 46 m in the January 2023 image), with a dark core and lobes to the east and south-south-east
HiRISE	none covers the spot (nearest footprint edge 14.6 km)
Nearest catalogued new impact	97 km (Daubar et al., 2022)

What changed. A dark spot about 45 m across, with a dark core and fainter lobes to the east and south-south-east, appeared on ridged plains 175 km south of Cobres crater between January 2010 and May 2014. It shows in three CTX images from three orbits and is still there, fainter, in January 2023.

Why it may be a new impact. The only earlier image, from 2010, was taken under a sun between those of the later images. It shows plain ground at the spot, while a dark feature 272 m away and a small pit 121 m away both show in it. A dark blast zone that fades slowly over a decade is how the site of a small new crater looks.

Caveats. Only one earlier image exists, so the time window in which the spot formed rests on that image alone, although it lines up well and a dark feature nearby shows in it, so it would have shown the spot. The latest image has a stripe defect passing just beside the spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 14.6 km away.

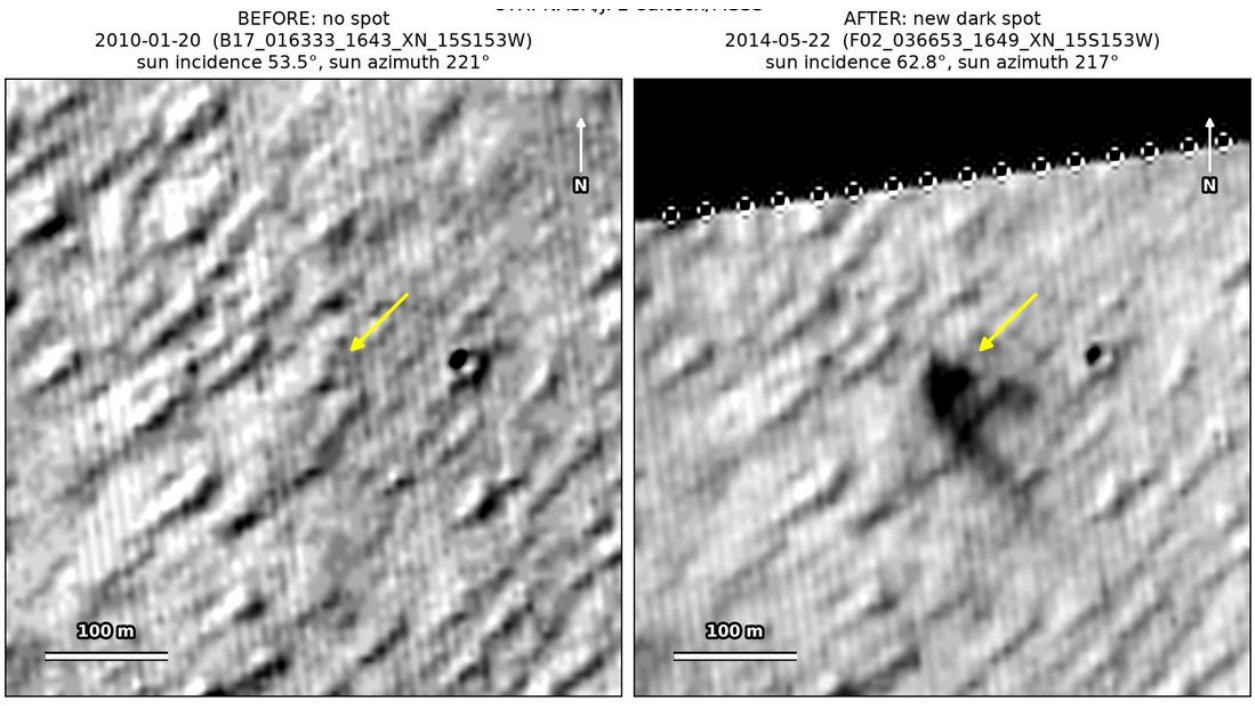

Figure 350: Candidate 148, before and after: the same 500 × 500 m of ground in 20 January 2010 (`B17_016333_1643_XN_15S153W`) and 22 May 2014 (`F02_036653_1649_XN_15S153W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

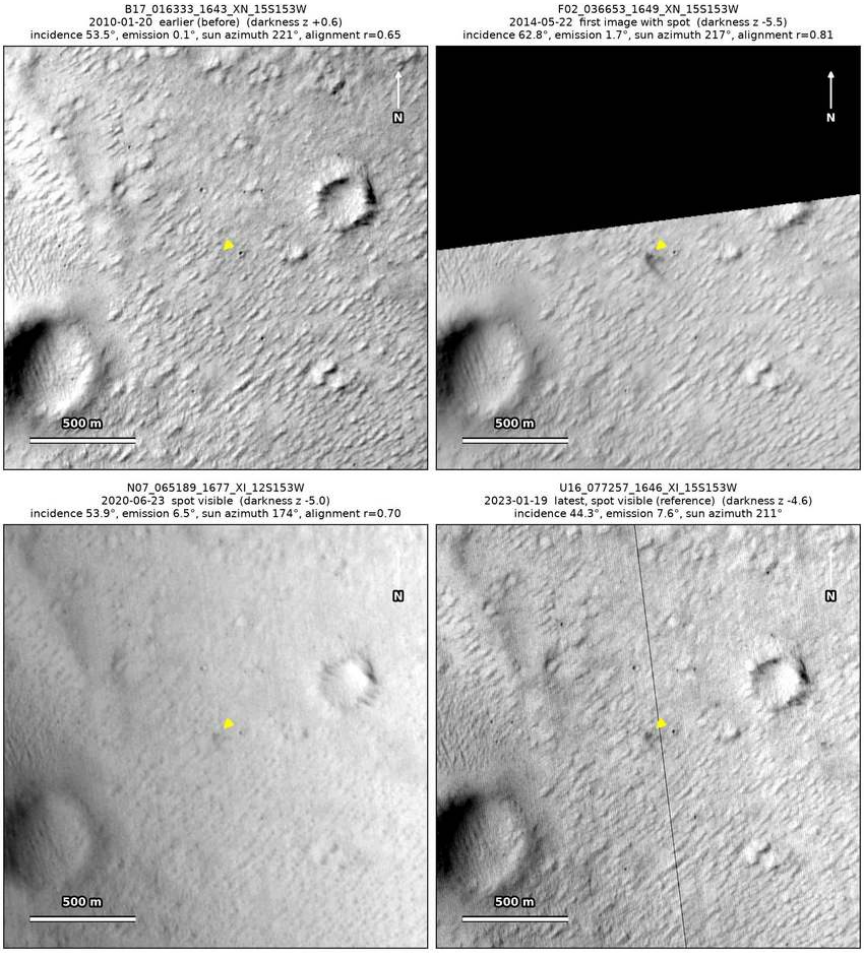

Figure 351: Candidate 148: four CTX images of the same 2×2 km patch, 2010 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 149: Wind-etched plains east of Medusae Fossae, 215 km north-west of Minio Vallis

Location	1.3076° S, 206.3880° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	23 February 2008 and 9 February 2023
Images	1 without the spot (absent: 23 February 2008); 3 with it (present: 9 February 2023, 1 April 2023 and 21 October 2024)
Size	Dark spot about 30 m across, elongated north-south (24 × 42 m in the February 2023 image, 26 × 44 m in the April 2023 image and 24 × 42 m in the October 2024 image)
HiRISE	none covers the spot (nearest footprint edge 2.6 km)
Nearest catalogued new impact	98 km (Daubar et al., 2022)

What changed. A compact dark spot about 30 m across appeared on smooth wind-etched plains east of Medusae Fossae, 215 km north-west of Minio Vallis, between February 2008 and February 2023, and is still there in October 2024. Three CTX images from three orbits, taken at three viewing angles, show it at the same size.

Why it may be a new impact. The only earlier image, from 2008, was taken under almost the same sun as the February 2023 image and shows plain ground there. A compact dark spot that stays in place for 20 months is how the blast zone of a small new crater looks.

Caveats. Only one earlier image exists, so the time window in which the spot formed spans 15 years and rests on that image alone. The plains have no strong dark feature nearby that could show whether that image would have caught a spot like it, although the image lines up well and is sharp. No image from the high-resolution HiRISE camera covers it; the nearest ends 2.6 km away.

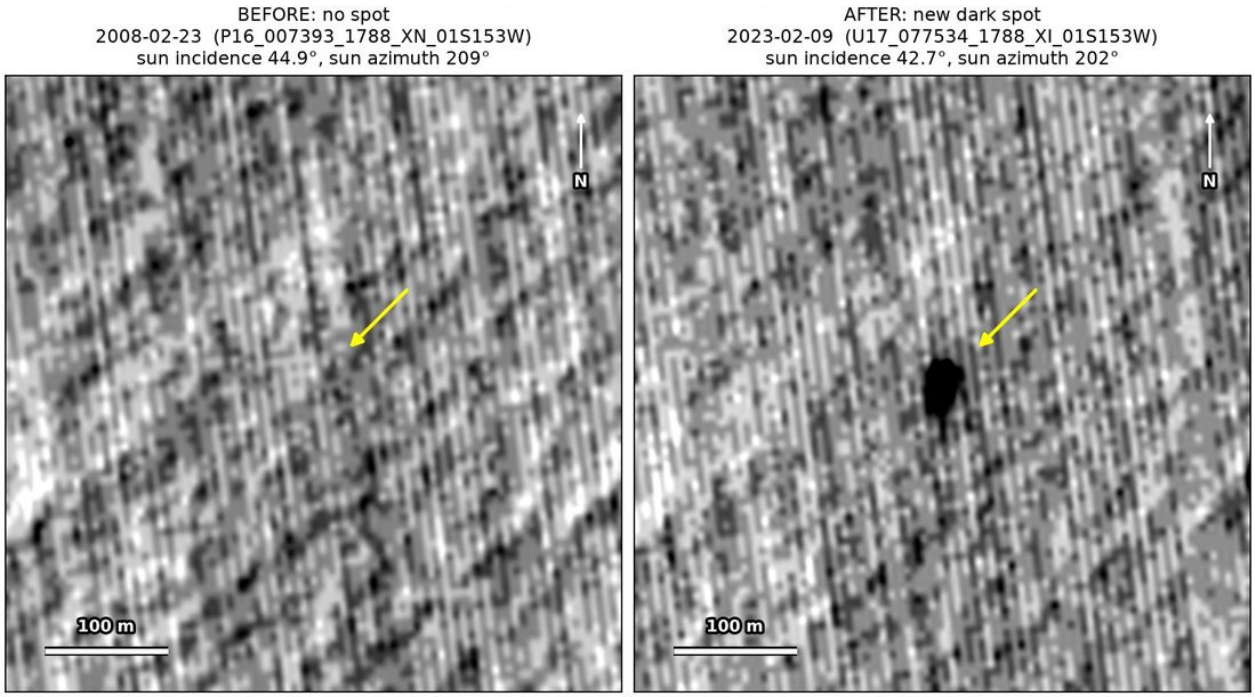

Figure 352: Candidate 149, before and after: the same 500 × 500 m of ground in 23 February 2008 (`P16_007393_1788_XN_01S153W`) and 9 February 2023 (`U17_077534_1788_XI_01S153W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

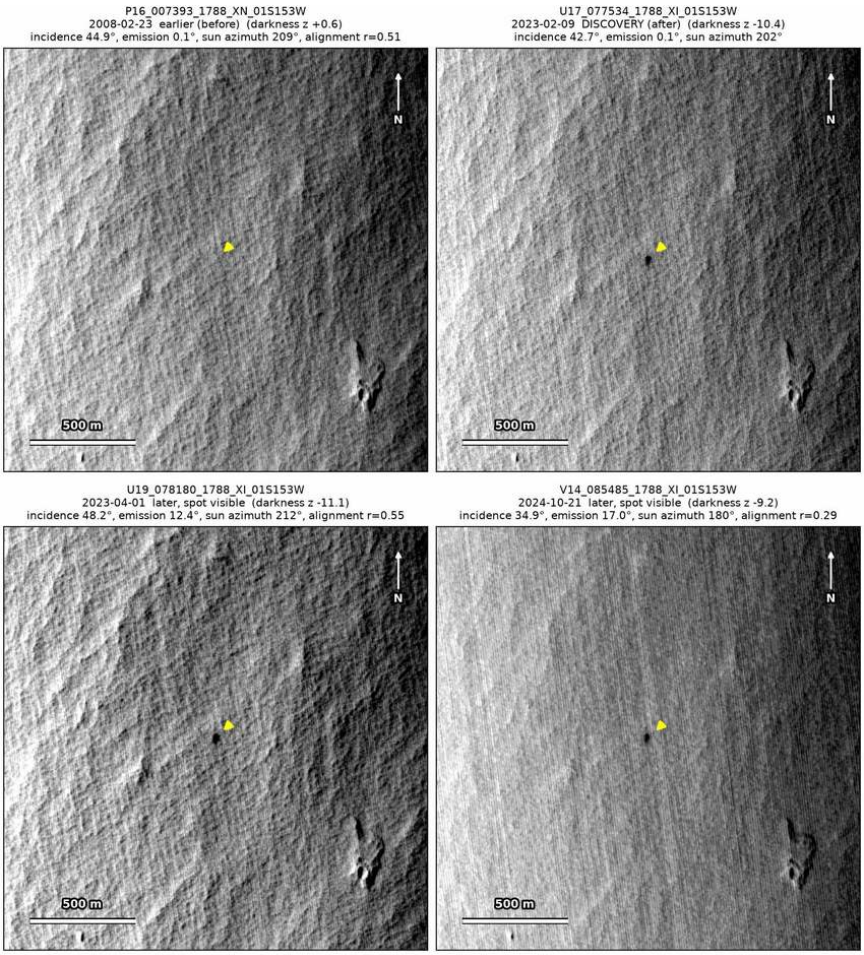

Figure 353: Candidate 149: four CTX images of the same 2×2 km patch, 2008 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 150: Knobby lava plains in western Daedalia Planum, 84 km north-west of candidate 42

Location	18.3646° S, 222.3114° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	13 March 2021 and 31 December 2022
Images	3 without the spot (absent: 14 December 2010, 20 July 2014 and 13 March 2021); 2 with it (present: 31 December 2022 and 22 January 2023)
Size	Dark spot about 40 m across, elongated north-south (40 × 64 m in the December 2022 image, 42 × 66 m in the January 2023 image)
HiRISE	none covers the spot (nearest footprint edge 10.4 km)
Nearest catalogued new impact	82 km (Daubar et al., 2022)

What changed. A dark spot about 40 m across appeared among low knobs on lava plains in western Daedalia Planum, 84 km north-west of candidate 42, between March 2021 and December 2022, and is still there three weeks later. Two CTX images from two orbits show it.

Why it may be a new impact. Three earlier images from 2010 to 2021 show only small knob shadows there, even though their lower sun makes shadows longer, and a dark feature 180 m away shows in all three, so they would have shown a new spot too. A compact new dark spot on dusty lava plains is how the blast zone of a small new crater looks.

Caveats. The two images with the spot are only three weeks apart, so they cannot show how long it lasted. The knobs around it cast small shadows, but a shadow would show more, not less, under the lower sun of the earlier images. No image from the high-resolution HiRISE camera covers it; the nearest ends 10.4 km away.

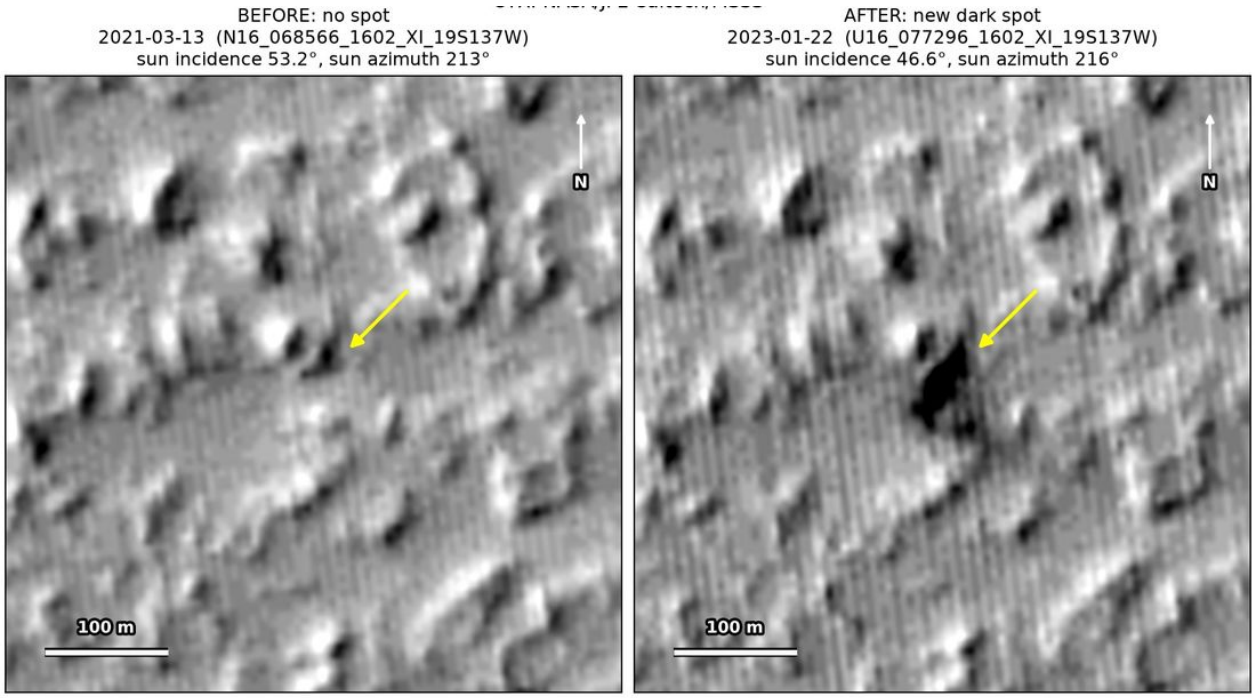

Figure 354: Candidate 150, before and after: the same 500 × 500 m of ground in 13 March 2021 (`N16_068566_1602_XI_19S137W`) and 22 January 2023 (`U16_077296_1602_XI_19S137W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

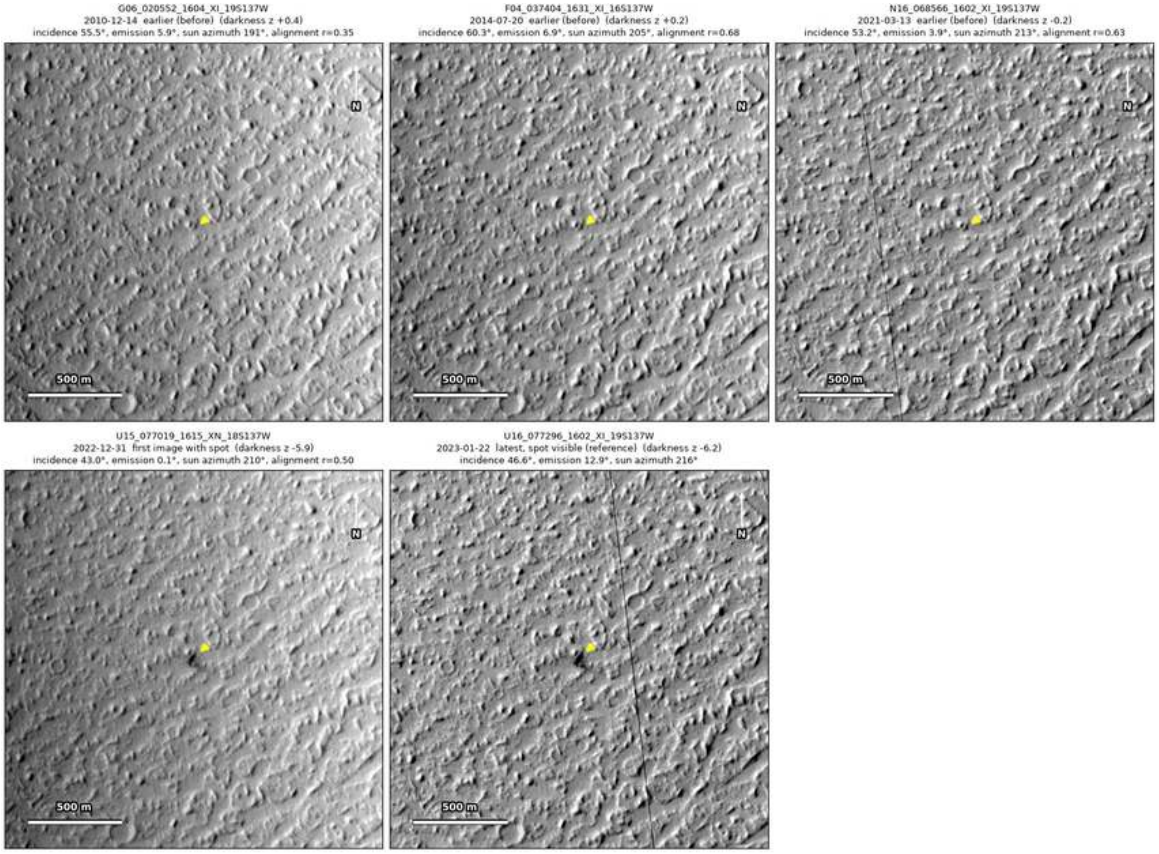

Figure 355: Candidate 150: five CTX images of the same 2×2 km patch, 2010 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 151: Plains in western Arabia Terra, 292 km south-west of Hiddekel Rupes

Location	13.1083° N, 13.0583° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	5 December 2014 and 6 July 2021
Images	3 without the spot (absent: 3 August 2014, 30 November 2014 and 5 December 2014); 1 with it (present: 6 July 2021)
Size	Dark spot about 40 m across (40 × 42 m in the July 2021 image), inside a fainter zone about 150 m wide
HiRISE	none covers the spot (nearest footprint edge 62.9 km)
Nearest catalogued new impact	81 km (Daubar et al., 2022)

What changed. A compact dark spot about 40 m across, inside a fainter dark zone about 150 m wide, appeared on smooth plains in western Arabia Terra, 292 km south-west of Hiddekel Rupes, between December 2014 and July 2021. One CTX image shows it.

Why it may be a new impact. Three earlier images from 2014 show plain ground there, while a dark feature 268 m away shows in two of them, so those two would have shown a new spot too. A compact dark spot with a halo on dusty plains is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, so there is no telling whether it lasts, and it is faint, with 8% less light at its darkest. The site lies near the edge of a 2010 image, which could not be lined up, so that image is not used to date the spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 62.9 km away.

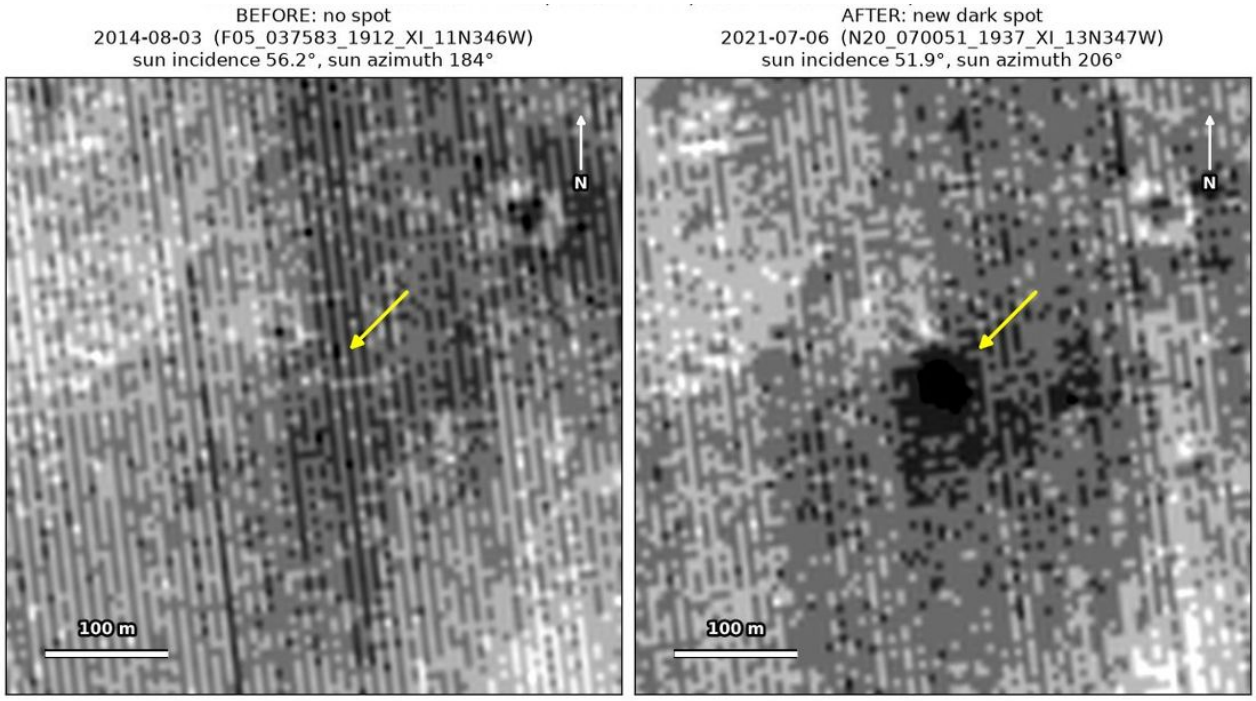

Figure 356: Candidate 151, before and after: the same 500 × 500 m of ground in 3 August 2014 (`F05_037583_1912_XI_11N346W`) and 6 July 2021 (`N20_070051_1937_XI_13N347W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

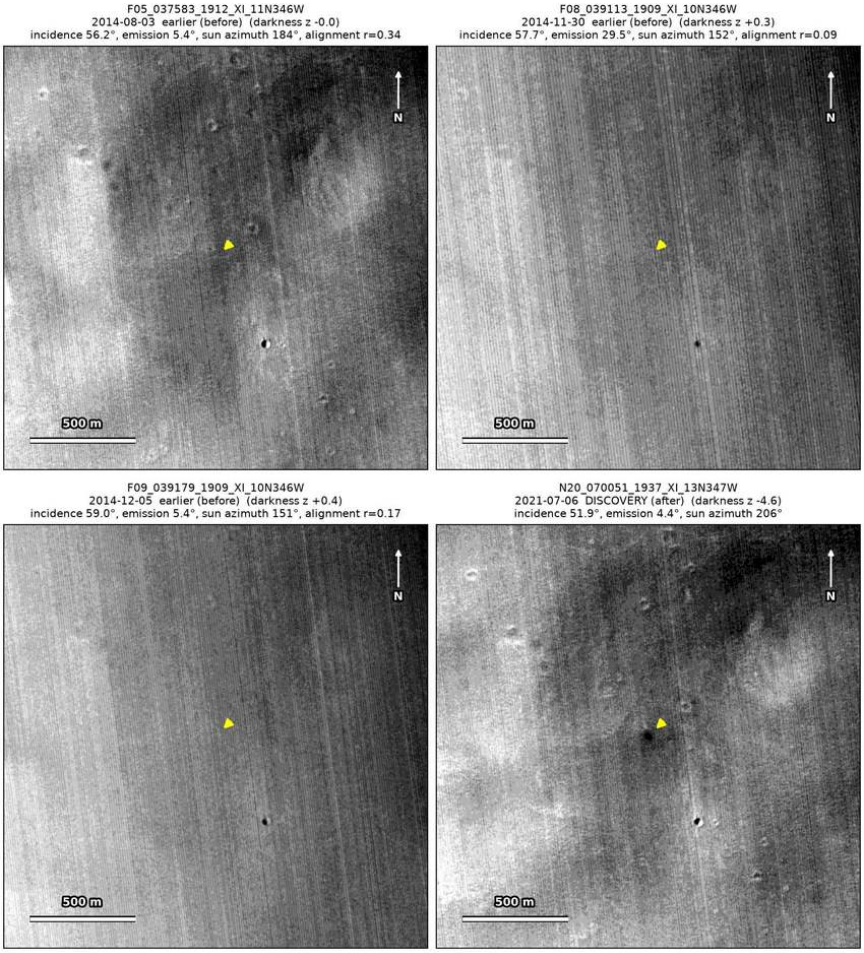

Figure 357: Candidate 151: four CTX images of the same 2×2 km patch, 2014 to 2021. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 152: Lava plains 295 km west of Fortuna Fossae

Location	4.5084° N, 262.3156° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	1 June 2020 and 14 August 2022
Images	2 without the spot (absent: 23 July 2014 and 1 June 2020); 2 with it (present: 14 August 2022 and 16 August 2024)
Size	Dark spot about 30 m across (32 × 26 m in the August 2022 image), smaller and fainter by August 2024 (20 × 18 m)
HiRISE	none covers the spot (nearest footprint edge 47.2 km)
Nearest catalogued new impact	54 km (Daubar et al., 2022)

What changed. A compact dark spot about 30 m across appeared on lava plains 295 km west of Fortuna Fossae between June 2020 and August 2022. Two CTX images from two orbits, two years apart, show it.

Why it may be a new impact. Two earlier images, from 2014 and 2020, show plain ground there under a lower sun, which would lengthen any shadow, and a dark feature 300 m away shows in both, so they would have shown a new spot too. A compact new dark spot on dusty lava plains is how the blast zone of a small new crater looks.

Caveats. It is smaller and fainter in the 2024 image, which fits a fading blast zone. In the difference between the earlier and later images, the shading of a crater at the edge of the frame stands out more than the spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 47.2 km away.

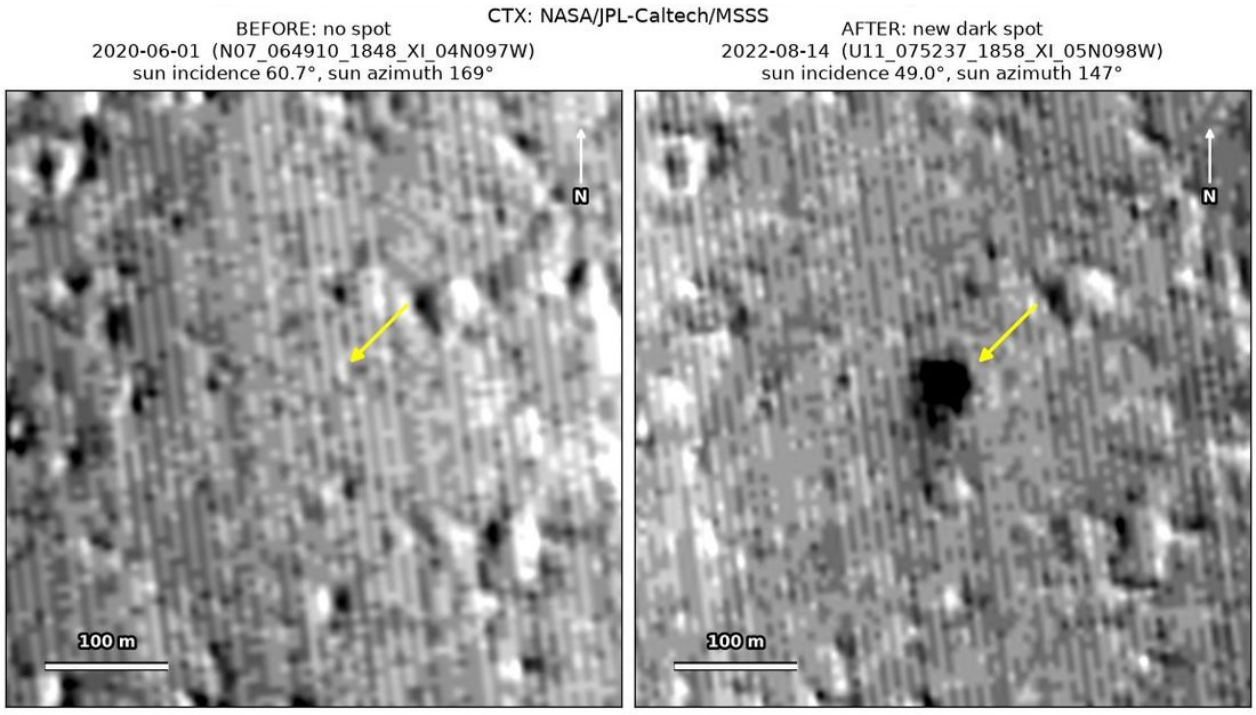

Figure 358: Candidate 152, before and after: the same 500 × 500 m of ground in 1 June 2020 (`N07_064910_1848_XI_04N097W`) and 14 August 2022 (`U11_075237_1858_XI_05N098W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

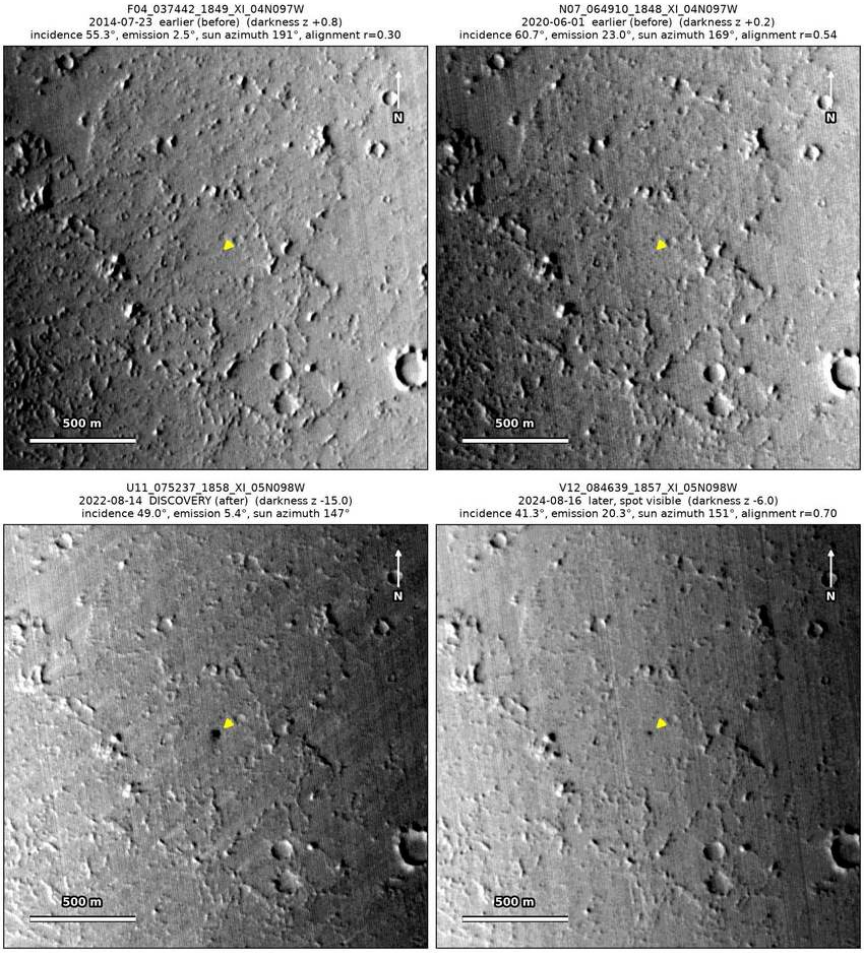

Figure 359: Candidate 152: four CTX images of the same 2×2 km patch, 2014 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 153: Wind-textured plains 309 km west of Arsia Sulci

Location	4.2734° S, 225.3506° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	28 December 2008 and 26 April 2023
Images	1 without the spot (absent: 28 December 2008); 1 with it (present: 26 April 2023)
Size	Dark spot about 35 m across (42 × 28 m in the April 2023 image)
HiRISE	none covers the spot (nearest footprint edge 39.4 km)
Nearest catalogued new impact	107 km (Daubar et al., 2022)

What changed. A compact dark spot about 35 m across appeared on wind-textured plains 309 km west of Arsia Sulci between December 2008 and April 2023. One CTX image shows it.

Why it may be a new impact. The only earlier image, from 2008, was taken under a lower sun than the 2023 image and shows plain ground there, while a small crater 180 m away shows in both, so the earlier image would have shown a new spot too. A compact dark spot on dusty plains is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, so there is no telling whether it lasts, and its only earlier image is 14 years older, so the date when it formed is only loosely known. No image from the high-resolution HiRISE camera covers it; the nearest ends 39.4 km away.

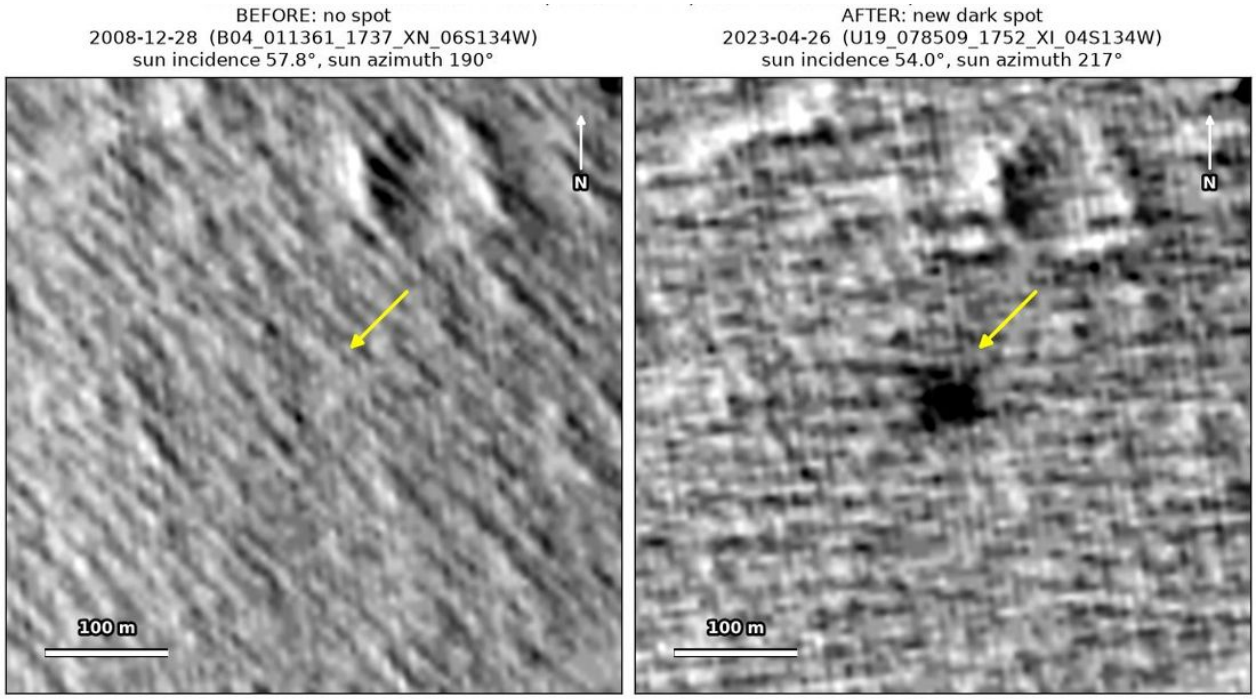

Figure 360: Candidate 153, before and after: the same 500 × 500 m of ground in 28 December 2008 (`B04_011361_1737_XN_06S134W`) and 26 April 2023 (`U19_078509_1752_XI_04S134W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

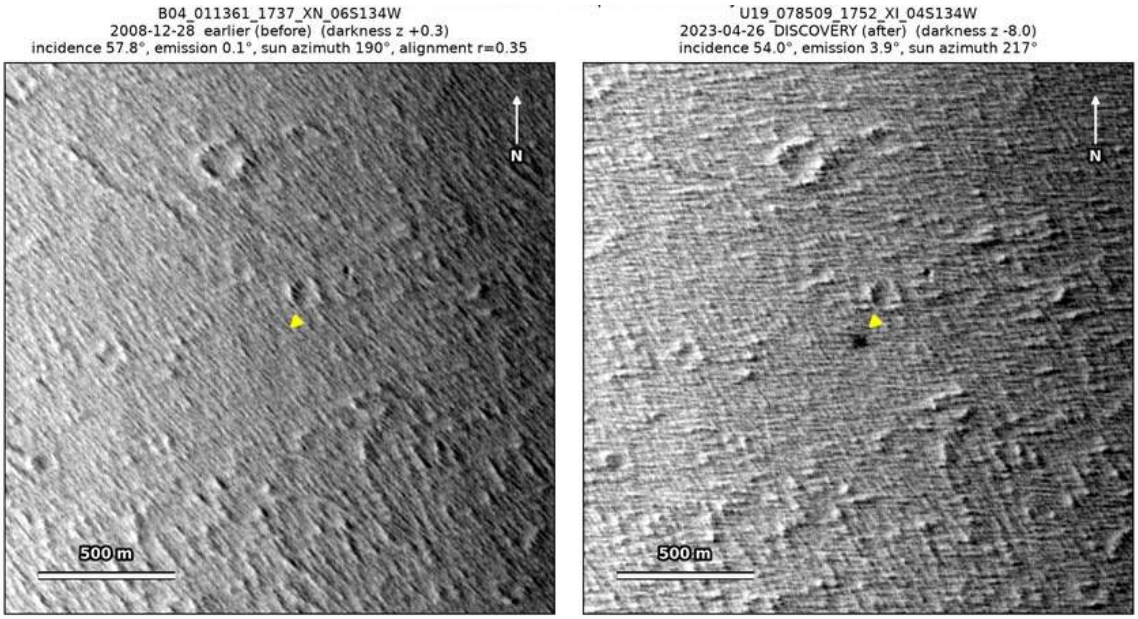

Figure 361: Candidate 153: both CTX images of the same 2×2 km patch, December 2008 and April 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 154: Crater floor in Arabia Terra, 267 km north of Indus Vallis

Location	23.2932° N, 40.1699° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 75%
Formed between	21 November 2020 and 7 July 2022
Images	5 without the spot (absent: 7 August 2010, 27 February 2012, 9 July 2020, 4 October 2020 and 21 November 2020); 3 with it (present: 7 July 2022, 6 September 2022 and 29 April 2023)
Size	Diffuse dark spot about 140 m across (132 × 180 m in the July 2022 image, 132 × 206 m in the September 2022 image and 128 × 194 m in the April 2023 image), darker at the centre
HiRISE	none covers the spot (nearest footprint edge 20.5 km)
Nearest catalogued new impact	26 km (Daubar et al., 2022)

What changed. A diffuse dark spot about 140 m across appeared on the floor of a worn-down crater in Arabia Terra, 267 km north of Indus Vallis, between November 2020 and July 2022. Three CTX images from three orbits over ten months show it.

Why it may be a new impact. Five earlier images from 2010 to 2020 show plain crater floor there, three of them each taken under nearly the same sun as one of the later images. A diffuse dark spot appearing on a dusty crater floor is how the blast zone of a small new crater looks.

Caveats. It is faint and diffuse, with no crater or core visible at the scale of the CTX images, so dust stripped from a patch of the floor by the wind is the main alternative. No feature nearby is dark enough to show whether the earlier images would have caught it. No image from the high-resolution HiRISE camera covers it; the nearest ends 20.5 km away.

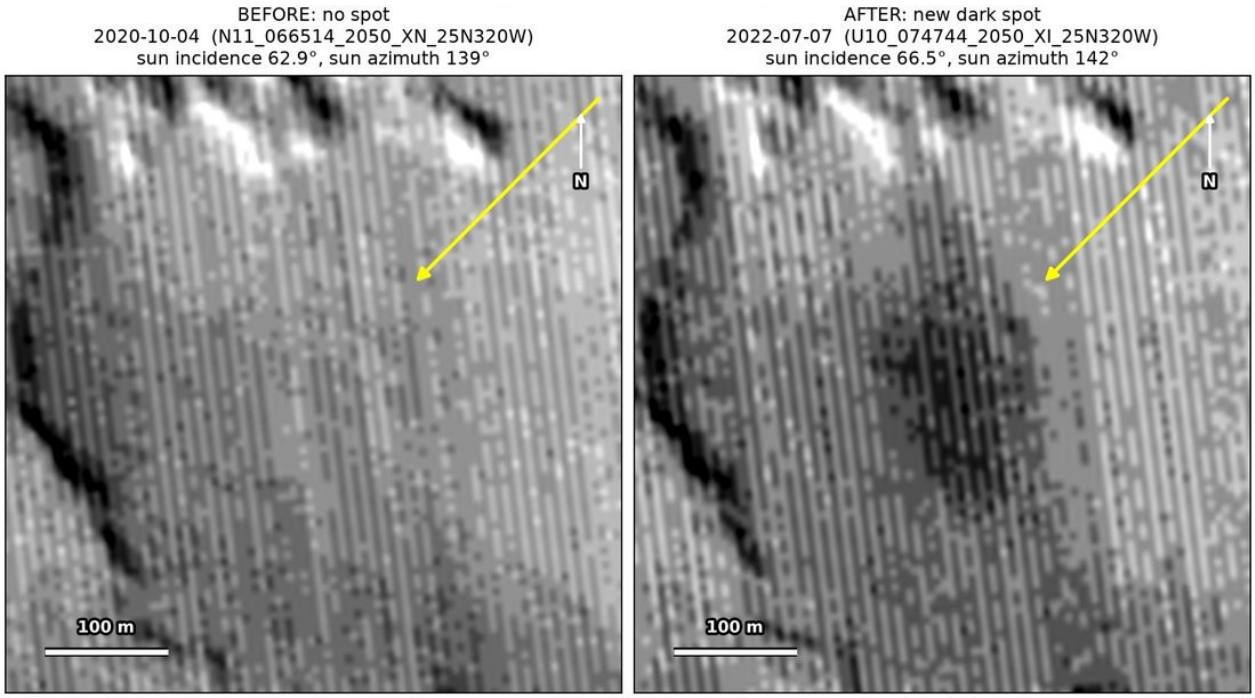

Figure 362: Candidate 154, before and after: the same 500 × 500 m of ground in 4 October 2020 (`N11_066514_2050_XN_25N320W`) and 7 July 2022 (`U10_074744_2050_XI_25N320W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

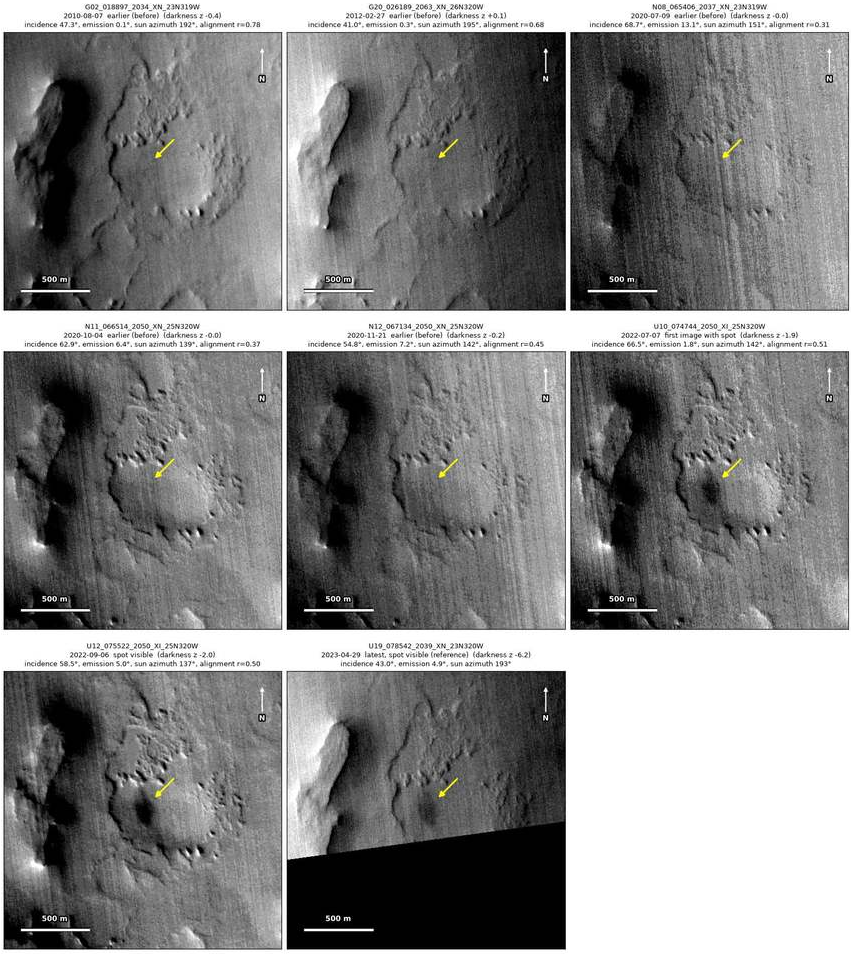

Figure 363: Candidate 154: eight CTX images of the same 2×2 km patch, 2010 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 155: North-west flank of Arsia Mons, 282 km east of Arsia Sulci

Location	4.5992° S, 234.6512° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	23 February 2007 and 29 January 2012
Images	1 without the spot (absent: 23 February 2007); 3 with it (present: 29 January 2012, 10 April 2012 and 25 June 2022)
Size	Dark spot about 120 m across, with a darker core (122 × 132 m in the January 2012 image, 130 × 132 m in the April 2012 image), smaller and fainter by June 2022 (64 × 128 m)
HiRISE	none covers the spot (nearest footprint edge 27.9 km)
Nearest catalogued new impact	63 km (Daubar et al., 2022)

What changed. A dark spot about 120 m across, with a darker core, appeared on the north-west flank of Arsia Mons, 282 km east of Arsia Sulci, between February 2007 and January 2012. Three CTX images from three orbits over ten years show it, smaller and fainter by 2022.

Why it may be a new impact. The only earlier image, from 2007, shows plain ground there, under a sun between those of the later images in height and direction. A compact dark spot with a halo that fades over the years on smooth dusty ground is how the blast zone of a small new crater looks.

Caveats. Only one earlier image exists, and no feature nearby is dark enough to show whether it would have caught a new spot, although the spot is large and dark and the image is clean and well aligned. No image from the high-resolution HiRISE camera covers it; the nearest ends 27.9 km away.

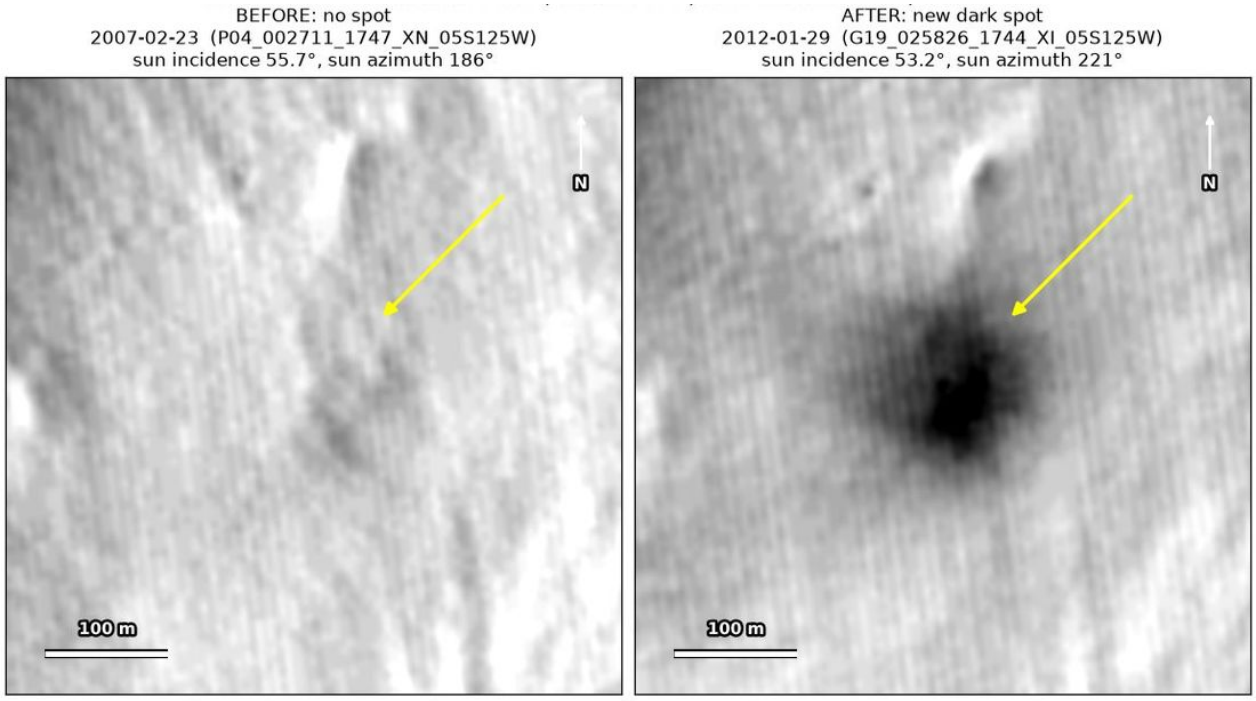

Figure 364: Candidate 155, before and after: the same 500 × 500 m of ground in 23 February 2007 (`P04_002711_1747_XN_05S125W`) and 29 January 2012 (`G19_025826_1744_XI_05S125W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

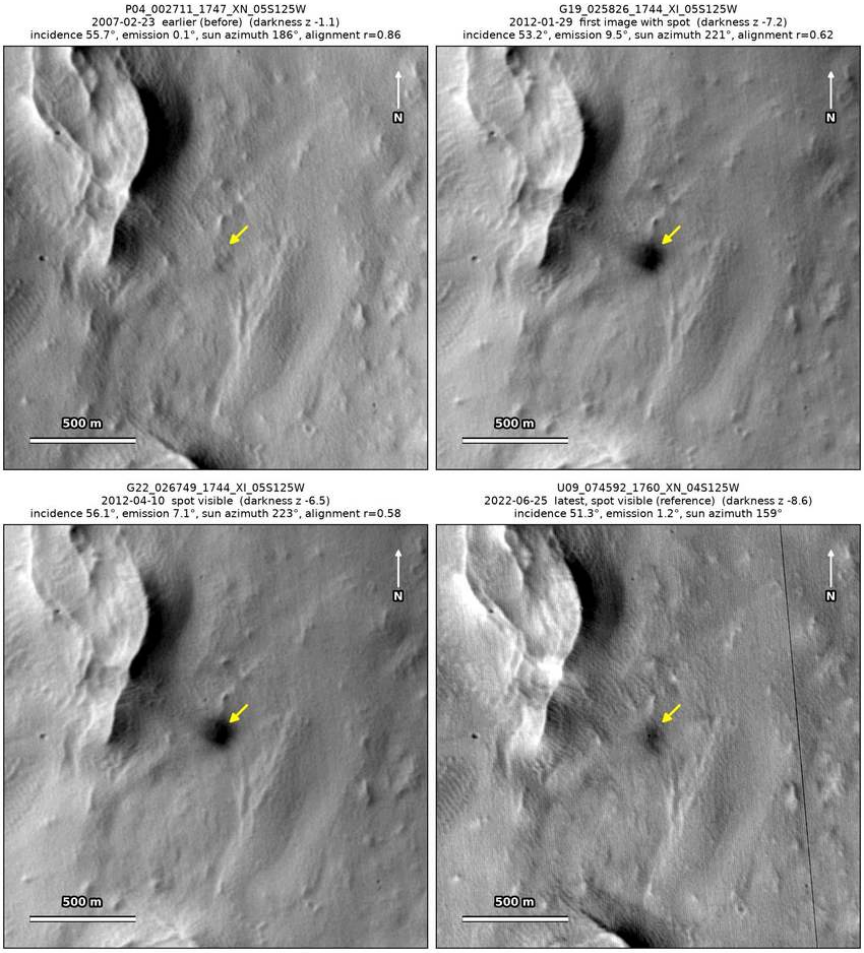

Figure 365: Candidate 155: four CTX images of the same 2×2 km patch, 2007 to 2022. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 156: Wind-etched plains in Daedalia Planum, 319 km south-west of Arsia Sulci

Location	8.4308° S, 225.2100° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	28 April 2014 and 24 May 2023
Images	4 without the spot (absent: 18 September 2010, 25 April 2012, 10 February 2014 and 28 April 2014); 1 with it (present: 24 May 2023)
Size	Dark spot about 30 m across (28 × 32 m in the May 2023 image)
HiRISE	none covers the spot (nearest footprint edge 10.8 km)
Nearest catalogued new impact	16 km (Daubar et al., 2022)

What changed. A compact dark spot about 30 m across appeared on wind-etched plains in Daedalia Planum, 319 km south-west of Arsia Sulci, between April 2014 and May 2023. One CTX image shows it.

Why it may be a new impact. Four earlier images from 2010 to 2014 show plain ground there, one of them under almost the same sun as the May 2023 image, while a dark feature 186 m away shows in all four. A compact dark spot on dusty plains is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, so there is no telling whether it lasts, and it is small, so a comparison of the images picks it out only weakly. No image from the high-resolution HiRISE camera covers it; the nearest ends 10.8 km away.

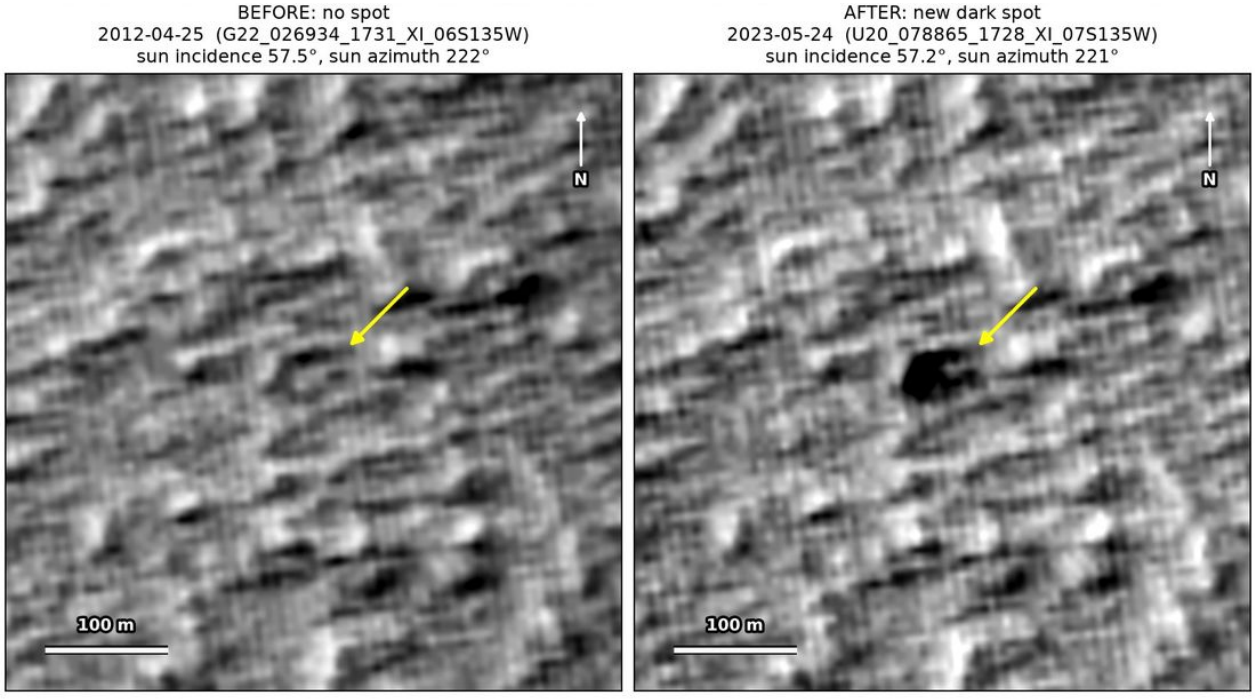

Figure 366: Candidate 156, before and after: the same 500 × 500 m of ground in 25 April 2012 (`G22_026934_1731_XI_06S135W`) and 24 May 2023 (`U20_078865_1728_XI_07S135W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

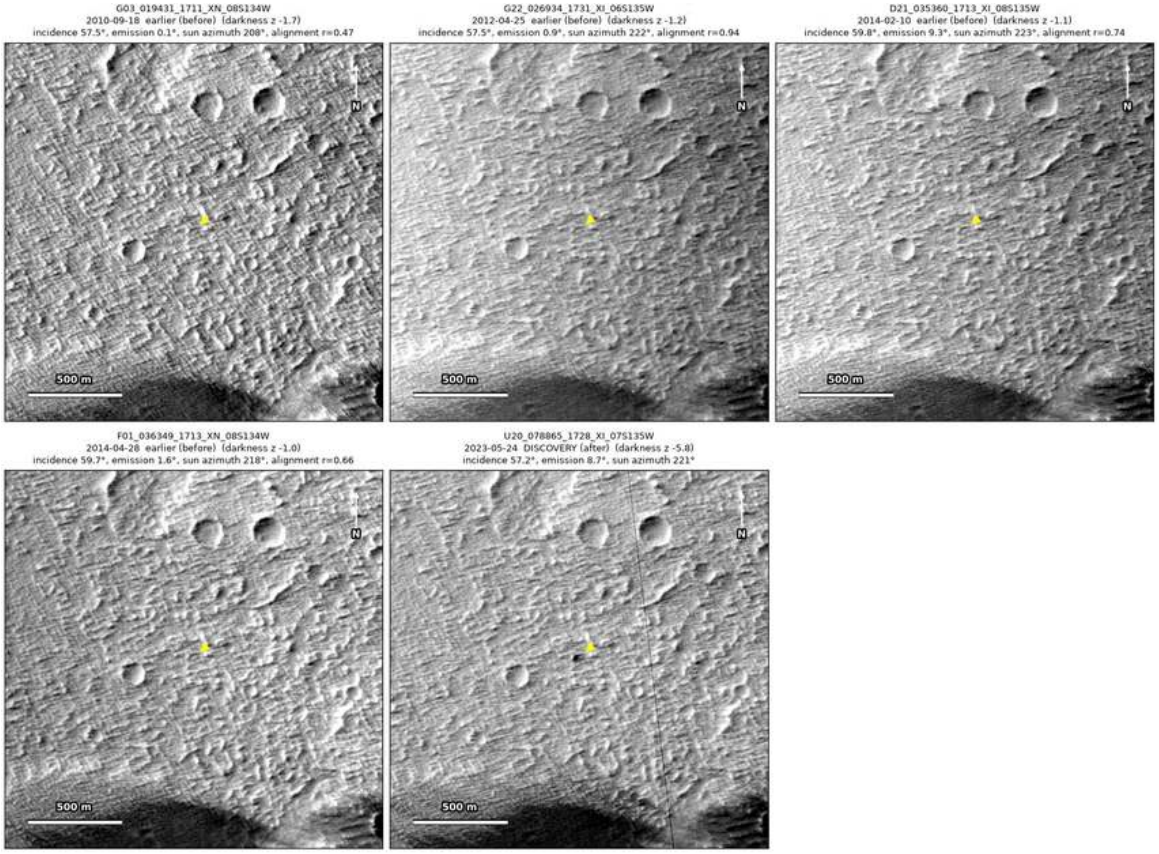

Figure 367: Candidate 156: five CTX images of the same 2×2 km patch, 2010 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 157: Fractured plains in the Cerberus Tholi region, 138 km west of Tombaugh crater

Location	3.2439° N, 159.6141° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	13 March 2008 and 9 December 2022
Images	1 without the spot (absent: 13 March 2008); 1 with it (present: 9 December 2022)
Size	Dark spot about 70 m across (78 × 74 m in the December 2022 image), with a darker core about 50 m wide
HiRISE	none covers the spot (nearest footprint edge 27.7 km)
Nearest catalogued new impact	46 km (Daubar et al., 2022)

What changed. A dark spot about 70 m across, with a darker core, appeared where narrow fractures meet in the Cerberus Tholi region, 138 km west of Tombaugh crater, between March 2008 and December 2022. One CTX image shows it.

Why it may be a new impact. The only earlier image, from 2008, shows the same fractures there but no spot, under a lower sun that would lengthen any shadow, while a dark feature 143 m away shows in it, so it would have shown a new spot too. A compact dark spot on dusty plains is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, and that image is striped, although the spot is a coherent blob spanning many stripes. The spot lies at a junction of fractures, where the wind may gather or expose dark material, or a pit could hide, and the only earlier image is 14 years older, with the sun shining from a direction 30° away. No image from the high-resolution HiRISE camera covers it; the nearest ends 27.7 km away.

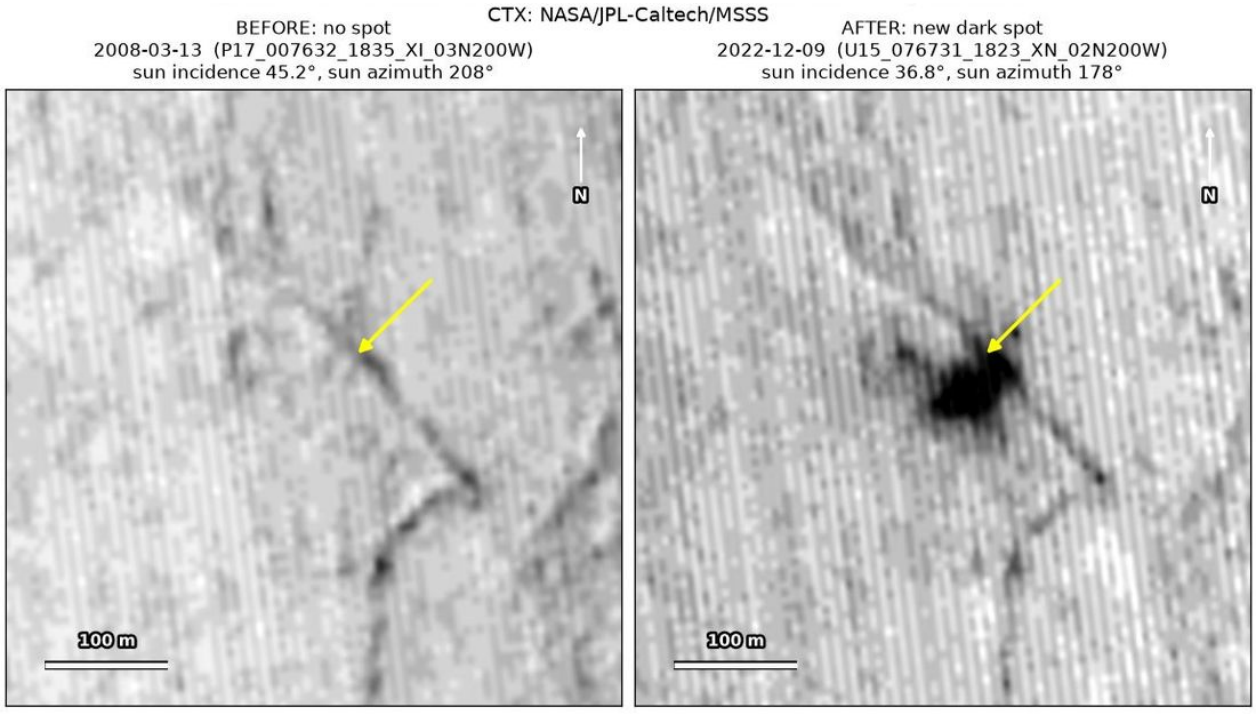

Figure 368: Candidate 157, before and after: the same 500 × 500 m of ground in 13 March 2008 (`P17_007632_1835_XI_03N200W`) and 9 December 2022 (`U15_076731_1823_XN_02N200W`), under a different sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

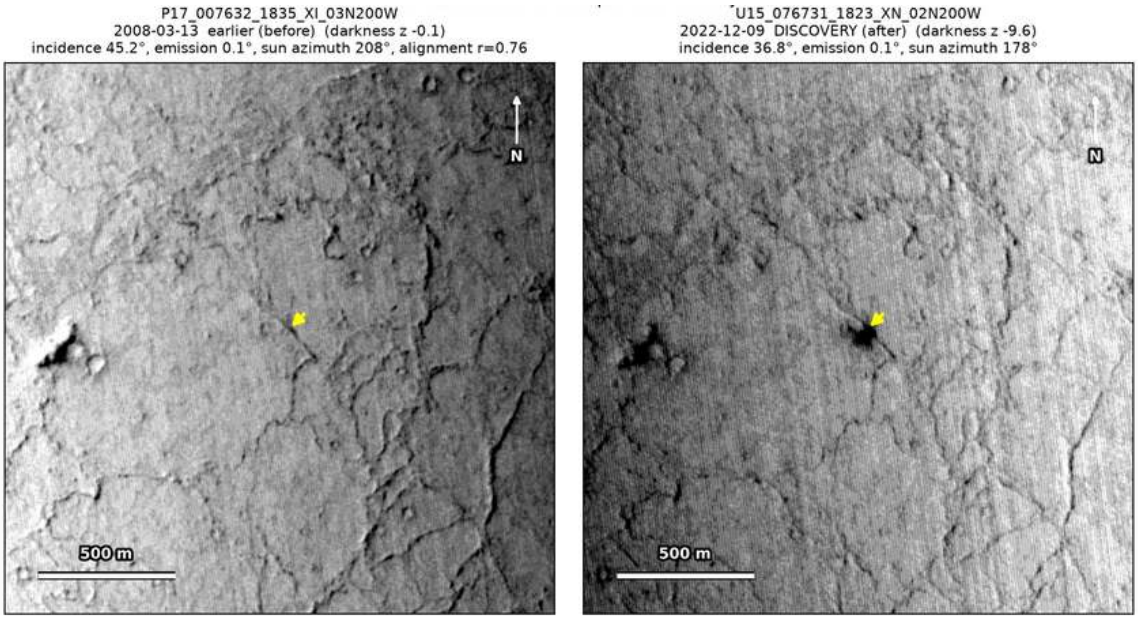

Figure 369: Candidate 157: both CTX images of the same 2×2 km patch, March 2008 and December 2022. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 158: Wind-etched plains in northern Daedalia Planum, 390 km south-east of Amazonis Sulci

Location	5.2335° S, 222.1199° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	2 November 2020 and 25 January 2021
Images	2 without the spot (absent: 3 July 2009 and 2 November 2020); 3 with it (present: 25 January 2021, 29 April 2021 and 13 September 2022)
Size	Dark spot about 35 m across (22 × 36 m in the January 2021 image, 44 × 46 m in the September 2022 image)
HiRISE	none covers the spot (nearest footprint edge 24.3 km)
Nearest catalogued new impact	95 km (Daubar et al., 2022)

What changed. A compact dark spot about 35 m across appeared on wind-etched plains in northern Daedalia Planum, 390 km south-east of Amazonis Sulci, between November 2020 and January 2021. Three CTX images from three orbits over 20 months show it.

Why it may be a new impact. Two earlier images, from 2009 and 2020, show plain ground there, the 2020 one under almost the same sun as the latest image. A compact dark spot that appears on dusty plains within 84 days and lasts is how the blast zone of a small new crater looks.

Caveats. A dark feature nearby cannot settle whether the earlier images would have shown the spot, because in them the wind-etched texture already darkens the ground at the spot slightly. Dark material moved by the wind is possible, although wind marks are usually streaks or diffuse patches, not a compact spot that appears within 84 days and stays. No image from the high-resolution HiRISE camera covers it; the nearest ends 24.3 km away.

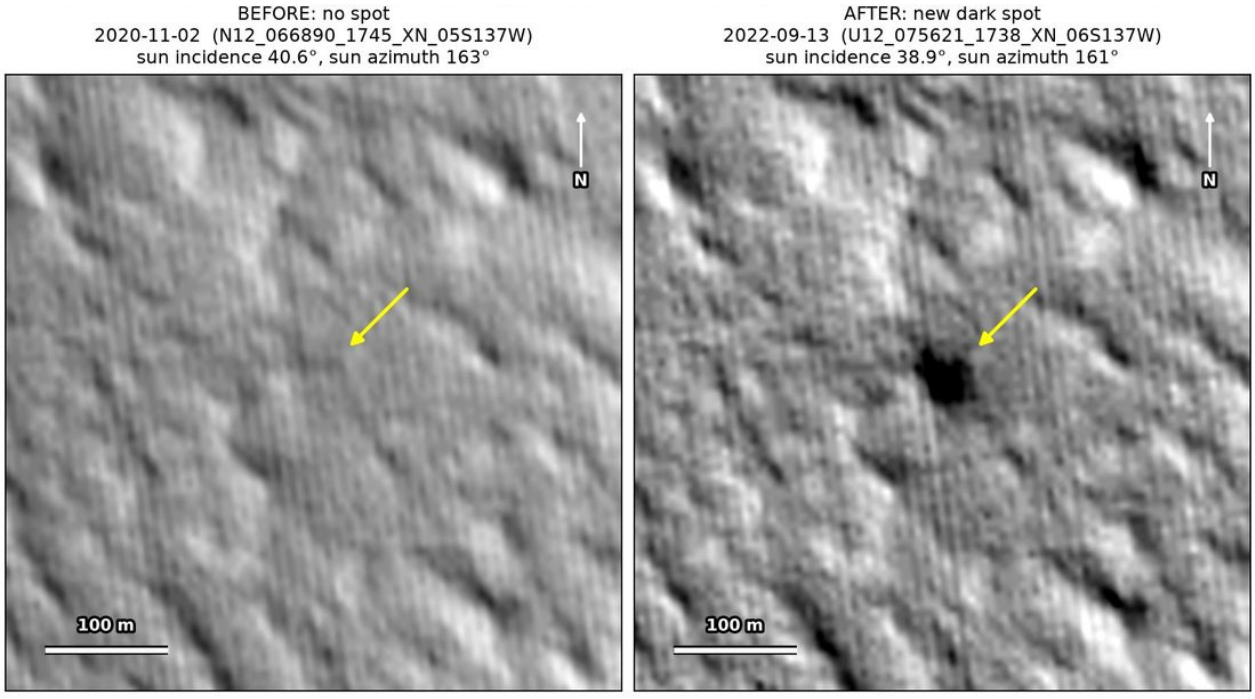

Figure 370: Candidate 158, before and after: the same 500 × 500 m of ground in 2 November 2020 (`N12_066890_1745_XN_05S137W`) and 13 September 2022 (`U12_075621_1738_XN_06S137W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

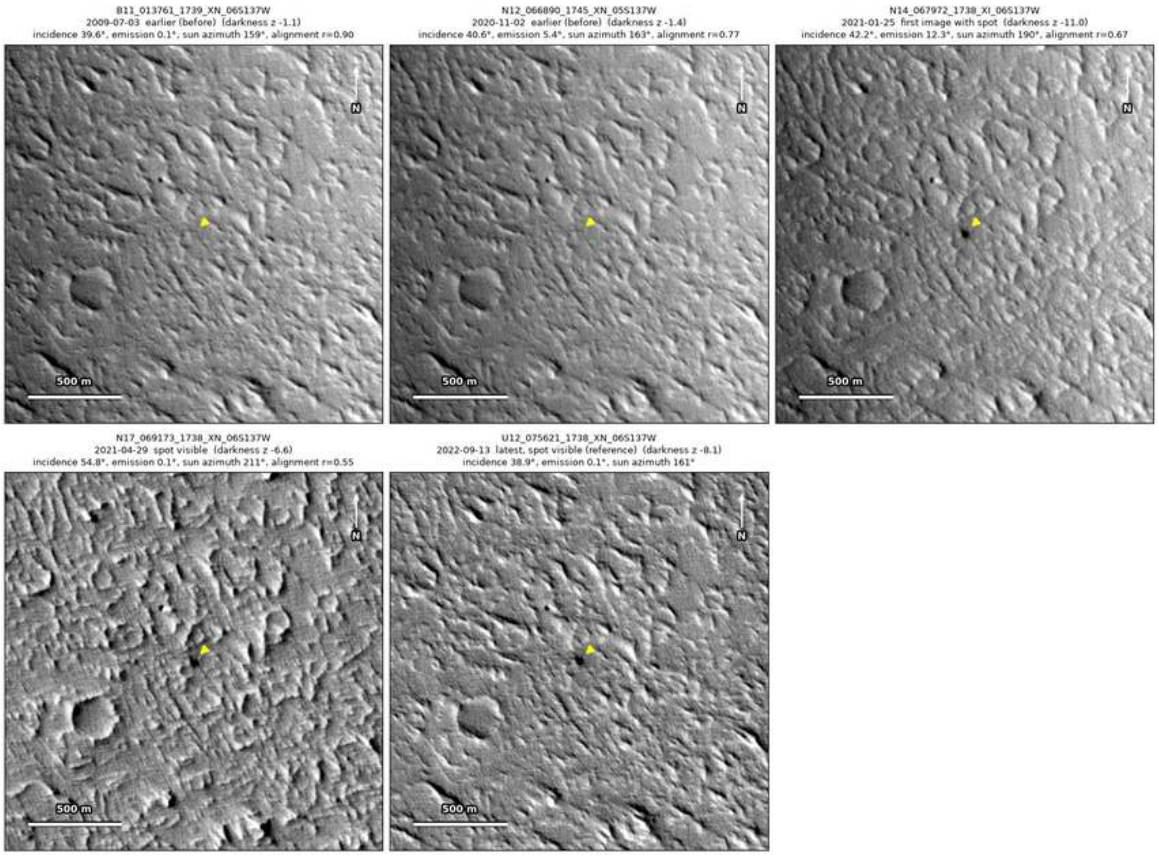

Figure 371: Candidate 158: five CTX images of the same 2×2 km patch, 2009 to 2022. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 159: Plains in northern Terra Cimmeria, 126 km west of Licus Vallis

Location	2.9275° S, 124.2268° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 80%
Formed between	6 July 2021 and 10 September 2022
Images	3 without the spot (absent: 14 June 2008, 6 December 2020 and 6 July 2021); 3 with it (present: 10 September 2022, 23 December 2022 and 31 May 2023)
Size	Dark spot about 50 m across (66 × 40 m in the May 2023 image), at the east edge of an older, faint diffuse dark zone
HiRISE	none covers the spot (nearest footprint edge 19.7 km)
Nearest catalogued new impact	481 km (Daubar et al., 2022)

What changed. A compact dark spot about 50 m across appeared on plains in northern Terra Cimmeria, 126 km west of Licus Vallis, between July 2021 and September 2022. Three CTX images from three orbits over nine months show it.

Why it may be a new impact. Three earlier images from 2008 to 2021 show only a faint, older diffuse dark zone there, one of them under the same sun as the first image with the spot. A dark feature 179 m away shows in two of them, so those would have shown a new spot too. A compact dark spot that appears on plains and lasts is how the blast zone of a small new crater looks.

Caveats. The spot sits at the edge of an older diffuse dark zone, so dark material moved by the wind is a possible alternative. The change between the images at the spot is also only modest, as two of the images are noisy, although nowhere else in the 2 × 2 km patch darkens as much. No image from the high-resolution HiRISE camera covers it; the nearest ends 19.7 km away.

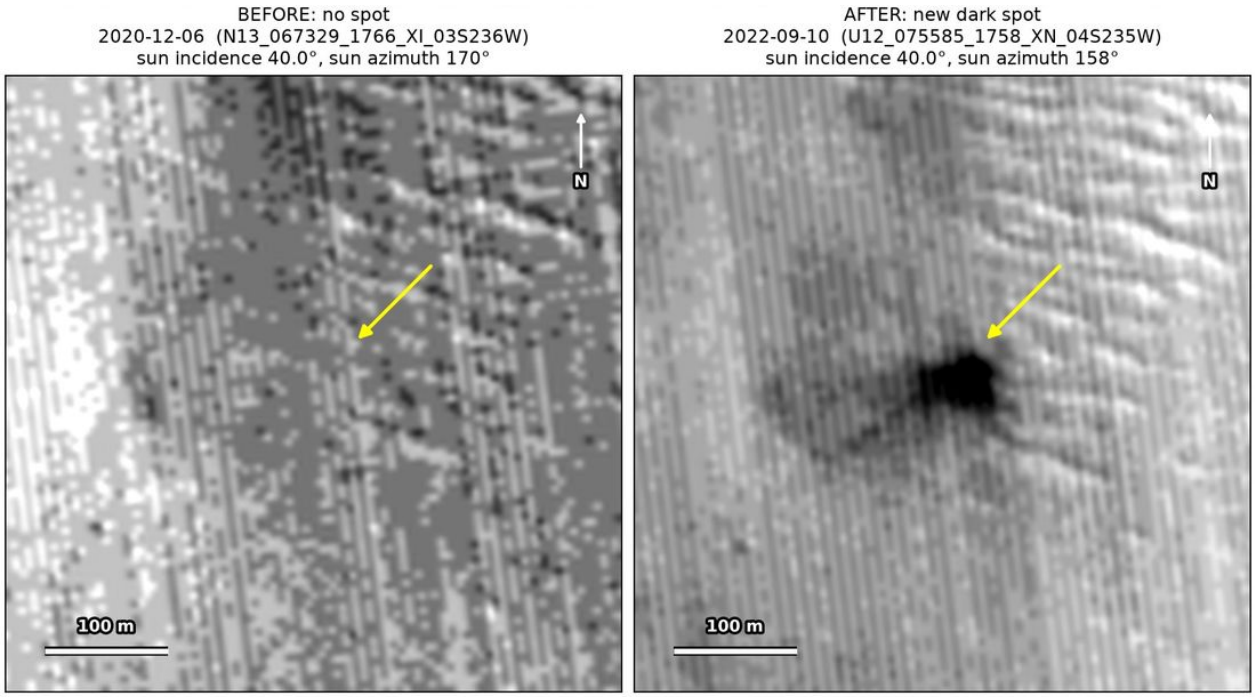

Figure 372: Candidate 159, before and after: the same 500 × 500 m of ground in 6 December 2020 (`N13_067329_1766_XI_03S236W`) and 10 September 2022 (`U12_075585_1758_XN_04S235W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

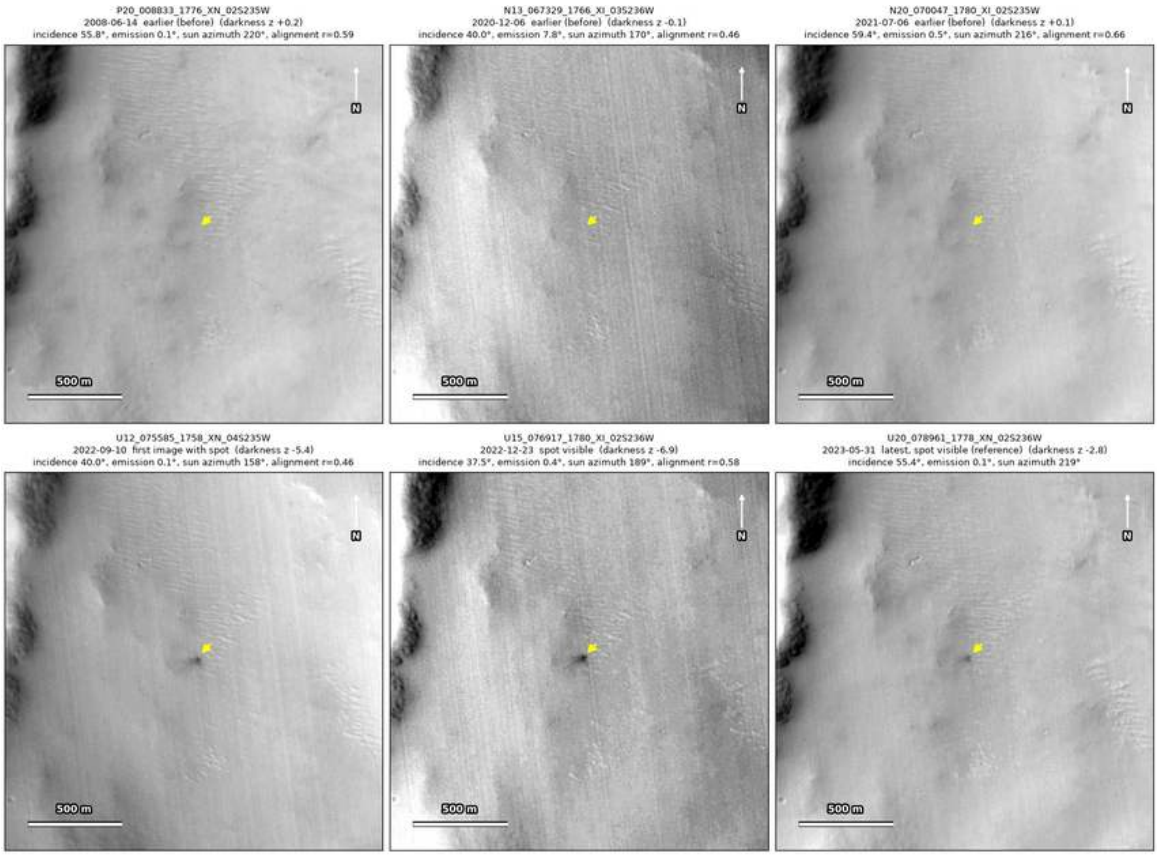

Figure 373: Candidate 159: six CTX images of the same 2×2 km patch, 2008 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 160: Knobby plains in southern Arabia Terra, 167 km south-east of Janssen crater

Location	0.8251° N, 39.7276° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 75%
Formed between	22 September 2020 and 14 August 2022
Images	2 without the spot (absent: 14 January 2011 and 22 September 2020); 2 with it (present: 14 August 2022 and 29 January 2023)
Size	Dark spot about 50 m across (34 × 66 m in the August 2022 image), in two lobes
HiRISE	none covers the spot (nearest footprint edge 53.0 km)
Nearest catalogued new impact	82 km (Daubar et al., 2022)

What changed. A compact dark spot about 50 m across, in two lobes, appeared on knobby plains in southern Arabia Terra, 167 km south-east of Janssen crater, between September 2020 and August 2022. Two CTX images from two orbits, five and a half months apart, show it.

Why it may be a new impact. Two earlier images, from 2011 and 2020, show only the knobby texture there. The 2020 one was taken under almost the same sun as the first image with the spot, and a dark feature 300 m away shows in it, so it would have shown a new spot too. A compact dark spot that appears on dusty plains and lasts is how the blast zone of a small new crater looks.

Caveats. Only two images show it, the second poorly aligned and taken at an angle. Comparing all four images picks out a slope 960 m away, whose shading changes with the sun, a little more strongly than the spot. It formed within a 23-month window; an image from 17 days before the first one with the spot could not be aligned and is not used. No image from the high-resolution HiRISE camera covers it; the nearest ends 53.0 km away.

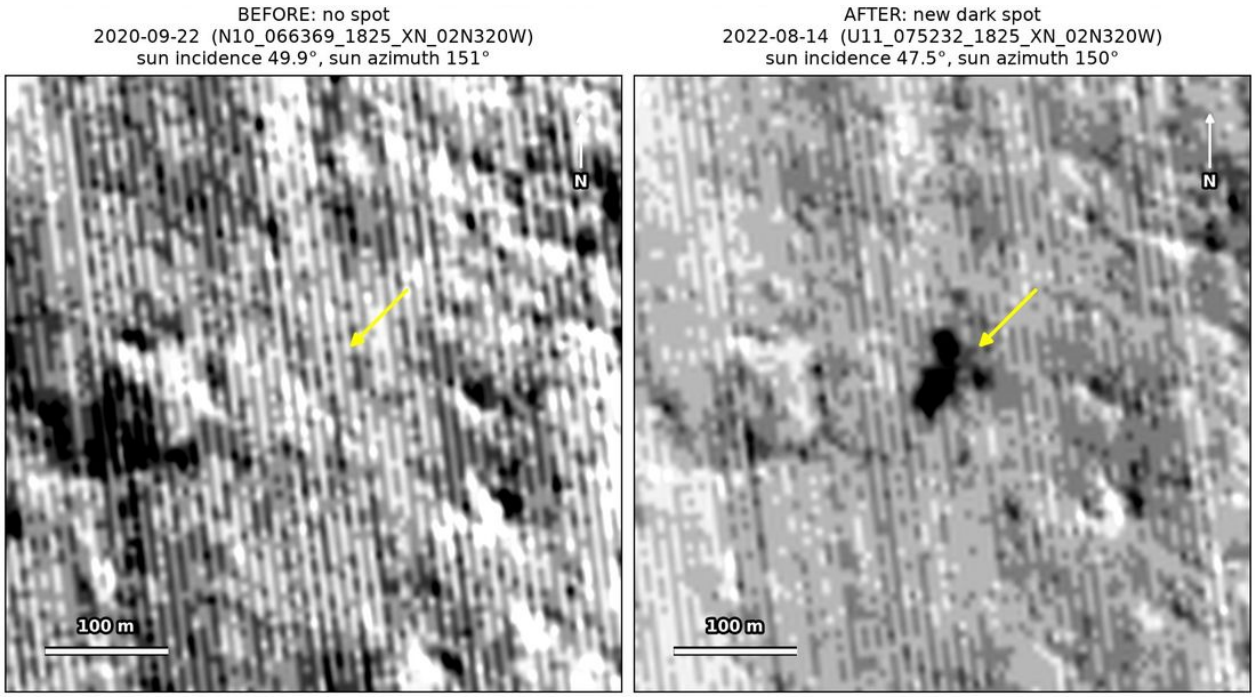

Figure 374: Candidate 160, before and after: the same 500 × 500 m of ground in 22 September 2020 (`N10_066369_1825_XN_02N320W`) and 14 August 2022 (`U11_075232_1825_XN_02N320W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

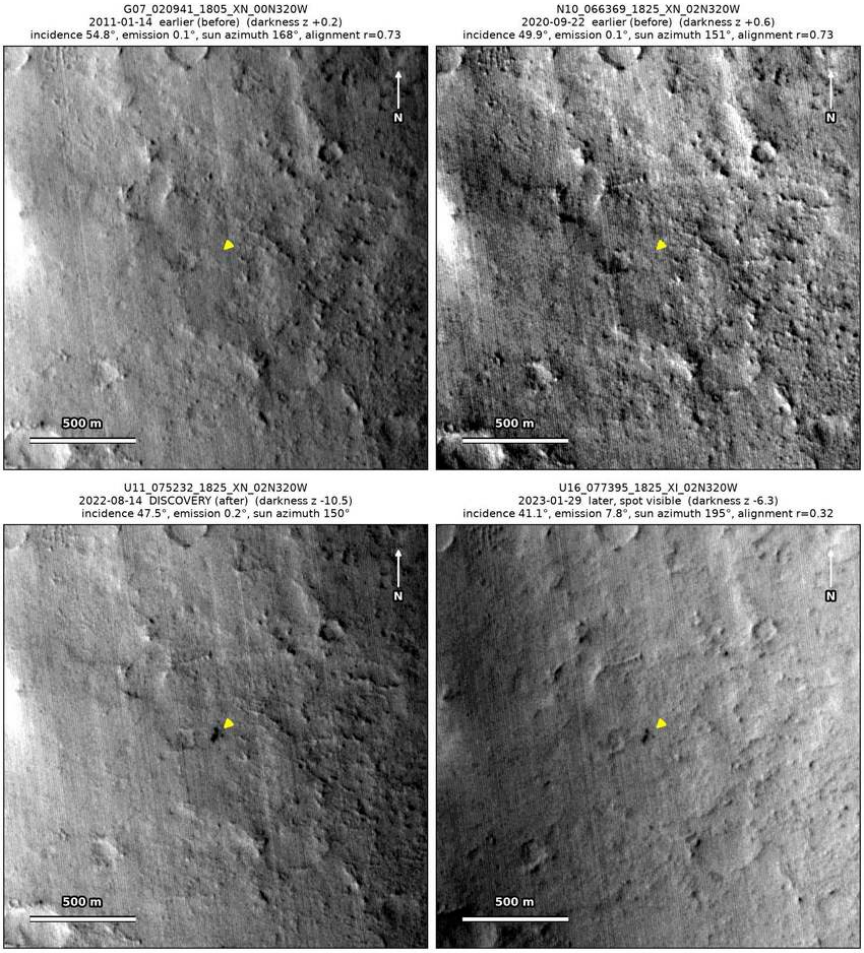

Figure 375: Candidate 160: four CTX images of the same 2×2 km patch, 2011 to 2023. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 161: Plains in Oti Fossae, 292 km south-east of Arsia Mons

Location	11.8158° S, 243.3738° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 50%
Formed between	19 February 2021 and 8 December 2022
Images	3 without the spot (absent: 31 October 2007, 12 March 2008 and 19 February 2021); 1 with it (present: 8 December 2022)
Size	Dark spot about 70 m across (90 × 60 m in the December 2022 image), with a tail to the east-south-east
HiRISE	none covers the spot (nearest footprint edge 15.3 km)
Nearest catalogued new impact	171 km (Daubar et al., 2022)

What changed. A dark spot about 70 m across, darkest at its west end with a fainter tail to the east-south-east, appeared on plains among low knobs in Oti Fossae, 292 km south-east of Arsia Mons, between February 2021 and December 2022. One CTX image shows it.

Why it may be a new impact. Three earlier images from 2007 to 2021 show only the knobby texture there. The 2007 one was taken under a sun only 3° higher than that of the December 2022 image and 10° away in direction. A dark spot appearing on dusty plains is how the blast zone of a small new crater looks.

Caveats. Only one image shows it, so there is no telling whether it lasts, and a dark feature nearby cannot settle whether the earlier images would have shown it. It is also elongated along the sun's direction and darkest at its sunward end, the shape the shadow of an active dust devil would have, although no dust column shows sunward of it. No image from the high-resolution HiRISE camera covers it; the nearest ends 15.3 km away.

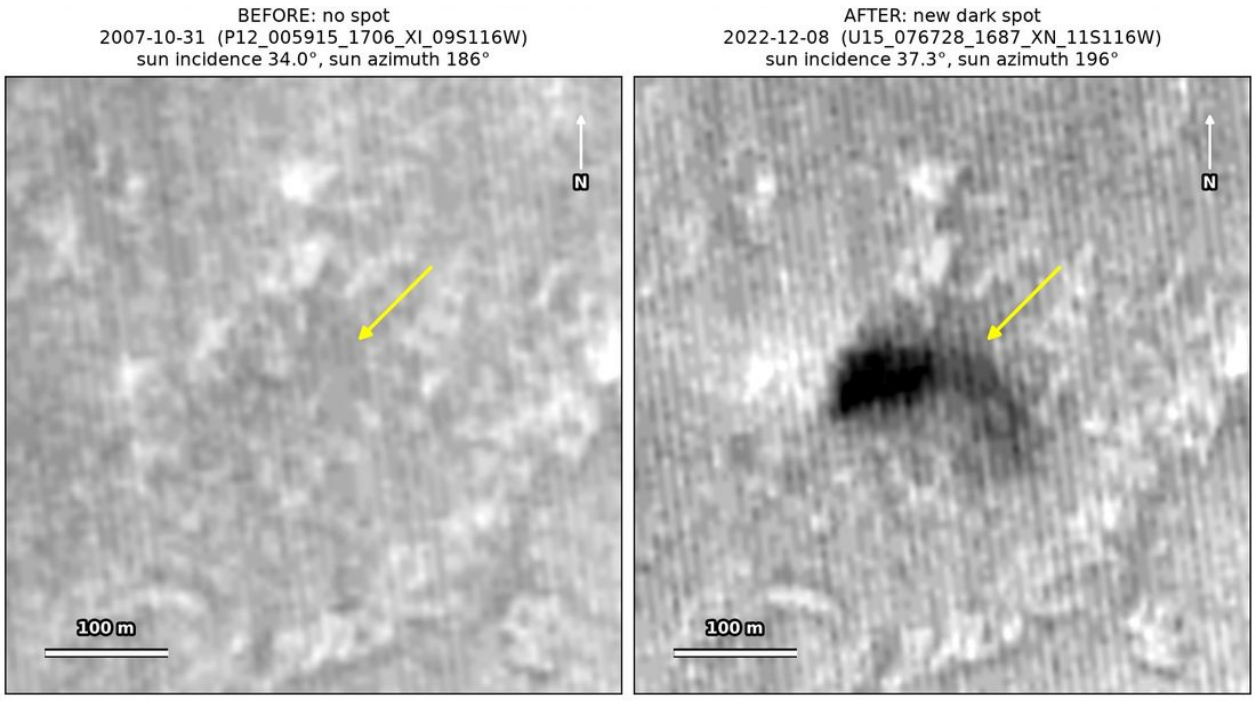

Figure 376: Candidate 161, before and after: the same 500 × 500 m of ground in 31 October 2007 (`P12_005915_1706_XI_09S116W`) and 8 December 2022 (`U15_076728_1687_XN_11S116W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

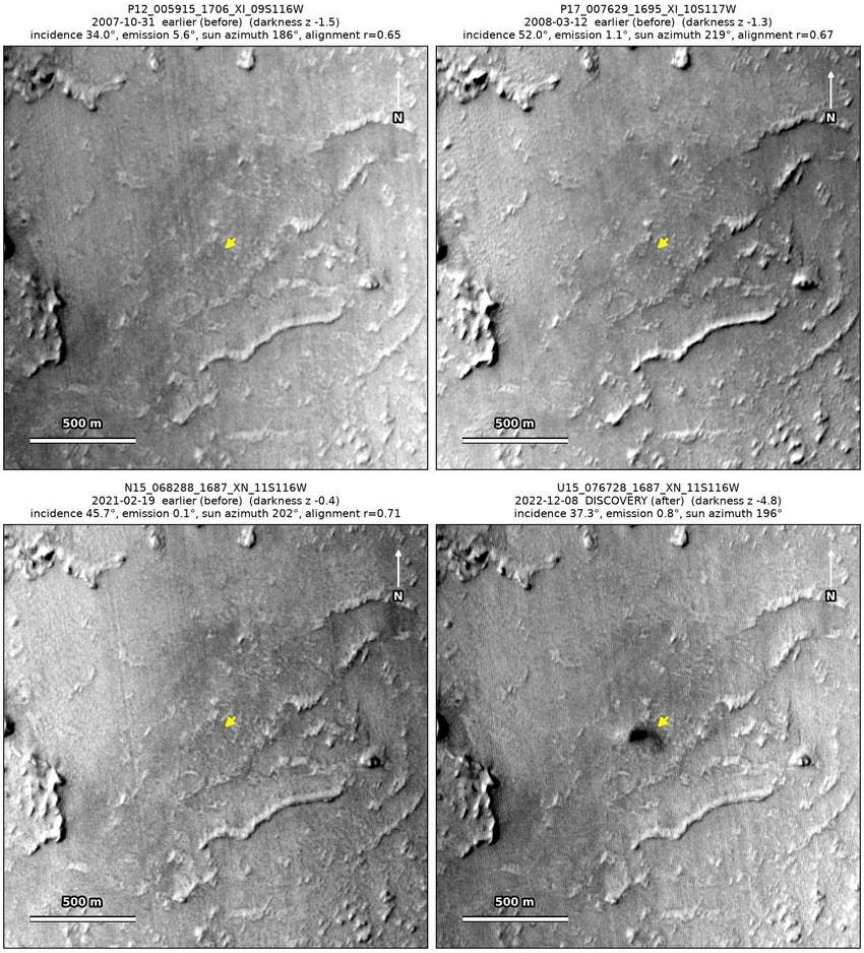

Figure 377: Candidate 161: four CTX images of the same 2×2 km patch, 2007 to 2022. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 162: Wind-streaked plains on south-eastern Amazonis Mensa, 113 km south-west of Amazonis Sulci

Location	2.9806° S, 214.5769° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 90%
Formed between	1 September 2008 and 18 January 2022
Images	1 without the spot (absent: 1 September 2008); 2 with it (present: 18 January 2022 and 12 November 2022)
Size	Dark core about 150 m across (78 × 148 m in the January 2022 image), inside a halo about 650 m across
HiRISE	none covers the spot (nearest footprint edge 41.5 km)
Nearest catalogued new impact	172 km (Daubar et al., 2022)

What changed. A dark spot appeared on wind-streaked plains on south-eastern Amazonis Mensa, 113 km south-west of Amazonis Sulci, between September 2008 and January 2022. Two CTX images from two orbits, ten months apart, show a dark core about 150 m across with short dark arms and a chain of smaller dark blobs to the east, inside a diffuse dark halo about 650 m across.

Why it may be a new impact. The only earlier image, from 2008, shows plain ground there under nearly the same sun as the first image with the spot. A dark core with arms and a halo appearing on dusty plains is how the blast zone of a new impact looks.

Caveats. There is only one earlier image, so the time window in which the spot formed is thirteen years long, and no comparable dark feature lies nearby to show whether that image would have caught a new spot. The second image with the spot is poorly aligned, although the spot is plain in it. No image from the high-resolution HiRISE camera covers it; the nearest ends 41.5 km away.

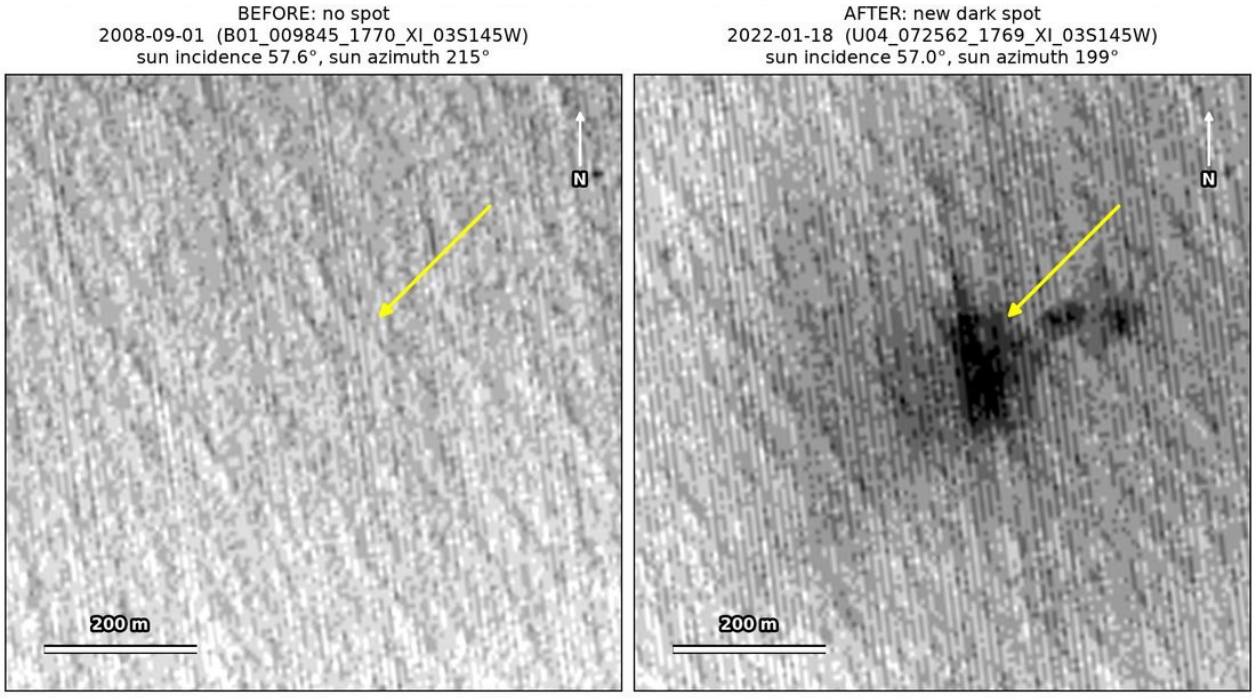

Figure 378: Candidate 162, before and after: the same 800 × 800 m of ground in 1 September 2008 (`B01_009845_1770_XI_03S145W`) and 18 January 2022 (`U04_072562_1769_XI_03S145W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

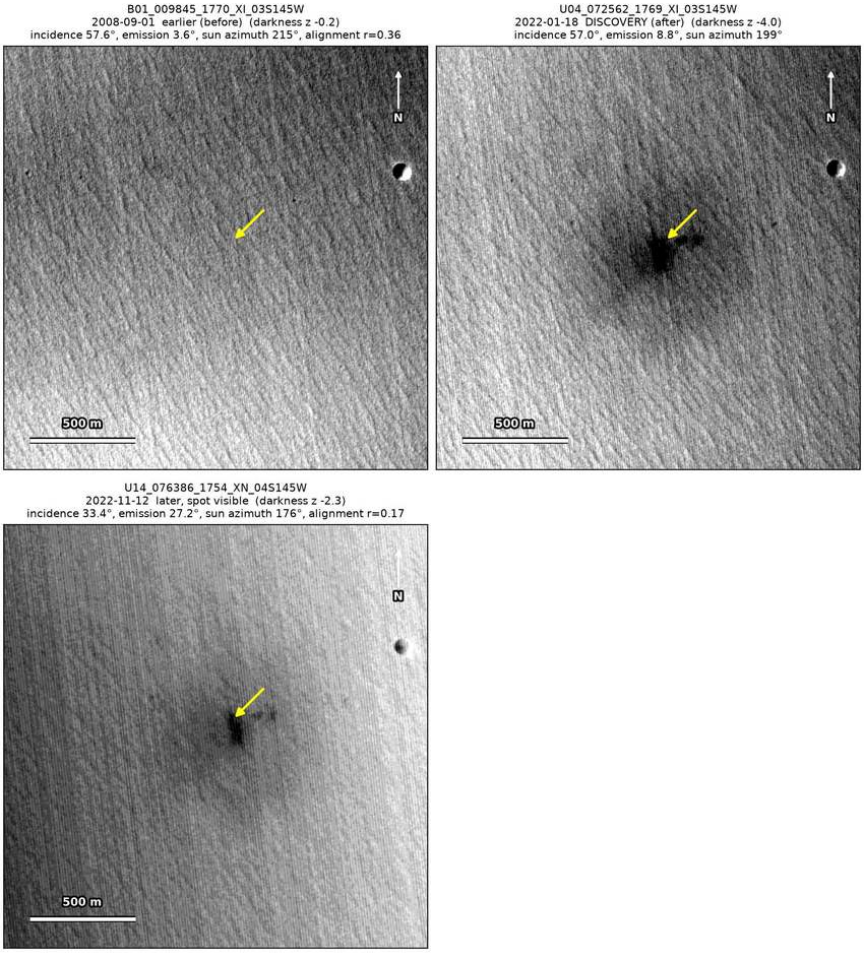

Figure 379: Candidate 162: three CTX images of the same 2×2 km patch, 2008 to 2022. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 163: Edge of knobby ground in the Medusae Fossae Formation, 116 km west of Medusae Sulci

Location	5.5014° S, 198.3926° E (planetocentric)
Assessment	promising candidate; probability of a new impact about 60%
Formed between	5 February 2008 and 2 March 2023
Images	1 without the spot (absent: 5 February 2008); 4 with it (present: 2 March 2023, 30 June 2024, 8 August 2024 and 30 December 2024)
Size	Diffuse dark patch about 250 m across, with a faint core about 30 m across (30 × 36 m in the August 2024 image)
HiRISE	none covers the spot (nearest footprint edge 9.0 km)
Nearest catalogued new impact	86 km (Daubar et al., 2022)

What changed. A diffuse dark patch about 250 m across, with a faint darker core, appeared at the edge of a field of knobs in the Medusae Fossae Formation, 116 km west of Medusae Sulci, between February 2008 and March 2023. Four CTX images from four orbits over 22 months show it.

Why it may be a new impact. The only usable earlier image, from 2008, shows plain ground there under the same sun as the latest image. The patch is a little fainter in December 2024 than in March 2023, under nearly the same sun, as a fading blast zone would be.

Caveats. The patch is irregular, with only a faint core, at the edge of a knob field where the wind can strip dust, and broad dark wind bands formed 2 km away between 2023 and 2024, so dark ground laid bare by the wind is the main alternative. Only one earlier image is usable, and no dark feature nearby can test it. No image from the high-resolution HiRISE camera covers it; the nearest ends 9.0 km away.

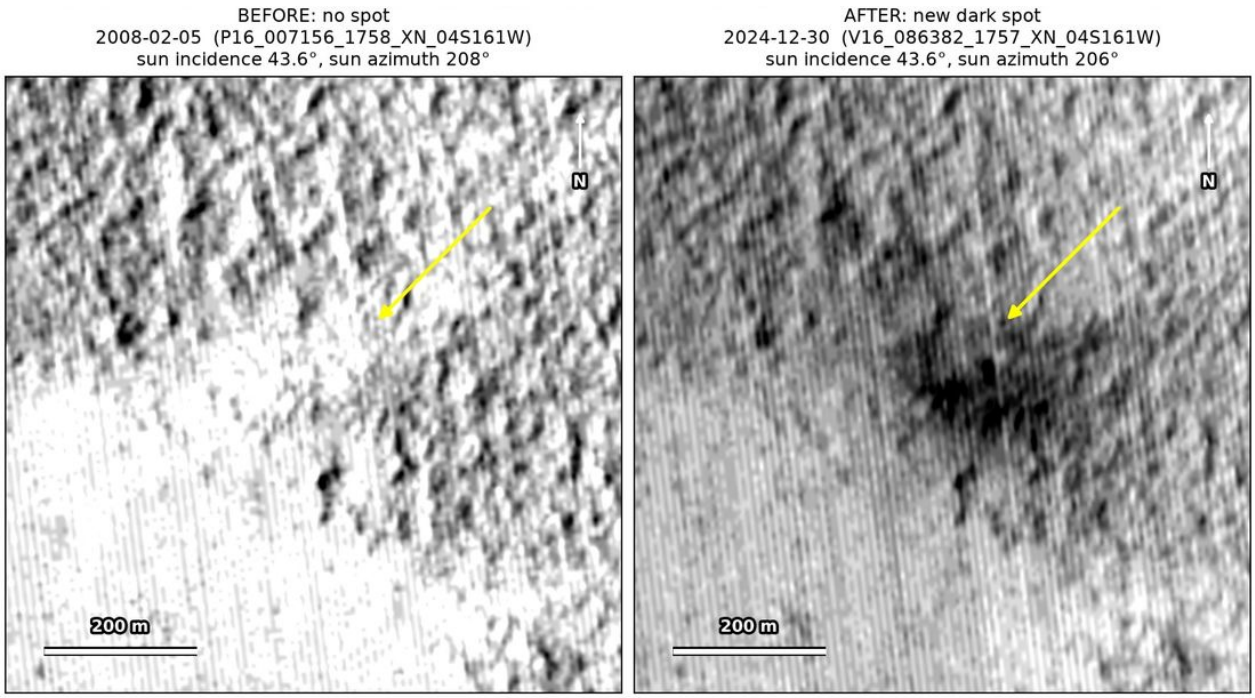

Figure 380: Candidate 163, before and after: the same 800 × 800 m of ground in 5 February 2008 (`P16_007156_1758_XN_04S161W`) and 30 December 2024 (`V16_086382_1757_XN_04S161W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

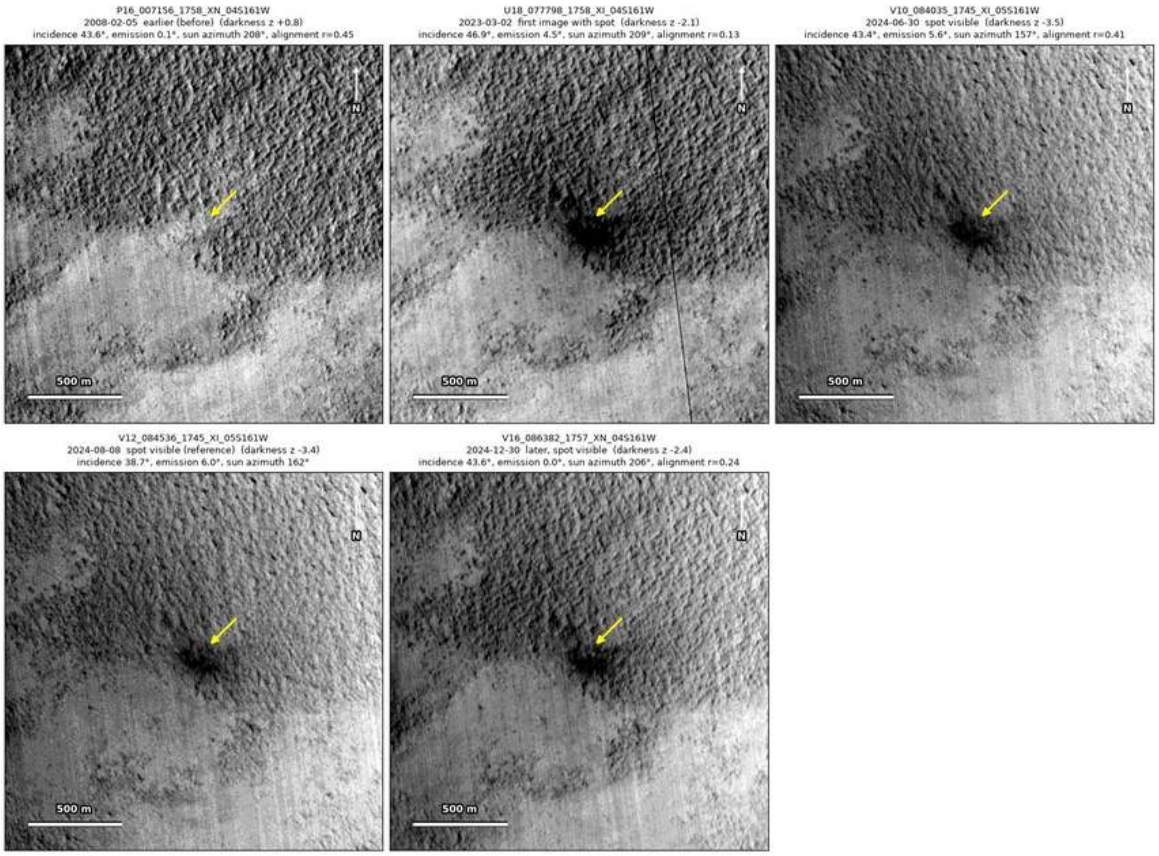

Figure 381: Candidate 163: five CTX images of the same 2×2 km patch, 2008 to 2024. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 164: Plains east of Licus Vallis, 93 km east of its centre

Location	2.5584° S, 127.8512° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 75%
Formed between	13 February 2021 and 21 May 2023
Images	5 without the spot (absent: 20 April 2008, 1 May 2008, 6 August 2020, 18 December 2020 and 13 February 2021); 2 with it (present: 21 May 2023 and 16 October 2025)
Size	Dark spot about 25 m across (24 × 32 m in the May 2023 image)
HiRISE	none covers the spot (nearest footprint edge 6.3 km)
Nearest catalogued new impact	303 km (Daubar et al., 2022)

What changed. A compact dark spot about 25 m across appeared on cratered plains east of Licus Vallis between February 2021 and May 2023. Two CTX images from two orbits, 29 months apart, show it.

Why it may be a new impact. Five earlier images from 2008 to 2021 show plain ground there. One of them was taken under the same sun as the first image with the spot, and a dark feature 220 m away shows in it, so it would have shown a new spot too. A compact dark spot that appears on dusty plains and lasts is how the blast zone of a small new crater looks.

Caveats. The spot is small, four to six pixels across, so a comparison of the images does not single it out, and it is fainter in the 2025 image. No image from the high-resolution HiRISE camera covers it; the nearest ends 6.3 km away.

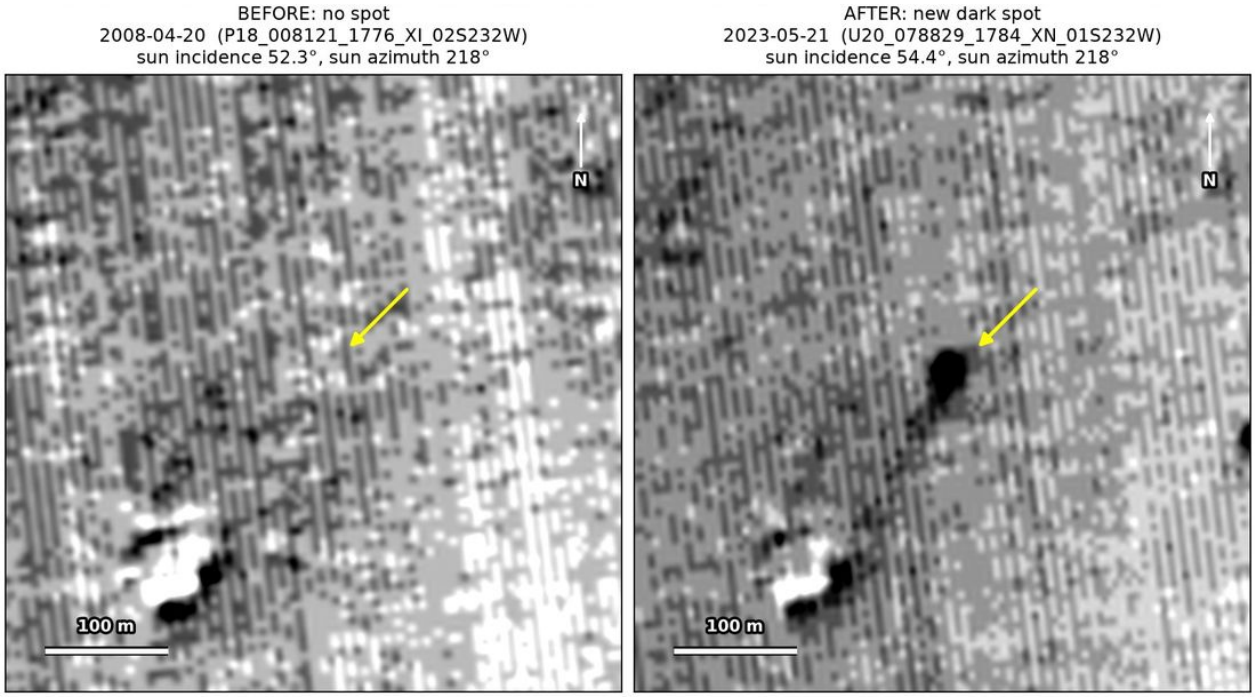

Figure 382: Candidate 164, before and after: the same 500 × 500 m of ground in 20 April 2008 (`P18_008121_1776_XI_02S232W`) and 21 May 2023 (`U20_078829_1784_XN_01S232W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

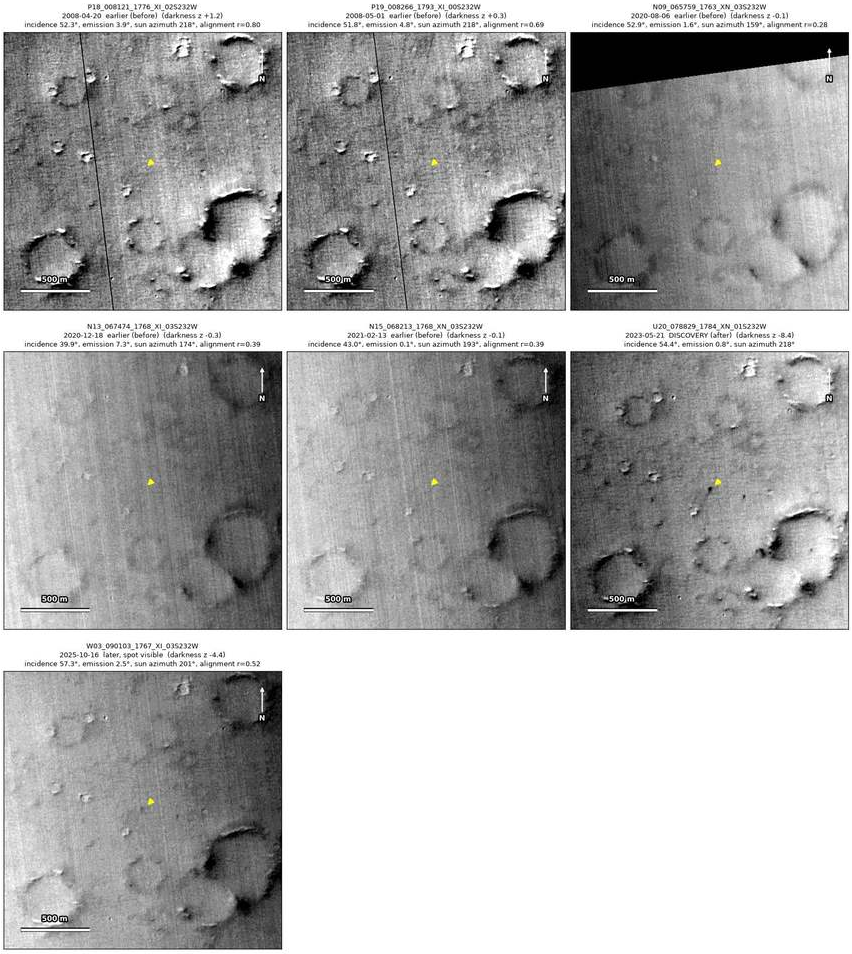

Figure 383: Candidate 164: seven CTX images of the same 2×2 km patch, 2008 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 165: Wind-ridged plains in Daedalia Planum, 299 km east of Arimanes Rupes

Location	10.8609° S, 217.3246° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	20 December 2007 and 6 April 2024
Images	2 without the spot (absent: 16 April 2007 and 20 December 2007); 3 with it (present: 6 April 2024, 2 November 2024 and 8 January 2025)
Size	Dark spot about 35 m across (52 × 38 m in the April 2024 image)
HiRISE	none covers the spot (nearest footprint edge 10.3 km)
Nearest catalogued new impact	83 km (Daubar et al., 2022)

What changed. A compact dark spot about 35 m across appeared on wind-ridged plains in Daedalia Planum between December 2007 and April 2024. Three CTX images from three orbits, taken over nine months, show it.

Why it may be a new impact. The two earlier images, from 2007, show plain ground there, one of them under the same sun as the first image with the spot. A compact dark spot that appears on dusty plains and lasts is how the blast zone of a small new crater looks.

Caveats. The last two images are blurred or noisy, and their measured alignment is poor; no dark feature nearby can show whether the earlier images would have caught a new spot; and the time window in which the spot formed is sixteen years long. No image from the high-resolution HiRISE camera covers it; the nearest ends 10.3 km away.

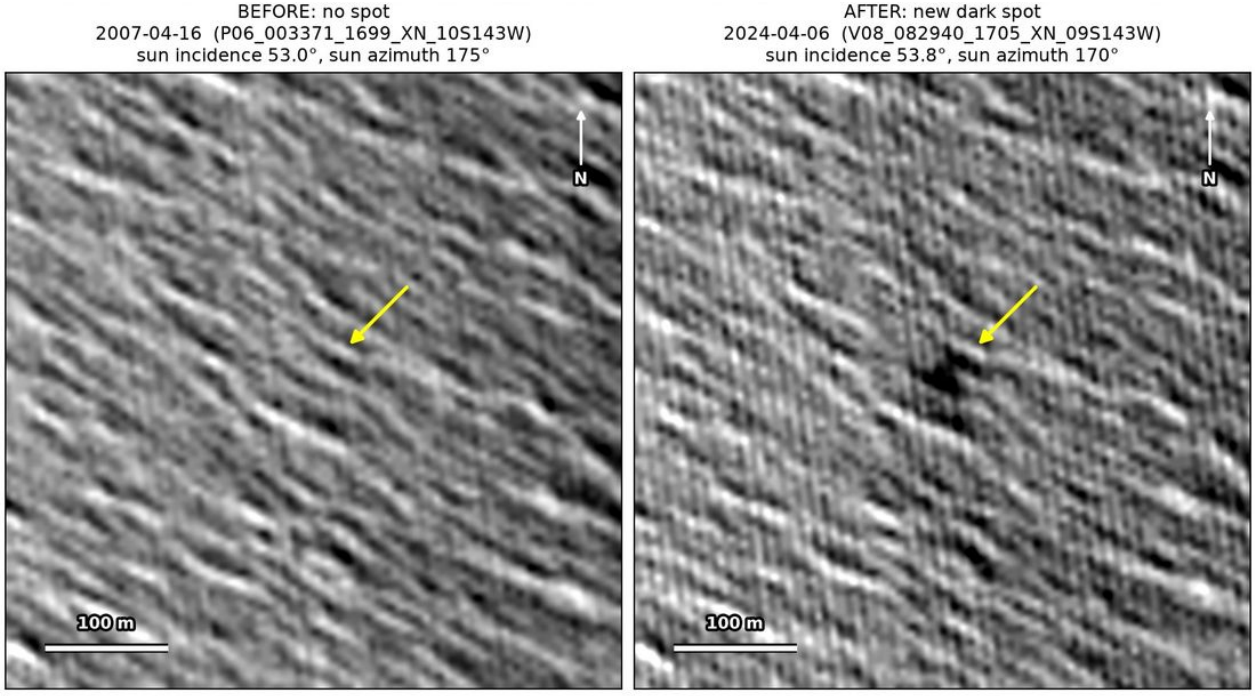

Figure 384: Candidate 165, before and after: the same 500 × 500 m of ground in 16 April 2007 (`P06_003371_1699_XN_10S143W`) and 6 April 2024 (`V08_082940_1705_XN_09S143W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

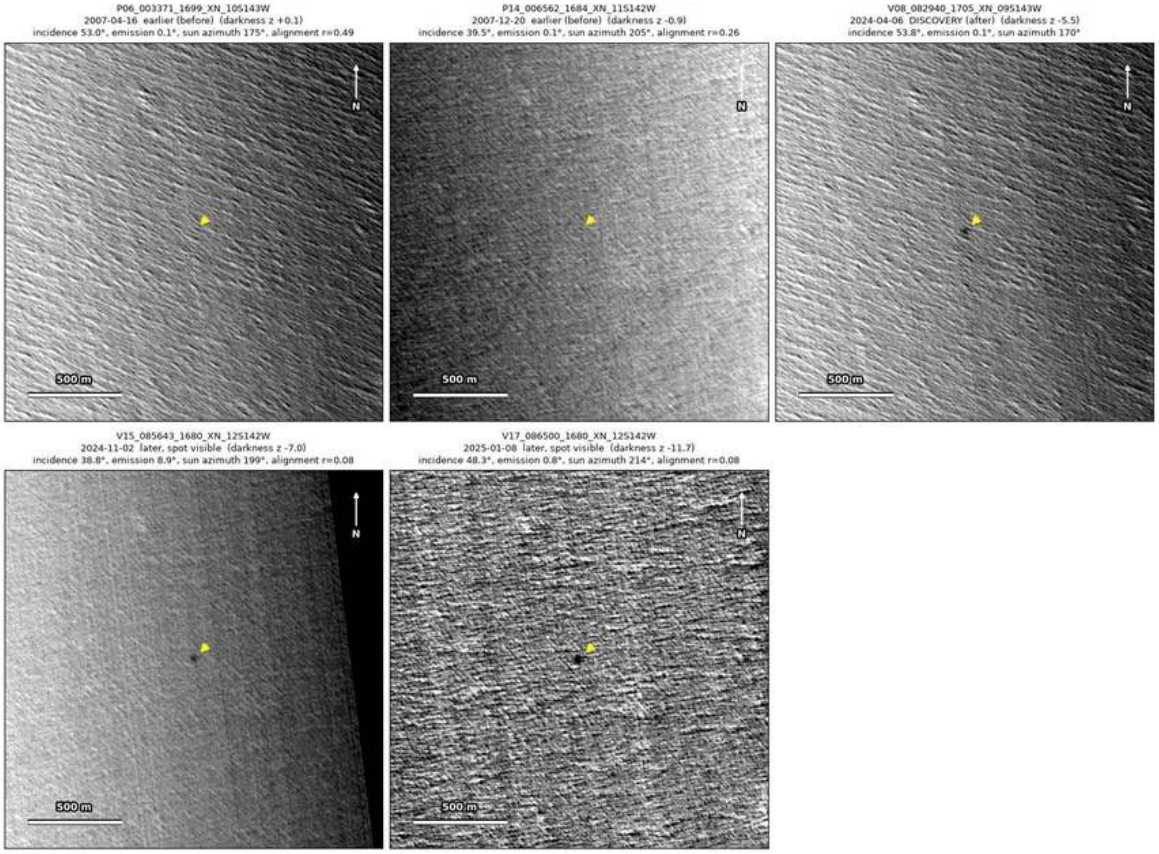

Figure 385: Candidate 165: five CTX images of the same 2×2 km patch, 2007 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

Candidate 166: Lava plains 350 km south-east of Fesenkov crater, 412 km north-east of Tharsis Tholus

Location	16.1984° N, 275.8236° E (planetocentric)
Assessment	strong candidate; probability of a new impact about 85%
Formed between	30 May 2020 and 12 June 2022
Images	2 without the spot (absent: 5 March 2019 and 30 May 2020); 2 with it (present: 12 June 2022 and 4 January 2025)
Size	Dark spot about 40 m across (46 × 38 m in the June 2022 image)
HiRISE	none covers the spot (nearest footprint edge 19.8 km)
Nearest catalogued new impact	148 km (Daubar et al., 2022)

What changed. A compact dark spot about 40 m across appeared on lava plains south-east of Fesenkov crater between May 2020 and June 2022. Two CTX images from two orbits, 31 months apart, show it.

Why it may be a new impact. The two earlier images, from 2019 and 2020, show plain ground there, one of them under nearly the same sun as the first image with the spot. A dark feature 200 m away shows in both, so they would have shown a new spot too. A compact dark spot that appears on dusty plains and lasts is how the blast zone of a small new crater looks.

Caveats. Only two images show it. When the aligned May 2020 image is subtracted from the June 2022 one, both taken under nearly the same sun, a place near a small crater 790 m away, whose shading varies with the direction of the sun, shows about as much change as the spot. No image from the high-resolution HiRISE camera covers it; the nearest ends 19.8 km away.

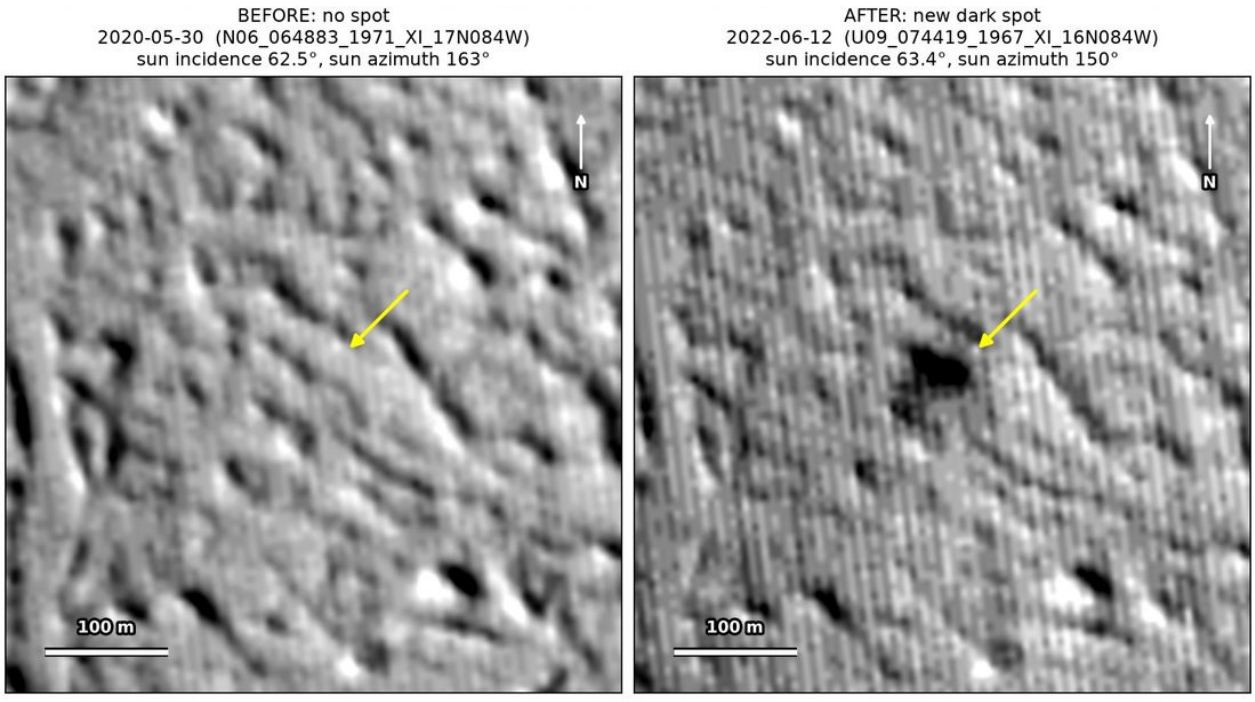

Figure 386: Candidate 166, before and after: the same 500 × 500 m of ground in 30 May 2020 (`N06_064883_1971_XI_17N084W`) and 12 June 2022 (`U09_074419_1967_XI_16N084W`), under a similar sun; north up, 4 m per pixel, brightness matched on the surroundings and one grey stretch. The arrow points at the spot, which is absent in the earlier image and present in the later one. Sun incidence and azimuth (CTX label convention, Section 3.3) are printed above each panel; the scale bar is 100 m. CTX: NASA/JPL-Caltech/MSSS.

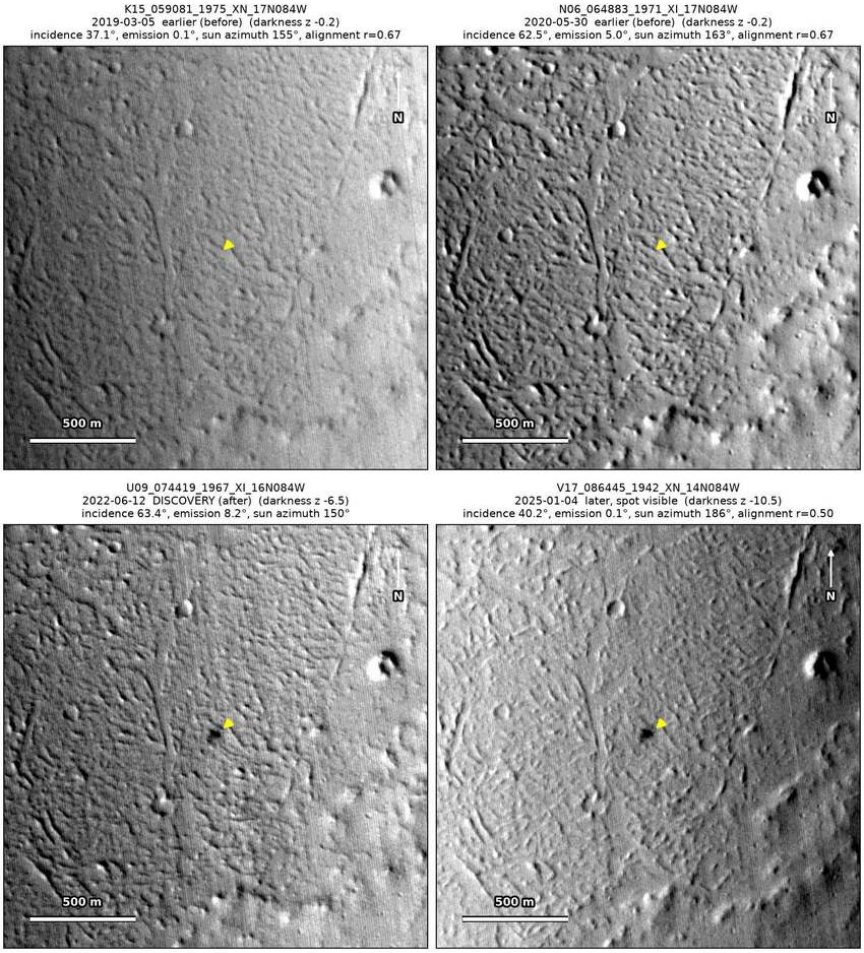

Figure 387: Candidate 166: four CTX images of the same 2×2 km patch, 2019 to 2025. Each panel is the same ground, north up, 4 m per pixel, with brightness and contrast matched on the surroundings and one grey stretch; the yellow arrow marks the spot. Above each panel: product identifier, date and role of the image, the darkness score z of the spot where measured, and the sun incidence, emission angle, sun azimuth and alignment score r (Section 3.3). Scale bar 500 m. CTX: NASA/JPL-Caltech/MSSS.

6 Discussion

6.1 Why these are candidates and not detections

For a new impact to be accepted into the catalogue of Daubar et al. (2022), a HiRISE image must resolve one or more craters at the centre of a blast zone whose formation is bracketed by earlier and later images. None of our 166 sites has been imaged by HiRISE on purpose, and only at candidates 21 and 57 does a chance HiRISE image show pits of the expected size at the centre of the dark zone. For the others the case rests on CTX images alone, in which a crater a few metres across cannot be resolved. The dark spots are consistent in size, shape, contrast and behaviour with the blast zones of known new impacts, but other processes can make similar spots:

- a small slope streak whose head is compact at CTX resolution, especially on the slopes of Arabia Terra, Tharsis and the Medusae Fossae Formation (for example candidates 25, 28, 131 and 134);
- wind stripping a thin layer of bright dust from a patch of ground, exposing darker material, which is the main alternative for diffuse or irregular spots and on soft, wind-eroded terrain (for example candidates 3, 20, 73, 83, 85, 109 and 163);
- a dust devil that stalled or was caught in the act (candidate 92, and possibly 161);
- a defect of a single image, which cannot be excluded for the 69 candidates seen in only one image, although their spots are smooth, span many lines and samples and lack the patterns of known defects;
- dark material gathering where a faint dark patch already lay (candidate 74).

Some blast zones may also have been made by meteoroids that broke up completely in the atmosphere, leaving no crater that HiRISE could resolve (Daubar et al., 2022). Even a candidate that is a genuine impact would therefore not necessarily enter a catalogue that requires a resolved crater.

6.2 Is each site really unrecorded?

We could search only public records. The CTX team examines new images in dusty regions for new dark spots (Daubar et al., 2022, 2013), and its working list is not public; HiRISE images released after August 2026 were not yet in the archive index; and the catalogue of Bickel et al. (2025) could not be retrieved. Sites first seen years ago (for example candidates 155, from 2012, 126 and 25, from 2016 and 2017, or 42, from 2018) and sites imaged by HiRISE (4, 21, 57, 69, 85 and 94) are the most likely to be known to the camera teams already. Several discovery images were taken together with HiRISE images of other targets nearby, and the CTX images of candidate 36 were taken alongside HiRISE monitoring of an older impact 9.7 km away. With roughly 240 to 400 impacts a year large enough to leave a crater of 4 m or more (Bickel et al., 2025; Daubar et al., 2013), the events themselves are not rare: what is new here is that these particular sites are not in the public record. The probabilities we give refer to whether each spot is a new impact, not to whether it is unknown.

6.3 How reliable are the probabilities?

The probabilities are careful judgements, not calibrated frequencies. Their sum, 116, would be the expected number of genuine new impacts among the 166 if the judgements were unbiased, but there is as yet no independent sample of confirmed outcomes with which to calibrate them. The independent re-examination of 38 sites agreed with them to within about 5 percentage points on average, but it disagreed strongly about candidate 92 and moderately about candidate 73, and agreement between two assessments of the same images is not a test against ground truth. Targeted HiRISE images of even a modest number of candidates, chosen across the range of probabilities, would show whether the estimates are too high or too low.

6.4 Completeness and biases

The catalogue is far from complete. The visual review proceeded from the darkest detections to the fainter ones and stopped with 259 fainter detections not yet inspected, so faint new spots are certainly missed. The method recovered only about 36% of the catalogued impacts inside its image pairs. The candidates are concentrated on dusty terrain and where CTX has imaged the same ground repeatedly, and the long formation windows of many of them reflect sparse earlier coverage rather than the age of the events. Several biases of this kind affect every search for new impacts (Bickel et al., 2025; Daubar et al., 2013; Wagstaff et al., 2022). For these reasons the catalogue should not be used to estimate an impact rate without a careful accounting of the area and time searched.

6.5 How the candidates could be confirmed

The decisive test is a targeted HiRISE image of each site, at 0.25–0.5 m per pixel. A fresh crater, or a cluster of craters, at the centre of the dark zone would confirm an impact; its diameter, together with the formation window, would add a dated crater to the record of the present impact flux. A HiRISE image showing a slope streak, a dust devil track or wind-sculpted ground without a crater would reject a candidate, although a missing crater cannot rule out an airburst. Because the positions given here can be in error by a few hundred metres (and by up to about 1.5 km for candidates 23 and 62), the re-measured positions of Table 3 should be used where they exist, and the image footprint should be large enough to include the alternatives. For candidates seen in only one CTX image, a second CTX image would already test whether the spot is persistent; images from other cameras, such as the Colour and Stereo Surface Imaging System on the ExoMars Trace Gas Orbiter, which we did not search, may already exist.

Natural priorities for follow-up are candidates with high probability and short formation windows, which add the most to the record of when impacts happen: for example candidates 40 (110 days, 90%), 158 (84 days, 90%), 19 (165 days, 85%), 140 (six months, 90%), 137 (eleven months, 90%), 68 (sixteen months, 90%) and 136 (22 months, 90%). Candidates in the InSight area whose formation windows overlap the lander's mission, from November 2018 to December 2022, could be compared with the catalogue of seismic events, as Bickel et al. (2025) did for their sites; among them candidate 159 formed between July 2021 and September 2022, entirely within the mission, 13.6° from the lander. Clusters such as candidates 59, 118 and 133 would add to the sample used to study the breakup of meteoroids in the atmosphere (Daubar et al., 2019).

6.6 Honest uncertainty

Every site in this catalogue is an unconfirmed candidate. The changes on the ground are well established for most of them, and in many cases beyond reasonable doubt, but whether each change is a new impact is a matter of probability. On the evidence collected here, most are probably ordinary small new impacts that are not yet in any public catalogue; some are probably not impacts at all. We have tried to state for each site what the images show, how the mundane alternatives were tested and what remains uncertain, so that each case can be checked against the original data and settled by new images.

7 Data availability

The public pages that present the 166 candidates, with their photographs, plain-language explanations and probabilities, are at `https://paatai.github.io/vibe-projects/mars/`; the candidates are numbered 1 to 166 as in this paper. The table behind those pages, with the evidence and alternatives for each candidate and the records of the 339 changes that were rejected or set aside, is available there as `candidates.csv`. The LaTeX source of this paper, its figures, the data tables from which its tables and statistics are generated (including the image list of every candidate) and the scripts that produce them are distributed with the paper.

All images used are public. CTX and HiRISE data are archived in NASA's Planetary Data System (`https://pds-imaging.jpl.nasa.gov` and `https://hirise-pds.lpl.arizona.edu`); map-projected CTX

images are available from Arizona State University (`https://viewer.mars.asu.edu`); footprints and MOC images from the PDS Geosciences Node's Orbital Data Explorer (`https://ode.rsl.wustl.edu/mars`); and MOLA topography from the PDS Geosciences Node. Image credits: CTX, NASA/JPL-Caltech/Malin Space Science Systems; HiRISE, NASA/JPL-Caltech/University of Arizona; MOC, NASA/JPL/Malin Space Science Systems. Contact: ivanishvili.paata@gmail.com.

8 Disclosure of AI assistance

The search described in this paper, the analysis of the images, the figures and much of the text were carried out with Grok Bot coordinating Cursor cloud agents, under the direction of the author. The author set the aims and rules of the search and reviewed and approved its results and the public descriptions of the candidates. The independent re-examination of 38 sites was carried out in the same way, by a separate agent instructed to be sceptical and not to rely on the original figures. All 166 sites are unconfirmed candidates.

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